Supplementary Material for "PRISM-XAI: Privacy-Accounted Semantic Explanation Scheduling for Smart-Grid Edge Intelligence"

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***Abstract*—This supplement provides the complete technical and experimental details that support the main paper. It expands the packet mechanism, privacy accountant, scheduler, evaluation protocol, matched-service analysis, sensitivity tests, and explanation-fidelity results. It also reports the complete nine-point public-score service–communication sweep, including every generated table and vector figure at aggregate, OPSD, and UCI scope. Labels are descriptive and identify scheduler score, release cap, and dataset scope.**

I. System Architecture and Notation

Figure 1 shows the division between trusted household processing and communication scheduling. Raw features and unperturbed attributions remain at the endpoint. The scheduler receives public operating state and metadata for privacy-admissible packet modes, but not the raw attribution residual used to construct those packets.

II. MAP-XAI Packet Construction and Accounting

The endpoint computes the attribution change relative to its last successfully delivered explanation and clips it as

$$\overline{\Delta\vec{\phi}}_{i,t} = \frac{\Delta\vec{\phi}_{i,t}}{\max\{1, \|\Delta\vec{\phi}_{i,t}\|_2/C_i\}}. \tag{1}$$

It then applies a Gaussian mechanism,

$$\widetilde{\Delta\vec{\phi}}_{i,t} = \overline{\Delta\vec{\phi}}_{i,t} + \vec{n}_{i,t}, \qquad \vec{n}_{i,t} \sim \mathcal{N}(0, \sigma_{i,t}^2 \Delta_i^2 I_d), \tag{2}$$

TABLE I
Core notation.

Symbol	Meaning
$\vec{\phi}_{i,t}$	Attribution vector for endpoint i in slot t
$\Delta\vec{\phi}_{i,t}$	Change from the last successfully delivered explanation
C_i	Endpoint-specific ℓ_2 clipping radius
$m = (k, q, \sigma)$	Packet mode: retained coordinates, quantization bits, noise multiplier
$S_i(m)$	Encoded packet length in bits
$A_{i,t}$	Age of Explanation (AoX)
$G_{i,t}(m)$	Public estimate of explanation-loss reduction
$\Gamma_{i,t}(m)$	Gain per transmitted bit, $G_{i,t}(m)/S_i(m)$
$a_{i,t}$	Indicator that a private packet is exposed outside the trusted endpoint

before Top-k selection and quantization. Because these two compression steps operate on the perturbed vector, they are post-processing. For d attribution coordinates, header size S_0, and q magnitude bits, the packet length is

$$S_i(m) = S_0 + k\big(\lceil \log_2 d \rceil + q + 1\big), \tag{3}$$

where one additional bit stores the sign.

At RDP order α, an admitted Gaussian release costs $\rho_{i,t}(\alpha) = \alpha/(2\sigma_{i,t}^2)$. Composition and conversion at the end of the protected horizon give

$$\epsilon_i(\delta_i) = \min_{\alpha} \left[\sum_t a_{i,t}\rho_{i,t}(\alpha) + \frac{\log(1/\delta_i)}{\alpha - 1} \right]. \tag{4}$$

The accountant rejects a candidate whose projected expenditure exceeds the declared horizon budget. A rejected candidate and an explicit abstention add no content release. An admitted packet is charged when exposed outside the trusted endpoint, even if its later radio delivery fails.

III. PRISM-XAI Scheduling Procedure

For each slot, MAP-XAI first enumerates packet modes that fit the remaining privacy budget. PRISM-XAI filters these modes using resource-block and deadline constraints, scores the survivors from public state, and selects at most one mode per endpoint. The multiple-choice knapsack core has complexity $O(NMB)$ for N endpoints, M candidate modes, and integer radio budget B.

IV. Formal Results and Proofs

Proposition 1 (Sensitivity of the clipped household contribution)**.** *Let $g(\vec{x}) = \Delta\vec{\phi}(\vec{x})/\max\{1, \|\Delta\vec{\phi}(\vec{x})\|_2/C_i\}$ and represent an absent household by the zero contribution. Under add/remove contribution adjacency, g has ℓ_2 sensitivity at most C_i.*

Proof. The clipping map projects every contribution into the closed ℓ_2 ball $\mathcal{B}_2(C_i) = \{v : \|v\|_2 \le C_i\}$. Under the stated add/remove adjacency, one dataset contains endpoint i's clipped contribution $g(\vec{x})$ and its neighbor contains the zero contribution. Therefore

$$\|g(\vec{x}) - g(\varnothing)\|_2 = \|g(\vec{x}) - 0\|_2 \le C_i. \tag{5}$$

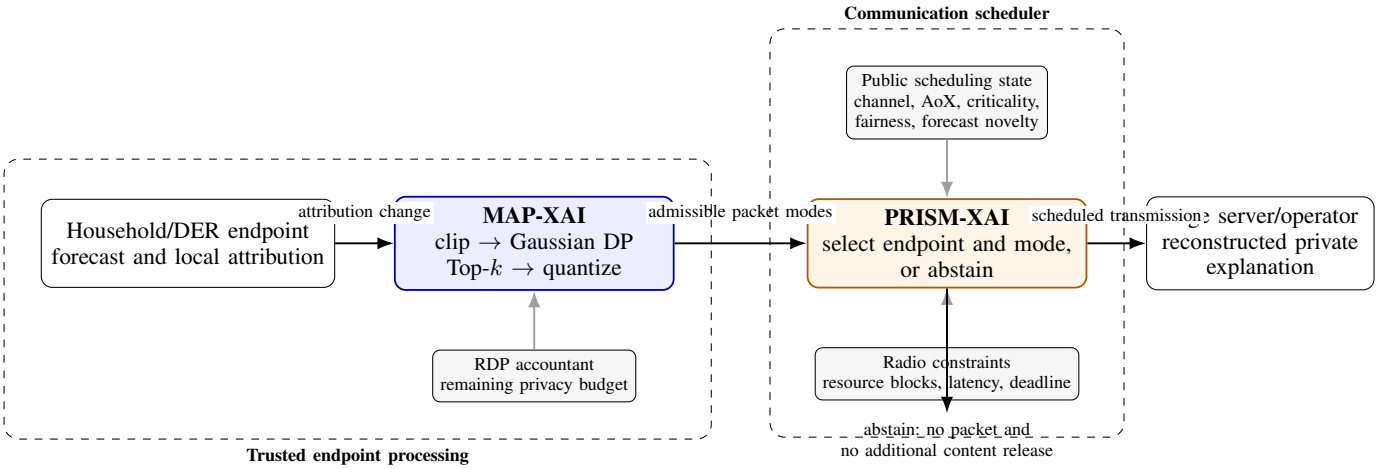


Fig. 1. Two-layer communication architecture. MAP-XAI constructs privacy-accounted packet modes locally. PRISM-XAI assigns a feasible mode to a shared uplink opportunity using declared public signals.

Taking the supremum over adjacent inputs gives sensitivity at most C_i. This is not a replacement-adjacency bound: if one household record could be replaced by an arbitrary second record, the triangle inequality would give the looser bound $2C_i$. The theorem below consequently uses the add/remove model stated in the main paper. □

Theorem 1 (Packet-content privacy over the protected horizon)**.** *Fix endpoint i and assume add/remove contribution adjacency, an ℓ_2 sensitivity bound Δ_i for every conditionally generated clipped delta, and Gaussian noise with standard deviation $\sigma_{i,t}\Delta_i$ at each exposed release. If the accountant admits only sequences satisfying*

$$\min_{\alpha>1}\left\{\sum_t a_{i,t}\frac{\alpha}{2\sigma_{i,t}^2}+\frac{\log(1/\delta_i)}{\alpha-1}\right\}\le\epsilon_i, \tag{6}$$

then the released explanation-packet contents for endpoint i satisfy (ϵ_i,δ_i)-differential privacy over the protected horizon. The statement remains valid when future noise levels and release decisions adapt to public state and previously released private outputs.

Proof. Condition on an arbitrary realized public-state history and on all previously released private outputs. By the sensitivity assumption, the next clipped delta differs by at most Δ_i between adjacent inputs. The Gaussian mechanism with noise standard deviation $\sigma_{i,t}\Delta_i$ is therefore $(\alpha,\rho_{i,t}(\alpha))$-RDP with $\rho_{i,t}(\alpha)=\alpha/(2\sigma_{i,t}^2)$. Adaptive composition of RDP mechanisms adds their conditional RDP parameters, so the complete sequence of exposed noisy deltas is $(\alpha,\sum_t a_{i,t}\rho_{i,t}(\alpha))$-RDP for every tracked order $\alpha>1$. Converting RDP to approximate DP gives

$$\epsilon_i(\delta_i)=\min_{\alpha>1}\left\{\sum_t a_{i,t}\rho_{i,t}(\alpha)+\frac{\log(1/\delta_i)}{\alpha-1}\right\}. \tag{7}$$

The admission rule ensures this value never exceeds ϵ_i. Finally, packet construction after perturbation—including Top-k selection, quantization, encoding, and reconstruction—is post-processing and cannot increase privacy loss. Thus the released packet contents satisfy the claimed horizon guarantee. The proof concerns packet contents; any additional data-dependent observable must be included in the mechanism before the same claim can be extended to it. □

Proposition 2 (Privacy preservation of packet formation and public-state scheduling)**.** *Assume a noisy attribution delta has already been released by a mechanism with RDP parameter $\rho(\alpha)$. Any randomized Top-k selection, quantization, encoding, reconstruction, or scheduling transformation whose additional inputs are public state and previously released private outputs remains $(\alpha,\rho(\alpha))$-RDP with respect to the protected contribution.*

Proof. Let P and Q be the distributions of the noisy delta under adjacent inputs, and let K denote the Markov kernel implementing all subsequent packet and scheduling transformations. Because K receives no additional unprotected function of the current private contribution, the output distributions are PK and QK. Rényi divergence obeys the data-processing inequality,

$$D_\alpha(PK\|QK)\le D_\alpha(P\|Q)\le\rho(\alpha). \tag{8}$$

Thus the transformations cannot increase the RDP parameter. Randomized transformations are included because a Markov kernel may contain independent randomness. Adaptive use of earlier private outputs is also included by conditioning on the realized history at each step and then applying adaptive composition. The condition on additional inputs is essential: a scheduler that directly reads an unprivatized current attribution residual is not covered by this post-processing argument. □

Proposition 3 (Zero additional content expenditure under abstention)**.** *If no new protected packet is exposed in slot t, then slot t contributes zero to the packet-content RDP sum.*

Proof. When $a_{i,t} = 0$, the slot emits no new packet content. Its contribution in the composition expression is $a_{i,t}\rho_{i,t}(\alpha) = 0$ for every α. Updating public controller state or retaining a previously released explanation is either independent of the current private delta or post-processing of already released information, so it creates no additional packet-content mechanism. This does not assert that silence itself is private if silence is an observable, data-dependent event; that question belongs to a transcript-level model. □

Proposition 4 (Complexity of the slot-level allocation)**.** *For N endpoints, at most M feasible modes per endpoint (including the zero-transmission choice), and integer resource budget B, the multiple-choice knapsack core is solvable in $O(NMB)$ time and $O(B)$ memory.*

Proof. Let $V_i(b)$ be the maximum accumulated utility after processing endpoints $1, \ldots, i$ with at most b resource units. For endpoint i, mode m has integer cost $c_{i,m}$ and utility $u_{i,m}$. The recurrence is

$$V_i(b) = \max_{m:\, c_{i,m} \le b} \{V_{i-1}(b - c_{i,m}) + u_{i,m}\}. \tag{9}$$

There are $N(B+1)$ states and at most M transitions per state, yielding $O(NMB)$ time. Only the preceding row is required to compute the next row, so two length-$(B+1)$ arrays—or one array with suitable buffering—give $O(B)$ memory. Feasibility filtering and score construction occur before this core and are not included in the bound, exactly as stated in the main paper. □

TABLE II
SCOPE OF THE PACKET-CONTENT PRIVACY STATEMENT.

Element	Assumption or treatment
Adjacency	Add/remove clipped household contribution; absent contribution is zero
Sensitivity	At most C_i when the reference is public, fixed, or derived only from prior private releases
Composition	Adaptive RDP costs are added for every exposed Gaussian release
Post-processing	Top-k, quantization, reconstruction, and decisions based on public state or prior private outputs
Accounted output	Released explanation-packet contents over the protected horizon
System metadata	Timing, silence, endpoint identity, and mode are communication metadata and require a separate deployment mechanism if sensitive

TABLE III
SLOT-LEVEL PROCEDURE AND STATE TRANSITION.

Step	Operation	Information used	State update
1	Update public state	AoX, channel, forecast-based criticality and novelty, fairness counters	No privacy charge
2	Enumerate MAP-XAI modes	Local clipped-and-perturbed attribution; packet parameters; accountant state	Remove modes that exceed remaining RDP allowance
3	Check radio feasibility	Packet size, resource blocks, channel-dependent latency, deadline	Remove infeasible modes
4	Score candidates	Public state, packet metadata, remaining allowance; no raw attribution residual	Compute $\Gamma_{i,t}(m) = G_{i,t}(m)/S_i(m)$
5	Allocate the shared uplink	Feasible candidate scores and slot resource budget	Select at most one mode per endpoint, or abstain
6	Account and transmit	Selected protected packet	Charge exposed release; reset AoX only after successful delivery

V. COMPLETE EXPERIMENTAL PROTOCOL

The evaluation was designed as a paired component study. Every PRISM-XAI observation is compared with the MAP-XAI-only layer under the same dataset, explainer, privacy budget, seed, and stochastic communication trace. This pairing removes between-trace variability from the incremental scheduler comparison. External fixed-DP policies are retained to show broader operating regions, but are not used for matched-service claims unless they satisfy the predeclared overlap rule in Table IV.

TABLE IV
EVALUATION MATRIX AND COMMUNICATION SETTING.

Item	Setting
Datasets	OPSD household electricity data; UCI ElectricityLoadDiagrams20112014
Endpoints and zones	OPSD: 9 endpoints in 3 synthetic zones; UCI: 32 endpoints in 4 synthetic zones
Forecast inputs	12 lag, rolling-statistic, and calendar features
Forecast error	OPSD RMSE/MAE $0.926/0.545$; UCI RMSE/MAE $1.289/0.221$
Explanation methods	TreeSHAP and feature ablation
Privacy settings	$\epsilon_{\max} \in \{8, 16, 32\}$; $\delta = 10^{-5}$; cumulative RDP accounting
Horizon	96 calibration slots followed by 512 evaluation slots
Wireless abstraction	Eight 180-kHz resource blocks, 1-ms slots, 5-ms deadlines, 3.5-GHz carrier, path loss, correlated shadowing, small-scale fading, edge delay, and bursty failures
Service guardrails	5% coverage target and minimum Jain fairness of 0.88
Primary matrix	2 datasets $\times 2$ explainers $\times 3$ privacy budgets $\times 6$ schedulers $\times 10$ paired seeds $= 720$ runs
Sensitivity matrices	Residual-score $M = 8$ and residual-score $M = 4$ repeat the 720-run matrix; 2,160 runs overall
Pairing unit	Same dataset, explanation method, privacy budget, and random seed
Matched region	Absolute coverage and recall gaps ≤ 0.02; relative spent-ϵ gap $\leq 5\%$
Inference	Paired mean difference and 95% CI half-width; paired t and Wilcoxon tests; Benjamini–Hochberg adjustment within comparison family

TABLE V
PRIMARY METRIC DEFINITIONS.

Metric	Definition
Traffic	Total transmitted packet bits divided by 10^3
Coverage	Critical events meeting both AoX and public-quality thresholds
Critical recall	Correct dominant feature group divided by all critical events
Top-K overlap	Fraction of selected private features shared with the non-private reference
Privacy spent	Maximum recorded endpoint expenditure after RDP conversion
Fairness	Jain index over endpoint service

VI. EXPANDED RESULTS

A. Primary matched component comparison

B. Scheduler sensitivity and ablations

The sensitivity analysis separates two design choices: the candidate score and the maximum number of private releases. The residual-score configurations show the behavior of the earlier scoring rule at two release caps. The primary public-score configuration replaces the private residual norm with declared forecast and communication state while retaining $M = 8$. The adaptive-mode ablation then compares the full mode choice with a fixed-resolution version of the same scheduler.

C. Matched-region eligibility of external operating points

The fixed-DP policies and the joint full-packet scheduler therefore describe different communication–service operating points. Their traffic totals are not used as matched-service evidence for the headline reduction.

D. Explanation-fidelity diagnostic

TABLE VI

COMPLETE METHOD MEANS FOR ALL THREE REPORTED CONFIGURATIONS. EACH ROW AGGREGATES 120 RUNS.

Configuration	Method	Traffic (kbit)	Top-K	Coverage	Critical recall	Max. ϵ
Residual $M = 8$	Periodic fixed-DP	4645.158	.2522	.3828	.1779	10.619
	Channel fixed-DP	4573.837	.2635	.3337	.1610	13.850
	AoX fixed-DP	4643.562	.2545	.8747	.3857	10.820
	Joint full packet	1655.413	.2596	.2484	.1166	18.665
	MAP-XAI-only	243.707	.2653	.0780	.0374	18.206
	PRISM-XAI	242.521	.2785	.0782	.0370	18.206
Residual $M = 4$	Periodic fixed-DP	4645.158	.2522	.3828	.1779	10.619
	Channel fixed-DP	4573.837	.2635	.3337	.1610	13.850
	AoX fixed-DP	4643.562	.2545	.8747	.3857	10.820
	Joint full packet	1655.413	.2596	.2484	.1166	18.665
	MAP-XAI-only	120.543	.3135	.0424	.0238	18.206
	PRISM-XAI	120.521	.3099	.0424	.0225	18.206
Public $M = 8$	Periodic fixed-DP	4645.158	.2522	.3828	.1779	10.619
	Channel fixed-DP	4573.837	.2635	.3337	.1610	13.850
	AoX fixed-DP	4643.562	.2545	.8747	.3857	10.820
	Joint full packet	1354.988	.2466	.1960	.0852	18.662
	MAP-XAI-only	243.707	.2653	.0780	.0374	18.206
	PRISM-XAI	**207.309**	.2693	.0779	.0366	18.206

TABLE VII

PUBLIC-SCORE $M = 8$ OPERATING POINTS, AVERAGED OVER 120 RUNS PER METHOD. ONLY MAP-XAI-ONLY SATISFIES THE DECLARED MATCHED-REGION RULE IN ALL CONDITIONS.

Method	Traffic (kbit)	Top-K overlap	Coverage	Critical recall	Max. ϵ spent
Periodic fixed-DP	4645.16	0.252	0.383	0.178	10.62
Channel fixed-DP	4573.84	0.264	0.334	0.161	13.85
AoX fixed-DP	4643.56	0.255	0.875	0.386	10.82
Joint full packet	1354.99	0.247	0.196	0.085	18.66
MAP-XAI-only (proposed packet layer)	243.71	0.265	0.078	0.037	18.21
PRISM-XAI (complete design)	**207.31**	**0.269**	0.078	0.037	18.21



Fig. 2. Matched component comparison across scheduler configurations. Panel (a) compares mean traffic; panel (b) gives relative traffic reduction; panel (c) shows PRISM-minus-MAP coverage (blue) and recall (orange) changes.

TABLE VIII
PRIMARY RESULTS BY DATASET (60 RUNS PER ROW).

Data	Method	kbit	Top-K	Coverage	Recall	$\epsilon_{\max}$
OPSD	MAP-only	106.15	.302	.0708	.0330	18.21
OPSD	PRISM	**87.98**	.294	.0710	.0324	18.21
UCI	MAP-only	381.26	.229	.0851	.0417	18.21
UCI	PRISM	**326.63**	.245	.0848	.0407	18.21

TABLE IX
PAIRED PRISM-XAI MINUS MAP-XAI-ONLY RESULTS FOR PUBLIC-SCORE $M = 8$ ($n = 120$). CI DENOTES THE 95% HALF-WIDTH.

Metric	Mean Δ	CI	q_t	q_W
Traffic (kbit)	-36.3979	3.3207	3.11×10^{-42}	1.76×10^{-20}
Top-K overlap	$+0.0040$	0.0155	0.832	0.653
Critical recall	-0.0008	0.0016	0.533	0.489
Intervention precision	-0.0170	0.0188	0.227	0.235

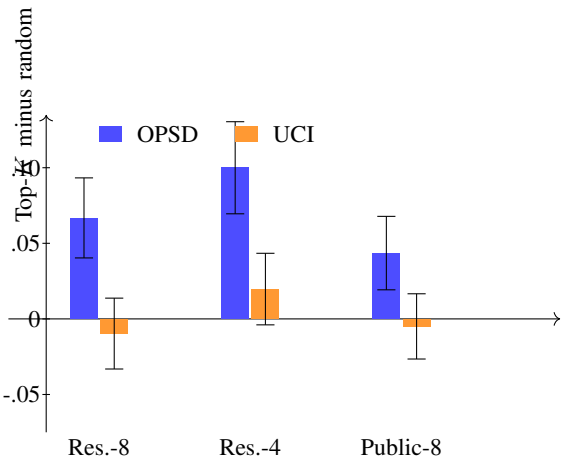


Fig. 3. Dataset-specific explanation-fidelity diagnostic. Bars show mean Top-K overlap above random; whiskers show 95% confidence intervals.

TABLE X
PAIRED PRISM-XAI MINUS MAP-XAI-ONLY SENSITIVITY ($n = 120$ PER CONFIGURATION). COVERAGE AND RECALL CHANGES ARE PERCENTAGE POINTS.

Score	Release cap	Δ traffic (kbit)	Relative change	Δ coverage	Δ critical recall
Residual	$M = 8$	-1.185 ± 0.411	-0.50%	$+0.021$	-0.039
Residual	$M = 4$	-0.022 ± 0.028	-0.00%	$+0.002$	-0.136
Public	$M = 8$	$\mathbf{-36.398 \pm 3.321}$	$\mathbf{-15.72\%}$	-0.010	-0.083

TABLE XI
ADAPTIVE-MODE ABLATION: PRISM-XAI MINUS FIXED-RESOLUTION PRISM ($n = 40$ PER CONFIGURATION). q_t AND q_W ARE BH-ADJUSTED PAIRED-TEST VALUES.

Score/cap	Metric	Mean Δ	95% CI half-width	Relative traffic change	q_t	q_W
Residual $M = 8$	Traffic (kbit)	-3.4384	0.8996	-1.464%	3.91×10^{-8}	6.42×10^{-7}
Residual $M = 4$	Traffic (kbit)	-0.0384	0.0329	-0.052%	0.252	0.371
Public $M = 8$	Traffic (kbit)	-35.3336	5.6256	-15.427%	1.05×10^{-13}	9.14×10^{-7}
Public $M = 8$	Top-K overlap	-0.0085	0.0268	–	1.000	0.907
Public $M = 8$	Critical recall	-0.0013	0.0029	–	1.000	0.907
Public $M = 8$	Intervention precision	-0.0119	0.0281	–	1.000	1.000

TABLE XII
CONDITIONS SATISFYING THE MATCHED-REGION RULE (OUT OF 12).

Comparator	Res.-8	Res.-4	Public-8
Periodic fixed-DP	0	0	0
Channel fixed-DP	0	0	0
AoX fixed-DP	0	0	0
Joint full packet	2	0	2
MAP-XAI-only	12	12	12

TABLE XIII

TOP-K OVERLAP ABOVE THE RANDOM-OVERLAP REFERENCE. POSITIVE VALUES INDICATE GREATER OVERLAP THAN RANDOM; q_t AND q_W ARE BH ADJUSTED.

Configuration	Scope	n	Mean difference	95% CI half-width	q_t	q_W
Residual $M = 8$	Combined	120	0.02855	0.01875	0.00471	0.02416
	OPSD	60	0.06681	0.02647	1.36×10^{-5}	1.01×10^{-4}
	UCI	60	-0.00971	0.02343	0.410	0.236
Residual $M = 4$	Combined	120	0.05985	0.02034	7.44×10^{-8}	9.88×10^{-7}
	OPSD	60	0.09997	0.03045	4.31×10^{-8}	9.88×10^{-7}
	UCI	60	0.01973	0.02366	0.100	0.143
Public $M = 8$	Combined	120	0.01929	0.01660	0.03475	0.08929
	OPSD	60	0.04354	0.02429	0.00205	0.00892
	UCI	60	-0.00496	0.02158	0.648	0.669

VII. Service–Communication Trade-off Sweep

This section reports the complete public-score sensitivity sweep supplied with the reproducibility package. Nine completed threshold/cap combinations are evaluated; every combination contains 120 runs per method (60 OPSD and 60 UCI), or 720 policy runs across the six methods and 6,480 policy runs over the full sweep. The coverage, recall, fidelity, fairness, traffic, and privacy-accounting definitions are unchanged. The threshold controls the public release score and M bounds private releases per endpoint. These are operating controls rather than coverage guarantees.

A. The sweep exposes a continuous service–traffic frontier

Increasing the release cap moves PRISM-XAI toward higher-traffic, higher-service operating points. Across the observed settings, mean PRISM-XAI traffic increases from 205.41–208.83 kbit at $M = 8$ to 342.50–345.42 kbit at $M = 16$ and 574.63–584.33 kbit at $M = 32$. Coverage rises from 0.044–0.078 to 0.086–0.131 and then 0.140–0.206. Critical-event recall follows the same broad pattern, rising from 0.021–0.037 to 0.040–0.059 and then 0.064–0.085. Thus, the low-traffic result is not a claim of uniformly superior monitoring service; the cap selects a point on the communication–service frontier.

The paired comparisons show traffic reductions at every tested setting, while the service changes are small in absolute terms. However, the declared descriptive-overlap rule is satisfied by MAP-XAI-only in all 12 dataset/explainer/privacy cells only for the four $M = 8$ settings. It is satisfied in 8 of 12 cells at $M = 16$, and in 4 of 12 cells for $(\tau, M) = (1, 32)$ and 5 of 12 for $(3, 32)$. The larger-cap points therefore describe the broader service frontier and are not additional matched-service confirmations of the headline claim.

TABLE XIV

OBSERVED PRISM-XAI OPERATING POINTS IN THE PUBLIC-SCORE SERVICE SWEEP. SAVINGS ARE THE RATIO OF AGGREGATE MEAN TRAFFIC RELATIVE TO MAP-XAI-ONLY; PAIRED CONFIDENCE INTERVALS ARE REPORTED IN TABLE XV.

τ	M	PRISM kbit	MAP kbit	Saving	Coverage	Recall	$\epsilon_{\max}$
0.5	8	208.83	243.80	14.35%	0.0444	0.0209	18.206
1	8	206.42	243.80	15.33%	0.0510	0.0230	18.206
2	8	205.41	243.37	15.60%	0.0737	0.0367	18.206
3	8	207.31	243.71	14.94%	0.0779	0.0366	18.206
1	16	342.50	407.81	16.01%	0.0856	0.0401	17.727
2	16	343.07	407.50	15.81%	0.1248	0.0594	17.745
3	16	345.42	407.78	15.29%	0.1312	0.0594	17.777
1	32	574.63	688.64	16.56%	0.1402	0.0638	17.196
3	32	584.33	688.42	15.12%	0.2064	0.0852	17.301

TABLE XV

PAIRED PRISM-XAI MINUS MAP-XAI-ONLY RESULTS FOR THE NINE SWEEP SETTINGS ($n = 120$ PAIRS PER ROW). COVERAGE AND RECALL ARE PROPORTIONS; CI IS THE 95% HALF-WIDTH.

τ	M	Δ kbit	CI	Δ coverage	CI	Δ recall	CI
0.5	8	-34.97	3.06	-0.00026	0.00046	-0.00049	0.00107
1	8	-37.38	3.49	-0.00082	0.00060	-0.00086	0.00107
2	8	-37.96	3.63	-0.00053	0.00046	-0.00061	0.00143
3	8	-36.40	3.32	-0.00010	0.00043	-0.00083	0.00156
1	16	-65.31	7.13	-0.00075	0.00093	0.00082	0.00162
2	16	-64.43	7.17	-0.00024	0.00023	0.00076	0.00208
3	16	-62.36	6.82	-0.00010	0.00034	0.00244	0.00227
1	32	-114.01	15.13	-0.00060	0.00111	0.00153	0.00178
3	32	-104.09	13.56	-0.00010	0.00026	0.00124	0.00307

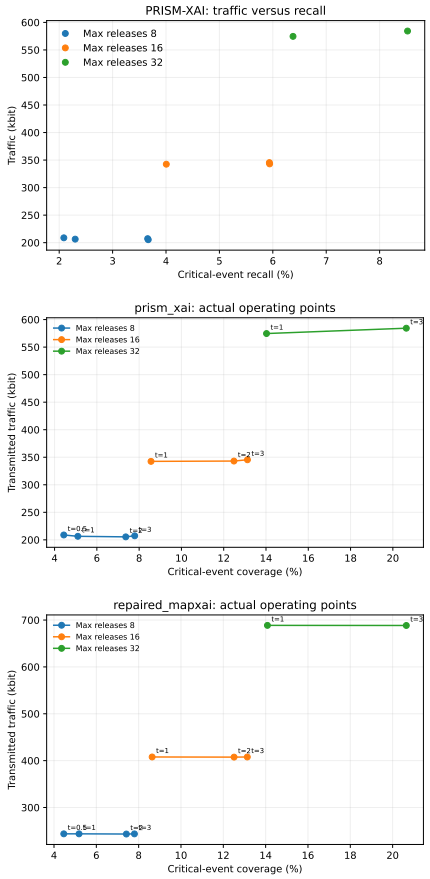


Fig. 4. Service–communication frontier from all completed sweep settings. Left: PRISM-XAI traffic versus critical-event recall. Center: PRISM-XAI traffic versus intervention coverage. Right: MAP-XAI-only traffic versus coverage at the corresponding cap/threshold configurations. Moving to larger release caps increases both communication and the measured service proxies; the figures therefore support operating-point selection, not a traffic-only ranking.

B. Complete generated sweep evidence

The following archive reproduces every generated vector figure and every generated LaTeX table for the nine settings at aggregate, OPSD, and UCI scope. Raster PNG counterparts, raw CSV files, JSON manifests, source hashes, and execution logs remain in the accompanying package. Each five-figure block is interpreted jointly: the trace shows temporal release behavior; fairness–reliability checks whether service is concentrated; privacy–fidelity relates recorded expenditure to attribution agreement; the Pareto plane displays communication cost against the fidelity proxy; and the ablation panel attributes changes to individual design elements. Tables preserve the complete method means, condition-level breakdowns, and paired contrasts.

1) Public-score threshold $\tau = 0.5$, release cap $M = 8$: At this operating point, PRISM-XAI transmits 208.83 kbit versus 243.80 kbit for MAP-XAI-only (aggregate-mean ratio saving 14.35%). The paired difference is -34.97$\pm$3.06 kbit. PRISM coverage is 0.0444, critical-event recall is 0.0209, and maximum recorded expenditure is 18.206. Relative to MAP-XAI-only, the paired coverage change is -0.00026$\pm$0.00046 and recall change is -0.00049$\pm$0.00107. The descriptive overlap rule holds in 12 of 12 cells. Accordingly, this point is read as a matched-service scheduler comparison.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XVI

PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-$\mathcal{K}$	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.227 ± 0.040	-0.036 ± 0.078	0.013 ± 0.005	0.013 ± 0.005	0.218 ± 0.087	0.964 ± 0.002	4.341 ± 1.447	87.697 ± 0.388	7.823 ± 0.000	0.018 ± 0.005	24929.927 ± 17099.604	0.993 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.198 ± 0.035	-0.021 ± 0.089	0.013 ± 0.005	0.013 ± 0.005	0.210 ± 0.097	0.964 ± 0.002	4.332 ± 1.445	87.461 ± 0.318	7.823 ± 0.000	0.019 ± 0.005	25702.427 ± 16465.572	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.254 ± 0.049	-0.070 ± 0.094	0.015 ± 0.005	0.015 ± 0.005	0.263 ± 0.086	0.964 ± 0.002	4.453 ± 1.383	105.817 ± 0.049	7.823 ± 0.000	0.016 ± 0.004	21748.429 ± 9602.332	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.237 ± 0.043	-0.048 ± 0.094	0.013 ± 0.006	0.013 ± 0.006	0.226 ± 0.081	0.964 ± 0.002	4.453 ± 1.383	105.675 ± 0.049	7.823 ± 0.000	0.017 ± 0.003	26069.368 ± 11426.445	0.993 ± 0.004
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.238 ± 0.091	0.008 ± 0.022	0.028 ± 0.004	0.028 ± 0.004	0.465 ± 0.035	0.940 ± 0.003	4.723 ± 1.732	330.248 ± 1.899	7.823 ± 0.000	0.022 ± 0.002	14725.609 ± 2267.563	0.991 ± 0.003
UCI	Tree	8	PRISM-XAI	0.233 ± 0.050	-0.032 ± 0.070	0.026 ± 0.003	0.026 ± 0.003	0.454 ± 0.085	0.945 ± 0.005	4.249 ± 1.633	327.398 ± 2.091	7.823 ± 0.000	0.022 ± 0.005	15910.590 ± 2766.316	0.991 ± 0.004
UCI	Tree	8	MAP-XAI fallback	0.224 ± 0.053	0.017 ± 0.017	0.027 ± 0.005	0.027 ± 0.005	0.459 ± 0.057	0.940 ± 0.003	4.895 ± 1.639	382.160 ± 0.072	7.823 ± 0.000	0.021 ± 0.002	17976.663 ± 3232.835	0.990 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.251 ± 0.037	-0.011 ± 0.080	0.026 ± 0.006	0.026 ± 0.006	0.462 ± 0.081	0.943 ± 0.006	5.591 ± 1.764	381.645 ± 0.090	7.823 ± 0.000	0.021 ± 0.005	18792.055 ± 3629.218	0.989 ± 0.004

TABLE XVII

PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.697 ± 0.388	0.227 ± 0.040	0.250 ± 0.000	-0.036 ± 0.078	0.013 ± 0.005	0.013 ± 0.005	0.218 ± 0.087	0.036 ± 0.002	0.964 ± 0.002	0.018 ± 0.005	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.461 ± 0.318	0.198 ± 0.035	0.250 ± 0.000	-0.021 ± 0.089	0.013 ± 0.005	0.013 ± 0.005	0.210 ± 0.097	0.036 ± 0.002	0.964 ± 0.002	0.019 ± 0.005	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.635 ± 0.376	0.225 ± 0.047	0.250 ± 0.000	0.065 ± 0.050	0.016 ± 0.004	0.016 ± 0.004	0.293 ± 0.081	0.036 ± 0.002	0.964 ± 0.002	0.015 ± 0.004	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.431 ± 0.305	0.188 ± 0.051	0.250 ± 0.000	0.103 ± 0.067	0.017 ± 0.005	0.017 ± 0.005	0.302 ± 0.076	0.036 ± 0.002	0.964 ± 0.002	0.015 ± 0.004	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.574 ± 0.349	0.335 ± 0.044	0.250 ± 0.000	0.205 ± 0.062	0.020 ± 0.003	0.020 ± 0.003	0.478 ± 0.125	0.036 ± 0.002	0.964 ± 0.002	0.010 ± 0.004	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.216 ± 0.281	0.311 ± 0.061	0.250 ± 0.000	0.292 ± 0.078	0.022 ± 0.004	0.022 ± 0.004	0.495 ± 0.133	0.036 ± 0.002	0.964 ± 0.002	0.009 ± 0.004	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	330.248 ± 1.899	0.238 ± 0.091	0.250 ± 0.000	0.008 ± 0.022	0.028 ± 0.004	0.028 ± 0.004	0.465 ± 0.035	0.060 ± 0.003	0.940 ± 0.003	0.022 ± 0.002	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	327.398 ± 2.091	0.233 ± 0.050	0.250 ± 0.000	-0.032 ± 0.070	0.026 ± 0.003	0.026 ± 0.003	0.454 ± 0.085	0.055 ± 0.005	0.945 ± 0.005	0.022 ± 0.005	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	330.309 ± 1.645	0.201 ± 0.066	0.250 ± 0.000	0.022 ± 0.015	0.027 ± 0.005	0.027 ± 0.005	0.573 ± 0.085	0.058 ± 0.004	0.942 ± 0.004	0.015 ± 0.004	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	329.517 ± 1.193	0.267 ± 0.020	0.250 ± 0.000	0.044 ± 0.071	0.022 ± 0.003	0.022 ± 0.003	0.521 ± 0.092	0.048 ± 0.004	0.952 ± 0.004	0.015 ± 0.005	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	332.151 ± 1.868	0.356 ± 0.056	0.250 ± 0.000	0.033 ± 0.016	0.023 ± 0.005	0.023 ± 0.005	0.969 ± 0.031	0.053 ± 0.005	0.947 ± 0.005	0.000 ± 0.001	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	331.266 ± 1.432	0.321 ± 0.046	0.250 ± 0.000	0.200 ± 0.047	0.024 ± 0.004	0.024 ± 0.004	0.943 ± 0.048	0.046 ± 0.003	0.954 ± 0.003	0.001 ± 0.001	31.175 ± 0.000

TABLE XVIII
PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.867 ± 0.388	-0.007 ± 0.067	-0.106 ± 0.030	-0.007 ± 0.003
OPSD	AoX-aware fixed-DP	-5012.277 ± 5.554	-0.029 ± 0.038	-0.385 ± 0.074	-0.007 ± 0.004
OPSD	AoX-aware fixed-DP	-5212.650 ± 0.349	0.082 ± 0.065	-0.392 ± 0.079	-0.007 ± 0.004
OPSD	AoX-aware fixed-DP	-1473.103 ± 0.318	-0.040 ± 0.037	-0.108 ± 0.027	-0.007 ± 0.003
OPSD	AoX-aware fixed-DP	-5012.482 ± 5.458	-0.064 ± 0.038	-0.397 ± 0.056	-0.007 ± 0.004
OPSD	AoX-aware fixed-DP	-5213.008 ± 0.281	0.060 ± 0.073	-0.403 ± 0.058	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-1472.867 ± 0.388	-0.066 ± 0.052	-0.094 ± 0.027	-0.007 ± 0.003
OPSD	Channel-aware fixed-DP	-4598.974 ± 13.023	-0.037 ± 0.042	-0.385 ± 0.089	0.055 ± 0.005
OPSD	Channel-aware fixed-DP	-5212.650 ± 0.349	0.077 ± 0.057	-0.414 ± 0.099	0.086 ± 0.006
OPSD	Channel-aware fixed-DP	-1473.103 ± 0.318	-0.094 ± 0.052	-0.098 ± 0.023	-0.007 ± 0.003
OPSD	Channel-aware fixed-DP	-4599.179 ± 12.890	-0.070 ± 0.045	-0.402 ± 0.087	0.056 ± 0.006
OPSD	Channel-aware fixed-DP	-5213.008 ± 0.281	0.054 ± 0.072	-0.434 ± 0.091	0.086 ± 0.007
OPSD	Joint, no compression	-102.203 ± 0.557	-0.061 ± 0.073	-0.003 ± 0.004	-0.003 ± 0.003
OPSD	Joint, no compression	-372.265 ± 1.026	-0.007 ± 0.078	-0.022 ± 0.008	-0.005 ± 0.004
OPSD	Joint, no compression	-1053.326 ± 8.018	0.100 ± 0.086	-0.053 ± 0.014	-0.006 ± 0.004
OPSD	Joint, no compression	-102.289 ± 0.585	-0.060 ± 0.065	-0.006 ± 0.004	-0.003 ± 0.003
OPSD	Joint, no compression	-372.619 ± 0.786	-0.052 ± 0.062	-0.022 ± 0.008	-0.004 ± 0.004
OPSD	Joint, no compression	-1053.684 ± 7.938	0.030 ± 0.075	-0.058 ± 0.011	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-1472.867 ± 0.388	-0.014 ± 0.067	-0.112 ± 0.022	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5024.959 ± 0.376	-0.024 ± 0.047	-0.434 ± 0.062	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5212.262 ± 0.599	0.086 ± 0.047	-0.438 ± 0.067	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-1473.103 ± 0.318	-0.044 ± 0.028	-0.122 ± 0.018	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5025.163 ± 0.305	-0.055 ± 0.045	-0.445 ± 0.049	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5212.620 ± 0.569	0.068 ± 0.064	-0.449 ± 0.049	-0.007 ± 0.004
OPSD	MAP-XAI fallback	-18.120 ± 0.382	-0.028 ± 0.044	-0.002 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.181 ± 0.373	-0.017 ± 0.079	-0.000 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.243 ± 0.359	-0.030 ± 0.076	0.001 ± 0.004	-0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.214 ± 0.313	-0.038 ± 0.052	-0.001 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.203 ± 0.325	-0.043 ± 0.051	0.001 ± 0.005	0.001 ± 0.001
OPSD	MAP-XAI fallback	-18.315 ± 0.369	-0.005 ± 0.066	0.002 ± 0.003	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4969.976 ± 1.899	-0.026 ± 0.105	-0.429 ± 0.032	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4969.915 ± 1.645	-0.063 ± 0.055	-0.430 ± 0.030	-0.007 ± 0.003
UCI	AoX-aware fixed-DP	-4968.073 ± 1.868	0.093 ± 0.060	-0.434 ± 0.032	-0.008 ± 0.003
UCI	AoX-aware fixed-DP	-4972.825 ± 2.091	-0.028 ± 0.058	-0.430 ± 0.048	-0.009 ± 0.004
UCI	AoX-aware fixed-DP	-4970.707 ± 1.193	0.007 ± 0.032	-0.433 ± 0.048	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4968.958 ± 1.432	0.060 ± 0.058	-0.430 ± 0.047	-0.007 ± 0.002
UCI	Channel-aware fixed-DP	-4964.930 ± 4.660	-0.011 ± 0.100	0.027 ± 0.004	-0.000 ± 0.003
UCI	Channel-aware fixed-DP	-4969.915 ± 1.645	-0.067 ± 0.076	0.027 ± 0.005	0.661 ± 0.003
UCI	Channel-aware fixed-DP	-4968.073 ± 1.868	0.088 ± 0.077	0.022 ± 0.005	0.736 ± 0.003
UCI	Channel-aware fixed-DP	-4967.779 ± 5.407	-0.006 ± 0.053	0.025 ± 0.003	-0.000 ± 0.004
UCI	Channel-aware fixed-DP	-4970.707 ± 1.193	0.007 ± 0.032	0.021 ± 0.003	0.660 ± 0.003
UCI	Channel-aware fixed-DP	-4968.958 ± 1.432	0.062 ± 0.059	0.024 ± 0.004	0.737 ± 0.002
UCI	Joint, no compression	-342.052 ± 2.226	-0.048 ± 0.116	-0.024 ± 0.006	-0.003 ± 0.005
UCI	Joint, no compression	-1301.841 ± 1.666	-0.016 ± 0.092	-0.067 ± 0.007	-0.006 ± 0.003
UCI	Joint, no compression	-3702.399 ± 5.796	0.110 ± 0.070	-0.205 ± 0.016	-0.007 ± 0.003
UCI	Joint, no compression	-344.902 ± 2.187	0.003 ± 0.081	-0.031 ± 0.010	-0.004 ± 0.004
UCI	Joint, no compression	-1303.083 ± 1.214	0.063 ± 0.025	-0.077 ± 0.017	-0.007 ± 0.003
UCI	Joint, no compression	-3703.284 ± 4.443	0.080 ± 0.051	-0.203 ± 0.030	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4969.071 ± 2.561	-0.022 ± 0.097	0.022 ± 0.004	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4969.009 ± 1.981	-0.058 ± 0.055	0.021 ± 0.005	-0.008 ± 0.003
UCI	Periodic fixed-DP	-4967.167 ± 2.181	0.097 ± 0.070	0.017 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4971.920 ± 2.875	-0.028 ± 0.074	0.020 ± 0.004	-0.009 ± 0.004
UCI	Periodic fixed-DP	-4969.801 ± 2.182	0.007 ± 0.049	0.016 ± 0.004	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4968.052 ± 1.956	0.060 ± 0.059	0.019 ± 0.004	-0.007 ± 0.002
UCI	MAP-XAI fallback	-51.912 ± 1.888	0.014 ± 0.091	0.001 ± 0.004	0.000 ± 0.001
UCI	MAP-XAI fallback	-51.851 ± 1.630	-0.019 ± 0.059	0.000 ± 0.005	0.001 ± 0.001
UCI	MAP-XAI fallback	-50.009 ± 1.863	-0.014 ± 0.071	-0.002 ± 0.004	0.001 ± 0.001
UCI	MAP-XAI fallback	-54.246 ± 2.117	-0.018 ± 0.061	-0.001 ± 0.006	0.002 ± 0.003
UCI	MAP-XAI fallback	-52.128 ± 1.236	-0.012 ± 0.083	-0.003 ± 0.003	0.000 ± 0.003
UCI	MAP-XAI fallback	-50.276 ± 1.397	-0.056 ± 0.050	-0.002 ± 0.003	0.002 ± 0.002

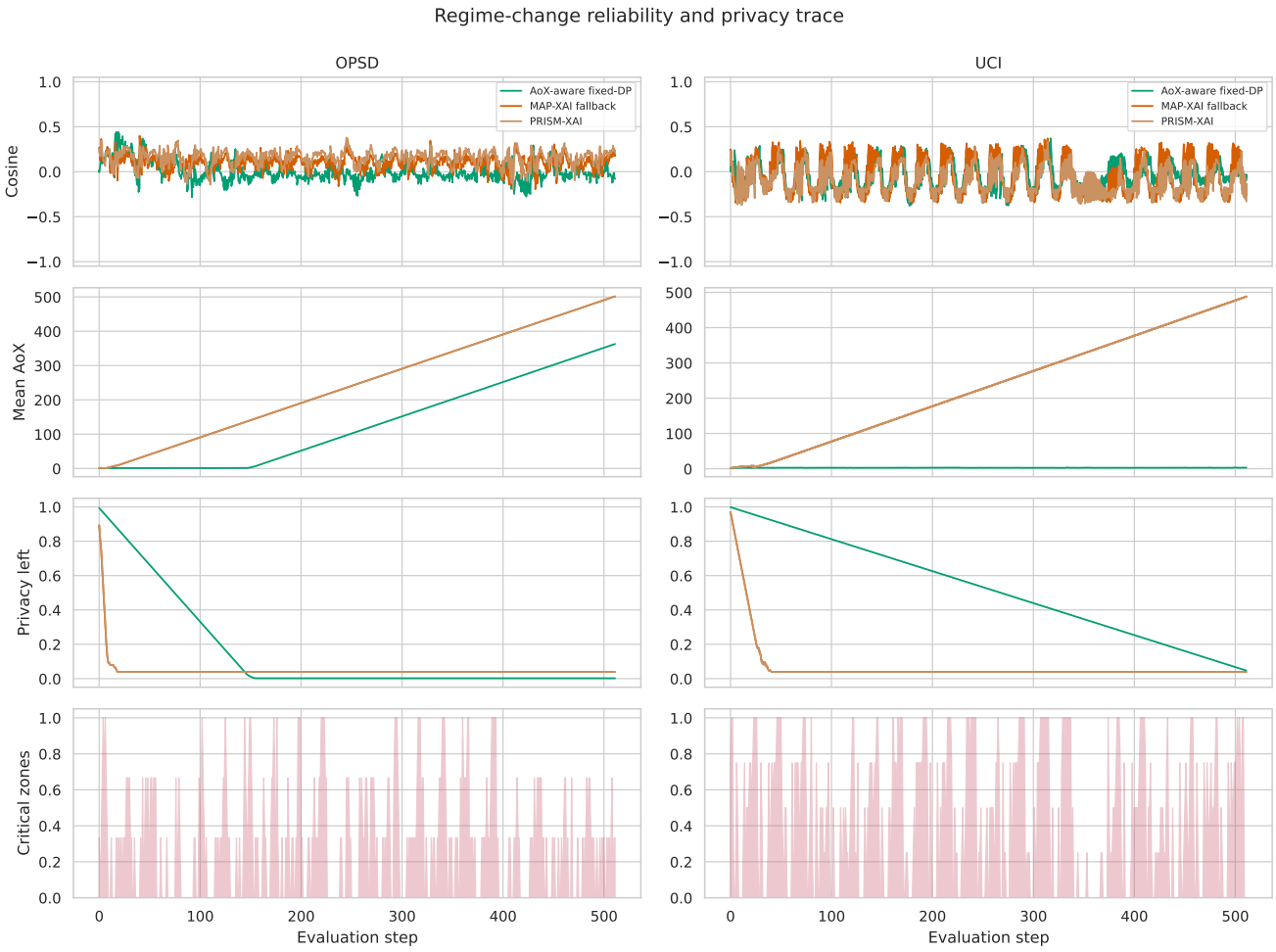


Fig. 5. Public-score threshold $\tau = 0.5$, release cap $M = 8$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

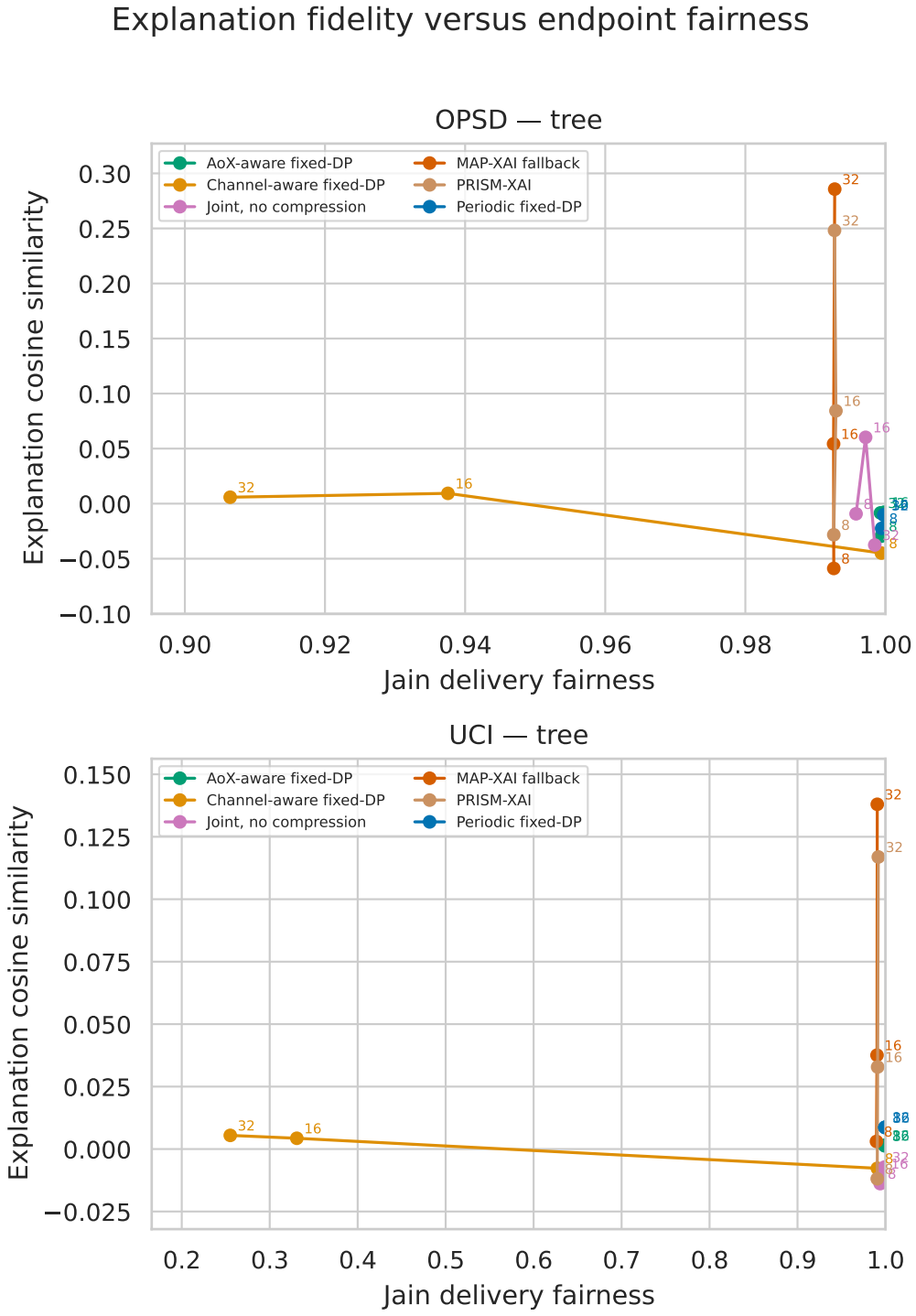


Fig. 6. Public-score threshold $\tau = 0.5$, release cap $M = 8$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

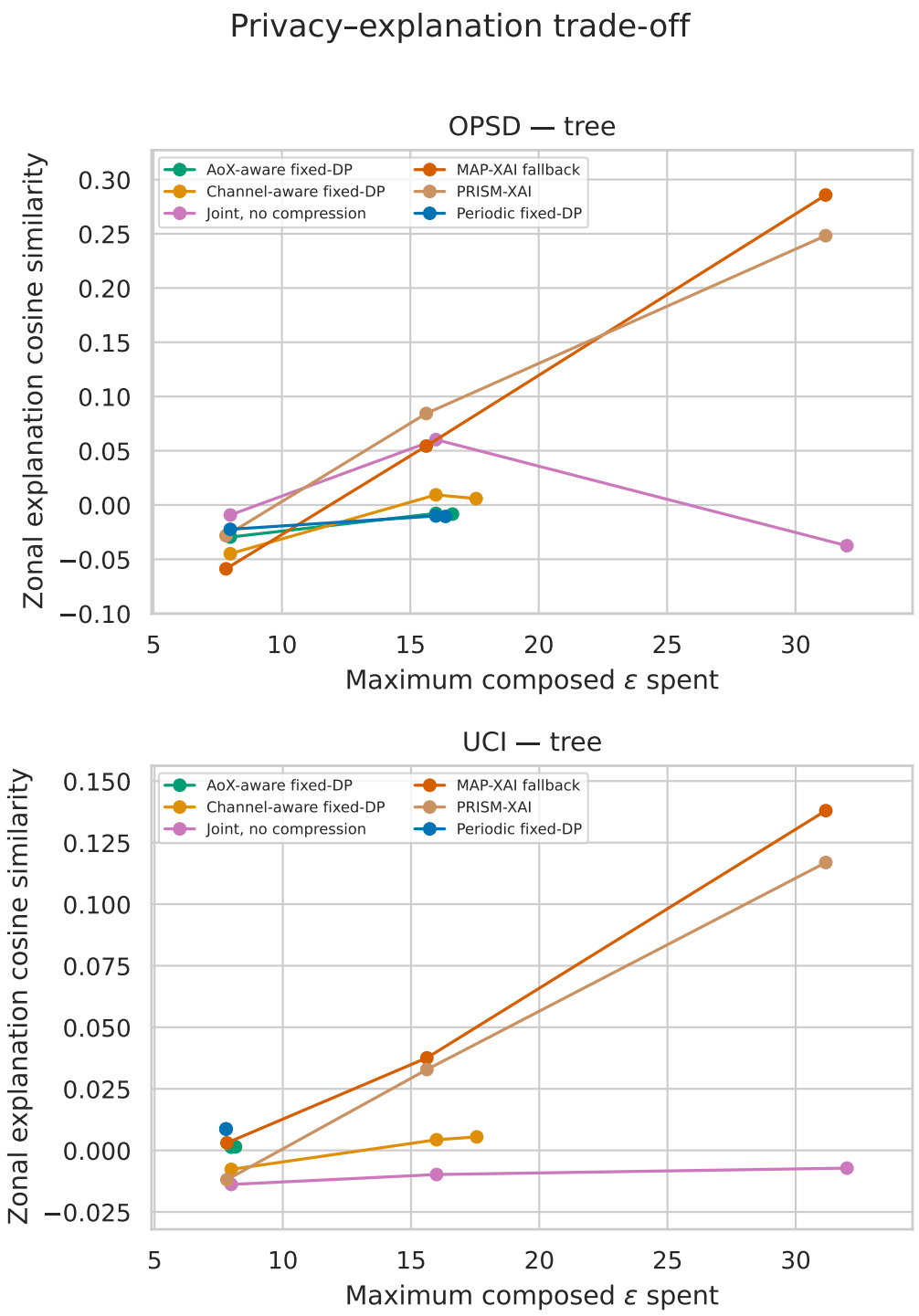


Fig. 7. Public-score threshold $\tau = 0.5$, release cap $M = 8$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

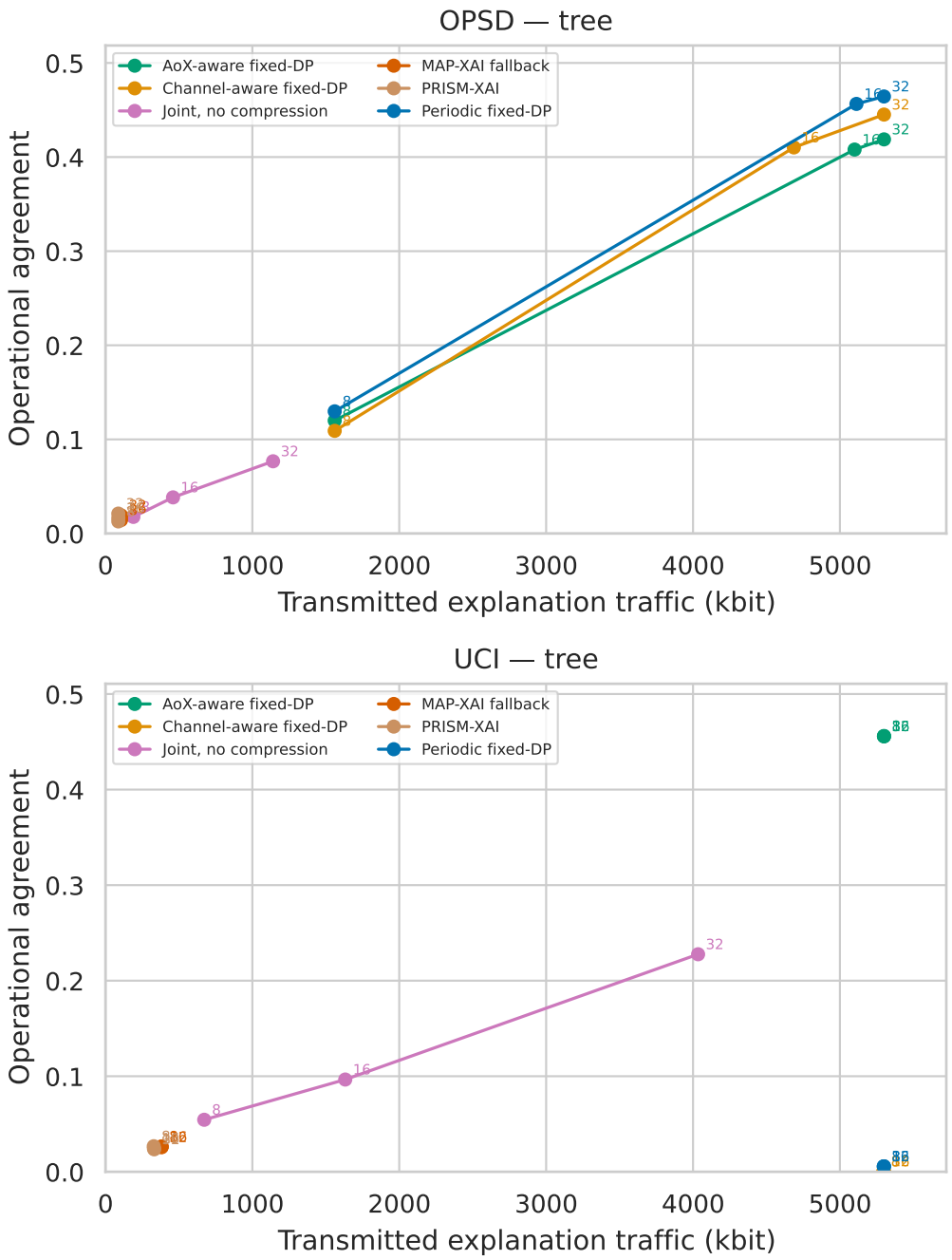


Fig. 8. Public-score threshold $\tau = 0.5$, release cap $M = 8$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

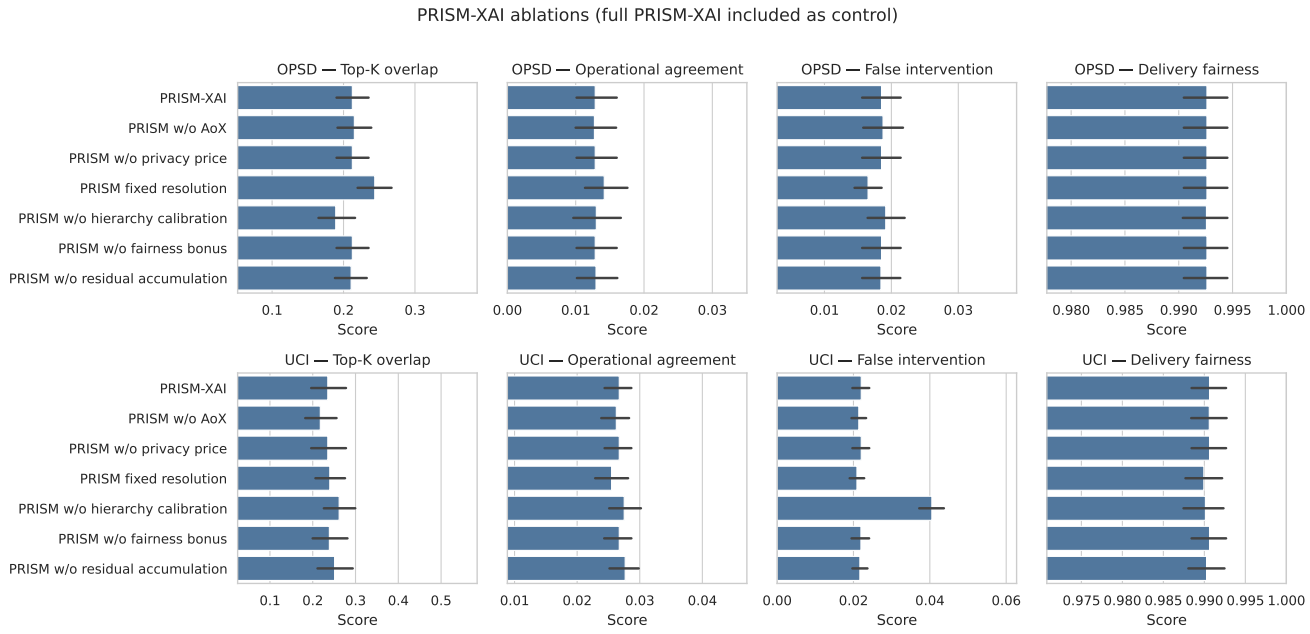


Fig. 9. Public-score threshold $\tau = 0.5$, release cap $M = 8$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XIX
PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.227 ± 0.040	-0.036 ± 0.078	0.013 ± 0.005	0.013 ± 0.005	0.218 ± 0.087	0.964 ± 0.002	4.341 ± 1.447	87.697 ± 0.388	7.823 ± 0.000	0.018 ± 0.005	24929.927 ± 17099.604	0.993 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.198 ± 0.035	-0.021 ± 0.089	0.013 ± 0.005	0.013 ± 0.005	0.210 ± 0.097	0.964 ± 0.002	4.332 ± 1.445	87.461 ± 0.318	7.823 ± 0.000	0.019 ± 0.005	25702.427 ± 16465.572	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.254 ± 0.049	-0.070 ± 0.094	0.015 ± 0.005	0.015 ± 0.005	0.263 ± 0.086	0.964 ± 0.002	4.453 ± 1.383	105.817 ± 0.049	7.823 ± 0.000	0.016 ± 0.004	21748.429 ± 9602.332	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.237 ± 0.043	-0.048 ± 0.094	0.013 ± 0.006	0.013 ± 0.006	0.226 ± 0.081	0.964 ± 0.002	4.453 ± 1.383	105.675 ± 0.049	7.823 ± 0.000	0.017 ± 0.003	26069.368 ± 11426.445	0.993 ± 0.004

TABLE XX

PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.697 ± 0.388	0.227 ± 0.040	0.250 ± 0.000	-0.036 ± 0.078	0.013 ± 0.005	0.013 ± 0.005	0.218 ± 0.087	0.036 ± 0.002	0.964 ± 0.002	0.018 ± 0.005	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.461 ± 0.318	0.198 ± 0.035	0.250 ± 0.000	-0.021 ± 0.089	0.013 ± 0.005	0.013 ± 0.005	0.210 ± 0.097	0.036 ± 0.002	0.964 ± 0.002	0.019 ± 0.005	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.635 ± 0.376	0.225 ± 0.047	0.250 ± 0.000	0.065 ± 0.050	0.016 ± 0.004	0.016 ± 0.004	0.293 ± 0.081	0.036 ± 0.002	0.964 ± 0.002	0.015 ± 0.004	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.431 ± 0.305	0.188 ± 0.051	0.250 ± 0.000	0.103 ± 0.067	0.017 ± 0.005	0.017 ± 0.005	0.302 ± 0.076	0.036 ± 0.002	0.964 ± 0.002	0.015 ± 0.004	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.574 ± 0.349	0.335 ± 0.044	0.250 ± 0.000	0.205 ± 0.062	0.020 ± 0.003	0.020 ± 0.003	0.478 ± 0.125	0.036 ± 0.002	0.964 ± 0.002	0.010 ± 0.004	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.216 ± 0.281	0.311 ± 0.061	0.250 ± 0.000	0.292 ± 0.078	0.022 ± 0.004	0.022 ± 0.004	0.495 ± 0.133	0.036 ± 0.002	0.964 ± 0.002	0.009 ± 0.004	31.175 ± 0.000

TABLE XXI
PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.867 ± 0.388	-0.007 ± 0.067	-0.106 ± 0.030	-0.007 ± 0.003
OPSD	AoX-aware fixed-DP	-5012.277 ± 5.554	-0.029 ± 0.038	-0.385 ± 0.074	-0.007 ± 0.004
OPSD	AoX-aware fixed-DP	-5212.650 ± 0.349	0.082 ± 0.065	-0.392 ± 0.079	-0.007 ± 0.004
OPSD	AoX-aware fixed-DP	-1473.103 ± 0.318	-0.040 ± 0.037	-0.108 ± 0.027	-0.007 ± 0.003
OPSD	AoX-aware fixed-DP	-5012.482 ± 5.458	-0.064 ± 0.038	-0.397 ± 0.056	-0.007 ± 0.004
OPSD	AoX-aware fixed-DP	-5213.008 ± 0.281	0.060 ± 0.073	-0.403 ± 0.058	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-1472.867 ± 0.388	-0.066 ± 0.052	-0.094 ± 0.027	-0.007 ± 0.003
OPSD	Channel-aware fixed-DP	-4598.974 ± 13.023	-0.037 ± 0.042	-0.385 ± 0.089	0.055 ± 0.005
OPSD	Channel-aware fixed-DP	-5212.650 ± 0.349	0.077 ± 0.057	-0.414 ± 0.099	0.086 ± 0.006
OPSD	Channel-aware fixed-DP	-1473.103 ± 0.318	-0.094 ± 0.052	-0.098 ± 0.023	-0.007 ± 0.003
OPSD	Channel-aware fixed-DP	-4599.179 ± 12.890	-0.070 ± 0.045	-0.402 ± 0.087	0.056 ± 0.006
OPSD	Channel-aware fixed-DP	-5213.008 ± 0.281	0.054 ± 0.072	-0.434 ± 0.091	0.086 ± 0.007
OPSD	Joint, no compression	-102.203 ± 0.557	-0.061 ± 0.073	-0.003 ± 0.004	-0.003 ± 0.003
OPSD	Joint, no compression	-372.265 ± 1.026	-0.007 ± 0.078	-0.022 ± 0.008	-0.005 ± 0.004
OPSD	Joint, no compression	-1053.326 ± 8.018	0.100 ± 0.086	-0.053 ± 0.014	-0.006 ± 0.004
OPSD	Joint, no compression	-102.289 ± 0.585	-0.060 ± 0.065	-0.006 ± 0.004	-0.003 ± 0.003
OPSD	Joint, no compression	-372.619 ± 0.786	-0.052 ± 0.062	-0.022 ± 0.008	-0.004 ± 0.004
OPSD	Joint, no compression	-1053.684 ± 7.938	0.030 ± 0.075	-0.058 ± 0.011	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-1472.867 ± 0.388	-0.014 ± 0.067	-0.112 ± 0.022	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5024.959 ± 0.376	-0.024 ± 0.047	-0.434 ± 0.062	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5212.262 ± 0.599	0.086 ± 0.047	-0.438 ± 0.067	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-1473.103 ± 0.318	-0.044 ± 0.028	-0.122 ± 0.018	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5025.163 ± 0.305	-0.055 ± 0.045	-0.445 ± 0.049	-0.007 ± 0.004
OPSD	Periodic fixed-DP	-5212.620 ± 0.569	0.068 ± 0.064	-0.449 ± 0.049	-0.007 ± 0.004
OPSD	MAP-XAI fallback	-18.120 ± 0.382	-0.028 ± 0.044	-0.002 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.181 ± 0.373	-0.017 ± 0.079	-0.000 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.243 ± 0.359	-0.030 ± 0.076	0.001 ± 0.004	-0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.214 ± 0.313	-0.038 ± 0.052	-0.001 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.203 ± 0.325	-0.043 ± 0.051	0.001 ± 0.005	0.001 ± 0.001
OPSD	MAP-XAI fallback	-18.315 ± 0.369	-0.005 ± 0.066	0.002 ± 0.003	0.000 ± 0.000

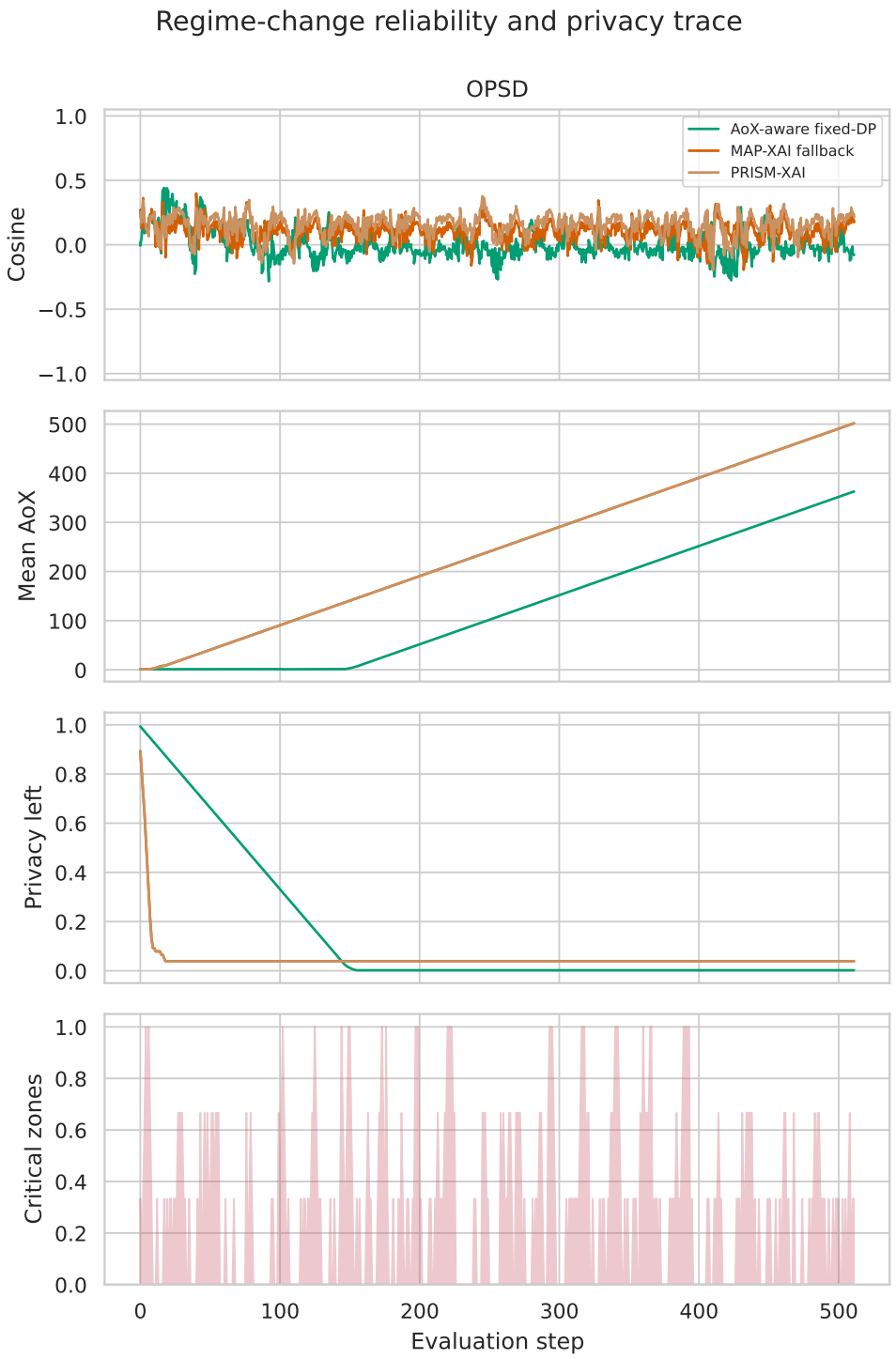


Fig. 10. Public-score threshold $\tau = 0.5$, release cap $M = 8$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

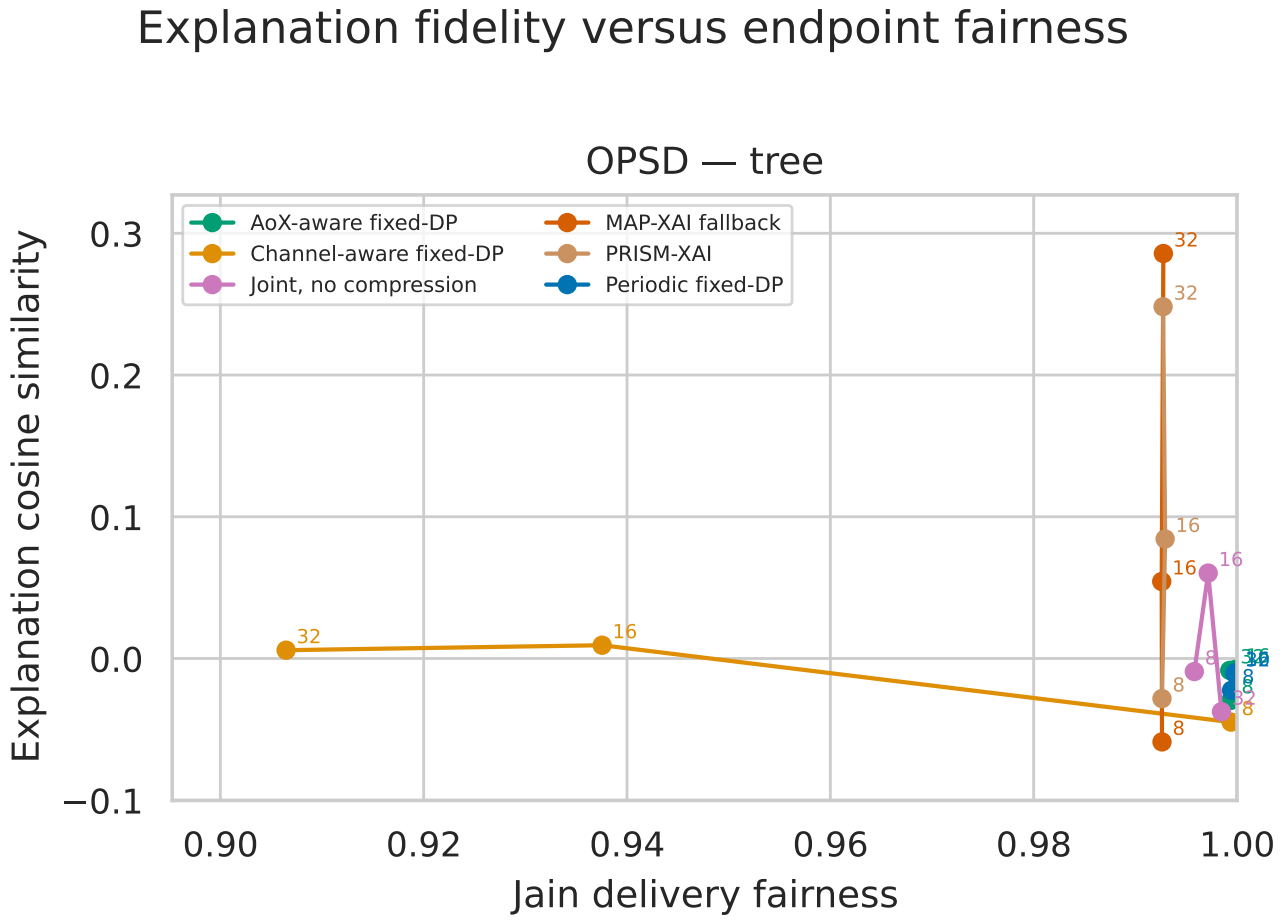


Fig. 11. Public-score threshold $\tau = 0.5$, release cap $M = 8$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

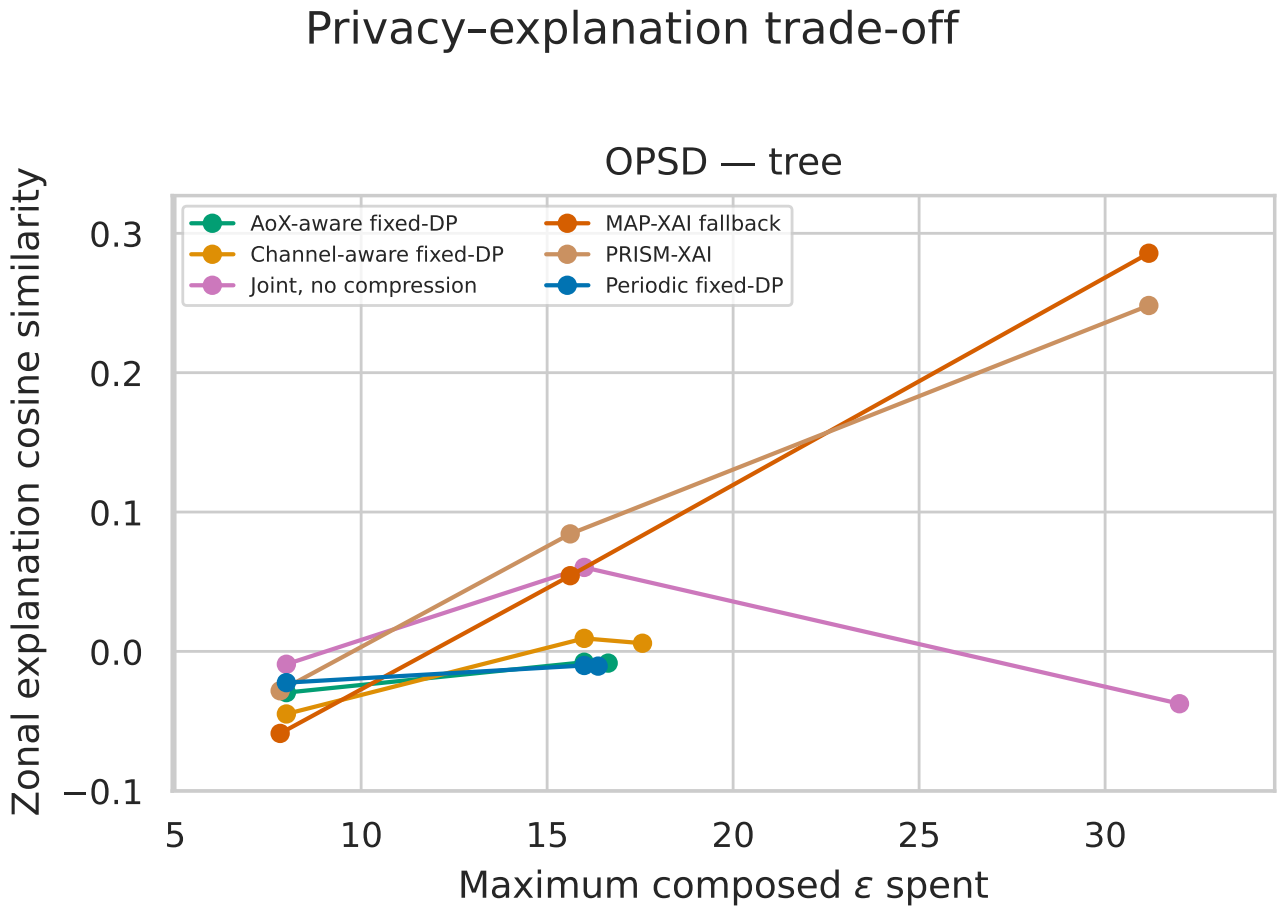


Fig. 12. Public-score threshold $\tau = 0.5$, release cap $M = 8$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

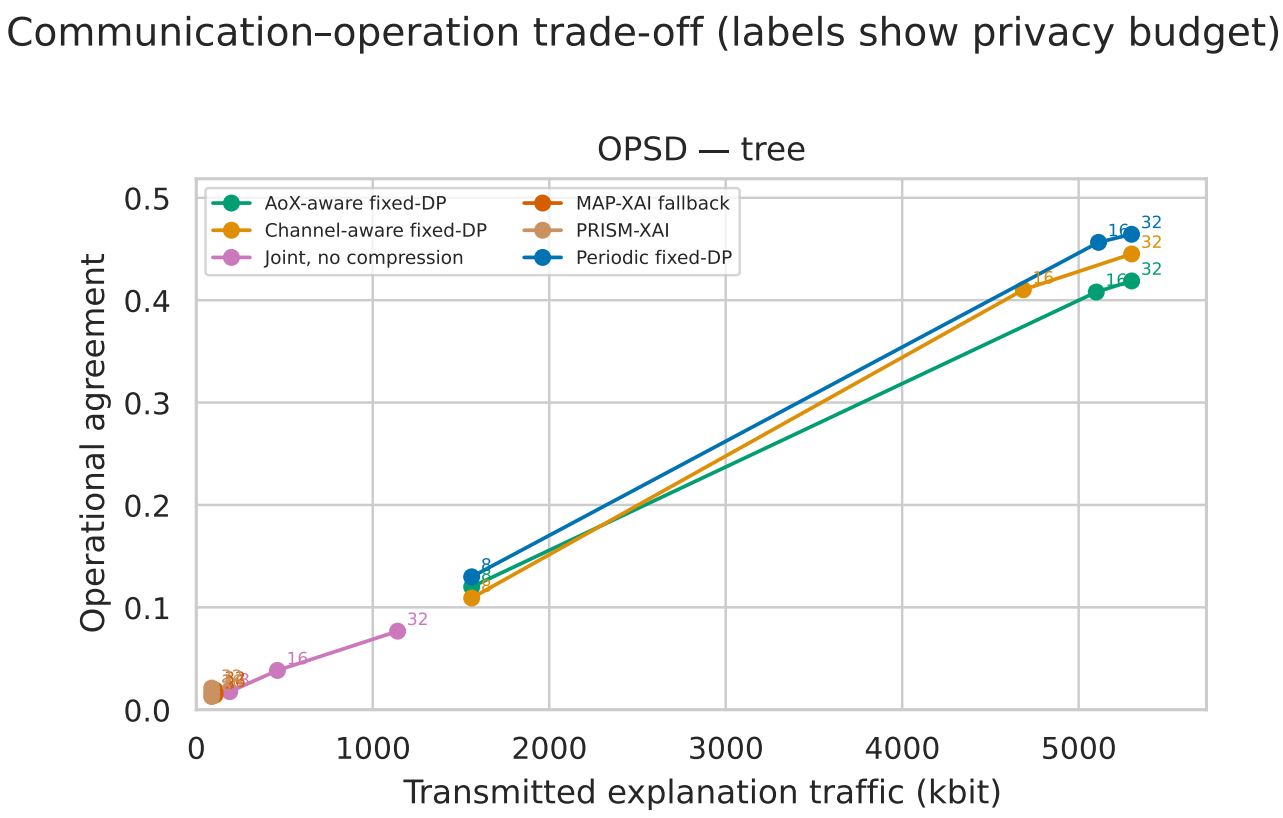


Fig. 13. Public-score threshold $\tau = 0.5$, release cap $M = 8$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

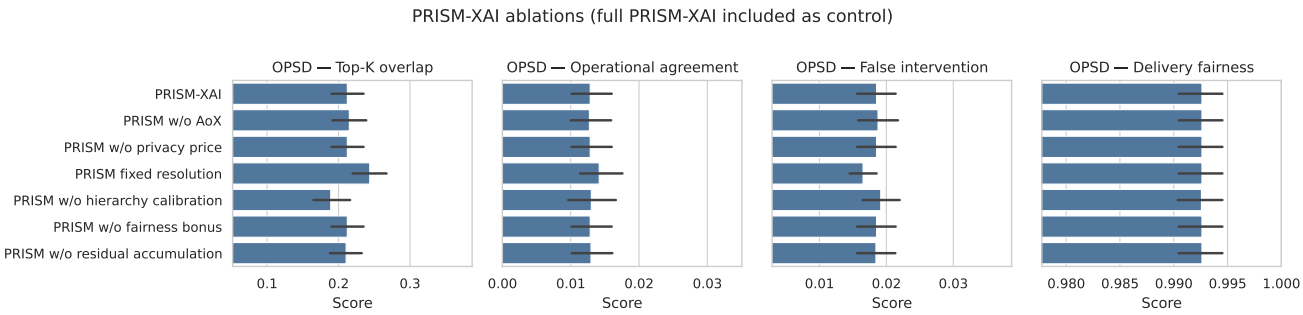


Fig. 14. Public-score threshold $\tau = 0.5$, release cap $M = 8$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XXII
PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.238 ± 0.091	0.008 ± 0.022	0.028 ± 0.004	0.028 ± 0.004	0.465 ± 0.035	0.940 ± 0.003	4.723 ± 1.732	330.248 ± 1.899	7.823 ± 0.000	0.022 ± 0.002	14725.609 ± 2267.563	0.991 ± 0.003
UCI	Tree	8	PRISM-XAI	0.233 ± 0.050	-0.032 ± 0.070	0.026 ± 0.003	0.026 ± 0.003	0.454 ± 0.085	0.945 ± 0.005	4.249 ± 1.633	327.398 ± 2.091	7.823 ± 0.000	0.022 ± 0.005	15910.590 ± 2766.316	0.991 ± 0.004
UCI	Tree	8	MAP-XAI fallback	0.224 ± 0.053	0.017 ± 0.017	0.027 ± 0.005	0.027 ± 0.005	0.459 ± 0.057	0.940 ± 0.003	4.895 ± 1.639	382.160 ± 0.072	7.823 ± 0.000	0.021 ± 0.002	17976.663 ± 3232.835	0.990 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.251 ± 0.037	-0.011 ± 0.080	0.026 ± 0.006	0.026 ± 0.006	0.462 ± 0.081	0.943 ± 0.006	5.591 ± 1.764	381.645 ± 0.090	7.823 ± 0.000	0.021 ± 0.005	18792.055 ± 3629.218	0.989 ± 0.004

TABLE XXIII
PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	330.248 ± 1.899	0.238 ± 0.091	0.250 ± 0.000	0.008 ± 0.022	0.028 ± 0.004	0.028 ± 0.004	0.465 ± 0.035	0.060 ± 0.003	0.940 ± 0.003	0.022 ± 0.002	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	327.398 ± 2.091	0.233 ± 0.050	0.250 ± 0.000	-0.032 ± 0.070	0.026 ± 0.003	0.026 ± 0.003	0.454 ± 0.085	0.055 ± 0.005	0.945 ± 0.005	0.022 ± 0.005	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	330.309 ± 1.645	0.201 ± 0.066	0.250 ± 0.000	0.022 ± 0.015	0.027 ± 0.005	0.027 ± 0.005	0.573 ± 0.085	0.058 ± 0.004	0.942 ± 0.004	0.015 ± 0.004	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	329.517 ± 1.193	0.267 ± 0.020	0.250 ± 0.000	0.044 ± 0.071	0.022 ± 0.003	0.022 ± 0.003	0.521 ± 0.092	0.048 ± 0.004	0.952 ± 0.004	0.015 ± 0.005	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	332.151 ± 1.868	0.356 ± 0.056	0.250 ± 0.000	0.033 ± 0.016	0.023 ± 0.005	0.023 ± 0.005	0.969 ± 0.031	0.053 ± 0.005	0.947 ± 0.005	0.000 ± 0.001	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	331.266 ± 1.432	0.321 ± 0.046	0.250 ± 0.000	0.200 ± 0.047	0.024 ± 0.004	0.024 ± 0.004	0.943 ± 0.048	0.046 ± 0.003	0.954 ± 0.003	0.001 ± 0.001	31.175 ± 0.000

TABLE XXIV
PUBLIC-SCORE THRESHOLD $\tau = 0.5$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4969.976 ± 1.899	-0.026 ± 0.105	-0.429 ± 0.032	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4969.915 ± 1.645	-0.063 ± 0.055	-0.430 ± 0.030	-0.007 ± 0.003
UCI	AoX-aware fixed-DP	-4968.073 ± 1.868	0.093 ± 0.060	-0.434 ± 0.032	-0.008 ± 0.003
UCI	AoX-aware fixed-DP	-4972.825 ± 2.091	-0.028 ± 0.058	-0.430 ± 0.048	-0.009 ± 0.004
UCI	AoX-aware fixed-DP	-4970.707 ± 1.193	0.007 ± 0.032	-0.433 ± 0.048	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4968.958 ± 1.432	0.060 ± 0.058	-0.430 ± 0.047	-0.007 ± 0.002
UCI	Channel-aware fixed-DP	-4964.930 ± 4.660	-0.011 ± 0.100	0.027 ± 0.004	-0.000 ± 0.003
UCI	Channel-aware fixed-DP	-4969.915 ± 1.645	-0.067 ± 0.076	0.027 ± 0.005	0.661 ± 0.003
UCI	Channel-aware fixed-DP	-4968.073 ± 1.868	0.088 ± 0.077	0.022 ± 0.005	0.736 ± 0.003
UCI	Channel-aware fixed-DP	-4967.779 ± 5.407	-0.006 ± 0.053	0.025 ± 0.003	-0.000 ± 0.004
UCI	Channel-aware fixed-DP	-4970.707 ± 1.193	0.007 ± 0.032	0.021 ± 0.003	0.660 ± 0.003
UCI	Channel-aware fixed-DP	-4968.958 ± 1.432	0.062 ± 0.059	0.024 ± 0.004	0.737 ± 0.002
UCI	Joint, no compression	-342.052 ± 2.226	-0.048 ± 0.116	-0.024 ± 0.006	-0.003 ± 0.005
UCI	Joint, no compression	-1301.841 ± 1.666	-0.016 ± 0.092	-0.067 ± 0.007	-0.006 ± 0.003
UCI	Joint, no compression	-3702.399 ± 5.796	0.110 ± 0.070	-0.205 ± 0.016	-0.007 ± 0.003
UCI	Joint, no compression	-344.902 ± 2.187	0.003 ± 0.081	-0.031 ± 0.010	-0.004 ± 0.004
UCI	Joint, no compression	-1303.083 ± 1.214	0.063 ± 0.025	-0.077 ± 0.017	-0.007 ± 0.003
UCI	Joint, no compression	-3703.284 ± 4.443	0.080 ± 0.051	-0.203 ± 0.030	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4969.071 ± 2.561	-0.022 ± 0.097	0.022 ± 0.004	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4969.009 ± 1.981	-0.058 ± 0.055	0.021 ± 0.005	-0.008 ± 0.003
UCI	Periodic fixed-DP	-4967.167 ± 2.181	0.097 ± 0.070	0.017 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4971.920 ± 2.875	-0.028 ± 0.074	0.020 ± 0.004	-0.009 ± 0.004
UCI	Periodic fixed-DP	-4969.801 ± 2.182	0.007 ± 0.049	0.016 ± 0.004	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4968.052 ± 1.956	0.060 ± 0.059	0.019 ± 0.004	-0.007 ± 0.002
UCI	MAP-XAI fallback	-51.912 ± 1.888	0.014 ± 0.091	0.001 ± 0.004	0.000 ± 0.001
UCI	MAP-XAI fallback	-51.851 ± 1.630	-0.019 ± 0.059	0.000 ± 0.005	0.001 ± 0.001
UCI	MAP-XAI fallback	-50.009 ± 1.863	-0.014 ± 0.071	-0.002 ± 0.004	0.001 ± 0.001
UCI	MAP-XAI fallback	-54.246 ± 2.117	-0.018 ± 0.061	-0.001 ± 0.006	0.002 ± 0.003
UCI	MAP-XAI fallback	-52.128 ± 1.236	-0.012 ± 0.083	-0.003 ± 0.003	0.000 ± 0.003
UCI	MAP-XAI fallback	-50.276 ± 1.397	-0.056 ± 0.050	-0.002 ± 0.003	0.002 ± 0.002

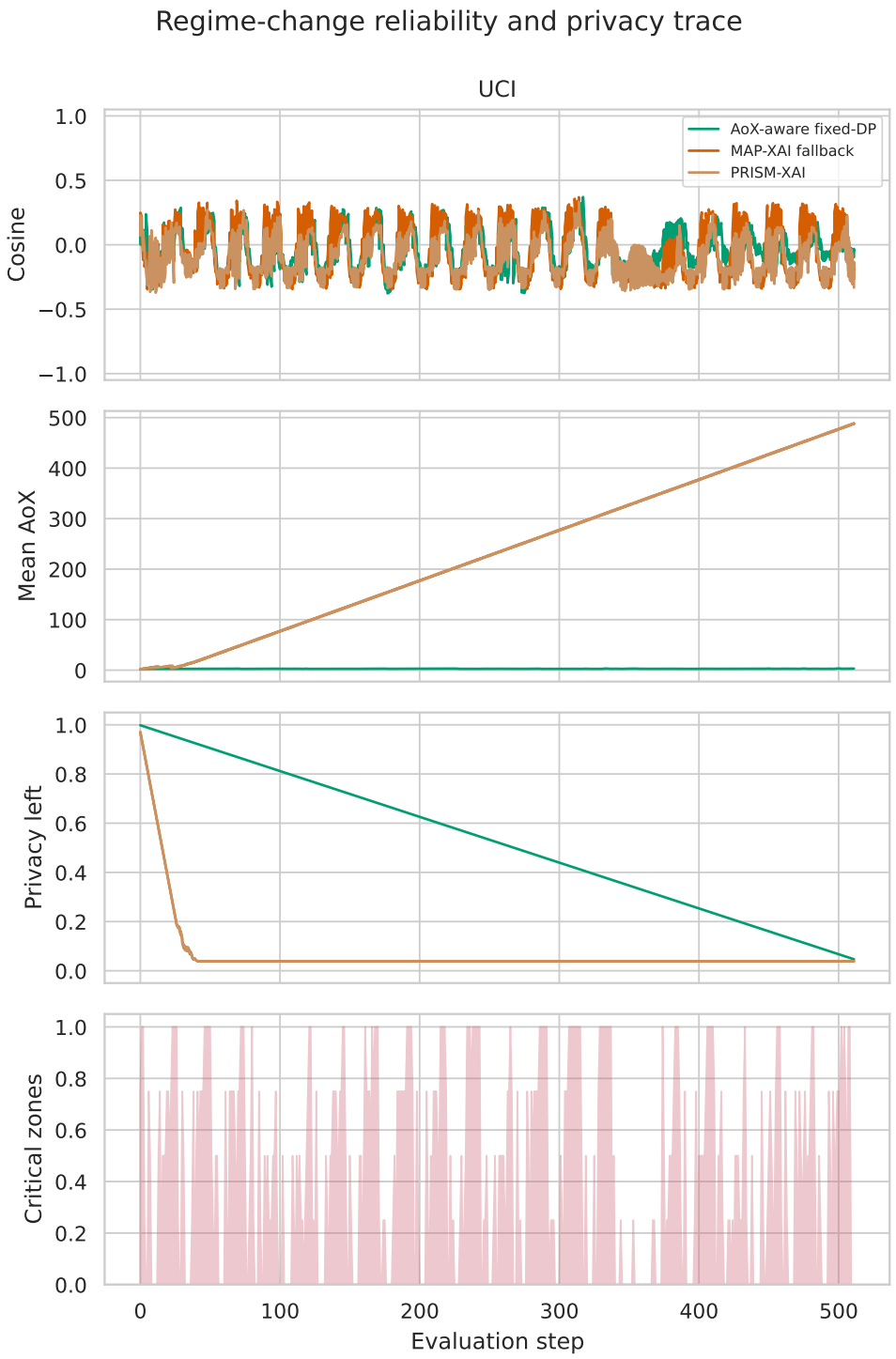


Fig. 15. Public-score threshold $\tau = 0.5$, release cap $M = 8$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

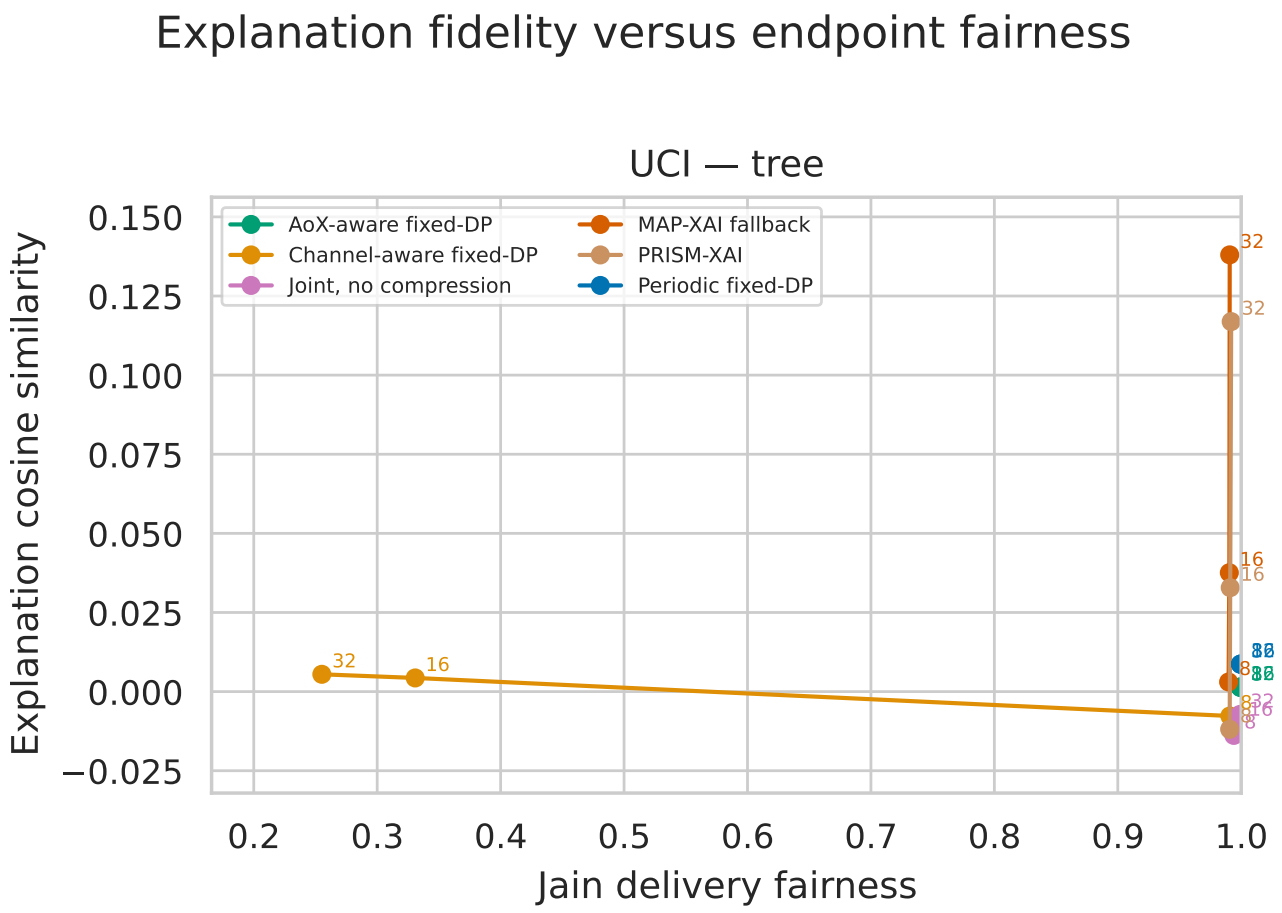


Fig. 16. Public-score threshold $\tau = 0.5$, release cap $M = 8$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

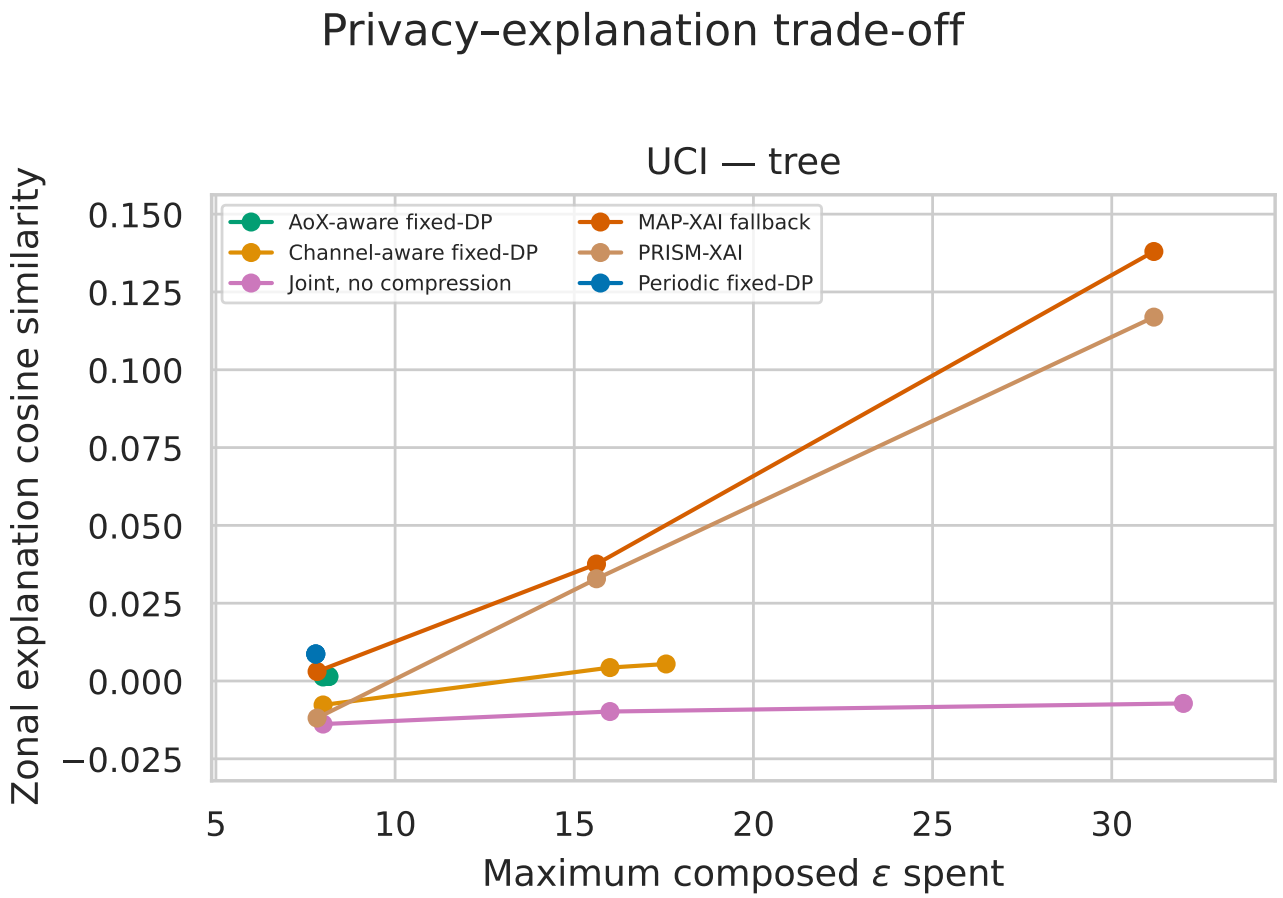


Fig. 17. Public-score threshold $\tau = 0.5$, release cap $M = 8$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

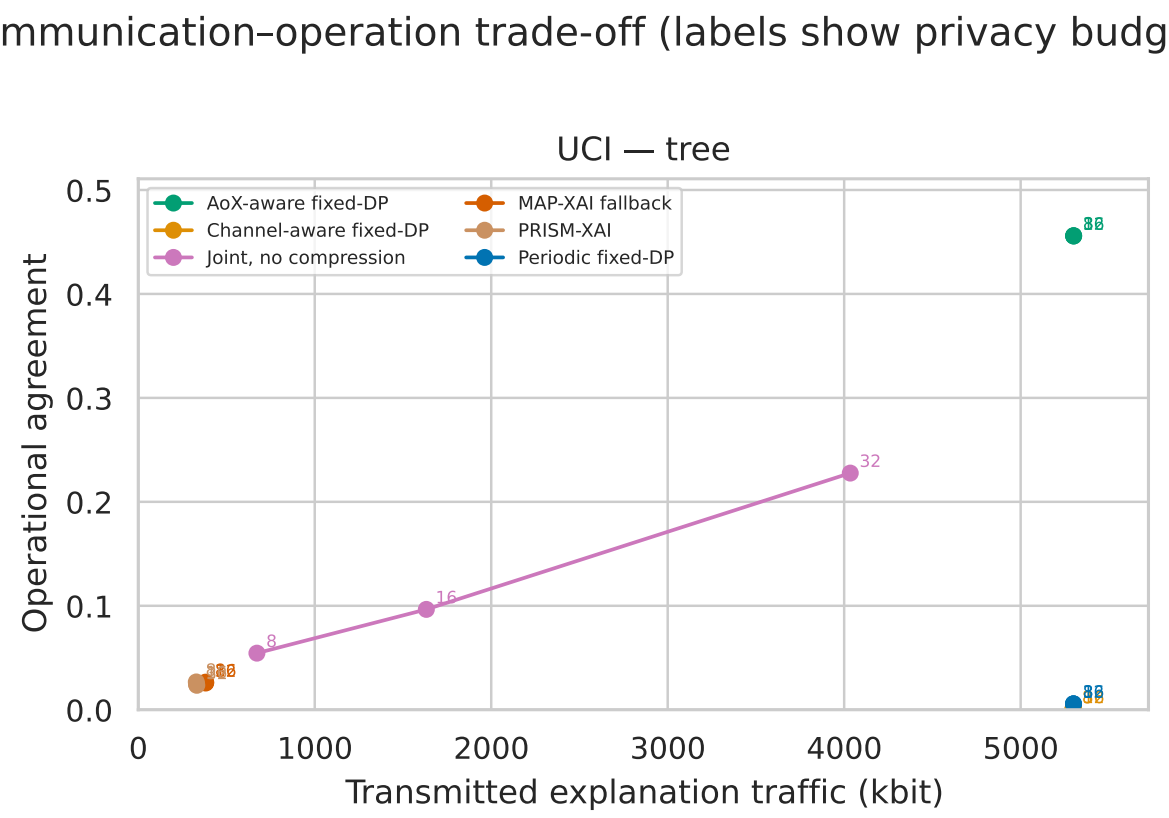


Fig. 18. Public-score threshold $\tau = 0.5$, release cap $M = 8$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

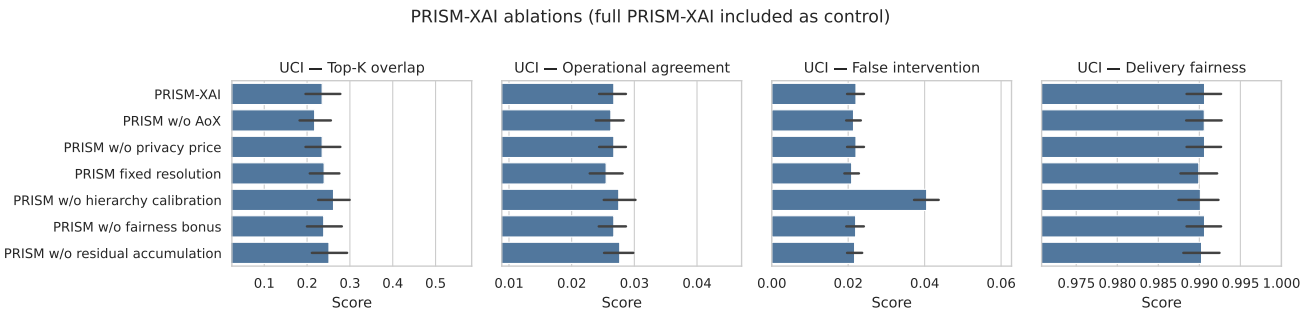


Fig. 19. Public-score threshold $\tau = 0.5$, release cap $M = 8$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

2) Public-score threshold $\tau = 1$, release cap $M = 8$: At this operating point, PRISM-XAI transmits 206.42 kbit versus 243.80 kbit for MAP-XAI-only (aggregate-mean ratio saving 15.33%). The paired difference is -37.38±3.49 kbit. PRISM coverage is 0.0510, critical-event recall is 0.0230, and maximum recorded expenditure is 18.206. Relative to MAP-XAI-only, the paired coverage change is -0.00082±0.00060 and recall change is -0.00086±0.00107. The descriptive overlap rule holds in 12 of 12 cells. Accordingly, this point is read as a matched-service scheduler comparison.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XXV

PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.271 ± 0.064	-0.008 ± 0.077	0.018 ± 0.004	0.018 ± 0.004	0.219 ± 0.073	0.956 ± 0.003	4.630 ± 1.839	87.543 ± 0.387	7.823 ± 0.000	0.026 ± 0.006	12978.492 ± 2874.959	0.993 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.269 ± 0.071	0.002 ± 0.071	0.017 ± 0.005	0.017 ± 0.005	0.217 ± 0.065	0.956 ± 0.003	4.622 ± 1.845	87.369 ± 0.346	7.823 ± 0.000	0.024 ± 0.005	13859.274 ± 3369.246	0.994 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.225 ± 0.063	-0.029 ± 0.071	0.015 ± 0.006	0.015 ± 0.006	0.190 ± 0.076	0.956 ± 0.003	4.732 ± 1.776	105.817 ± 0.049	7.823 ± 0.000	0.026 ± 0.006	22620.844 ± 10124.929	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.251 ± 0.064	-0.005 ± 0.077	0.017 ± 0.006	0.017 ± 0.006	0.217 ± 0.076	0.956 ± 0.003	4.732 ± 1.776	105.675 ± 0.049	7.823 ± 0.000	0.024 ± 0.006	18265.481 ± 6339.471	0.993 ± 0.004
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.228 ± 0.051	-0.001 ± 0.021	0.031 ± 0.005	0.031 ± 0.005	0.433 ± 0.052	0.933 ± 0.005	4.578 ± 1.800	323.463 ± 1.280	7.823 ± 0.000	0.029 ± 0.005	12910.343 ± 2005.939	0.992 ± 0.002
UCI	Tree	8	PRISM-XAI	0.219 ± 0.034	-0.052 ± 0.048	0.025 ± 0.003	0.025 ± 0.003	0.390 ± 0.053	0.940 ± 0.005	4.798 ± 1.710	322.302 ± 1.580	7.823 ± 0.000	0.028 ± 0.004	15856.808 ± 2743.852	0.990 ± 0.005
UCI	Tree	8	MAP-XAI fallback	0.220 ± 0.055	-0.006 ± 0.023	0.032 ± 0.005	0.032 ± 0.005	0.453 ± 0.058	0.932 ± 0.004	4.643 ± 1.763	382.099 ± 0.159	7.823 ± 0.000	0.027 ± 0.004	14883.660 ± 3313.232	0.992 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.250 ± 0.054	0.006 ± 0.060	0.030 ± 0.005	0.030 ± 0.005	0.453 ± 0.060	0.935 ± 0.006	5.564 ± 1.788	381.645 ± 0.090	7.823 ± 0.000	0.025 ± 0.004	15932.448 ± 2964.700	0.990 ± 0.005

TABLE XXVI

PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.543 ± 0.387	0.271 ± 0.064	0.250 ± 0.000	-0.008 ± 0.077	0.018 ± 0.004	0.018 ± 0.004	0.219 ± 0.073	0.044 ± 0.003	0.956 ± 0.003	0.026 ± 0.006	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.369 ± 0.346	0.269 ± 0.071	0.250 ± 0.000	0.002 ± 0.071	0.017 ± 0.005	0.017 ± 0.005	0.217 ± 0.065	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.005	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.513 ± 0.398	0.282 ± 0.051	0.250 ± 0.000	0.088 ± 0.071	0.018 ± 0.005	0.018 ± 0.005	0.230 ± 0.080	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.006	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.369 ± 0.313	0.275 ± 0.076	0.250 ± 0.000	0.136 ± 0.053	0.018 ± 0.006	0.018 ± 0.006	0.236 ± 0.080	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.006	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.482 ± 0.366	0.365 ± 0.069	0.250 ± 0.000	0.225 ± 0.051	0.022 ± 0.005	0.022 ± 0.005	0.373 ± 0.132	0.044 ± 0.003	0.956 ± 0.003	0.015 ± 0.005	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.185 ± 0.219	0.398 ± 0.060	0.250 ± 0.000	0.291 ± 0.044	0.022 ± 0.005	0.022 ± 0.005	0.365 ± 0.152	0.044 ± 0.003	0.956 ± 0.003	0.017 ± 0.006	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	323.463 ± 1.280	0.228 ± 0.051	0.250 ± 0.000	-0.001 ± 0.021	0.031 ± 0.005	0.031 ± 0.005	0.433 ± 0.052	0.067 ± 0.005	0.933 ± 0.005	0.029 ± 0.005	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	322.302 ± 1.580	0.219 ± 0.034	0.250 ± 0.000	-0.052 ± 0.048	0.025 ± 0.003	0.025 ± 0.003	0.390 ± 0.053	0.060 ± 0.005	0.940 ± 0.005	0.028 ± 0.004	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	323.309 ± 1.557	0.217 ± 0.067	0.250 ± 0.000	0.003 ± 0.026	0.032 ± 0.005	0.032 ± 0.005	0.526 ± 0.039	0.066 ± 0.005	0.934 ± 0.005	0.020 ± 0.005	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	326.600 ± 1.887	0.258 ± 0.051	0.250 ± 0.000	0.042 ± 0.057	0.024 ± 0.003	0.024 ± 0.003	0.530 ± 0.088	0.051 ± 0.004	0.949 ± 0.004	0.016 ± 0.005	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	328.006 ± 2.663	0.282 ± 0.083	0.250 ± 0.000	0.020 ± 0.013	0.026 ± 0.005	0.026 ± 0.005	0.974 ± 0.028	0.057 ± 0.006	0.943 ± 0.006	0.000 ± 0.001	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	328.933 ± 1.490	0.315 ± 0.048	0.250 ± 0.000	0.135 ± 0.042	0.023 ± 0.002	0.023 ± 0.002	0.909 ± 0.094	0.047 ± 0.002	0.953 ± 0.002	0.002 ± 0.002	31.175 ± 0.000

TABLE XXVII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1473.021 ± 0.387	0.037 ± 0.076	-0.102 ± 0.029	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5012.400 ± 5.552	0.029 ± 0.064	-0.383 ± 0.074	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5212.742 ± 0.366	0.112 ± 0.088	-0.391 ± 0.079	-0.006 ± 0.005
OPSD	AoX-aware fixed-DP	-1473.195 ± 0.346	0.030 ± 0.077	-0.104 ± 0.026	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5012.544 ± 5.514	0.023 ± 0.077	-0.396 ± 0.057	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5213.039 ± 0.219	0.147 ± 0.078	-0.403 ± 0.059	-0.006 ± 0.005
OPSD	Channel-aware fixed-DP	-1473.021 ± 0.387	-0.021 ± 0.081	-0.089 ± 0.026	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4599.097 ± 12.954	0.021 ± 0.063	-0.383 ± 0.090	0.056 ± 0.006
OPSD	Channel-aware fixed-DP	-5212.742 ± 0.366	0.107 ± 0.084	-0.413 ± 0.099	0.087 ± 0.007
OPSD	Channel-aware fixed-DP	-1473.195 ± 0.346	-0.024 ± 0.075	-0.094 ± 0.023	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4599.240 ± 12.910	0.017 ± 0.079	-0.401 ± 0.088	0.056 ± 0.006
OPSD	Channel-aware fixed-DP	-5213.039 ± 0.219	0.141 ± 0.071	-0.434 ± 0.092	0.087 ± 0.007
OPSD	Joint, no compression	-102.357 ± 0.507	-0.017 ± 0.096	0.001 ± 0.004	-0.002 ± 0.003
OPSD	Joint, no compression	-372.387 ± 1.050	0.051 ± 0.071	-0.020 ± 0.010	-0.004 ± 0.004
OPSD	Joint, no compression	-1053.418 ± 7.970	0.129 ± 0.123	-0.052 ± 0.014	-0.005 ± 0.005
OPSD	Joint, no compression	-102.381 ± 0.421	0.010 ± 0.097	-0.002 ± 0.003	-0.002 ± 0.003
OPSD	Joint, no compression	-372.681 ± 0.901	0.035 ± 0.072	-0.020 ± 0.008	-0.004 ± 0.004
OPSD	Joint, no compression	-1053.715 ± 7.867	0.116 ± 0.104	-0.058 ± 0.012	-0.005 ± 0.005
OPSD	Periodic fixed-DP	-1473.021 ± 0.387	0.031 ± 0.076	-0.107 ± 0.021	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5025.081 ± 0.398	0.033 ± 0.059	-0.432 ± 0.063	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5212.354 ± 0.641	0.116 ± 0.083	-0.436 ± 0.068	-0.007 ± 0.005
OPSD	Periodic fixed-DP	-1473.195 ± 0.346	0.027 ± 0.083	-0.118 ± 0.016	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5025.225 ± 0.313	0.032 ± 0.072	-0.444 ± 0.049	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5212.651 ± 0.592	0.155 ± 0.065	-0.449 ± 0.050	-0.007 ± 0.005
OPSD	MAP-XAI fallback	-18.273 ± 0.401	0.046 ± 0.070	0.002 ± 0.003	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.304 ± 0.409	0.038 ± 0.045	0.001 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.335 ± 0.374	0.013 ± 0.060	-0.002 ± 0.005	-0.000 ± 0.001
OPSD	MAP-XAI fallback	-18.306 ± 0.346	0.018 ± 0.054	0.000 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.285 ± 0.309	0.011 ± 0.065	-0.000 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.367 ± 0.250	0.022 ± 0.068	-0.002 ± 0.003	-0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4976.761 ± 1.280	-0.036 ± 0.068	-0.426 ± 0.033	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4976.915 ± 1.557	-0.047 ± 0.088	-0.425 ± 0.031	-0.007 ± 0.003
UCI	AoX-aware fixed-DP	-4972.218 ± 2.663	0.018 ± 0.072	-0.431 ± 0.033	-0.008 ± 0.004
UCI	AoX-aware fixed-DP	-4977.922 ± 1.580	-0.041 ± 0.037	-0.430 ± 0.047	-0.009 ± 0.005
UCI	AoX-aware fixed-DP	-4973.624 ± 1.887	-0.002 ± 0.064	-0.430 ± 0.047	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4971.291 ± 1.490	0.054 ± 0.058	-0.431 ± 0.047	-0.007 ± 0.002
UCI	Channel-aware fixed-DP	-4971.715 ± 4.160	-0.020 ± 0.068	0.031 ± 0.005	0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4976.915 ± 1.557	-0.051 ± 0.075	0.031 ± 0.005	0.662 ± 0.003
UCI	Channel-aware fixed-DP	-4972.218 ± 2.663	0.013 ± 0.098	0.025 ± 0.005	0.736 ± 0.004
UCI	Channel-aware fixed-DP	-4972.875 ± 4.681	-0.020 ± 0.051	0.025 ± 0.003	-0.001 ± 0.005
UCI	Channel-aware fixed-DP	-4973.624 ± 1.887	-0.002 ± 0.063	0.024 ± 0.003	0.661 ± 0.002
UCI	Channel-aware fixed-DP	-4971.291 ± 1.490	0.055 ± 0.057	0.023 ± 0.002	0.737 ± 0.003
UCI	Joint, no compression	-348.837 ± 1.583	-0.058 ± 0.074	-0.021 ± 0.005	-0.002 ± 0.004
UCI	Joint, no compression	-1308.841 ± 1.426	0.000 ± 0.093	-0.062 ± 0.006	-0.005 ± 0.003
UCI	Joint, no compression	-3706.543 ± 6.544	0.035 ± 0.097	-0.202 ± 0.016	-0.008 ± 0.004
UCI	Joint, no compression	-349.998 ± 1.687	-0.010 ± 0.045	-0.031 ± 0.008	-0.004 ± 0.005
UCI	Joint, no compression	-1306.000 ± 2.008	0.054 ± 0.052	-0.075 ± 0.017	-0.006 ± 0.002
UCI	Joint, no compression	-3705.617 ± 4.600	0.073 ± 0.055	-0.204 ± 0.029	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4975.855 ± 1.998	-0.032 ± 0.088	0.025 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4976.009 ± 2.472	-0.043 ± 0.092	0.026 ± 0.005	-0.007 ± 0.003
UCI	Periodic fixed-DP	-4971.312 ± 3.350	0.022 ± 0.066	0.020 ± 0.005	-0.008 ± 0.004
UCI	Periodic fixed-DP	-4977.016 ± 2.100	-0.041 ± 0.062	0.020 ± 0.004	-0.009 ± 0.005
UCI	Periodic fixed-DP	-4972.718 ± 2.453	-0.002 ± 0.064	0.019 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4970.385 ± 1.709	0.054 ± 0.055	0.018 ± 0.002	-0.008 ± 0.002
UCI	MAP-XAI fallback	-58.636 ± 1.168	0.008 ± 0.063	-0.001 ± 0.003	-0.000 ± 0.001
UCI	MAP-XAI fallback	-58.851 ± 1.541	0.016 ± 0.083	0.000 ± 0.005	0.001 ± 0.001
UCI	MAP-XAI fallback	-54.154 ± 2.660	-0.011 ± 0.130	-0.000 ± 0.003	0.000 ± 0.002
UCI	MAP-XAI fallback	-59.343 ± 1.639	-0.031 ± 0.059	-0.005 ± 0.004	-0.000 ± 0.003
UCI	MAP-XAI fallback	-55.045 ± 1.932	0.026 ± 0.053	-0.000 ± 0.005	0.002 ± 0.003
UCI	MAP-XAI fallback	-52.609 ± 1.452	-0.000 ± 0.070	-0.003 ± 0.003	0.001 ± 0.001

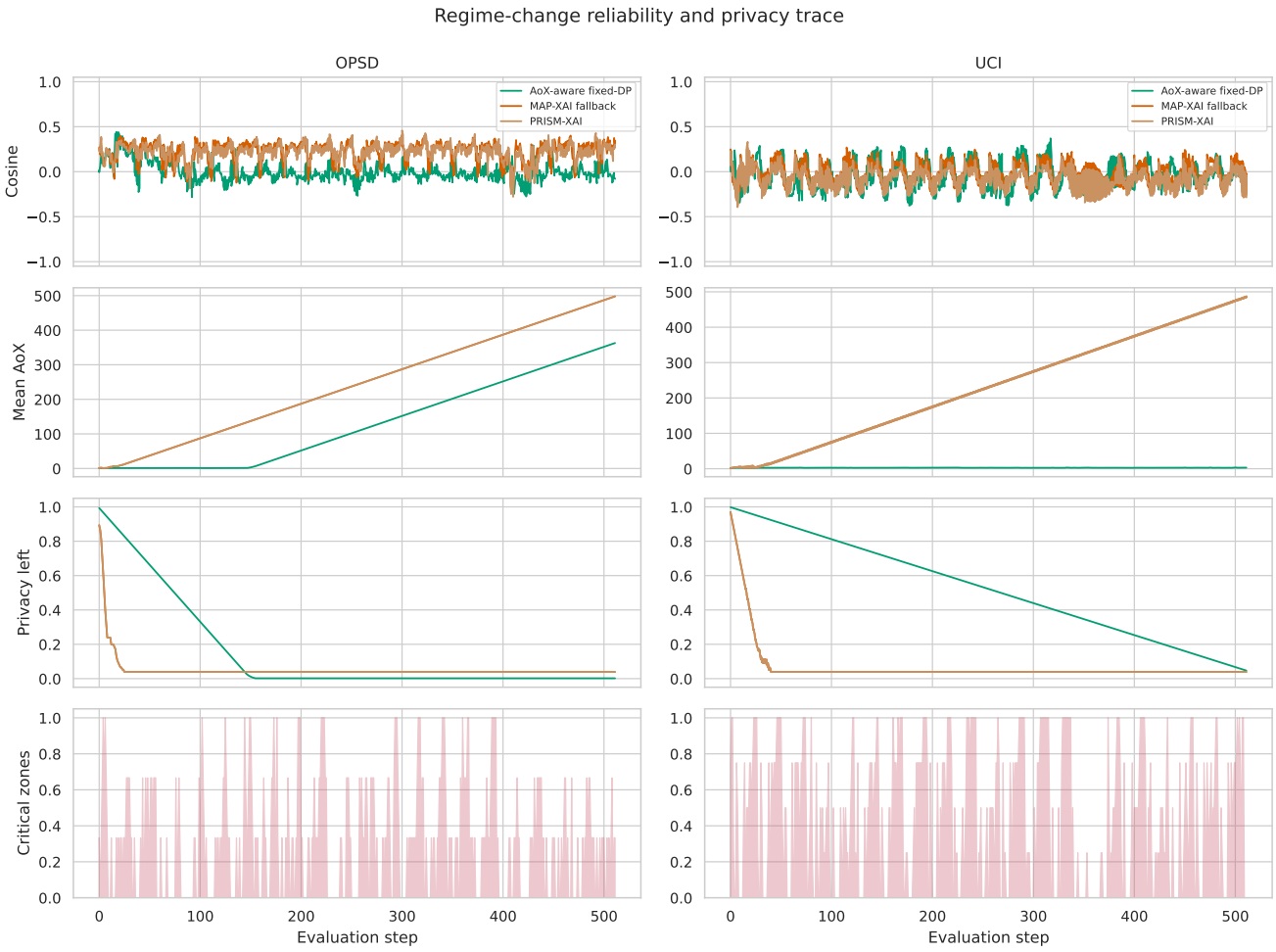


Fig. 20. Public-score threshold $\tau = 1$, release cap $M = 8$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

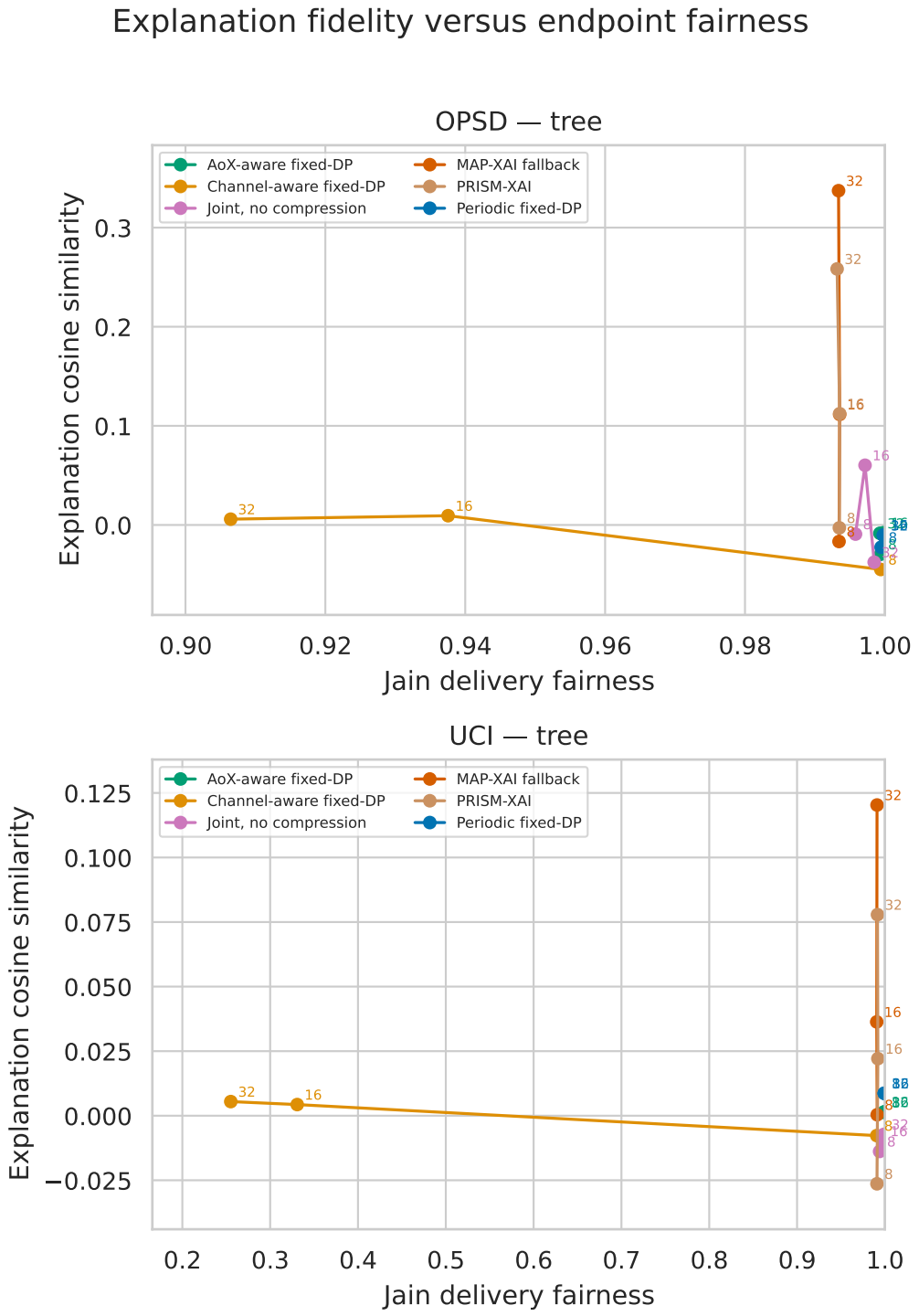


Fig. 21. Public-score threshold $\tau = 1$, release cap $M = 8$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

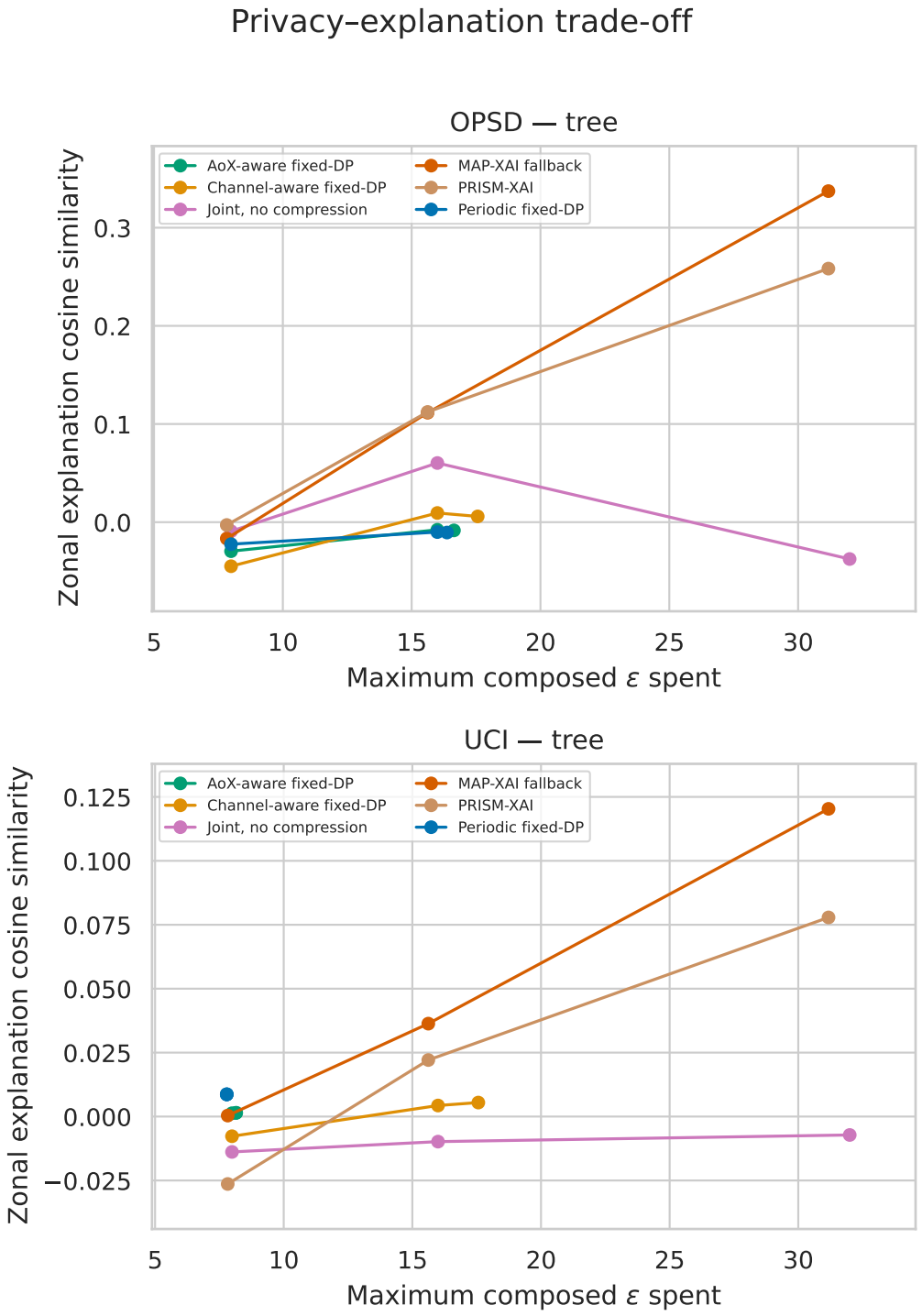


Fig. 22. Public-score threshold $\tau = 1$, release cap $M = 8$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

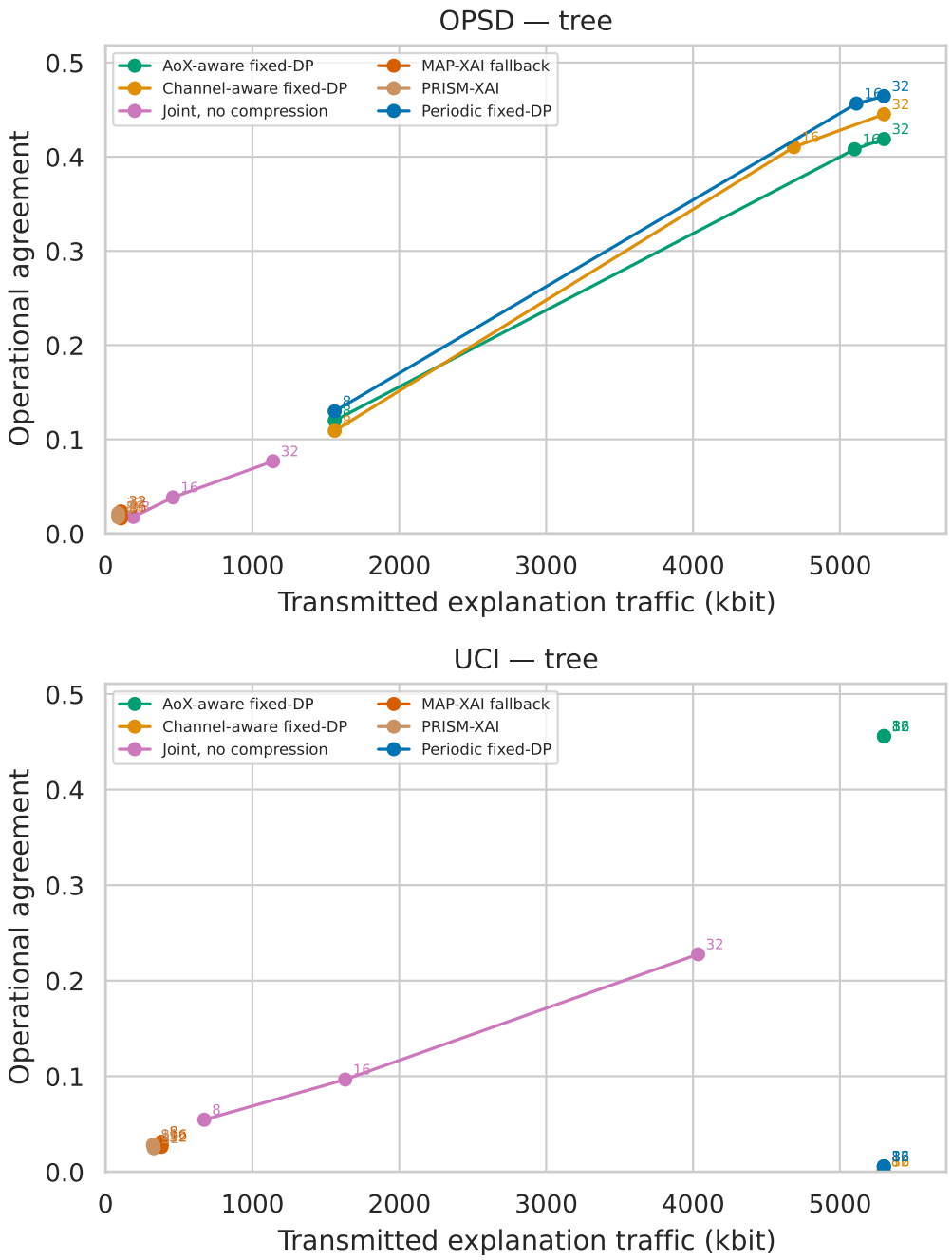


Fig. 23. Public-score threshold $\tau = 1$, release cap $M = 8$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

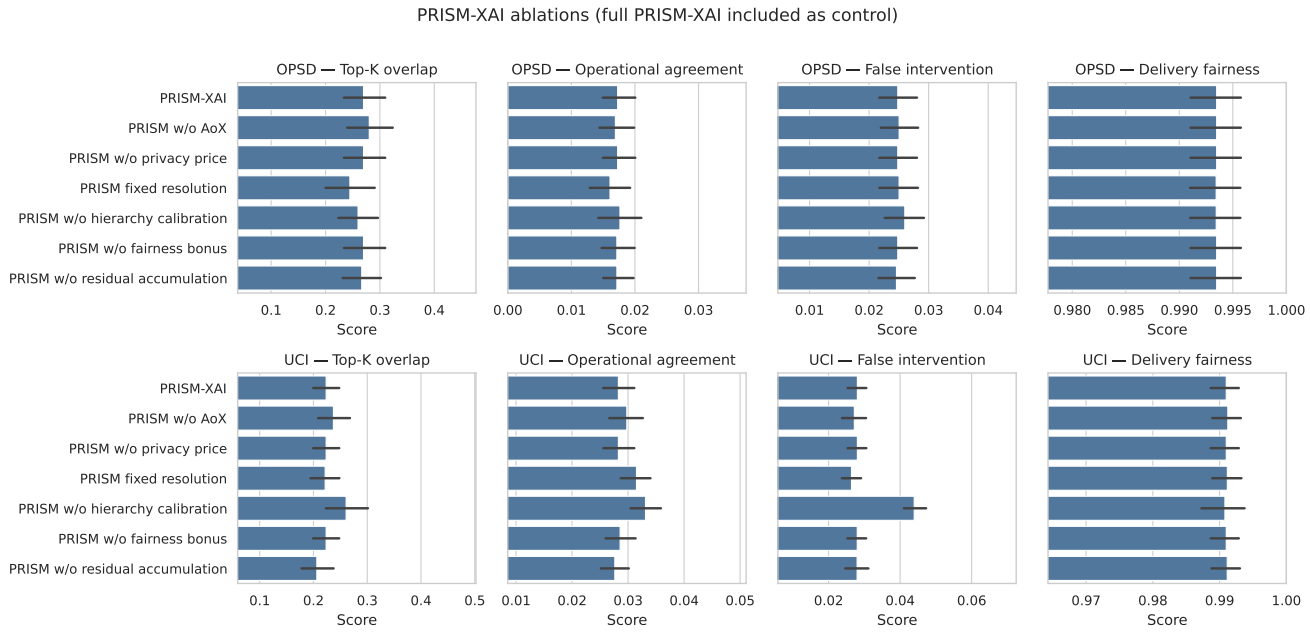


Fig. 24. Public-score threshold $\tau = 1$, release cap $M = 8$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XXVIII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-$\mathcal{K}$	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.271 ± 0.064	-0.008 ± 0.077	0.018 ± 0.004	0.018 ± 0.004	0.219 ± 0.073	0.956 ± 0.003	4.630 ± 1.839	87.543 ± 0.387	7.823 ± 0.000	0.026 ± 0.006	12978.492 ± 2874.959	0.993 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.269 ± 0.071	0.002 ± 0.071	0.017 ± 0.005	0.017 ± 0.005	0.217 ± 0.065	0.956 ± 0.003	4.622 ± 1.845	87.369 ± 0.346	7.823 ± 0.000	0.024 ± 0.005	13859.274 ± 3369.246	0.994 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.225 ± 0.063	-0.029 ± 0.071	0.015 ± 0.006	0.015 ± 0.006	0.190 ± 0.076	0.956 ± 0.003	4.732 ± 1.776	105.817 ± 0.049	7.823 ± 0.000	0.026 ± 0.006	22620.844 ± 10124.929	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.251 ± 0.064	-0.005 ± 0.077	0.017 ± 0.006	0.017 ± 0.006	0.217 ± 0.076	0.956 ± 0.003	4.732 ± 1.776	105.675 ± 0.049	7.823 ± 0.000	0.024 ± 0.006	18265.481 ± 6339.471	0.993 ± 0.004

TABLE XXIX
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.543 ± 0.387	0.271 ± 0.064	0.250 ± 0.000	-0.008 ± 0.077	0.018 ± 0.004	0.018 ± 0.004	0.219 ± 0.073	0.044 ± 0.003	0.956 ± 0.003	0.026 ± 0.006	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.369 ± 0.346	0.269 ± 0.071	0.250 ± 0.000	0.002 ± 0.071	0.017 ± 0.005	0.017 ± 0.005	0.217 ± 0.065	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.005	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.513 ± 0.398	0.282 ± 0.051	0.250 ± 0.000	0.088 ± 0.071	0.018 ± 0.005	0.018 ± 0.005	0.230 ± 0.080	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.006	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.369 ± 0.313	0.275 ± 0.076	0.250 ± 0.000	0.136 ± 0.053	0.018 ± 0.006	0.018 ± 0.006	0.236 ± 0.080	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.006	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.482 ± 0.366	0.365 ± 0.069	0.250 ± 0.000	0.225 ± 0.051	0.022 ± 0.005	0.022 ± 0.005	0.373 ± 0.132	0.044 ± 0.003	0.956 ± 0.003	0.015 ± 0.005	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.185 ± 0.219	0.398 ± 0.060	0.250 ± 0.000	0.291 ± 0.044	0.022 ± 0.005	0.022 ± 0.005	0.365 ± 0.152	0.044 ± 0.003	0.956 ± 0.003	0.017 ± 0.006	31.175 ± 0.000

TABLE XXX
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1473.021 ± 0.387	0.037 ± 0.076	-0.102 ± 0.029	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5012.400 ± 5.552	0.029 ± 0.064	-0.383 ± 0.074	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5212.742 ± 0.366	0.112 ± 0.088	-0.391 ± 0.079	-0.006 ± 0.005
OPSD	AoX-aware fixed-DP	-1473.195 ± 0.346	0.030 ± 0.077	-0.104 ± 0.026	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5012.544 ± 5.514	0.023 ± 0.077	-0.396 ± 0.057	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-5213.039 ± 0.219	0.147 ± 0.078	-0.403 ± 0.059	-0.006 ± 0.005
OPSD	Channel-aware fixed-DP	-1473.021 ± 0.387	-0.021 ± 0.081	-0.089 ± 0.026	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4599.097 ± 12.954	0.021 ± 0.063	-0.383 ± 0.090	0.056 ± 0.006
OPSD	Channel-aware fixed-DP	-5212.742 ± 0.366	0.107 ± 0.084	-0.413 ± 0.099	0.087 ± 0.007
OPSD	Channel-aware fixed-DP	-1473.195 ± 0.346	-0.024 ± 0.075	-0.094 ± 0.023	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4599.240 ± 12.910	0.017 ± 0.079	-0.401 ± 0.088	0.056 ± 0.006
OPSD	Channel-aware fixed-DP	-5213.039 ± 0.219	0.141 ± 0.071	-0.434 ± 0.092	0.087 ± 0.007
OPSD	Joint, no compression	-102.357 ± 0.507	-0.017 ± 0.096	0.001 ± 0.004	-0.002 ± 0.003
OPSD	Joint, no compression	-372.387 ± 1.050	0.051 ± 0.071	-0.020 ± 0.010	-0.004 ± 0.004
OPSD	Joint, no compression	-1053.418 ± 7.970	0.129 ± 0.123	-0.052 ± 0.014	-0.005 ± 0.005
OPSD	Joint, no compression	-102.381 ± 0.421	0.010 ± 0.097	-0.002 ± 0.003	-0.002 ± 0.003
OPSD	Joint, no compression	-372.681 ± 0.901	0.035 ± 0.072	-0.020 ± 0.008	-0.004 ± 0.004
OPSD	Joint, no compression	-1053.715 ± 7.867	0.116 ± 0.104	-0.058 ± 0.012	-0.005 ± 0.005
OPSD	Periodic fixed-DP	-1473.021 ± 0.387	0.031 ± 0.076	-0.107 ± 0.021	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5025.081 ± 0.398	0.033 ± 0.059	-0.432 ± 0.063	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5212.354 ± 0.641	0.116 ± 0.083	-0.436 ± 0.068	-0.007 ± 0.005
OPSD	Periodic fixed-DP	-1473.195 ± 0.346	0.027 ± 0.083	-0.118 ± 0.016	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5025.225 ± 0.313	0.032 ± 0.072	-0.444 ± 0.049	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-5212.651 ± 0.592	0.155 ± 0.065	-0.449 ± 0.050	-0.007 ± 0.005
OPSD	MAP-XAI fallback	-18.273 ± 0.401	0.046 ± 0.070	0.002 ± 0.003	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.304 ± 0.409	0.038 ± 0.045	0.001 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.335 ± 0.374	0.013 ± 0.060	-0.002 ± 0.005	-0.000 ± 0.001
OPSD	MAP-XAI fallback	-18.306 ± 0.346	0.018 ± 0.054	0.000 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.285 ± 0.309	0.011 ± 0.065	-0.000 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.367 ± 0.250	0.022 ± 0.068	-0.002 ± 0.003	-0.000 ± 0.000

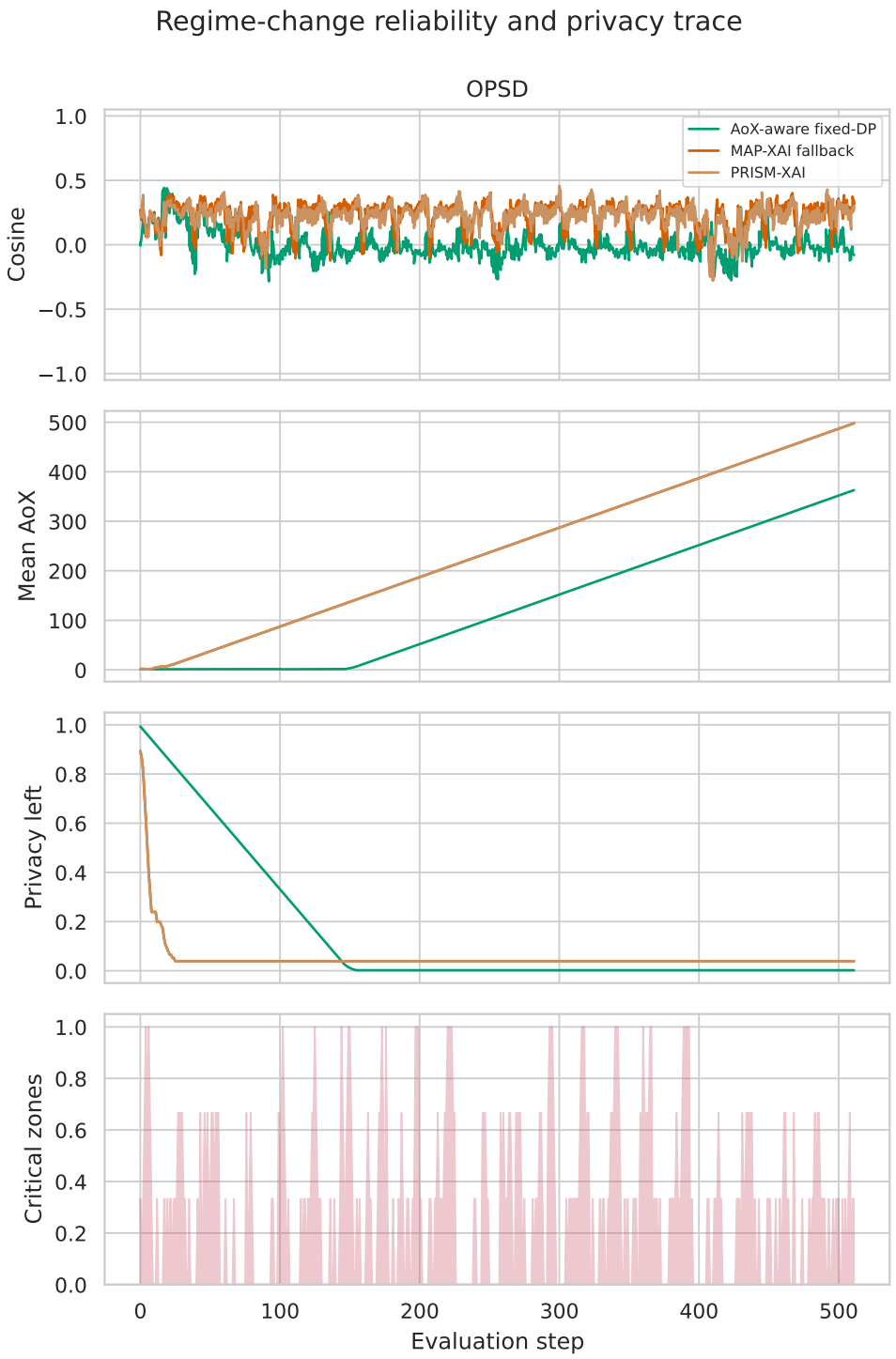


Fig. 25. Public-score threshold $\tau = 1$, release cap $M = 8$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

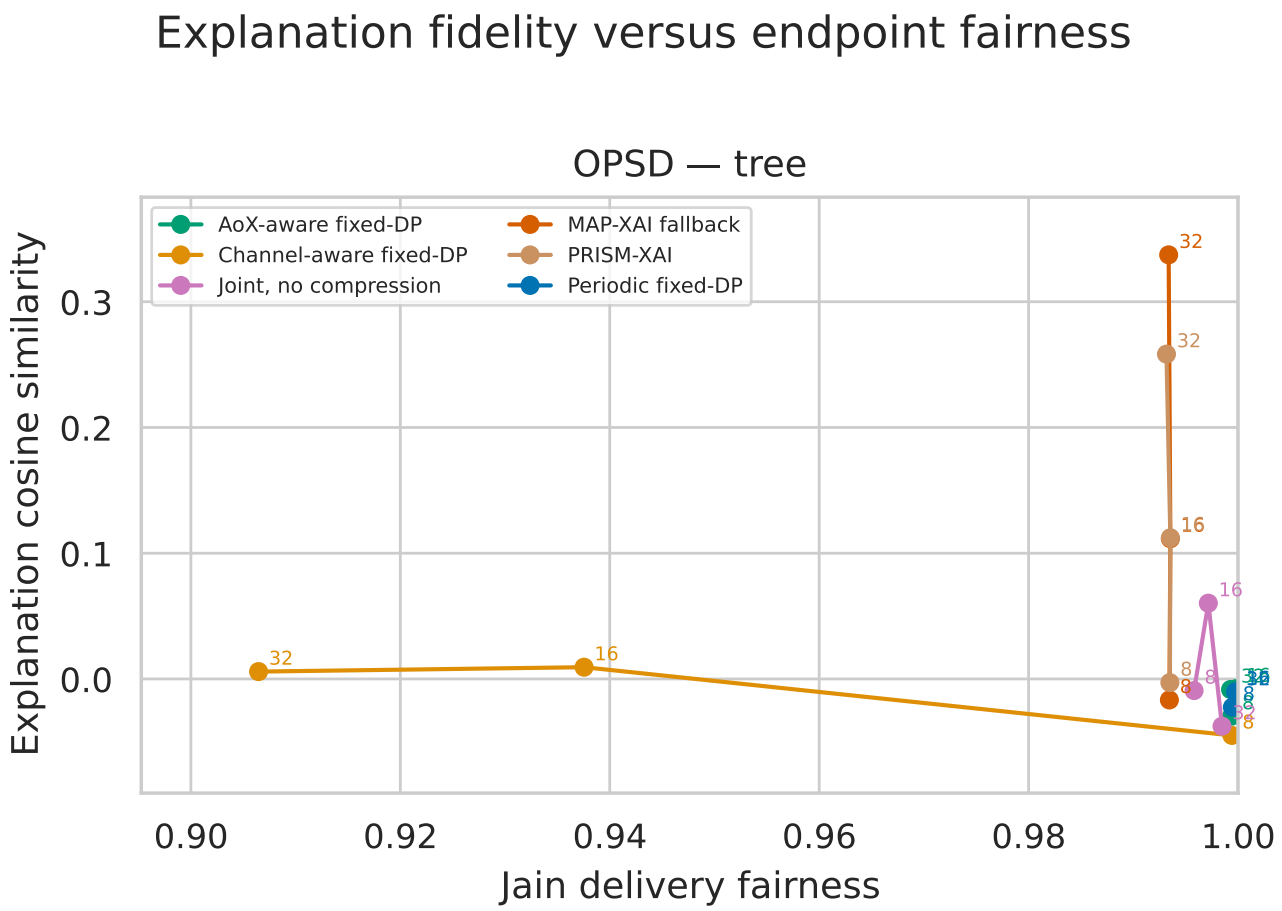


Fig. 26. Public-score threshold $\tau = 1$, release cap $M = 8$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

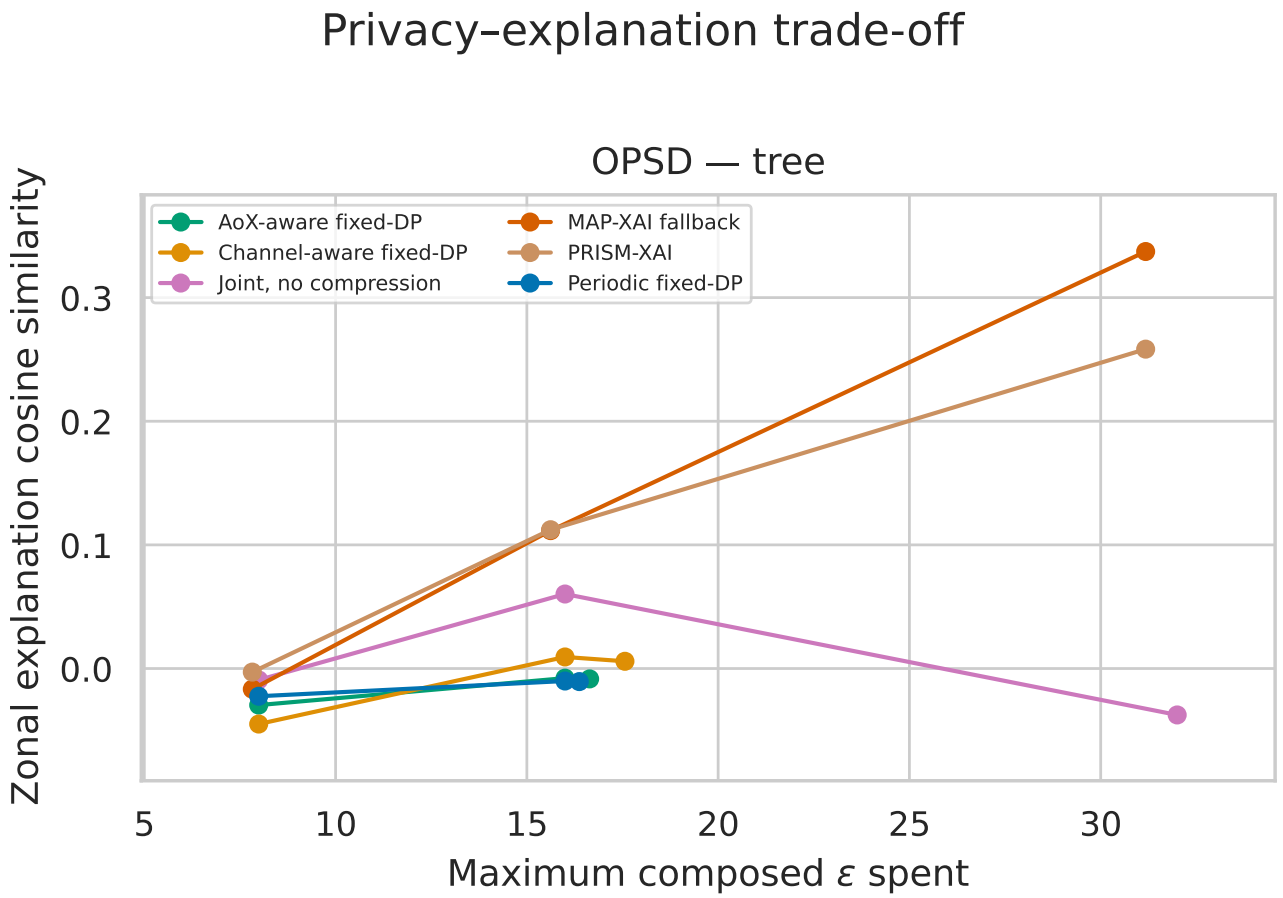


Fig. 27. Public-score threshold $\tau = 1$, release cap $M = 8$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

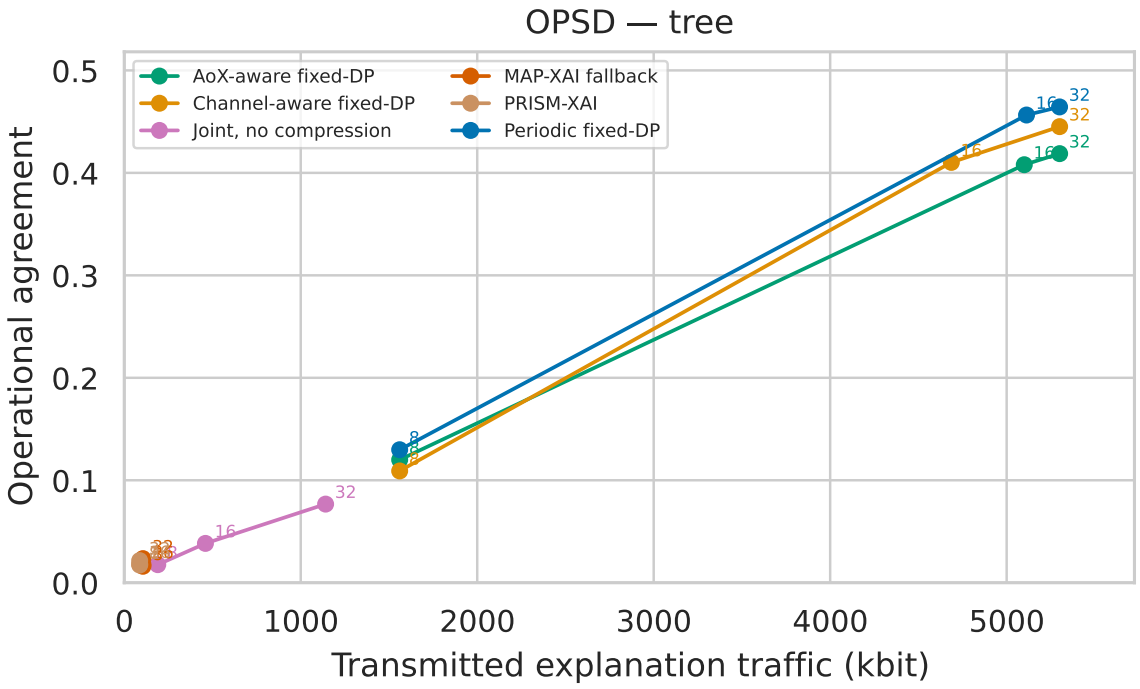


Fig. 28. Public-score threshold $\tau = 1$, release cap $M = 8$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

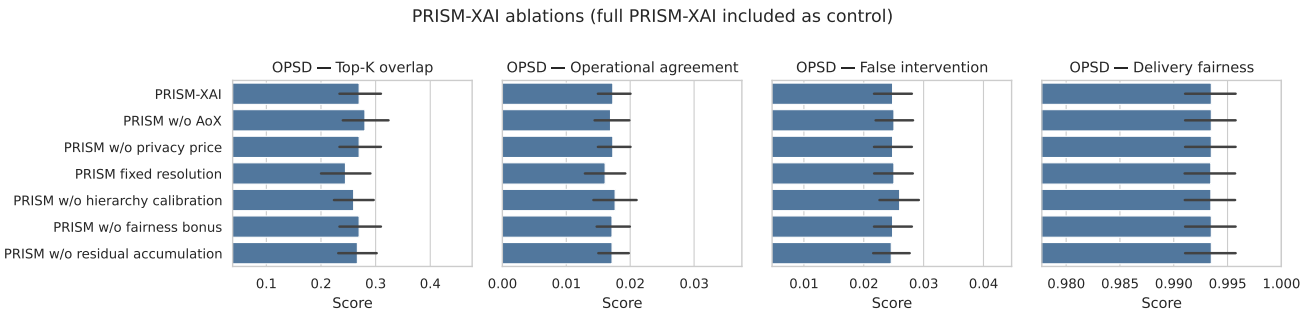


Fig. 29. Public-score threshold $\tau = 1$, release cap $M = 8$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XXXI
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.228 ± 0.051	-0.001 ± 0.021	0.031 ± 0.005	0.031 ± 0.005	0.433 ± 0.052	0.933 ± 0.005	4.578 ± 1.800	323.463 ± 1.280	7.823 ± 0.000	0.029 ± 0.005	12910.343 ± 2005.939	0.992 ± 0.002
UCI	Tree	8	PRISM-XAI	0.219 ± 0.034	-0.052 ± 0.048	0.025 ± 0.003	0.025 ± 0.003	0.390 ± 0.053	0.940 ± 0.005	4.798 ± 1.710	322.302 ± 1.580	7.823 ± 0.000	0.028 ± 0.004	15856.808 ± 2743.852	0.990 ± 0.005
UCI	Tree	8	MAP-XAI fallback	0.220 ± 0.055	-0.006 ± 0.023	0.032 ± 0.005	0.032 ± 0.005	0.453 ± 0.058	0.932 ± 0.004	4.643 ± 1.763	382.099 ± 0.159	7.823 ± 0.000	0.027 ± 0.004	14883.660 ± 3313.232	0.992 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.250 ± 0.054	0.006 ± 0.060	0.030 ± 0.005	0.030 ± 0.005	0.453 ± 0.060	0.935 ± 0.006	5.564 ± 1.788	381.645 ± 0.090	7.823 ± 0.000	0.025 ± 0.004	15932.448 ± 2964.700	0.990 ± 0.005

TABLE XXXII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	323.463 ± 1.280	0.228 ± 0.051	0.250 ± 0.000	-0.001 ± 0.021	0.031 ± 0.005	0.031 ± 0.005	0.433 ± 0.052	0.067 ± 0.005	0.933 ± 0.005	0.029 ± 0.005	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	322.302 ± 1.580	0.219 ± 0.034	0.250 ± 0.000	-0.052 ± 0.048	0.025 ± 0.003	0.025 ± 0.003	0.390 ± 0.053	0.060 ± 0.005	0.940 ± 0.005	0.028 ± 0.004	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	323.309 ± 1.557	0.217 ± 0.067	0.250 ± 0.000	0.003 ± 0.026	0.032 ± 0.005	0.032 ± 0.005	0.526 ± 0.039	0.066 ± 0.005	0.934 ± 0.005	0.020 ± 0.005	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	326.600 ± 1.887	0.258 ± 0.051	0.250 ± 0.000	0.042 ± 0.057	0.024 ± 0.003	0.024 ± 0.003	0.530 ± 0.088	0.051 ± 0.004	0.949 ± 0.004	0.016 ± 0.005	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	328.006 ± 2.663	0.282 ± 0.083	0.250 ± 0.000	0.020 ± 0.013	0.026 ± 0.005	0.026 ± 0.005	0.974 ± 0.028	0.057 ± 0.006	0.943 ± 0.006	0.000 ± 0.001	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	328.933 ± 1.490	0.315 ± 0.048	0.250 ± 0.000	0.135 ± 0.042	0.023 ± 0.002	0.023 ± 0.002	0.909 ± 0.094	0.047 ± 0.002	0.953 ± 0.002	0.002 ± 0.002	31.175 ± 0.000

TABLE XXXIII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4976.761 ± 1.280	-0.036 ± 0.068	-0.426 ± 0.033	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4976.915 ± 1.557	-0.047 ± 0.088	-0.425 ± 0.031	-0.007 ± 0.003
UCI	AoX-aware fixed-DP	-4972.218 ± 2.663	0.018 ± 0.072	-0.431 ± 0.033	-0.008 ± 0.004
UCI	AoX-aware fixed-DP	-4977.922 ± 1.580	-0.041 ± 0.037	-0.430 ± 0.047	-0.009 ± 0.005
UCI	AoX-aware fixed-DP	-4973.624 ± 1.887	-0.002 ± 0.064	-0.430 ± 0.047	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4971.291 ± 1.490	0.054 ± 0.058	-0.431 ± 0.047	-0.007 ± 0.002
UCI	Channel-aware fixed-DP	-4971.715 ± 4.160	-0.020 ± 0.068	0.031 ± 0.005	0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4976.915 ± 1.557	-0.051 ± 0.075	0.031 ± 0.005	0.662 ± 0.003
UCI	Channel-aware fixed-DP	-4972.218 ± 2.663	0.013 ± 0.098	0.025 ± 0.005	0.736 ± 0.004
UCI	Channel-aware fixed-DP	-4972.875 ± 4.681	-0.020 ± 0.051	0.025 ± 0.003	-0.001 ± 0.005
UCI	Channel-aware fixed-DP	-4973.624 ± 1.887	-0.002 ± 0.063	0.024 ± 0.003	0.661 ± 0.002
UCI	Channel-aware fixed-DP	-4971.291 ± 1.490	0.055 ± 0.057	0.023 ± 0.002	0.737 ± 0.003
UCI	Joint, no compression	-348.837 ± 1.583	-0.058 ± 0.074	-0.021 ± 0.005	-0.002 ± 0.004
UCI	Joint, no compression	-1308.841 ± 1.426	0.000 ± 0.093	-0.062 ± 0.006	-0.005 ± 0.003
UCI	Joint, no compression	-3706.543 ± 6.544	0.035 ± 0.097	-0.202 ± 0.016	-0.008 ± 0.004
UCI	Joint, no compression	-349.998 ± 1.687	-0.010 ± 0.045	-0.031 ± 0.008	-0.004 ± 0.005
UCI	Joint, no compression	-1306.000 ± 2.008	0.054 ± 0.052	-0.075 ± 0.017	-0.006 ± 0.002
UCI	Joint, no compression	-3705.617 ± 4.600	0.073 ± 0.055	-0.204 ± 0.029	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4975.855 ± 1.998	-0.032 ± 0.088	0.025 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4976.009 ± 2.472	-0.043 ± 0.092	0.026 ± 0.005	-0.007 ± 0.003
UCI	Periodic fixed-DP	-4971.312 ± 3.350	0.022 ± 0.066	0.020 ± 0.005	-0.008 ± 0.004
UCI	Periodic fixed-DP	-4977.016 ± 2.100	-0.041 ± 0.062	0.020 ± 0.004	-0.009 ± 0.005
UCI	Periodic fixed-DP	-4972.718 ± 2.453	-0.002 ± 0.064	0.019 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4970.385 ± 1.709	0.054 ± 0.055	0.018 ± 0.002	-0.008 ± 0.002
UCI	MAP-XAI fallback	-58.636 ± 1.168	0.008 ± 0.063	-0.001 ± 0.003	-0.000 ± 0.001
UCI	MAP-XAI fallback	-58.851 ± 1.541	0.016 ± 0.083	0.000 ± 0.005	0.001 ± 0.001
UCI	MAP-XAI fallback	-54.154 ± 2.660	-0.011 ± 0.130	-0.000 ± 0.003	0.000 ± 0.002
UCI	MAP-XAI fallback	-59.343 ± 1.639	-0.031 ± 0.059	-0.005 ± 0.004	-0.000 ± 0.003
UCI	MAP-XAI fallback	-55.045 ± 1.932	0.026 ± 0.053	-0.000 ± 0.005	0.002 ± 0.003
UCI	MAP-XAI fallback	-52.609 ± 1.452	-0.000 ± 0.070	-0.003 ± 0.003	0.001 ± 0.001

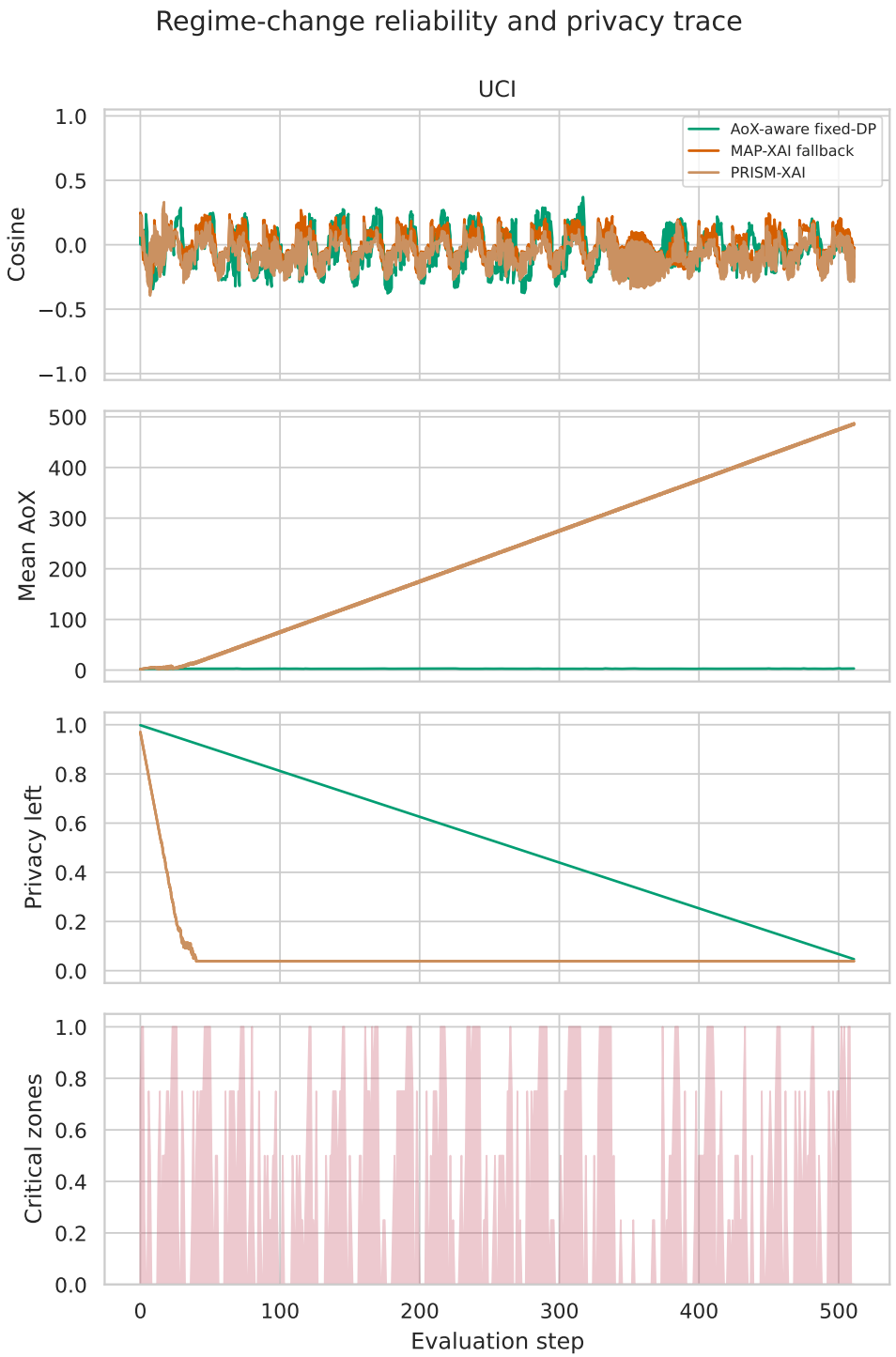


Fig. 30. Public-score threshold $\tau = 1$, release cap $M = 8$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

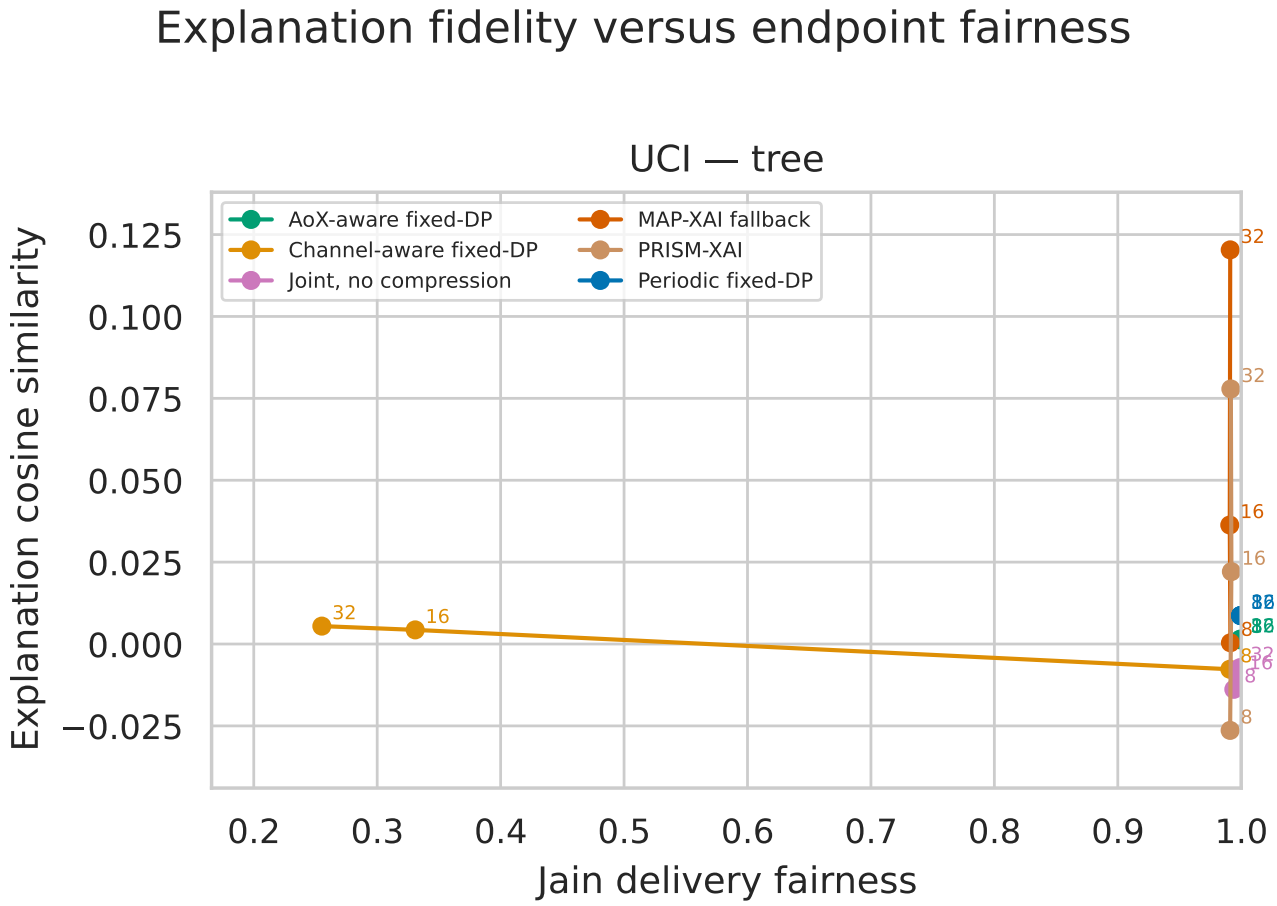


Fig. 31. Public-score threshold $\tau = 1$, release cap $M = 8$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

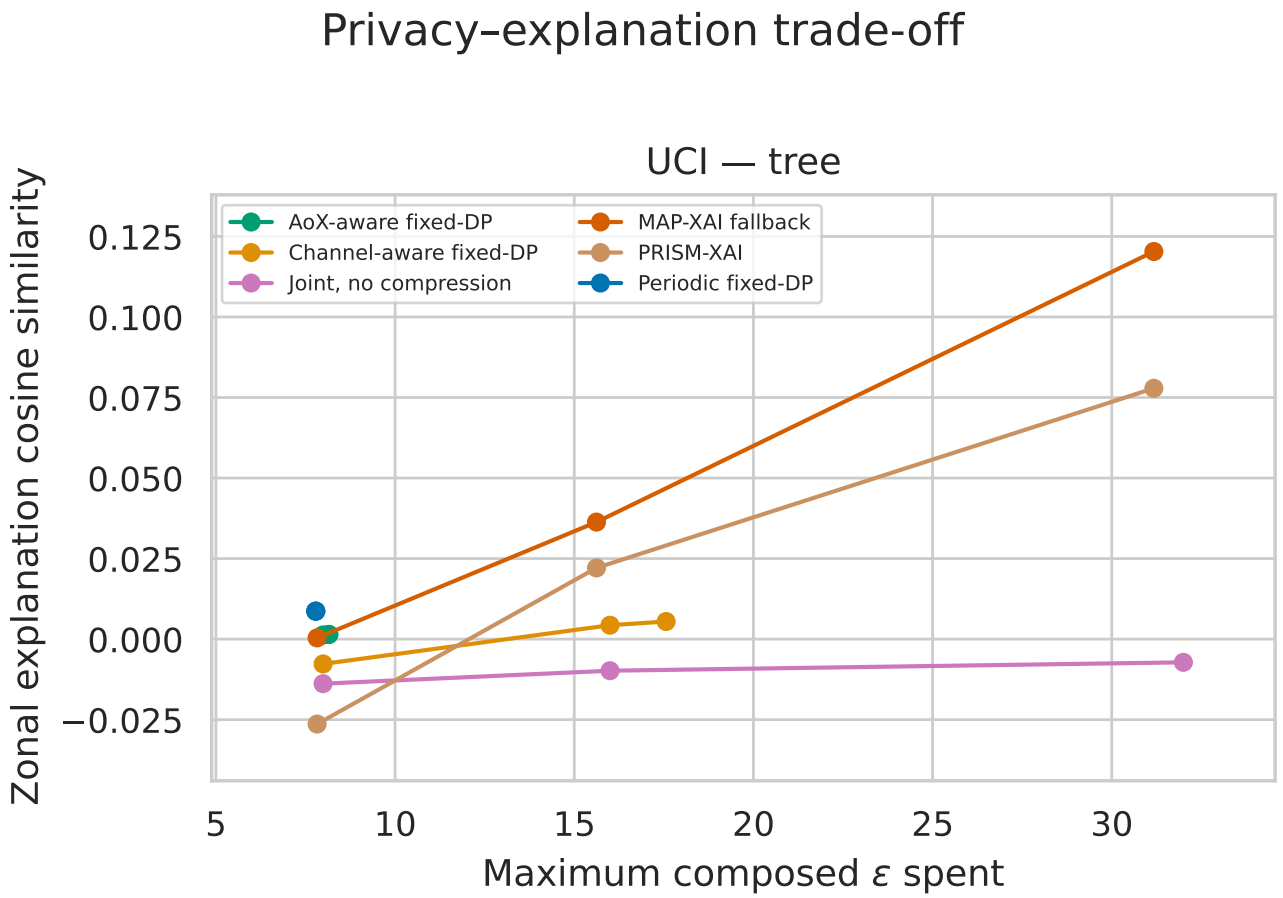


Fig. 32. Public-score threshold $\tau = 1$, release cap $M = 8$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

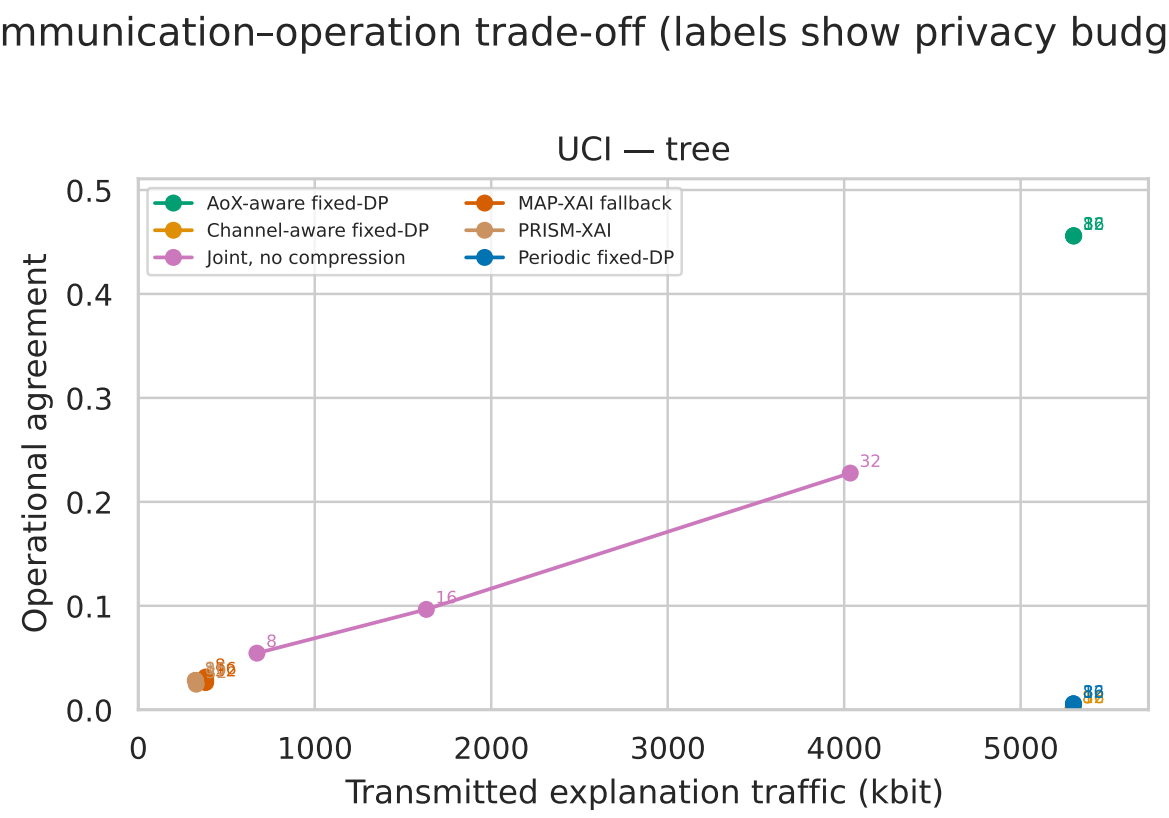


Fig. 33. Public-score threshold $\tau = 1$, release cap $M = 8$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

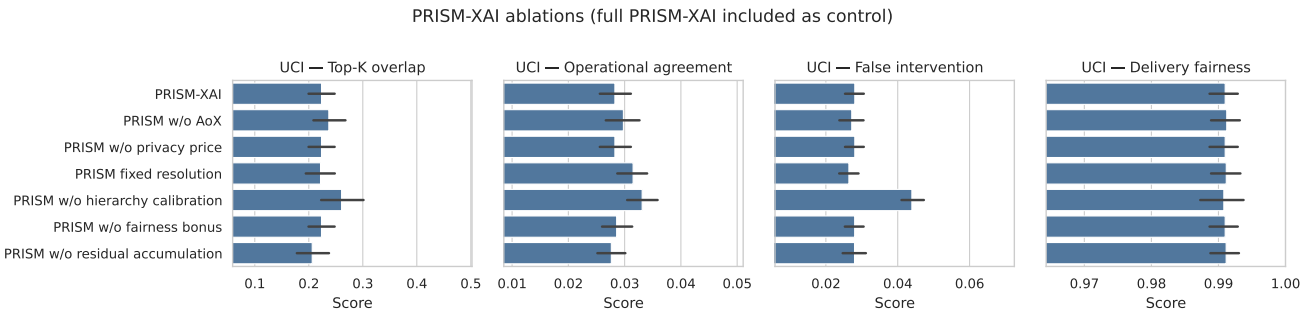


Fig. 34. Public-score threshold $\tau = 1$, release cap $M = 8$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

3) Public-score threshold $\tau = 2$, release cap $M = 8$: At this operating point, PRISM-XAI transmits 205.41 kbit versus 243.37 kbit for MAP-XAI-only (aggregate-mean ratio saving 15.60%). The paired difference is -37.96±3.63 kbit. PRISM coverage is 0.0737, critical-event recall is 0.0367, and maximum recorded expenditure is 18.206. Relative to MAP-XAI-only, the paired coverage change is -0.00053±0.00046 and recall change is -0.00061±0.00143. The descriptive overlap rule holds in 12 of 12 cells. Accordingly, this point is read as a matched-service scheduler comparison.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XXXIV

PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.181 ± 0.038	0.036 ± 0.059	0.031 ± 0.010	0.031 ± 0.010	0.347 ± 0.128	0.938 ± 0.004	4.881 ± 1.730	87.922 ± 0.587	7.823 ± 0.000	0.023 ± 0.007	8276.543 ± 2832.262	0.991 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.204 ± 0.062	0.043 ± 0.077	0.029 ± 0.009	0.029 ± 0.009	0.330 ± 0.128	0.938 ± 0.004	4.876 ± 1.737	87.707 ± 0.512	7.823 ± 0.000	0.025 ± 0.007	8335.040 ± 2320.748	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.221 ± 0.065	0.030 ± 0.075	0.029 ± 0.009	0.029 ± 0.009	0.319 ± 0.121	0.938 ± 0.004	4.960 ± 1.678	105.889 ± 0.169	7.823 ± 0.000	0.024 ± 0.006	10646.987 ± 3613.342	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.192 ± 0.066	0.031 ± 0.076	0.027 ± 0.007	0.027 ± 0.007	0.297 ± 0.095	0.938 ± 0.004	4.960 ± 1.678	105.737 ± 0.152	7.823 ± 0.000	0.026 ± 0.006	10712.923 ± 2768.192	0.991 ± 0.006
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.226 ± 0.039	-0.007 ± 0.014	0.045 ± 0.006	0.045 ± 0.006	0.448 ± 0.069	0.907 ± 0.003	4.627 ± 1.490	325.204 ± 1.099	7.823 ± 0.000	0.038 ± 0.006	9052.203 ± 1462.054	0.991 ± 0.002
UCI	Tree	8	PRISM-XAI	0.218 ± 0.040	-0.021 ± 0.077	0.041 ± 0.008	0.041 ± 0.008	0.442 ± 0.091	0.912 ± 0.003	4.663 ± 1.766	322.006 ± 0.783	7.823 ± 0.000	0.036 ± 0.008	10546.447 ± 3413.298	0.991 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.167 ± 0.072	-0.002 ± 0.015	0.047 ± 0.007	0.047 ± 0.007	0.444 ± 0.055	0.907 ± 0.003	4.955 ± 1.626	381.383 ± 0.421	7.823 ± 0.000	0.040 ± 0.005	10267.244 ± 1941.397	0.991 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.241 ± 0.074	0.039 ± 0.065	0.046 ± 0.006	0.046 ± 0.006	0.475 ± 0.053	0.909 ± 0.003	5.078 ± 1.827	381.010 ± 0.445	7.823 ± 0.000	0.035 ± 0.005	10274.502 ± 1367.610	0.991 ± 0.003

TABLE XXXV

PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.922 ± 0.587	0.181 ± 0.038	0.250 ± 0.000	0.036 ± 0.059	0.031 ± 0.010	0.031 ± 0.010	0.347 ± 0.128	0.062 ± 0.004	0.938 ± 0.004	0.023 ± 0.007	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.707 ± 0.512	0.204 ± 0.062	0.250 ± 0.000	0.043 ± 0.077	0.029 ± 0.009	0.029 ± 0.009	0.330 ± 0.128	0.062 ± 0.004	0.938 ± 0.004	0.025 ± 0.007	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.922 ± 0.597	0.221 ± 0.066	0.250 ± 0.000	0.141 ± 0.056	0.029 ± 0.010	0.029 ± 0.010	0.385 ± 0.148	0.062 ± 0.004	0.938 ± 0.004	0.020 ± 0.007	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.585 ± 0.454	0.246 ± 0.046	0.250 ± 0.000	0.149 ± 0.072	0.030 ± 0.008	0.030 ± 0.008	0.354 ± 0.120	0.062 ± 0.004	0.938 ± 0.004	0.022 ± 0.006	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.891 ± 0.540	0.397 ± 0.067	0.250 ± 0.000	0.306 ± 0.044	0.040 ± 0.008	0.040 ± 0.008	0.627 ± 0.190	0.063 ± 0.004	0.937 ± 0.004	0.011 ± 0.007	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.431 ± 0.439	0.369 ± 0.054	0.250 ± 0.000	0.347 ± 0.057	0.041 ± 0.006	0.041 ± 0.006	0.574 ± 0.099	0.062 ± 0.004	0.938 ± 0.004	0.012 ± 0.004	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	325.204 ± 1.099	0.226 ± 0.039	0.250 ± 0.000	-0.007 ± 0.014	0.045 ± 0.006	0.045 ± 0.006	0.448 ± 0.069	0.093 ± 0.003	0.907 ± 0.003	0.038 ± 0.006	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	322.006 ± 0.783	0.218 ± 0.040	0.250 ± 0.000	-0.021 ± 0.077	0.041 ± 0.008	0.041 ± 0.008	0.442 ± 0.091	0.088 ± 0.003	0.912 ± 0.003	0.036 ± 0.008	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	324.692 ± 0.734	0.260 ± 0.069	0.250 ± 0.000	-0.003 ± 0.016	0.041 ± 0.006	0.041 ± 0.006	0.561 ± 0.104	0.090 ± 0.004	0.910 ± 0.004	0.023 ± 0.006	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	321.197 ± 1.037	0.247 ± 0.060	0.250 ± 0.000	-0.017 ± 0.061	0.038 ± 0.007	0.038 ± 0.007	0.561 ± 0.070	0.081 ± 0.003	0.919 ± 0.003	0.021 ± 0.004	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	324.272 ± 0.853	0.285 ± 0.057	0.250 ± 0.000	0.008 ± 0.016	0.039 ± 0.004	0.039 ± 0.004	0.892 ± 0.052	0.081 ± 0.003	0.919 ± 0.003	0.004 ± 0.002	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	321.095 ± 0.887	0.246 ± 0.074	0.250 ± 0.000	0.042 ± 0.069	0.036 ± 0.005	0.036 ± 0.005	0.872 ± 0.111	0.076 ± 0.004	0.924 ± 0.004	0.004 ± 0.004	31.175 ± 0.000

TABLE XXXVI
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.642 ± 0.587	-0.052 ± 0.056	-0.089 ± 0.033	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-5011.991 ± 5.546	-0.033 ± 0.086	-0.372 ± 0.079	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.333 ± 0.540	0.144 ± 0.080	-0.373 ± 0.080	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-1472.857 ± 0.512	-0.034 ± 0.064	-0.091 ± 0.028	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-5012.328 ± 5.478	-0.006 ± 0.062	-0.385 ± 0.059	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.793 ± 0.439	0.117 ± 0.064	-0.384 ± 0.059	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.642 ± 0.587	-0.111 ± 0.049	-0.076 ± 0.029	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4598.687 ± 12.788	-0.041 ± 0.083	-0.372 ± 0.091	0.054 ± 0.008
OPSD	Channel-aware fixed-DP	-5212.333 ± 0.540	0.139 ± 0.074	-0.394 ± 0.099	0.083 ± 0.008
OPSD	Channel-aware fixed-DP	-1472.857 ± 0.512	-0.088 ± 0.072	-0.082 ± 0.026	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4599.025 ± 12.782	-0.012 ± 0.057	-0.389 ± 0.089	0.054 ± 0.008
OPSD	Channel-aware fixed-DP	-5212.793 ± 0.439	0.112 ± 0.064	-0.415 ± 0.091	0.085 ± 0.008
OPSD	Joint, no compression	-101.978 ± 0.649	-0.106 ± 0.082	0.014 ± 0.009	-0.005 ± 0.005
OPSD	Joint, no compression	-371.978 ± 1.235	-0.011 ± 0.086	-0.009 ± 0.014	-0.006 ± 0.006
OPSD	Joint, no compression	-1053.008 ± 7.902	0.162 ± 0.100	-0.034 ± 0.017	-0.009 ± 0.006
OPSD	Joint, no compression	-102.043 ± 0.542	-0.054 ± 0.092	0.010 ± 0.008	-0.005 ± 0.005
OPSD	Joint, no compression	-372.466 ± 0.983	0.006 ± 0.073	-0.009 ± 0.011	-0.006 ± 0.006
OPSD	Joint, no compression	-1053.469 ± 8.014	0.087 ± 0.081	-0.039 ± 0.013	-0.007 ± 0.006
OPSD	Periodic fixed-DP	-1472.642 ± 0.587	-0.059 ± 0.047	-0.094 ± 0.024	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-5024.672 ± 0.597	-0.028 ± 0.077	-0.421 ± 0.066	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5211.944 ± 0.546	0.148 ± 0.067	-0.418 ± 0.066	-0.010 ± 0.006
OPSD	Periodic fixed-DP	-1472.857 ± 0.512	-0.038 ± 0.054	-0.105 ± 0.018	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-5025.010 ± 0.454	0.003 ± 0.052	-0.433 ± 0.050	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5212.405 ± 0.471	0.125 ± 0.063	-0.430 ± 0.050	-0.009 ± 0.006
OPSD	MAP-XAI fallback	-17.966 ± 0.682	-0.040 ± 0.046	0.002 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-17.966 ± 0.687	-0.043 ± 0.089	0.000 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-17.997 ± 0.643	-0.000 ± 0.067	-0.002 ± 0.005	-0.001 ± 0.003
OPSD	MAP-XAI fallback	-18.030 ± 0.583	0.013 ± 0.040	0.003 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.153 ± 0.544	0.003 ± 0.052	0.002 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.121 ± 0.487	0.009 ± 0.066	0.004 ± 0.005	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4975.020 ± 1.099	-0.037 ± 0.046	-0.412 ± 0.033	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4975.532 ± 0.734	-0.003 ± 0.089	-0.416 ± 0.028	-0.007 ± 0.003
UCI	AoX-aware fixed-DP	-4975.952 ± 0.853	0.021 ± 0.068	-0.418 ± 0.031	-0.007 ± 0.002
UCI	AoX-aware fixed-DP	-4978.218 ± 0.783	-0.043 ± 0.038	-0.414 ± 0.047	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4979.027 ± 1.037	-0.013 ± 0.061	-0.416 ± 0.044	-0.007 ± 0.002
UCI	AoX-aware fixed-DP	-4979.130 ± 0.887	-0.014 ± 0.081	-0.419 ± 0.045	-0.008 ± 0.002
UCI	Channel-aware fixed-DP	-4969.974 ± 3.637	-0.022 ± 0.056	0.044 ± 0.006	0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4975.532 ± 0.734	-0.007 ± 0.079	0.040 ± 0.006	0.661 ± 0.003
UCI	Channel-aware fixed-DP	-4975.952 ± 0.853	0.016 ± 0.078	0.039 ± 0.004	0.737 ± 0.002
UCI	Channel-aware fixed-DP	-4973.172 ± 3.993	-0.022 ± 0.051	0.041 ± 0.008	-0.000 ± 0.003
UCI	Channel-aware fixed-DP	-4979.027 ± 1.037	-0.013 ± 0.056	0.038 ± 0.007	0.661 ± 0.002
UCI	Channel-aware fixed-DP	-4979.130 ± 0.887	-0.013 ± 0.077	0.035 ± 0.005	0.736 ± 0.002
UCI	Joint, no compression	-347.096 ± 1.294	-0.060 ± 0.070	-0.007 ± 0.008	-0.002 ± 0.003
UCI	Joint, no compression	-1307.458 ± 0.943	0.044 ± 0.088	-0.053 ± 0.006	-0.006 ± 0.002
UCI	Joint, no compression	-3710.278 ± 4.537	0.038 ± 0.072	-0.188 ± 0.014	-0.007 ± 0.002
UCI	Joint, no compression	-350.294 ± 0.927	-0.012 ± 0.080	-0.016 ± 0.011	-0.004 ± 0.004
UCI	Joint, no compression	-1311.403 ± 0.582	0.044 ± 0.058	-0.061 ± 0.019	-0.006 ± 0.002
UCI	Joint, no compression	-3713.455 ± 5.127	0.005 ± 0.053	-0.192 ± 0.027	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4974.114 ± 1.995	-0.033 ± 0.077	0.039 ± 0.007	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4974.626 ± 1.848	0.001 ± 0.099	0.035 ± 0.007	-0.008 ± 0.003
UCI	Periodic fixed-DP	-4975.046 ± 1.596	0.025 ± 0.092	0.033 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4977.312 ± 1.437	-0.043 ± 0.073	0.036 ± 0.007	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4978.121 ± 1.405	-0.013 ± 0.077	0.033 ± 0.007	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4978.224 ± 1.527	-0.014 ± 0.101	0.030 ± 0.005	-0.008 ± 0.002
UCI	MAP-XAI fallback	-56.179 ± 0.880	0.059 ± 0.050	-0.002 ± 0.004	0.000 ± 0.000
UCI	MAP-XAI fallback	-56.609 ± 0.650	0.023 ± 0.053	-0.001 ± 0.006	0.000 ± 0.001
UCI	MAP-XAI fallback	-56.864 ± 0.771	0.066 ± 0.045	-0.003 ± 0.005	-0.000 ± 0.001
UCI	MAP-XAI fallback	-59.004 ± 0.825	-0.024 ± 0.054	-0.005 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-59.566 ± 1.156	0.043 ± 0.054	-0.003 ± 0.007	-0.000 ± 0.001
UCI	MAP-XAI fallback	-59.092 ± 1.030	0.025 ± 0.105	-0.002 ± 0.007	0.001 ± 0.001

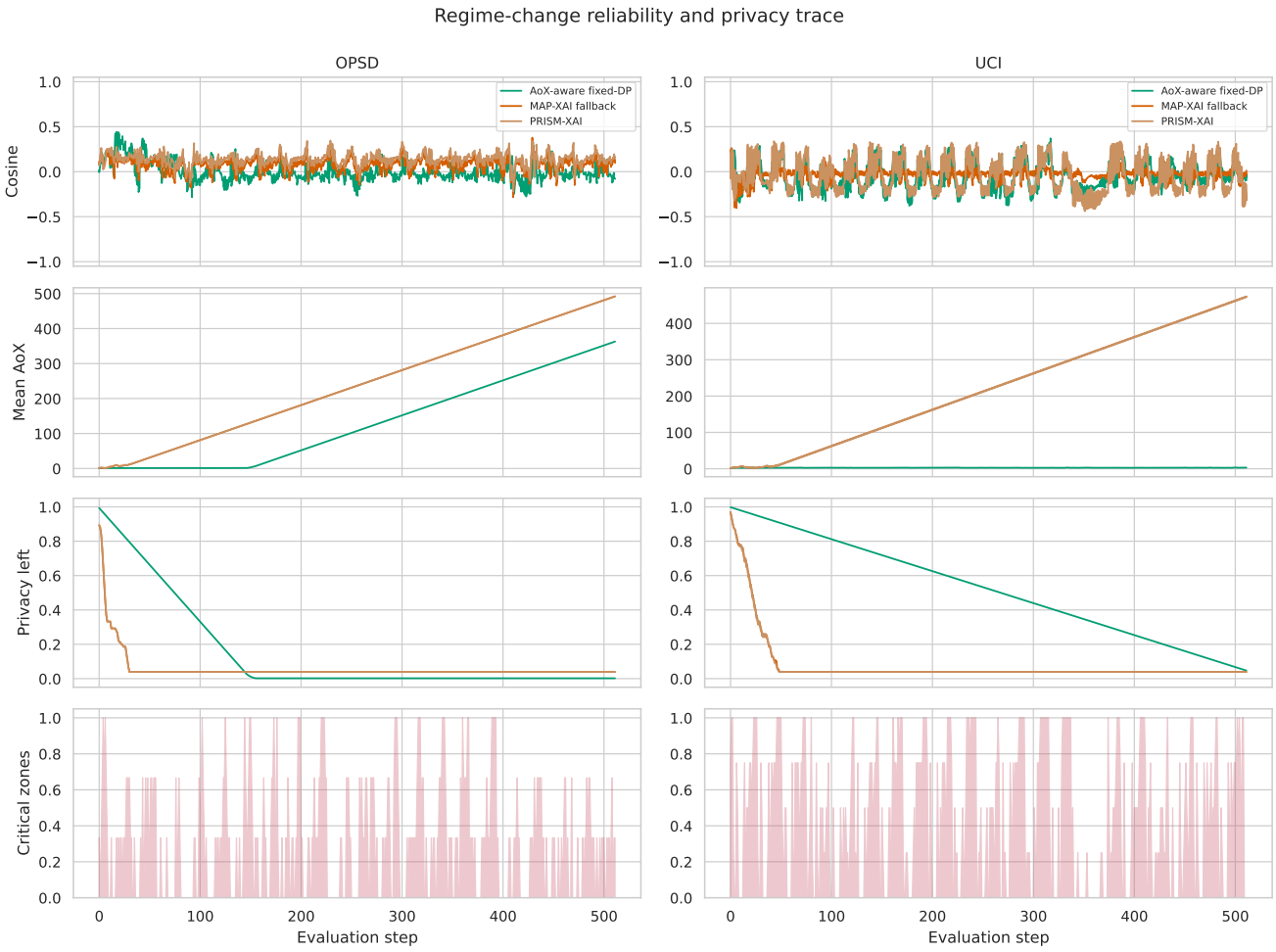


Fig. 35. Public-score threshold $\tau = 2$, release cap $M = 8$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

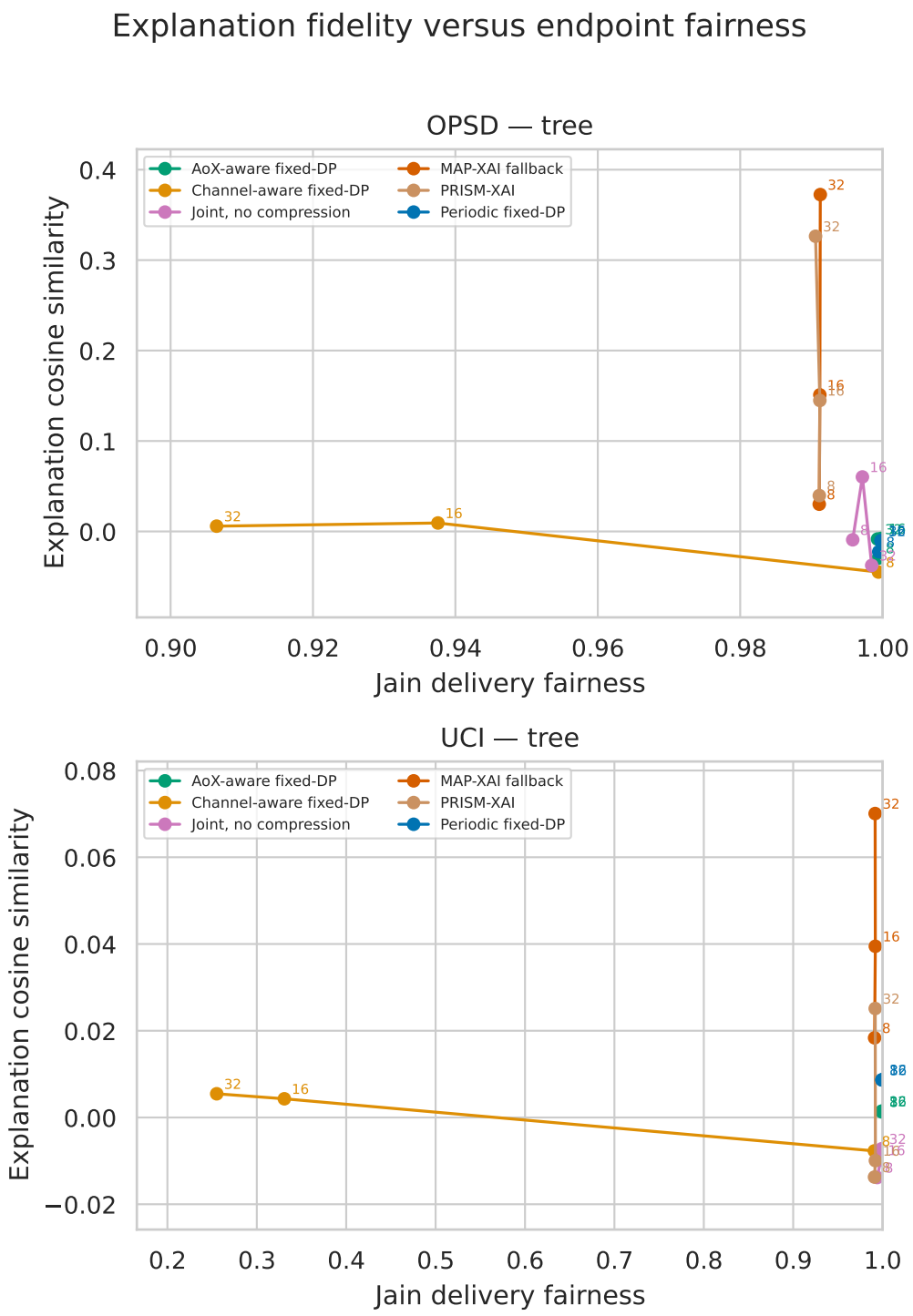


Fig. 36. Public-score threshold $\tau = 2$, release cap $M = 8$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

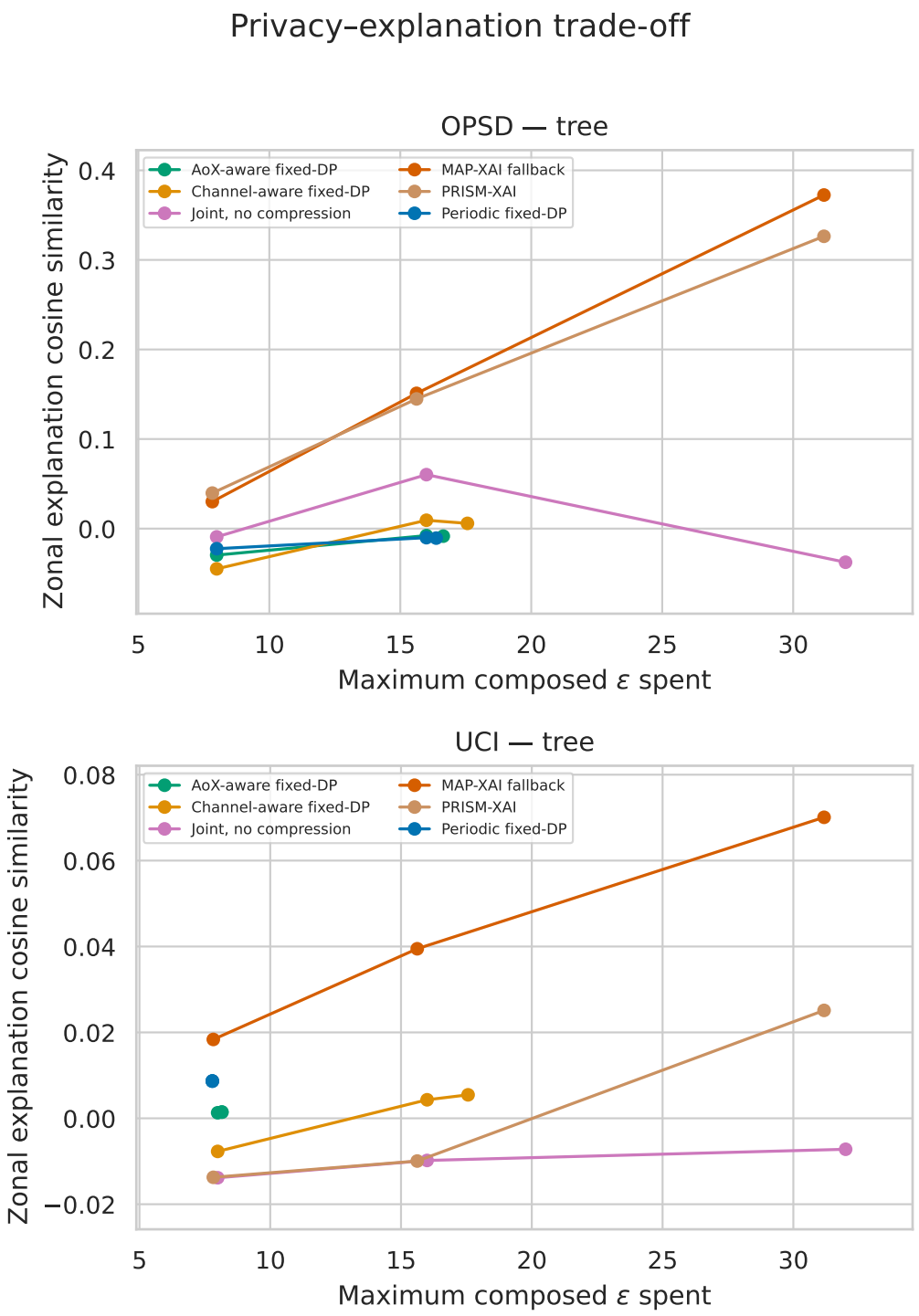


Fig. 37. Public-score threshold $\tau = 2$, release cap $M = 8$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

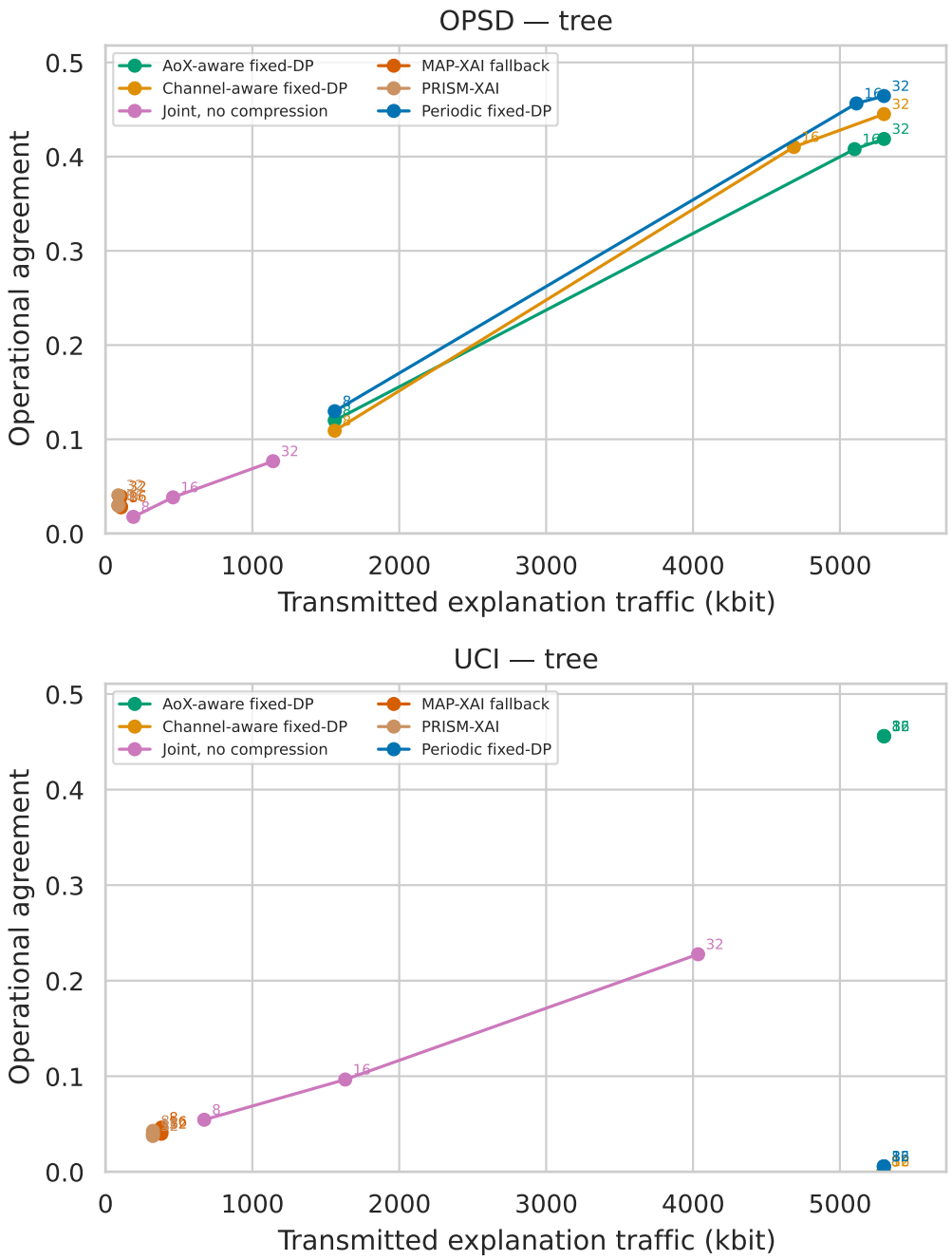


Fig. 38. Public-score threshold $\tau = 2$, release cap $M = 8$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

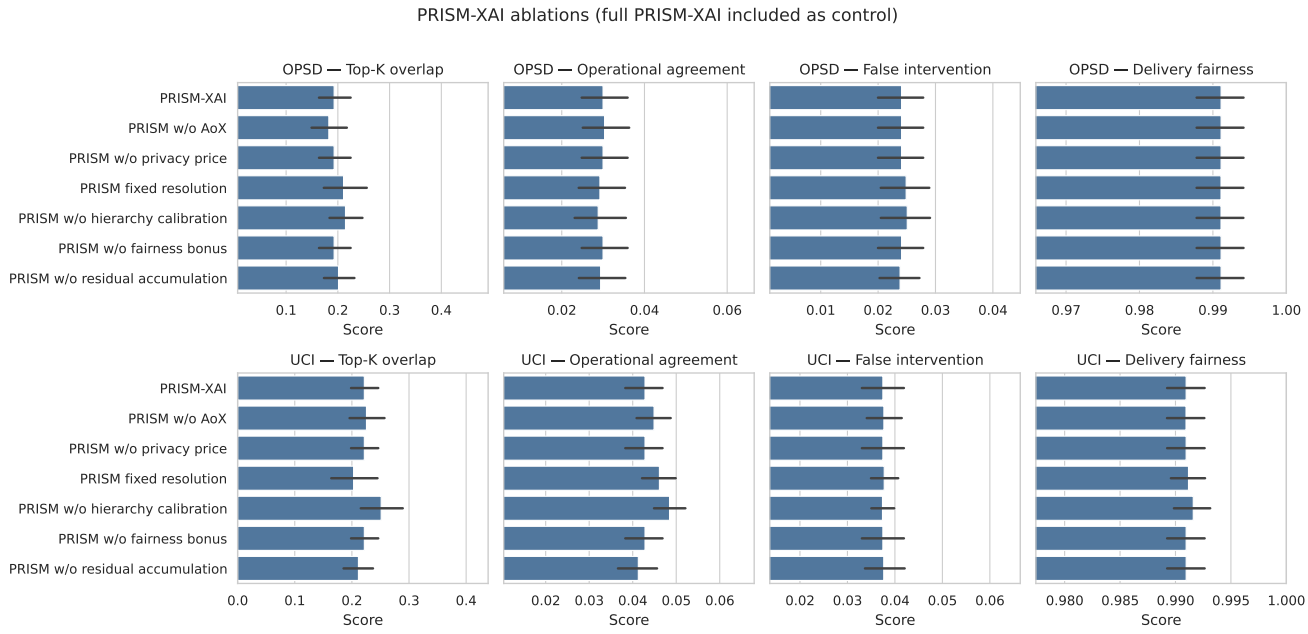


Fig. 39. Public-score threshold $\tau = 2$, release cap $M = 8$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XXXVII
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.181 ± 0.038	0.036 ± 0.059	0.031 ± 0.010	0.031 ± 0.010	0.347 ± 0.128	0.938 ± 0.004	4.881 ± 1.730	87.922 ± 0.587	7.823 ± 0.000	0.023 ± 0.007	8276.543 ± 2832.262	0.991 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.204 ± 0.062	0.043 ± 0.077	0.029 ± 0.009	0.029 ± 0.009	0.330 ± 0.128	0.938 ± 0.004	4.876 ± 1.737	87.707 ± 0.512	7.823 ± 0.000	0.025 ± 0.007	8335.040 ± 2320.748	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.221 ± 0.065	0.030 ± 0.075	0.029 ± 0.009	0.029 ± 0.009	0.319 ± 0.121	0.938 ± 0.004	4.960 ± 1.678	105.889 ± 0.169	7.823 ± 0.000	0.024 ± 0.006	10646.987 ± 3613.342	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.192 ± 0.066	0.031 ± 0.076	0.027 ± 0.007	0.027 ± 0.007	0.297 ± 0.095	0.938 ± 0.004	4.960 ± 1.678	105.737 ± 0.152	7.823 ± 0.000	0.026 ± 0.006	10712.923 ± 2768.192	0.991 ± 0.006

TABLE XXXVIII
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.922 ± 0.587	0.181 ± 0.038	0.250 ± 0.000	0.036 ± 0.059	0.031 ± 0.010	0.031 ± 0.010	0.347 ± 0.128	0.062 ± 0.004	0.938 ± 0.004	0.023 ± 0.007	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.707 ± 0.512	0.204 ± 0.062	0.250 ± 0.000	0.043 ± 0.077	0.029 ± 0.009	0.029 ± 0.009	0.330 ± 0.128	0.062 ± 0.004	0.938 ± 0.004	0.025 ± 0.007	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.922 ± 0.597	0.221 ± 0.066	0.250 ± 0.000	0.141 ± 0.056	0.029 ± 0.010	0.029 ± 0.010	0.385 ± 0.148	0.062 ± 0.004	0.938 ± 0.004	0.020 ± 0.007	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.585 ± 0.454	0.246 ± 0.046	0.250 ± 0.000	0.149 ± 0.072	0.030 ± 0.008	0.030 ± 0.008	0.354 ± 0.120	0.062 ± 0.004	0.938 ± 0.004	0.022 ± 0.006	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.891 ± 0.540	0.397 ± 0.067	0.250 ± 0.000	0.306 ± 0.044	0.040 ± 0.008	0.040 ± 0.008	0.627 ± 0.190	0.063 ± 0.004	0.937 ± 0.004	0.011 ± 0.007	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.431 ± 0.439	0.369 ± 0.054	0.250 ± 0.000	0.347 ± 0.057	0.041 ± 0.006	0.041 ± 0.006	0.574 ± 0.099	0.062 ± 0.004	0.938 ± 0.004	0.012 ± 0.004	31.175 ± 0.000

TABLE XXXIX
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.642 ± 0.587	-0.052 ± 0.056	-0.089 ± 0.033	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-5011.991 ± 5.546	-0.033 ± 0.086	-0.372 ± 0.079	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.333 ± 0.540	0.144 ± 0.080	-0.373 ± 0.080	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-1472.857 ± 0.512	-0.034 ± 0.064	-0.091 ± 0.028	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-5012.328 ± 5.478	-0.006 ± 0.062	-0.385 ± 0.059	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.793 ± 0.439	0.117 ± 0.064	-0.384 ± 0.059	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.642 ± 0.587	-0.111 ± 0.049	-0.076 ± 0.029	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4598.687 ± 12.788	-0.041 ± 0.083	-0.372 ± 0.091	0.054 ± 0.008
OPSD	Channel-aware fixed-DP	-5212.333 ± 0.540	0.139 ± 0.074	-0.394 ± 0.099	0.083 ± 0.008
OPSD	Channel-aware fixed-DP	-1472.857 ± 0.512	-0.088 ± 0.072	-0.082 ± 0.026	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4599.025 ± 12.782	-0.012 ± 0.057	-0.389 ± 0.089	0.054 ± 0.008
OPSD	Channel-aware fixed-DP	-5212.793 ± 0.439	0.112 ± 0.064	-0.415 ± 0.091	0.085 ± 0.008
OPSD	Joint, no compression	-101.978 ± 0.649	-0.106 ± 0.082	0.014 ± 0.009	-0.005 ± 0.005
OPSD	Joint, no compression	-371.978 ± 1.235	-0.011 ± 0.086	-0.009 ± 0.014	-0.006 ± 0.006
OPSD	Joint, no compression	-1053.008 ± 7.902	0.162 ± 0.100	-0.034 ± 0.017	-0.009 ± 0.006
OPSD	Joint, no compression	-102.043 ± 0.542	-0.054 ± 0.092	0.010 ± 0.008	-0.005 ± 0.005
OPSD	Joint, no compression	-372.466 ± 0.983	0.006 ± 0.073	-0.009 ± 0.011	-0.006 ± 0.006
OPSD	Joint, no compression	-1053.469 ± 8.014	0.087 ± 0.081	-0.039 ± 0.013	-0.007 ± 0.006
OPSD	Periodic fixed-DP	-1472.642 ± 0.587	-0.059 ± 0.047	-0.094 ± 0.024	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-5024.672 ± 0.597	-0.028 ± 0.077	-0.421 ± 0.066	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5211.944 ± 0.546	0.148 ± 0.067	-0.418 ± 0.066	-0.010 ± 0.006
OPSD	Periodic fixed-DP	-1472.857 ± 0.512	-0.038 ± 0.054	-0.105 ± 0.018	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-5025.010 ± 0.454	0.003 ± 0.052	-0.433 ± 0.050	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5212.405 ± 0.471	0.125 ± 0.063	-0.430 ± 0.050	-0.009 ± 0.006
OPSD	MAP-XAI fallback	-17.966 ± 0.682	-0.040 ± 0.046	0.002 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-17.966 ± 0.687	-0.043 ± 0.089	0.000 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-17.997 ± 0.643	-0.000 ± 0.067	-0.002 ± 0.005	-0.001 ± 0.003
OPSD	MAP-XAI fallback	-18.030 ± 0.583	0.013 ± 0.040	0.003 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.153 ± 0.544	0.003 ± 0.052	0.002 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.121 ± 0.487	0.009 ± 0.066	0.004 ± 0.005	0.000 ± 0.000

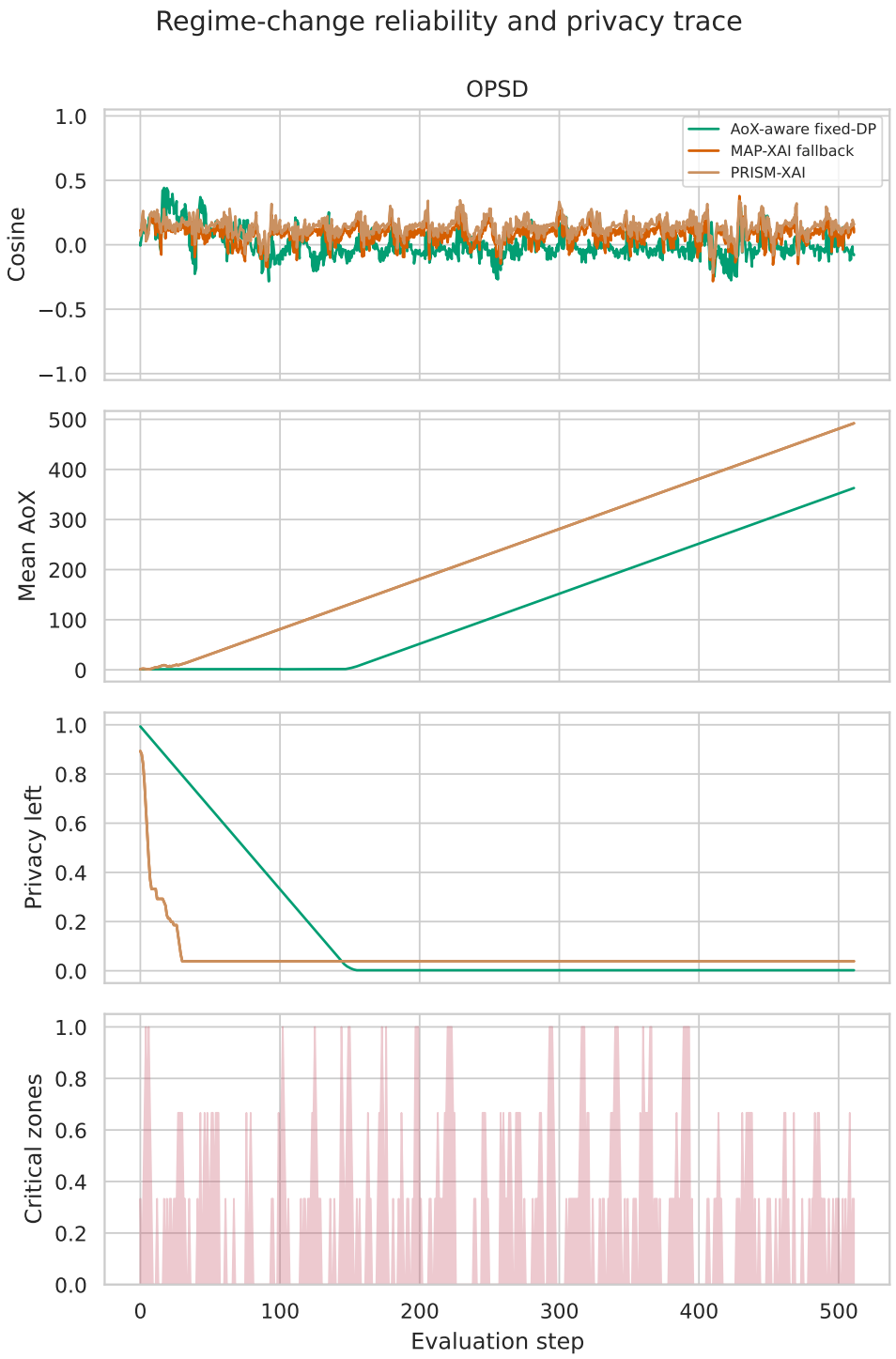


Fig. 40. Public-score threshold $\tau = 2$, release cap $M = 8$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

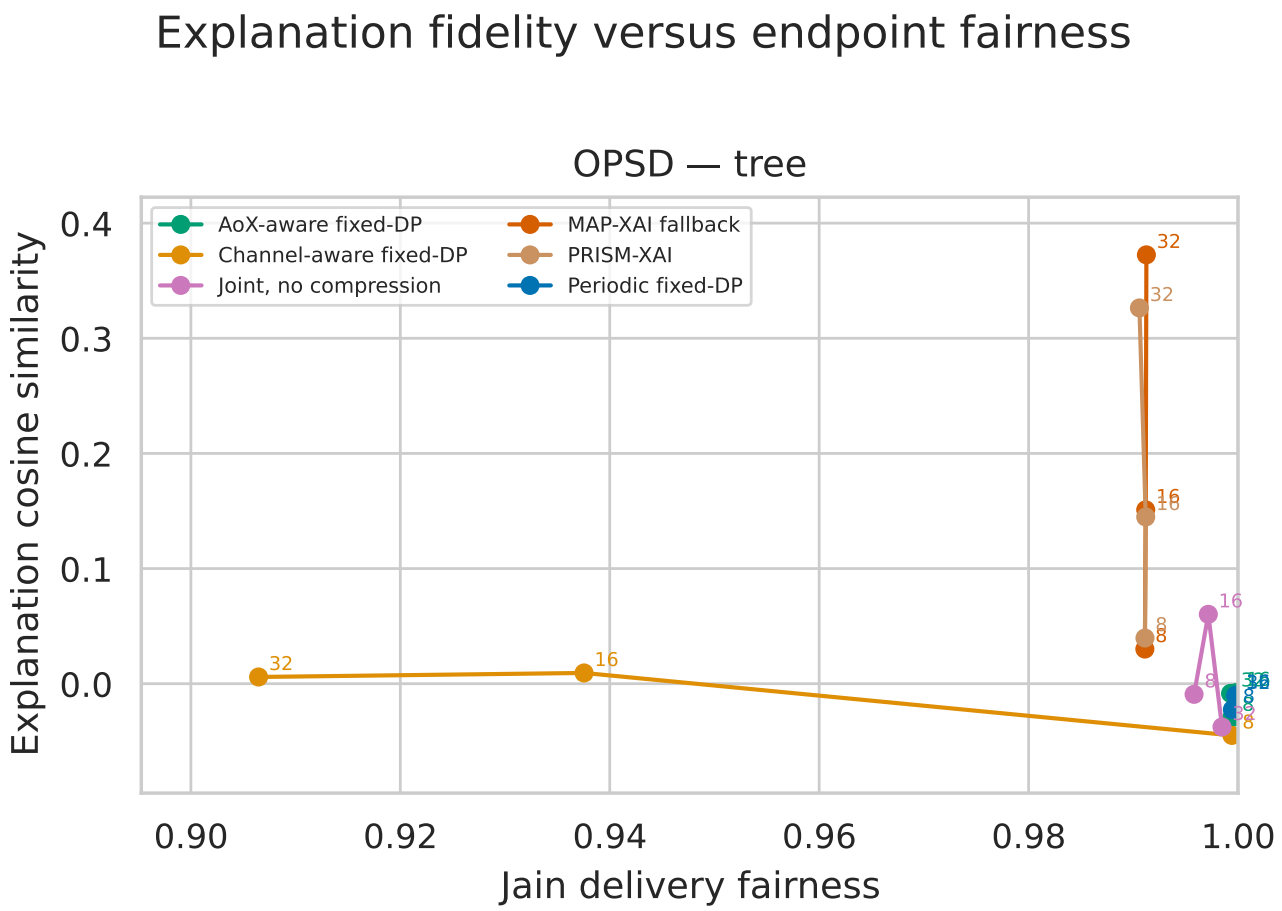


Fig. 41. Public-score threshold $\tau = 2$, release cap $M = 8$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

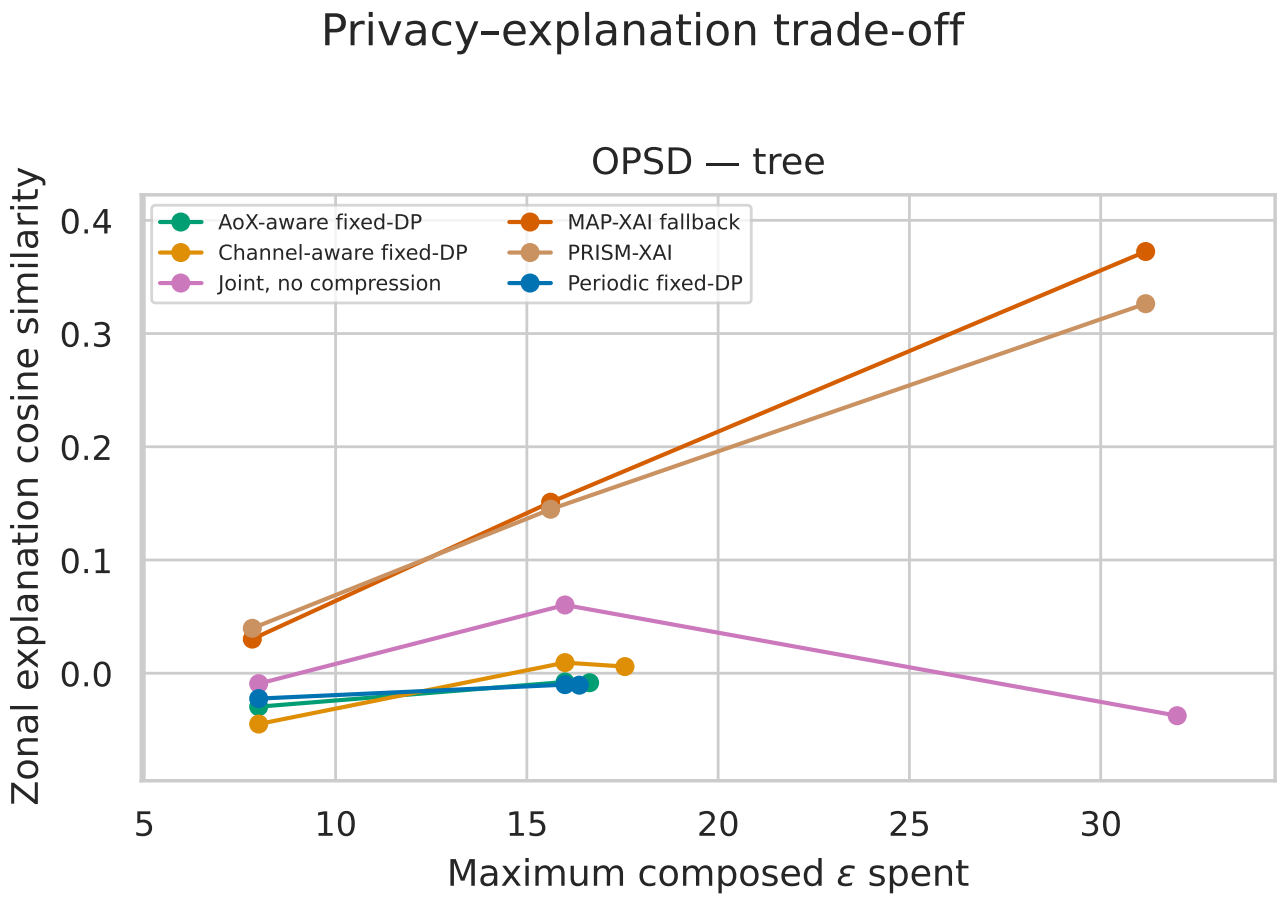


Fig. 42. Public-score threshold $\tau = 2$, release cap $M = 8$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

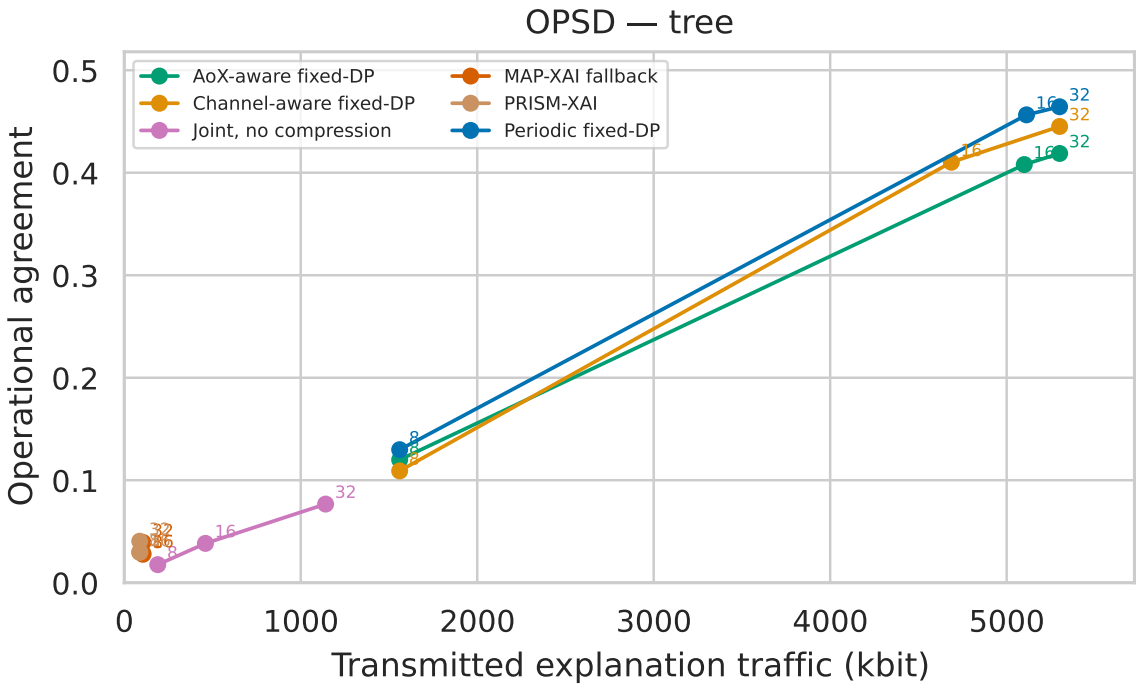


Fig. 43. Public-score threshold $\tau = 2$, release cap $M = 8$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

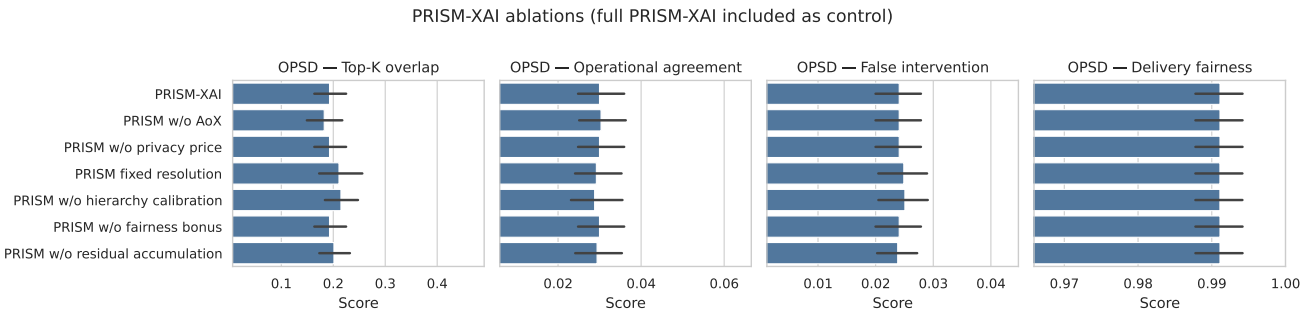


Fig. 44. Public-score threshold $\tau = 2$, release cap $M = 8$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XL
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.226 ± 0.039	-0.007 ± 0.014	0.045 ± 0.006	0.045 ± 0.006	0.448 ± 0.069	0.907 ± 0.003	4.627 ± 1.490	325.204 ± 1.099	7.823 ± 0.000	0.038 ± 0.006	9052.203 ± 1462.054	0.991 ± 0.002
UCI	Tree	8	PRISM-XAI	0.218 ± 0.040	-0.021 ± 0.077	0.041 ± 0.008	0.041 ± 0.008	0.442 ± 0.091	0.912 ± 0.003	4.663 ± 1.766	322.006 ± 0.783	7.823 ± 0.000	0.036 ± 0.008	10546.447 ± 3413.298	0.991 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.167 ± 0.072	-0.002 ± 0.015	0.047 ± 0.007	0.047 ± 0.007	0.444 ± 0.055	0.907 ± 0.003	4.955 ± 1.626	381.383 ± 0.421	7.823 ± 0.000	0.040 ± 0.005	10267.244 ± 1941.397	0.991 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.241 ± 0.074	0.039 ± 0.065	0.046 ± 0.006	0.046 ± 0.006	0.475 ± 0.053	0.909 ± 0.003	5.078 ± 1.827	381.010 ± 0.445	7.823 ± 0.000	0.035 ± 0.005	10274.502 ± 1367.610	0.991 ± 0.003

TABLE XLI
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	325.204 ± 1.099	0.226 ± 0.039	0.250 ± 0.000	-0.007 ± 0.014	0.045 ± 0.006	0.045 ± 0.006	0.448 ± 0.069	0.093 ± 0.003	0.907 ± 0.003	0.038 ± 0.006	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	322.006 ± 0.783	0.218 ± 0.040	0.250 ± 0.000	-0.021 ± 0.077	0.041 ± 0.008	0.041 ± 0.008	0.442 ± 0.091	0.088 ± 0.003	0.912 ± 0.003	0.036 ± 0.008	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	324.692 ± 0.734	0.260 ± 0.069	0.250 ± 0.000	-0.003 ± 0.016	0.041 ± 0.006	0.041 ± 0.006	0.561 ± 0.104	0.090 ± 0.004	0.910 ± 0.004	0.023 ± 0.006	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	321.197 ± 1.037	0.247 ± 0.060	0.250 ± 0.000	-0.017 ± 0.061	0.038 ± 0.007	0.038 ± 0.007	0.561 ± 0.070	0.081 ± 0.003	0.919 ± 0.003	0.021 ± 0.004	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	324.272 ± 0.853	0.285 ± 0.057	0.250 ± 0.000	0.008 ± 0.016	0.039 ± 0.004	0.039 ± 0.004	0.892 ± 0.052	0.081 ± 0.003	0.919 ± 0.003	0.004 ± 0.002	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	321.095 ± 0.887	0.246 ± 0.074	0.250 ± 0.000	0.042 ± 0.069	0.036 ± 0.005	0.036 ± 0.005	0.872 ± 0.111	0.076 ± 0.004	0.924 ± 0.004	0.004 ± 0.004	31.175 ± 0.000

TABLE XLII
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4975.020 ± 1.099	-0.037 ± 0.046	-0.412 ± 0.033	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4975.532 ± 0.734	-0.003 ± 0.089	-0.416 ± 0.028	-0.007 ± 0.003
UCI	AoX-aware fixed-DP	-4975.952 ± 0.853	0.021 ± 0.068	-0.418 ± 0.031	-0.007 ± 0.002
UCI	AoX-aware fixed-DP	-4978.218 ± 0.783	-0.043 ± 0.038	-0.414 ± 0.047	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4979.027 ± 1.037	-0.013 ± 0.061	-0.416 ± 0.044	-0.007 ± 0.002
UCI	AoX-aware fixed-DP	-4979.130 ± 0.887	-0.014 ± 0.081	-0.419 ± 0.045	-0.008 ± 0.002
UCI	Channel-aware fixed-DP	-4969.974 ± 3.637	-0.022 ± 0.056	0.044 ± 0.006	0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4975.532 ± 0.734	-0.007 ± 0.079	0.040 ± 0.006	0.661 ± 0.003
UCI	Channel-aware fixed-DP	-4975.952 ± 0.853	0.016 ± 0.078	0.039 ± 0.004	0.737 ± 0.002
UCI	Channel-aware fixed-DP	-4973.172 ± 3.993	-0.022 ± 0.051	0.041 ± 0.008	-0.000 ± 0.003
UCI	Channel-aware fixed-DP	-4979.027 ± 1.037	-0.013 ± 0.056	0.038 ± 0.007	0.661 ± 0.002
UCI	Channel-aware fixed-DP	-4979.130 ± 0.887	-0.013 ± 0.077	0.035 ± 0.005	0.736 ± 0.002
UCI	Joint, no compression	-347.096 ± 1.294	-0.060 ± 0.070	-0.007 ± 0.008	-0.002 ± 0.003
UCI	Joint, no compression	-1307.458 ± 0.943	0.044 ± 0.088	-0.053 ± 0.006	-0.006 ± 0.002
UCI	Joint, no compression	-3710.278 ± 4.537	0.038 ± 0.072	-0.188 ± 0.014	-0.007 ± 0.002
UCI	Joint, no compression	-350.294 ± 0.927	-0.012 ± 0.080	-0.016 ± 0.011	-0.004 ± 0.004
UCI	Joint, no compression	-1311.403 ± 0.582	0.044 ± 0.058	-0.061 ± 0.019	-0.006 ± 0.002
UCI	Joint, no compression	-3713.455 ± 5.127	0.005 ± 0.053	-0.192 ± 0.027	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4974.114 ± 1.995	-0.033 ± 0.077	0.039 ± 0.007	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4974.626 ± 1.848	0.001 ± 0.099	0.035 ± 0.007	-0.008 ± 0.003
UCI	Periodic fixed-DP	-4975.046 ± 1.596	0.025 ± 0.092	0.033 ± 0.005	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4977.312 ± 1.437	-0.043 ± 0.073	0.036 ± 0.007	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4978.121 ± 1.405	-0.013 ± 0.077	0.033 ± 0.007	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4978.224 ± 1.527	-0.014 ± 0.101	0.030 ± 0.005	-0.008 ± 0.002
UCI	MAP-XAI fallback	-56.179 ± 0.880	0.059 ± 0.050	-0.002 ± 0.004	0.000 ± 0.000
UCI	MAP-XAI fallback	-56.609 ± 0.650	0.023 ± 0.053	-0.001 ± 0.006	0.000 ± 0.001
UCI	MAP-XAI fallback	-56.864 ± 0.771	0.066 ± 0.045	-0.003 ± 0.005	-0.000 ± 0.001
UCI	MAP-XAI fallback	-59.004 ± 0.825	-0.024 ± 0.054	-0.005 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-59.566 ± 1.156	0.043 ± 0.054	-0.003 ± 0.007	-0.000 ± 0.001
UCI	MAP-XAI fallback	-59.092 ± 1.030	0.025 ± 0.105	-0.002 ± 0.007	0.001 ± 0.001

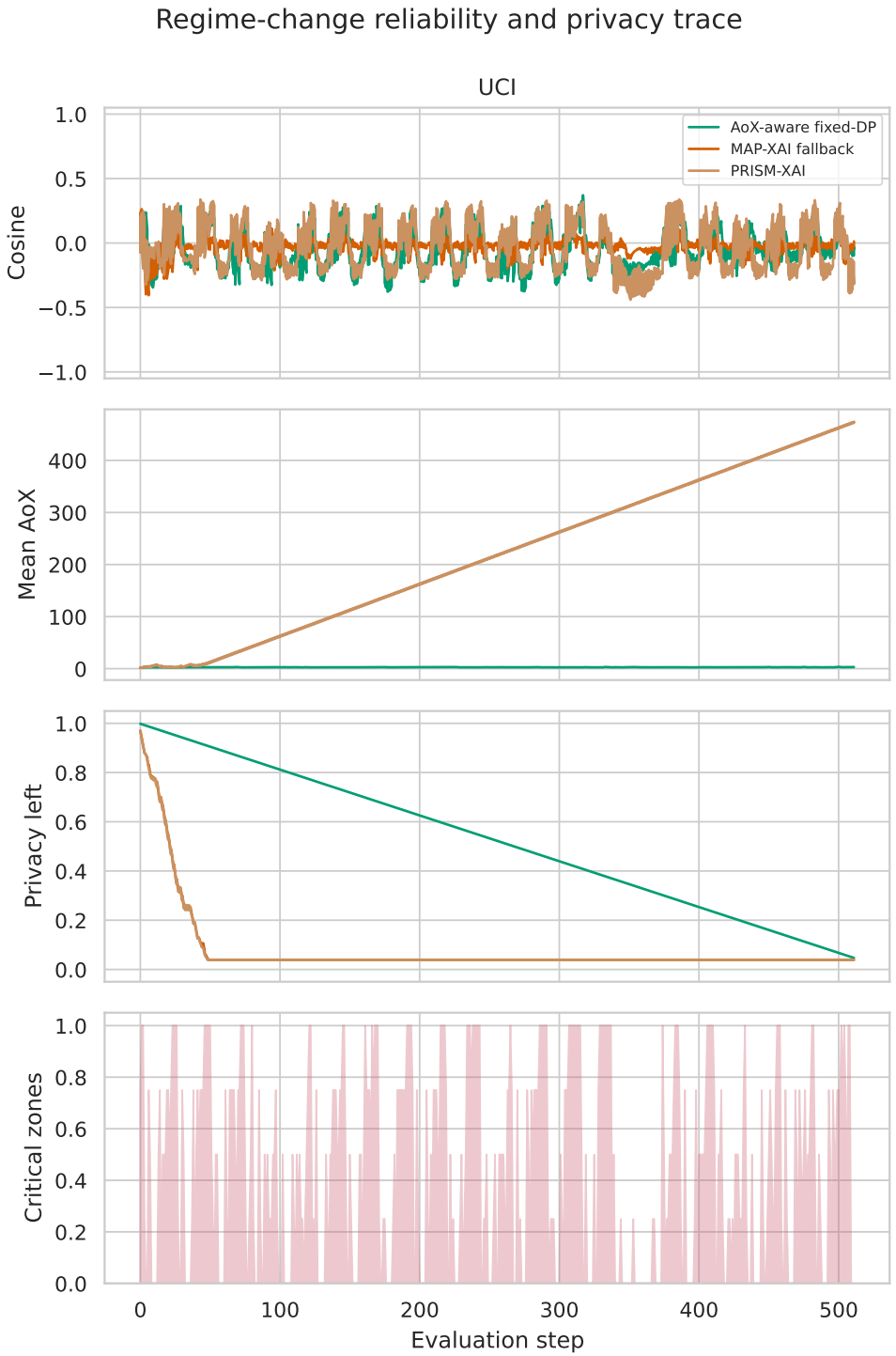


Fig. 45. Public-score threshold $\tau = 2$, release cap $M = 8$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

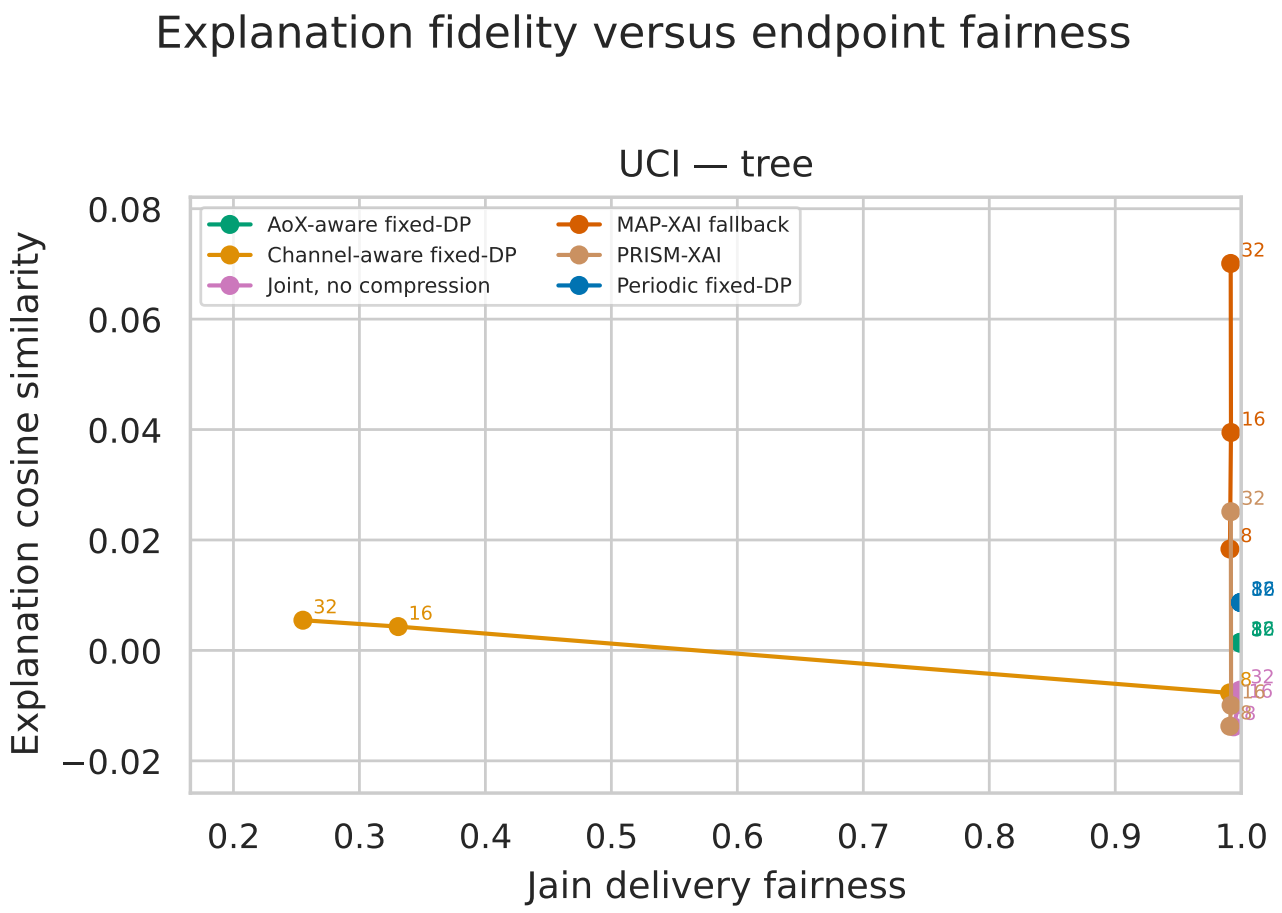


Fig. 46. Public-score threshold $\tau = 2$, release cap $M = 8$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

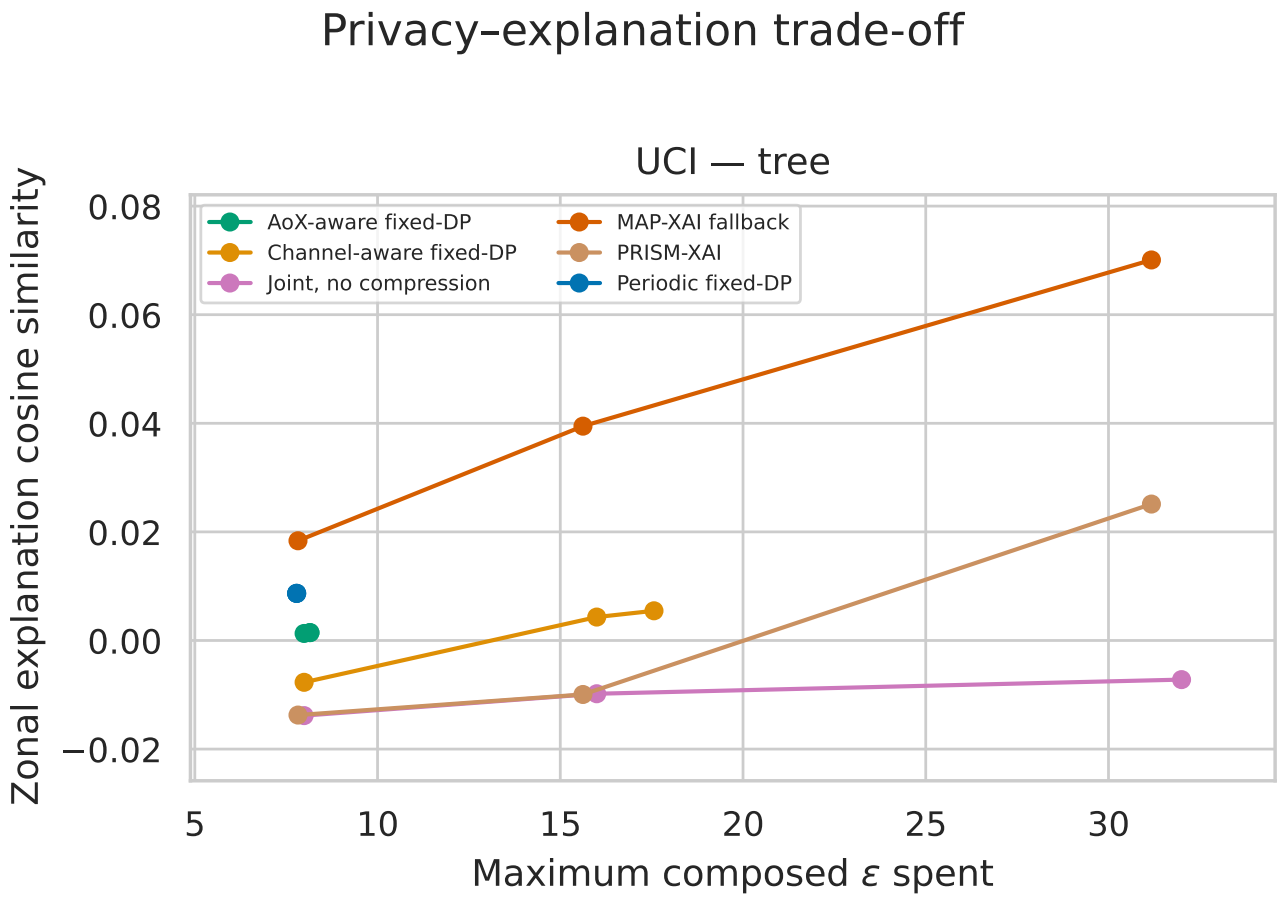


Fig. 47. Public-score threshold $\tau = 2$, release cap $M = 8$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

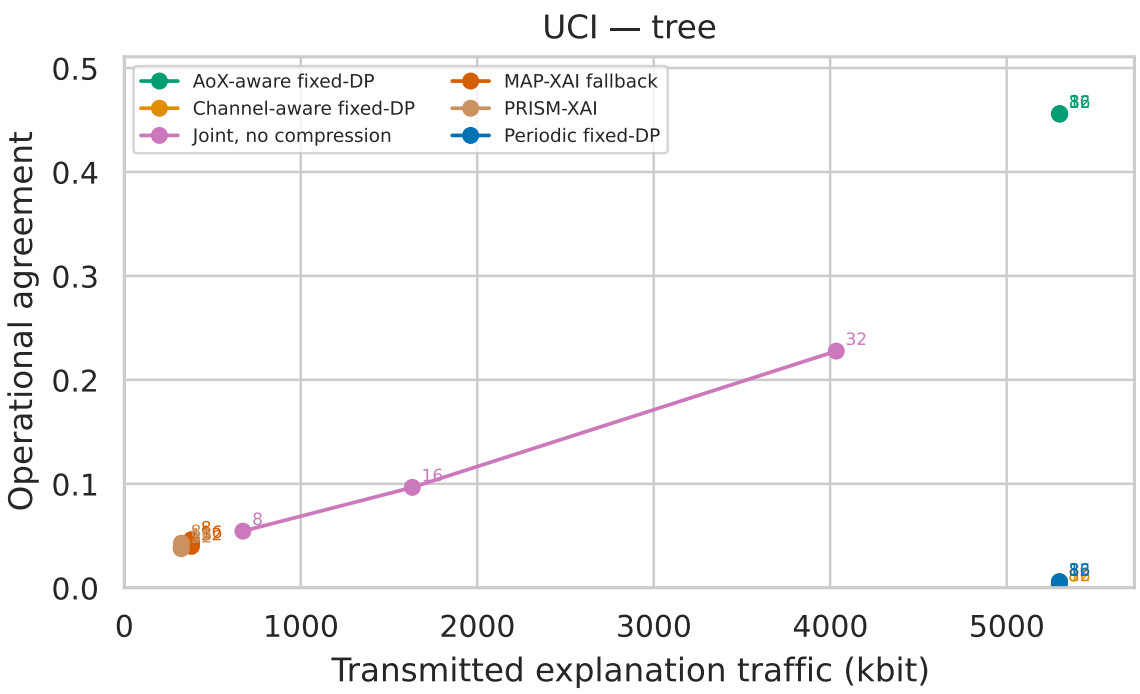


Fig. 48. Public-score threshold $\tau = 2$, release cap $M = 8$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

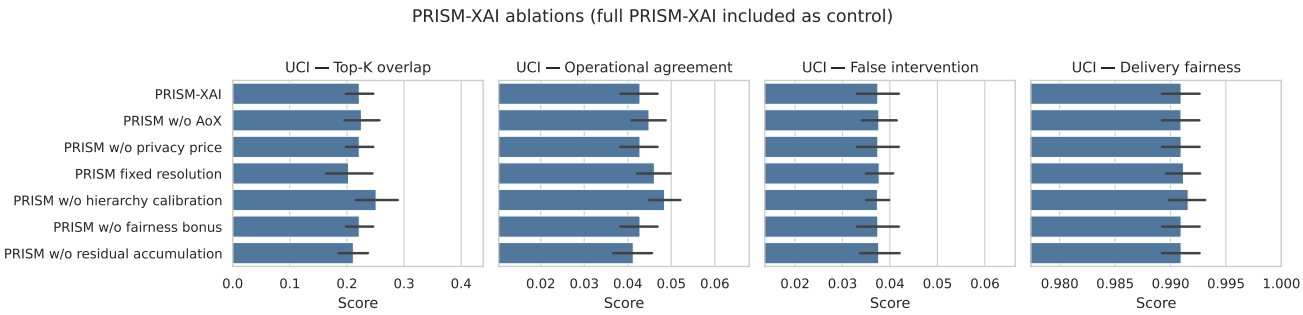


Fig. 49. Public-score threshold $\tau = 2$, release cap $M = 8$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

4) Public-score threshold $\tau = 3$, release cap $M = 8$: At this operating point, PRISM-XAI transmits 207.31 kbit versus 243.71 kbit for MAP-XAI-only (aggregate-mean ratio saving 14.94%). The paired difference is -36.40±3.32 kbit. PRISM coverage is 0.0779, critical-event recall is 0.0366, and maximum recorded expenditure is 18.206. Relative to MAP-XAI-only, the paired coverage change is -0.00010±0.00043 and recall change is -0.00083±0.00156. The descriptive overlap rule holds in 12 of 12 cells. Accordingly, this point is read as a matched-service scheduler comparison.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XLIII

PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.229 ± 0.033	0.036 ± 0.077	0.027 ± 0.010	0.027 ± 0.010	0.299 ± 0.119	0.929 ± 0.005	4.879 ± 1.736	88.291 ± 0.526	7.823 ± 0.000	0.024 ± 0.005	9900.309 ± 3435.464	0.990 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.220 ± 0.042	0.038 ± 0.084	0.029 ± 0.011	0.029 ± 0.011	0.301 ± 0.126	0.929 ± 0.004	5.164 ± 1.832	87.953 ± 0.538	7.823 ± 0.000	0.026 ± 0.006	10259.898 ± 5527.518	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.286 ± 0.074	0.022 ± 0.080	0.028 ± 0.007	0.028 ± 0.007	0.313 ± 0.084	0.929 ± 0.005	4.945 ± 1.699	106.298 ± 0.291	7.823 ± 0.000	0.023 ± 0.004	10008.188 ± 1905.763	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.232 ± 0.058	0.033 ± 0.078	0.028 ± 0.007	0.028 ± 0.007	0.293 ± 0.078	0.929 ± 0.004	5.230 ± 1.785	106.157 ± 0.283	7.823 ± 0.000	0.026 ± 0.004	9811.388 ± 1913.898	0.990 ± 0.006
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.224 ± 0.051	-0.005 ± 0.014	0.040 ± 0.007	0.040 ± 0.007	0.469 ± 0.065	0.910 ± 0.005	5.647 ± 1.735	328.571 ± 1.335	7.823 ± 0.000	0.031 ± 0.005	10347.978 ± 1665.157	0.991 ± 0.002
UCI	Tree	8	PRISM-XAI	0.228 ± 0.055	-0.004 ± 0.076	0.044 ± 0.009	0.044 ± 0.009	0.477 ± 0.093	0.911 ± 0.005	4.164 ± 1.674	325.598 ± 1.298	7.823 ± 0.000	0.033 ± 0.006	9624.204 ± 2175.413	0.992 ± 0.001
UCI	Tree	8	MAP-XAI fallback	0.178 ± 0.062	0.011 ± 0.013	0.040 ± 0.005	0.040 ± 0.005	0.458 ± 0.054	0.910 ± 0.005	5.670 ± 1.711	381.640 ± 0.105	7.823 ± 0.000	0.033 ± 0.004	11809.010 ± 1479.313	0.991 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.211 ± 0.046	0.055 ± 0.046	0.048 ± 0.009	0.048 ± 0.009	0.519 ± 0.078	0.910 ± 0.005	5.202 ± 1.753	381.153 ± 0.449	7.823 ± 0.000	0.031 ± 0.005	10053.490 ± 2134.182	0.992 ± 0.002

TABLE XLIV

PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	88.291 ± 0.526	0.229 ± 0.033	0.250 ± 0.000	0.036 ± 0.077	0.027 ± 0.010	0.027 ± 0.010	0.299 ± 0.119	0.071 ± 0.005	0.929 ± 0.005	0.024 ± 0.005	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.953 ± 0.538	0.220 ± 0.042	0.250 ± 0.000	0.038 ± 0.084	0.029 ± 0.011	0.029 ± 0.011	0.301 ± 0.126	0.071 ± 0.004	0.929 ± 0.004	0.026 ± 0.006	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	88.198 ± 0.520	0.240 ± 0.048	0.250 ± 0.000	0.143 ± 0.061	0.028 ± 0.008	0.028 ± 0.008	0.339 ± 0.094	0.071 ± 0.004	0.929 ± 0.004	0.021 ± 0.004	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.769 ± 0.479	0.284 ± 0.040	0.250 ± 0.000	0.171 ± 0.069	0.030 ± 0.009	0.030 ± 0.009	0.339 ± 0.095	0.071 ± 0.004	0.929 ± 0.004	0.022 ± 0.004	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	88.137 ± 0.486	0.399 ± 0.059	0.250 ± 0.000	0.306 ± 0.070	0.039 ± 0.009	0.039 ± 0.009	0.559 ± 0.157	0.071 ± 0.004	0.929 ± 0.004	0.013 ± 0.005	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.554 ± 0.429	0.388 ± 0.030	0.250 ± 0.000	0.359 ± 0.065	0.041 ± 0.007	0.041 ± 0.007	0.531 ± 0.102	0.071 ± 0.004	0.929 ± 0.004	0.014 ± 0.004	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	328.571 ± 1.335	0.224 ± 0.051	0.250 ± 0.000	-0.005 ± 0.014	0.040 ± 0.007	0.040 ± 0.007	0.469 ± 0.065	0.090 ± 0.005	0.910 ± 0.005	0.031 ± 0.005	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	325.598 ± 1.298	0.228 ± 0.055	0.250 ± 0.000	-0.004 ± 0.076	0.044 ± 0.009	0.044 ± 0.009	0.477 ± 0.093	0.089 ± 0.005	0.911 ± 0.005	0.033 ± 0.006	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	327.906 ± 1.114	0.246 ± 0.072	0.250 ± 0.000	0.001 ± 0.014	0.043 ± 0.005	0.043 ± 0.005	0.544 ± 0.057	0.090 ± 0.005	0.910 ± 0.005	0.025 ± 0.004	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	324.636 ± 1.010	0.225 ± 0.054	0.250 ± 0.000	-0.001 ± 0.073	0.043 ± 0.008	0.043 ± 0.008	0.586 ± 0.085	0.084 ± 0.004	0.916 ± 0.004	0.021 ± 0.005	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	328.131 ± 1.290	0.274 ± 0.061	0.250 ± 0.000	0.011 ± 0.017	0.038 ± 0.004	0.038 ± 0.004	0.873 ± 0.070	0.083 ± 0.003	0.917 ± 0.003	0.004 ± 0.003	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	324.963 ± 1.575	0.273 ± 0.067	0.250 ± 0.000	0.016 ± 0.096	0.037 ± 0.005	0.037 ± 0.005	0.936 ± 0.069	0.073 ± 0.003	0.927 ± 0.003	0.002 ± 0.003	31.175 ± 0.000

TABLE XLV
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.273 ± 0.526	-0.004 ± 0.050	-0.092 ± 0.035	-0.009 ± 0.005
OPSD	AoX-aware fixed-DP	-5011.714 ± 5.451	-0.013 ± 0.080	-0.373 ± 0.076	-0.010 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.087 ± 0.486	0.146 ± 0.079	-0.373 ± 0.080	-0.010 ± 0.006
OPSD	AoX-aware fixed-DP	-1472.611 ± 0.538	-0.018 ± 0.064	-0.091 ± 0.034	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-5012.144 ± 5.460	0.032 ± 0.058	-0.384 ± 0.058	-0.010 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.670 ± 0.429	0.136 ± 0.044	-0.384 ± 0.058	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.273 ± 0.526	-0.063 ± 0.047	-0.080 ± 0.032	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4598.411 ± 12.815	-0.021 ± 0.068	-0.373 ± 0.089	0.053 ± 0.007
OPSD	Channel-aware fixed-DP	-5212.087 ± 0.486	0.141 ± 0.068	-0.395 ± 0.100	0.083 ± 0.007
OPSD	Channel-aware fixed-DP	-1472.611 ± 0.538	-0.072 ± 0.070	-0.082 ± 0.030	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4598.841 ± 12.832	0.026 ± 0.054	-0.389 ± 0.088	0.053 ± 0.007
OPSD	Channel-aware fixed-DP	-5212.670 ± 0.429	0.131 ± 0.041	-0.415 ± 0.091	0.084 ± 0.008
OPSD	Joint, no compression	-101.609 ± 0.684	-0.058 ± 0.082	0.011 ± 0.010	-0.005 ± 0.005
OPSD	Joint, no compression	-371.702 ± 1.119	0.009 ± 0.094	-0.010 ± 0.013	-0.007 ± 0.006
OPSD	Joint, no compression	-1052.763 ± 7.970	0.164 ± 0.086	-0.035 ± 0.021	-0.009 ± 0.006
OPSD	Joint, no compression	-101.797 ± 0.692	-0.038 ± 0.066	0.010 ± 0.011	-0.006 ± 0.005
OPSD	Joint, no compression	-372.281 ± 1.026	0.044 ± 0.067	-0.008 ± 0.013	-0.007 ± 0.006
OPSD	Joint, no compression	-1053.346 ± 8.125	0.106 ± 0.065	-0.039 ± 0.016	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-1472.273 ± 0.526	-0.011 ± 0.046	-0.098 ± 0.026	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5024.396 ± 0.520	-0.009 ± 0.063	-0.422 ± 0.063	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5211.699 ± 0.561	0.150 ± 0.058	-0.419 ± 0.067	-0.010 ± 0.006
OPSD	Periodic fixed-DP	-1472.611 ± 0.538	-0.022 ± 0.072	-0.106 ± 0.022	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5024.825 ± 0.479	0.041 ± 0.045	-0.432 ± 0.050	-0.010 ± 0.006
OPSD	Periodic fixed-DP	-5212.282 ± 0.583	0.144 ± 0.042	-0.430 ± 0.050	-0.010 ± 0.006
OPSD	MAP-XAI fallback	-18.008 ± 0.539	-0.057 ± 0.060	-0.001 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.100 ± 0.509	-0.057 ± 0.073	-0.001 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.058 ± 0.492	0.014 ± 0.057	-0.003 ± 0.007	-0.001 ± 0.002
OPSD	MAP-XAI fallback	-18.204 ± 0.553	-0.012 ± 0.070	0.001 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.367 ± 0.475	0.029 ± 0.046	0.002 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.273 ± 0.432	0.033 ± 0.045	-0.001 ± 0.006	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4971.653 ± 1.335	-0.039 ± 0.084	-0.417 ± 0.031	-0.009 ± 0.002
UCI	AoX-aware fixed-DP	-4972.318 ± 1.114	-0.018 ± 0.091	-0.415 ± 0.030	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4972.093 ± 1.290	0.010 ± 0.061	-0.419 ± 0.032	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4974.626 ± 1.298	-0.032 ± 0.056	-0.411 ± 0.049	-0.007 ± 0.001
UCI	AoX-aware fixed-DP	-4975.588 ± 1.010	-0.035 ± 0.047	-0.411 ± 0.047	-0.006 ± 0.002
UCI	AoX-aware fixed-DP	-4975.261 ± 1.575	0.013 ± 0.058	-0.417 ± 0.047	-0.006 ± 0.002
UCI	Channel-aware fixed-DP	-4966.606 ± 3.827	-0.024 ± 0.060	0.039 ± 0.006	-0.000 ± 0.002
UCI	Channel-aware fixed-DP	-4972.318 ± 1.114	-0.022 ± 0.079	0.042 ± 0.005	0.661 ± 0.002
UCI	Channel-aware fixed-DP	-4972.093 ± 1.290	0.006 ± 0.063	0.037 ± 0.005	0.736 ± 0.002
UCI	Channel-aware fixed-DP	-4969.580 ± 4.222	-0.011 ± 0.068	0.044 ± 0.010	0.001 ± 0.001
UCI	Channel-aware fixed-DP	-4975.588 ± 1.010	-0.035 ± 0.050	0.043 ± 0.008	0.662 ± 0.002
UCI	Channel-aware fixed-DP	-4975.261 ± 1.575	0.014 ± 0.069	0.037 ± 0.005	0.738 ± 0.003
UCI	Joint, no compression	-343.729 ± 1.578	-0.061 ± 0.086	-0.012 ± 0.008	-0.003 ± 0.003
UCI	Joint, no compression	-1304.244 ± 1.271	0.029 ± 0.092	-0.052 ± 0.006	-0.006 ± 0.002
UCI	Joint, no compression	-3706.419 ± 4.695	0.028 ± 0.058	-0.190 ± 0.013	-0.007 ± 0.002
UCI	Joint, no compression	-346.702 ± 1.528	-0.002 ± 0.075	-0.013 ± 0.012	-0.002 ± 0.003
UCI	Joint, no compression	-1307.964 ± 0.799	0.021 ± 0.051	-0.056 ± 0.018	-0.005 ± 0.002
UCI	Joint, no compression	-3709.587 ± 5.063	0.032 ± 0.078	-0.191 ± 0.030	-0.006 ± 0.002
UCI	Periodic fixed-DP	-4970.747 ± 1.956	-0.035 ± 0.082	0.034 ± 0.007	-0.009 ± 0.002
UCI	Periodic fixed-DP	-4971.412 ± 1.872	-0.014 ± 0.077	0.037 ± 0.006	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4971.187 ± 1.698	0.015 ± 0.099	0.032 ± 0.004	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4973.721 ± 1.578	-0.033 ± 0.093	0.039 ± 0.008	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4974.682 ± 1.543	-0.035 ± 0.074	0.038 ± 0.007	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4974.355 ± 1.546	0.013 ± 0.096	0.032 ± 0.005	-0.006 ± 0.002
UCI	MAP-XAI fallback	-53.069 ± 1.324	0.047 ± 0.064	0.000 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-53.673 ± 1.109	0.023 ± 0.057	0.001 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-53.427 ± 1.194	0.041 ± 0.046	-0.004 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-55.556 ± 1.430	0.017 ± 0.054	-0.004 ± 0.007	0.000 ± 0.000
UCI	MAP-XAI fallback	-56.375 ± 1.113	-0.005 ± 0.048	0.001 ± 0.007	0.001 ± 0.002
UCI	MAP-XAI fallback	-55.666 ± 1.169	-0.026 ± 0.079	-0.001 ± 0.006	0.000 ± 0.001

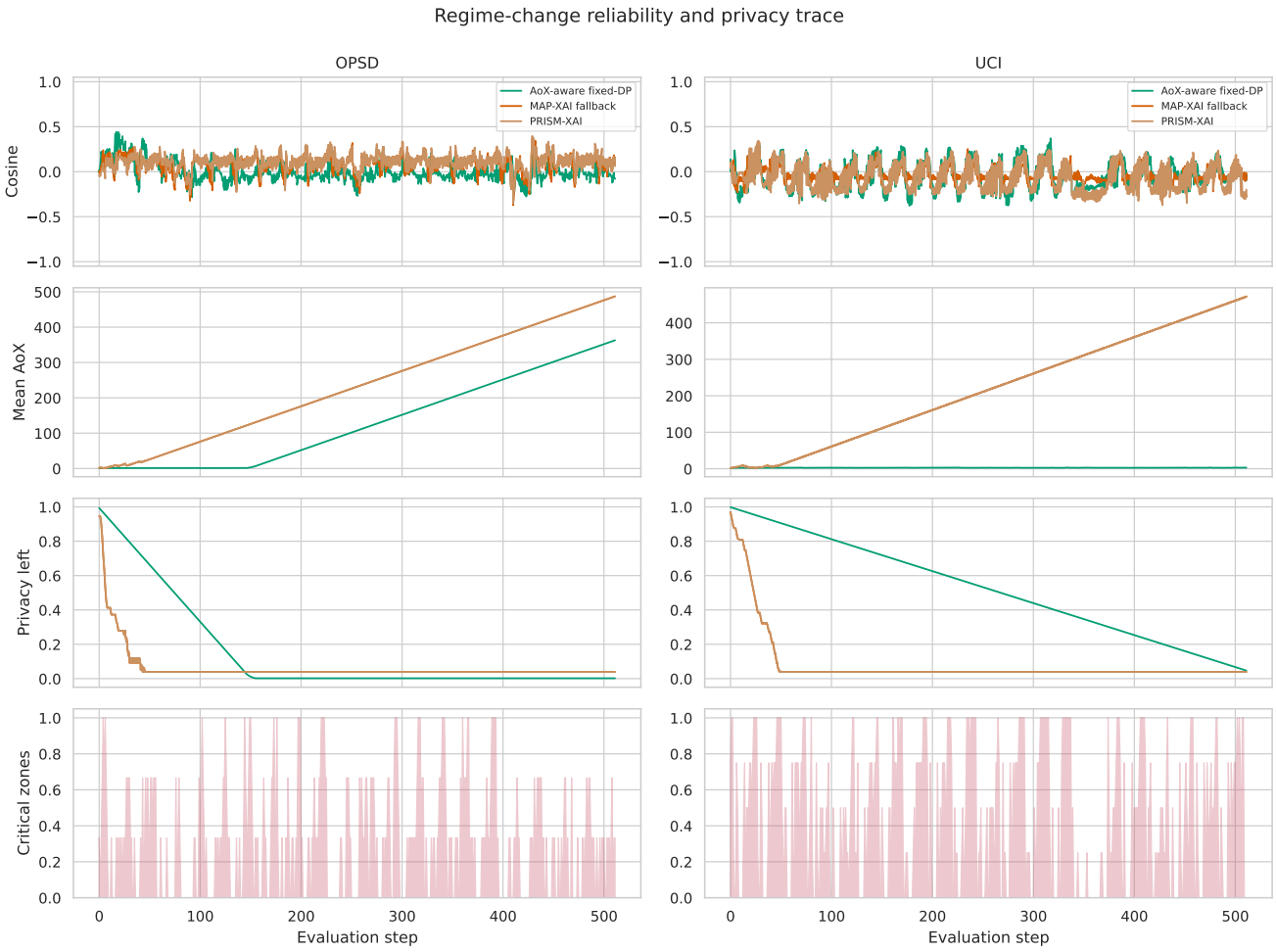


Fig. 50. Public-score threshold $\tau = 3$, release cap $M = 8$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

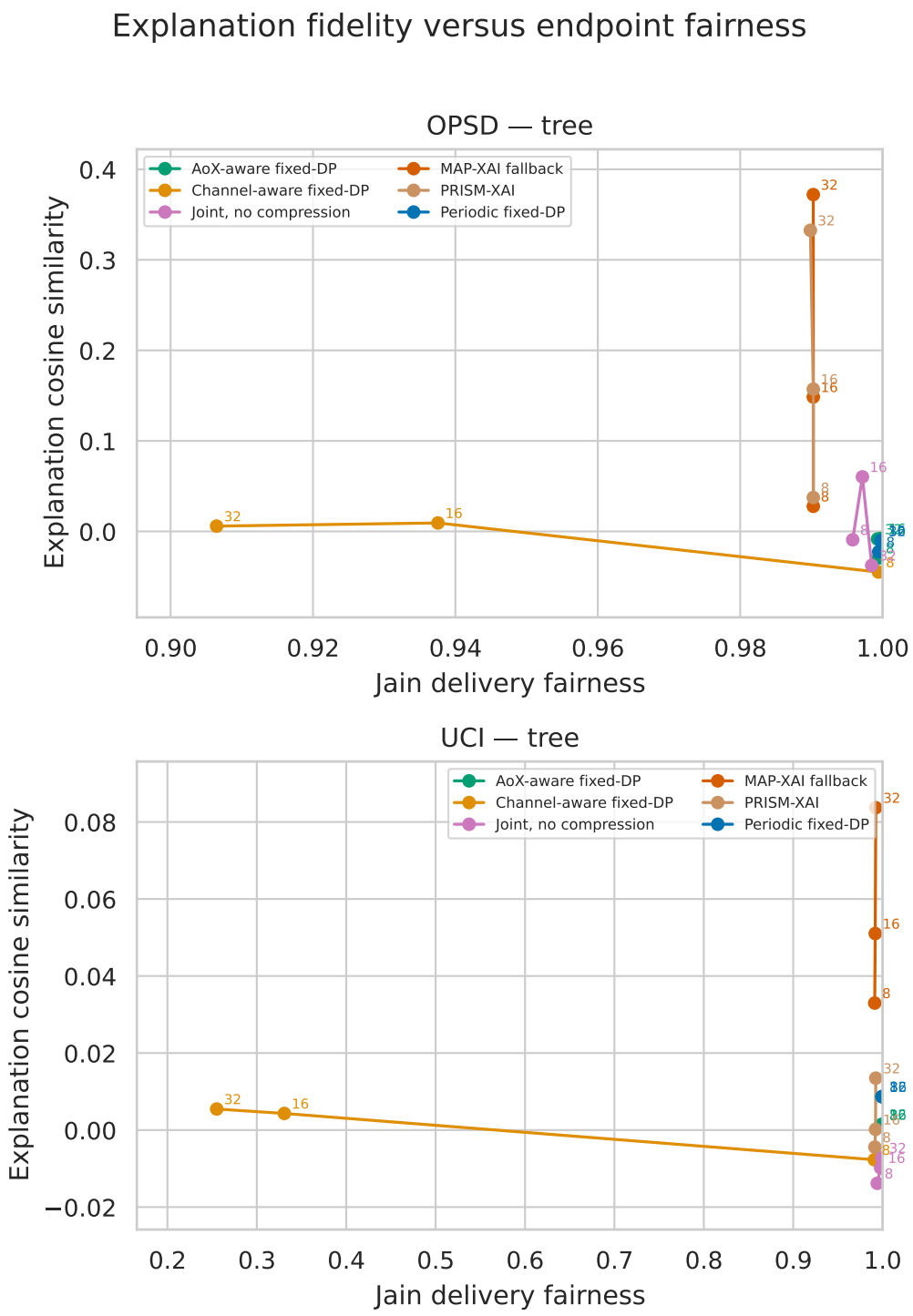


Fig. 51. Public-score threshold $\tau = 3$, release cap $M = 8$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

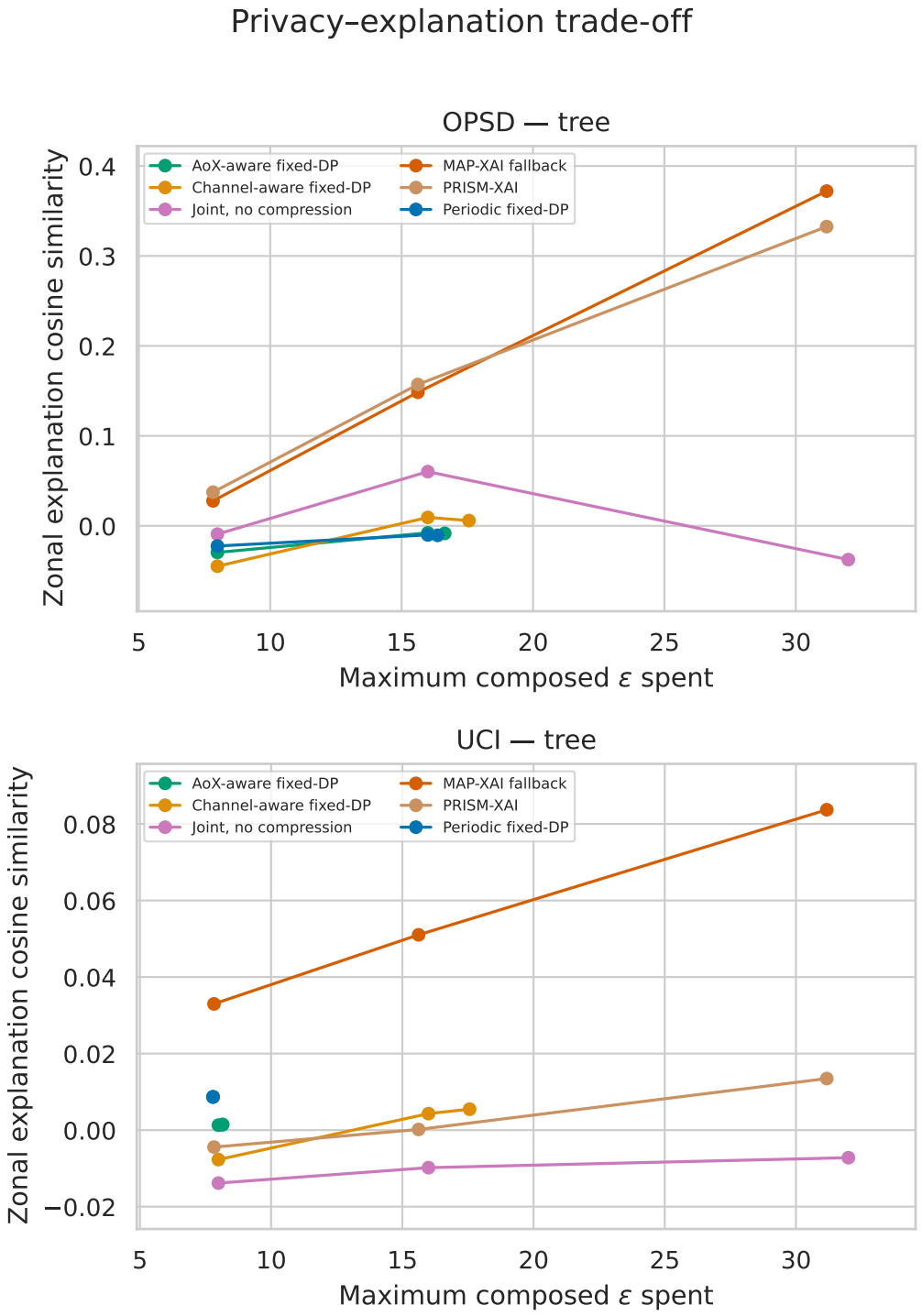


Fig. 52. Public-score threshold $\tau = 3$, release cap $M = 8$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

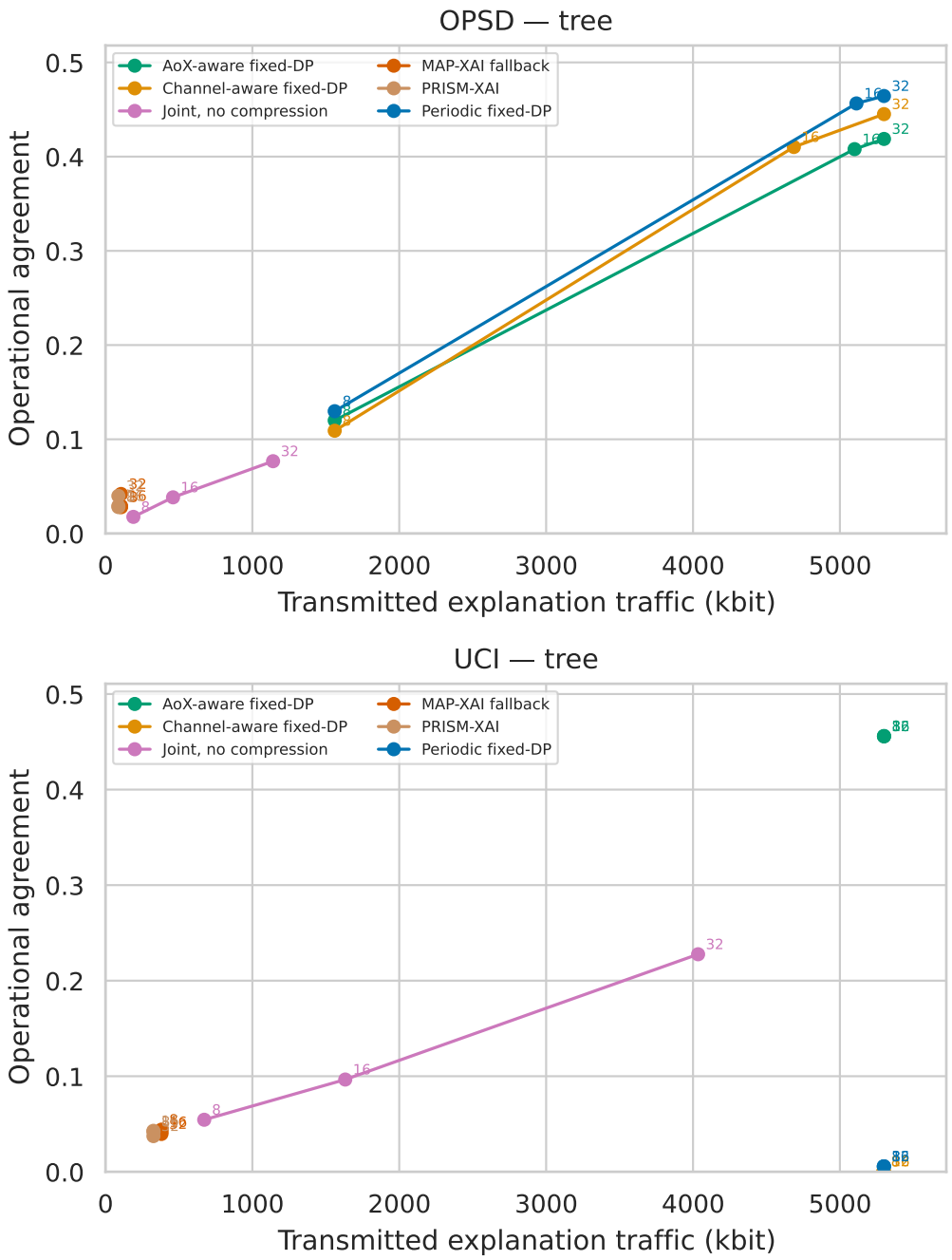


Fig. 53. Public-score threshold $\tau = 3$, release cap $M = 8$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

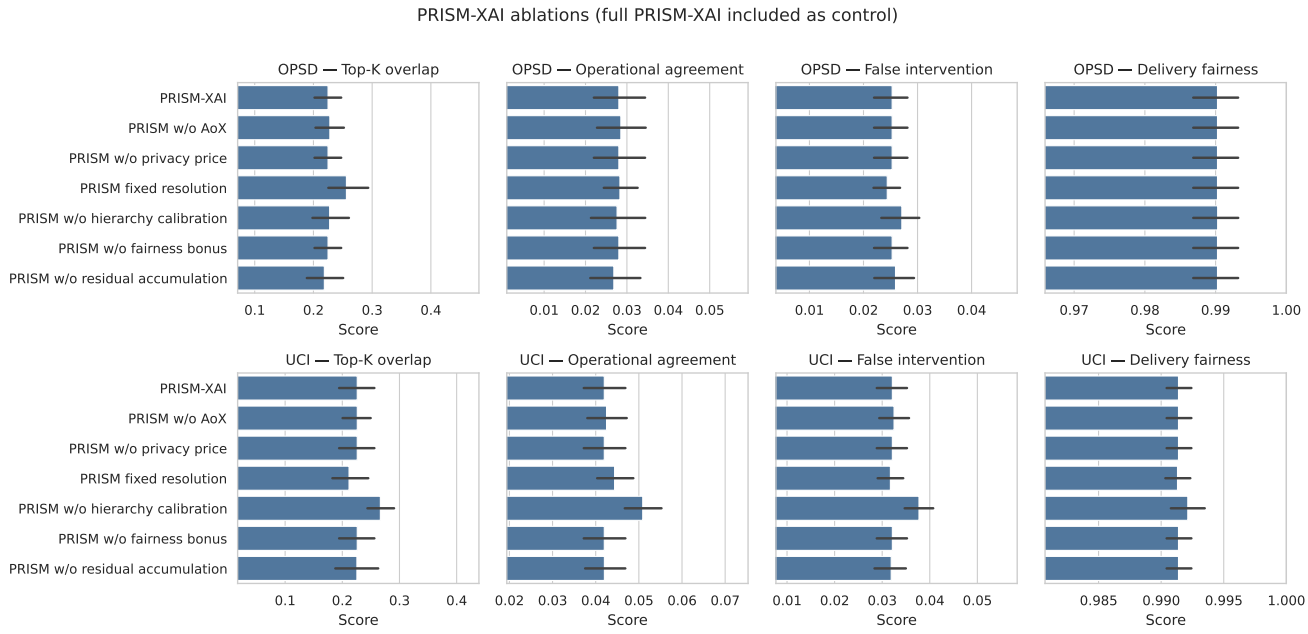


Fig. 54. Public-score threshold $\tau = 3$, release cap $M = 8$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XLVI
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.229 ± 0.033	0.036 ± 0.077	0.027 ± 0.010	0.027 ± 0.010	0.299 ± 0.119	0.929 ± 0.005	4.879 ± 1.736	88.291 ± 0.526	7.823 ± 0.000	0.024 ± 0.005	9900.309 ± 3435.464	0.990 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.220 ± 0.042	0.038 ± 0.084	0.029 ± 0.011	0.029 ± 0.011	0.301 ± 0.126	0.929 ± 0.004	5.164 ± 1.832	87.953 ± 0.538	7.823 ± 0.000	0.026 ± 0.006	10259.898 ± 5527.518	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.286 ± 0.074	0.022 ± 0.080	0.028 ± 0.007	0.028 ± 0.007	0.313 ± 0.084	0.929 ± 0.005	4.945 ± 1.699	106.298 ± 0.291	7.823 ± 0.000	0.023 ± 0.004	10008.188 ± 1905.763	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.232 ± 0.058	0.033 ± 0.078	0.028 ± 0.007	0.028 ± 0.007	0.293 ± 0.078	0.929 ± 0.004	5.230 ± 1.785	106.157 ± 0.283	7.823 ± 0.000	0.026 ± 0.004	9811.388 ± 1913.898	0.990 ± 0.006

TABLE XLVII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	88.291 ± 0.526	0.229 ± 0.033	0.250 ± 0.000	0.036 ± 0.077	0.027 ± 0.010	0.027 ± 0.010	0.299 ± 0.119	0.071 ± 0.005	0.929 ± 0.005	0.024 ± 0.005	7.823 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.953 ± 0.538	0.220 ± 0.042	0.250 ± 0.000	0.038 ± 0.084	0.029 ± 0.011	0.029 ± 0.011	0.301 ± 0.126	0.071 ± 0.004	0.929 ± 0.004	0.026 ± 0.006	7.823 ± 0.000
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	88.198 ± 0.520	0.240 ± 0.048	0.250 ± 0.000	0.143 ± 0.061	0.028 ± 0.008	0.028 ± 0.008	0.339 ± 0.094	0.071 ± 0.004	0.929 ± 0.004	0.021 ± 0.004	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	87.769 ± 0.479	0.284 ± 0.040	0.250 ± 0.000	0.171 ± 0.069	0.030 ± 0.009	0.030 ± 0.009	0.339 ± 0.095	0.071 ± 0.004	0.929 ± 0.004	0.022 ± 0.004	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	88.137 ± 0.486	0.399 ± 0.059	0.250 ± 0.000	0.306 ± 0.070	0.039 ± 0.009	0.039 ± 0.009	0.559 ± 0.157	0.071 ± 0.004	0.929 ± 0.004	0.013 ± 0.005	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	87.554 ± 0.429	0.388 ± 0.030	0.250 ± 0.000	0.359 ± 0.065	0.041 ± 0.007	0.041 ± 0.007	0.531 ± 0.102	0.071 ± 0.004	0.929 ± 0.004	0.014 ± 0.004	31.175 ± 0.000

TABLE XLVIII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.273 ± 0.526	-0.004 ± 0.050	-0.092 ± 0.035	-0.009 ± 0.005
OPSD	AoX-aware fixed-DP	-5011.714 ± 5.451	-0.013 ± 0.080	-0.373 ± 0.076	-0.010 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.087 ± 0.486	0.146 ± 0.079	-0.373 ± 0.080	-0.010 ± 0.006
OPSD	AoX-aware fixed-DP	-1472.611 ± 0.538	-0.018 ± 0.064	-0.091 ± 0.034	-0.009 ± 0.006
OPSD	AoX-aware fixed-DP	-5012.144 ± 5.460	0.032 ± 0.058	-0.384 ± 0.058	-0.010 ± 0.006
OPSD	AoX-aware fixed-DP	-5212.670 ± 0.429	0.136 ± 0.044	-0.384 ± 0.058	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.273 ± 0.526	-0.063 ± 0.047	-0.080 ± 0.032	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4598.411 ± 12.815	-0.021 ± 0.068	-0.373 ± 0.089	0.053 ± 0.007
OPSD	Channel-aware fixed-DP	-5212.087 ± 0.486	0.141 ± 0.068	-0.395 ± 0.100	0.083 ± 0.007
OPSD	Channel-aware fixed-DP	-1472.611 ± 0.538	-0.072 ± 0.070	-0.082 ± 0.030	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4598.841 ± 12.832	0.026 ± 0.054	-0.389 ± 0.088	0.053 ± 0.007
OPSD	Channel-aware fixed-DP	-5212.670 ± 0.429	0.131 ± 0.041	-0.415 ± 0.091	0.084 ± 0.008
OPSD	Joint, no compression	-101.609 ± 0.684	-0.058 ± 0.082	0.011 ± 0.010	-0.005 ± 0.005
OPSD	Joint, no compression	-371.702 ± 1.119	0.009 ± 0.094	-0.010 ± 0.013	-0.007 ± 0.006
OPSD	Joint, no compression	-1052.763 ± 7.970	0.164 ± 0.086	-0.035 ± 0.021	-0.009 ± 0.006
OPSD	Joint, no compression	-101.797 ± 0.692	-0.038 ± 0.066	0.010 ± 0.011	-0.006 ± 0.005
OPSD	Joint, no compression	-372.281 ± 1.026	0.044 ± 0.067	-0.008 ± 0.013	-0.007 ± 0.006
OPSD	Joint, no compression	-1053.346 ± 8.125	0.106 ± 0.065	-0.039 ± 0.016	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-1472.273 ± 0.526	-0.011 ± 0.046	-0.098 ± 0.026	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5024.396 ± 0.520	-0.009 ± 0.063	-0.422 ± 0.063	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5211.699 ± 0.561	0.150 ± 0.058	-0.419 ± 0.067	-0.010 ± 0.006
OPSD	Periodic fixed-DP	-1472.611 ± 0.538	-0.022 ± 0.072	-0.106 ± 0.022	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-5024.825 ± 0.479	0.041 ± 0.045	-0.432 ± 0.050	-0.010 ± 0.006
OPSD	Periodic fixed-DP	-5212.282 ± 0.583	0.144 ± 0.042	-0.430 ± 0.050	-0.010 ± 0.006
OPSD	MAP-XAI fallback	-18.008 ± 0.539	-0.057 ± 0.060	-0.001 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.100 ± 0.509	-0.057 ± 0.073	-0.001 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.058 ± 0.492	0.014 ± 0.057	-0.003 ± 0.007	-0.001 ± 0.002
OPSD	MAP-XAI fallback	-18.204 ± 0.553	-0.012 ± 0.070	0.001 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.367 ± 0.475	0.029 ± 0.046	0.002 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.273 ± 0.432	0.033 ± 0.045	-0.001 ± 0.006	0.000 ± 0.000

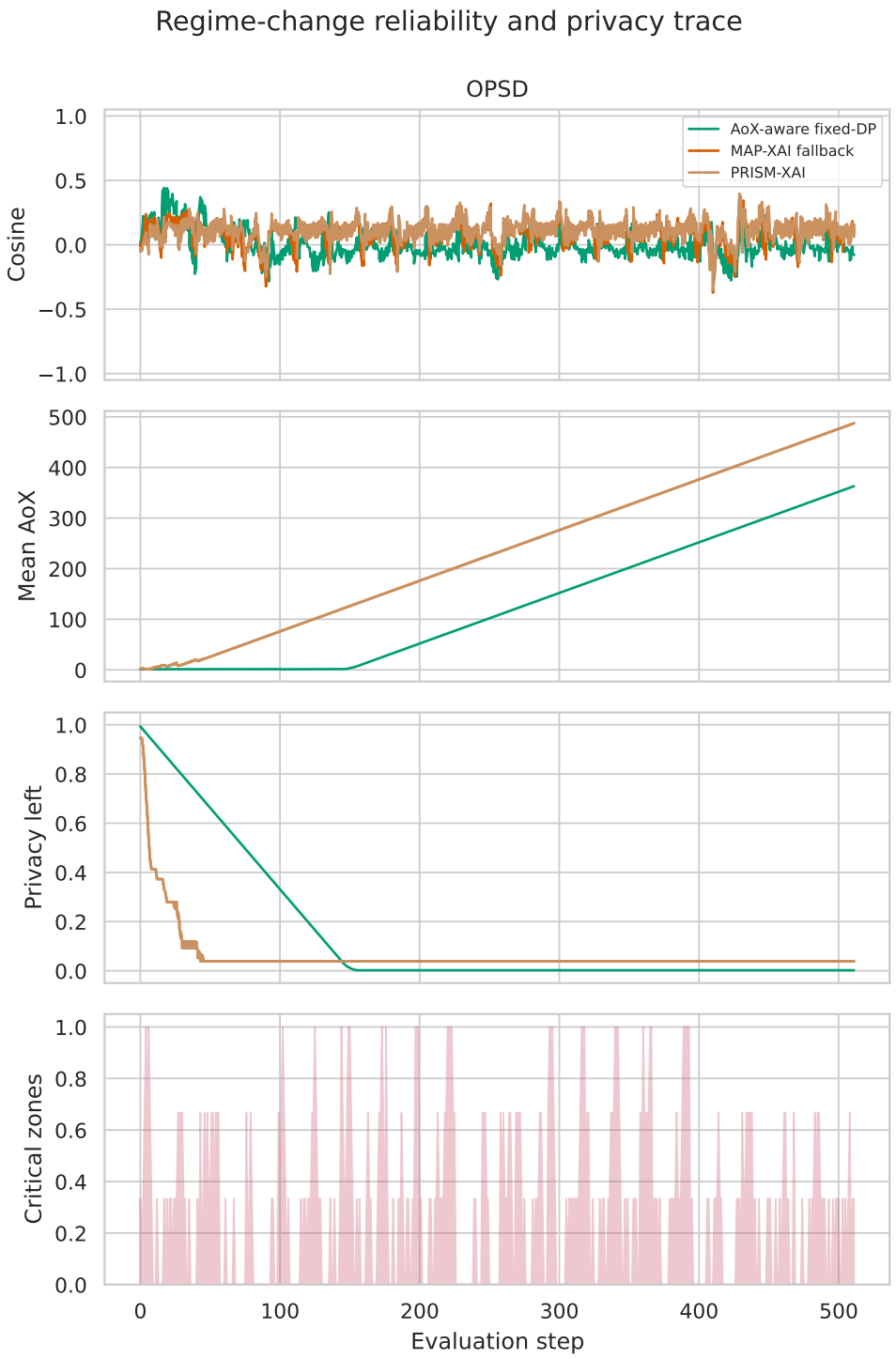


Fig. 55. Public-score threshold $\tau = 3$, release cap $M = 8$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

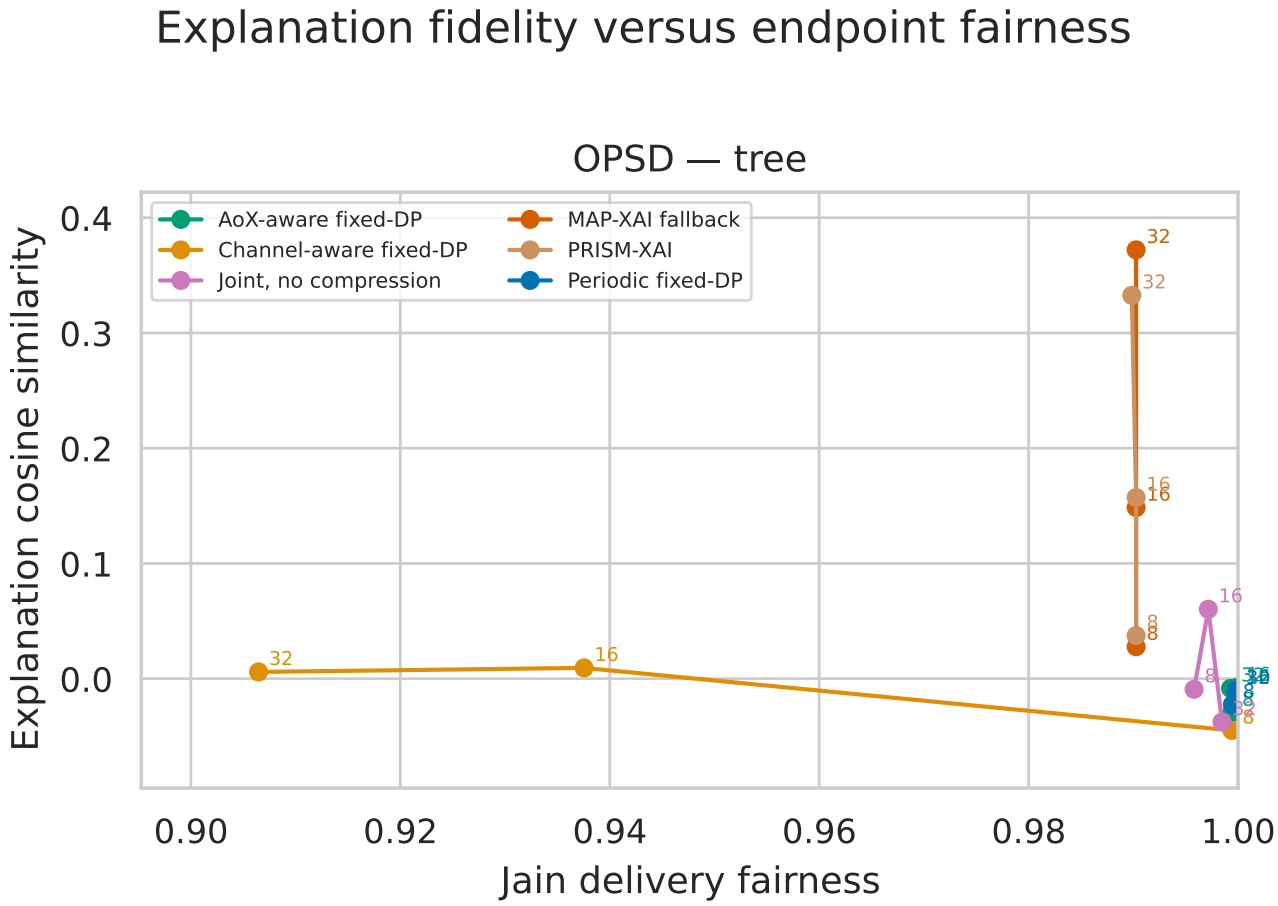


Fig. 56. Public-score threshold $\tau = 3$, release cap $M = 8$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

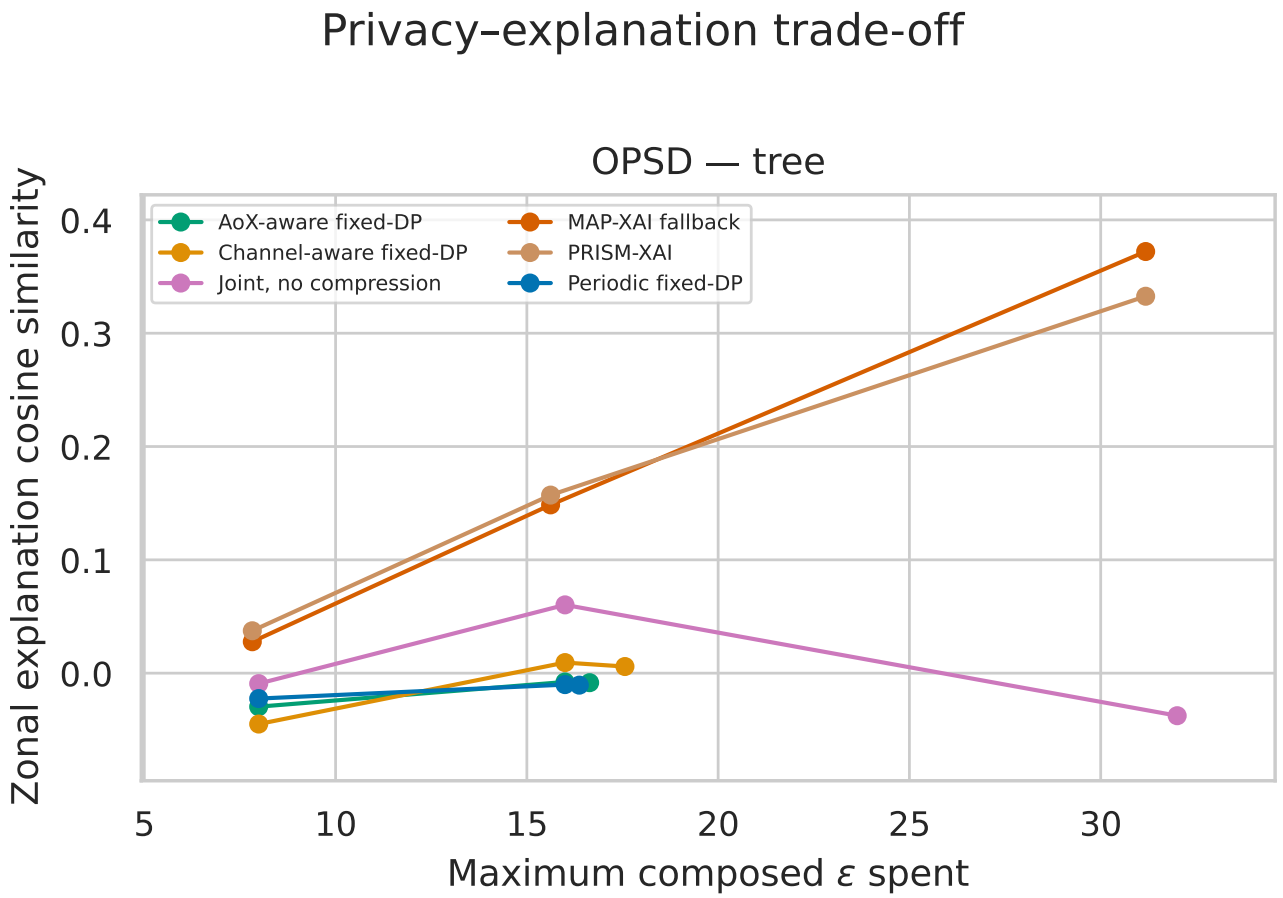


Fig. 57. Public-score threshold $\tau = 3$, release cap $M = 8$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

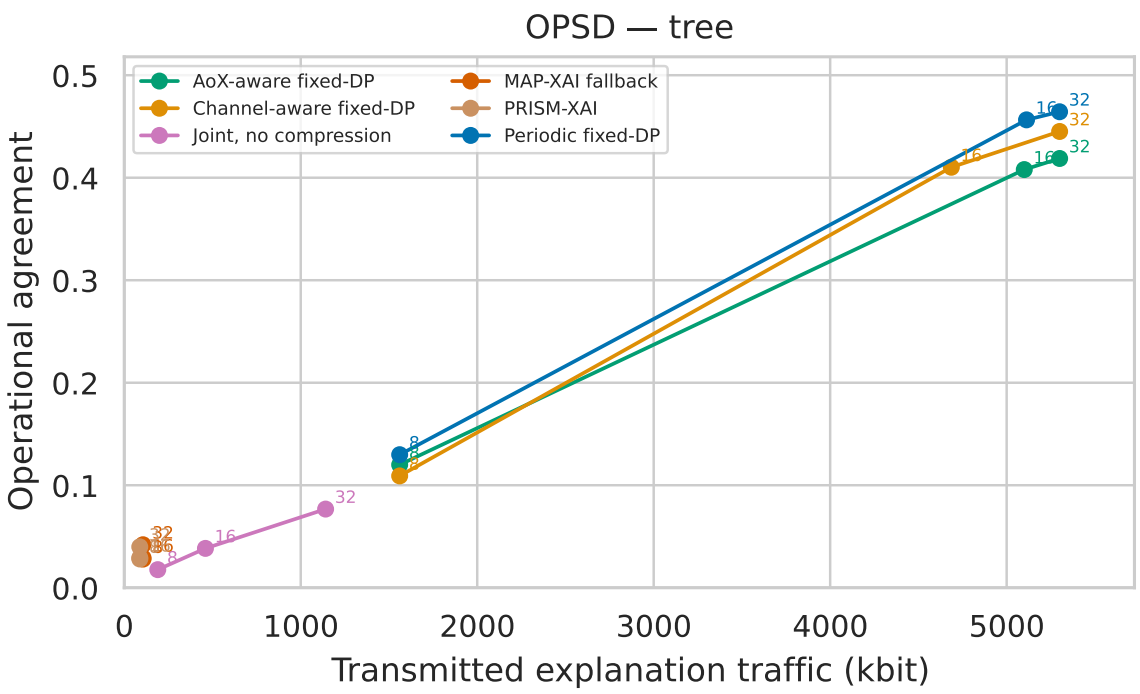


Fig. 58. Public-score threshold $\tau = 3$, release cap $M = 8$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

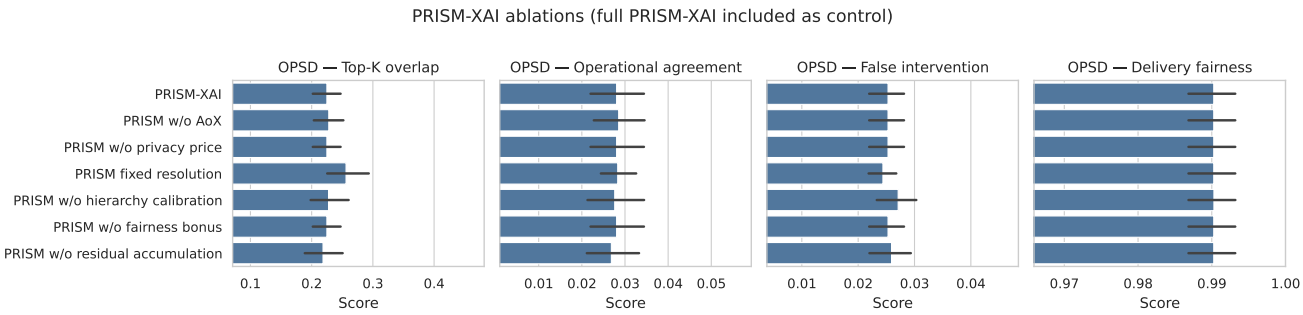


Fig. 59. Public-score threshold $\tau = 3$, release cap $M = 8$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XLIX
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.224 ± 0.051	-0.005 ± 0.014	0.040 ± 0.007	0.040 ± 0.007	0.469 ± 0.065	0.910 ± 0.005	5.647 ± 1.735	328.571 ± 1.335	7.823 ± 0.000	0.031 ± 0.005	10347.978 ± 1665.157	0.991 ± 0.002
UCI	Tree	8	PRISM-XAI	0.228 ± 0.055	-0.004 ± 0.076	0.044 ± 0.009	0.044 ± 0.009	0.477 ± 0.093	0.911 ± 0.005	4.164 ± 1.674	325.598 ± 1.298	7.823 ± 0.000	0.033 ± 0.006	9624.204 ± 2175.413	0.992 ± 0.001
UCI	Tree	8	MAP-XAI fallback	0.178 ± 0.062	0.011 ± 0.013	0.040 ± 0.005	0.040 ± 0.005	0.458 ± 0.054	0.910 ± 0.005	5.670 ± 1.711	381.640 ± 0.105	7.823 ± 0.000	0.033 ± 0.004	11809.010 ± 1479.313	0.991 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.211 ± 0.046	0.055 ± 0.046	0.048 ± 0.009	0.048 ± 0.009	0.519 ± 0.078	0.910 ± 0.005	5.202 ± 1.753	381.153 ± 0.449	7.823 ± 0.000	0.031 ± 0.005	10053.490 ± 2134.182	0.992 ± 0.002

TABLE L

PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	328.571 ± 1.335	0.224 ± 0.051	0.250 ± 0.000	-0.005 ± 0.014	0.040 ± 0.007	0.040 ± 0.007	0.469 ± 0.065	0.090 ± 0.005	0.910 ± 0.005	0.031 ± 0.005	7.823 ± 0.000
UCI	Tree	8	PRISM-XAI	325.598 ± 1.298	0.228 ± 0.055	0.250 ± 0.000	-0.004 ± 0.076	0.044 ± 0.009	0.044 ± 0.009	0.477 ± 0.093	0.089 ± 0.005	0.911 ± 0.005	0.033 ± 0.006	7.823 ± 0.000
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	327.906 ± 1.114	0.246 ± 0.072	0.250 ± 0.000	0.001 ± 0.014	0.043 ± 0.005	0.043 ± 0.005	0.544 ± 0.057	0.090 ± 0.005	0.910 ± 0.005	0.025 ± 0.004	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	324.636 ± 1.010	0.225 ± 0.054	0.250 ± 0.000	-0.001 ± 0.073	0.043 ± 0.008	0.043 ± 0.008	0.586 ± 0.085	0.084 ± 0.004	0.916 ± 0.004	0.021 ± 0.005	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	328.131 ± 1.290	0.274 ± 0.061	0.250 ± 0.000	0.011 ± 0.017	0.038 ± 0.004	0.038 ± 0.004	0.873 ± 0.070	0.083 ± 0.003	0.917 ± 0.003	0.004 ± 0.003	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	324.963 ± 1.575	0.273 ± 0.067	0.250 ± 0.000	0.016 ± 0.096	0.037 ± 0.005	0.037 ± 0.005	0.936 ± 0.069	0.073 ± 0.003	0.927 ± 0.003	0.002 ± 0.003	31.175 ± 0.000

TABLE LI
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 8$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4971.653 ± 1.335	-0.039 ± 0.084	-0.417 ± 0.031	-0.009 ± 0.002
UCI	AoX-aware fixed-DP	-4972.318 ± 1.114	-0.018 ± 0.091	-0.415 ± 0.030	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4972.093 ± 1.290	0.010 ± 0.061	-0.419 ± 0.032	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4974.626 ± 1.298	-0.032 ± 0.056	-0.411 ± 0.049	-0.007 ± 0.001
UCI	AoX-aware fixed-DP	-4975.588 ± 1.010	-0.035 ± 0.047	-0.411 ± 0.047	-0.006 ± 0.002
UCI	AoX-aware fixed-DP	-4975.261 ± 1.575	0.013 ± 0.058	-0.417 ± 0.047	-0.006 ± 0.002
UCI	Channel-aware fixed-DP	-4966.606 ± 3.827	-0.024 ± 0.060	0.039 ± 0.006	-0.000 ± 0.002
UCI	Channel-aware fixed-DP	-4972.318 ± 1.114	-0.022 ± 0.079	0.042 ± 0.005	0.661 ± 0.002
UCI	Channel-aware fixed-DP	-4972.093 ± 1.290	0.006 ± 0.063	0.037 ± 0.005	0.736 ± 0.002
UCI	Channel-aware fixed-DP	-4969.580 ± 4.222	-0.011 ± 0.068	0.044 ± 0.010	0.001 ± 0.001
UCI	Channel-aware fixed-DP	-4975.588 ± 1.010	-0.035 ± 0.050	0.043 ± 0.008	0.662 ± 0.002
UCI	Channel-aware fixed-DP	-4975.261 ± 1.575	0.014 ± 0.069	0.037 ± 0.005	0.738 ± 0.003
UCI	Joint, no compression	-343.729 ± 1.578	-0.061 ± 0.086	-0.012 ± 0.008	-0.003 ± 0.003
UCI	Joint, no compression	-1304.244 ± 1.271	0.029 ± 0.092	-0.052 ± 0.006	-0.006 ± 0.002
UCI	Joint, no compression	-3706.419 ± 4.695	0.028 ± 0.058	-0.190 ± 0.013	-0.007 ± 0.002
UCI	Joint, no compression	-346.702 ± 1.528	-0.002 ± 0.075	-0.013 ± 0.012	-0.002 ± 0.003
UCI	Joint, no compression	-1307.964 ± 0.799	0.021 ± 0.051	-0.056 ± 0.018	-0.005 ± 0.002
UCI	Joint, no compression	-3709.587 ± 5.063	0.032 ± 0.078	-0.191 ± 0.030	-0.006 ± 0.002
UCI	Periodic fixed-DP	-4970.747 ± 1.956	-0.035 ± 0.082	0.034 ± 0.007	-0.009 ± 0.002
UCI	Periodic fixed-DP	-4971.412 ± 1.872	-0.014 ± 0.077	0.037 ± 0.006	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4971.187 ± 1.698	0.015 ± 0.099	0.032 ± 0.004	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4973.721 ± 1.578	-0.033 ± 0.093	0.039 ± 0.008	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4974.682 ± 1.543	-0.035 ± 0.074	0.038 ± 0.007	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4974.355 ± 1.546	0.013 ± 0.096	0.032 ± 0.005	-0.006 ± 0.002
UCI	MAP-XAI fallback	-53.069 ± 1.324	0.047 ± 0.064	0.000 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-53.673 ± 1.109	0.023 ± 0.057	0.001 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-53.427 ± 1.194	0.041 ± 0.046	-0.004 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-55.556 ± 1.430	0.017 ± 0.054	-0.004 ± 0.007	0.000 ± 0.000
UCI	MAP-XAI fallback	-56.375 ± 1.113	-0.005 ± 0.048	0.001 ± 0.007	0.001 ± 0.002
UCI	MAP-XAI fallback	-55.666 ± 1.169	-0.026 ± 0.079	-0.001 ± 0.006	0.000 ± 0.001

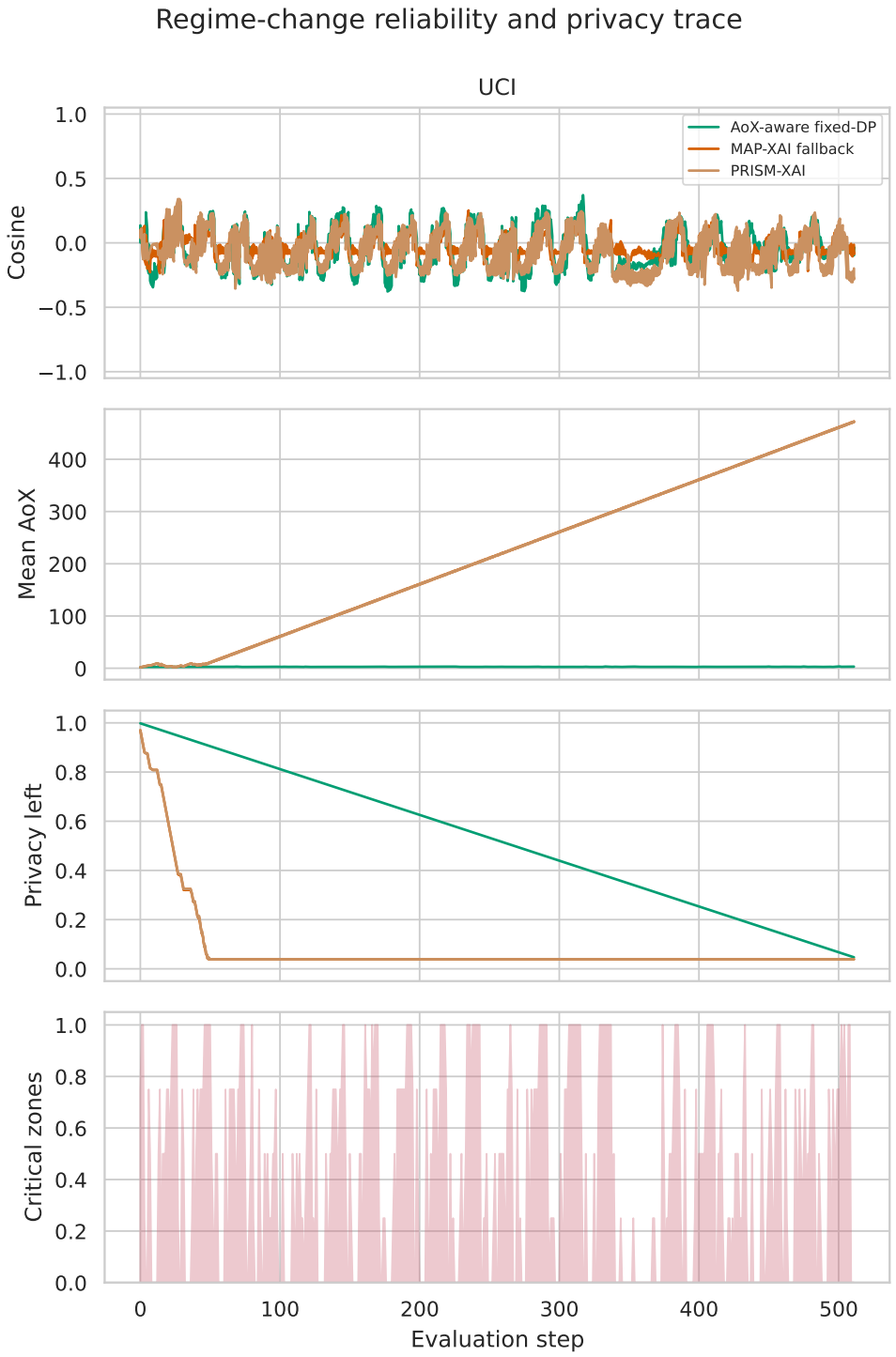


Fig. 60. Public-score threshold $\tau = 3$, release cap $M = 8$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

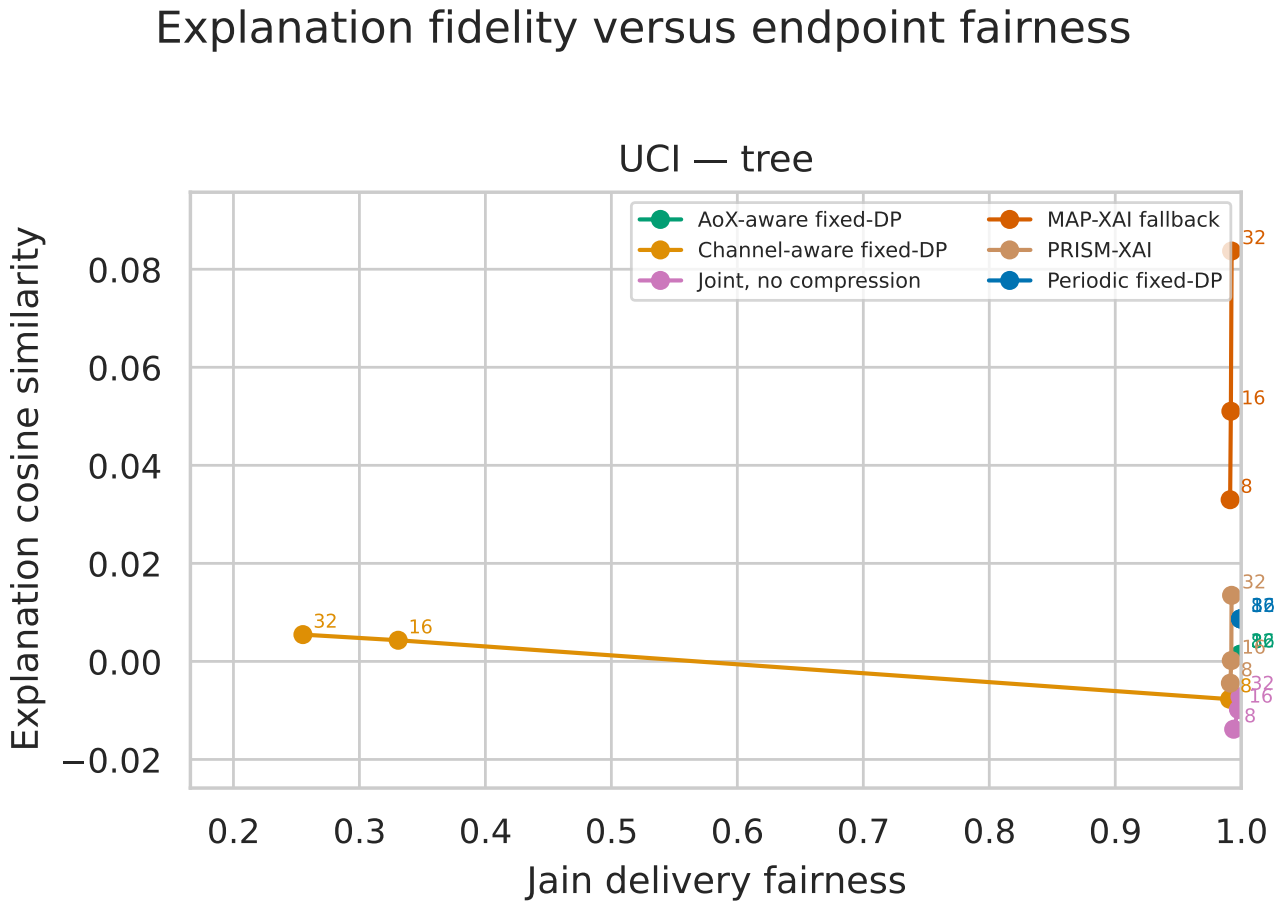


Fig. 61. Public-score threshold $\tau = 3$, release cap $M = 8$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

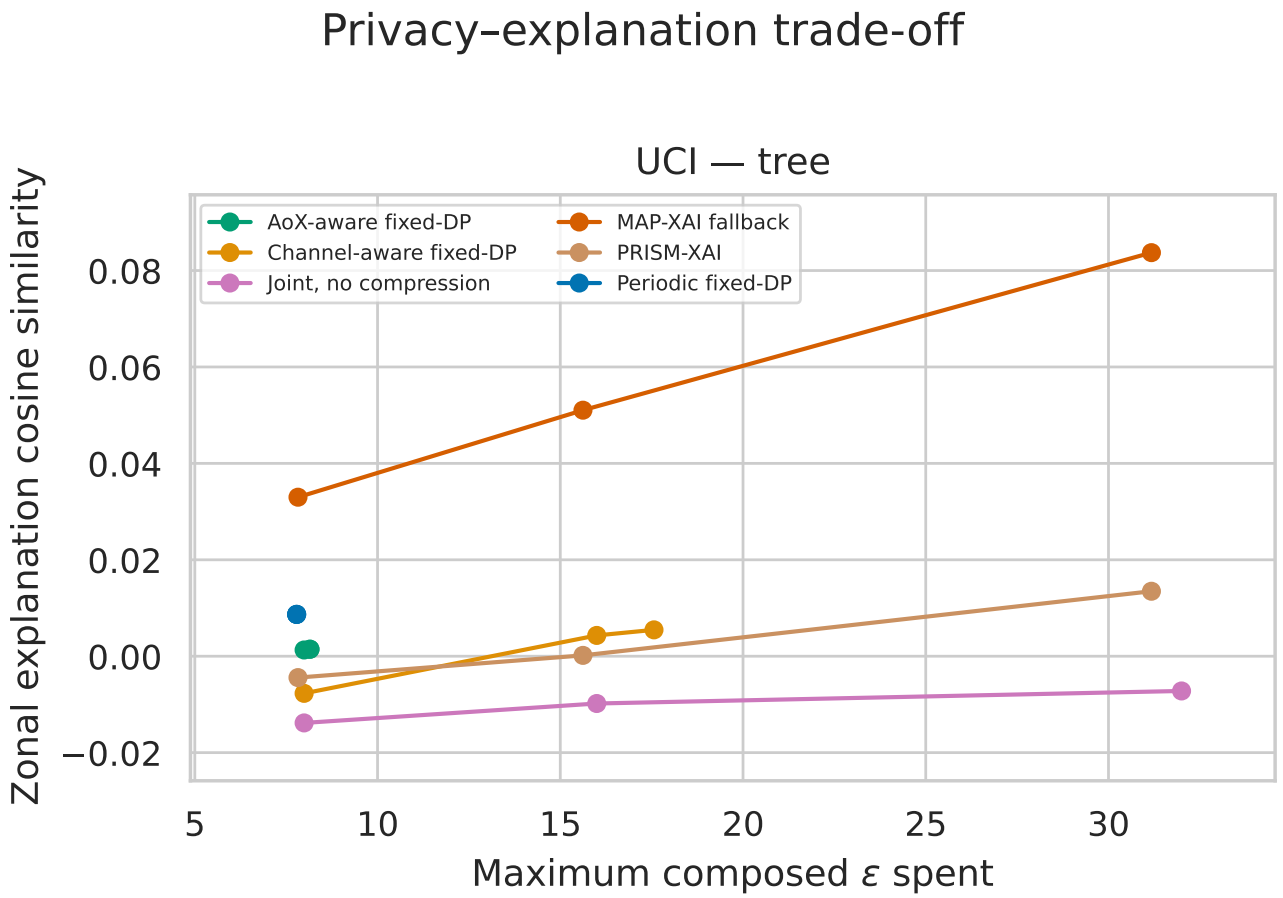


Fig. 62. Public-score threshold $\tau = 3$, release cap $M = 8$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

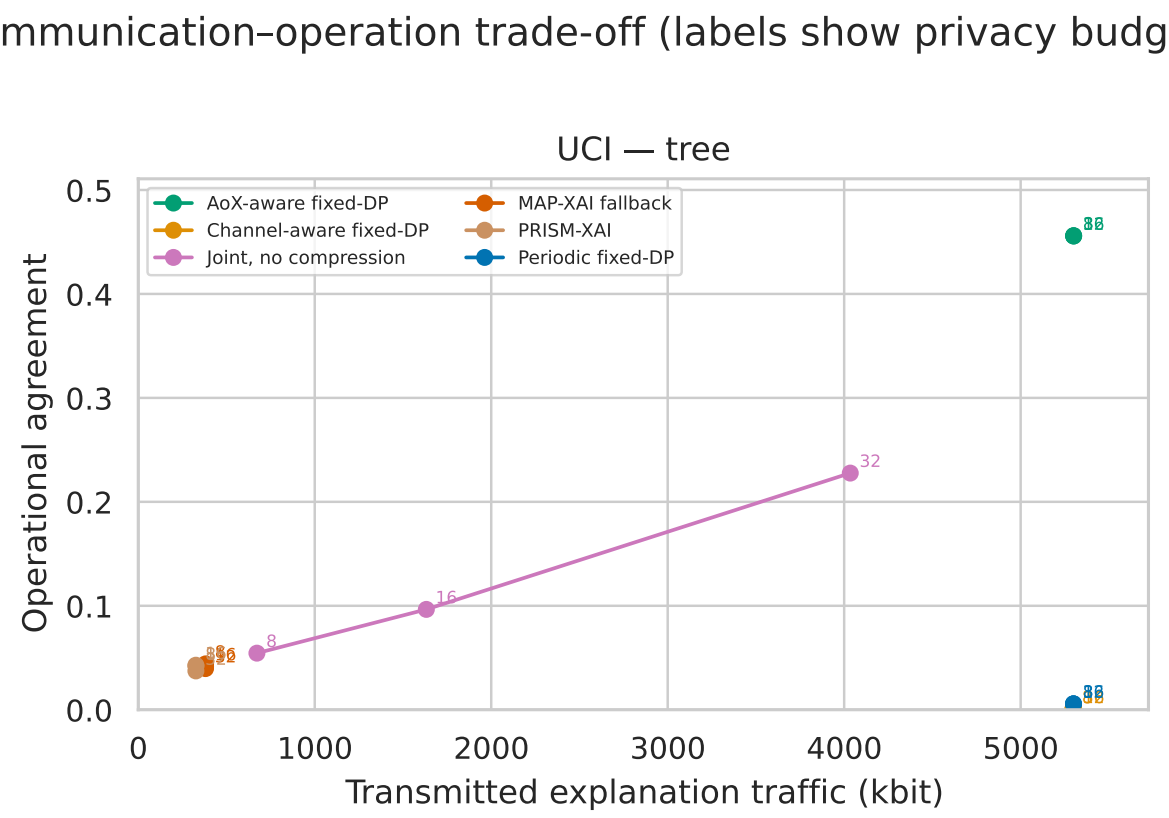


Fig. 63. Public-score threshold $\tau = 3$, release cap $M = 8$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

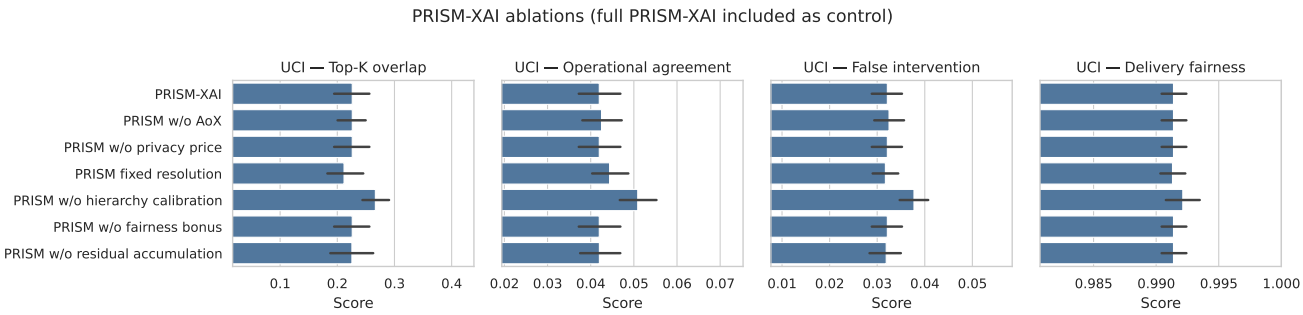


Fig. 64. Public-score threshold $\tau = 3$, release cap $M = 8$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

5) Public-score threshold $\tau = 1$, release cap $M = 16$: At this operating point, PRISM-XAI transmits 342.50 kbit versus 407.81 kbit for MAP-XAI-only (aggregate-mean ratio saving 16.01%). The paired difference is -65.31$\pm$7.13 kbit. PRISM coverage is 0.0856, critical-event recall is 0.0401, and maximum recorded expenditure is 17.727. Relative to MAP-XAI-only, the paired coverage change is -0.00075$\pm$0.00093 and recall change is +0.00082$\pm$0.00162. The descriptive overlap rule holds in 8 of 12 cells. Accordingly, this point is read as a sensitivity point on the service frontier, not a complete matched-service confirmation.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LII

PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.270 ± 0.066	-0.025 ± 0.093	0.015 ± 0.004	0.015 ± 0.004	0.200 ± 0.078	0.956 ± 0.003	4.630 ± 1.839	87.697 ± 0.410	6.034 ± 0.230	0.025 ± 0.005	15451.736 ± 3728.991	0.993 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.283 ± 0.054	-0.005 ± 0.086	0.017 ± 0.005	0.017 ± 0.005	0.222 ± 0.080	0.956 ± 0.003	4.630 ± 1.839	87.431 ± 0.364	5.936 ± 0.213	0.024 ± 0.005	14279.307 ± 3768.065	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.234 ± 0.073	-0.037 ± 0.075	0.016 ± 0.005	0.016 ± 0.005	0.191 ± 0.080	0.956 ± 0.003	4.732 ± 1.776	105.817 ± 0.049	7.786 ± 0.000	0.027 ± 0.006	21385.963 ± 9729.377	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.230 ± 0.067	-0.012 ± 0.085	0.017 ± 0.005	0.017 ± 0.005	0.219 ± 0.073	0.956 ± 0.003	4.732 ± 1.776	105.675 ± 0.049	7.786 ± 0.000	0.024 ± 0.006	19083.433 ± 9057.454	0.993 ± 0.004
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.240 ± 0.089	-0.008 ± 0.019	0.031 ± 0.004	0.031 ± 0.004	0.422 ± 0.039	0.931 ± 0.004	6.267 ± 1.448	323.739 ± 1.239	6.844 ± 0.203	0.030 ± 0.005	13090.115 ± 2761.149	0.990 ± 0.003
UCI	Tree	8	PRISM-XAI	0.264 ± 0.044	-0.056 ± 0.057	0.027 ± 0.005	0.027 ± 0.005	0.404 ± 0.063	0.938 ± 0.003	4.814 ± 1.710	321.228 ± 1.066	6.726 ± 0.146	0.027 ± 0.004	15136.774 ± 2629.908	0.989 ± 0.006
UCI	Tree	8	MAP-XAI fallback	0.240 ± 0.077	-0.011 ± 0.019	0.032 ± 0.005	0.032 ± 0.005	0.446 ± 0.058	0.931 ± 0.004	5.672 ± 1.709	382.099 ± 0.159	7.786 ± 0.000	0.027 ± 0.003	15488.351 ± 4957.026	0.991 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.265 ± 0.046	-0.009 ± 0.057	0.029 ± 0.005	0.029 ± 0.005	0.424 ± 0.053	0.937 ± 0.004	5.276 ± 1.728	381.645 ± 0.090	7.786 ± 0.000	0.027 ± 0.005	16573.020 ± 2779.080	0.990 ± 0.004

TABLE LIII

PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.697 ± 0.410	0.270 ± 0.066	0.250 ± 0.000	-0.025 ± 0.093	0.015 ± 0.004	0.015 ± 0.004	0.200 ± 0.078	0.044 ± 0.003	0.956 ± 0.003	0.025 ± 0.005	6.034 ± 0.230
OPSD	Tree	8	PRISM-XAI	87.431 ± 0.364	0.283 ± 0.054	0.250 ± 0.000	-0.005 ± 0.086	0.017 ± 0.005	0.017 ± 0.005	0.222 ± 0.080	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.005	5.936 ± 0.213
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	175.005 ± 0.564	0.242 ± 0.064	0.250 ± 0.000	0.085 ± 0.083	0.039 ± 0.008	0.039 ± 0.008	0.287 ± 0.077	0.081 ± 0.003	0.919 ± 0.003	0.038 ± 0.008	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	174.493 ± 0.509	0.213 ± 0.052	0.250 ± 0.000	0.102 ± 0.085	0.037 ± 0.009	0.037 ± 0.009	0.264 ± 0.059	0.081 ± 0.003	0.919 ± 0.003	0.039 ± 0.007	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	174.944 ± 0.552	0.309 ± 0.062	0.250 ± 0.000	0.203 ± 0.080	0.043 ± 0.009	0.043 ± 0.009	0.336 ± 0.099	0.081 ± 0.003	0.919 ± 0.003	0.033 ± 0.008	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	174.309 ± 0.461	0.326 ± 0.085	0.250 ± 0.000	0.247 ± 0.077	0.042 ± 0.009	0.042 ± 0.009	0.294 ± 0.073	0.081 ± 0.003	0.919 ± 0.003	0.039 ± 0.007	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	323.739 ± 1.239	0.240 ± 0.089	0.250 ± 0.000	-0.008 ± 0.019	0.031 ± 0.004	0.031 ± 0.004	0.422 ± 0.039	0.069 ± 0.004	0.931 ± 0.004	0.030 ± 0.005	6.844 ± 0.203
UCI	Tree	8	PRISM-XAI	321.228 ± 1.066	0.264 ± 0.044	0.250 ± 0.000	-0.056 ± 0.057	0.027 ± 0.005	0.027 ± 0.005	0.404 ± 0.063	0.062 ± 0.003	0.938 ± 0.003	0.027 ± 0.004	6.726 ± 0.146
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	647.230 ± 2.247	0.247 ± 0.082	0.250 ± 0.000	0.013 ± 0.024	0.064 ± 0.006	0.064 ± 0.006	0.454 ± 0.035	0.138 ± 0.010	0.862 ± 0.010	0.054 ± 0.009	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	644.319 ± 3.516	0.227 ± 0.040	0.250 ± 0.000	-0.008 ± 0.057	0.055 ± 0.012	0.055 ± 0.012	0.428 ± 0.061	0.123 ± 0.012	0.877 ± 0.012	0.051 ± 0.012	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	649.317 ± 2.732	0.249 ± 0.061	0.250 ± 0.000	0.022 ± 0.019	0.063 ± 0.009	0.063 ± 0.009	0.602 ± 0.060	0.128 ± 0.016	0.872 ± 0.016	0.030 ± 0.009	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	650.306 ± 2.517	0.221 ± 0.026	0.250 ± 0.000	0.064 ± 0.065	0.048 ± 0.007	0.048 ± 0.007	0.624 ± 0.077	0.095 ± 0.011	0.905 ± 0.011	0.021 ± 0.006	31.175 ± 0.000

TABLE LIV
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.867 ± 0.410	0.036 ± 0.086	-0.104 ± 0.029	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-4924.908 ± 5.576	-0.011 ± 0.052	-0.362 ± 0.075	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5125.280 ± 0.552	0.056 ± 0.077	-0.370 ± 0.075	-0.003 ± 0.002
OPSD	AoX-aware fixed-DP	-1473.133 ± 0.364	0.045 ± 0.066	-0.104 ± 0.027	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-4925.420 ± 5.602	-0.039 ± 0.045	-0.378 ± 0.056	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5125.915 ± 0.461	0.074 ± 0.085	-0.384 ± 0.056	-0.004 ± 0.002
OPSD	Channel-aware fixed-DP	-1472.867 ± 0.410	-0.023 ± 0.091	-0.092 ± 0.025	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4511.604 ± 12.966	-0.019 ± 0.055	-0.362 ± 0.090	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5125.280 ± 0.552	0.051 ± 0.072	-0.392 ± 0.096	0.089 ± 0.006
OPSD	Channel-aware fixed-DP	-1473.133 ± 0.364	-0.010 ± 0.076	-0.094 ± 0.024	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4512.116 ± 12.805	-0.045 ± 0.039	-0.382 ± 0.087	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5125.915 ± 0.461	0.069 ± 0.081	-0.414 ± 0.089	0.089 ± 0.006
OPSD	Joint, no compression	-102.203 ± 0.525	-0.018 ± 0.097	-0.001 ± 0.004	-0.002 ± 0.003
OPSD	Joint, no compression	-284.895 ± 1.260	0.011 ± 0.071	0.001 ± 0.011	-0.001 ± 0.002
OPSD	Joint, no compression	-965.956 ± 8.155	0.073 ± 0.062	-0.031 ± 0.015	-0.003 ± 0.002
OPSD	Joint, no compression	-102.319 ± 0.436	0.024 ± 0.089	-0.002 ± 0.003	-0.002 ± 0.003
OPSD	Joint, no compression	-285.557 ± 1.134	-0.027 ± 0.062	-0.002 ± 0.012	-0.001 ± 0.002
OPSD	Joint, no compression	-966.591 ± 8.036	0.044 ± 0.085	-0.038 ± 0.011	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-1472.867 ± 0.410	0.029 ± 0.092	-0.110 ± 0.020	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-4937.589 ± 0.564	-0.007 ± 0.060	-0.411 ± 0.063	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5124.892 ± 0.754	0.060 ± 0.073	-0.415 ± 0.064	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-1473.133 ± 0.364	0.041 ± 0.073	-0.118 ± 0.016	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-4938.101 ± 0.509	-0.029 ± 0.052	-0.426 ± 0.048	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5125.527 ± 0.718	0.082 ± 0.090	-0.429 ± 0.048	-0.004 ± 0.002
OPSD	MAP-XAI fallback	-18.120 ± 0.422	0.035 ± 0.088	-0.000 ± 0.003	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.812 ± 0.567	-0.011 ± 0.064	0.004 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.873 ± 0.554	-0.004 ± 0.061	0.003 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.245 ± 0.372	0.053 ± 0.042	-0.000 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-39.182 ± 0.498	-0.034 ± 0.034	0.004 ± 0.009	0.000 ± 0.000
OPSD	MAP-XAI fallback	-39.346 ± 0.444	0.055 ± 0.050	0.001 ± 0.005	-0.000 ± 0.001
UCI	AoX-aware fixed-DP	-4976.485 ± 1.239	-0.023 ± 0.103	-0.426 ± 0.034	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4652.994 ± 2.247	-0.017 ± 0.102	-0.393 ± 0.032	-0.005 ± 0.002
UCI	AoX-aware fixed-DP	-4650.907 ± 2.732	-0.015 ± 0.080	-0.394 ± 0.036	-0.004 ± 0.001
UCI	AoX-aware fixed-DP	-4978.996 ± 1.066	0.003 ± 0.059	-0.428 ± 0.050	-0.011 ± 0.006
UCI	AoX-aware fixed-DP	-4655.905 ± 3.516	-0.034 ± 0.057	-0.399 ± 0.048	-0.004 ± 0.002
UCI	AoX-aware fixed-DP	-4649.918 ± 2.517	-0.039 ± 0.038	-0.407 ± 0.047	-0.004 ± 0.001
UCI	Channel-aware fixed-DP	-4971.438 ± 4.206	-0.008 ± 0.107	0.031 ± 0.005	-0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4652.994 ± 2.247	-0.020 ± 0.100	0.064 ± 0.006	0.664 ± 0.002
UCI	Channel-aware fixed-DP	-4650.907 ± 2.732	-0.020 ± 0.069	0.062 ± 0.009	0.740 ± 0.001
UCI	Channel-aware fixed-DP	-4973.950 ± 4.295	0.024 ± 0.064	0.027 ± 0.005	-0.002 ± 0.006
UCI	Channel-aware fixed-DP	-4655.905 ± 3.516	-0.034 ± 0.053	0.055 ± 0.012	0.664 ± 0.002
UCI	Channel-aware fixed-DP	-4649.918 ± 2.517	-0.038 ± 0.034	0.048 ± 0.007	0.740 ± 0.002
UCI	Joint, no compression	-348.561 ± 1.547	-0.045 ± 0.099	-0.021 ± 0.006	-0.004 ± 0.004
UCI	Joint, no compression	-984.920 ± 2.348	0.031 ± 0.081	-0.030 ± 0.007	-0.003 ± 0.002
UCI	Joint, no compression	-3385.233 ± 6.863	0.002 ± 0.071	-0.165 ± 0.019	-0.004 ± 0.001
UCI	Joint, no compression	-351.072 ± 1.196	0.034 ± 0.062	-0.030 ± 0.008	-0.006 ± 0.006
UCI	Joint, no compression	-988.281 ± 3.390	0.023 ± 0.047	-0.044 ± 0.018	-0.003 ± 0.002
UCI	Joint, no compression	-3384.244 ± 6.390	-0.020 ± 0.038	-0.180 ± 0.031	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4975.579 ± 1.762	-0.019 ± 0.103	0.025 ± 0.005	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4652.089 ± 3.147	-0.012 ± 0.080	0.058 ± 0.005	-0.005 ± 0.002
UCI	Periodic fixed-DP	-4650.001 ± 3.755	-0.011 ± 0.060	0.057 ± 0.009	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4978.090 ± 1.619	0.003 ± 0.071	0.022 ± 0.005	-0.011 ± 0.006
UCI	Periodic fixed-DP	-4654.999 ± 3.804	-0.034 ± 0.048	0.050 ± 0.012	-0.005 ± 0.002
UCI	Periodic fixed-DP	-4649.013 ± 3.233	-0.039 ± 0.037	0.042 ± 0.008	-0.004 ± 0.001
UCI	MAP-XAI fallback	-58.359 ± 1.137	-0.000 ± 0.105	-0.001 ± 0.002	-0.001 ± 0.001
UCI	MAP-XAI fallback	-118.869 ± 2.191	0.006 ± 0.053	0.004 ± 0.008	-0.000 ± 0.001
UCI	MAP-XAI fallback	-116.843 ± 2.730	0.009 ± 0.084	0.003 ± 0.004	0.000 ± 0.001
UCI	MAP-XAI fallback	-60.417 ± 1.098	-0.002 ± 0.046	-0.002 ± 0.006	-0.001 ± 0.002
UCI	MAP-XAI fallback	-121.326 ± 3.553	0.006 ± 0.056	-0.004 ± 0.010	-0.000 ± 0.001
UCI	MAP-XAI fallback	-115.339 ± 2.574	-0.015 ± 0.037	-0.002 ± 0.008	0.000 ± 0.001

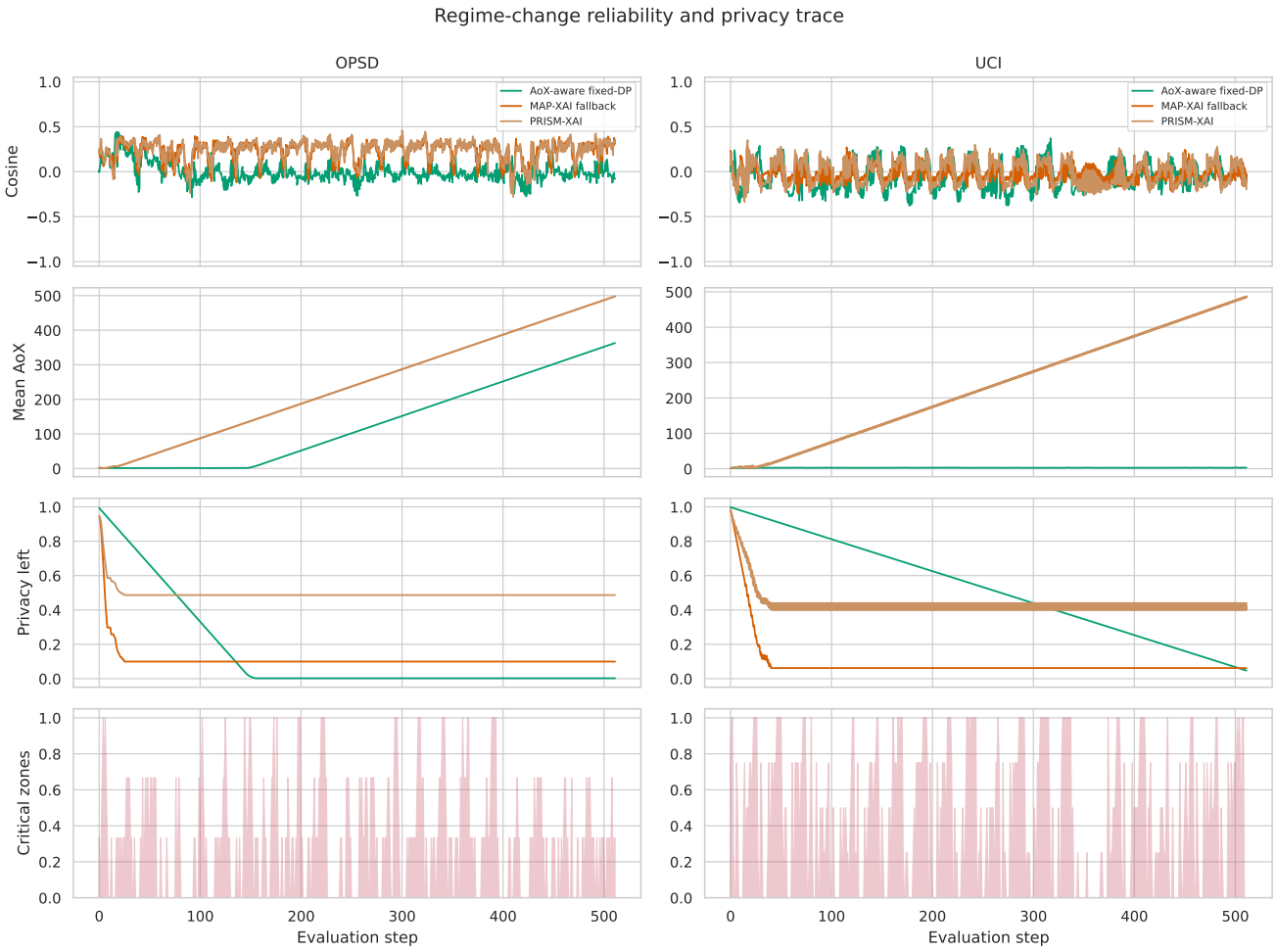


Fig. 65. Public-score threshold $\tau = 1$, release cap $M = 16$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

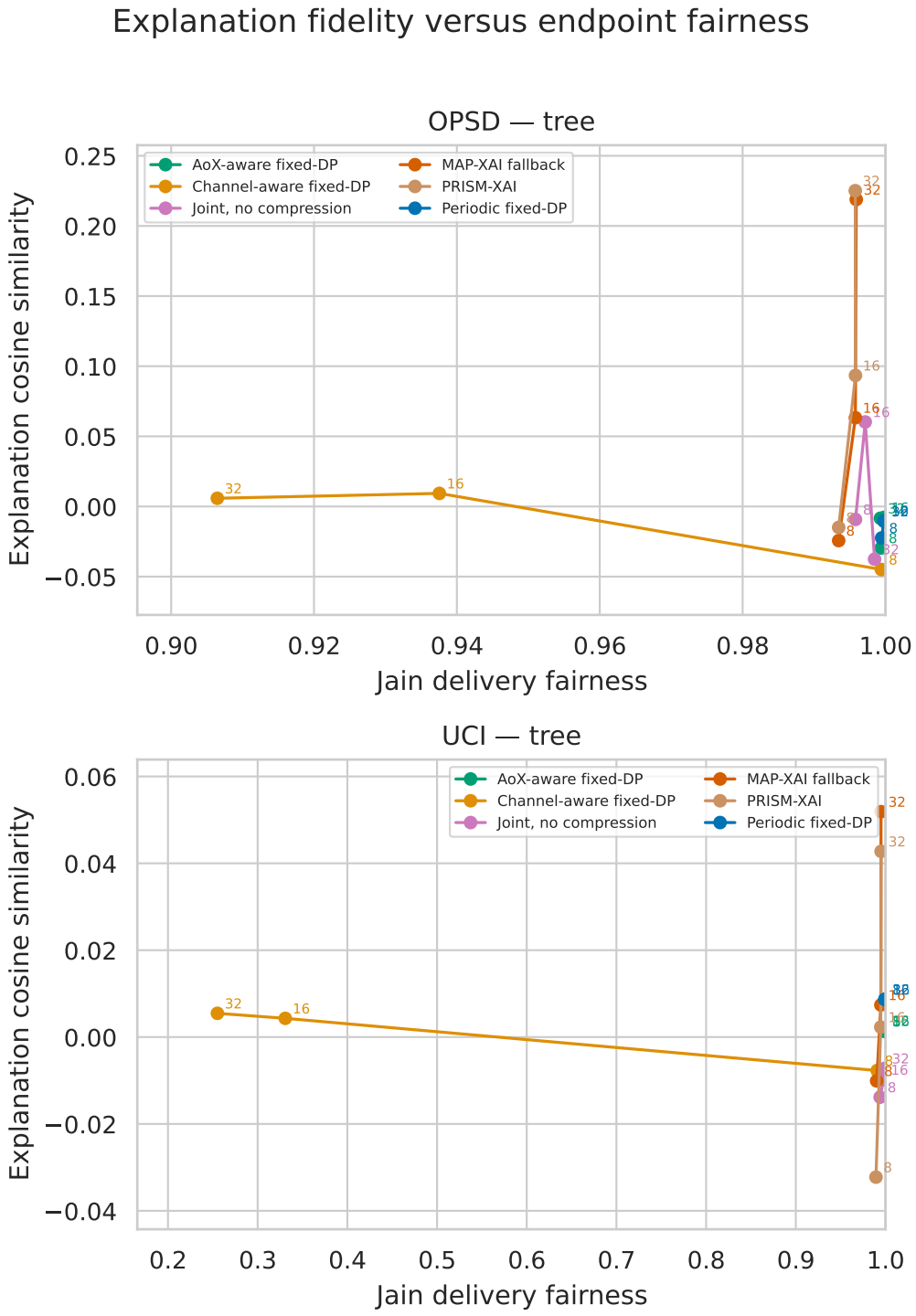


Fig. 66. Public-score threshold $\tau = 1$, release cap $M = 16$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

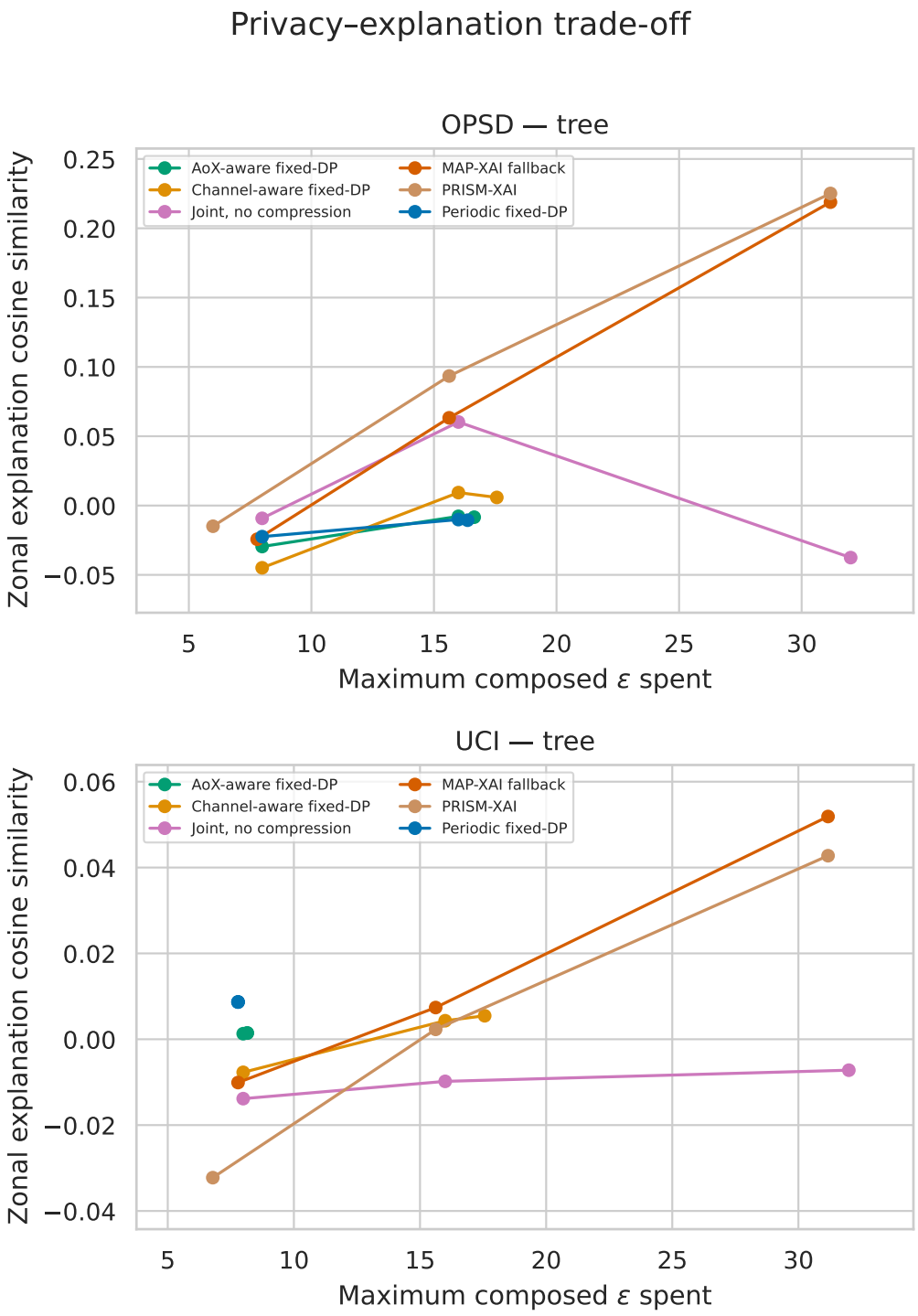


Fig. 67. Public-score threshold $\tau = 1$, release cap $M = 16$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

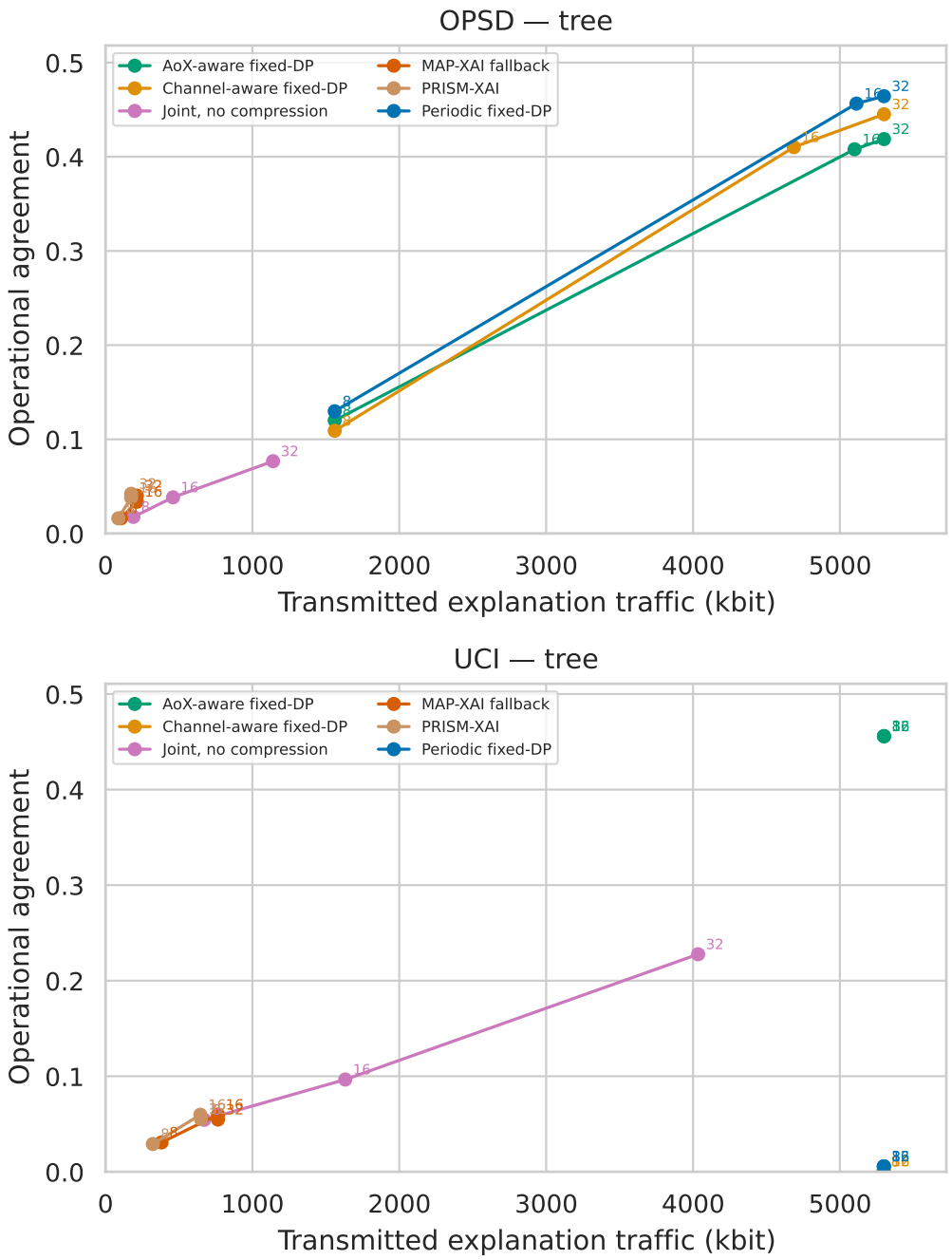


Fig. 68. Public-score threshold $\tau = 1$, release cap $M = 16$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

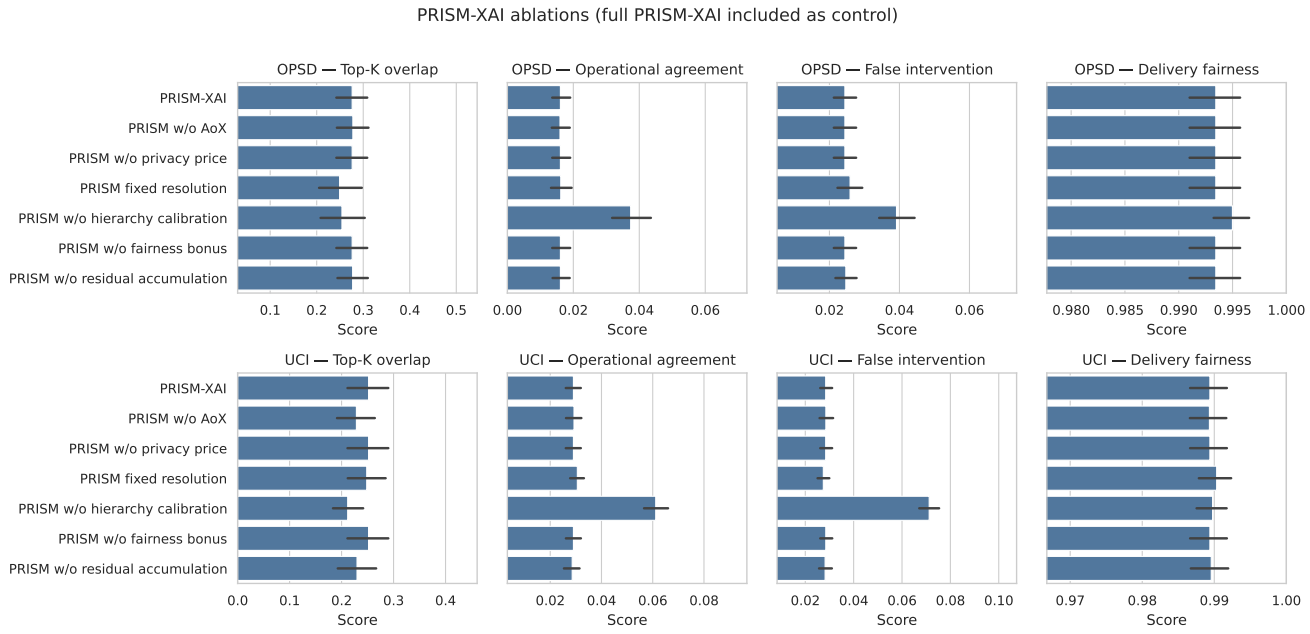


Fig. 69. Public-score threshold $\tau = 1$, release cap $M = 16$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LV
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.270 ± 0.066	-0.025 ± 0.093	0.015 ± 0.004	0.015 ± 0.004	0.200 ± 0.078	0.956 ± 0.003	4.630 ± 1.839	87.697 ± 0.410	6.034 ± 0.230	0.025 ± 0.005	15451.736 ± 3728.991	0.993 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.283 ± 0.054	-0.005 ± 0.086	0.017 ± 0.005	0.017 ± 0.005	0.222 ± 0.080	0.956 ± 0.003	4.630 ± 1.839	87.431 ± 0.364	5.936 ± 0.213	0.024 ± 0.005	14279.307 ± 3768.065	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.234 ± 0.073	-0.037 ± 0.075	0.016 ± 0.005	0.016 ± 0.005	0.191 ± 0.080	0.956 ± 0.003	4.732 ± 1.776	105.817 ± 0.049	7.786 ± 0.000	0.027 ± 0.006	21385.963 ± 9729.377	0.993 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.230 ± 0.067	-0.012 ± 0.085	0.017 ± 0.005	0.017 ± 0.005	0.219 ± 0.073	0.956 ± 0.003	4.732 ± 1.776	105.675 ± 0.049	7.786 ± 0.000	0.024 ± 0.006	19083.433 ± 9057.454	0.993 ± 0.004

TABLE LVI
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	87.697 ± 0.410	0.270 ± 0.066	0.250 ± 0.000	-0.025 ± 0.093	0.015 ± 0.004	0.015 ± 0.004	0.200 ± 0.078	0.044 ± 0.003	0.956 ± 0.003	0.025 ± 0.005	6.034 ± 0.230
OPSD	Tree	8	PRISM-XAI	87.431 ± 0.364	0.283 ± 0.054	0.250 ± 0.000	-0.005 ± 0.086	0.017 ± 0.005	0.017 ± 0.005	0.222 ± 0.080	0.044 ± 0.003	0.956 ± 0.003	0.024 ± 0.005	5.936 ± 0.213
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	175.005 ± 0.564	0.242 ± 0.064	0.250 ± 0.000	0.085 ± 0.083	0.039 ± 0.008	0.039 ± 0.008	0.287 ± 0.077	0.081 ± 0.003	0.919 ± 0.003	0.038 ± 0.008	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	174.493 ± 0.509	0.213 ± 0.052	0.250 ± 0.000	0.102 ± 0.085	0.037 ± 0.009	0.037 ± 0.009	0.264 ± 0.059	0.081 ± 0.003	0.919 ± 0.003	0.039 ± 0.007	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	174.944 ± 0.552	0.309 ± 0.062	0.250 ± 0.000	0.203 ± 0.080	0.043 ± 0.009	0.043 ± 0.009	0.336 ± 0.099	0.081 ± 0.003	0.919 ± 0.003	0.033 ± 0.008	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	174.309 ± 0.461	0.326 ± 0.085	0.250 ± 0.000	0.247 ± 0.077	0.042 ± 0.009	0.042 ± 0.009	0.294 ± 0.073	0.081 ± 0.003	0.919 ± 0.003	0.039 ± 0.007	31.175 ± 0.000

TABLE LVII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.867 ± 0.410	0.036 ± 0.086	-0.104 ± 0.029	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-4924.908 ± 5.576	-0.011 ± 0.052	-0.362 ± 0.075	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5125.280 ± 0.552	0.056 ± 0.077	-0.370 ± 0.075	-0.003 ± 0.002
OPSD	AoX-aware fixed-DP	-1473.133 ± 0.364	0.045 ± 0.066	-0.104 ± 0.027	-0.006 ± 0.004
OPSD	AoX-aware fixed-DP	-4925.420 ± 5.602	-0.039 ± 0.045	-0.378 ± 0.056	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5125.915 ± 0.461	0.074 ± 0.085	-0.384 ± 0.056	-0.004 ± 0.002
OPSD	Channel-aware fixed-DP	-1472.867 ± 0.410	-0.023 ± 0.091	-0.092 ± 0.025	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4511.604 ± 12.966	-0.019 ± 0.055	-0.362 ± 0.090	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5125.280 ± 0.552	0.051 ± 0.072	-0.392 ± 0.096	0.089 ± 0.006
OPSD	Channel-aware fixed-DP	-1473.133 ± 0.364	-0.010 ± 0.076	-0.094 ± 0.024	-0.006 ± 0.004
OPSD	Channel-aware fixed-DP	-4512.116 ± 12.805	-0.045 ± 0.039	-0.382 ± 0.087	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5125.915 ± 0.461	0.069 ± 0.081	-0.414 ± 0.089	0.089 ± 0.006
OPSD	Joint, no compression	-102.203 ± 0.525	-0.018 ± 0.097	-0.001 ± 0.004	-0.002 ± 0.003
OPSD	Joint, no compression	-284.895 ± 1.260	0.011 ± 0.071	0.001 ± 0.011	-0.001 ± 0.002
OPSD	Joint, no compression	-965.956 ± 8.155	0.073 ± 0.062	-0.031 ± 0.015	-0.003 ± 0.002
OPSD	Joint, no compression	-102.319 ± 0.436	0.024 ± 0.089	-0.002 ± 0.003	-0.002 ± 0.003
OPSD	Joint, no compression	-285.557 ± 1.134	-0.027 ± 0.062	-0.002 ± 0.012	-0.001 ± 0.002
OPSD	Joint, no compression	-966.591 ± 8.036	0.044 ± 0.085	-0.038 ± 0.011	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-1472.867 ± 0.410	0.029 ± 0.092	-0.110 ± 0.020	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-4937.589 ± 0.564	-0.007 ± 0.060	-0.411 ± 0.063	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5124.892 ± 0.754	0.060 ± 0.073	-0.415 ± 0.064	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-1473.133 ± 0.364	0.041 ± 0.073	-0.118 ± 0.016	-0.006 ± 0.004
OPSD	Periodic fixed-DP	-4938.101 ± 0.509	-0.029 ± 0.052	-0.426 ± 0.048	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5125.527 ± 0.718	0.082 ± 0.090	-0.429 ± 0.048	-0.004 ± 0.002
OPSD	MAP-XAI fallback	-18.120 ± 0.422	0.035 ± 0.088	-0.000 ± 0.003	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.812 ± 0.567	-0.011 ± 0.064	0.004 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.873 ± 0.554	-0.004 ± 0.061	0.003 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.245 ± 0.372	0.053 ± 0.042	-0.000 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-39.182 ± 0.498	-0.034 ± 0.034	0.004 ± 0.009	0.000 ± 0.000
OPSD	MAP-XAI fallback	-39.346 ± 0.444	0.055 ± 0.050	0.001 ± 0.005	-0.000 ± 0.001

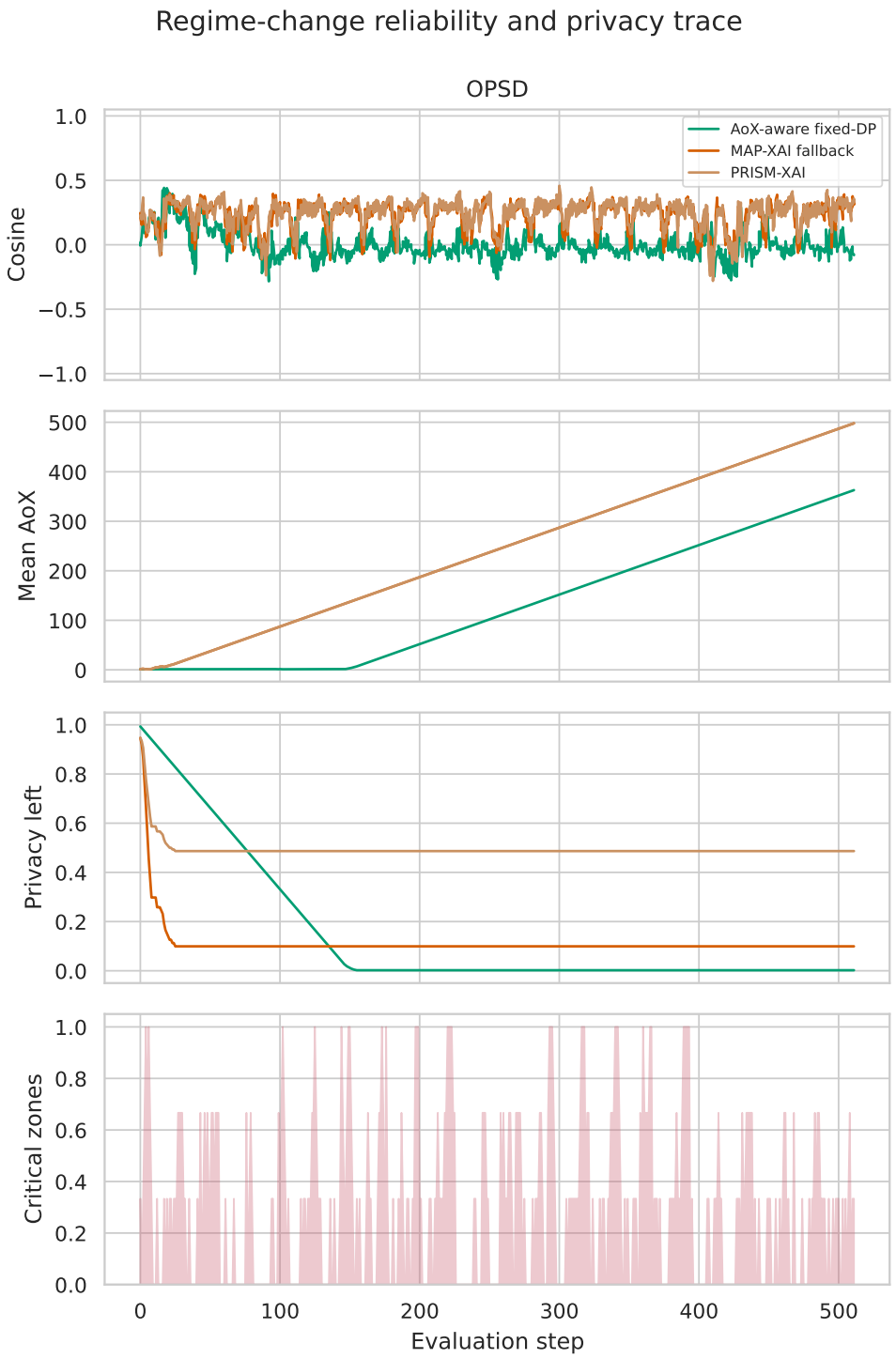


Fig. 70. Public-score threshold $\tau = 1$, release cap $M = 16$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

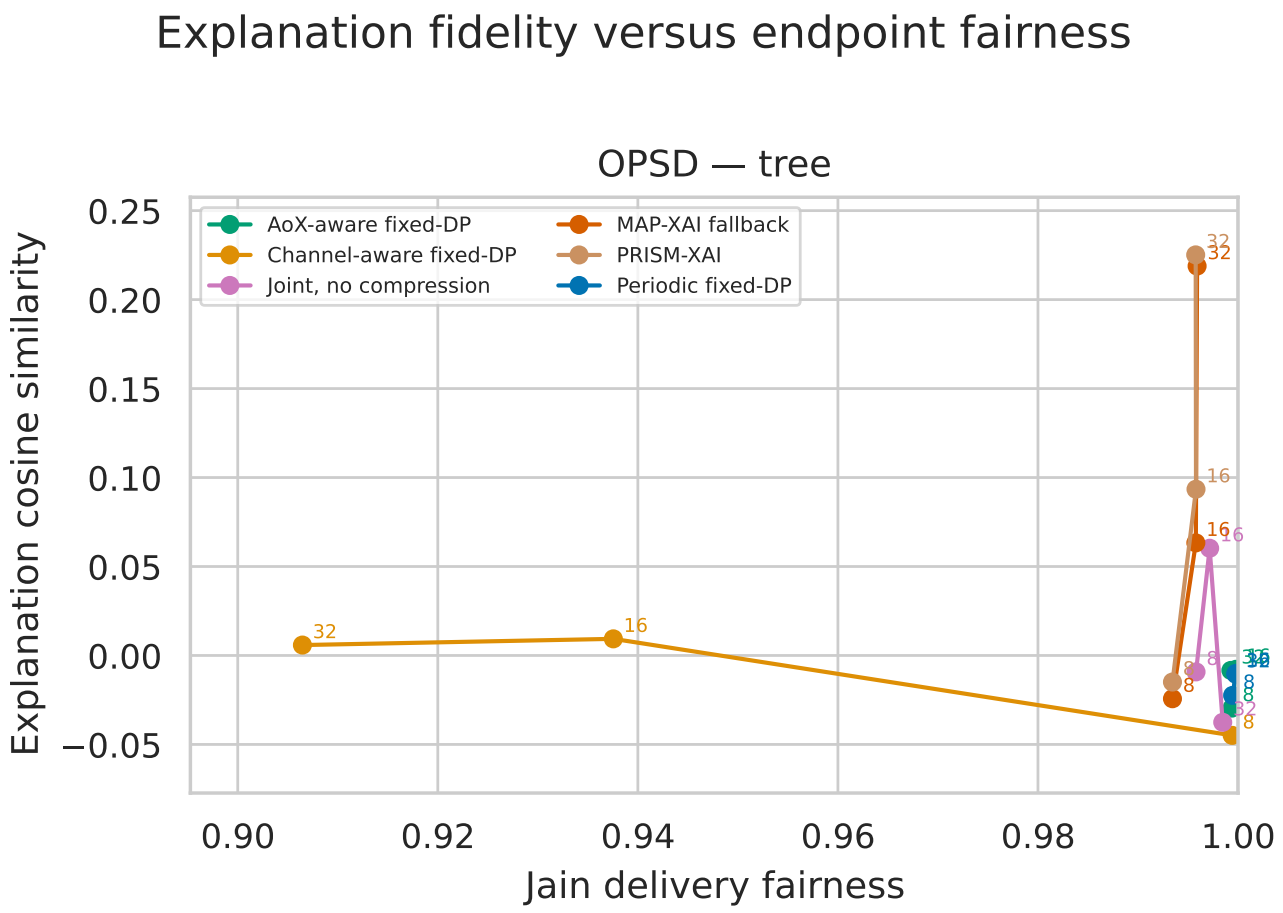


Fig. 71. Public-score threshold $\tau = 1$, release cap $M = 16$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

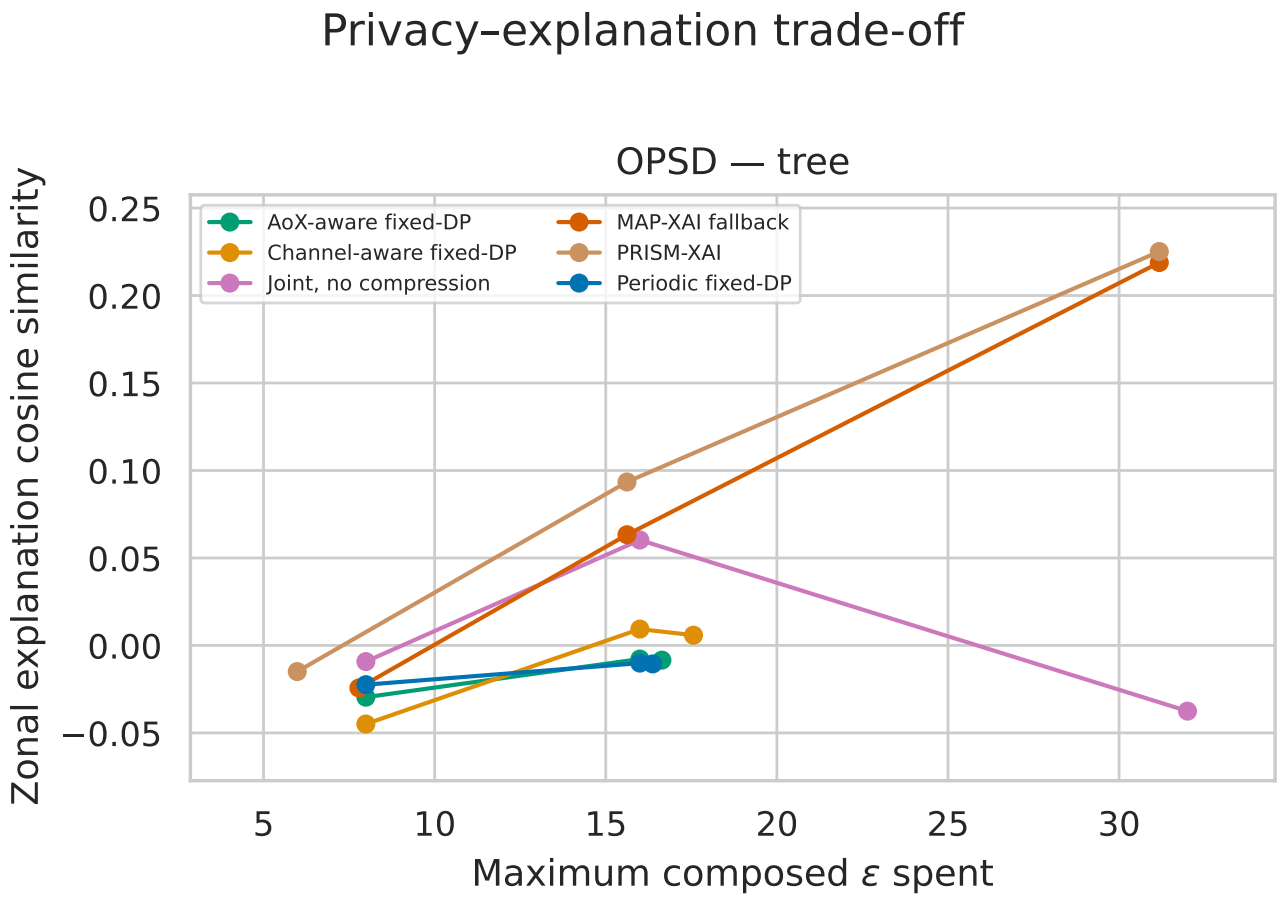


Fig. 72. Public-score threshold $\tau = 1$, release cap $M = 16$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

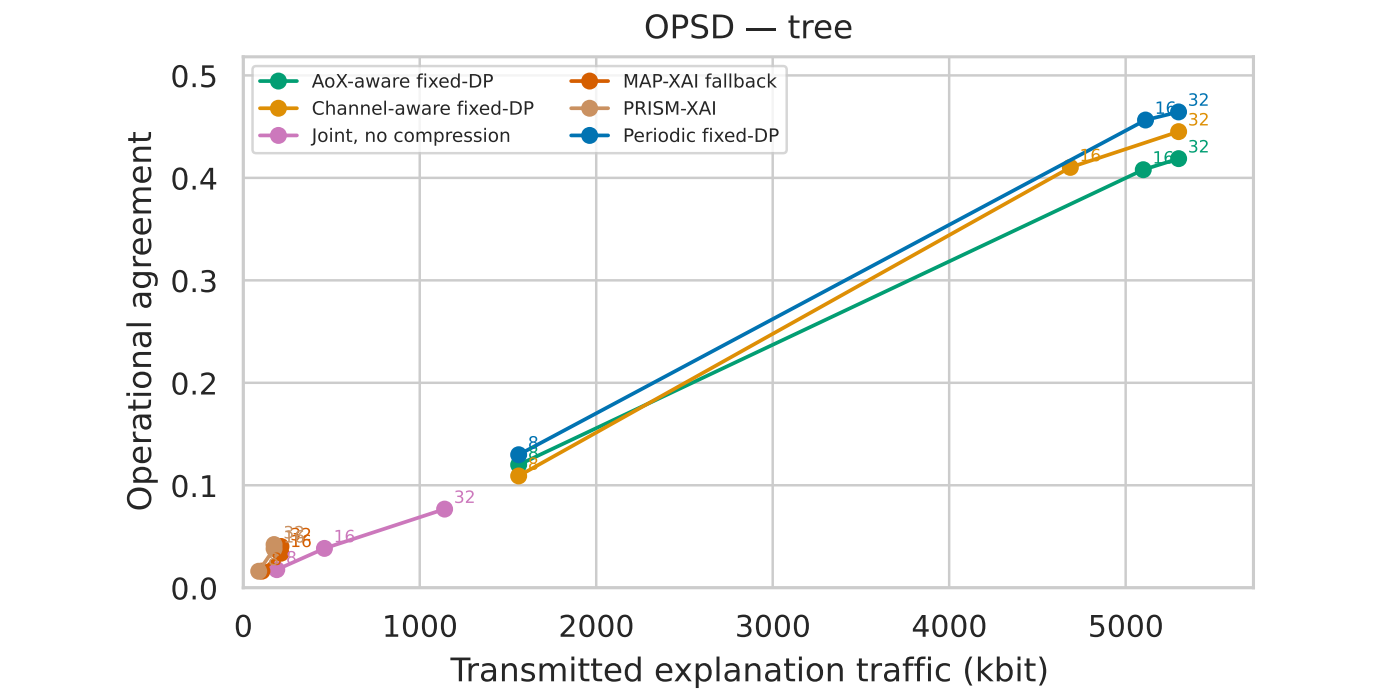


Fig. 73. Public-score threshold $\tau = 1$, release cap $M = 16$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

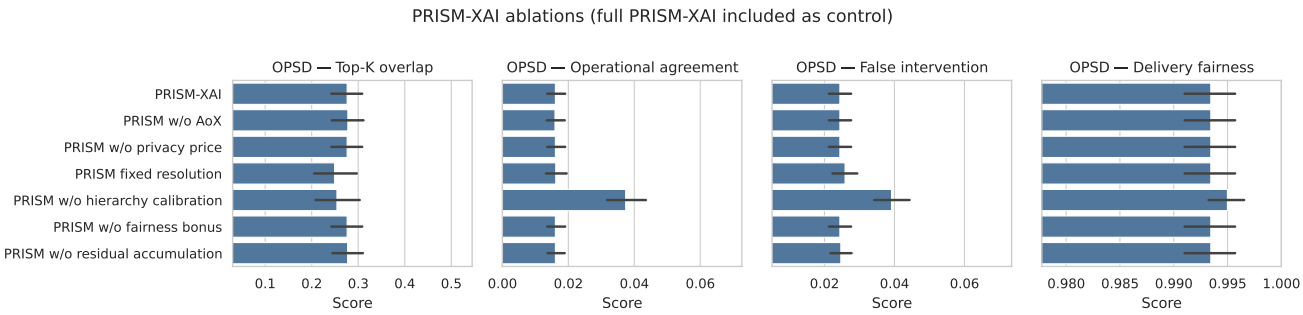


Fig. 74. Public-score threshold $\tau = 1$, release cap $M = 16$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LVIII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.240 ± 0.089	-0.008 ± 0.019	0.031 ± 0.004	0.031 ± 0.004	0.422 ± 0.039	0.931 ± 0.004	6.267 ± 1.448	323.739 ± 1.239	6.844 ± 0.203	0.030 ± 0.005	13090.115 ± 2761.149	0.990 ± 0.003
UCI	Tree	8	PRISM-XAI	0.264 ± 0.044	-0.056 ± 0.057	0.027 ± 0.005	0.027 ± 0.005	0.404 ± 0.063	0.938 ± 0.003	4.814 ± 1.710	321.228 ± 1.066	6.726 ± 0.146	0.027 ± 0.004	15136.774 ± 2629.908	0.989 ± 0.006
UCI	Tree	8	MAP-XAI fallback	0.240 ± 0.077	-0.011 ± 0.019	0.032 ± 0.005	0.032 ± 0.005	0.446 ± 0.058	0.931 ± 0.004	5.672 ± 1.709	382.099 ± 0.159	7.786 ± 0.000	0.027 ± 0.003	15488.351 ± 4957.026	0.991 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.265 ± 0.046	-0.009 ± 0.057	0.029 ± 0.005	0.029 ± 0.005	0.424 ± 0.053	0.937 ± 0.004	5.276 ± 1.728	381.645 ± 0.090	7.786 ± 0.000	0.027 ± 0.005	16573.020 ± 2779.080	0.990 ± 0.004

TABLE LIX
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	323.739 ± 1.239	0.240 ± 0.089	0.250 ± 0.000	-0.008 ± 0.019	0.031 ± 0.004	0.031 ± 0.004	0.422 ± 0.039	0.069 ± 0.004	0.931 ± 0.004	0.030 ± 0.005	6.844 ± 0.203
UCI	Tree	8	PRISM-XAI	321.228 ± 1.066	0.264 ± 0.044	0.250 ± 0.000	-0.056 ± 0.057	0.027 ± 0.005	0.027 ± 0.005	0.404 ± 0.063	0.062 ± 0.003	0.938 ± 0.003	0.027 ± 0.004	6.726 ± 0.146
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	647.230 ± 2.247	0.247 ± 0.082	0.250 ± 0.000	0.013 ± 0.024	0.064 ± 0.006	0.064 ± 0.006	0.454 ± 0.035	0.138 ± 0.010	0.862 ± 0.010	0.054 ± 0.009	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	644.319 ± 3.516	0.227 ± 0.040	0.250 ± 0.000	-0.008 ± 0.057	0.055 ± 0.012	0.055 ± 0.012	0.428 ± 0.061	0.123 ± 0.012	0.877 ± 0.012	0.051 ± 0.012	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	649.317 ± 2.732	0.249 ± 0.061	0.250 ± 0.000	0.022 ± 0.019	0.063 ± 0.009	0.063 ± 0.009	0.602 ± 0.060	0.128 ± 0.016	0.872 ± 0.016	0.030 ± 0.009	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	650.306 ± 2.517	0.221 ± 0.026	0.250 ± 0.000	0.064 ± 0.065	0.048 ± 0.007	0.048 ± 0.007	0.624 ± 0.077	0.095 ± 0.011	0.905 ± 0.011	0.021 ± 0.006	31.175 ± 0.000

TABLE LX
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 16$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4976.485 ± 1.239	-0.023 ± 0.103	-0.426 ± 0.034	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4652.994 ± 2.247	-0.017 ± 0.102	-0.393 ± 0.032	-0.005 ± 0.002
UCI	AoX-aware fixed-DP	-4650.907 ± 2.732	-0.015 ± 0.080	-0.394 ± 0.036	-0.004 ± 0.001
UCI	AoX-aware fixed-DP	-4978.996 ± 1.066	0.003 ± 0.059	-0.428 ± 0.050	-0.011 ± 0.006
UCI	AoX-aware fixed-DP	-4655.905 ± 3.516	-0.034 ± 0.057	-0.399 ± 0.048	-0.004 ± 0.002
UCI	AoX-aware fixed-DP	-4649.918 ± 2.517	-0.039 ± 0.038	-0.407 ± 0.047	-0.004 ± 0.001
UCI	Channel-aware fixed-DP	-4971.438 ± 4.206	-0.008 ± 0.107	0.031 ± 0.005	-0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4652.994 ± 2.247	-0.020 ± 0.100	0.064 ± 0.006	0.664 ± 0.002
UCI	Channel-aware fixed-DP	-4650.907 ± 2.732	-0.020 ± 0.069	0.062 ± 0.009	0.740 ± 0.001
UCI	Channel-aware fixed-DP	-4973.950 ± 4.295	0.024 ± 0.064	0.027 ± 0.005	-0.002 ± 0.006
UCI	Channel-aware fixed-DP	-4655.905 ± 3.516	-0.034 ± 0.053	0.055 ± 0.012	0.664 ± 0.002
UCI	Channel-aware fixed-DP	-4649.918 ± 2.517	-0.038 ± 0.034	0.048 ± 0.007	0.740 ± 0.002
UCI	Joint, no compression	-348.561 ± 1.547	-0.045 ± 0.099	-0.021 ± 0.006	-0.004 ± 0.004
UCI	Joint, no compression	-984.920 ± 2.348	0.031 ± 0.081	-0.030 ± 0.007	-0.003 ± 0.002
UCI	Joint, no compression	-3385.233 ± 6.863	0.002 ± 0.071	-0.165 ± 0.019	-0.004 ± 0.001
UCI	Joint, no compression	-351.072 ± 1.196	0.034 ± 0.062	-0.030 ± 0.008	-0.006 ± 0.006
UCI	Joint, no compression	-988.281 ± 3.390	0.023 ± 0.047	-0.044 ± 0.018	-0.003 ± 0.002
UCI	Joint, no compression	-3384.244 ± 6.390	-0.020 ± 0.038	-0.180 ± 0.031	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4975.579 ± 1.762	-0.019 ± 0.103	0.025 ± 0.005	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4652.089 ± 3.147	-0.012 ± 0.080	0.058 ± 0.005	-0.005 ± 0.002
UCI	Periodic fixed-DP	-4650.001 ± 3.755	-0.011 ± 0.060	0.057 ± 0.009	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4978.090 ± 1.619	0.003 ± 0.071	0.022 ± 0.005	-0.011 ± 0.006
UCI	Periodic fixed-DP	-4654.999 ± 3.804	-0.034 ± 0.048	0.050 ± 0.012	-0.005 ± 0.002
UCI	Periodic fixed-DP	-4649.013 ± 3.233	-0.039 ± 0.037	0.042 ± 0.008	-0.004 ± 0.001
UCI	MAP-XAI fallback	-58.359 ± 1.137	-0.000 ± 0.105	-0.001 ± 0.002	-0.001 ± 0.001
UCI	MAP-XAI fallback	-118.869 ± 2.191	0.006 ± 0.053	0.004 ± 0.008	-0.000 ± 0.001
UCI	MAP-XAI fallback	-116.843 ± 2.730	0.009 ± 0.084	0.003 ± 0.004	0.000 ± 0.001
UCI	MAP-XAI fallback	-60.417 ± 1.098	-0.002 ± 0.046	-0.002 ± 0.006	-0.001 ± 0.002
UCI	MAP-XAI fallback	-121.326 ± 3.553	0.006 ± 0.056	-0.004 ± 0.010	-0.000 ± 0.001
UCI	MAP-XAI fallback	-115.339 ± 2.574	-0.015 ± 0.037	-0.002 ± 0.008	0.000 ± 0.001

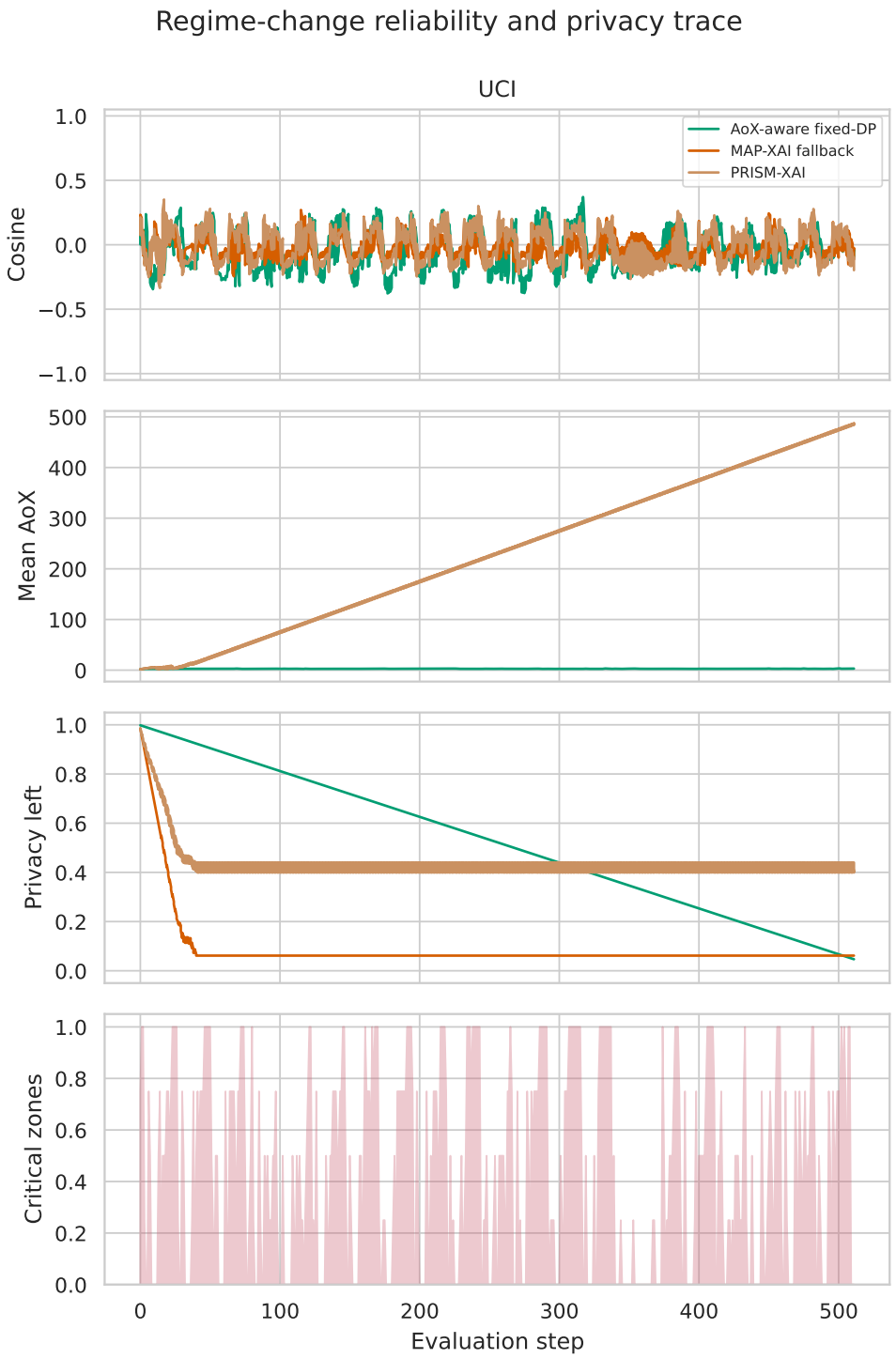


Fig. 75. Public-score threshold $\tau = 1$, release cap $M = 16$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

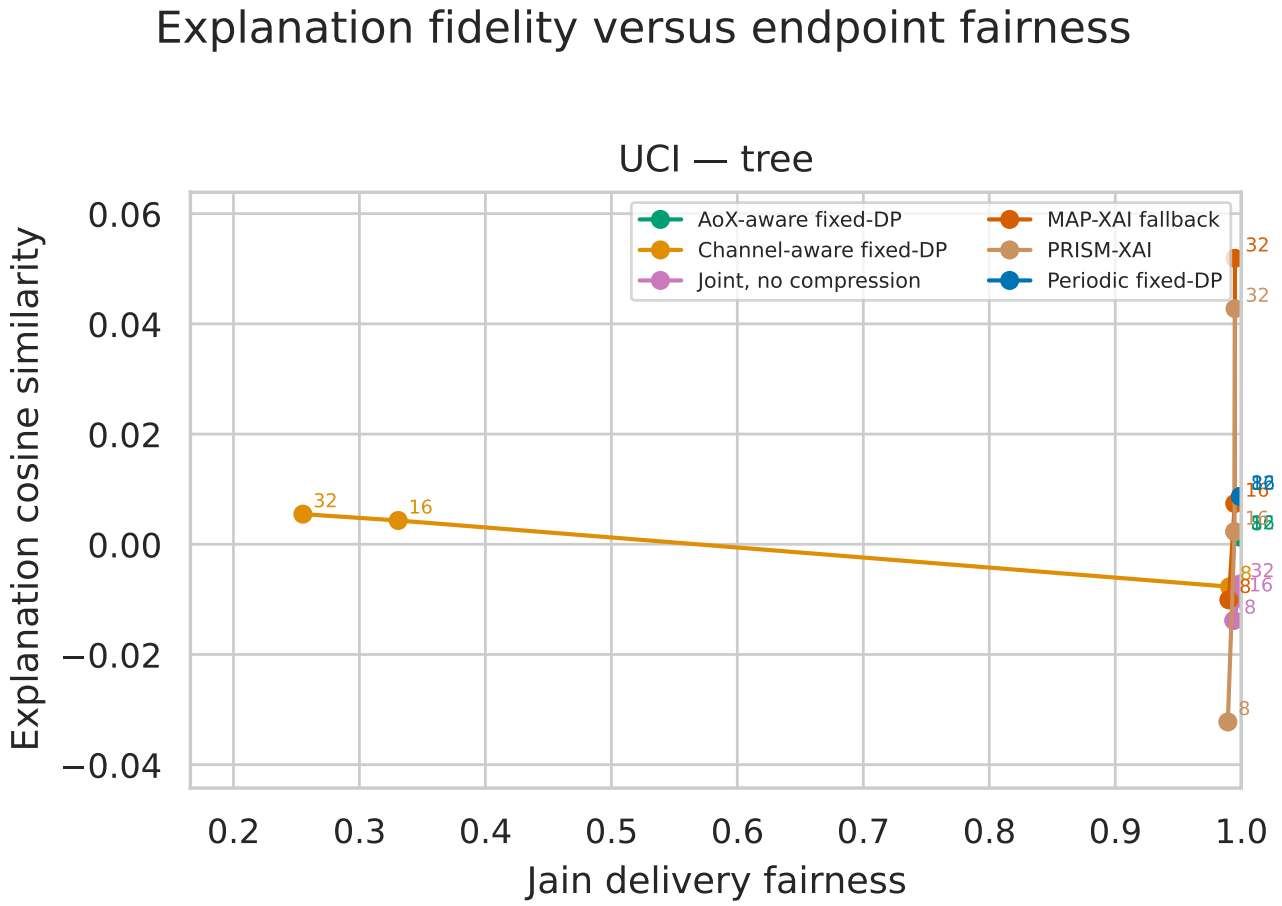


Fig. 76. Public-score threshold $\tau = 1$, release cap $M = 16$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

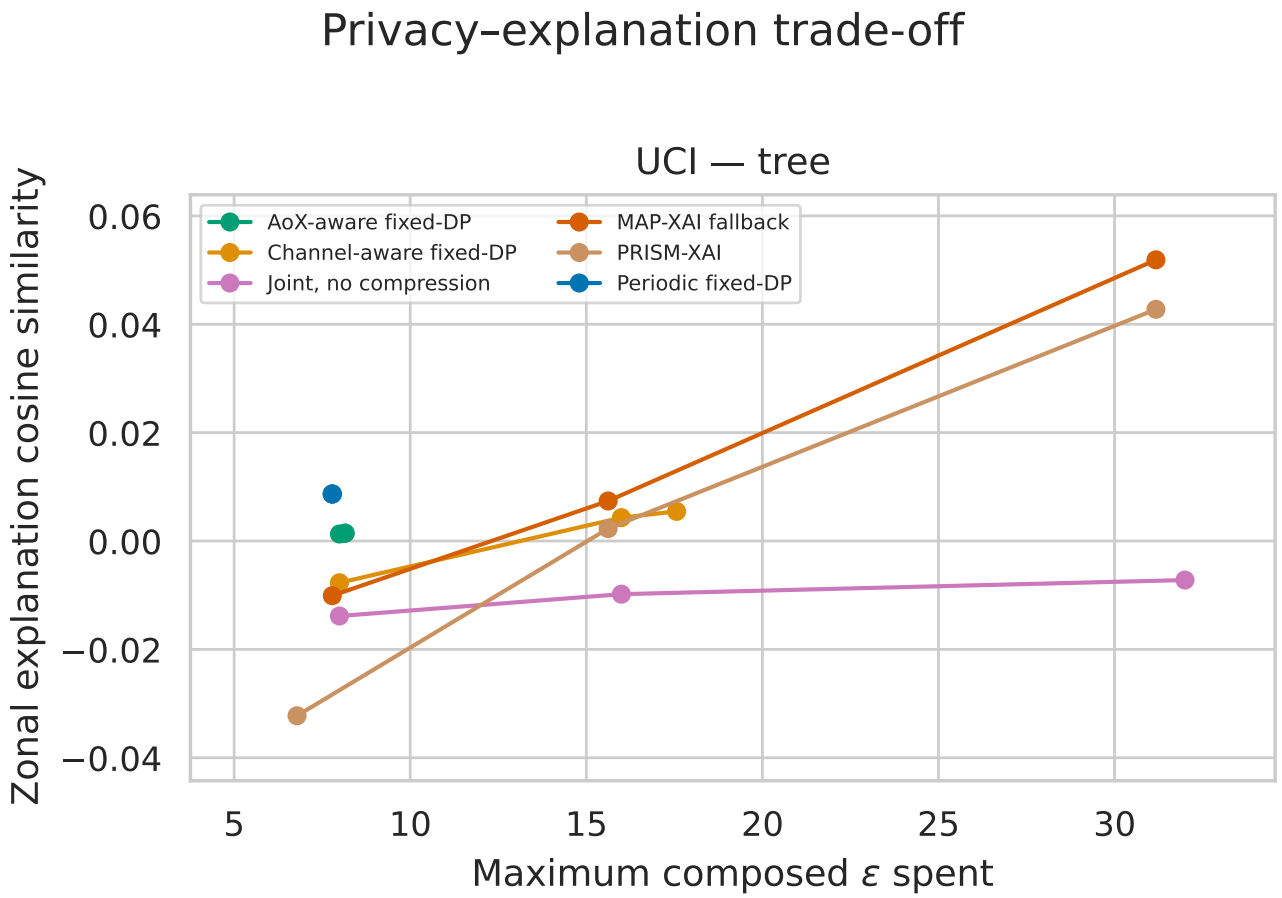


Fig. 77. Public-score threshold $\tau = 1$, release cap $M = 16$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

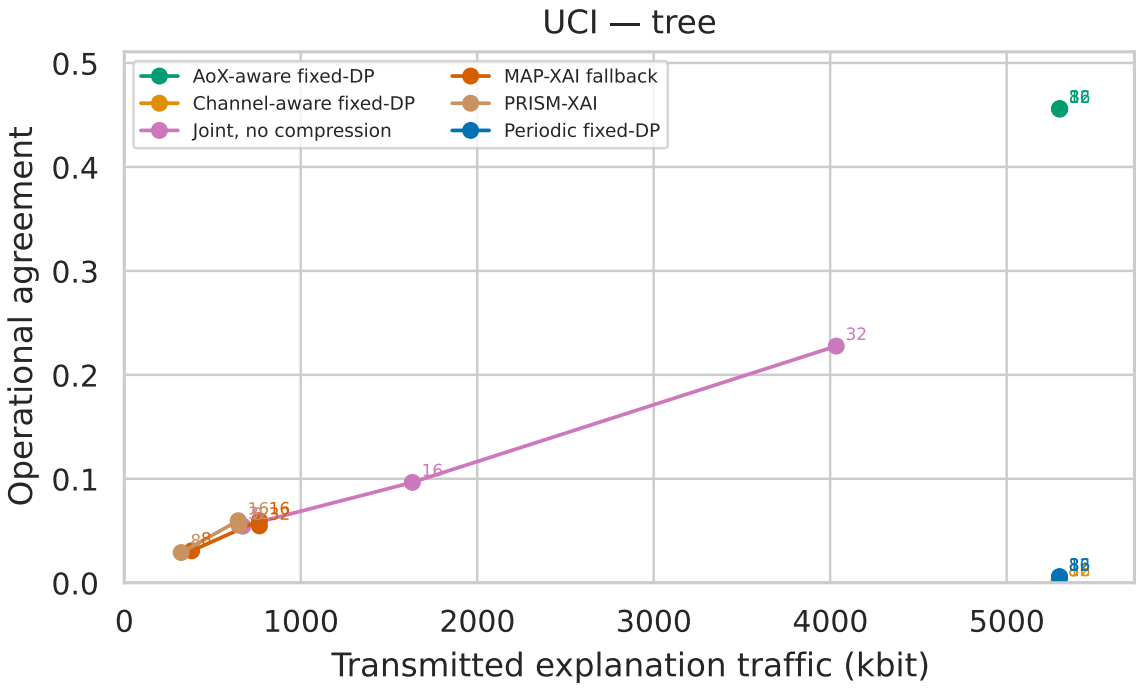


Fig. 78. Public-score threshold $\tau = 1$, release cap $M = 16$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

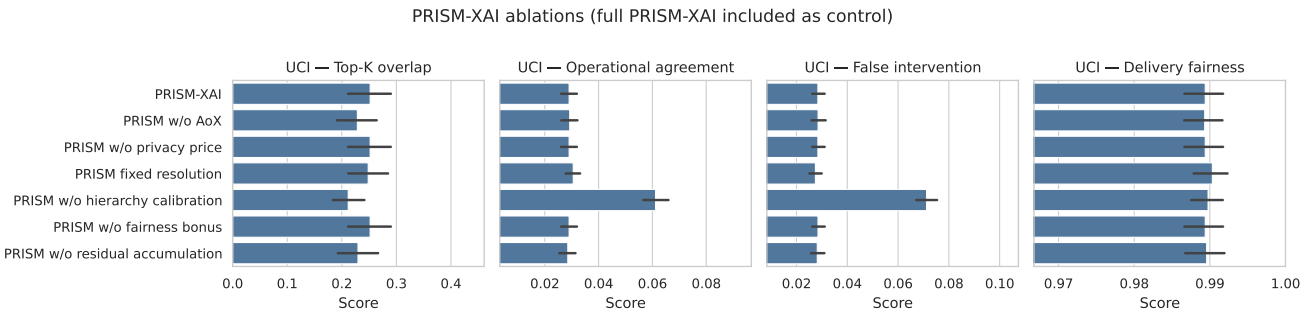


Fig. 79. Public-score threshold $\tau = 1$, release cap $M = 16$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

6) Public-score threshold $\tau = 2$, release cap $M = 16$: At this operating point, PRISM-XAI transmits 343.07 kbit versus 407.50 kbit for MAP-XAI-only (aggregate-mean ratio saving 15.81%). The paired difference is -64.43±7.17 kbit. PRISM coverage is 0.1248, critical-event recall is 0.0594, and maximum recorded expenditure is 17.745. Relative to MAP-XAI-only, the paired coverage change is -0.00024±0.00023 and recall change is +0.00076±0.00208. The descriptive overlap rule holds in 8 of 12 cells. Accordingly, this point is read as a sensitivity point on the service frontier, not a complete matched-service confirmation.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXI
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.170 ± 0.039	0.027 ± 0.072	0.029 ± 0.012	0.029 ± 0.012	0.326 ± 0.147	0.938 ± 0.004	4.881 ± 1.730	88.137 ± 0.577	6.066 ± 0.239	0.024 ± 0.007	10754.619 ± 5615.749	0.991 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.171 ± 0.046	0.026 ± 0.082	0.031 ± 0.010	0.031 ± 0.010	0.343 ± 0.130	0.938 ± 0.004	4.873 ± 1.737	87.707 ± 0.532	5.936 ± 0.213	0.024 ± 0.007	8388.650 ± 2960.712	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.203 ± 0.058	0.019 ± 0.075	0.029 ± 0.011	0.029 ± 0.011	0.324 ± 0.148	0.938 ± 0.004	4.960 ± 1.678	105.889 ± 0.169	7.786 ± 0.000	0.026 ± 0.008	11344.761 ± 4319.900	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.181 ± 0.054	0.025 ± 0.081	0.029 ± 0.009	0.029 ± 0.009	0.310 ± 0.118	0.938 ± 0.004	4.960 ± 1.678	105.737 ± 0.152	7.786 ± 0.000	0.026 ± 0.007	10402.432 ± 2994.804	0.991 ± 0.006
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.220 ± 0.061	-0.009 ± 0.015	0.045 ± 0.006	0.045 ± 0.006	0.442 ± 0.072	0.906 ± 0.004	4.504 ± 1.537	325.842 ± 1.091	6.877 ± 0.090	0.040 ± 0.007	8917.519 ± 1309.048	0.992 ± 0.002
UCI	Tree	8	PRISM-XAI	0.217 ± 0.043	-0.008 ± 0.077	0.045 ± 0.007	0.045 ± 0.007	0.447 ± 0.078	0.908 ± 0.002	5.058 ± 1.846	322.524 ± 0.793	6.877 ± 0.090	0.039 ± 0.008	9128.968 ± 1604.303	0.991 ± 0.004
UCI	Tree	8	MAP-XAI fallback	0.155 ± 0.082	0.000 ± 0.016	0.047 ± 0.007	0.047 ± 0.007	0.456 ± 0.051	0.905 ± 0.004	4.584 ± 1.508	381.745 ± 0.387	7.787 ± 0.001	0.038 ± 0.003	10122.948 ± 1695.832	0.992 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.221 ± 0.054	0.040 ± 0.080	0.046 ± 0.008	0.046 ± 0.008	0.469 ± 0.072	0.907 ± 0.003	4.763 ± 1.714	381.283 ± 0.251	7.787 ± 0.001	0.036 ± 0.006	10525.744 ± 1825.173	0.991 ± 0.004

TABLE LXII

PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	88.137 ± 0.577	0.170 ± 0.039	0.250 ± 0.000	0.027 ± 0.072	0.029 ± 0.012	0.029 ± 0.012	0.326 ± 0.147	0.062 ± 0.004	0.938 ± 0.004	0.024 ± 0.007	6.066 ± 0.239
OPSD	Tree	8	PRISM-XAI	87.707 ± 0.532	0.171 ± 0.046	0.250 ± 0.000	0.026 ± 0.082	0.031 ± 0.010	0.031 ± 0.010	0.343 ± 0.130	0.062 ± 0.004	0.938 ± 0.004	0.024 ± 0.007	5.936 ± 0.213
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	176.428 ± 0.771	0.259 ± 0.050	0.250 ± 0.000	0.054 ± 0.064	0.053 ± 0.015	0.053 ± 0.015	0.339 ± 0.133	0.120 ± 0.006	0.880 ± 0.006	0.042 ± 0.011	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	175.629 ± 0.704	0.256 ± 0.054	0.250 ± 0.000	0.071 ± 0.073	0.050 ± 0.013	0.050 ± 0.013	0.291 ± 0.104	0.120 ± 0.006	0.880 ± 0.006	0.049 ± 0.010	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	176.336 ± 0.759	0.315 ± 0.088	0.250 ± 0.000	0.193 ± 0.062	0.061 ± 0.013	0.061 ± 0.013	0.397 ± 0.114	0.120 ± 0.006	0.880 ± 0.006	0.037 ± 0.010	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	175.415 ± 0.593	0.314 ± 0.062	0.250 ± 0.000	0.240 ± 0.062	0.060 ± 0.013	0.060 ± 0.013	0.365 ± 0.089	0.120 ± 0.006	0.880 ± 0.006	0.040 ± 0.008	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	325.842 ± 1.091	0.220 ± 0.061	0.250 ± 0.000	-0.009 ± 0.015	0.045 ± 0.006	0.045 ± 0.006	0.442 ± 0.072	0.094 ± 0.004	0.906 ± 0.004	0.040 ± 0.007	6.877 ± 0.090
UCI	Tree	8	PRISM-XAI	322.524 ± 0.793	0.217 ± 0.043	0.250 ± 0.000	-0.008 ± 0.077	0.045 ± 0.007	0.045 ± 0.007	0.447 ± 0.078	0.092 ± 0.002	0.908 ± 0.002	0.039 ± 0.008	6.877 ± 0.090
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	650.843 ± 1.226	0.297 ± 0.062	0.250 ± 0.000	0.008 ± 0.020	0.087 ± 0.008	0.087 ± 0.008	0.454 ± 0.046	0.180 ± 0.003	0.820 ± 0.003	0.073 ± 0.007	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	644.135 ± 1.590	0.251 ± 0.053	0.250 ± 0.000	0.011 ± 0.046	0.084 ± 0.009	0.084 ± 0.009	0.463 ± 0.061	0.177 ± 0.004	0.823 ± 0.004	0.069 ± 0.011	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	649.871 ± 1.443	0.332 ± 0.056	0.250 ± 0.000	0.010 ± 0.024	0.085 ± 0.007	0.085 ± 0.007	0.572 ± 0.040	0.177 ± 0.004	0.823 ± 0.004	0.044 ± 0.006	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	644.002 ± 1.573	0.310 ± 0.057	0.250 ± 0.000	0.016 ± 0.064	0.083 ± 0.012	0.083 ± 0.012	0.561 ± 0.063	0.173 ± 0.006	0.827 ± 0.006	0.045 ± 0.010	31.175 ± 0.000

TABLE LXIII
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.427 ± 0.577	-0.064 ± 0.057	-0.091 ± 0.033	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-4923.485 ± 5.595	0.006 ± 0.053	-0.349 ± 0.082	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5123.888 ± 0.759	0.062 ± 0.093	-0.351 ± 0.083	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-1472.857 ± 0.532	-0.068 ± 0.063	-0.090 ± 0.028	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-4924.283 ± 5.582	0.004 ± 0.062	-0.364 ± 0.062	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5124.809 ± 0.593	0.062 ± 0.071	-0.366 ± 0.060	-0.004 ± 0.002
OPSD	Channel-aware fixed-DP	-1472.427 ± 0.577	-0.122 ± 0.049	-0.078 ± 0.029	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4510.181 ± 12.999	-0.002 ± 0.048	-0.349 ± 0.095	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5123.888 ± 0.759	0.057 ± 0.093	-0.373 ± 0.102	0.089 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.857 ± 0.532	-0.122 ± 0.074	-0.080 ± 0.025	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4510.980 ± 13.099	-0.002 ± 0.051	-0.369 ± 0.093	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5124.809 ± 0.593	0.057 ± 0.070	-0.396 ± 0.090	0.089 ± 0.006
OPSD	Joint, no compression	-101.763 ± 0.687	-0.118 ± 0.078	0.012 ± 0.011	-0.005 ± 0.005
OPSD	Joint, no compression	-283.472 ± 1.424	0.028 ± 0.096	0.015 ± 0.017	-0.002 ± 0.002
OPSD	Joint, no compression	-964.564 ± 8.170	0.080 ± 0.106	-0.013 ± 0.022	-0.003 ± 0.002
OPSD	Joint, no compression	-102.043 ± 0.513	-0.088 ± 0.080	0.012 ± 0.009	-0.005 ± 0.005
OPSD	Joint, no compression	-284.421 ± 1.294	0.015 ± 0.063	0.011 ± 0.014	-0.002 ± 0.002
OPSD	Joint, no compression	-965.485 ± 8.153	0.032 ± 0.101	-0.020 ± 0.016	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-1472.427 ± 0.577	-0.070 ± 0.050	-0.096 ± 0.024	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-4936.166 ± 0.771	0.010 ± 0.060	-0.398 ± 0.067	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5123.500 ± 0.672	0.066 ± 0.097	-0.397 ± 0.070	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-1472.857 ± 0.532	-0.071 ± 0.050	-0.104 ± 0.018	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-4936.965 ± 0.704	0.013 ± 0.062	-0.412 ± 0.051	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5124.421 ± 0.602	0.070 ± 0.071	-0.411 ± 0.048	-0.004 ± 0.002
OPSD	MAP-XAI fallback	-17.751 ± 0.664	-0.033 ± 0.057	-0.000 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.461 ± 0.817	0.030 ± 0.036	0.009 ± 0.010	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.553 ± 0.814	0.025 ± 0.073	0.005 ± 0.009	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.030 ± 0.589	-0.010 ± 0.051	0.002 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.108 ± 0.765	0.032 ± 0.027	0.007 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.323 ± 0.668	0.046 ± 0.040	0.003 ± 0.008	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4974.382 ± 1.091	-0.043 ± 0.084	-0.411 ± 0.033	-0.007 ± 0.002
UCI	AoX-aware fixed-DP	-4649.381 ± 1.226	0.034 ± 0.071	-0.370 ± 0.032	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4650.353 ± 1.443	0.069 ± 0.077	-0.372 ± 0.035	-0.004 ± 0.002
UCI	AoX-aware fixed-DP	-4977.700 ± 0.793	-0.043 ± 0.042	-0.411 ± 0.047	-0.008 ± 0.004
UCI	AoX-aware fixed-DP	-4656.089 ± 1.590	-0.009 ± 0.054	-0.370 ± 0.044	-0.004 ± 0.001
UCI	AoX-aware fixed-DP	-4656.222 ± 1.573	0.050 ± 0.050	-0.371 ± 0.047	-0.003 ± 0.001
UCI	Channel-aware fixed-DP	-4969.336 ± 3.532	-0.028 ± 0.075	0.045 ± 0.006	0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4649.381 ± 1.226	0.030 ± 0.079	0.087 ± 0.009	0.665 ± 0.001
UCI	Channel-aware fixed-DP	-4650.353 ± 1.443	0.064 ± 0.076	0.084 ± 0.007	0.740 ± 0.002
UCI	Channel-aware fixed-DP	-4972.653 ± 4.225	-0.022 ± 0.057	0.044 ± 0.008	0.000 ± 0.004
UCI	Channel-aware fixed-DP	-4656.089 ± 1.590	-0.009 ± 0.072	0.084 ± 0.009	0.665 ± 0.002
UCI	Channel-aware fixed-DP	-4656.222 ± 1.573	0.051 ± 0.058	0.083 ± 0.012	0.741 ± 0.002
UCI	Joint, no compression	-346.458 ± 1.330	-0.065 ± 0.094	-0.007 ± 0.008	-0.002 ± 0.002
UCI	Joint, no compression	-981.307 ± 1.189	0.081 ± 0.077	-0.007 ± 0.007	-0.002 ± 0.001
UCI	Joint, no compression	-3384.679 ± 5.661	0.086 ± 0.072	-0.143 ± 0.011	-0.004 ± 0.001
UCI	Joint, no compression	-349.776 ± 1.051	-0.012 ± 0.075	-0.012 ± 0.009	-0.003 ± 0.004
UCI	Joint, no compression	-988.465 ± 1.244	0.048 ± 0.054	-0.015 ± 0.021	-0.002 ± 0.001
UCI	Joint, no compression	-3390.548 ± 4.834	0.069 ± 0.071	-0.144 ± 0.034	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4973.477 ± 1.902	-0.039 ± 0.091	0.039 ± 0.007	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4648.475 ± 2.065	0.038 ± 0.074	0.081 ± 0.009	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4649.447 ± 2.451	0.073 ± 0.052	0.079 ± 0.007	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4976.794 ± 1.677	-0.043 ± 0.067	0.039 ± 0.006	-0.009 ± 0.004
UCI	Periodic fixed-DP	-4655.183 ± 2.320	-0.009 ± 0.059	0.079 ± 0.009	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4655.316 ± 2.411	0.050 ± 0.049	0.078 ± 0.011	-0.004 ± 0.001
UCI	MAP-XAI fallback	-55.904 ± 0.864	0.065 ± 0.067	-0.002 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-114.499 ± 1.474	0.053 ± 0.062	0.001 ± 0.009	-0.000 ± 0.000
UCI	MAP-XAI fallback	-115.399 ± 1.659	0.042 ± 0.048	-0.007 ± 0.007	-0.000 ± 0.001
UCI	MAP-XAI fallback	-58.759 ± 0.856	-0.003 ± 0.033	-0.001 ± 0.006	0.000 ± 0.001
UCI	MAP-XAI fallback	-120.762 ± 1.805	0.010 ± 0.056	-0.006 ± 0.013	0.000 ± 0.001
UCI	MAP-XAI fallback	-120.556 ± 1.752	0.001 ± 0.070	-0.003 ± 0.007	0.000 ± 0.000

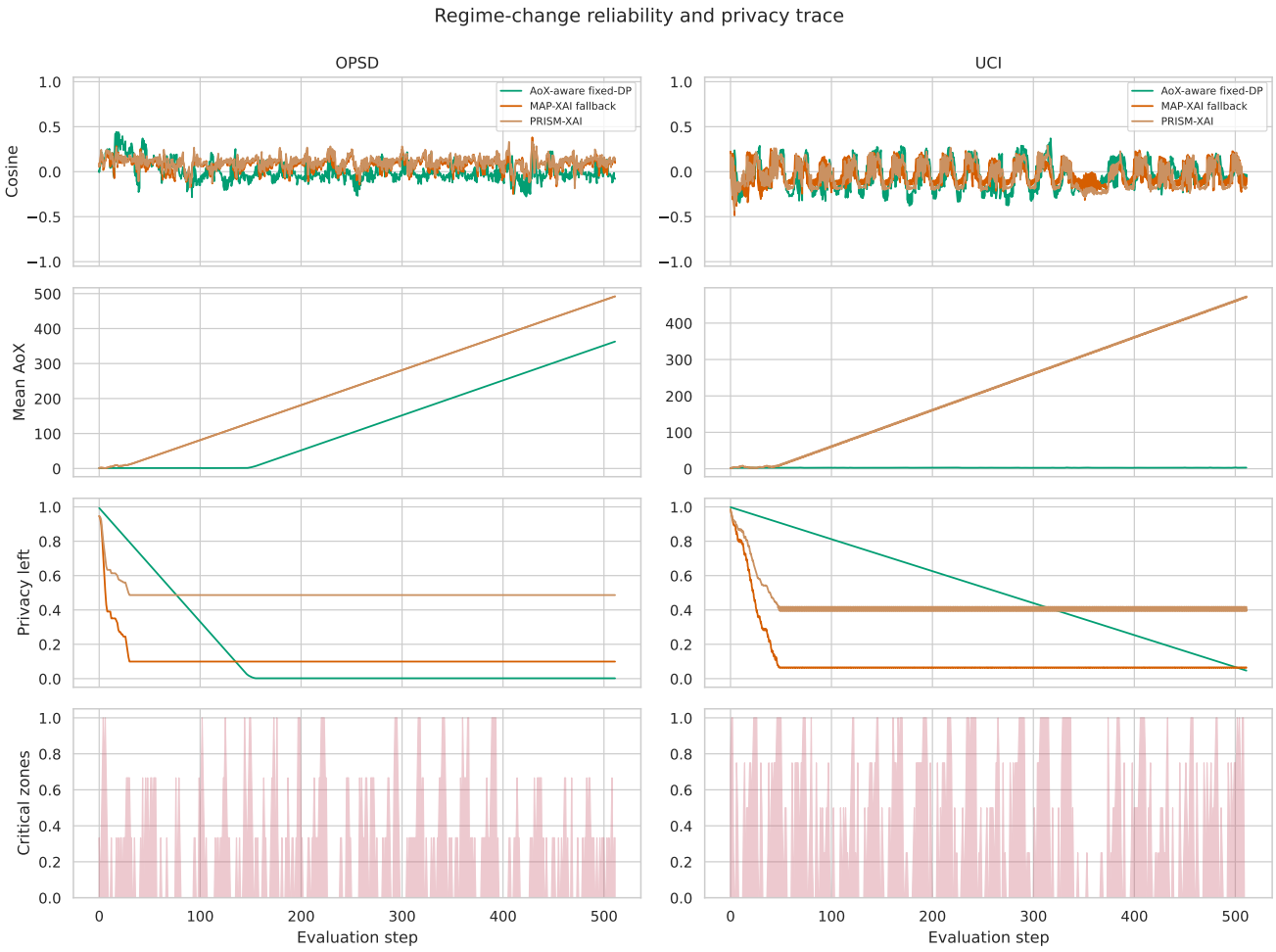


Fig. 80. Public-score threshold $\tau = 2$, release cap $M = 16$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

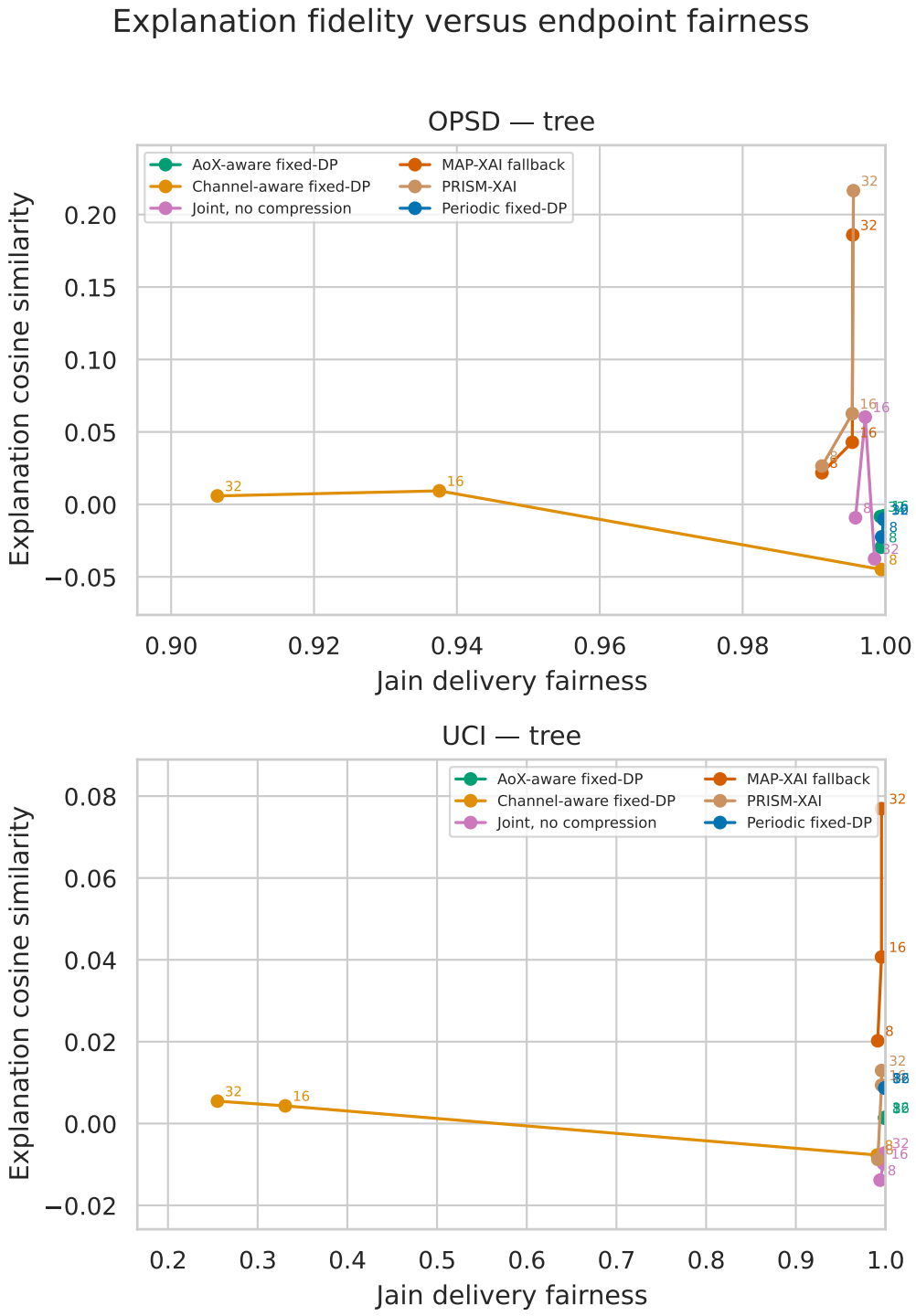


Fig. 81. Public-score threshold $\tau = 2$, release cap $M = 16$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

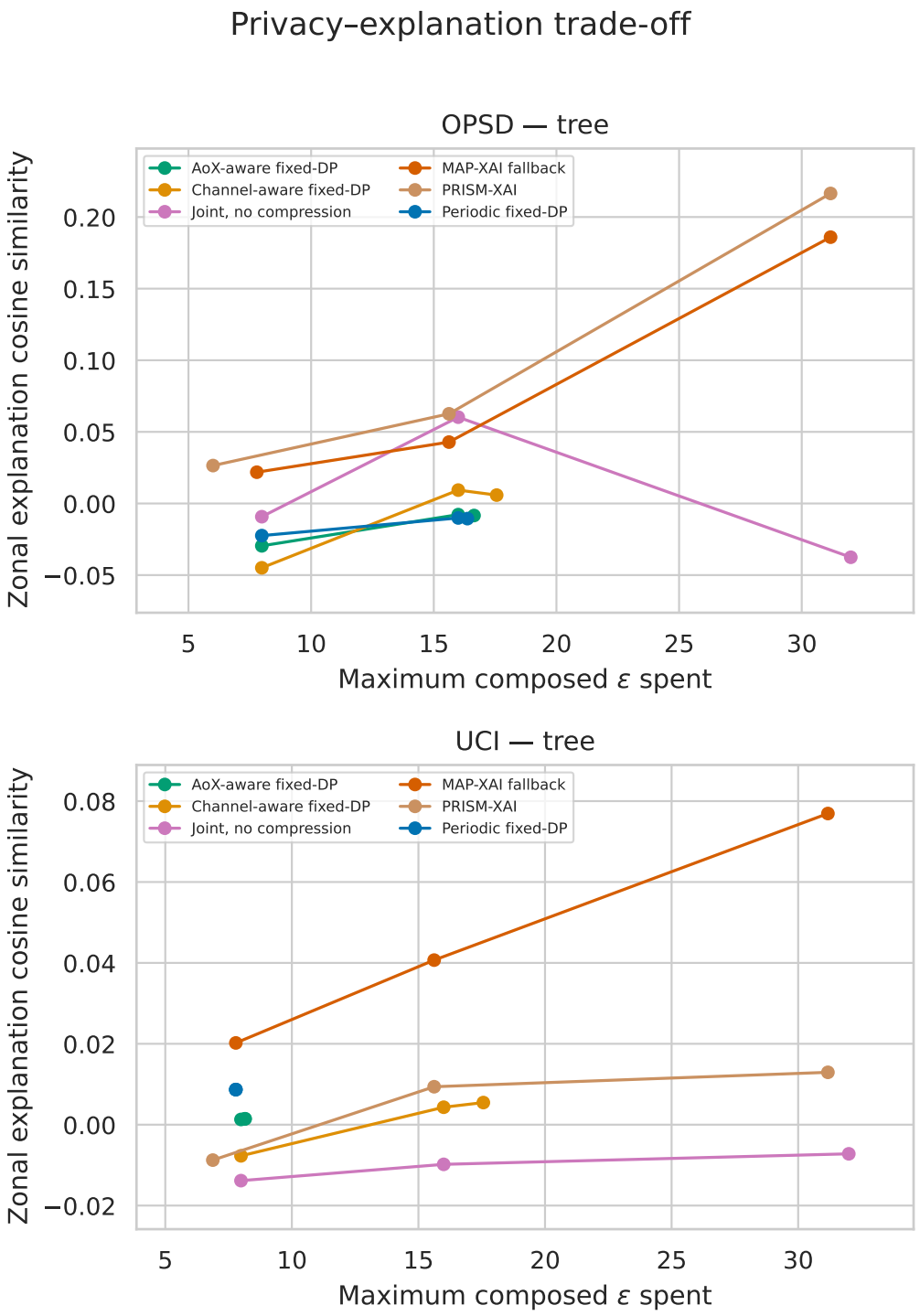


Fig. 82. Public-score threshold $\tau = 2$, release cap $M = 16$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

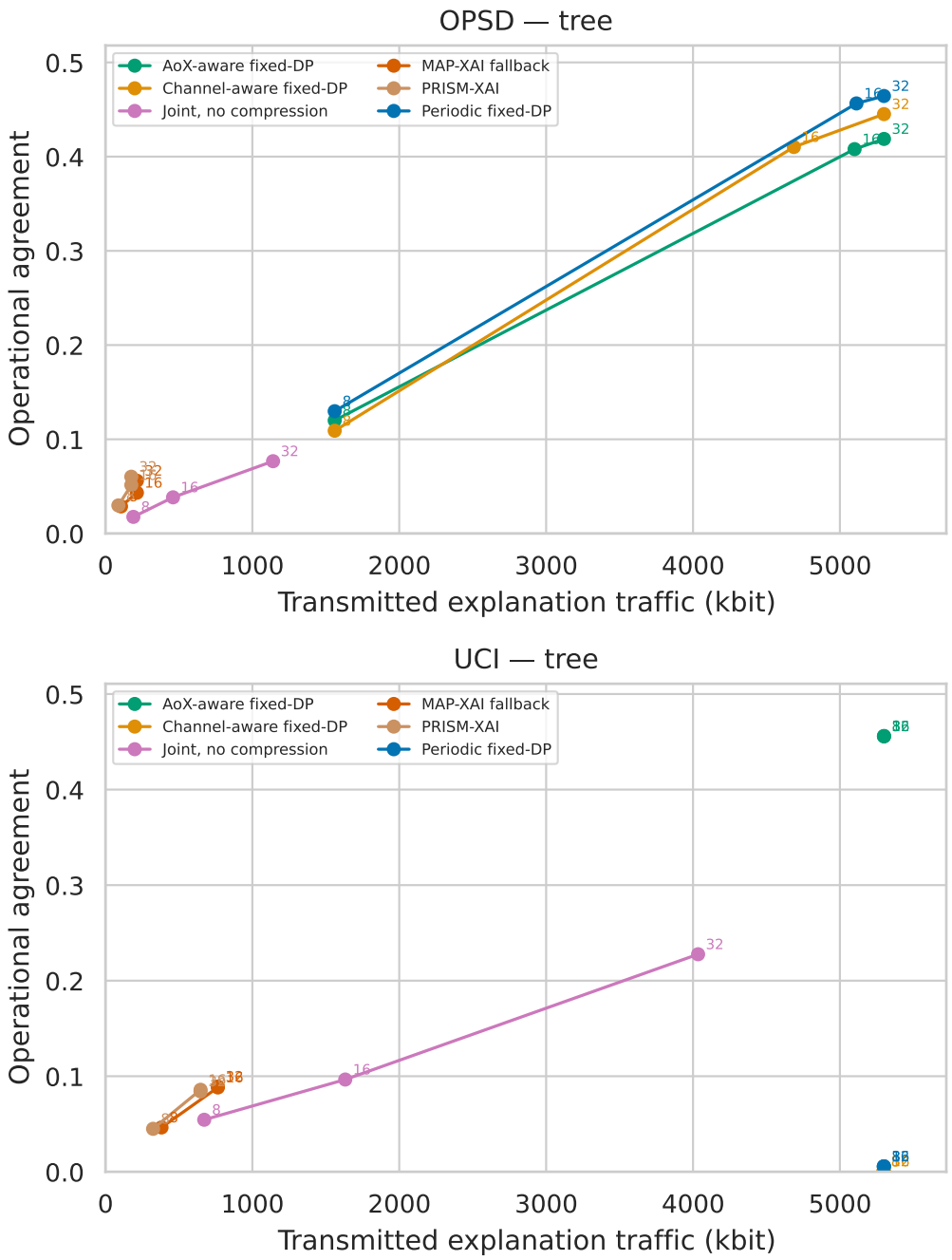


Fig. 83. Public-score threshold $\tau = 2$, release cap $M = 16$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

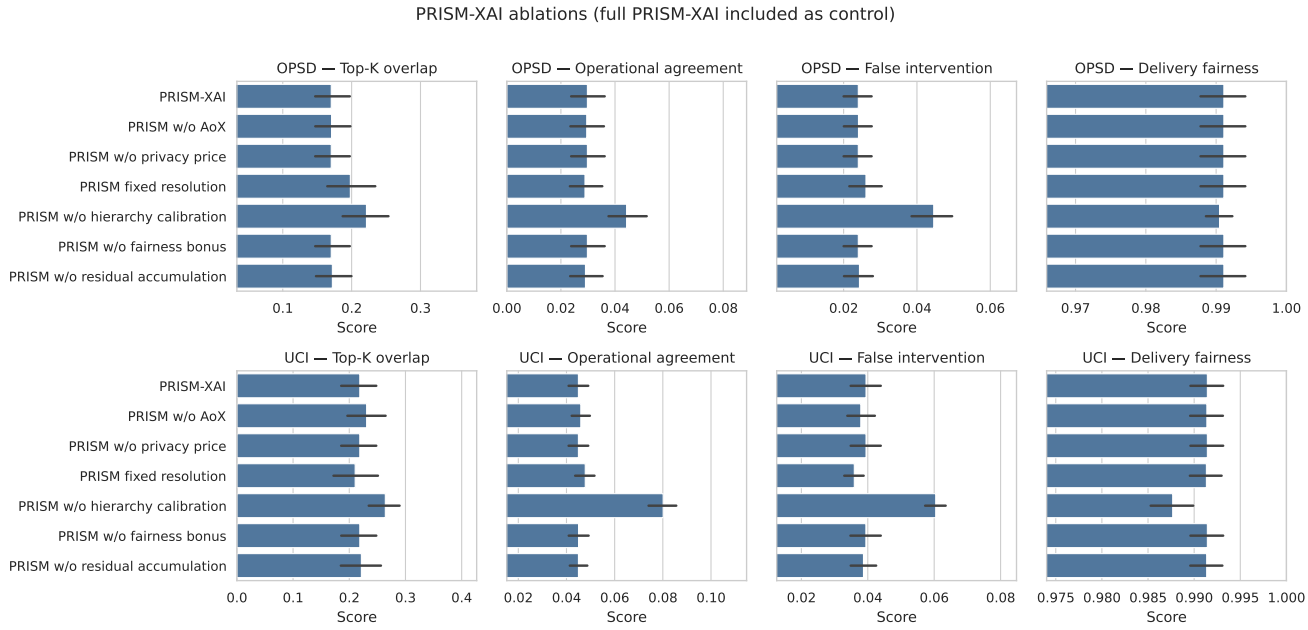


Fig. 84. Public-score threshold $\tau = 2$, release cap $M = 16$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXIV
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.170 ± 0.039	0.027 ± 0.072	0.029 ± 0.012	0.029 ± 0.012	0.326 ± 0.147	0.938 ± 0.004	4.881 ± 1.730	88.137 ± 0.577	6.066 ± 0.239	0.024 ± 0.007	10754.619 ± 5615.749	0.991 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.171 ± 0.046	0.026 ± 0.082	0.031 ± 0.010	0.031 ± 0.010	0.343 ± 0.130	0.938 ± 0.004	4.873 ± 1.737	87.707 ± 0.532	5.936 ± 0.213	0.024 ± 0.007	8388.650 ± 2960.712	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.203 ± 0.058	0.019 ± 0.075	0.029 ± 0.011	0.029 ± 0.011	0.324 ± 0.148	0.938 ± 0.004	4.960 ± 1.678	105.889 ± 0.169	7.786 ± 0.000	0.026 ± 0.008	11344.761 ± 4319.900	0.991 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.181 ± 0.054	0.025 ± 0.081	0.029 ± 0.009	0.029 ± 0.009	0.310 ± 0.118	0.938 ± 0.004	4.960 ± 1.678	105.737 ± 0.152	7.786 ± 0.000	0.026 ± 0.007	10402.432 ± 2994.804	0.991 ± 0.006

TABLE LXV
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	88.137 ± 0.577	0.170 ± 0.039	0.250 ± 0.000	0.027 ± 0.072	0.029 ± 0.012	0.029 ± 0.012	0.326 ± 0.147	0.062 ± 0.004	0.938 ± 0.004	0.024 ± 0.007	6.066 ± 0.239
OPSD	Tree	8	PRISM-XAI	87.707 ± 0.532	0.171 ± 0.046	0.250 ± 0.000	0.026 ± 0.082	0.031 ± 0.010	0.031 ± 0.010	0.343 ± 0.130	0.062 ± 0.004	0.938 ± 0.004	0.024 ± 0.007	5.936 ± 0.213
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	176.428 ± 0.771	0.259 ± 0.050	0.250 ± 0.000	0.054 ± 0.064	0.053 ± 0.015	0.053 ± 0.015	0.339 ± 0.133	0.120 ± 0.006	0.880 ± 0.006	0.042 ± 0.011	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	175.629 ± 0.704	0.256 ± 0.054	0.250 ± 0.000	0.071 ± 0.073	0.050 ± 0.013	0.050 ± 0.013	0.291 ± 0.104	0.120 ± 0.006	0.880 ± 0.006	0.049 ± 0.010	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	176.336 ± 0.759	0.315 ± 0.088	0.250 ± 0.000	0.193 ± 0.062	0.061 ± 0.013	0.061 ± 0.013	0.397 ± 0.114	0.120 ± 0.006	0.880 ± 0.006	0.037 ± 0.010	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	175.415 ± 0.593	0.314 ± 0.062	0.250 ± 0.000	0.240 ± 0.062	0.060 ± 0.013	0.060 ± 0.013	0.365 ± 0.089	0.120 ± 0.006	0.880 ± 0.006	0.040 ± 0.008	31.175 ± 0.000

TABLE LXVI
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.427 ± 0.577	-0.064 ± 0.057	-0.091 ± 0.033	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-4923.485 ± 5.595	0.006 ± 0.053	-0.349 ± 0.082	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5123.888 ± 0.759	0.062 ± 0.093	-0.351 ± 0.083	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-1472.857 ± 0.532	-0.068 ± 0.063	-0.090 ± 0.028	-0.008 ± 0.006
OPSD	AoX-aware fixed-DP	-4924.283 ± 5.582	0.004 ± 0.062	-0.364 ± 0.062	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5124.809 ± 0.593	0.062 ± 0.071	-0.366 ± 0.060	-0.004 ± 0.002
OPSD	Channel-aware fixed-DP	-1472.427 ± 0.577	-0.122 ± 0.049	-0.078 ± 0.029	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4510.181 ± 12.999	-0.002 ± 0.048	-0.349 ± 0.095	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5123.888 ± 0.759	0.057 ± 0.093	-0.373 ± 0.102	0.089 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.857 ± 0.532	-0.122 ± 0.074	-0.080 ± 0.025	-0.008 ± 0.006
OPSD	Channel-aware fixed-DP	-4510.980 ± 13.099	-0.002 ± 0.051	-0.369 ± 0.093	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5124.809 ± 0.593	0.057 ± 0.070	-0.396 ± 0.090	0.089 ± 0.006
OPSD	Joint, no compression	-101.763 ± 0.687	-0.118 ± 0.078	0.012 ± 0.011	-0.005 ± 0.005
OPSD	Joint, no compression	-283.472 ± 1.424	0.028 ± 0.096	0.015 ± 0.017	-0.002 ± 0.002
OPSD	Joint, no compression	-964.564 ± 8.170	0.080 ± 0.106	-0.013 ± 0.022	-0.003 ± 0.002
OPSD	Joint, no compression	-102.043 ± 0.513	-0.088 ± 0.080	0.012 ± 0.009	-0.005 ± 0.005
OPSD	Joint, no compression	-284.421 ± 1.294	0.015 ± 0.063	0.011 ± 0.014	-0.002 ± 0.002
OPSD	Joint, no compression	-965.485 ± 8.153	0.032 ± 0.101	-0.020 ± 0.016	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-1472.427 ± 0.577	-0.070 ± 0.050	-0.096 ± 0.024	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-4936.166 ± 0.771	0.010 ± 0.060	-0.398 ± 0.067	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5123.500 ± 0.672	0.066 ± 0.097	-0.397 ± 0.070	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-1472.857 ± 0.532	-0.071 ± 0.050	-0.104 ± 0.018	-0.008 ± 0.006
OPSD	Periodic fixed-DP	-4936.965 ± 0.704	0.013 ± 0.062	-0.412 ± 0.051	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5124.421 ± 0.602	0.070 ± 0.071	-0.411 ± 0.048	-0.004 ± 0.002
OPSD	MAP-XAI fallback	-17.751 ± 0.664	-0.033 ± 0.057	-0.000 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.461 ± 0.817	0.030 ± 0.036	0.009 ± 0.010	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.553 ± 0.814	0.025 ± 0.073	0.005 ± 0.009	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.030 ± 0.589	-0.010 ± 0.051	0.002 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.108 ± 0.765	0.032 ± 0.027	0.007 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.323 ± 0.668	0.046 ± 0.040	0.003 ± 0.008	0.000 ± 0.000

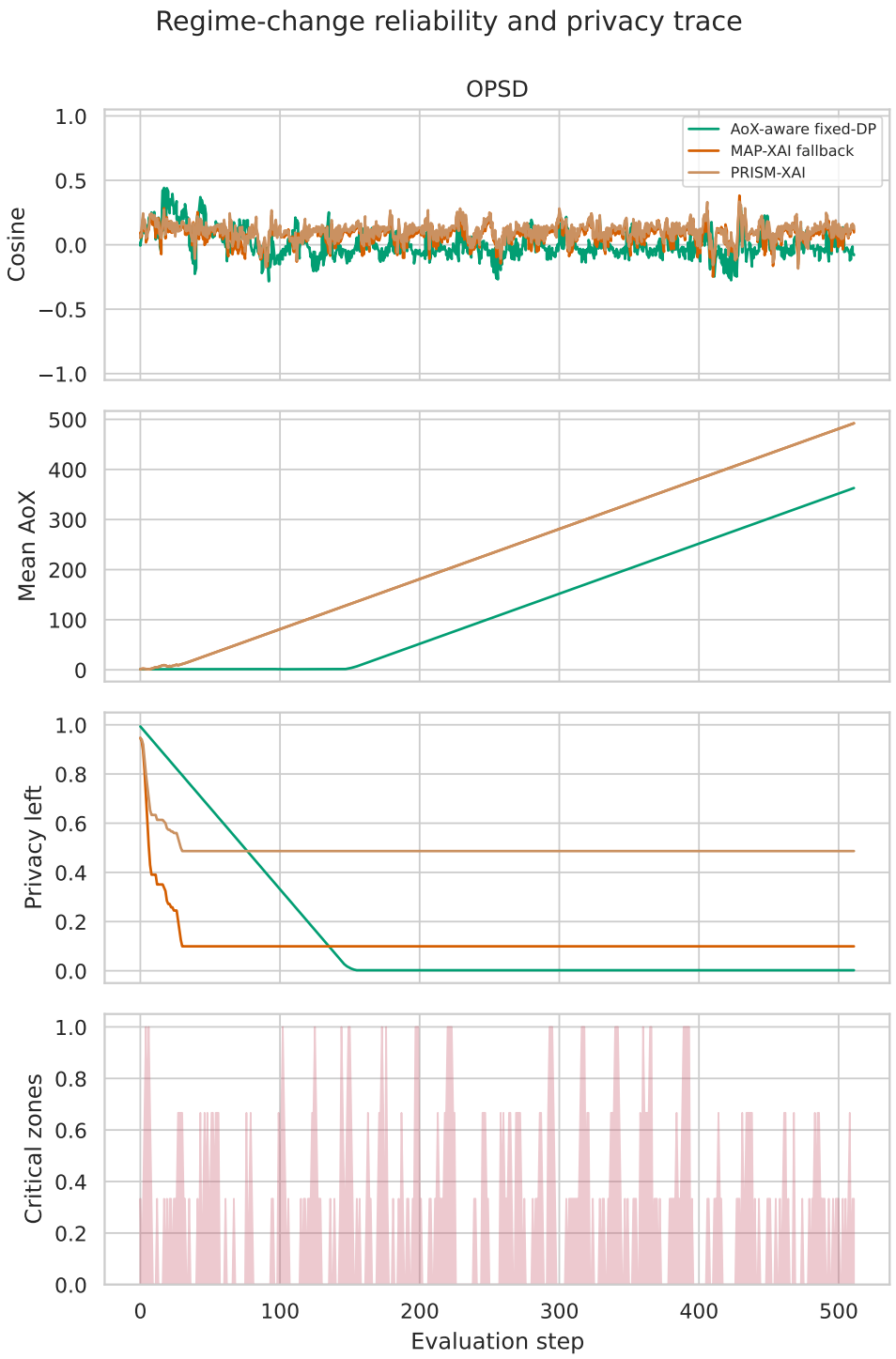


Fig. 85. Public-score threshold $\tau = 2$, release cap $M = 16$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

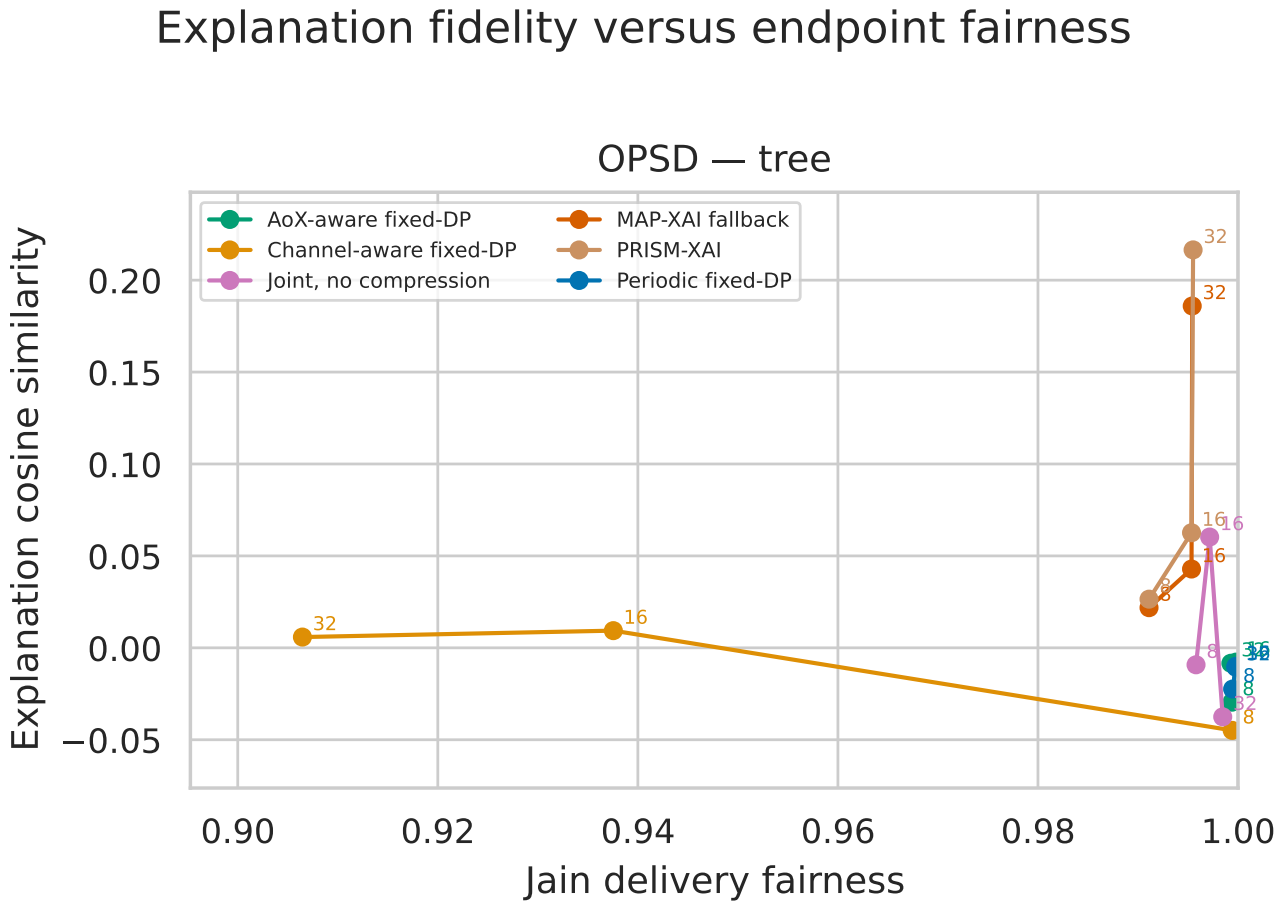


Fig. 86. Public-score threshold $\tau = 2$, release cap $M = 16$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

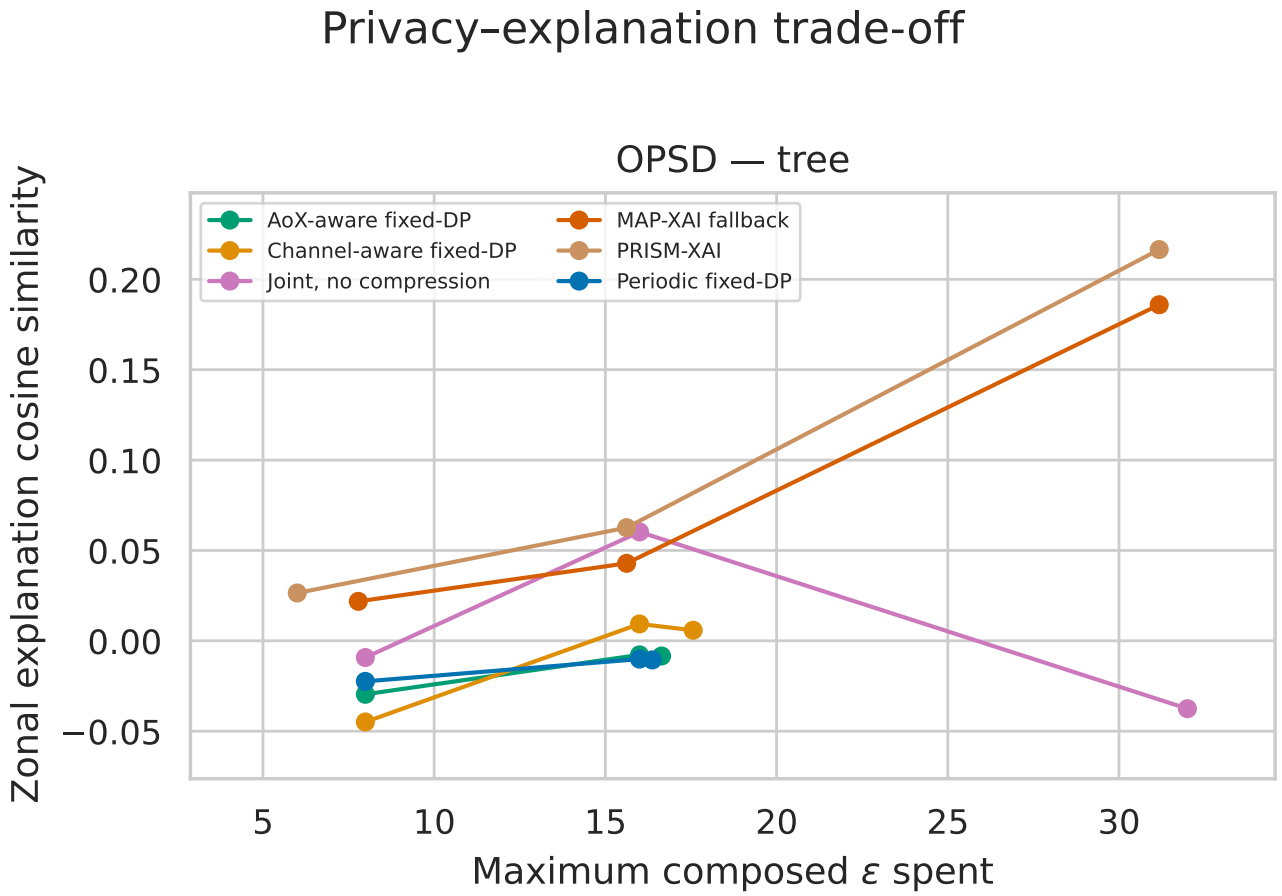


Fig. 87. Public-score threshold $\tau = 2$, release cap $M = 16$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

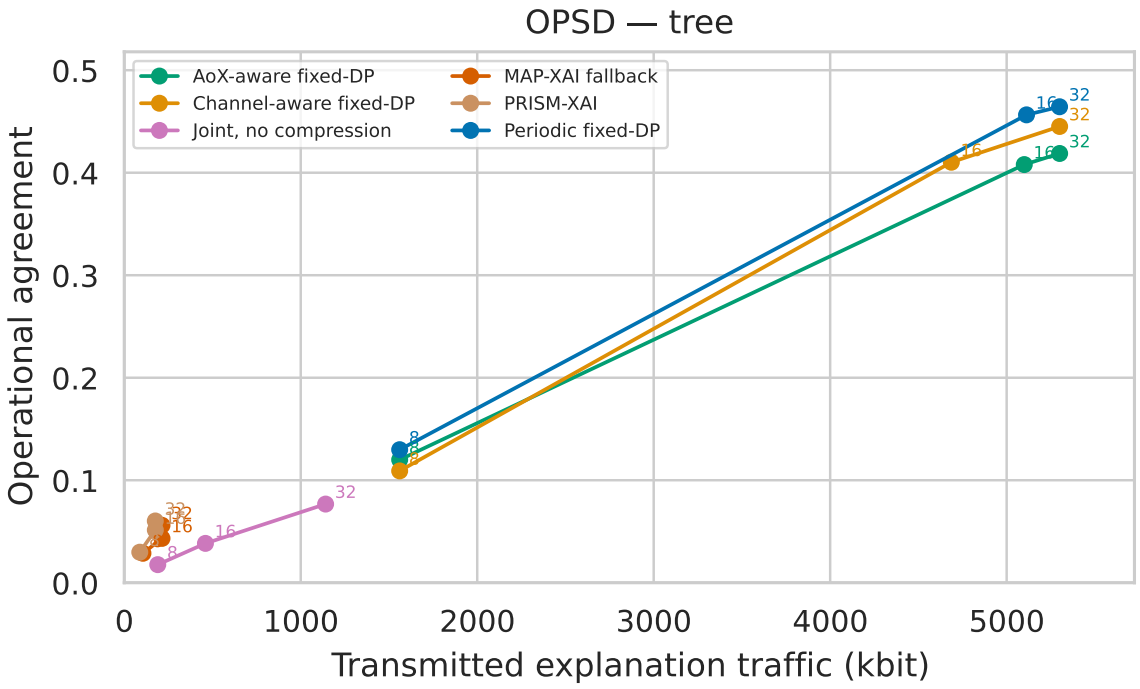


Fig. 88. Public-score threshold $\tau = 2$, release cap $M = 16$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

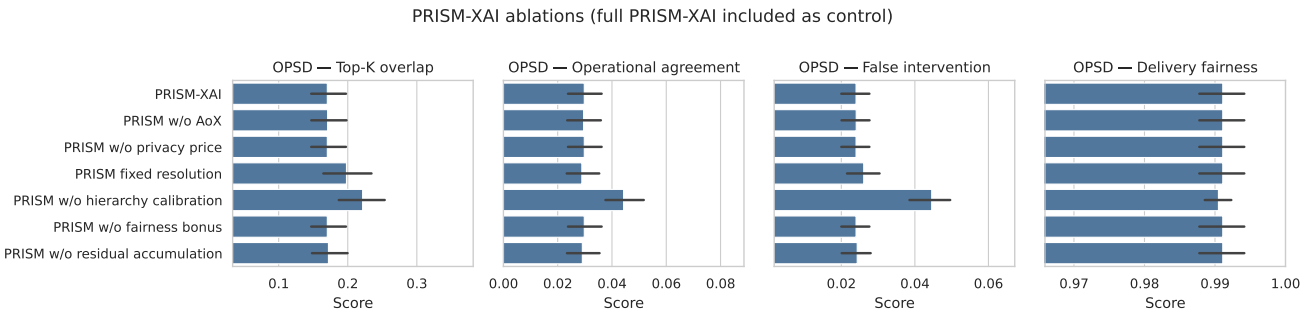


Fig. 89. Public-score threshold $\tau = 2$, release cap $M = 16$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXVII
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.220 ± 0.061	-0.009 ± 0.015	0.045 ± 0.006	0.045 ± 0.006	0.442 ± 0.072	0.906 ± 0.004	4.504 ± 1.537	325.842 ± 1.091	6.877 ± 0.090	0.040 ± 0.007	8917.519 ± 1309.048	0.992 ± 0.002
UCI	Tree	8	PRISM-XAI	0.217 ± 0.043	-0.008 ± 0.077	0.045 ± 0.007	0.045 ± 0.007	0.447 ± 0.078	0.908 ± 0.002	5.058 ± 1.846	322.524 ± 0.793	6.877 ± 0.090	0.039 ± 0.008	9128.968 ± 1604.303	0.991 ± 0.004
UCI	Tree	8	MAP-XAI fallback	0.155 ± 0.082	0.000 ± 0.016	0.047 ± 0.007	0.047 ± 0.007	0.456 ± 0.051	0.905 ± 0.004	4.584 ± 1.508	381.745 ± 0.387	7.787 ± 0.001	0.038 ± 0.003	10122.948 ± 1695.832	0.992 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.221 ± 0.054	0.040 ± 0.080	0.046 ± 0.008	0.046 ± 0.008	0.469 ± 0.072	0.907 ± 0.003	4.763 ± 1.714	381.283 ± 0.251	7.787 ± 0.001	0.036 ± 0.006	10525.744 ± 1825.173	0.991 ± 0.004

TABLE LXVIII
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	325.842 ± 1.091	0.220 ± 0.061	0.250 ± 0.000	-0.009 ± 0.015	0.045 ± 0.006	0.045 ± 0.006	0.442 ± 0.072	0.094 ± 0.004	0.906 ± 0.004	0.040 ± 0.007	6.877 ± 0.090
UCI	Tree	8	PRISM-XAI	322.524 ± 0.793	0.217 ± 0.043	0.250 ± 0.000	-0.008 ± 0.077	0.045 ± 0.007	0.045 ± 0.007	0.447 ± 0.078	0.092 ± 0.002	0.908 ± 0.002	0.039 ± 0.008	6.877 ± 0.090
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	650.843 ± 1.226	0.297 ± 0.062	0.250 ± 0.000	0.008 ± 0.020	0.087 ± 0.008	0.087 ± 0.008	0.454 ± 0.046	0.180 ± 0.003	0.820 ± 0.003	0.073 ± 0.007	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	644.135 ± 1.590	0.251 ± 0.053	0.250 ± 0.000	0.011 ± 0.046	0.084 ± 0.009	0.084 ± 0.009	0.463 ± 0.061	0.177 ± 0.004	0.823 ± 0.004	0.069 ± 0.011	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	649.871 ± 1.443	0.332 ± 0.056	0.250 ± 0.000	0.010 ± 0.024	0.085 ± 0.007	0.085 ± 0.007	0.572 ± 0.040	0.177 ± 0.004	0.823 ± 0.004	0.044 ± 0.006	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	644.002 ± 1.573	0.310 ± 0.057	0.250 ± 0.000	0.016 ± 0.064	0.083 ± 0.012	0.083 ± 0.012	0.561 ± 0.063	0.173 ± 0.006	0.827 ± 0.006	0.045 ± 0.010	31.175 ± 0.000

TABLE LXIX
PUBLIC-SCORE THRESHOLD $\tau = 2$, RELEASE CAP $M = 16$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4974.382 ± 1.091	-0.043 ± 0.084	-0.411 ± 0.033	-0.007 ± 0.002
UCI	AoX-aware fixed-DP	-4649.381 ± 1.226	0.034 ± 0.071	-0.370 ± 0.032	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4650.353 ± 1.443	0.069 ± 0.077	-0.372 ± 0.035	-0.004 ± 0.002
UCI	AoX-aware fixed-DP	-4977.700 ± 0.793	-0.043 ± 0.042	-0.411 ± 0.047	-0.008 ± 0.004
UCI	AoX-aware fixed-DP	-4656.089 ± 1.590	-0.009 ± 0.054	-0.370 ± 0.044	-0.004 ± 0.001
UCI	AoX-aware fixed-DP	-4656.222 ± 1.573	0.050 ± 0.050	-0.371 ± 0.047	-0.003 ± 0.001
UCI	Channel-aware fixed-DP	-4969.336 ± 3.532	-0.028 ± 0.075	0.045 ± 0.006	0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4649.381 ± 1.226	0.030 ± 0.079	0.087 ± 0.009	0.665 ± 0.001
UCI	Channel-aware fixed-DP	-4650.353 ± 1.443	0.064 ± 0.076	0.084 ± 0.007	0.740 ± 0.002
UCI	Channel-aware fixed-DP	-4972.653 ± 4.225	-0.022 ± 0.057	0.044 ± 0.008	0.000 ± 0.004
UCI	Channel-aware fixed-DP	-4656.089 ± 1.590	-0.009 ± 0.072	0.084 ± 0.009	0.665 ± 0.002
UCI	Channel-aware fixed-DP	-4656.222 ± 1.573	0.051 ± 0.058	0.083 ± 0.012	0.741 ± 0.002
UCI	Joint, no compression	-346.458 ± 1.330	-0.065 ± 0.094	-0.007 ± 0.008	-0.002 ± 0.002
UCI	Joint, no compression	-981.307 ± 1.189	0.081 ± 0.077	-0.007 ± 0.007	-0.002 ± 0.001
UCI	Joint, no compression	-3384.679 ± 5.661	0.086 ± 0.072	-0.143 ± 0.011	-0.004 ± 0.001
UCI	Joint, no compression	-349.776 ± 1.051	-0.012 ± 0.075	-0.012 ± 0.009	-0.003 ± 0.004
UCI	Joint, no compression	-988.465 ± 1.244	0.048 ± 0.054	-0.015 ± 0.021	-0.002 ± 0.001
UCI	Joint, no compression	-3390.548 ± 4.834	0.069 ± 0.071	-0.144 ± 0.034	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4973.477 ± 1.902	-0.039 ± 0.091	0.039 ± 0.007	-0.007 ± 0.002
UCI	Periodic fixed-DP	-4648.475 ± 2.065	0.038 ± 0.074	0.081 ± 0.009	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4649.447 ± 2.451	0.073 ± 0.052	0.079 ± 0.007	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4976.794 ± 1.677	-0.043 ± 0.067	0.039 ± 0.006	-0.009 ± 0.004
UCI	Periodic fixed-DP	-4655.183 ± 2.320	-0.009 ± 0.059	0.079 ± 0.009	-0.004 ± 0.001
UCI	Periodic fixed-DP	-4655.316 ± 2.411	0.050 ± 0.049	0.078 ± 0.011	-0.004 ± 0.001
UCI	MAP-XAI fallback	-55.904 ± 0.864	0.065 ± 0.067	-0.002 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-114.499 ± 1.474	0.053 ± 0.062	0.001 ± 0.009	-0.000 ± 0.000
UCI	MAP-XAI fallback	-115.399 ± 1.659	0.042 ± 0.048	-0.007 ± 0.007	-0.000 ± 0.001
UCI	MAP-XAI fallback	-58.759 ± 0.856	-0.003 ± 0.033	-0.001 ± 0.006	0.000 ± 0.001
UCI	MAP-XAI fallback	-120.762 ± 1.805	0.010 ± 0.056	-0.006 ± 0.013	0.000 ± 0.001
UCI	MAP-XAI fallback	-120.556 ± 1.752	0.001 ± 0.070	-0.003 ± 0.007	0.000 ± 0.000

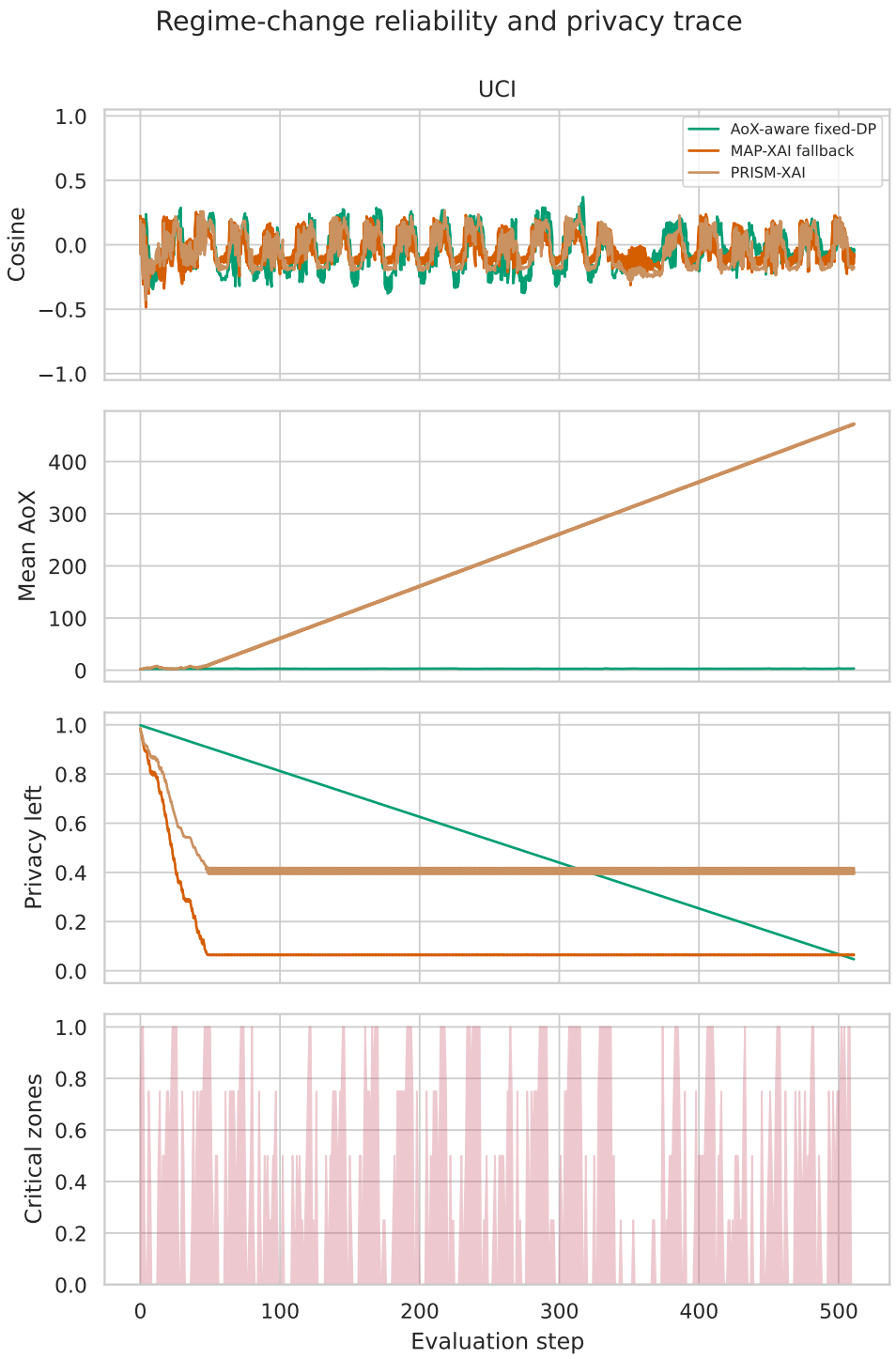


Fig. 90. Public-score threshold $\tau = 2$, release cap $M = 16$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

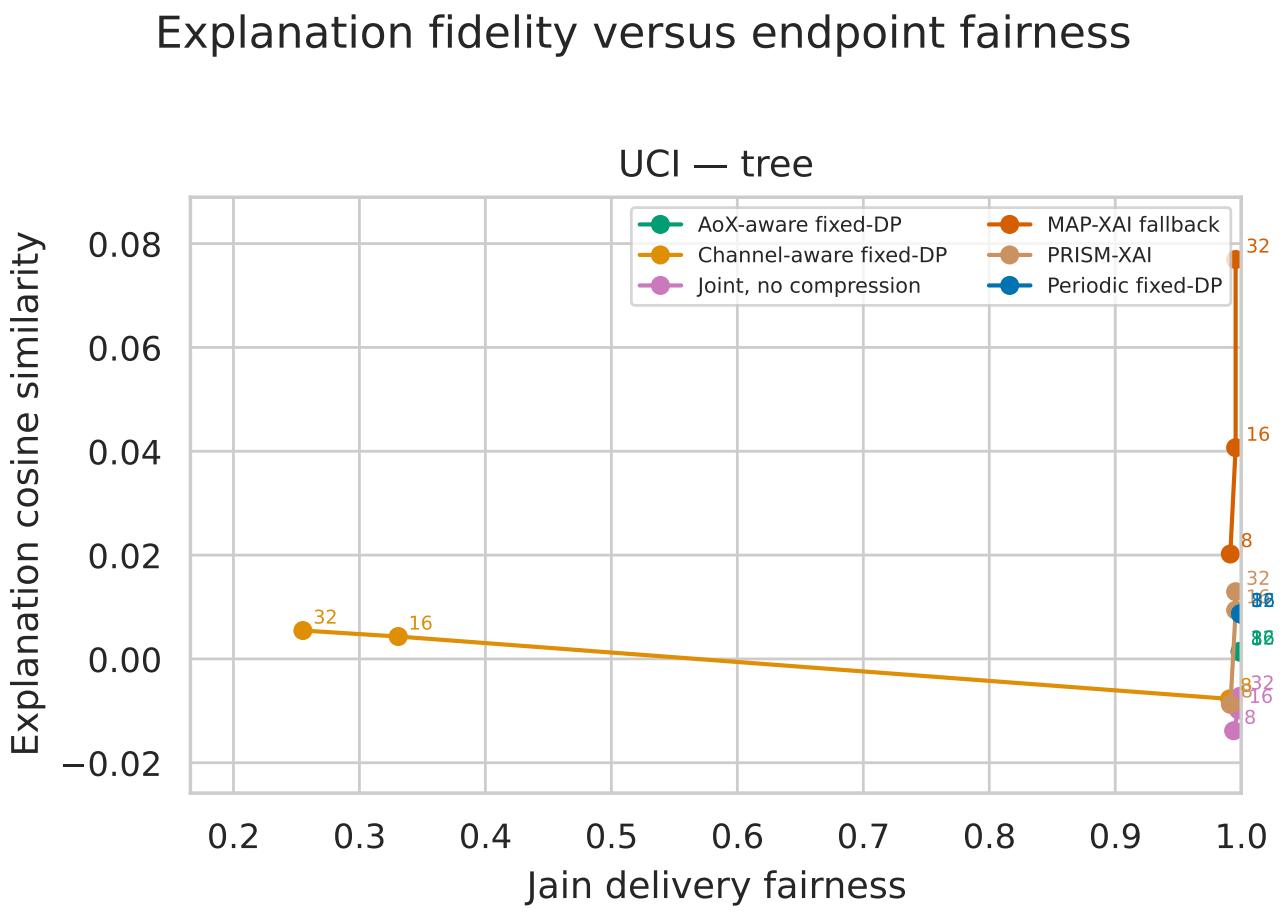


Fig. 91. Public-score threshold $\tau = 2$, release cap $M = 16$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

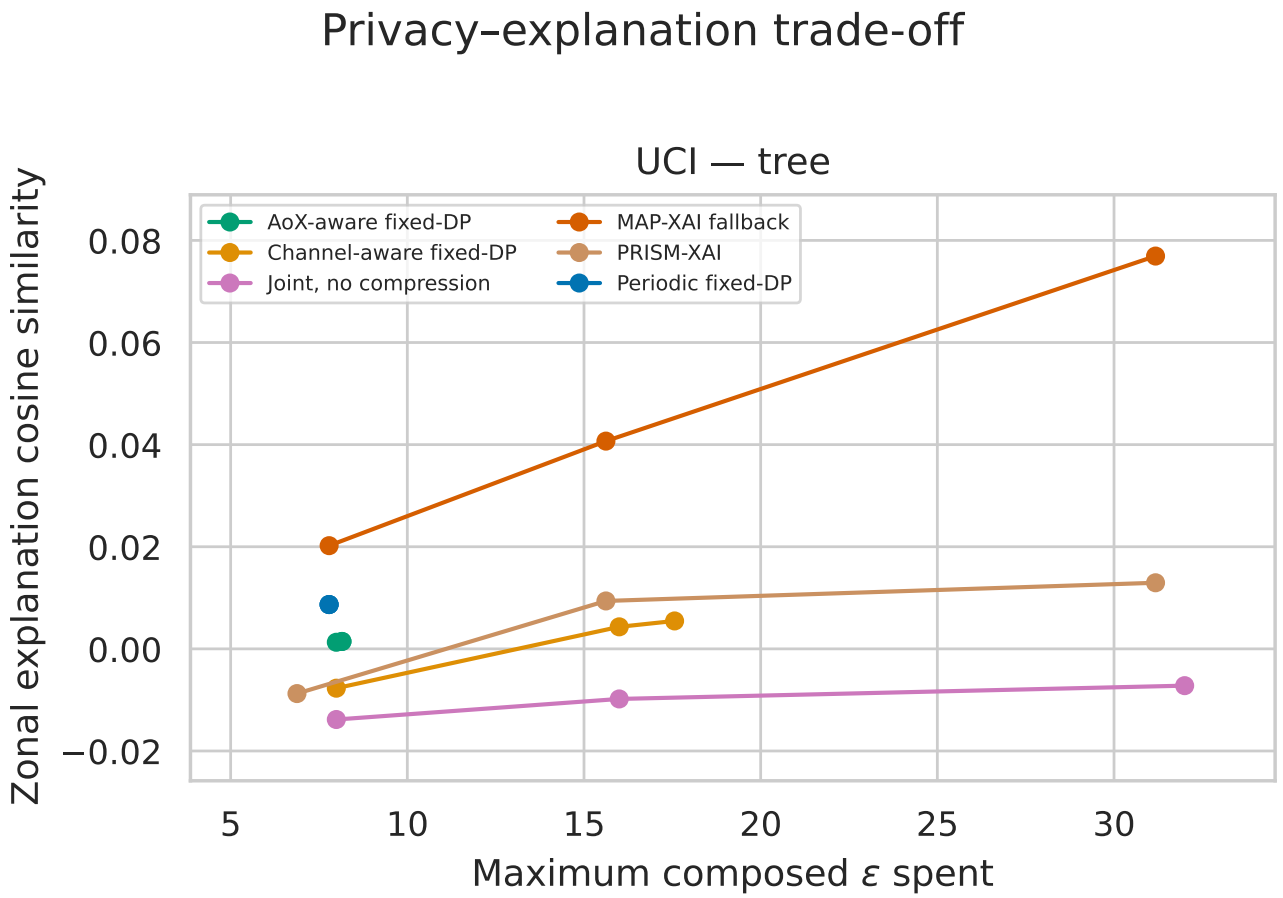


Fig. 92. Public-score threshold $\tau = 2$, release cap $M = 16$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

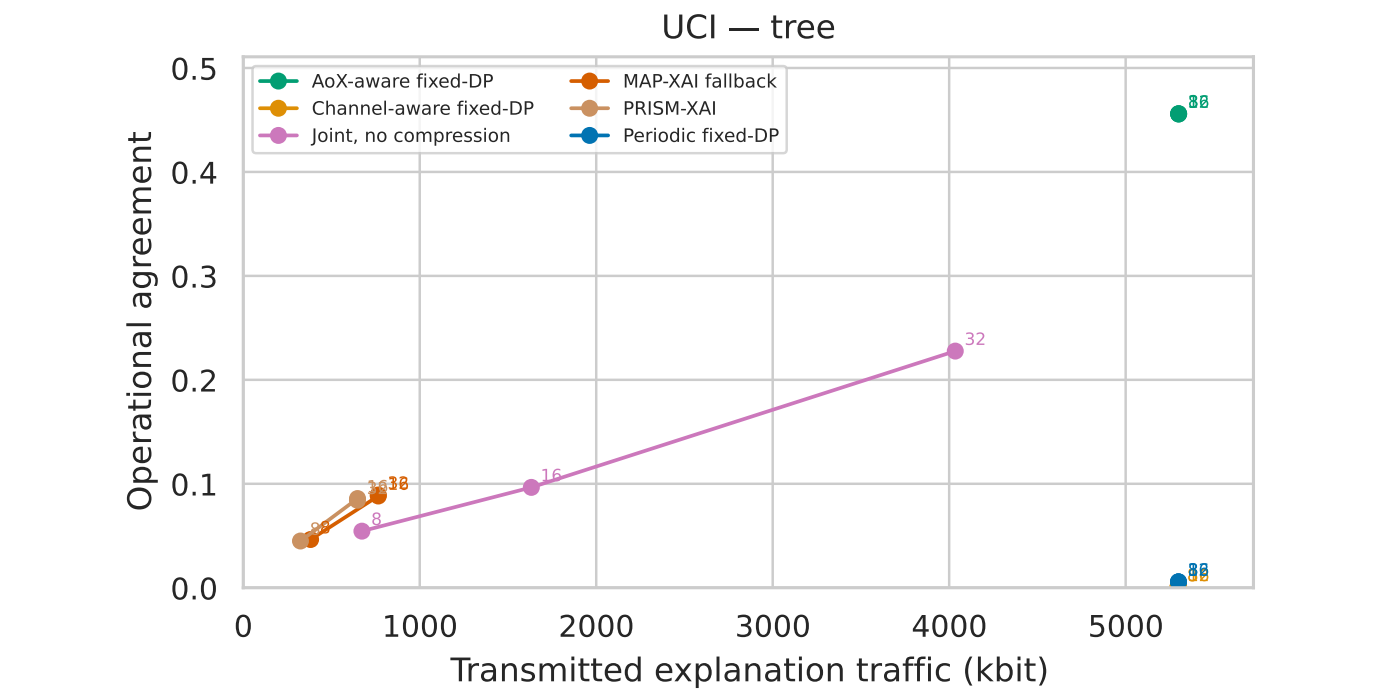


Fig. 93. Public-score threshold $\tau = 2$, release cap $M = 16$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

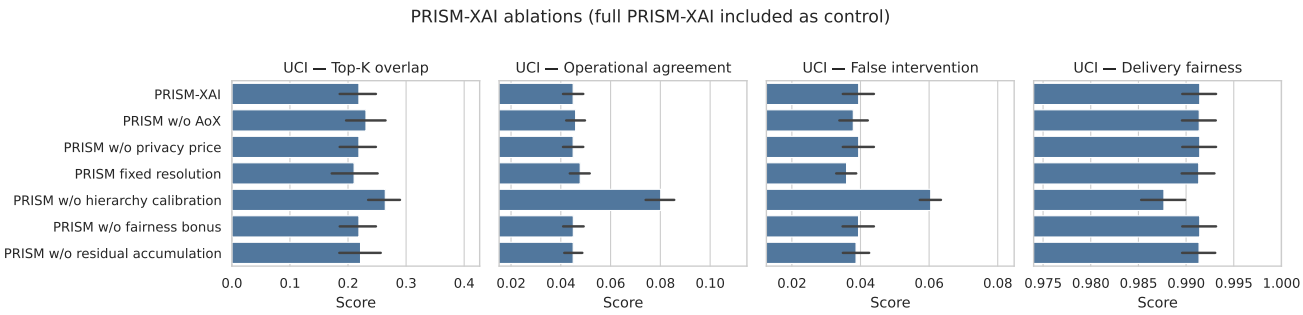


Fig. 94. Public-score threshold $\tau = 2$, release cap $M = 16$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

7) Public-score threshold $\tau = 3$, release cap $M = 16$: At this operating point, PRISM-XAI transmits 345.42 kbit versus 407.78 kbit for MAP-XAI-only (aggregate-mean ratio saving 15.29%). The paired difference is -62.36±6.82 kbit. PRISM coverage is 0.1312, critical-event recall is 0.0594, and maximum recorded expenditure is 17.777. Relative to MAP-XAI-only, the paired coverage change is -0.00010±0.00034 and recall change is +0.00244±0.00227. The descriptive overlap rule holds in 8 of 12 cells. Accordingly, this point is read as a sensitivity point on the service frontier, not a complete matched-service confirmation.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXX

PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.231 ± 0.042	0.028 ± 0.075	0.025 ± 0.011	0.025 ± 0.011	0.268 ± 0.127	0.929 ± 0.005	4.879 ± 1.736	88.444 ± 0.568	6.067 ± 0.212	0.026 ± 0.005	12196.035 ± 5705.209	0.990 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.244 ± 0.039	0.030 ± 0.085	0.026 ± 0.012	0.026 ± 0.012	0.274 ± 0.138	0.929 ± 0.004	4.869 ± 1.744	87.922 ± 0.520	5.936 ± 0.213	0.027 ± 0.007	11503.042 ± 5462.280	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.289 ± 0.065	0.023 ± 0.079	0.028 ± 0.008	0.028 ± 0.008	0.296 ± 0.082	0.929 ± 0.005	4.945 ± 1.699	106.298 ± 0.291	7.786 ± 0.000	0.025 ± 0.004	10222.108 ± 2661.883	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.267 ± 0.070	0.045 ± 0.077	0.027 ± 0.007	0.027 ± 0.007	0.257 ± 0.051	0.929 ± 0.005	4.948 ± 1.696	106.157 ± 0.283	7.786 ± 0.000	0.028 ± 0.003	10802.437 ± 2867.714	0.990 ± 0.006
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.231 ± 0.061	-0.001 ± 0.013	0.043 ± 0.009	0.043 ± 0.009	0.466 ± 0.081	0.909 ± 0.005	5.656 ± 1.725	328.663 ± 1.360	7.081 ± 0.177	0.033 ± 0.005	10019.834 ± 2488.756	0.990 ± 0.003
UCI	Tree	8	PRISM-XAI	0.245 ± 0.047	-0.007 ± 0.082	0.045 ± 0.007	0.045 ± 0.007	0.507 ± 0.071	0.910 ± 0.005	5.163 ± 1.785	325.833 ± 1.364	7.052 ± 0.147	0.030 ± 0.006	9029.695 ± 1367.291	0.991 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.202 ± 0.058	0.011 ± 0.012	0.041 ± 0.005	0.041 ± 0.005	0.453 ± 0.045	0.909 ± 0.005	5.677 ± 1.704	381.702 ± 0.092	7.786 ± 0.000	0.034 ± 0.003	11391.295 ± 1487.072	0.990 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.206 ± 0.051	0.062 ± 0.058	0.048 ± 0.010	0.048 ± 0.010	0.505 ± 0.102	0.910 ± 0.005	4.861 ± 1.663	381.256 ± 0.293	7.786 ± 0.000	0.033 ± 0.007	10444.880 ± 2673.464	0.991 ± 0.002

TABLE LXXI

PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	88.444 ± 0.568	0.231 ± 0.042	0.250 ± 0.000	0.028 ± 0.075	0.025 ± 0.011	0.025 ± 0.011	0.268 ± 0.127	0.071 ± 0.005	0.929 ± 0.005	0.026 ± 0.005	6.067 ± 0.212
OPSD	Tree	8	PRISM-XAI	87.922 ± 0.520	0.244 ± 0.039	0.250 ± 0.000	0.030 ± 0.085	0.026 ± 0.012	0.026 ± 0.012	0.274 ± 0.138	0.071 ± 0.004	0.929 ± 0.004	0.027 ± 0.007	5.936 ± 0.213
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	177.011 ± 0.621	0.255 ± 0.065	0.250 ± 0.000	0.062 ± 0.075	0.049 ± 0.016	0.049 ± 0.016	0.290 ± 0.110	0.133 ± 0.003	0.867 ± 0.003	0.047 ± 0.010	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	176.182 ± 0.595	0.244 ± 0.039	0.250 ± 0.000	0.080 ± 0.081	0.049 ± 0.013	0.049 ± 0.013	0.287 ± 0.097	0.132 ± 0.004	0.868 ± 0.004	0.048 ± 0.009	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	176.950 ± 0.602	0.317 ± 0.078	0.250 ± 0.000	0.217 ± 0.056	0.065 ± 0.016	0.065 ± 0.016	0.389 ± 0.094	0.133 ± 0.003	0.867 ± 0.003	0.038 ± 0.005	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	175.967 ± 0.530	0.283 ± 0.060	0.250 ± 0.000	0.255 ± 0.060	0.069 ± 0.013	0.069 ± 0.013	0.404 ± 0.078	0.132 ± 0.004	0.868 ± 0.004	0.038 ± 0.006	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	328.663 ± 1.360	0.231 ± 0.061	0.250 ± 0.000	-0.001 ± 0.013	0.043 ± 0.009	0.043 ± 0.009	0.466 ± 0.081	0.091 ± 0.005	0.909 ± 0.005	0.033 ± 0.005	7.081 ± 0.177
UCI	Tree	8	PRISM-XAI	325.833 ± 1.364	0.245 ± 0.047	0.250 ± 0.000	-0.007 ± 0.082	0.045 ± 0.007	0.045 ± 0.007	0.507 ± 0.071	0.090 ± 0.005	0.910 ± 0.005	0.030 ± 0.006	7.052 ± 0.147
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	656.329 ± 1.493	0.313 ± 0.040	0.250 ± 0.000	0.015 ± 0.023	0.081 ± 0.007	0.081 ± 0.007	0.445 ± 0.042	0.183 ± 0.005	0.817 ± 0.005	0.070 ± 0.007	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	648.965 ± 1.563	0.238 ± 0.050	0.250 ± 0.000	-0.027 ± 0.047	0.082 ± 0.013	0.082 ± 0.013	0.471 ± 0.071	0.182 ± 0.004	0.818 ± 0.004	0.064 ± 0.010	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	654.773 ± 1.193	0.343 ± 0.070	0.250 ± 0.000	0.019 ± 0.021	0.085 ± 0.008	0.085 ± 0.008	0.556 ± 0.045	0.181 ± 0.003	0.819 ± 0.003	0.048 ± 0.007	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	648.024 ± 1.912	0.281 ± 0.061	0.250 ± 0.000	-0.049 ± 0.057	0.092 ± 0.010	0.092 ± 0.010	0.595 ± 0.069	0.176 ± 0.005	0.824 ± 0.005	0.044 ± 0.009	31.175 ± 0.000

TABLE LXXII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.120 ± 0.568	-0.002 ± 0.037	-0.094 ± 0.036	-0.009 ± 0.005
OPSD	AoX-aware fixed-DP	-4922.902 ± 5.453	0.002 ± 0.065	-0.353 ± 0.085	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5123.274 ± 0.602	0.064 ± 0.095	-0.348 ± 0.082	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-1472.642 ± 0.520	0.006 ± 0.074	-0.094 ± 0.032	-0.009 ± 0.005
OPSD	AoX-aware fixed-DP	-4923.731 ± 5.465	-0.009 ± 0.028	-0.365 ± 0.064	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5124.257 ± 0.530	0.031 ± 0.068	-0.356 ± 0.063	-0.004 ± 0.002
OPSD	Channel-aware fixed-DP	-1472.120 ± 0.568	-0.061 ± 0.040	-0.082 ± 0.032	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4509.598 ± 13.076	-0.006 ± 0.066	-0.353 ± 0.099	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5123.274 ± 0.602	0.059 ± 0.086	-0.370 ± 0.105	0.089 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.642 ± 0.520	-0.048 ± 0.081	-0.085 ± 0.029	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4510.427 ± 13.230	-0.014 ± 0.034	-0.370 ± 0.095	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5124.257 ± 0.530	0.026 ± 0.060	-0.387 ± 0.094	0.089 ± 0.006
OPSD	Joint, no compression	-101.456 ± 0.717	-0.057 ± 0.086	0.009 ± 0.011	-0.005 ± 0.005
OPSD	Joint, no compression	-282.889 ± 1.277	0.024 ± 0.088	0.011 ± 0.020	-0.002 ± 0.002
OPSD	Joint, no compression	-963.950 ± 8.300	0.081 ± 0.110	-0.009 ± 0.026	-0.003 ± 0.002
OPSD	Joint, no compression	-101.828 ± 0.658	-0.014 ± 0.072	0.007 ± 0.012	-0.006 ± 0.005
OPSD	Joint, no compression	-283.868 ± 1.164	0.003 ± 0.051	0.011 ± 0.015	-0.002 ± 0.002
OPSD	Joint, no compression	-964.933 ± 8.317	0.001 ± 0.091	-0.011 ± 0.019	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-1472.120 ± 0.568	-0.009 ± 0.039	-0.100 ± 0.027	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-4935.583 ± 0.621	0.006 ± 0.073	-0.402 ± 0.071	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5122.886 ± 0.681	0.068 ± 0.082	-0.393 ± 0.071	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-1472.642 ± 0.520	0.002 ± 0.063	-0.108 ± 0.022	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-4936.412 ± 0.595	0.001 ± 0.039	-0.413 ± 0.054	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5123.869 ± 0.738	0.039 ± 0.068	-0.402 ± 0.053	-0.004 ± 0.002
OPSD	MAP-XAI fallback	-17.854 ± 0.558	-0.058 ± 0.062	-0.003 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.287 ± 0.637	0.025 ± 0.049	0.001 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.328 ± 0.611	0.037 ± 0.050	0.007 ± 0.010	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.235 ± 0.524	-0.023 ± 0.076	-0.000 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.954 ± 0.593	0.018 ± 0.059	0.003 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.087 ± 0.533	0.046 ± 0.041	0.011 ± 0.010	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4971.561 ± 1.360	-0.033 ± 0.085	-0.414 ± 0.028	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4643.895 ± 1.493	0.049 ± 0.050	-0.376 ± 0.029	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4645.451 ± 1.193	0.079 ± 0.059	-0.372 ± 0.030	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4974.391 ± 1.364	-0.016 ± 0.045	-0.410 ± 0.049	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4651.259 ± 1.563	-0.023 ± 0.046	-0.372 ± 0.044	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4652.200 ± 1.912	0.020 ± 0.052	-0.362 ± 0.051	-0.003 ± 0.001
UCI	Channel-aware fixed-DP	-4966.514 ± 3.919	-0.018 ± 0.079	0.042 ± 0.009	-0.000 ± 0.002
UCI	Channel-aware fixed-DP	-4643.895 ± 1.493	0.045 ± 0.059	0.080 ± 0.007	0.665 ± 0.001
UCI	Channel-aware fixed-DP	-4645.451 ± 1.193	0.074 ± 0.071	0.085 ± 0.009	0.741 ± 0.001
UCI	Channel-aware fixed-DP	-4969.344 ± 4.185	0.006 ± 0.055	0.045 ± 0.007	0.000 ± 0.002
UCI	Channel-aware fixed-DP	-4651.259 ± 1.563	-0.022 ± 0.058	0.082 ± 0.013	0.666 ± 0.001
UCI	Channel-aware fixed-DP	-4652.200 ± 1.912	0.021 ± 0.068	0.092 ± 0.011	0.741 ± 0.001
UCI	Joint, no compression	-343.637 ± 1.641	-0.055 ± 0.093	-0.009 ± 0.010	-0.003 ± 0.003
UCI	Joint, no compression	-975.821 ± 1.564	0.096 ± 0.035	-0.013 ± 0.007	-0.002 ± 0.001
UCI	Joint, no compression	-3379.777 ± 5.671	0.097 ± 0.081	-0.142 ± 0.012	-0.003 ± 0.001
UCI	Joint, no compression	-346.467 ± 1.610	0.015 ± 0.070	-0.012 ± 0.010	-0.003 ± 0.003
UCI	Joint, no compression	-983.635 ± 1.427	0.034 ± 0.053	-0.017 ± 0.024	-0.002 ± 0.001
UCI	Joint, no compression	-3386.526 ± 6.147	0.040 ± 0.071	-0.135 ± 0.029	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4970.655 ± 1.984	-0.029 ± 0.088	0.037 ± 0.010	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4642.989 ± 2.460	0.053 ± 0.059	0.075 ± 0.007	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4644.545 ± 2.331	0.083 ± 0.082	0.080 ± 0.008	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4973.485 ± 1.727	-0.016 ± 0.075	0.040 ± 0.006	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4650.353 ± 1.894	-0.022 ± 0.038	0.077 ± 0.012	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4651.295 ± 2.358	0.020 ± 0.078	0.087 ± 0.009	-0.003 ± 0.001
UCI	MAP-XAI fallback	-53.039 ± 1.350	0.029 ± 0.074	0.002 ± 0.006	0.000 ± 0.000
UCI	MAP-XAI fallback	-109.291 ± 1.508	0.053 ± 0.050	-0.000 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-110.785 ± 1.251	0.052 ± 0.090	0.001 ± 0.009	0.000 ± 0.000
UCI	MAP-XAI fallback	-55.422 ± 1.449	0.039 ± 0.039	-0.003 ± 0.007	0.000 ± 0.000
UCI	MAP-XAI fallback	-116.127 ± 1.715	0.008 ± 0.046	-0.002 ± 0.009	0.000 ± 0.000
UCI	MAP-XAI fallback	-116.946 ± 2.015	0.012 ± 0.040	0.014 ± 0.012	-0.000 ± 0.001

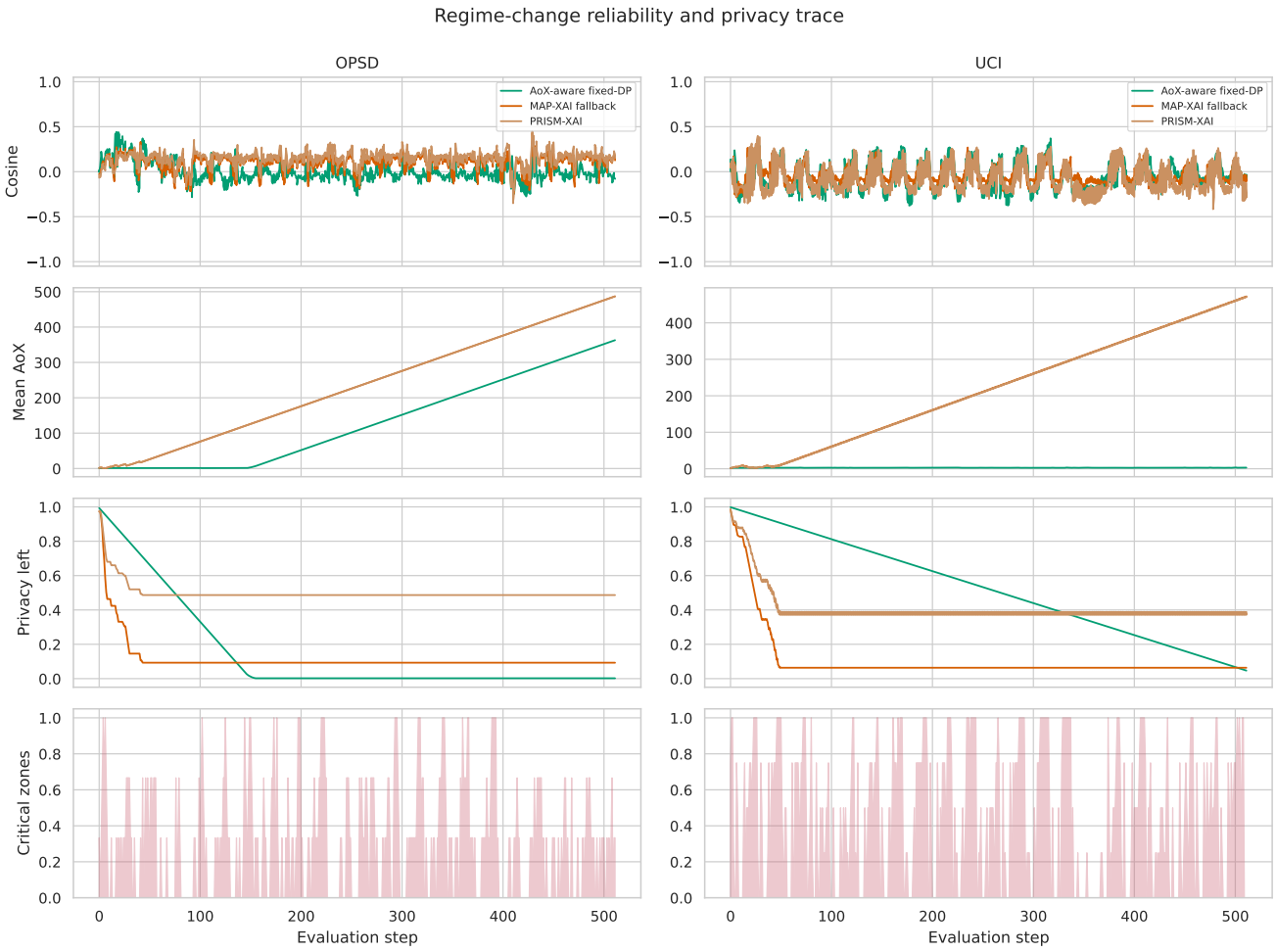


Fig. 95. Public-score threshold $\tau = 3$, release cap $M = 16$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

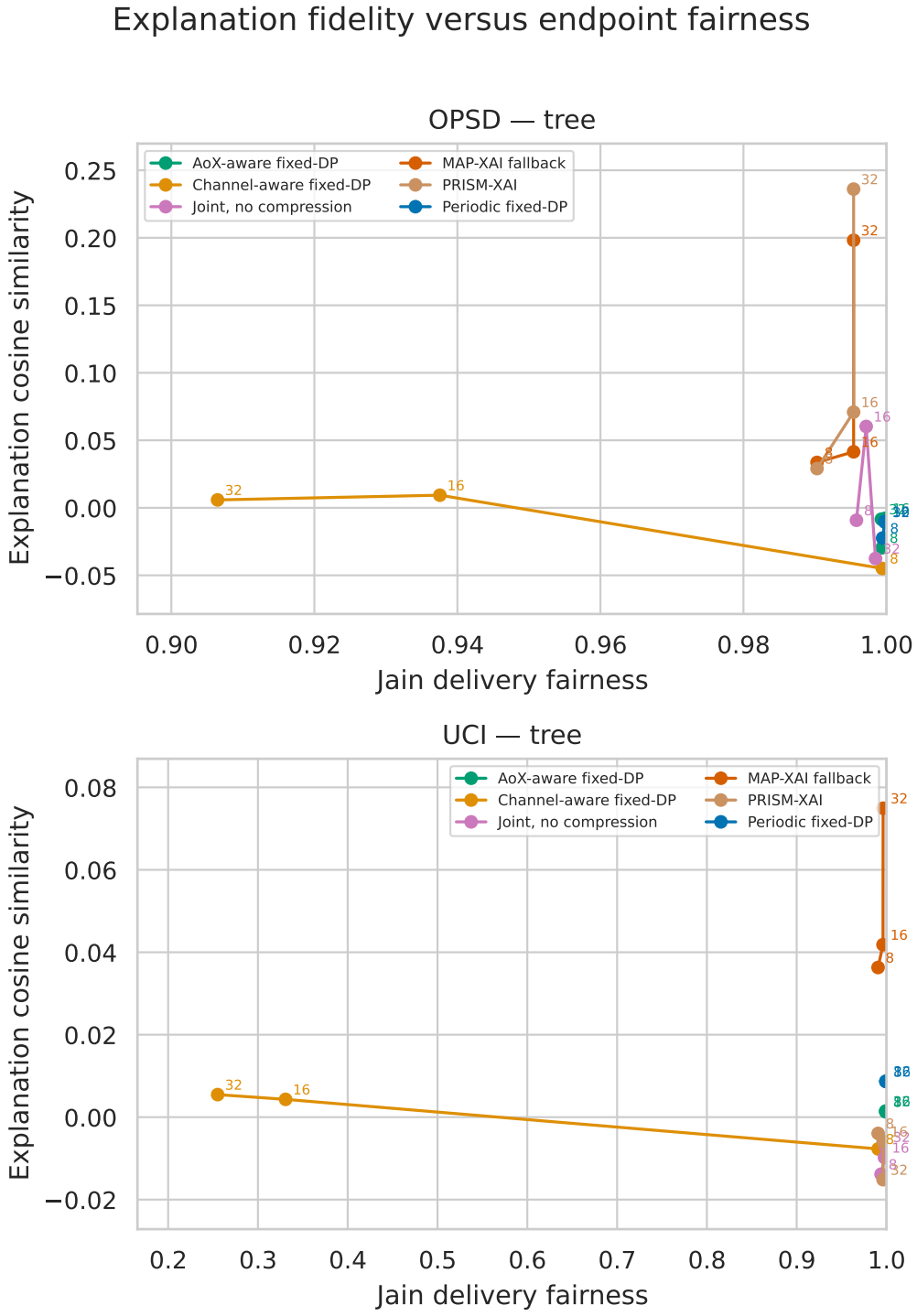


Fig. 96. Public-score threshold $\tau = 3$, release cap $M = 16$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

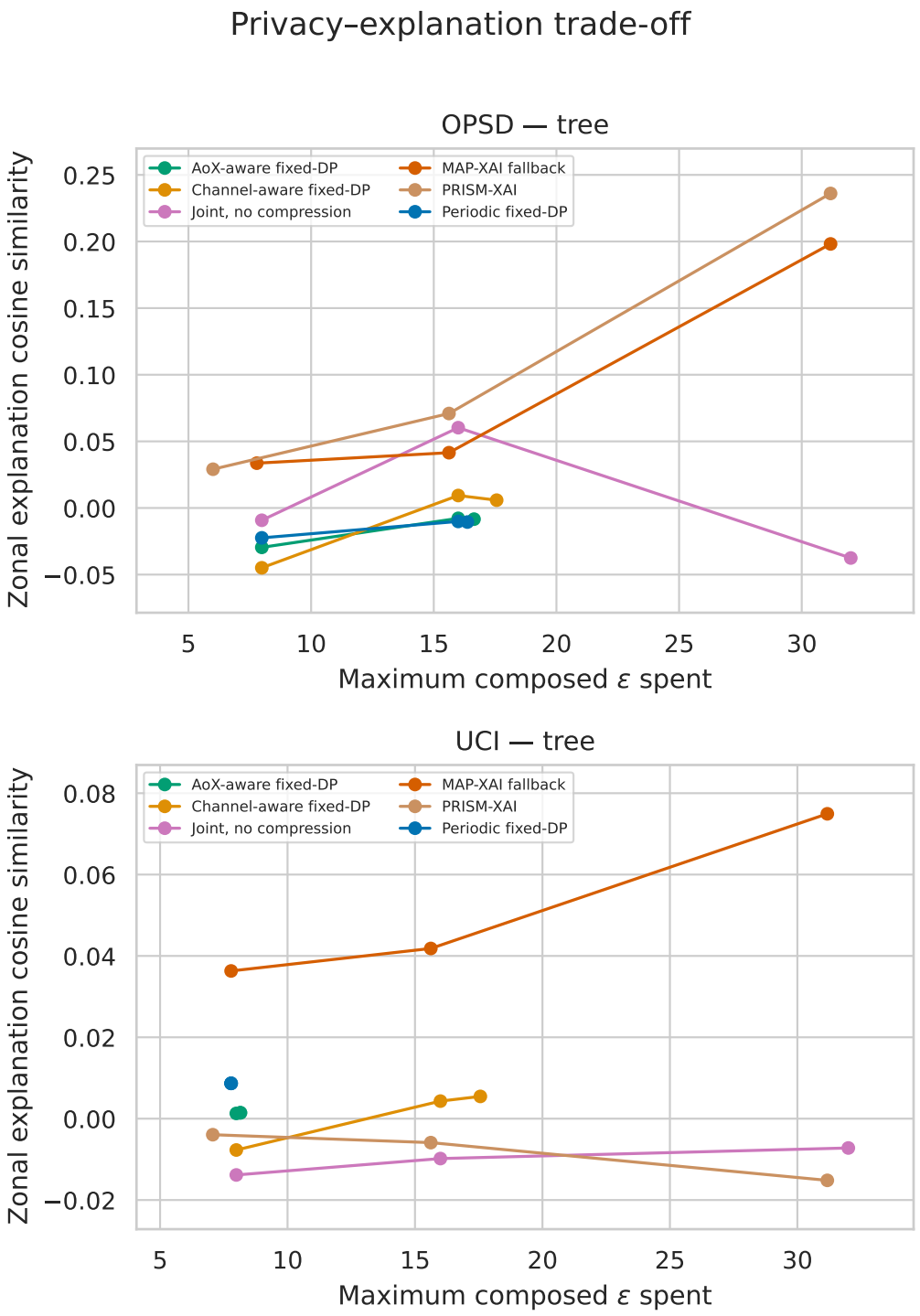


Fig. 97. Public-score threshold $\tau = 3$, release cap $M = 16$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

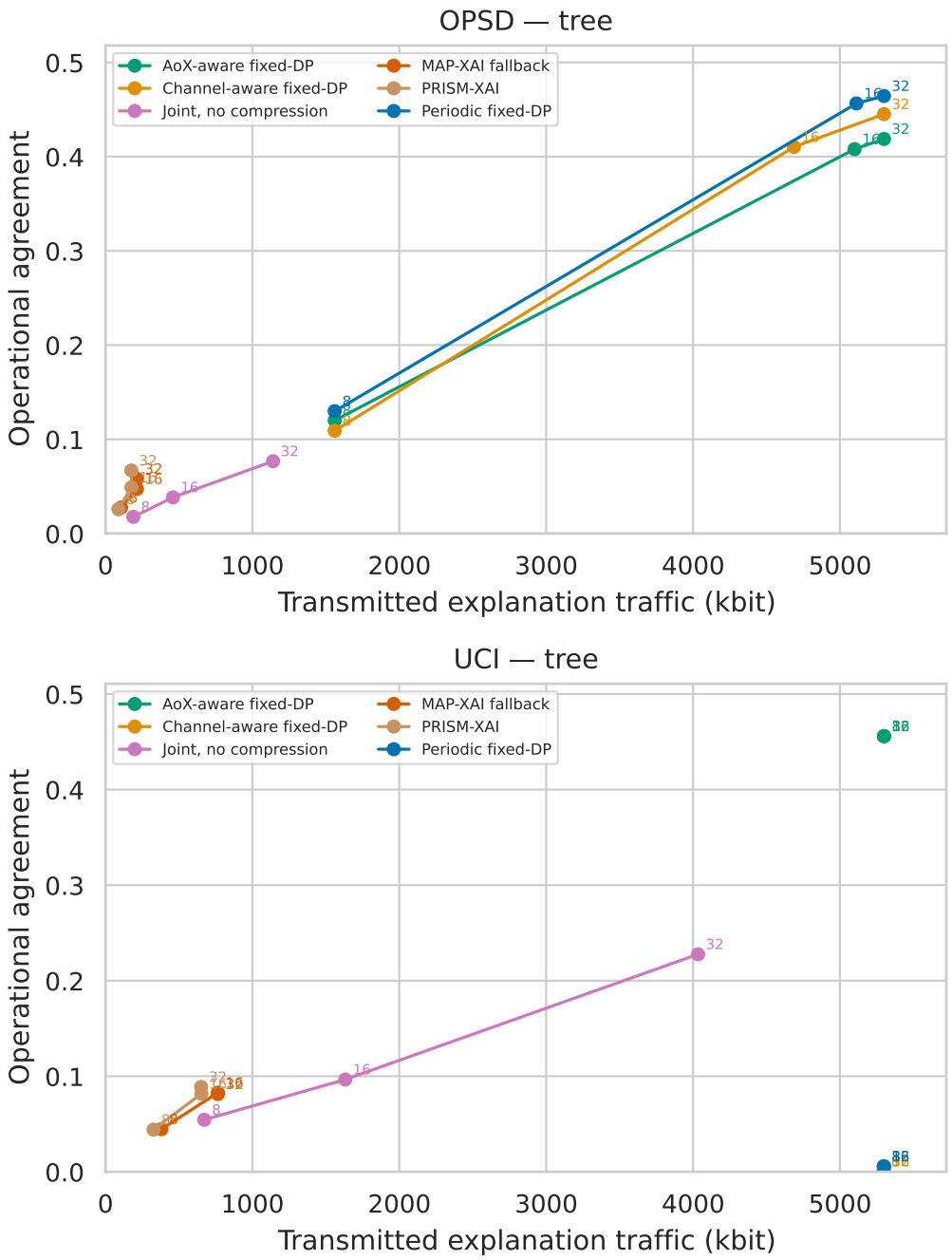


Fig. 98. Public-score threshold $\tau = 3$, release cap $M = 16$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

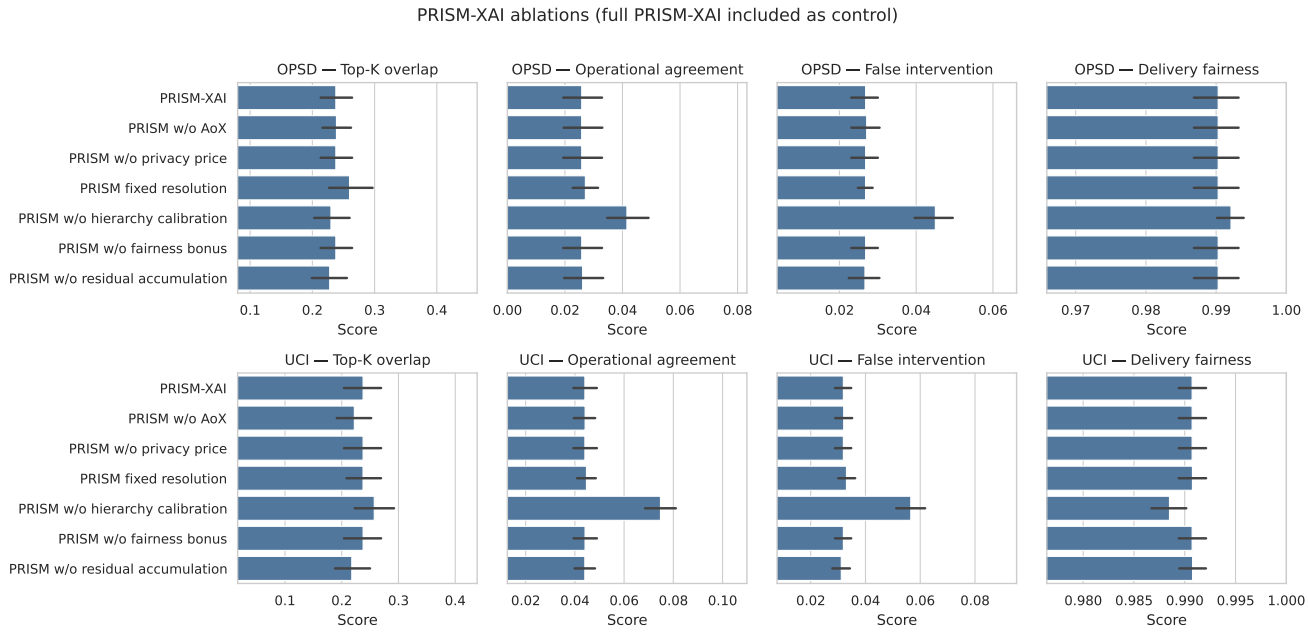


Fig. 99. Public-score threshold $\tau = 3$, release cap $M = 16$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXXIII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.231 ± 0.042	0.028 ± 0.075	0.025 ± 0.011	0.025 ± 0.011	0.268 ± 0.127	0.929 ± 0.005	4.879 ± 1.736	88.444 ± 0.568	6.067 ± 0.212	0.026 ± 0.005	12196.035 ± 5705.209	0.990 ± 0.006
OPSD	Tree	8	PRISM-XAI	0.244 ± 0.039	0.030 ± 0.085	0.026 ± 0.012	0.026 ± 0.012	0.274 ± 0.138	0.929 ± 0.004	4.869 ± 1.744	87.922 ± 0.520	5.936 ± 0.213	0.027 ± 0.007	11503.042 ± 5462.280	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.289 ± 0.065	0.023 ± 0.079	0.028 ± 0.008	0.028 ± 0.008	0.296 ± 0.082	0.929 ± 0.005	4.945 ± 1.699	106.298 ± 0.291	7.786 ± 0.000	0.025 ± 0.004	10222.108 ± 2661.883	0.990 ± 0.006
OPSD	Tree	8	MAP-XAI fallback	0.267 ± 0.070	0.045 ± 0.077	0.027 ± 0.007	0.027 ± 0.007	0.257 ± 0.051	0.929 ± 0.005	4.948 ± 1.696	106.157 ± 0.283	7.786 ± 0.000	0.028 ± 0.003	10802.437 ± 2867.714	0.990 ± 0.006

TABLE LXXIV
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	88.444 ± 0.568	0.231 ± 0.042	0.250 ± 0.000	0.028 ± 0.075	0.025 ± 0.011	0.025 ± 0.011	0.268 ± 0.127	0.071 ± 0.005	0.929 ± 0.005	0.026 ± 0.005	6.067 ± 0.212
OPSD	Tree	8	PRISM-XAI	87.922 ± 0.520	0.244 ± 0.039	0.250 ± 0.000	0.030 ± 0.085	0.026 ± 0.012	0.026 ± 0.012	0.274 ± 0.138	0.071 ± 0.004	0.929 ± 0.004	0.027 ± 0.007	5.936 ± 0.213
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	177.011 ± 0.621	0.255 ± 0.065	0.250 ± 0.000	0.062 ± 0.075	0.049 ± 0.016	0.049 ± 0.016	0.290 ± 0.110	0.133 ± 0.003	0.867 ± 0.003	0.047 ± 0.010	15.620 ± 0.000
OPSD	Tree	16	PRISM-XAI	176.182 ± 0.595	0.244 ± 0.039	0.250 ± 0.000	0.080 ± 0.081	0.049 ± 0.013	0.049 ± 0.013	0.287 ± 0.097	0.132 ± 0.004	0.868 ± 0.004	0.048 ± 0.009	15.620 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	176.950 ± 0.602	0.317 ± 0.078	0.250 ± 0.000	0.217 ± 0.056	0.065 ± 0.016	0.065 ± 0.016	0.389 ± 0.094	0.133 ± 0.003	0.867 ± 0.003	0.038 ± 0.005	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	175.967 ± 0.530	0.283 ± 0.060	0.250 ± 0.000	0.255 ± 0.060	0.069 ± 0.013	0.069 ± 0.013	0.404 ± 0.078	0.132 ± 0.004	0.868 ± 0.004	0.038 ± 0.006	31.175 ± 0.000

TABLE LXXV
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1472.120 ± 0.568	-0.002 ± 0.037	-0.094 ± 0.036	-0.009 ± 0.005
OPSD	AoX-aware fixed-DP	-4922.902 ± 5.453	0.002 ± 0.065	-0.353 ± 0.085	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5123.274 ± 0.602	0.064 ± 0.095	-0.348 ± 0.082	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-1472.642 ± 0.520	0.006 ± 0.074	-0.094 ± 0.032	-0.009 ± 0.005
OPSD	AoX-aware fixed-DP	-4923.731 ± 5.465	-0.009 ± 0.028	-0.365 ± 0.064	-0.004 ± 0.002
OPSD	AoX-aware fixed-DP	-5124.257 ± 0.530	0.031 ± 0.068	-0.356 ± 0.063	-0.004 ± 0.002
OPSD	Channel-aware fixed-DP	-1472.120 ± 0.568	-0.061 ± 0.040	-0.082 ± 0.032	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4509.598 ± 13.076	-0.006 ± 0.066	-0.353 ± 0.099	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5123.274 ± 0.602	0.059 ± 0.086	-0.370 ± 0.105	0.089 ± 0.006
OPSD	Channel-aware fixed-DP	-1472.642 ± 0.520	-0.048 ± 0.081	-0.085 ± 0.029	-0.009 ± 0.006
OPSD	Channel-aware fixed-DP	-4510.427 ± 13.230	-0.014 ± 0.034	-0.370 ± 0.095	0.058 ± 0.005
OPSD	Channel-aware fixed-DP	-5124.257 ± 0.530	0.026 ± 0.060	-0.387 ± 0.094	0.089 ± 0.006
OPSD	Joint, no compression	-101.456 ± 0.717	-0.057 ± 0.086	0.009 ± 0.011	-0.005 ± 0.005
OPSD	Joint, no compression	-282.889 ± 1.277	0.024 ± 0.088	0.011 ± 0.020	-0.002 ± 0.002
OPSD	Joint, no compression	-963.950 ± 8.300	0.081 ± 0.110	-0.009 ± 0.026	-0.003 ± 0.002
OPSD	Joint, no compression	-101.828 ± 0.658	-0.014 ± 0.072	0.007 ± 0.012	-0.006 ± 0.005
OPSD	Joint, no compression	-283.868 ± 1.164	0.003 ± 0.051	0.011 ± 0.015	-0.002 ± 0.002
OPSD	Joint, no compression	-964.933 ± 8.317	0.001 ± 0.091	-0.011 ± 0.019	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-1472.120 ± 0.568	-0.009 ± 0.039	-0.100 ± 0.027	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-4935.583 ± 0.621	0.006 ± 0.073	-0.402 ± 0.071	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5122.886 ± 0.681	0.068 ± 0.082	-0.393 ± 0.071	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-1472.642 ± 0.520	0.002 ± 0.063	-0.108 ± 0.022	-0.009 ± 0.006
OPSD	Periodic fixed-DP	-4936.412 ± 0.595	0.001 ± 0.039	-0.413 ± 0.054	-0.004 ± 0.002
OPSD	Periodic fixed-DP	-5123.869 ± 0.738	0.039 ± 0.068	-0.402 ± 0.053	-0.004 ± 0.002
OPSD	MAP-XAI fallback	-17.854 ± 0.558	-0.058 ± 0.062	-0.003 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.287 ± 0.637	0.025 ± 0.049	0.001 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.328 ± 0.611	0.037 ± 0.050	0.007 ± 0.010	0.000 ± 0.000
OPSD	MAP-XAI fallback	-18.235 ± 0.524	-0.023 ± 0.076	-0.000 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-37.954 ± 0.593	0.018 ± 0.059	0.003 ± 0.008	0.000 ± 0.000
OPSD	MAP-XAI fallback	-38.087 ± 0.533	0.046 ± 0.041	0.011 ± 0.010	0.000 ± 0.000

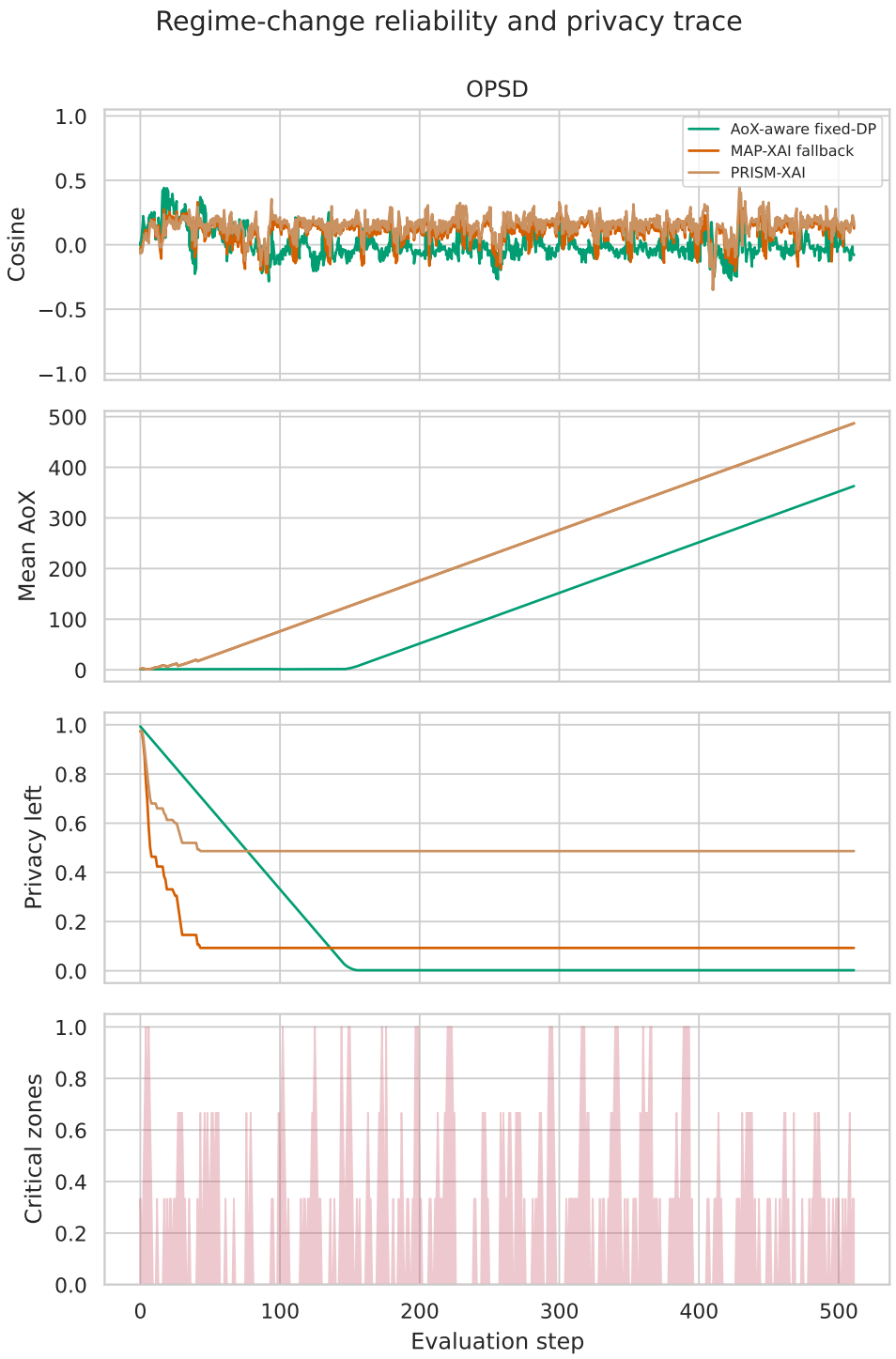


Fig. 100. Public-score threshold $\tau = 3$, release cap $M = 16$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

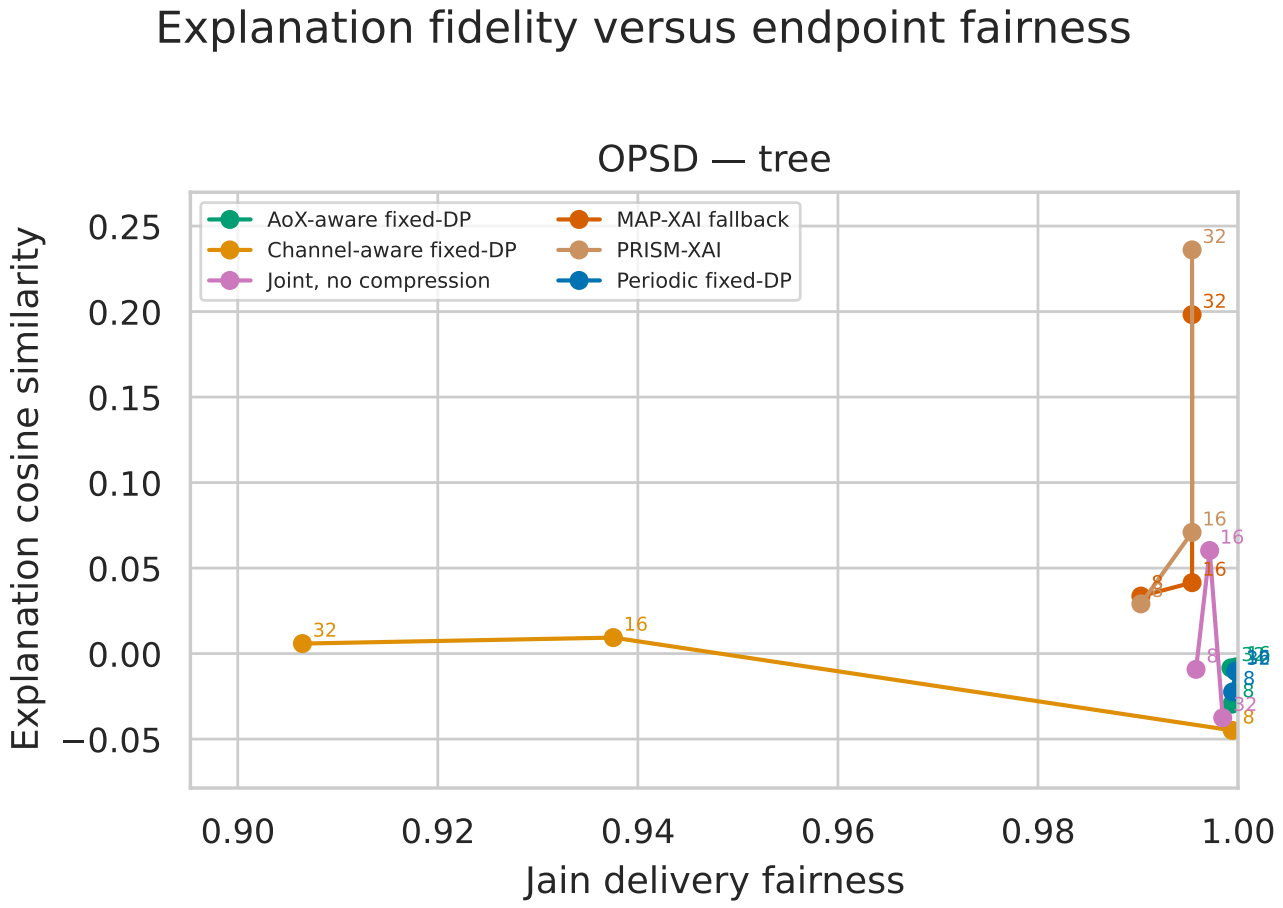


Fig. 101. Public-score threshold $\tau = 3$, release cap $M = 16$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

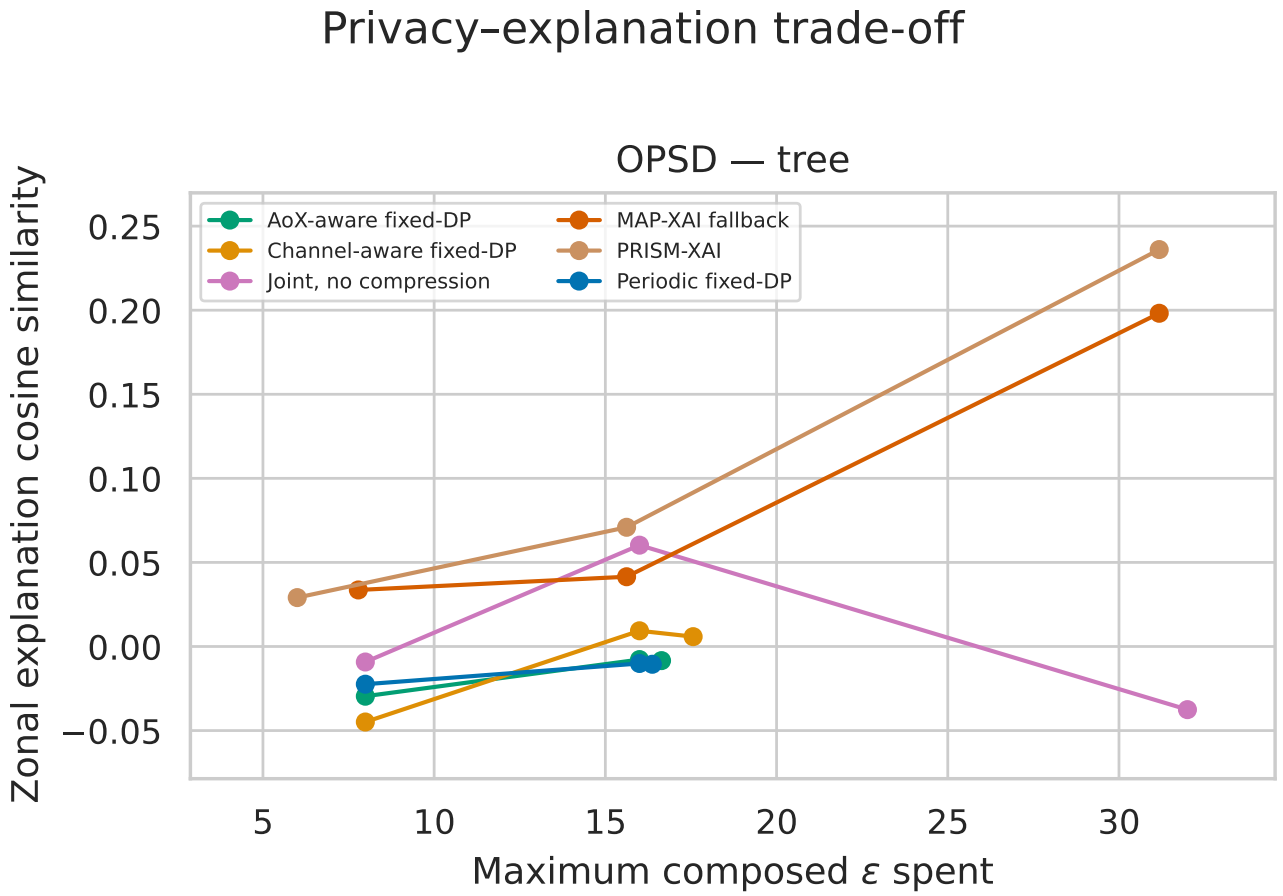


Fig. 102. Public-score threshold $\tau = 3$, release cap $M = 16$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

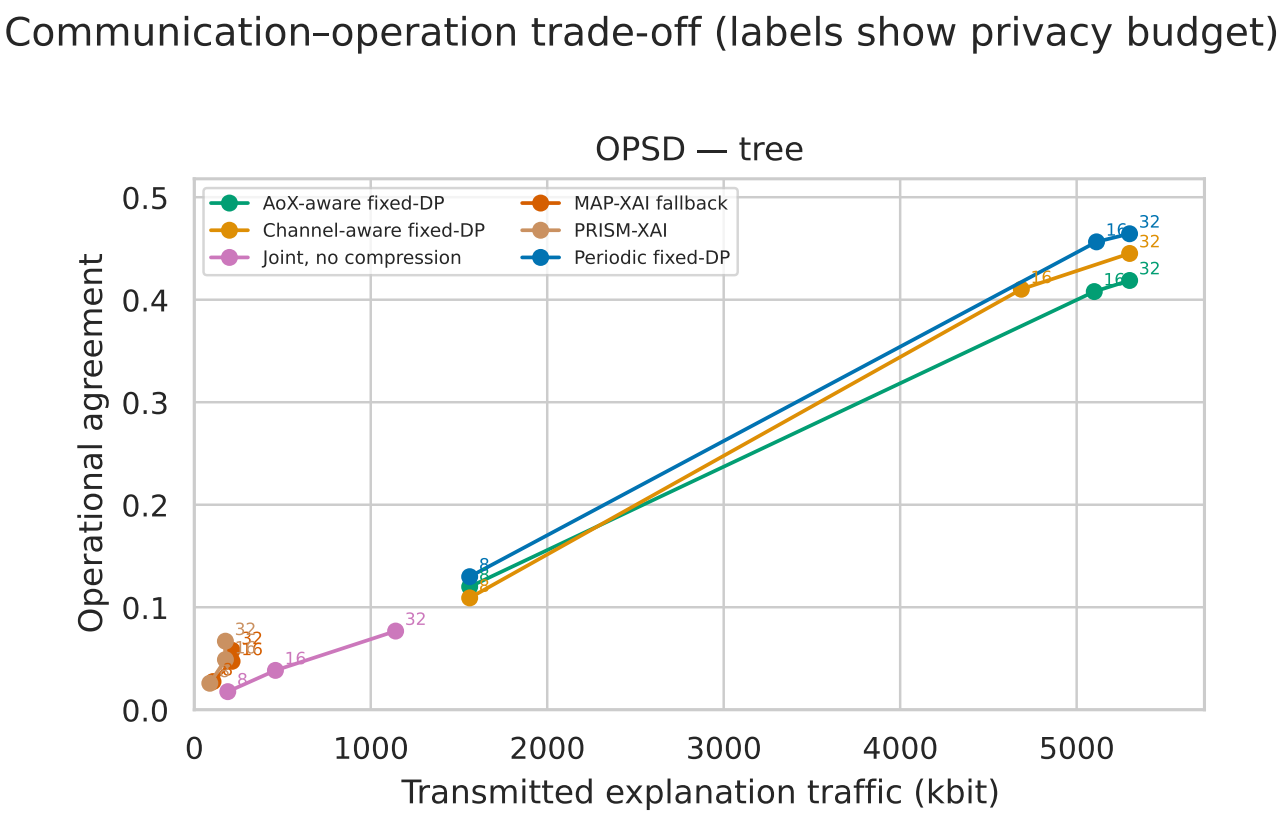


Fig. 103. Public-score threshold $\tau = 3$, release cap $M = 16$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

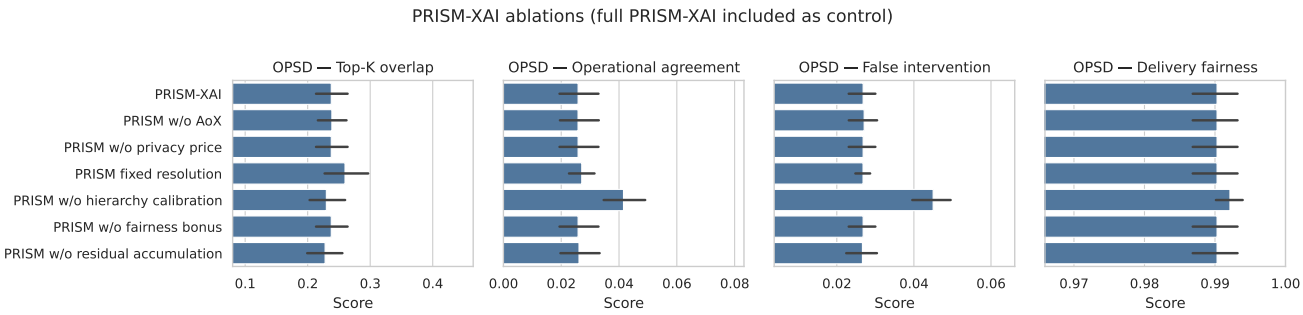


Fig. 104. Public-score threshold $\tau = 3$, release cap $M = 16$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXXVI
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.231 ± 0.061	-0.001 ± 0.013	0.043 ± 0.009	0.043 ± 0.009	0.466 ± 0.081	0.909 ± 0.005	5.656 ± 1.725	328.663 ± 1.360	7.081 ± 0.177	0.033 ± 0.005	10019.834 ± 2488.756	0.990 ± 0.003
UCI	Tree	8	PRISM-XAI	0.245 ± 0.047	-0.007 ± 0.082	0.045 ± 0.007	0.045 ± 0.007	0.507 ± 0.071	0.910 ± 0.005	5.163 ± 1.785	325.833 ± 1.364	7.052 ± 0.147	0.030 ± 0.006	9029.695 ± 1367.291	0.991 ± 0.002
UCI	Tree	8	MAP-XAI fallback	0.202 ± 0.058	0.011 ± 0.012	0.041 ± 0.005	0.041 ± 0.005	0.453 ± 0.045	0.909 ± 0.005	5.677 ± 1.704	381.702 ± 0.092	7.786 ± 0.000	0.034 ± 0.003	11391.295 ± 1487.072	0.990 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.206 ± 0.051	0.062 ± 0.058	0.048 ± 0.010	0.048 ± 0.010	0.505 ± 0.102	0.910 ± 0.005	4.861 ± 1.663	381.256 ± 0.293	7.786 ± 0.000	0.033 ± 0.007	10444.880 ± 2673.464	0.991 ± 0.002

TABLE LXXVII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	328.663 ± 1.360	0.231 ± 0.061	0.250 ± 0.000	-0.001 ± 0.013	0.043 ± 0.009	0.043 ± 0.009	0.466 ± 0.081	0.091 ± 0.005	0.909 ± 0.005	0.033 ± 0.005	7.081 ± 0.177
UCI	Tree	8	PRISM-XAI	325.833 ± 1.364	0.245 ± 0.047	0.250 ± 0.000	-0.007 ± 0.082	0.045 ± 0.007	0.045 ± 0.007	0.507 ± 0.071	0.090 ± 0.005	0.910 ± 0.005	0.030 ± 0.006	7.052 ± 0.147
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	656.329 ± 1.493	0.313 ± 0.040	0.250 ± 0.000	0.015 ± 0.023	0.081 ± 0.007	0.081 ± 0.007	0.445 ± 0.042	0.183 ± 0.005	0.817 ± 0.005	0.070 ± 0.007	15.620 ± 0.000
UCI	Tree	16	PRISM-XAI	648.965 ± 1.563	0.238 ± 0.050	0.250 ± 0.000	-0.027 ± 0.047	0.082 ± 0.013	0.082 ± 0.013	0.471 ± 0.071	0.182 ± 0.004	0.818 ± 0.004	0.064 ± 0.010	15.620 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	654.773 ± 1.193	0.343 ± 0.070	0.250 ± 0.000	0.019 ± 0.021	0.085 ± 0.008	0.085 ± 0.008	0.556 ± 0.045	0.181 ± 0.003	0.819 ± 0.003	0.048 ± 0.007	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	648.024 ± 1.912	0.281 ± 0.061	0.250 ± 0.000	-0.049 ± 0.057	0.092 ± 0.010	0.092 ± 0.010	0.595 ± 0.069	0.176 ± 0.005	0.824 ± 0.005	0.044 ± 0.009	31.175 ± 0.000

TABLE LXXVIII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 16$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4971.561 ± 1.360	-0.033 ± 0.085	-0.414 ± 0.028	-0.009 ± 0.003
UCI	AoX-aware fixed-DP	-4643.895 ± 1.493	0.049 ± 0.050	-0.376 ± 0.029	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4645.451 ± 1.193	0.079 ± 0.059	-0.372 ± 0.030	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4974.391 ± 1.364	-0.016 ± 0.045	-0.410 ± 0.049	-0.008 ± 0.002
UCI	AoX-aware fixed-DP	-4651.259 ± 1.563	-0.023 ± 0.046	-0.372 ± 0.044	-0.003 ± 0.001
UCI	AoX-aware fixed-DP	-4652.200 ± 1.912	0.020 ± 0.052	-0.362 ± 0.051	-0.003 ± 0.001
UCI	Channel-aware fixed-DP	-4966.514 ± 3.919	-0.018 ± 0.079	0.042 ± 0.009	-0.000 ± 0.002
UCI	Channel-aware fixed-DP	-4643.895 ± 1.493	0.045 ± 0.059	0.080 ± 0.007	0.665 ± 0.001
UCI	Channel-aware fixed-DP	-4645.451 ± 1.193	0.074 ± 0.071	0.085 ± 0.009	0.741 ± 0.001
UCI	Channel-aware fixed-DP	-4969.344 ± 4.185	0.006 ± 0.055	0.045 ± 0.007	0.000 ± 0.002
UCI	Channel-aware fixed-DP	-4651.259 ± 1.563	-0.022 ± 0.058	0.082 ± 0.013	0.666 ± 0.001
UCI	Channel-aware fixed-DP	-4652.200 ± 1.912	0.021 ± 0.068	0.092 ± 0.011	0.741 ± 0.001
UCI	Joint, no compression	-343.637 ± 1.641	-0.055 ± 0.093	-0.009 ± 0.010	-0.003 ± 0.003
UCI	Joint, no compression	-975.821 ± 1.564	0.096 ± 0.035	-0.013 ± 0.007	-0.002 ± 0.001
UCI	Joint, no compression	-3379.777 ± 5.671	0.097 ± 0.081	-0.142 ± 0.012	-0.003 ± 0.001
UCI	Joint, no compression	-346.467 ± 1.610	0.015 ± 0.070	-0.012 ± 0.010	-0.003 ± 0.003
UCI	Joint, no compression	-983.635 ± 1.427	0.034 ± 0.053	-0.017 ± 0.024	-0.002 ± 0.001
UCI	Joint, no compression	-3386.526 ± 6.147	0.040 ± 0.071	-0.135 ± 0.029	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4970.655 ± 1.984	-0.029 ± 0.088	0.037 ± 0.010	-0.009 ± 0.003
UCI	Periodic fixed-DP	-4642.989 ± 2.460	0.053 ± 0.059	0.075 ± 0.007	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4644.545 ± 2.331	0.083 ± 0.082	0.080 ± 0.008	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4973.485 ± 1.727	-0.016 ± 0.075	0.040 ± 0.006	-0.008 ± 0.002
UCI	Periodic fixed-DP	-4650.353 ± 1.894	-0.022 ± 0.038	0.077 ± 0.012	-0.003 ± 0.001
UCI	Periodic fixed-DP	-4651.295 ± 2.358	0.020 ± 0.078	0.087 ± 0.009	-0.003 ± 0.001
UCI	MAP-XAI fallback	-53.039 ± 1.350	0.029 ± 0.074	0.002 ± 0.006	0.000 ± 0.000
UCI	MAP-XAI fallback	-109.291 ± 1.508	0.053 ± 0.050	-0.000 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-110.785 ± 1.251	0.052 ± 0.090	0.001 ± 0.009	0.000 ± 0.000
UCI	MAP-XAI fallback	-55.422 ± 1.449	0.039 ± 0.039	-0.003 ± 0.007	0.000 ± 0.000
UCI	MAP-XAI fallback	-116.127 ± 1.715	0.008 ± 0.046	-0.002 ± 0.009	0.000 ± 0.000
UCI	MAP-XAI fallback	-116.946 ± 2.015	0.012 ± 0.040	0.014 ± 0.012	-0.000 ± 0.001

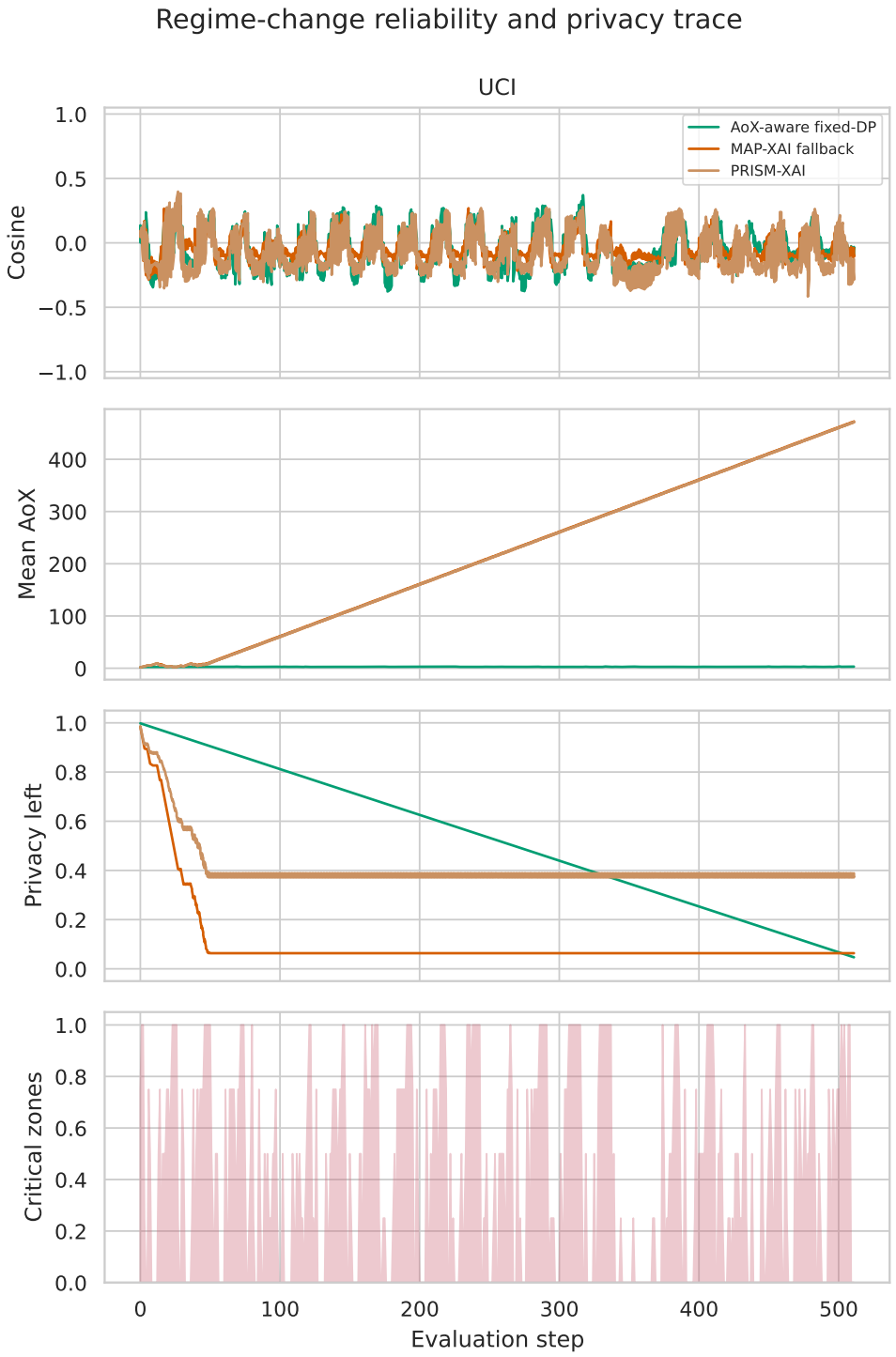


Fig. 105. Public-score threshold $\tau = 3$, release cap $M = 16$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

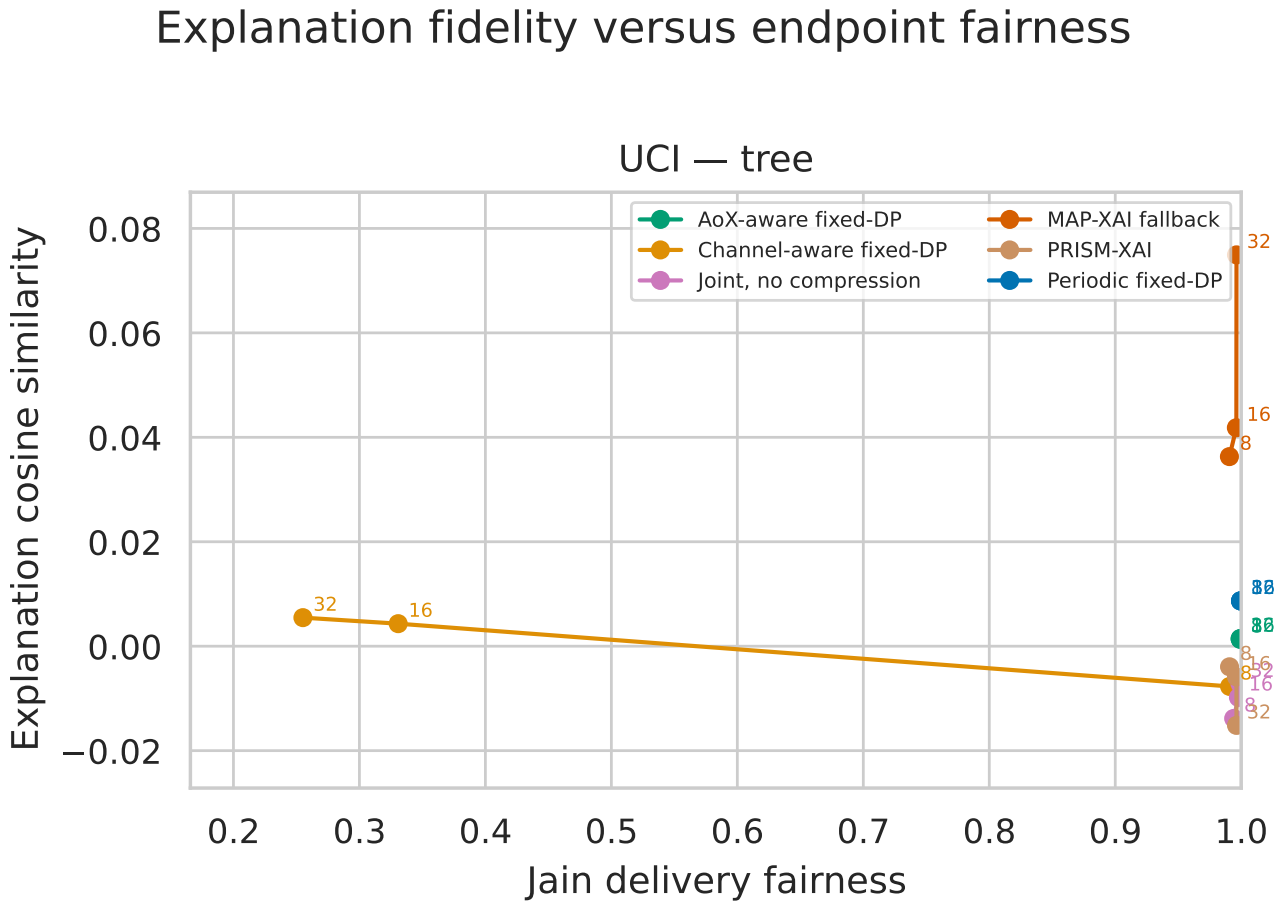


Fig. 106. Public-score threshold $\tau = 3$, release cap $M = 16$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

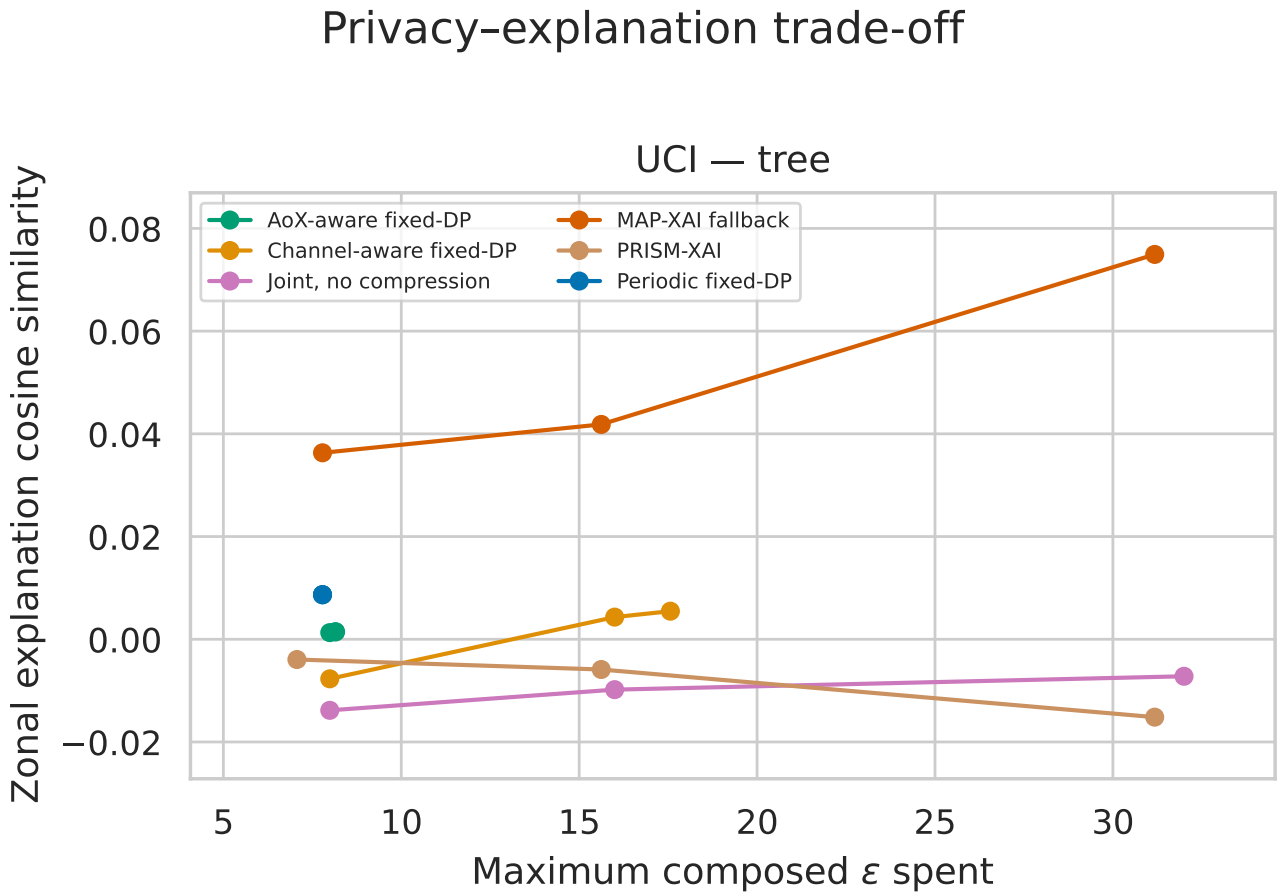


Fig. 107. Public-score threshold $\tau = 3$, release cap $M = 16$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

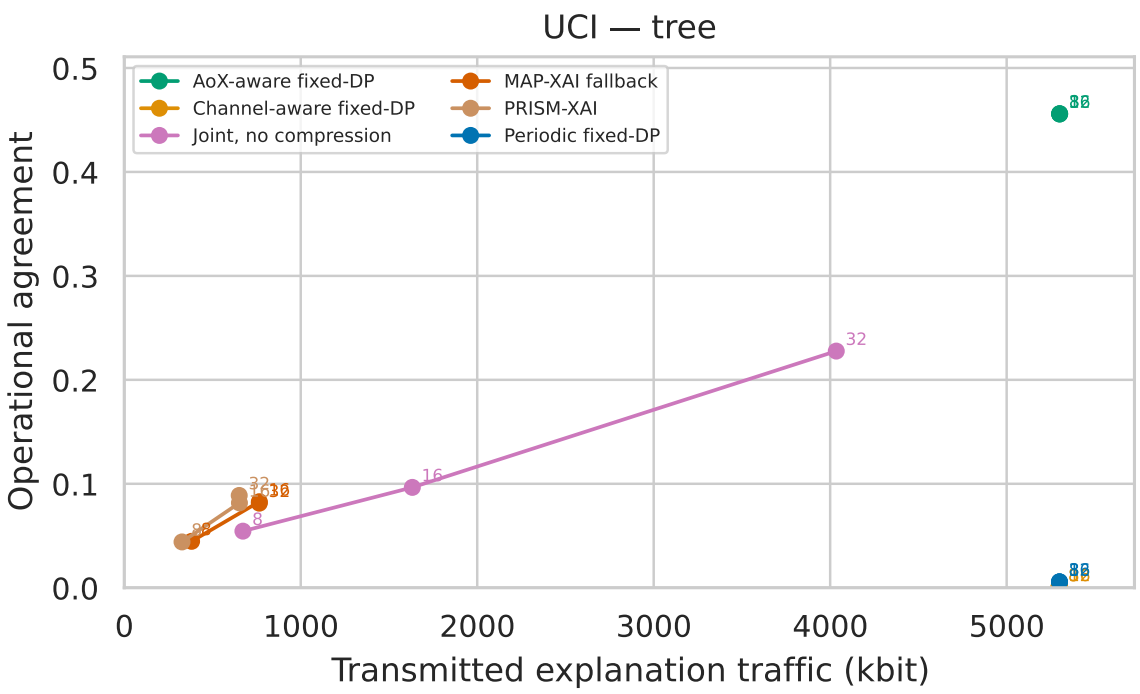


Fig. 108. Public-score threshold $\tau = 3$, release cap $M = 16$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

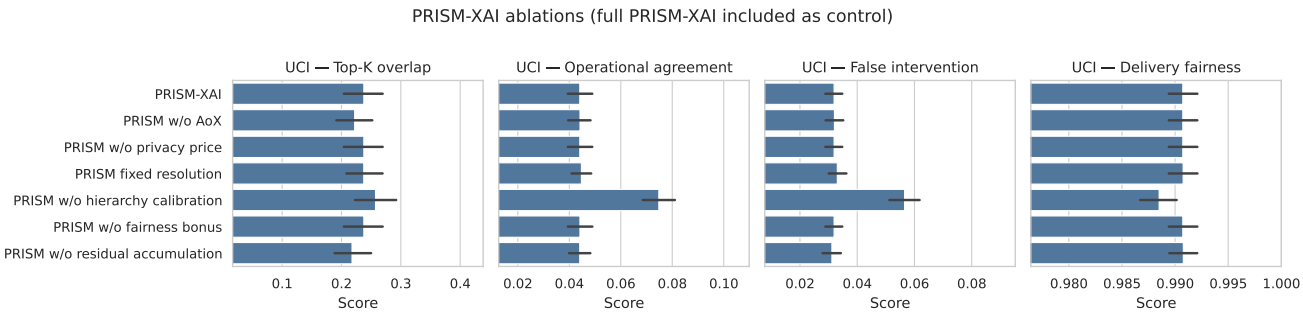


Fig. 109. Public-score threshold $\tau = 3$, release cap $M = 16$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

8) Public-score threshold $\tau = 1$, release cap $M = 32$: At this operating point, PRISM-XAI transmits 574.63 kbit versus 688.64 kbit for MAP-XAI-only (aggregate-mean ratio saving 16.56%). The paired difference is -114.01$\pm$15.13 kbit. PRISM coverage is 0.1402, critical-event recall is 0.0638, and maximum recorded expenditure is 17.196. Relative to MAP-XAI-only, the paired coverage change is -0.00060$\pm$0.00111 and recall change is +0.00153$\pm$0.00178. The descriptive overlap rule holds in 4 of 12 cells. Accordingly, this point is read as a sensitivity point on the service frontier, not a complete matched-service confirmation.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXXIX
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.261 ± 0.053	-0.019 ± 0.080	0.021 ± 0.005	0.021 ± 0.005	0.246 ± 0.094	0.947 ± 0.003	4.452 ± 1.878	97.640 ± 0.539	5.377 ± 0.416	0.027 ± 0.007	13250.048 ± 5126.243	0.994 ± 0.003
OPSD	Tree	8	PRISM-XAI	0.284 ± 0.064	-0.006 ± 0.076	0.022 ± 0.005	0.022 ± 0.005	0.253 ± 0.080	0.947 ± 0.003	4.452 ± 1.878	97.251 ± 0.416	5.108 ± 0.367	0.026 ± 0.006	12180.416 ± 3594.328	0.994 ± 0.003
OPSD	Tree	8	MAP-XAI fallback	0.249 ± 0.062	-0.025 ± 0.053	0.019 ± 0.007	0.019 ± 0.007	0.212 ± 0.098	0.947 ± 0.003	4.568 ± 1.808	117.817 ± 0.049	7.924 ± 0.000	0.030 ± 0.007	25854.893 ± 23642.714	0.994 ± 0.003
OPSD	Tree	8	MAP-XAI fallback	0.233 ± 0.051	0.005 ± 0.068	0.022 ± 0.007	0.022 ± 0.007	0.257 ± 0.098	0.947 ± 0.003	4.568 ± 1.808	117.675 ± 0.049	7.924 ± 0.000	0.026 ± 0.007	18081.688 ± 11158.136	0.994 ± 0.003
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.270 ± 0.078	-0.004 ± 0.021	0.029 ± 0.006	0.029 ± 0.006	0.346 ± 0.060	0.928 ± 0.003	5.789 ± 1.601	333.552 ± 0.722	6.549 ± 0.228	0.037 ± 0.005	15343.847 ± 4327.572	0.989 ± 0.003
UCI	Tree	8	PRISM-XAI	0.262 ± 0.032	-0.011 ± 0.061	0.031 ± 0.004	0.031 ± 0.004	0.416 ± 0.078	0.931 ± 0.004	4.104 ± 1.678	331.348 ± 1.358	6.250 ± 0.232	0.031 ± 0.006	13176.006 ± 1853.692	0.988 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.256 ± 0.104	-0.001 ± 0.023	0.030 ± 0.005	0.030 ± 0.005	0.390 ± 0.058	0.929 ± 0.004	5.428 ± 1.682	394.757 ± 0.724	7.924 ± 0.000	0.033 ± 0.004	17064.051 ± 5665.496	0.990 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.215 ± 0.046	0.034 ± 0.038	0.031 ± 0.003	0.031 ± 0.003	0.437 ± 0.060	0.932 ± 0.004	5.132 ± 1.788	393.884 ± 0.595	7.924 ± 0.000	0.029 ± 0.006	15461.186 ± 1718.511	0.988 ± 0.003

TABLE LXXX

PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	97.640 ± 0.539	0.261 ± 0.053	0.250 ± 0.000	-0.019 ± 0.080	0.021 ± 0.005	0.021 ± 0.005	0.246 ± 0.094	0.053 ± 0.003	0.947 ± 0.003	0.027 ± 0.007	5.377 ± 0.416
OPSD	Tree	8	PRISM-XAI	97.251 ± 0.416	0.284 ± 0.064	0.250 ± 0.000	-0.006 ± 0.076	0.022 ± 0.005	0.022 ± 0.005	0.253 ± 0.080	0.053 ± 0.003	0.947 ± 0.003	0.026 ± 0.006	5.108 ± 0.367
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	296.336 ± 0.924	0.239 ± 0.051	0.250 ± 0.000	0.018 ± 0.080	0.059 ± 0.016	0.059 ± 0.016	0.259 ± 0.080	0.134 ± 0.004	0.866 ± 0.004	0.066 ± 0.012	14.393 ± 0.046
OPSD	Tree	16	PRISM-XAI	295.240 ± 0.855	0.226 ± 0.027	0.250 ± 0.000	0.001 ± 0.087	0.061 ± 0.015	0.061 ± 0.015	0.274 ± 0.080	0.134 ± 0.004	0.866 ± 0.004	0.065 ± 0.014	14.360 ± 0.035
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	350.911 ± 0.837	0.222 ± 0.050	0.250 ± 0.000	0.012 ± 0.068	0.062 ± 0.017	0.062 ± 0.017	0.228 ± 0.070	0.140 ± 0.005	0.860 ± 0.005	0.082 ± 0.015	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	349.693 ± 0.728	0.243 ± 0.064	0.250 ± 0.000	0.019 ± 0.085	0.060 ± 0.016	0.060 ± 0.016	0.233 ± 0.084	0.140 ± 0.005	0.860 ± 0.005	0.081 ± 0.019	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	333.552 ± 0.722	0.270 ± 0.078	0.250 ± 0.000	-0.004 ± 0.021	0.029 ± 0.006	0.029 ± 0.006	0.346 ± 0.060	0.072 ± 0.003	0.928 ± 0.003	0.037 ± 0.005	6.549 ± 0.228
UCI	Tree	8	PRISM-XAI	331.348 ± 1.358	0.262 ± 0.032	0.250 ± 0.000	-0.011 ± 0.061	0.031 ± 0.004	0.031 ± 0.004	0.416 ± 0.078	0.069 ± 0.004	0.931 ± 0.004	0.031 ± 0.006	6.250 ± 0.232
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	1088.410 ± 2.255	0.252 ± 0.062	0.250 ± 0.000	0.007 ± 0.013	0.106 ± 0.014	0.106 ± 0.014	0.397 ± 0.042	0.218 ± 0.007	0.782 ± 0.007	0.110 ± 0.007	14.811 ± 0.029
UCI	Tree	16	PRISM-XAI	1081.416 ± 2.529	0.236 ± 0.022	0.250 ± 0.000	0.006 ± 0.081	0.098 ± 0.011	0.098 ± 0.011	0.394 ± 0.027	0.208 ± 0.013	0.792 ± 0.013	0.104 ± 0.012	14.805 ± 0.044
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	1289.912 ± 2.162	0.259 ± 0.042	0.250 ± 0.000	0.016 ± 0.015	0.118 ± 0.011	0.118 ± 0.011	0.389 ± 0.026	0.247 ± 0.009	0.753 ± 0.009	0.128 ± 0.011	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	1283.870 ± 3.710	0.235 ± 0.035	0.250 ± 0.000	0.020 ± 0.034	0.100 ± 0.015	0.100 ± 0.015	0.395 ± 0.041	0.216 ± 0.019	0.784 ± 0.019	0.105 ± 0.010	31.175 ± 0.000

TABLE LXXXI
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1462.924 ± 0.539	0.027 ± 0.083	-0.099 ± 0.029	-0.006 ± 0.003
OPSD	AoX-aware fixed-DP	-4803.577 ± 5.551	-0.015 ± 0.045	-0.343 ± 0.075	-0.003 ± 0.002
OPSD	AoX-aware fixed-DP	-4949.313 ± 0.837	-0.030 ± 0.066	-0.350 ± 0.078	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-1463.313 ± 0.416	0.046 ± 0.054	-0.099 ± 0.026	-0.006 ± 0.003
OPSD	AoX-aware fixed-DP	-4804.672 ± 5.519	-0.026 ± 0.034	-0.353 ± 0.061	-0.003 ± 0.002
OPSD	AoX-aware fixed-DP	-4950.531 ± 0.728	-0.009 ± 0.071	-0.365 ± 0.060	-0.002 ± 0.001
OPSD	Channel-aware fixed-DP	-1462.924 ± 0.539	-0.032 ± 0.060	-0.086 ± 0.026	-0.006 ± 0.003
OPSD	Channel-aware fixed-DP	-4390.274 ± 13.308	-0.023 ± 0.044	-0.343 ± 0.094	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4949.313 ± 0.837	-0.035 ± 0.059	-0.372 ± 0.099	0.091 ± 0.005
OPSD	Channel-aware fixed-DP	-1463.313 ± 0.416	-0.008 ± 0.077	-0.090 ± 0.024	-0.006 ± 0.003
OPSD	Channel-aware fixed-DP	-4391.369 ± 13.318	-0.032 ± 0.032	-0.358 ± 0.092	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4950.531 ± 0.728	-0.014 ± 0.068	-0.396 ± 0.091	0.091 ± 0.005
OPSD	Joint, no compression	-92.260 ± 0.532	-0.027 ± 0.058	0.004 ± 0.006	-0.002 ± 0.002
OPSD	Joint, no compression	-163.565 ± 1.527	0.007 ± 0.064	0.021 ± 0.012	0.000 ± 0.001
OPSD	Joint, no compression	-789.989 ± 8.405	-0.013 ± 0.095	-0.012 ± 0.017	-0.001 ± 0.001
OPSD	Joint, no compression	-92.499 ± 0.556	0.026 ± 0.104	0.003 ± 0.003	-0.002 ± 0.002
OPSD	Joint, no compression	-164.810 ± 1.425	-0.014 ± 0.044	0.022 ± 0.014	0.000 ± 0.001
OPSD	Joint, no compression	-791.207 ± 8.350	-0.039 ± 0.094	-0.020 ± 0.015	-0.001 ± 0.001
OPSD	Periodic fixed-DP	-1462.924 ± 0.539	0.020 ± 0.076	-0.104 ± 0.021	-0.006 ± 0.003
OPSD	Periodic fixed-DP	-4816.259 ± 0.924	-0.011 ± 0.051	-0.392 ± 0.064	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-4948.925 ± 0.865	-0.027 ± 0.061	-0.396 ± 0.066	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-1463.313 ± 0.416	0.043 ± 0.057	-0.113 ± 0.016	-0.006 ± 0.003
OPSD	Periodic fixed-DP	-4817.354 ± 0.855	-0.017 ± 0.029	-0.402 ± 0.055	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-4950.143 ± 0.776	-0.000 ± 0.065	-0.411 ± 0.052	-0.002 ± 0.001
OPSD	MAP-XAI fallback	-20.177 ± 0.549	0.012 ± 0.081	0.001 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-65.981 ± 0.926	0.031 ± 0.045	0.002 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-78.906 ± 0.839	0.007 ± 0.030	0.002 ± 0.009	0.000 ± 0.000
OPSD	MAP-XAI fallback	-20.424 ± 0.409	0.051 ± 0.064	-0.000 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-66.935 ± 0.867	0.017 ± 0.028	0.005 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-79.982 ± 0.736	0.019 ± 0.017	0.001 ± 0.009	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4966.672 ± 0.722	0.007 ± 0.099	-0.428 ± 0.034	-0.010 ± 0.003
UCI	AoX-aware fixed-DP	-4211.814 ± 2.255	-0.012 ± 0.049	-0.351 ± 0.042	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-4010.312 ± 2.162	-0.004 ± 0.066	-0.339 ± 0.039	-0.001 ± 0.000
UCI	AoX-aware fixed-DP	-4968.876 ± 1.358	0.002 ± 0.043	-0.424 ± 0.048	-0.011 ± 0.003
UCI	AoX-aware fixed-DP	-4218.808 ± 2.529	-0.024 ± 0.028	-0.357 ± 0.049	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-4016.354 ± 3.710	-0.025 ± 0.048	-0.354 ± 0.051	-0.001 ± 0.001
UCI	Channel-aware fixed-DP	-4961.625 ± 3.959	0.022 ± 0.103	0.028 ± 0.006	-0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4211.814 ± 2.255	-0.016 ± 0.045	0.105 ± 0.014	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-4010.312 ± 2.162	-0.009 ± 0.044	0.117 ± 0.011	0.743 ± 0.001
UCI	Channel-aware fixed-DP	-4963.829 ± 4.621	0.023 ± 0.025	0.031 ± 0.004	-0.003 ± 0.003
UCI	Channel-aware fixed-DP	-4218.808 ± 2.529	-0.024 ± 0.023	0.097 ± 0.011	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-4016.354 ± 3.710	-0.024 ± 0.030	0.100 ± 0.015	0.743 ± 0.001
UCI	Joint, no compression	-338.748 ± 0.953	-0.016 ± 0.089	-0.024 ± 0.007	-0.004 ± 0.004
UCI	Joint, no compression	-543.740 ± 2.301	0.035 ± 0.083	0.012 ± 0.013	-0.000 ± 0.000
UCI	Joint, no compression	-2744.638 ± 6.413	0.013 ± 0.055	-0.110 ± 0.017	-0.001 ± 0.001
UCI	Joint, no compression	-340.952 ± 1.422	0.033 ± 0.058	-0.026 ± 0.008	-0.006 ± 0.004
UCI	Joint, no compression	-551.184 ± 2.335	0.032 ± 0.017	-0.001 ± 0.022	-0.000 ± 0.000
UCI	Joint, no compression	-2750.680 ± 6.897	-0.006 ± 0.045	-0.127 ± 0.025	-0.001 ± 0.001
UCI	Periodic fixed-DP	-4965.766 ± 1.640	0.010 ± 0.116	0.023 ± 0.006	-0.010 ± 0.003
UCI	Periodic fixed-DP	-4210.908 ± 2.896	-0.008 ± 0.088	0.100 ± 0.014	-0.002 ± 0.000
UCI	Periodic fixed-DP	-4009.406 ± 2.846	-0.000 ± 0.080	0.112 ± 0.011	-0.002 ± 0.000
UCI	Periodic fixed-DP	-4967.970 ± 2.045	0.002 ± 0.046	0.026 ± 0.005	-0.012 ± 0.003
UCI	Periodic fixed-DP	-4217.902 ± 2.998	-0.024 ± 0.056	0.092 ± 0.011	-0.002 ± 0.001
UCI	Periodic fixed-DP	-4015.448 ± 4.177	-0.025 ± 0.069	0.095 ± 0.015	-0.002 ± 0.001
UCI	MAP-XAI fallback	-61.204 ± 1.019	0.014 ± 0.094	-0.002 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-205.689 ± 2.231	-0.053 ± 0.065	0.006 ± 0.009	-0.000 ± 0.000
UCI	MAP-XAI fallback	-244.248 ± 2.169	0.031 ± 0.072	0.001 ± 0.004	0.000 ± 0.000
UCI	MAP-XAI fallback	-62.535 ± 1.585	0.047 ± 0.049	-0.000 ± 0.003	-0.000 ± 0.002
UCI	MAP-XAI fallback	-212.229 ± 2.556	-0.010 ± 0.038	0.000 ± 0.008	-0.000 ± 0.000
UCI	MAP-XAI fallback	-249.775 ± 3.761	-0.032 ± 0.052	0.001 ± 0.013	0.000 ± 0.000

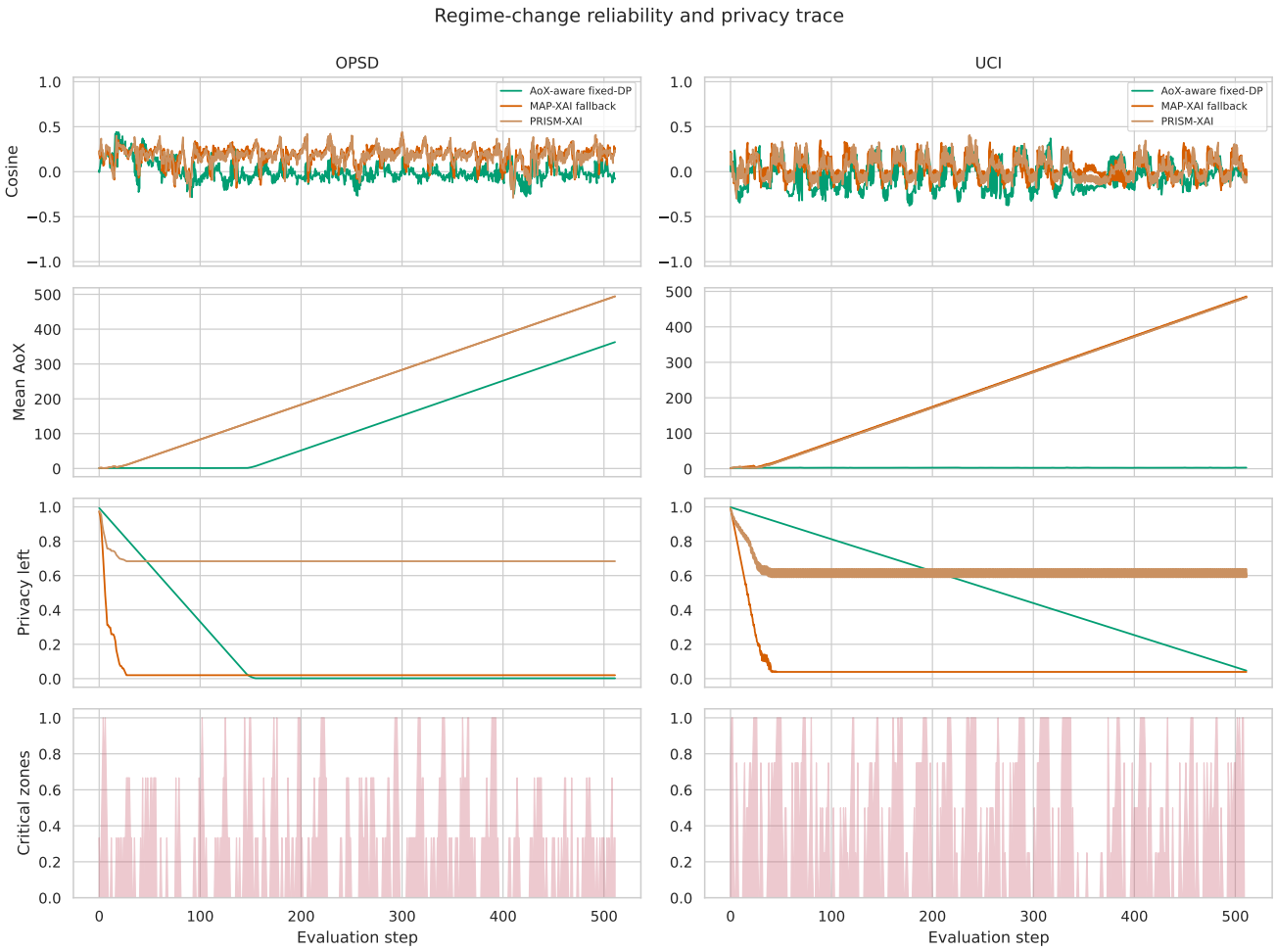


Fig. 110. Public-score threshold $\tau = 1$, release cap $M = 32$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

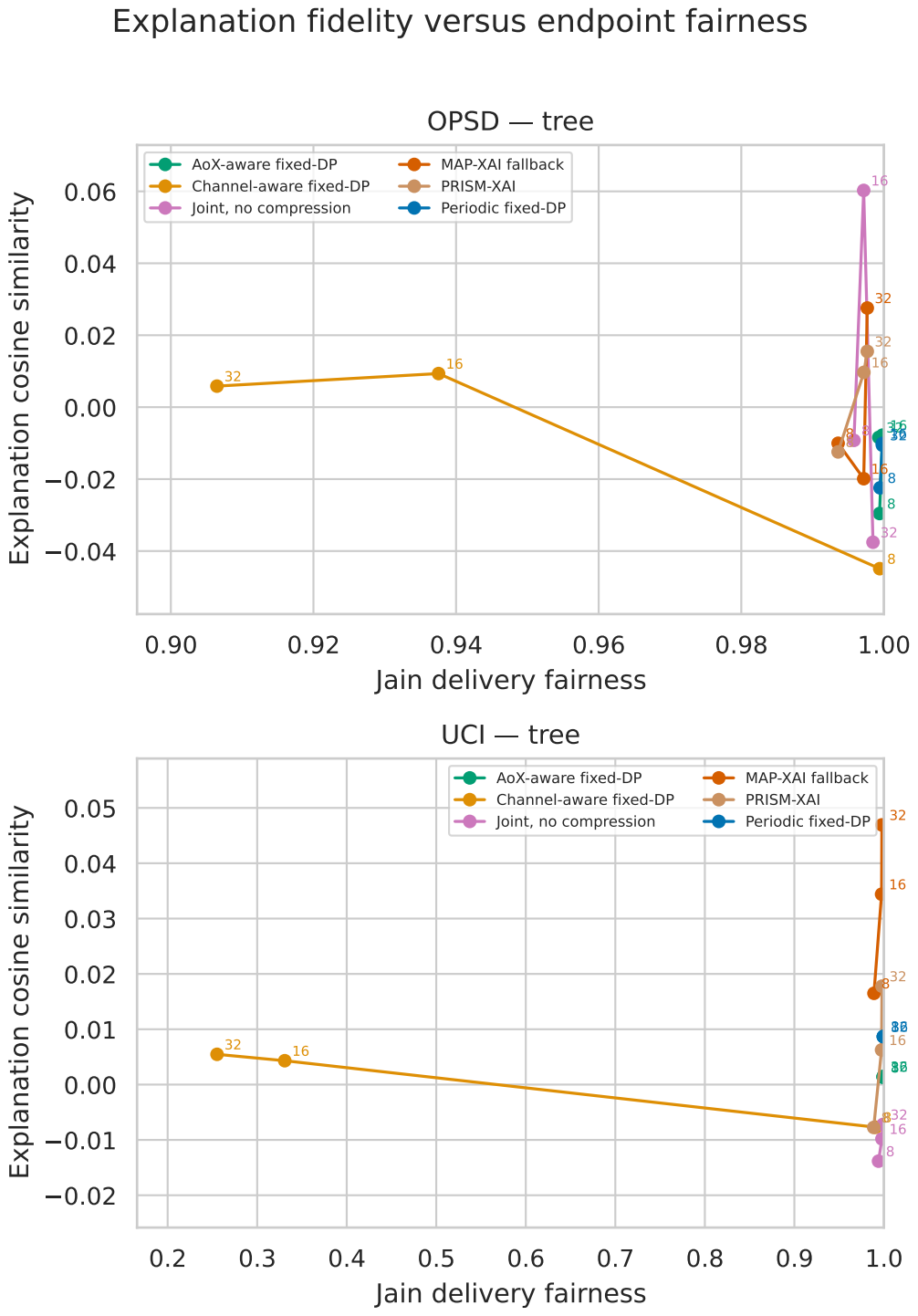


Fig. 111. Public-score threshold $\tau = 1$, release cap $M = 32$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

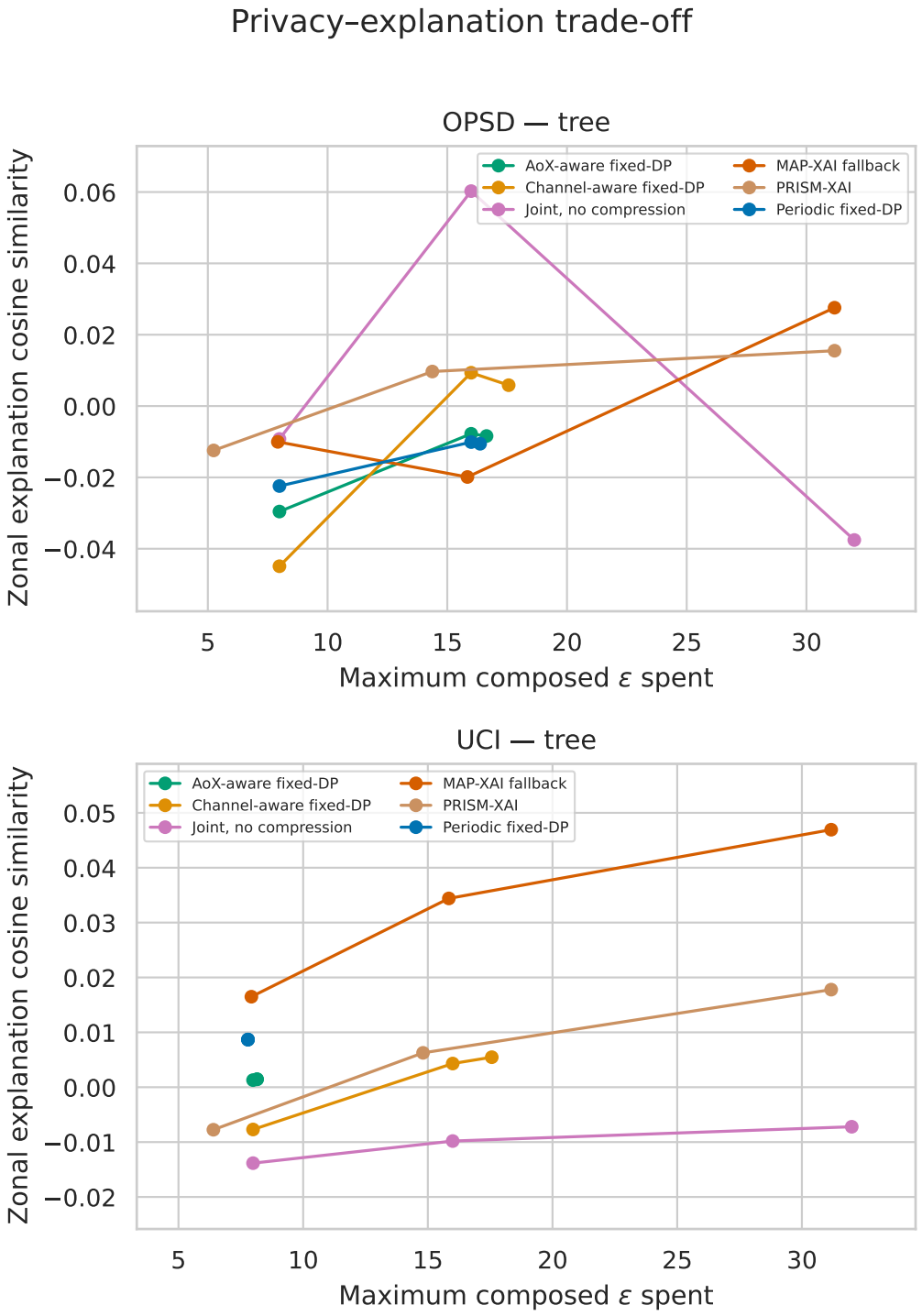


Fig. 112. Public-score threshold $\tau = 1$, release cap $M = 32$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

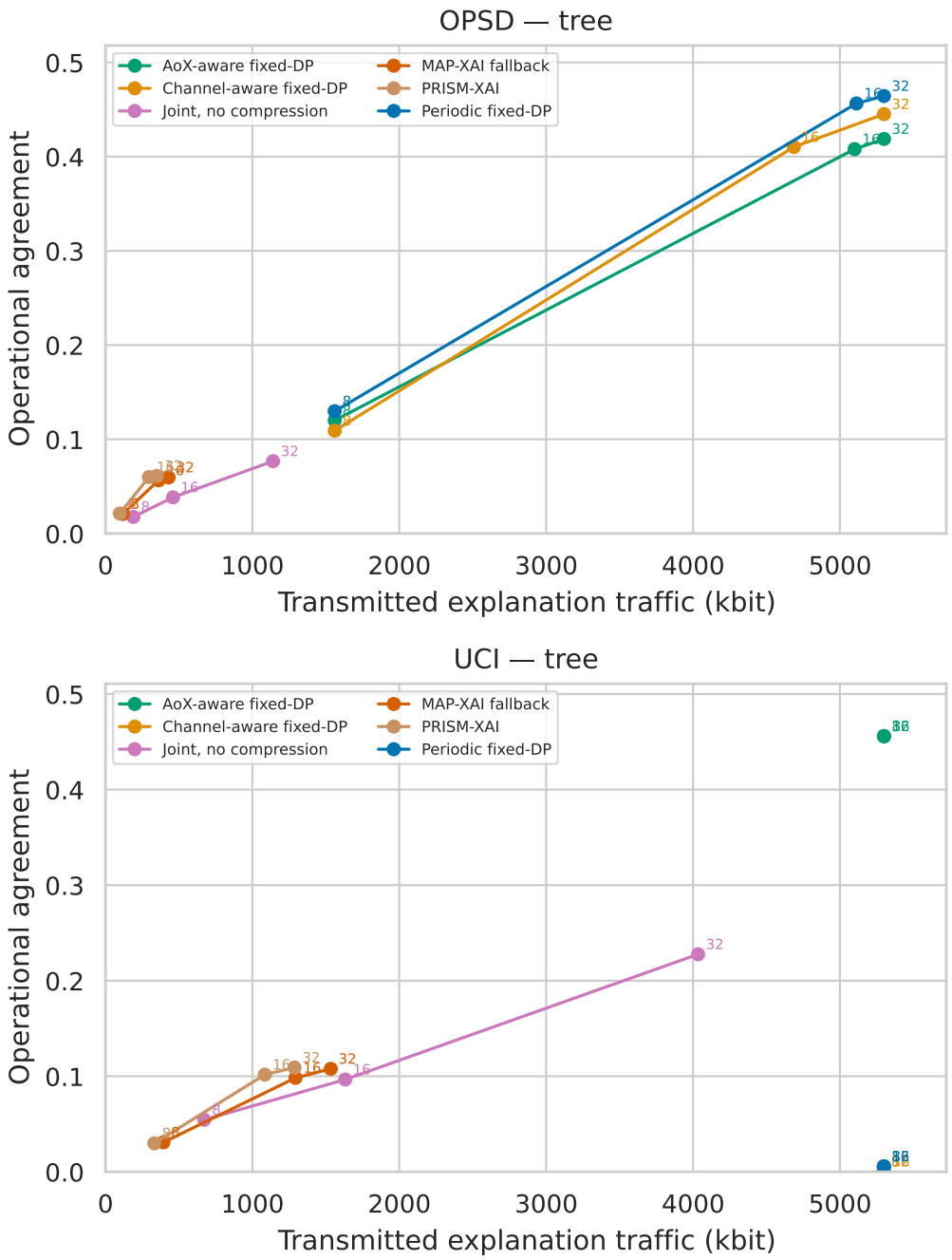


Fig. 113. Public-score threshold $\tau = 1$, release cap $M = 32$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

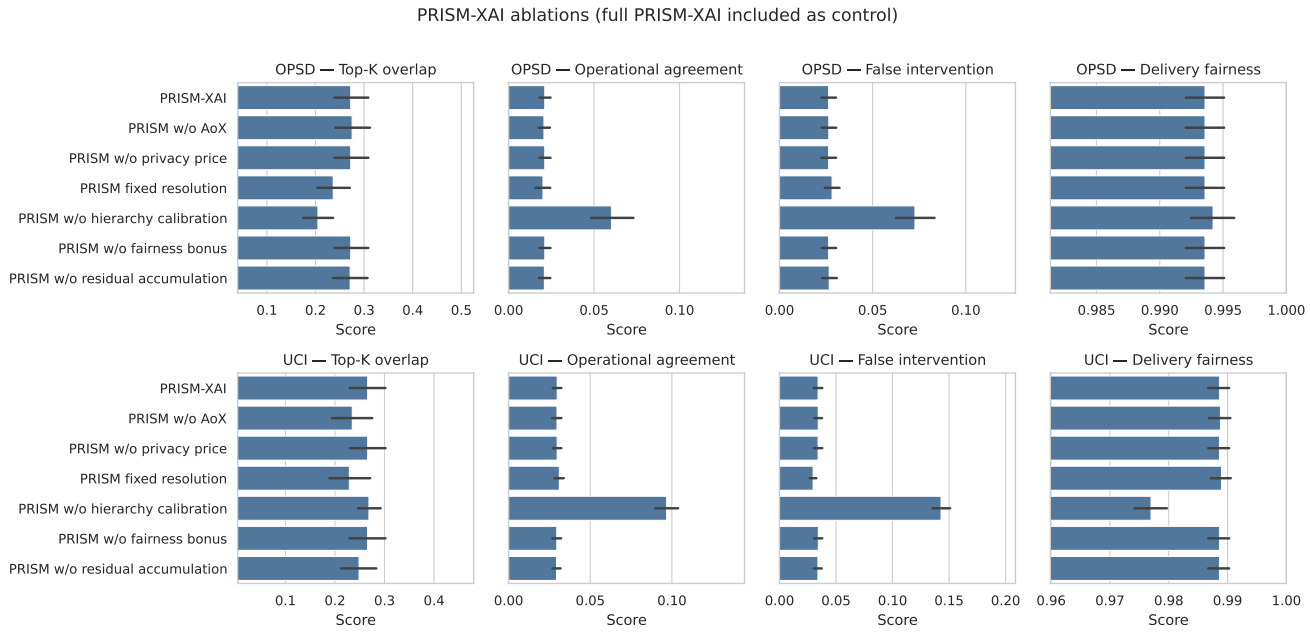


Fig. 114. Public-score threshold $\tau = 1$, release cap $M = 32$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXXXII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.261 ± 0.053	-0.019 ± 0.080	0.021 ± 0.005	0.021 ± 0.005	0.246 ± 0.094	0.947 ± 0.003	4.452 ± 1.878	97.640 ± 0.539	5.377 ± 0.416	0.027 ± 0.007	13250.048 ± 5126.243	0.994 ± 0.003
OPSD	Tree	8	PRISM-XAI	0.284 ± 0.064	-0.006 ± 0.076	0.022 ± 0.005	0.022 ± 0.005	0.253 ± 0.080	0.947 ± 0.003	4.452 ± 1.878	97.251 ± 0.416	5.108 ± 0.367	0.026 ± 0.006	12180.416 ± 3594.328	0.994 ± 0.003
OPSD	Tree	8	MAP-XAI fallback	0.249 ± 0.062	-0.025 ± 0.053	0.019 ± 0.007	0.019 ± 0.007	0.212 ± 0.098	0.947 ± 0.003	4.568 ± 1.808	117.817 ± 0.049	7.924 ± 0.000	0.030 ± 0.007	25854.893 ± 23642.714	0.994 ± 0.003
OPSD	Tree	8	MAP-XAI fallback	0.233 ± 0.051	0.005 ± 0.068	0.022 ± 0.007	0.022 ± 0.007	0.257 ± 0.098	0.947 ± 0.003	4.568 ± 1.808	117.675 ± 0.049	7.924 ± 0.000	0.026 ± 0.007	18081.688 ± 11158.136	0.994 ± 0.003

TABLE LXXXIII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	97.640 ± 0.539	0.261 ± 0.053	0.250 ± 0.000	-0.019 ± 0.080	0.021 ± 0.005	0.021 ± 0.005	0.246 ± 0.094	0.053 ± 0.003	0.947 ± 0.003	0.027 ± 0.007	5.377 ± 0.416
OPSD	Tree	8	PRISM-XAI	97.251 ± 0.416	0.284 ± 0.064	0.250 ± 0.000	-0.006 ± 0.076	0.022 ± 0.005	0.022 ± 0.005	0.253 ± 0.080	0.053 ± 0.003	0.947 ± 0.003	0.026 ± 0.006	5.108 ± 0.367
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	296.336 ± 0.924	0.239 ± 0.051	0.250 ± 0.000	0.018 ± 0.080	0.059 ± 0.016	0.059 ± 0.016	0.259 ± 0.080	0.134 ± 0.004	0.866 ± 0.004	0.066 ± 0.012	14.393 ± 0.046
OPSD	Tree	16	PRISM-XAI	295.240 ± 0.855	0.226 ± 0.027	0.250 ± 0.000	0.001 ± 0.087	0.061 ± 0.015	0.061 ± 0.015	0.274 ± 0.080	0.134 ± 0.004	0.866 ± 0.004	0.065 ± 0.014	14.360 ± 0.035
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	350.911 ± 0.837	0.222 ± 0.050	0.250 ± 0.000	0.012 ± 0.068	0.062 ± 0.017	0.062 ± 0.017	0.228 ± 0.070	0.140 ± 0.005	0.860 ± 0.005	0.082 ± 0.015	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	349.693 ± 0.728	0.243 ± 0.064	0.250 ± 0.000	0.019 ± 0.085	0.060 ± 0.016	0.060 ± 0.016	0.233 ± 0.084	0.140 ± 0.005	0.860 ± 0.005	0.081 ± 0.019	31.175 ± 0.000

TABLE LXXXIV
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1462.924 ± 0.539	0.027 ± 0.083	-0.099 ± 0.029	-0.006 ± 0.003
OPSD	AoX-aware fixed-DP	-4803.577 ± 5.551	-0.015 ± 0.045	-0.343 ± 0.075	-0.003 ± 0.002
OPSD	AoX-aware fixed-DP	-4949.313 ± 0.837	-0.030 ± 0.066	-0.350 ± 0.078	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-1463.313 ± 0.416	0.046 ± 0.054	-0.099 ± 0.026	-0.006 ± 0.003
OPSD	AoX-aware fixed-DP	-4804.672 ± 5.519	-0.026 ± 0.034	-0.353 ± 0.061	-0.003 ± 0.002
OPSD	AoX-aware fixed-DP	-4950.531 ± 0.728	-0.009 ± 0.071	-0.365 ± 0.060	-0.002 ± 0.001
OPSD	Channel-aware fixed-DP	-1462.924 ± 0.539	-0.032 ± 0.060	-0.086 ± 0.026	-0.006 ± 0.003
OPSD	Channel-aware fixed-DP	-4390.274 ± 13.308	-0.023 ± 0.044	-0.343 ± 0.094	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4949.313 ± 0.837	-0.035 ± 0.059	-0.372 ± 0.099	0.091 ± 0.005
OPSD	Channel-aware fixed-DP	-1463.313 ± 0.416	-0.008 ± 0.077	-0.090 ± 0.024	-0.006 ± 0.003
OPSD	Channel-aware fixed-DP	-4391.369 ± 13.318	-0.032 ± 0.032	-0.358 ± 0.092	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4950.531 ± 0.728	-0.014 ± 0.068	-0.396 ± 0.091	0.091 ± 0.005
OPSD	Joint, no compression	-92.260 ± 0.532	-0.027 ± 0.058	0.004 ± 0.006	-0.002 ± 0.002
OPSD	Joint, no compression	-163.565 ± 1.527	0.007 ± 0.064	0.021 ± 0.012	0.000 ± 0.001
OPSD	Joint, no compression	-789.989 ± 8.405	-0.013 ± 0.095	-0.012 ± 0.017	-0.001 ± 0.001
OPSD	Joint, no compression	-92.499 ± 0.556	0.026 ± 0.104	0.003 ± 0.003	-0.002 ± 0.002
OPSD	Joint, no compression	-164.810 ± 1.425	-0.014 ± 0.044	0.022 ± 0.014	0.000 ± 0.001
OPSD	Joint, no compression	-791.207 ± 8.350	-0.039 ± 0.094	-0.020 ± 0.015	-0.001 ± 0.001
OPSD	Periodic fixed-DP	-1462.924 ± 0.539	0.020 ± 0.076	-0.104 ± 0.021	-0.006 ± 0.003
OPSD	Periodic fixed-DP	-4816.259 ± 0.924	-0.011 ± 0.051	-0.392 ± 0.064	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-4948.925 ± 0.865	-0.027 ± 0.061	-0.396 ± 0.066	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-1463.313 ± 0.416	0.043 ± 0.057	-0.113 ± 0.016	-0.006 ± 0.003
OPSD	Periodic fixed-DP	-4817.354 ± 0.855	-0.017 ± 0.029	-0.402 ± 0.055	-0.003 ± 0.002
OPSD	Periodic fixed-DP	-4950.143 ± 0.776	-0.000 ± 0.065	-0.411 ± 0.052	-0.002 ± 0.001
OPSD	MAP-XAI fallback	-20.177 ± 0.549	0.012 ± 0.081	0.001 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-65.981 ± 0.926	0.031 ± 0.045	0.002 ± 0.004	0.000 ± 0.000
OPSD	MAP-XAI fallback	-78.906 ± 0.839	0.007 ± 0.030	0.002 ± 0.009	0.000 ± 0.000
OPSD	MAP-XAI fallback	-20.424 ± 0.409	0.051 ± 0.064	-0.000 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-66.935 ± 0.867	0.017 ± 0.028	0.005 ± 0.005	0.000 ± 0.000
OPSD	MAP-XAI fallback	-79.982 ± 0.736	0.019 ± 0.017	0.001 ± 0.009	0.000 ± 0.000

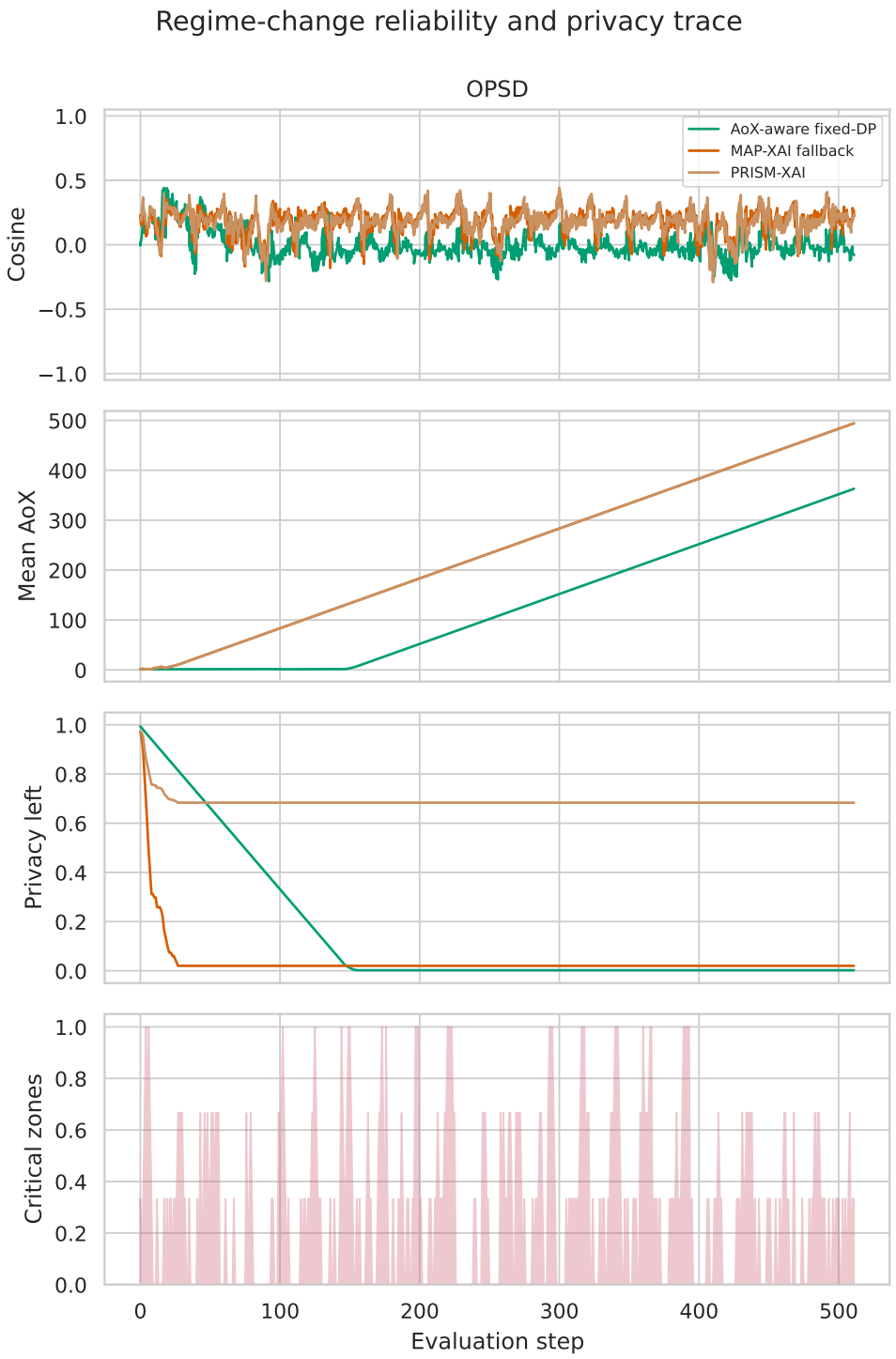


Fig. 115. Public-score threshold $\tau = 1$, release cap $M = 32$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

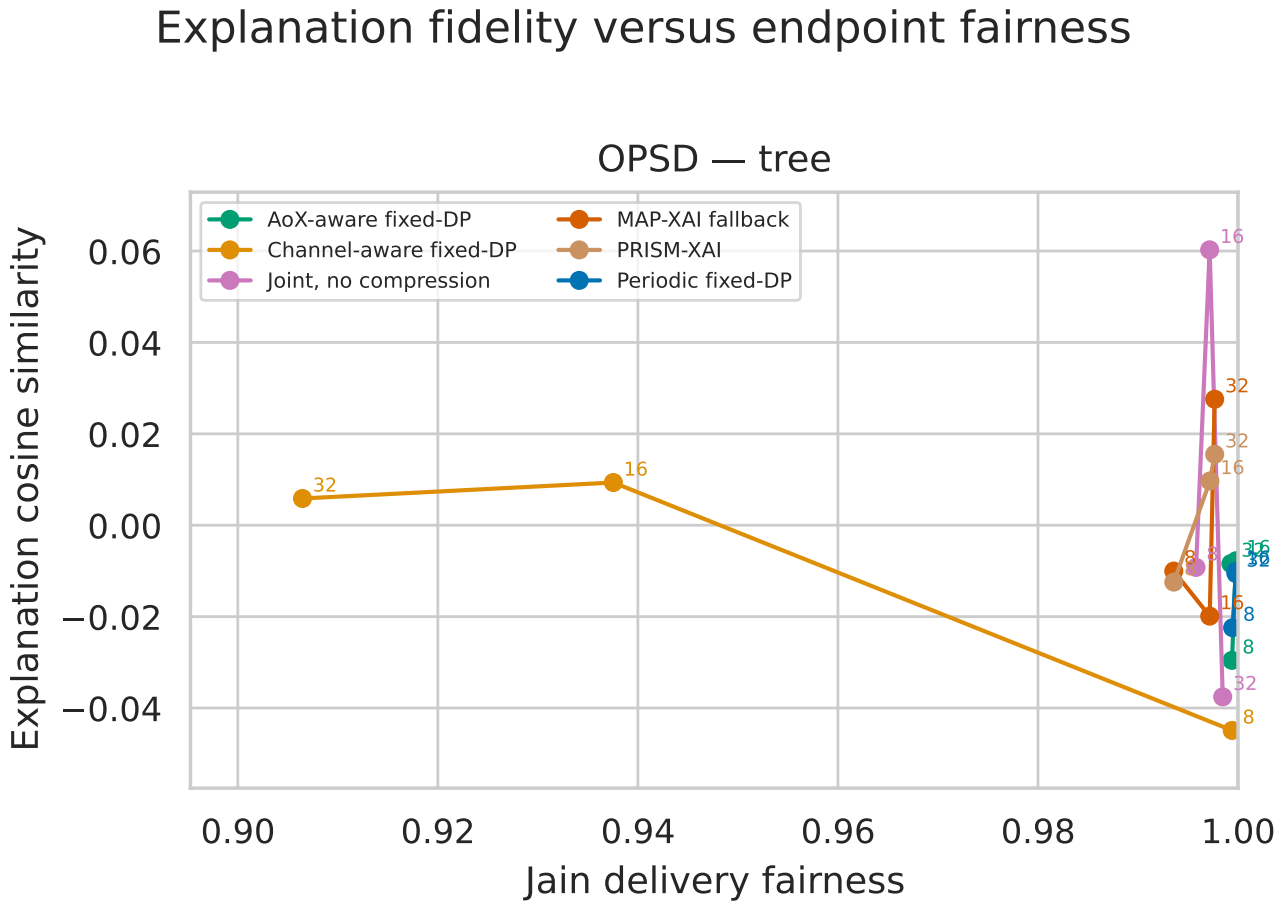


Fig. 116. Public-score threshold $\tau = 1$, release cap $M = 32$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

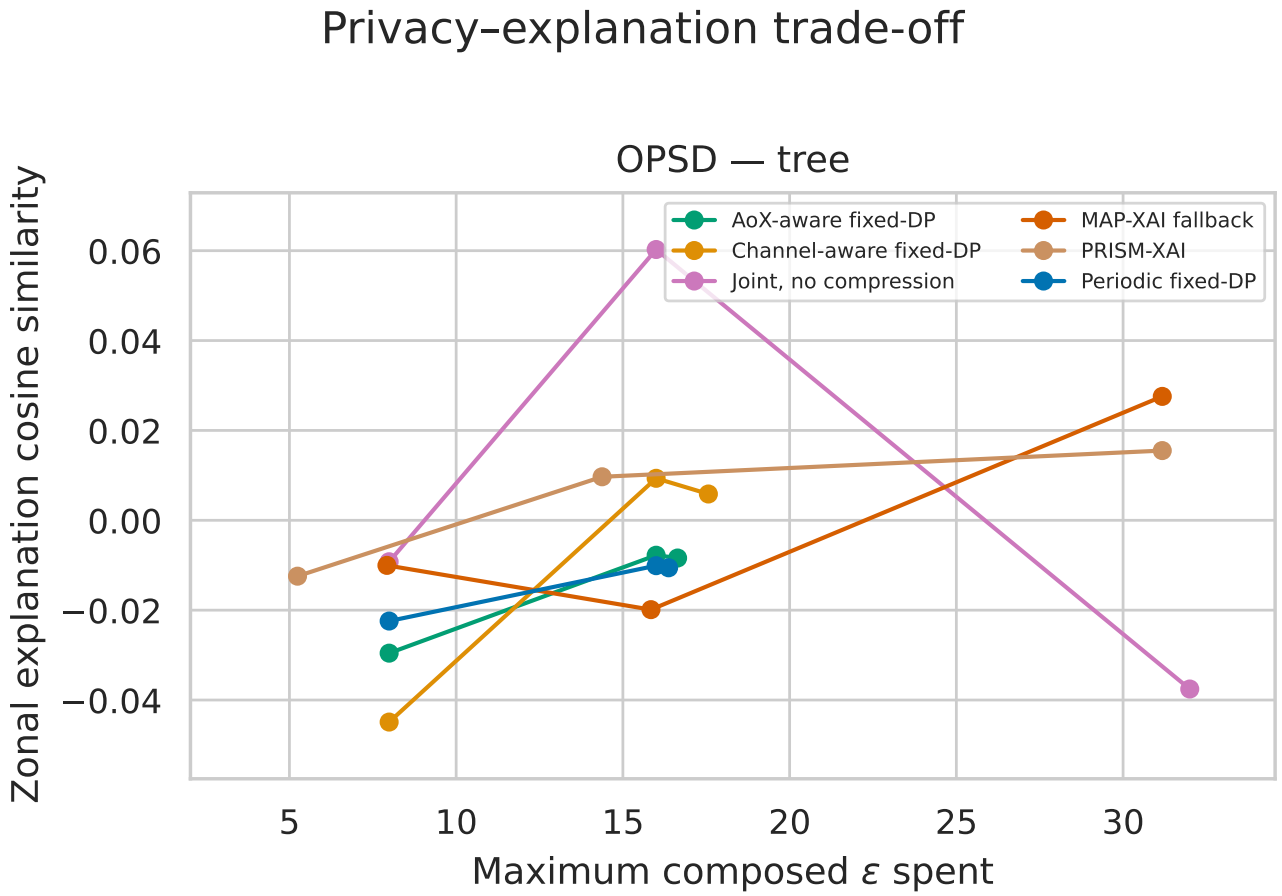


Fig. 117. Public-score threshold $\tau = 1$, release cap $M = 32$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

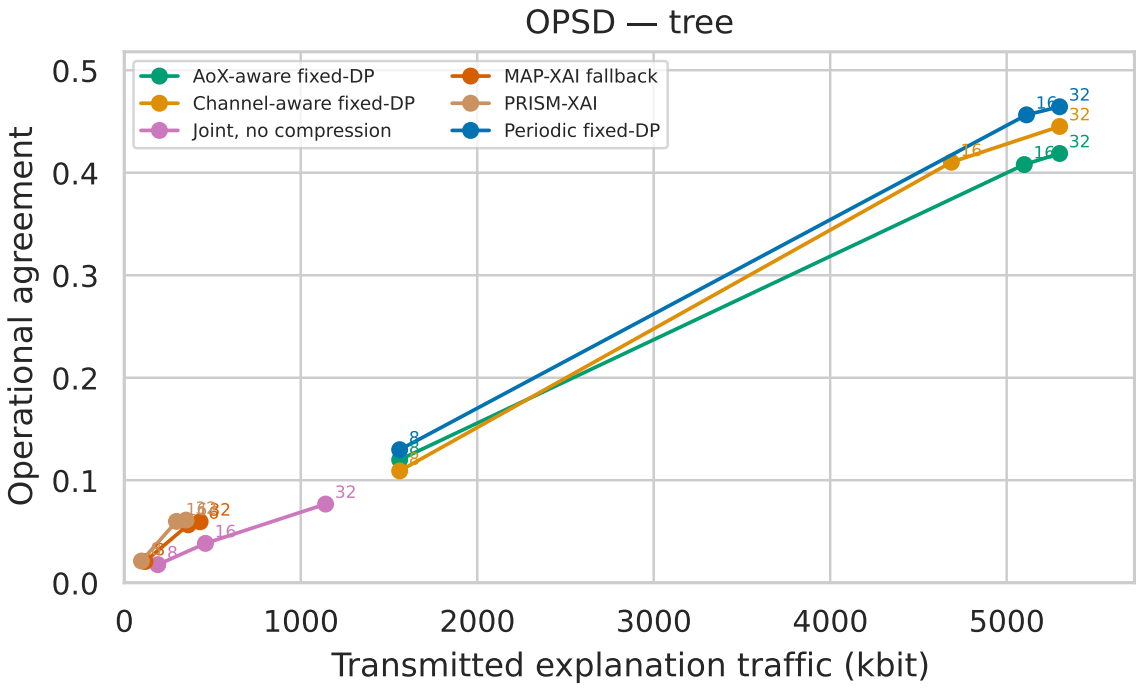


Fig. 118. Public-score threshold $\tau = 1$, release cap $M = 32$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

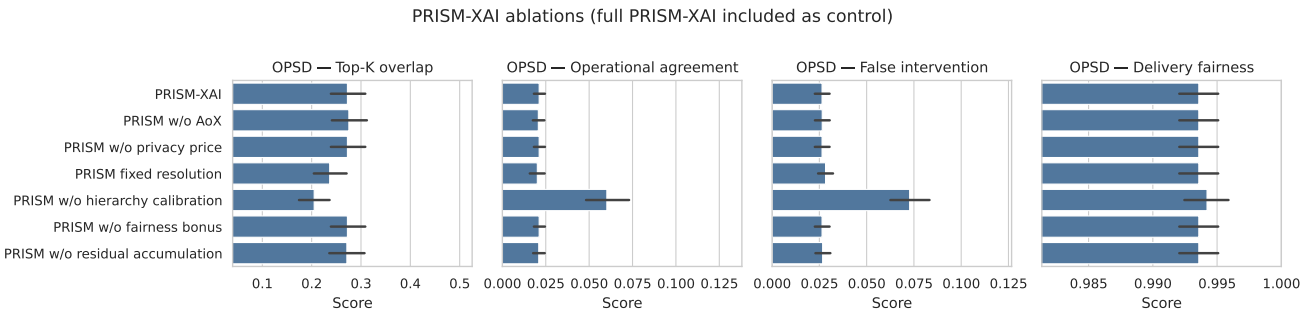


Fig. 119. Public-score threshold $\tau = 1$, release cap $M = 32$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXXXV
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.270 ± 0.078	-0.004 ± 0.021	0.029 ± 0.006	0.029 ± 0.006	0.346 ± 0.060	0.928 ± 0.003	5.789 ± 1.601	333.552 ± 0.722	6.549 ± 0.228	0.037 ± 0.005	15343.847 ± 4327.572	0.989 ± 0.003
UCI	Tree	8	PRISM-XAI	0.262 ± 0.032	-0.011 ± 0.061	0.031 ± 0.004	0.031 ± 0.004	0.416 ± 0.078	0.931 ± 0.004	4.104 ± 1.678	331.348 ± 1.358	6.250 ± 0.232	0.031 ± 0.006	13176.006 ± 1853.692	0.988 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.256 ± 0.104	-0.001 ± 0.023	0.030 ± 0.005	0.030 ± 0.005	0.390 ± 0.058	0.929 ± 0.004	5.428 ± 1.682	394.757 ± 0.724	7.924 ± 0.000	0.033 ± 0.004	17064.051 ± 5665.496	0.990 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.215 ± 0.046	0.034 ± 0.038	0.031 ± 0.003	0.031 ± 0.003	0.437 ± 0.060	0.932 ± 0.004	5.132 ± 1.788	393.884 ± 0.595	7.924 ± 0.000	0.029 ± 0.006	15461.186 ± 1718.511	0.988 ± 0.003

TABLE LXXXVI
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	333.552 ± 0.722	0.270 ± 0.078	0.250 ± 0.000	-0.004 ± 0.021	0.029 ± 0.006	0.029 ± 0.006	0.346 ± 0.060	0.072 ± 0.003	0.928 ± 0.003	0.037 ± 0.005	6.549 ± 0.228
UCI	Tree	8	PRISM-XAI	331.348 ± 1.358	0.262 ± 0.032	0.250 ± 0.000	-0.011 ± 0.061	0.031 ± 0.004	0.031 ± 0.004	0.416 ± 0.078	0.069 ± 0.004	0.931 ± 0.004	0.031 ± 0.006	6.250 ± 0.232
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	1088.410 ± 2.255	0.252 ± 0.062	0.250 ± 0.000	0.007 ± 0.013	0.106 ± 0.014	0.106 ± 0.014	0.397 ± 0.042	0.218 ± 0.007	0.782 ± 0.007	0.110 ± 0.007	14.811 ± 0.029
UCI	Tree	16	PRISM-XAI	1081.416 ± 2.529	0.236 ± 0.022	0.250 ± 0.000	0.006 ± 0.081	0.098 ± 0.011	0.098 ± 0.011	0.394 ± 0.027	0.208 ± 0.013	0.792 ± 0.013	0.104 ± 0.012	14.805 ± 0.044
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	1289.912 ± 2.162	0.259 ± 0.042	0.250 ± 0.000	0.016 ± 0.015	0.118 ± 0.011	0.118 ± 0.011	0.389 ± 0.026	0.247 ± 0.009	0.753 ± 0.009	0.128 ± 0.011	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	1283.870 ± 3.710	0.235 ± 0.035	0.250 ± 0.000	0.020 ± 0.034	0.100 ± 0.015	0.100 ± 0.015	0.395 ± 0.041	0.216 ± 0.019	0.784 ± 0.019	0.105 ± 0.010	31.175 ± 0.000

TABLE LXXXVII
PUBLIC-SCORE THRESHOLD $\tau = 1$, RELEASE CAP $M = 32$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4966.672 ± 0.722	0.007 ± 0.099	-0.428 ± 0.034	-0.010 ± 0.003
UCI	AoX-aware fixed-DP	-4211.814 ± 2.255	-0.012 ± 0.049	-0.351 ± 0.042	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-4010.312 ± 2.162	-0.004 ± 0.066	-0.339 ± 0.039	-0.001 ± 0.000
UCI	AoX-aware fixed-DP	-4968.876 ± 1.358	0.002 ± 0.043	-0.424 ± 0.048	-0.011 ± 0.003
UCI	AoX-aware fixed-DP	-4218.808 ± 2.529	-0.024 ± 0.028	-0.357 ± 0.049	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-4016.354 ± 3.710	-0.025 ± 0.048	-0.354 ± 0.051	-0.001 ± 0.001
UCI	Channel-aware fixed-DP	-4961.625 ± 3.959	0.022 ± 0.103	0.028 ± 0.006	-0.001 ± 0.002
UCI	Channel-aware fixed-DP	-4211.814 ± 2.255	-0.016 ± 0.045	0.105 ± 0.014	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-4010.312 ± 2.162	-0.009 ± 0.044	0.117 ± 0.011	0.743 ± 0.001
UCI	Channel-aware fixed-DP	-4963.829 ± 4.621	0.023 ± 0.025	0.031 ± 0.004	-0.003 ± 0.003
UCI	Channel-aware fixed-DP	-4218.808 ± 2.529	-0.024 ± 0.023	0.097 ± 0.011	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-4016.354 ± 3.710	-0.024 ± 0.030	0.100 ± 0.015	0.743 ± 0.001
UCI	Joint, no compression	-338.748 ± 0.953	-0.016 ± 0.089	-0.024 ± 0.007	-0.004 ± 0.004
UCI	Joint, no compression	-543.740 ± 2.301	0.035 ± 0.083	0.012 ± 0.013	-0.000 ± 0.000
UCI	Joint, no compression	-2744.638 ± 6.413	0.013 ± 0.055	-0.110 ± 0.017	-0.001 ± 0.001
UCI	Joint, no compression	-340.952 ± 1.422	0.033 ± 0.058	-0.026 ± 0.008	-0.006 ± 0.004
UCI	Joint, no compression	-551.184 ± 2.335	0.032 ± 0.017	-0.001 ± 0.022	-0.000 ± 0.000
UCI	Joint, no compression	-2750.680 ± 6.897	-0.006 ± 0.045	-0.127 ± 0.025	-0.001 ± 0.001
UCI	Periodic fixed-DP	-4965.766 ± 1.640	0.010 ± 0.116	0.023 ± 0.006	-0.010 ± 0.003
UCI	Periodic fixed-DP	-4210.908 ± 2.896	-0.008 ± 0.088	0.100 ± 0.014	-0.002 ± 0.000
UCI	Periodic fixed-DP	-4009.406 ± 2.846	-0.000 ± 0.080	0.112 ± 0.011	-0.002 ± 0.000
UCI	Periodic fixed-DP	-4967.970 ± 2.045	0.002 ± 0.046	0.026 ± 0.005	-0.012 ± 0.003
UCI	Periodic fixed-DP	-4217.902 ± 2.998	-0.024 ± 0.056	0.092 ± 0.011	-0.002 ± 0.001
UCI	Periodic fixed-DP	-4015.448 ± 4.177	-0.025 ± 0.069	0.095 ± 0.015	-0.002 ± 0.001
UCI	MAP-XAI fallback	-61.204 ± 1.019	0.014 ± 0.094	-0.002 ± 0.004	-0.000 ± 0.001
UCI	MAP-XAI fallback	-205.689 ± 2.231	-0.053 ± 0.065	0.006 ± 0.009	-0.000 ± 0.000
UCI	MAP-XAI fallback	-244.248 ± 2.169	0.031 ± 0.072	0.001 ± 0.004	0.000 ± 0.000
UCI	MAP-XAI fallback	-62.535 ± 1.585	0.047 ± 0.049	-0.000 ± 0.003	-0.000 ± 0.002
UCI	MAP-XAI fallback	-212.229 ± 2.556	-0.010 ± 0.038	0.000 ± 0.008	-0.000 ± 0.000
UCI	MAP-XAI fallback	-249.775 ± 3.761	-0.032 ± 0.052	0.001 ± 0.013	0.000 ± 0.000

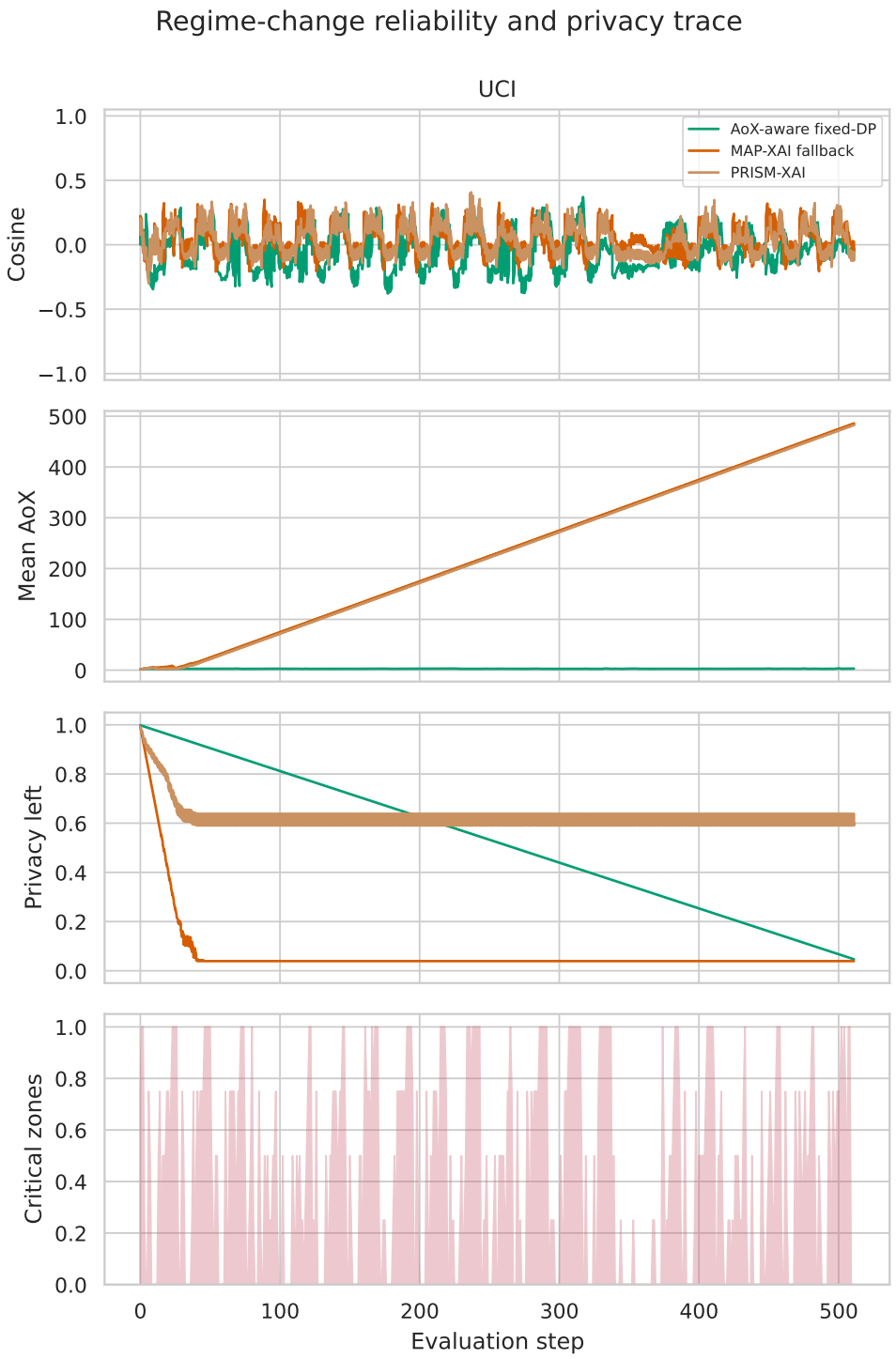


Fig. 120. Public-score threshold $\tau = 1$, release cap $M = 32$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

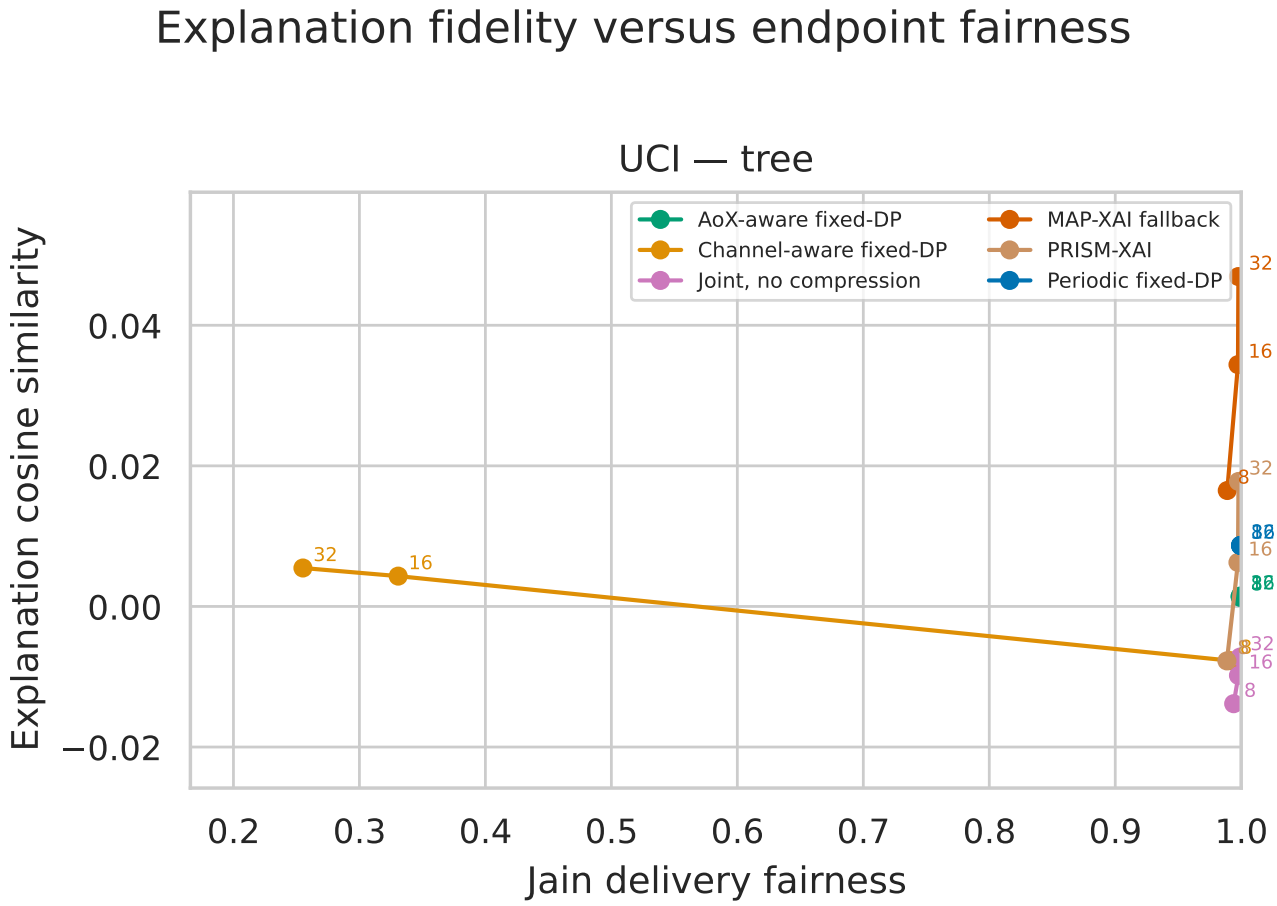


Fig. 121. Public-score threshold $\tau = 1$, release cap $M = 32$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

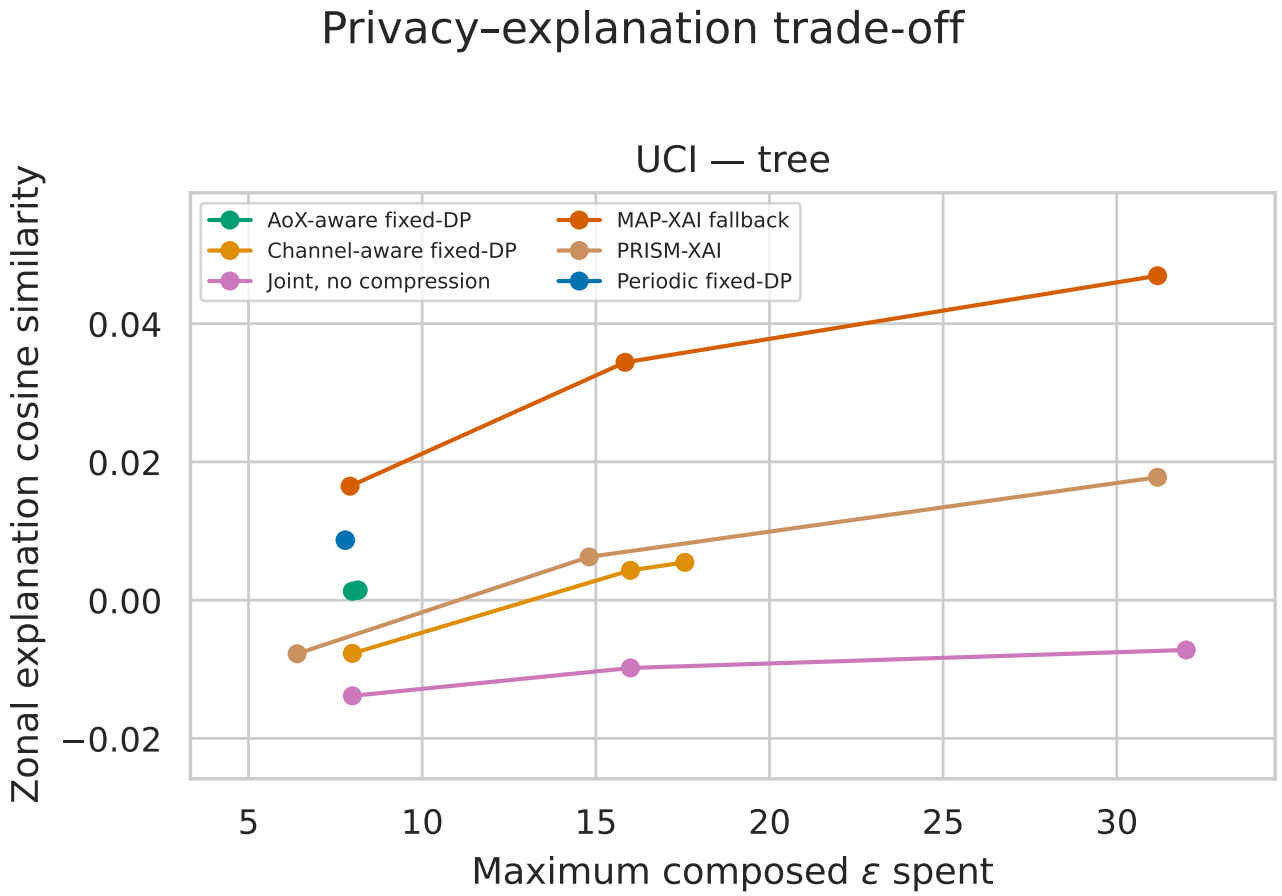


Fig. 122. Public-score threshold $\tau = 1$, release cap $M = 32$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

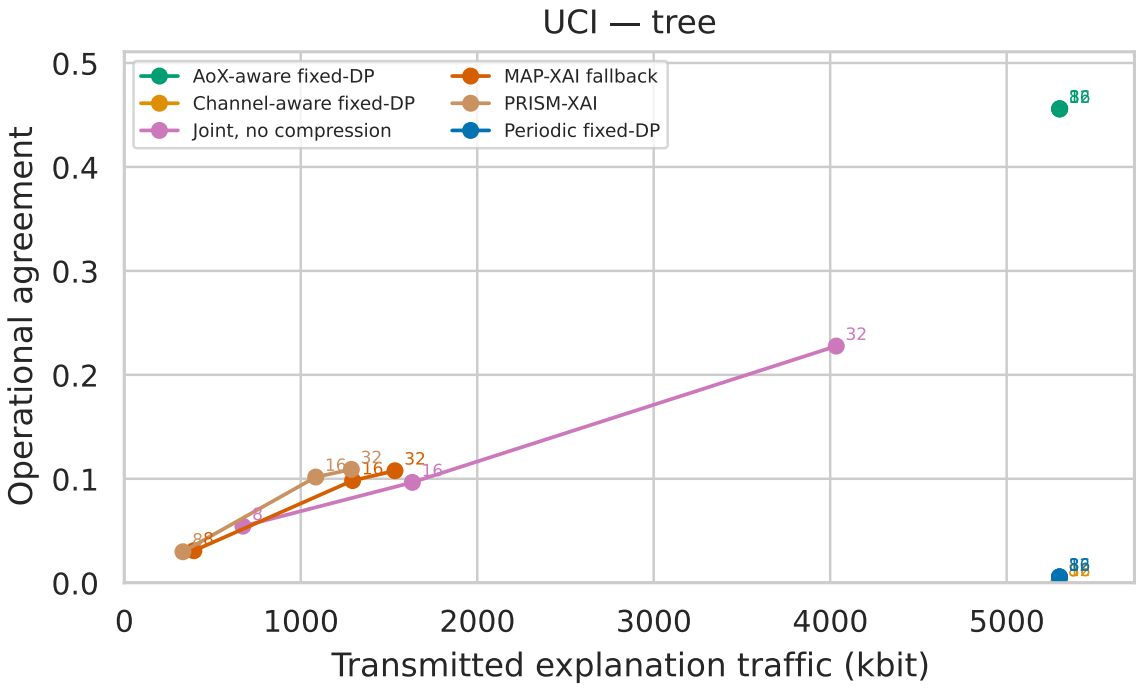


Fig. 123. Public-score threshold $\tau = 1$, release cap $M = 32$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

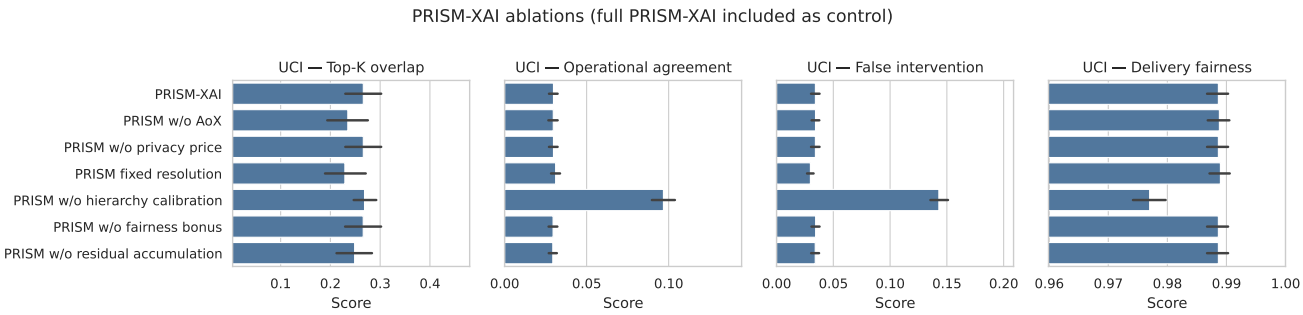


Fig. 124. Public-score threshold $\tau = 1$, release cap $M = 32$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

9) Public-score threshold $\tau = 3$, release cap $M = 32$: At this operating point, PRISM-XAI transmits 584.33 kbit versus 688.42 kbit for MAP-XAI-only (aggregate-mean ratio saving 15.12%). The paired difference is -104.09$\pm$13.56 kbit. PRISM coverage is 0.2064, critical-event recall is 0.0852, and maximum recorded expenditure is 17.301. Relative to MAP-XAI-only, the paired coverage change is -0.00010$\pm$0.00026 and recall change is +0.00124$\pm$0.00307. The descriptive overlap rule holds in 5 of 12 cells. Accordingly, this point is read as a sensitivity point on the service frontier, not a complete matched-service confirmation.

a) Combined datasets.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE LXXXVIII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, COMBINED DATASETS: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.257 ± 0.067	0.028 ± 0.059	0.028 ± 0.011	0.028 ± 0.011	0.273 ± 0.131	0.923 ± 0.004	4.713 ± 1.733	96.298 ± 1.228	5.340 ± 0.419	0.030 ± 0.007	11906.519 ± 6206.672	0.989 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.220 ± 0.047	0.027 ± 0.062	0.029 ± 0.013	0.029 ± 0.013	0.278 ± 0.137	0.923 ± 0.004	4.708 ± 1.737	95.592 ± 1.248	5.012 ± 0.388	0.030 ± 0.007	13471.090 ± 9725.652	0.989 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.240 ± 0.060	0.022 ± 0.057	0.030 ± 0.008	0.030 ± 0.008	0.271 ± 0.091	0.923 ± 0.004	4.795 ± 1.684	115.748 ± 1.149	7.924 ± 0.000	0.031 ± 0.006	10380.916 ± 2295.453	0.989 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.242 ± 0.055	0.038 ± 0.059	0.028 ± 0.007	0.028 ± 0.007	0.248 ± 0.070	0.923 ± 0.004	4.795 ± 1.684	115.607 ± 1.156	7.924 ± 0.000	0.032 ± 0.005	11185.765 ± 2730.237	0.989 ± 0.004
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.207 ± 0.052	0.004 ± 0.012	0.046 ± 0.010	0.046 ± 0.010	0.469 ± 0.081	0.901 ± 0.005	5.035 ± 1.860	341.133 ± 1.293	6.779 ± 0.208	0.035 ± 0.005	9780.191 ± 2585.599	0.988 ± 0.003
UCI	Tree	8	PRISM-XAI	0.256 ± 0.043	0.000 ± 0.081	0.047 ± 0.008	0.047 ± 0.008	0.503 ± 0.086	0.902 ± 0.005	5.524 ± 1.825	338.064 ± 2.047	6.657 ± 0.214	0.032 ± 0.007	9159.950 ± 1670.303	0.989 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.193 ± 0.046	0.014 ± 0.010	0.046 ± 0.007	0.046 ± 0.007	0.479 ± 0.053	0.901 ± 0.005	5.106 ± 1.806	395.052 ± 0.314	7.924 ± 0.000	0.035 ± 0.004	10596.392 ± 1554.907	0.988 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.246 ± 0.061	0.064 ± 0.067	0.049 ± 0.007	0.049 ± 0.007	0.510 ± 0.076	0.902 ± 0.005	5.409 ± 1.720	395.226 ± 1.353	7.924 ± 0.000	0.033 ± 0.007	10216.728 ± 2063.280	0.989 ± 0.003

TABLE LXXXIX

PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, COMBINED DATASETS: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	96.298 ± 1.228	0.257 ± 0.067	0.250 ± 0.000	0.028 ± 0.059	0.028 ± 0.011	0.028 ± 0.011	0.273 ± 0.131	0.077 ± 0.004	0.923 ± 0.004	0.030 ± 0.007	5.340 ± 0.419
OPSD	Tree	8	PRISM-XAI	95.592 ± 1.248	0.220 ± 0.047	0.250 ± 0.000	0.027 ± 0.062	0.029 ± 0.013	0.029 ± 0.013	0.278 ± 0.137	0.077 ± 0.004	0.923 ± 0.004	0.030 ± 0.007	5.012 ± 0.388
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	300.705 ± 0.565	0.278 ± 0.056	0.250 ± 0.000	0.047 ± 0.080	0.066 ± 0.020	0.066 ± 0.020	0.224 ± 0.088	0.200 ± 0.005	0.800 ± 0.005	0.090 ± 0.015	14.564 ± 0.022
OPSD	Tree	16	PRISM-XAI	298.955 ± 0.657	0.289 ± 0.047	0.250 ± 0.000	0.055 ± 0.090	0.063 ± 0.021	0.063 ± 0.021	0.207 ± 0.090	0.199 ± 0.004	0.801 ± 0.004	0.097 ± 0.018	14.498 ± 0.044
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	356.570 ± 0.841	0.321 ± 0.068	0.250 ± 0.000	0.150 ± 0.077	0.105 ± 0.023	0.105 ± 0.023	0.301 ± 0.084	0.243 ± 0.007	0.757 ± 0.007	0.094 ± 0.017	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	354.636 ± 0.782	0.326 ± 0.048	0.250 ± 0.000	0.146 ± 0.091	0.101 ± 0.022	0.101 ± 0.022	0.283 ± 0.077	0.243 ± 0.008	0.757 ± 0.008	0.099 ± 0.016	31.175 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	341.133 ± 1.293	0.207 ± 0.052	0.250 ± 0.000	0.004 ± 0.012	0.046 ± 0.010	0.046 ± 0.010	0.469 ± 0.081	0.099 ± 0.005	0.901 ± 0.005	0.035 ± 0.005	6.779 ± 0.208
UCI	Tree	8	PRISM-XAI	338.064 ± 2.047	0.256 ± 0.043	0.250 ± 0.000	0.000 ± 0.081	0.047 ± 0.008	0.047 ± 0.008	0.503 ± 0.086	0.098 ± 0.005	0.902 ± 0.005	0.032 ± 0.007	6.657 ± 0.214
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	1110.802 ± 3.293	0.269 ± 0.046	0.250 ± 0.000	0.018 ± 0.010	0.126 ± 0.013	0.126 ± 0.013	0.407 ± 0.047	0.285 ± 0.006	0.715 ± 0.006	0.127 ± 0.012	15.063 ± 0.065
UCI	Tree	16	PRISM-XAI	1099.581 ± 2.815	0.240 ± 0.048	0.250 ± 0.000	-0.063 ± 0.054	0.115 ± 0.016	0.115 ± 0.016	0.379 ± 0.064	0.282 ± 0.007	0.718 ± 0.007	0.132 ± 0.019	14.998 ± 0.062
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	1317.135 ± 2.878	0.286 ± 0.041	0.250 ± 0.000	0.016 ± 0.014	0.150 ± 0.019	0.150 ± 0.019	0.423 ± 0.062	0.338 ± 0.007	0.662 ± 0.007	0.143 ± 0.016	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	1302.454 ± 3.178	0.249 ± 0.044	0.250 ± 0.000	-0.087 ± 0.048	0.148 ± 0.011	0.148 ± 0.011	0.437 ± 0.045	0.335 ± 0.007	0.665 ± 0.007	0.132 ± 0.013	31.175 ± 0.000

TABLE XC
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, COMBINED DATASETS: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1464.266 ± 1.228	0.024 ± 0.053	-0.092 ± 0.036	-0.010 ± 0.004
OPSD	AoX-aware fixed-DP	-4799.207 ± 5.399	0.025 ± 0.072	-0.335 ± 0.087	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-4943.654 ± 0.841	0.068 ± 0.080	-0.308 ± 0.086	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-1464.972 ± 1.248	-0.018 ± 0.064	-0.092 ± 0.035	-0.010 ± 0.004
OPSD	AoX-aware fixed-DP	-4800.957 ± 5.327	0.036 ± 0.052	-0.351 ± 0.071	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-4945.588 ± 0.782	0.074 ± 0.055	-0.325 ± 0.067	-0.002 ± 0.001
OPSD	Channel-aware fixed-DP	-1464.266 ± 1.228	-0.035 ± 0.061	-0.079 ± 0.032	-0.010 ± 0.004
OPSD	Channel-aware fixed-DP	-4385.904 ± 13.238	0.016 ± 0.064	-0.335 ± 0.100	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4943.654 ± 0.841	0.063 ± 0.076	-0.330 ± 0.104	0.091 ± 0.005
OPSD	Channel-aware fixed-DP	-1464.972 ± 1.248	-0.072 ± 0.074	-0.082 ± 0.031	-0.010 ± 0.004
OPSD	Channel-aware fixed-DP	-4387.654 ± 13.405	0.031 ± 0.047	-0.356 ± 0.098	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4945.588 ± 0.782	0.069 ± 0.051	-0.355 ± 0.094	0.091 ± 0.005
OPSD	Joint, no compression	-93.602 ± 1.154	-0.031 ± 0.103	0.011 ± 0.012	-0.007 ± 0.004
OPSD	Joint, no compression	-159.195 ± 1.122	0.047 ± 0.099	0.028 ± 0.023	0.000 ± 0.001
OPSD	Joint, no compression	-784.330 ± 7.944	0.086 ± 0.101	0.031 ± 0.032	-0.001 ± 0.001
OPSD	Joint, no compression	-94.158 ± 1.241	-0.038 ± 0.069	0.010 ± 0.013	-0.007 ± 0.004
OPSD	Joint, no compression	-161.095 ± 1.115	0.048 ± 0.086	0.025 ± 0.021	0.000 ± 0.001
OPSD	Joint, no compression	-786.264 ± 8.092	0.044 ± 0.080	0.021 ± 0.024	-0.001 ± 0.001
OPSD	Periodic fixed-DP	-1464.266 ± 1.228	0.017 ± 0.044	-0.097 ± 0.025	-0.010 ± 0.004
OPSD	Periodic fixed-DP	-4811.889 ± 0.565	0.029 ± 0.053	-0.384 ± 0.072	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-4943.266 ± 0.807	0.072 ± 0.074	-0.353 ± 0.071	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-1464.972 ± 1.248	-0.022 ± 0.054	-0.106 ± 0.022	-0.010 ± 0.004
OPSD	Periodic fixed-DP	-4813.639 ± 0.657	0.046 ± 0.040	-0.399 ± 0.060	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-4945.200 ± 0.800	0.082 ± 0.043	-0.370 ± 0.053	-0.002 ± 0.001
OPSD	MAP-XAI fallback	-19.450 ± 0.713	0.017 ± 0.052	-0.002 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-62.093 ± 0.598	-0.013 ± 0.032	0.003 ± 0.015	0.000 ± 0.000
OPSD	MAP-XAI fallback	-73.728 ± 0.773	-0.007 ± 0.048	0.011 ± 0.016	0.000 ± 0.000
OPSD	MAP-XAI fallback	-20.015 ± 0.643	-0.022 ± 0.056	0.001 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-63.701 ± 0.679	-0.004 ± 0.046	-0.001 ± 0.018	0.000 ± 0.000
OPSD	MAP-XAI fallback	-75.500 ± 0.722	0.005 ± 0.038	0.012 ± 0.013	0.000 ± 0.000
UCI	AoX-aware fixed-DP	-4959.091 ± 1.293	-0.056 ± 0.063	-0.411 ± 0.026	-0.011 ± 0.003
UCI	AoX-aware fixed-DP	-4189.422 ± 3.293	0.005 ± 0.058	-0.332 ± 0.035	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-3983.089 ± 2.878	0.023 ± 0.064	-0.307 ± 0.043	-0.001 ± 0.001
UCI	AoX-aware fixed-DP	-4962.160 ± 2.047	-0.004 ± 0.042	-0.409 ± 0.046	-0.010 ± 0.003
UCI	AoX-aware fixed-DP	-4200.643 ± 2.815	-0.020 ± 0.050	-0.340 ± 0.048	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-3997.770 ± 3.178	-0.012 ± 0.032	-0.307 ± 0.052	-0.001 ± 0.001
UCI	Channel-aware fixed-DP	-4954.044 ± 4.047	-0.041 ± 0.066	0.046 ± 0.010	-0.002 ± 0.003
UCI	Channel-aware fixed-DP	-4189.422 ± 3.293	0.001 ± 0.058	0.125 ± 0.013	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-3983.089 ± 2.878	0.018 ± 0.049	0.150 ± 0.019	0.743 ± 0.001
UCI	Channel-aware fixed-DP	-4957.113 ± 4.745	0.017 ± 0.045	0.046 ± 0.008	-0.002 ± 0.003
UCI	Channel-aware fixed-DP	-4200.643 ± 2.815	-0.020 ± 0.052	0.115 ± 0.016	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-3997.770 ± 3.178	-0.011 ± 0.043	0.148 ± 0.011	0.743 ± 0.001
UCI	Joint, no compression	-331.167 ± 1.523	-0.079 ± 0.085	-0.006 ± 0.011	-0.006 ± 0.003
UCI	Joint, no compression	-521.348 ± 3.369	0.052 ± 0.038	0.031 ± 0.015	-0.000 ± 0.001
UCI	Joint, no compression	-2717.415 ± 6.000	0.040 ± 0.049	-0.077 ± 0.014	-0.001 ± 0.001
UCI	Joint, no compression	-334.236 ± 2.296	0.026 ± 0.066	-0.010 ± 0.011	-0.005 ± 0.003
UCI	Joint, no compression	-533.019 ± 2.619	0.037 ± 0.048	0.016 ± 0.022	-0.000 ± 0.001
UCI	Joint, no compression	-2732.096 ± 5.519	0.007 ± 0.058	-0.080 ± 0.032	-0.001 ± 0.001
UCI	Periodic fixed-DP	-4958.185 ± 2.040	-0.053 ± 0.077	0.040 ± 0.010	-0.011 ± 0.003
UCI	Periodic fixed-DP	-4188.516 ± 3.590	0.009 ± 0.089	0.120 ± 0.014	-0.002 ± 0.001
UCI	Periodic fixed-DP	-3982.184 ± 3.108	0.027 ± 0.068	0.145 ± 0.020	-0.002 ± 0.001
UCI	Periodic fixed-DP	-4961.254 ± 2.973	-0.005 ± 0.068	0.041 ± 0.007	-0.011 ± 0.003
UCI	Periodic fixed-DP	-4199.738 ± 3.048	-0.020 ± 0.078	0.109 ± 0.015	-0.002 ± 0.001
UCI	Periodic fixed-DP	-3996.864 ± 3.261	-0.012 ± 0.073	0.142 ± 0.011	-0.002 ± 0.001
UCI	MAP-XAI fallback	-53.919 ± 1.196	0.014 ± 0.036	-0.000 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-182.879 ± 3.287	0.006 ± 0.059	-0.002 ± 0.010	0.000 ± 0.000
UCI	MAP-XAI fallback	-216.465 ± 2.875	0.076 ± 0.044	0.002 ± 0.015	0.000 ± 0.000
UCI	MAP-XAI fallback	-57.162 ± 1.160	0.010 ± 0.051	-0.002 ± 0.008	-0.000 ± 0.001
UCI	MAP-XAI fallback	-193.593 ± 2.996	0.006 ± 0.034	-0.010 ± 0.011	-0.000 ± 0.000
UCI	MAP-XAI fallback	-230.577 ± 3.247	0.018 ± 0.059	0.003 ± 0.011	-0.000 ± 0.000

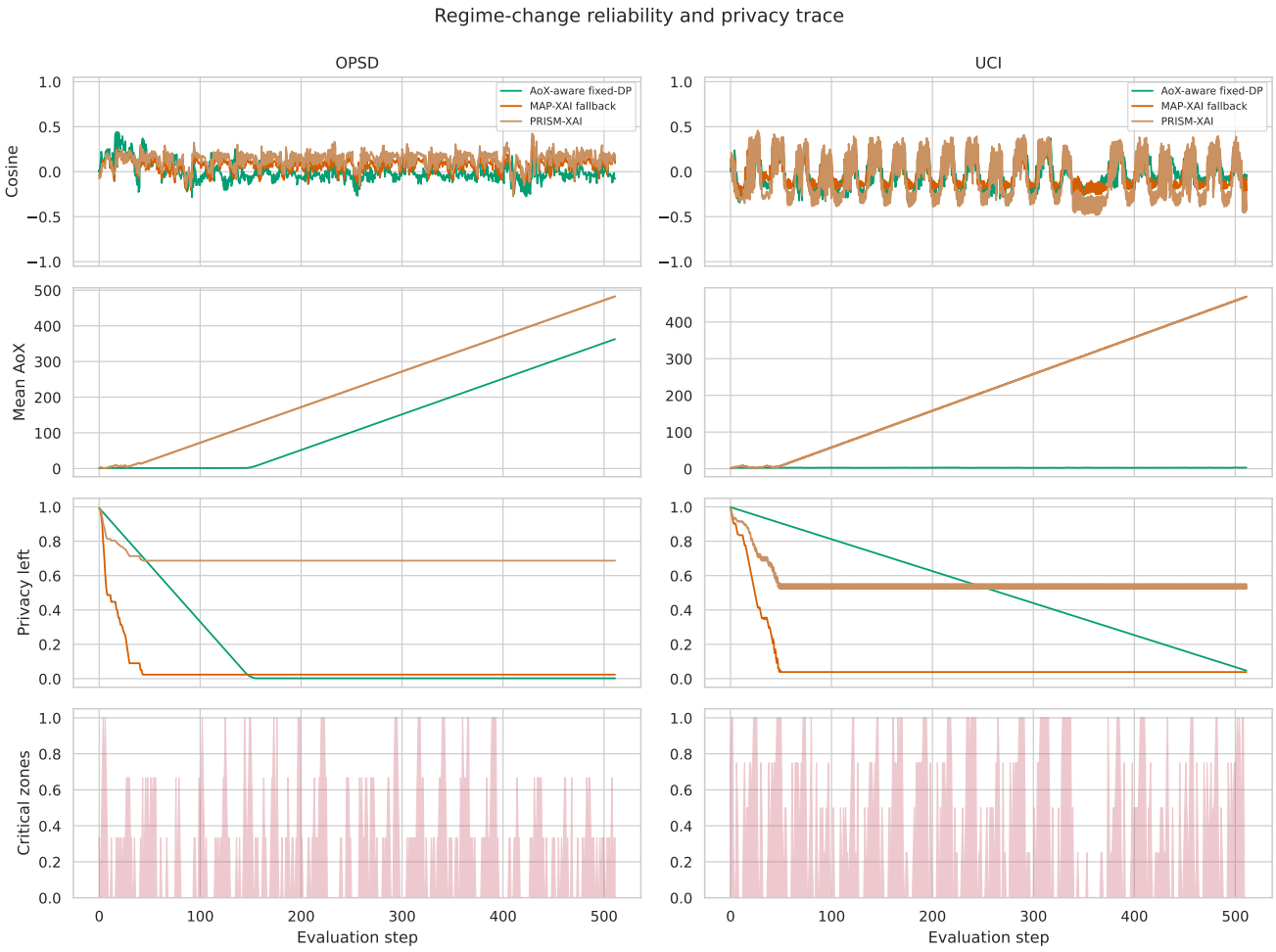


Fig. 125. Public-score threshold $\tau = 3$, release cap $M = 32$, Combined datasets: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

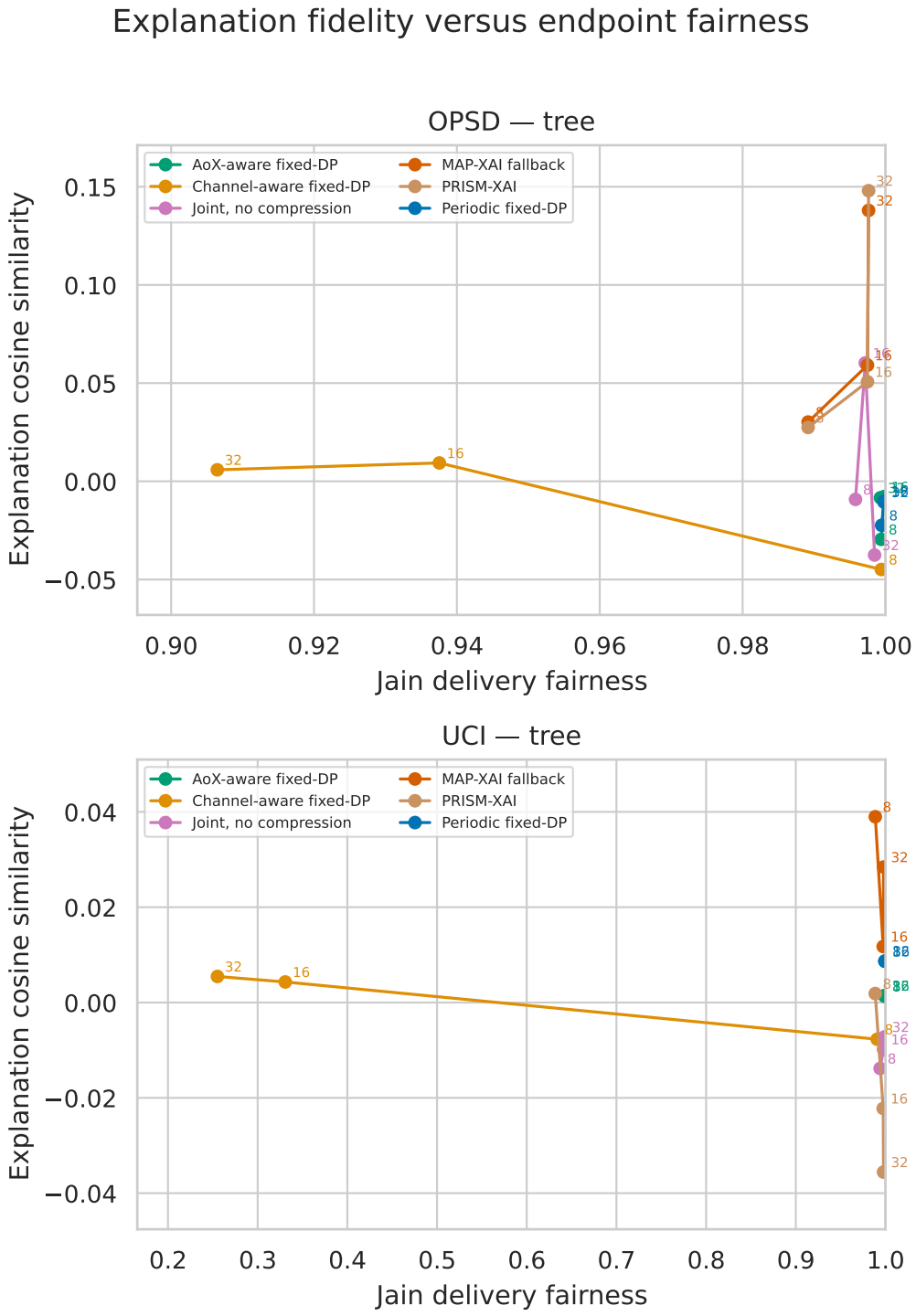


Fig. 126. Public-score threshold $\tau = 3$, release cap $M = 32$, Combined datasets: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

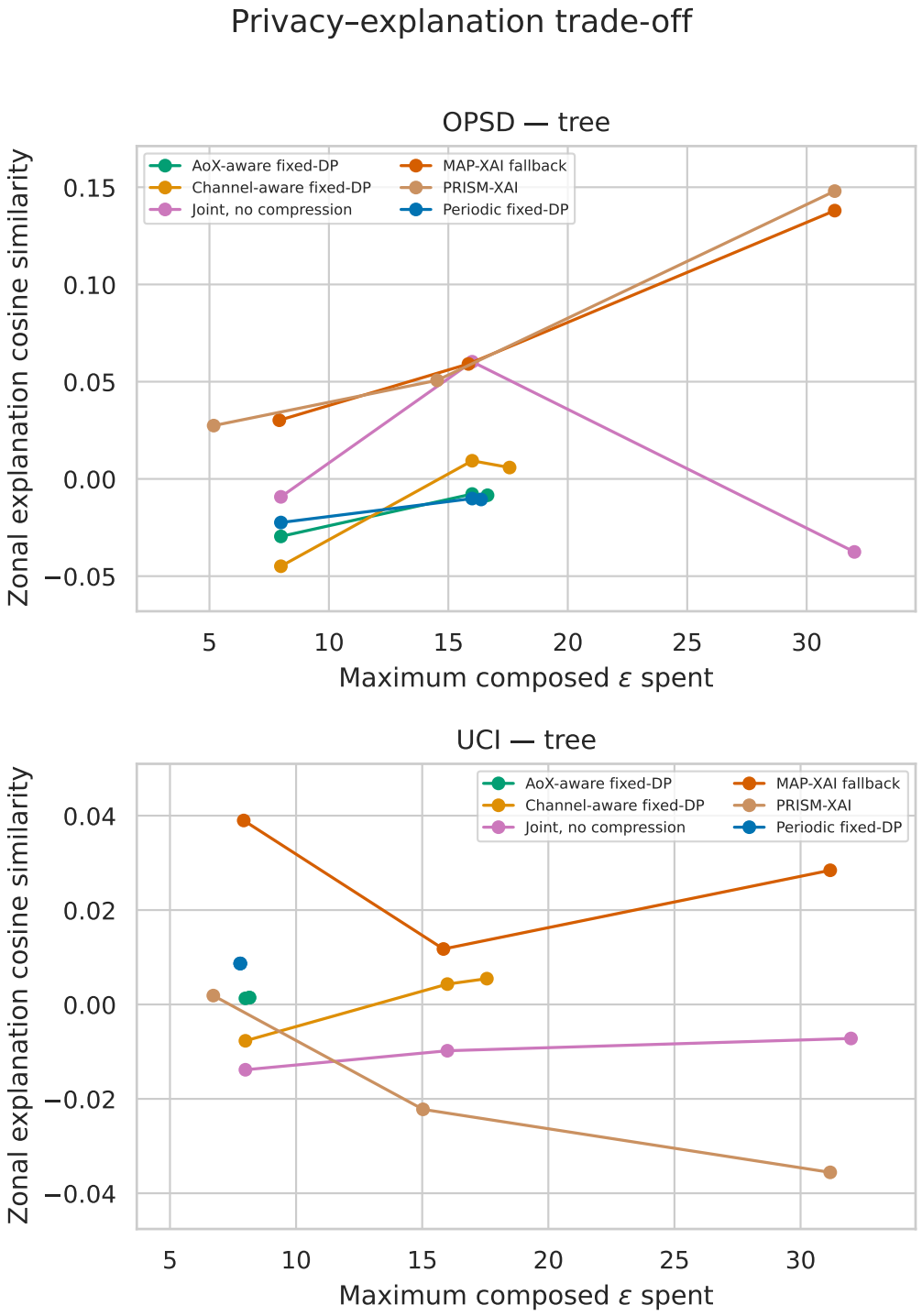


Fig. 127. Public-score threshold $\tau = 3$, release cap $M = 32$, Combined datasets: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

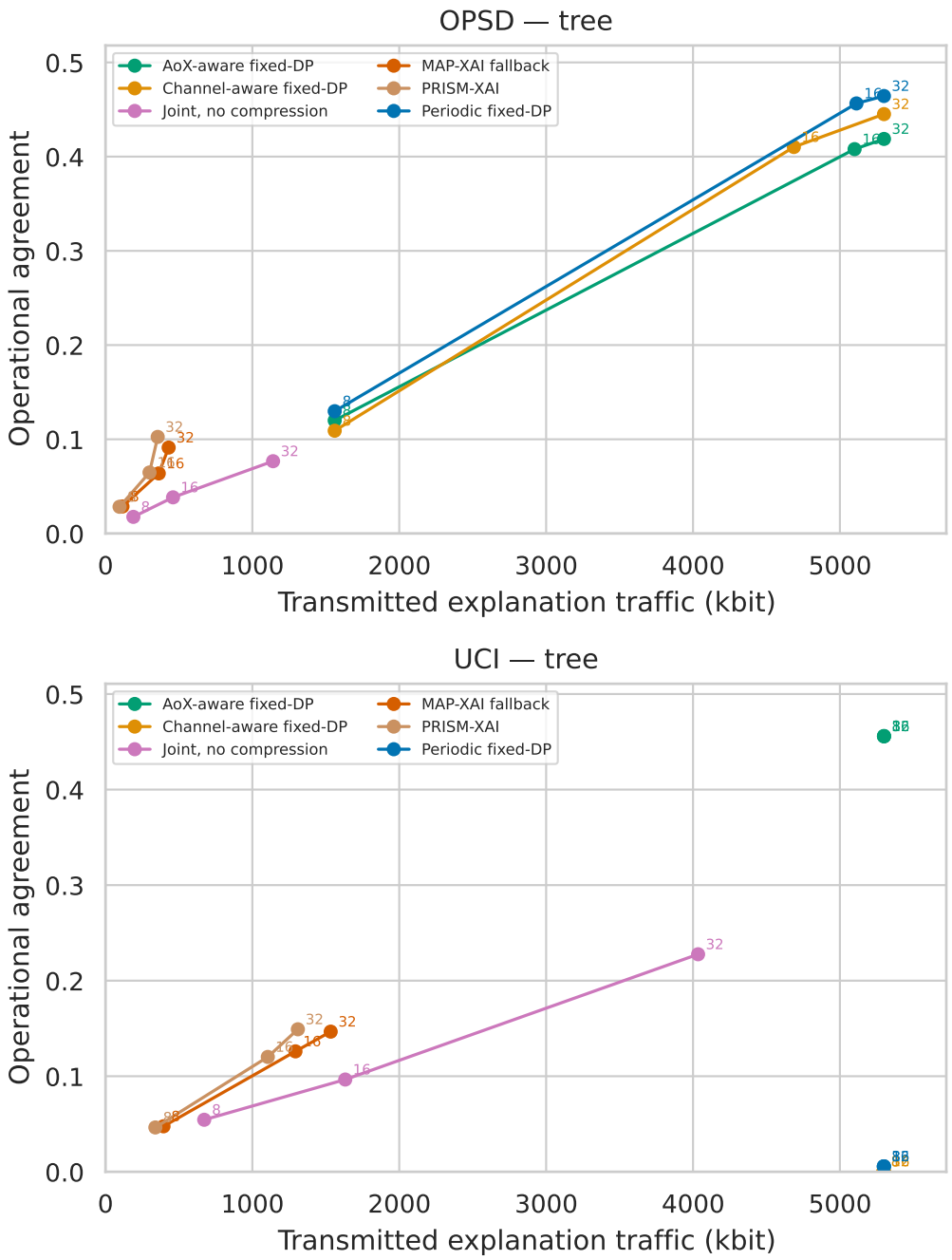


Fig. 128. Public-score threshold $\tau = 3$, release cap $M = 32$, Combined datasets: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

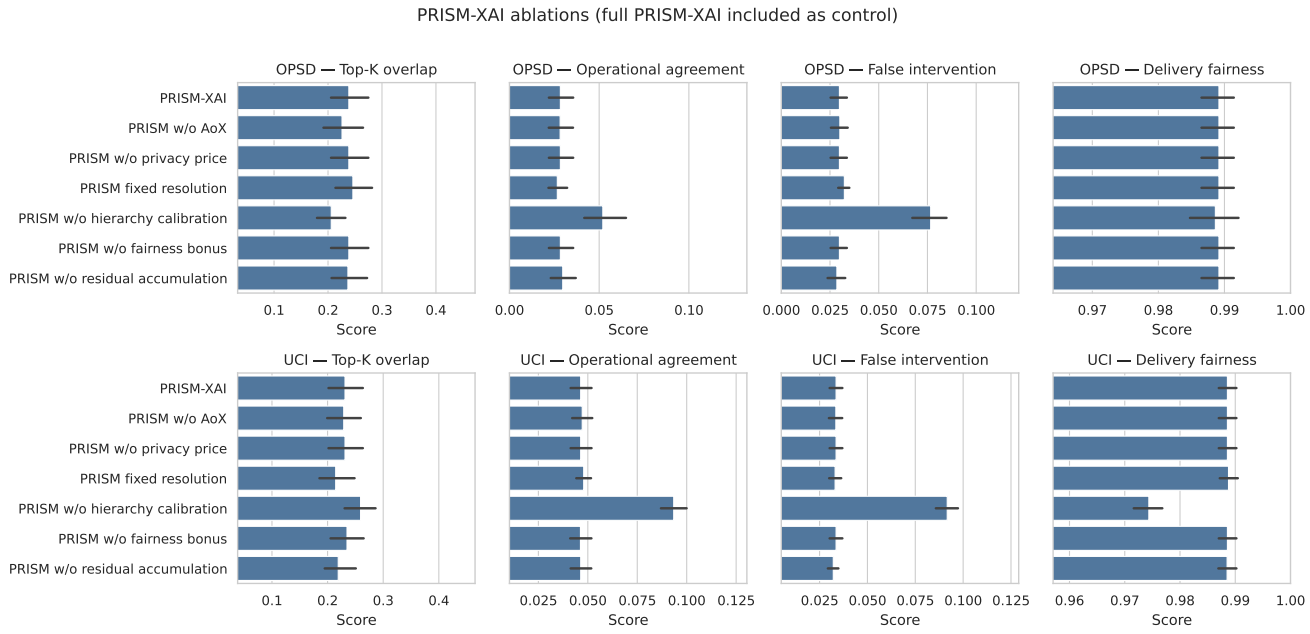


Fig. 129. Public-score threshold $\tau = 3$, release cap $M = 32$, Combined datasets: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

b) OPSD.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XCI
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, OPSD: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
OPSD	Tree	8	AoX-aware fixed-DP	0.234 ± 0.048	-0.033 ± 0.028	0.119 ± 0.028	0.119 ± 0.028	0.198 ± 0.054	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.183 ± 0.020	35567.482 ± 10569.818	0.999 ± 0.000
OPSD	Tree	8	AoX-aware fixed-DP	0.238 ± 0.049	-0.026 ± 0.055	0.121 ± 0.026	0.121 ± 0.026	0.205 ± 0.053	0.721 ± 0.005	6.903 ± 1.126	1560.564 ± 0.000	7.990 ± 0.000	0.179 ± 0.023	34123.214 ± 8376.542	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.037	-0.041 ± 0.051	0.107 ± 0.026	0.107 ± 0.026	0.217 ± 0.068	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.154 ± 0.029	40860.756 ± 14126.920	0.999 ± 0.000
OPSD	Tree	8	Channel-aware fixed-DP	0.292 ± 0.054	-0.049 ± 0.064	0.111 ± 0.023	0.111 ± 0.023	0.228 ± 0.066	0.779 ± 0.011	5.571 ± 1.784	1560.564 ± 0.000	7.990 ± 0.000	0.149 ± 0.026	37980.855 ± 12338.094	0.999 ± 0.000
OPSD	Tree	8	Joint, no compression	0.288 ± 0.064	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.957 ± 0.003	4.560 ± 1.811	189.900 ± 0.554	7.990 ± 0.000	0.027 ± 0.005	32490.357 ± 10149.755	0.996 ± 0.002
OPSD	Tree	8	Joint, no compression	0.258 ± 0.055	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.957 ± 0.003	4.562 ± 1.809	189.750 ± 0.566	7.990 ± 0.000	0.025 ± 0.003	26079.167 ± 6335.386	0.996 ± 0.002
OPSD	Tree	8	Periodic fixed-DP	0.241 ± 0.049	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.181 ± 0.021	31183.082 ± 4356.378	0.999 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	0.242 ± 0.048	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.722 ± 0.006	6.303 ± 1.438	1560.564 ± 0.000	7.990 ± 0.000	0.169 ± 0.017	28520.235 ± 3511.595	0.999 ± 0.000
OPSD	Tree	8	PRISM-XAI	0.257 ± 0.067	0.028 ± 0.059	0.028 ± 0.011	0.028 ± 0.011	0.273 ± 0.131	0.923 ± 0.004	4.713 ± 1.733	96.298 ± 1.228	5.340 ± 0.419	0.030 ± 0.007	11906.519 ± 6206.672	0.989 ± 0.004
OPSD	Tree	8	PRISM-XAI	0.220 ± 0.047	0.027 ± 0.062	0.029 ± 0.013	0.029 ± 0.013	0.278 ± 0.137	0.923 ± 0.004	4.708 ± 1.737	95.592 ± 1.248	5.012 ± 0.388	0.030 ± 0.007	13471.090 ± 9725.652	0.989 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.240 ± 0.060	0.022 ± 0.057	0.030 ± 0.008	0.030 ± 0.008	0.271 ± 0.091	0.923 ± 0.004	4.795 ± 1.684	115.748 ± 1.149	7.924 ± 0.000	0.031 ± 0.006	10380.916 ± 2295.453	0.989 ± 0.004
OPSD	Tree	8	MAP-XAI fallback	0.242 ± 0.055	0.038 ± 0.059	0.028 ± 0.007	0.028 ± 0.007	0.248 ± 0.070	0.923 ± 0.004	4.795 ± 1.684	115.607 ± 1.156	7.924 ± 0.000	0.032 ± 0.005	11185.765 ± 2730.237	0.989 ± 0.004

TABLE XCII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, OPSD: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
OPSD	Tree	8	Joint, no compression	189.900 ± 0.554	0.288 ± 0.064	0.250 ± 0.000	-0.030 ± 0.079	0.016 ± 0.005	0.016 ± 0.005	0.193 ± 0.059	0.043 ± 0.003	0.957 ± 0.003	0.027 ± 0.005	7.990 ± 0.000
OPSD	Tree	8	Joint, no compression	189.750 ± 0.566	0.258 ± 0.055	0.250 ± 0.000	0.011 ± 0.094	0.019 ± 0.004	0.019 ± 0.004	0.224 ± 0.044	0.043 ± 0.003	0.957 ± 0.003	0.025 ± 0.003	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.241 ± 0.049	0.250 ± 0.000	-0.019 ± 0.061	0.125 ± 0.020	0.125 ± 0.020	0.209 ± 0.044	0.278 ± 0.006	0.722 ± 0.006	0.181 ± 0.021	7.990 ± 0.000
OPSD	Tree	8	Periodic fixed-DP	1560.564 ± 0.000	0.242 ± 0.048	0.250 ± 0.000	-0.026 ± 0.082	0.135 ± 0.016	0.135 ± 0.016	0.231 ± 0.037	0.278 ± 0.006	0.722 ± 0.006	0.169 ± 0.017	7.990 ± 0.000
OPSD	Tree	8	PRISM-XAI	96.298 ± 1.228	0.257 ± 0.067	0.250 ± 0.000	0.028 ± 0.059	0.028 ± 0.011	0.028 ± 0.011	0.273 ± 0.131	0.077 ± 0.004	0.923 ± 0.004	0.030 ± 0.007	5.340 ± 0.419
OPSD	Tree	8	PRISM-XAI	95.592 ± 1.248	0.220 ± 0.047	0.250 ± 0.000	0.027 ± 0.062	0.029 ± 0.013	0.029 ± 0.013	0.278 ± 0.137	0.077 ± 0.004	0.923 ± 0.004	0.030 ± 0.007	5.012 ± 0.388
OPSD	Tree	16	Joint, no compression	459.900 ± 0.750	0.232 ± 0.065	0.250 ± 0.000	0.059 ± 0.048	0.038 ± 0.009	0.038 ± 0.009	0.222 ± 0.049	0.088 ± 0.004	0.912 ± 0.004	0.049 ± 0.005	15.997 ± 0.000
OPSD	Tree	16	Joint, no compression	460.050 ± 0.724	0.241 ± 0.055	0.250 ± 0.000	0.061 ± 0.052	0.039 ± 0.008	0.039 ± 0.008	0.241 ± 0.042	0.088 ± 0.004	0.912 ± 0.004	0.045 ± 0.004	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.045	0.450 ± 0.062	0.450 ± 0.062	0.237 ± 0.042	0.985 ± 0.002	0.015 ± 0.002	0.551 ± 0.057	15.997 ± 0.000
OPSD	Tree	16	Periodic fixed-DP	5112.594 ± 0.000	0.243 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.463 ± 0.047	0.463 ± 0.047	0.240 ± 0.028	0.985 ± 0.002	0.015 ± 0.002	0.547 ± 0.036	15.997 ± 0.000
OPSD	Tree	16	PRISM-XAI	300.705 ± 0.565	0.278 ± 0.056	0.250 ± 0.000	0.047 ± 0.080	0.066 ± 0.020	0.066 ± 0.020	0.224 ± 0.088	0.200 ± 0.005	0.800 ± 0.005	0.090 ± 0.015	14.564 ± 0.022
OPSD	Tree	16	PRISM-XAI	298.955 ± 0.657	0.289 ± 0.047	0.250 ± 0.000	0.055 ± 0.090	0.063 ± 0.021	0.063 ± 0.021	0.207 ± 0.090	0.199 ± 0.004	0.801 ± 0.004	0.097 ± 0.018	14.498 ± 0.044
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.235 ± 0.064	0.250 ± 0.000	-0.031 ± 0.089	0.074 ± 0.015	0.074 ± 0.015	0.205 ± 0.039	0.157 ± 0.006	0.843 ± 0.006	0.107 ± 0.009	31.999 ± 0.000
OPSD	Tree	32	Joint, no compression	1140.900 ± 7.953	0.282 ± 0.054	0.250 ± 0.000	-0.044 ± 0.076	0.080 ± 0.011	0.080 ± 0.011	0.227 ± 0.052	0.158 ± 0.006	0.842 ± 0.006	0.105 ± 0.015	31.999 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.249 ± 0.022	0.250 ± 0.000	-0.009 ± 0.044	0.458 ± 0.066	0.458 ± 0.066	0.237 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.562 ± 0.062	16.371 ± 0.000
OPSD	Tree	32	Periodic fixed-DP	5299.836 ± 0.625	0.244 ± 0.019	0.250 ± 0.000	-0.012 ± 0.050	0.471 ± 0.049	0.471 ± 0.049	0.240 ± 0.029	1.000 ± 0.000	0.000 ± 0.000	0.559 ± 0.039	16.371 ± 0.000
OPSD	Tree	32	PRISM-XAI	356.570 ± 0.841	0.321 ± 0.068	0.250 ± 0.000	0.150 ± 0.077	0.105 ± 0.023	0.105 ± 0.023	0.301 ± 0.084	0.243 ± 0.007	0.757 ± 0.007	0.094 ± 0.017	31.175 ± 0.000
OPSD	Tree	32	PRISM-XAI	354.636 ± 0.782	0.326 ± 0.048	0.250 ± 0.000	0.146 ± 0.091	0.101 ± 0.022	0.101 ± 0.022	0.283 ± 0.077	0.243 ± 0.008	0.757 ± 0.008	0.099 ± 0.016	31.175 ± 0.000

TABLE XCIII
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, OPSD: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
OPSD	AoX-aware fixed-DP	-1464.266 ± 1.228	0.024 ± 0.053	-0.092 ± 0.036	-0.010 ± 0.004
OPSD	AoX-aware fixed-DP	-4799.207 ± 5.399	0.025 ± 0.072	-0.335 ± 0.087	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-4943.654 ± 0.841	0.068 ± 0.080	-0.308 ± 0.086	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-1464.972 ± 1.248	-0.018 ± 0.064	-0.092 ± 0.035	-0.010 ± 0.004
OPSD	AoX-aware fixed-DP	-4800.957 ± 5.327	0.036 ± 0.052	-0.351 ± 0.071	-0.002 ± 0.001
OPSD	AoX-aware fixed-DP	-4945.588 ± 0.782	0.074 ± 0.055	-0.325 ± 0.067	-0.002 ± 0.001
OPSD	Channel-aware fixed-DP	-1464.266 ± 1.228	-0.035 ± 0.061	-0.079 ± 0.032	-0.010 ± 0.004
OPSD	Channel-aware fixed-DP	-4385.904 ± 13.238	0.016 ± 0.064	-0.335 ± 0.100	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4943.654 ± 0.841	0.063 ± 0.076	-0.330 ± 0.104	0.091 ± 0.005
OPSD	Channel-aware fixed-DP	-1464.972 ± 1.248	-0.072 ± 0.074	-0.082 ± 0.031	-0.010 ± 0.004
OPSD	Channel-aware fixed-DP	-4387.654 ± 13.405	0.031 ± 0.047	-0.356 ± 0.098	0.060 ± 0.005
OPSD	Channel-aware fixed-DP	-4945.588 ± 0.782	0.069 ± 0.051	-0.355 ± 0.094	0.091 ± 0.005
OPSD	Joint, no compression	-93.602 ± 1.154	-0.031 ± 0.103	0.011 ± 0.012	-0.007 ± 0.004
OPSD	Joint, no compression	-159.195 ± 1.122	0.047 ± 0.099	0.028 ± 0.023	0.000 ± 0.001
OPSD	Joint, no compression	-784.330 ± 7.944	0.086 ± 0.101	0.031 ± 0.032	-0.001 ± 0.001
OPSD	Joint, no compression	-94.158 ± 1.241	-0.038 ± 0.069	0.010 ± 0.013	-0.007 ± 0.004
OPSD	Joint, no compression	-161.095 ± 1.115	0.048 ± 0.086	0.025 ± 0.021	0.000 ± 0.001
OPSD	Joint, no compression	-786.264 ± 8.092	0.044 ± 0.080	0.021 ± 0.024	-0.001 ± 0.001
OPSD	Periodic fixed-DP	-1464.266 ± 1.228	0.017 ± 0.044	-0.097 ± 0.025	-0.010 ± 0.004
OPSD	Periodic fixed-DP	-4811.889 ± 0.565	0.029 ± 0.053	-0.384 ± 0.072	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-4943.266 ± 0.807	0.072 ± 0.074	-0.353 ± 0.071	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-1464.972 ± 1.248	-0.022 ± 0.054	-0.106 ± 0.022	-0.010 ± 0.004
OPSD	Periodic fixed-DP	-4813.639 ± 0.657	0.046 ± 0.040	-0.399 ± 0.060	-0.002 ± 0.001
OPSD	Periodic fixed-DP	-4945.200 ± 0.800	0.082 ± 0.043	-0.370 ± 0.053	-0.002 ± 0.001
OPSD	MAP-XAI fallback	-19.450 ± 0.713	0.017 ± 0.052	-0.002 ± 0.006	0.000 ± 0.000
OPSD	MAP-XAI fallback	-62.093 ± 0.598	-0.013 ± 0.032	0.003 ± 0.015	0.000 ± 0.000
OPSD	MAP-XAI fallback	-73.728 ± 0.773	-0.007 ± 0.048	0.011 ± 0.016	0.000 ± 0.000
OPSD	MAP-XAI fallback	-20.015 ± 0.643	-0.022 ± 0.056	0.001 ± 0.007	0.000 ± 0.000
OPSD	MAP-XAI fallback	-63.701 ± 0.679	-0.004 ± 0.046	-0.001 ± 0.018	0.000 ± 0.000
OPSD	MAP-XAI fallback	-75.500 ± 0.722	0.005 ± 0.038	0.012 ± 0.013	0.000 ± 0.000

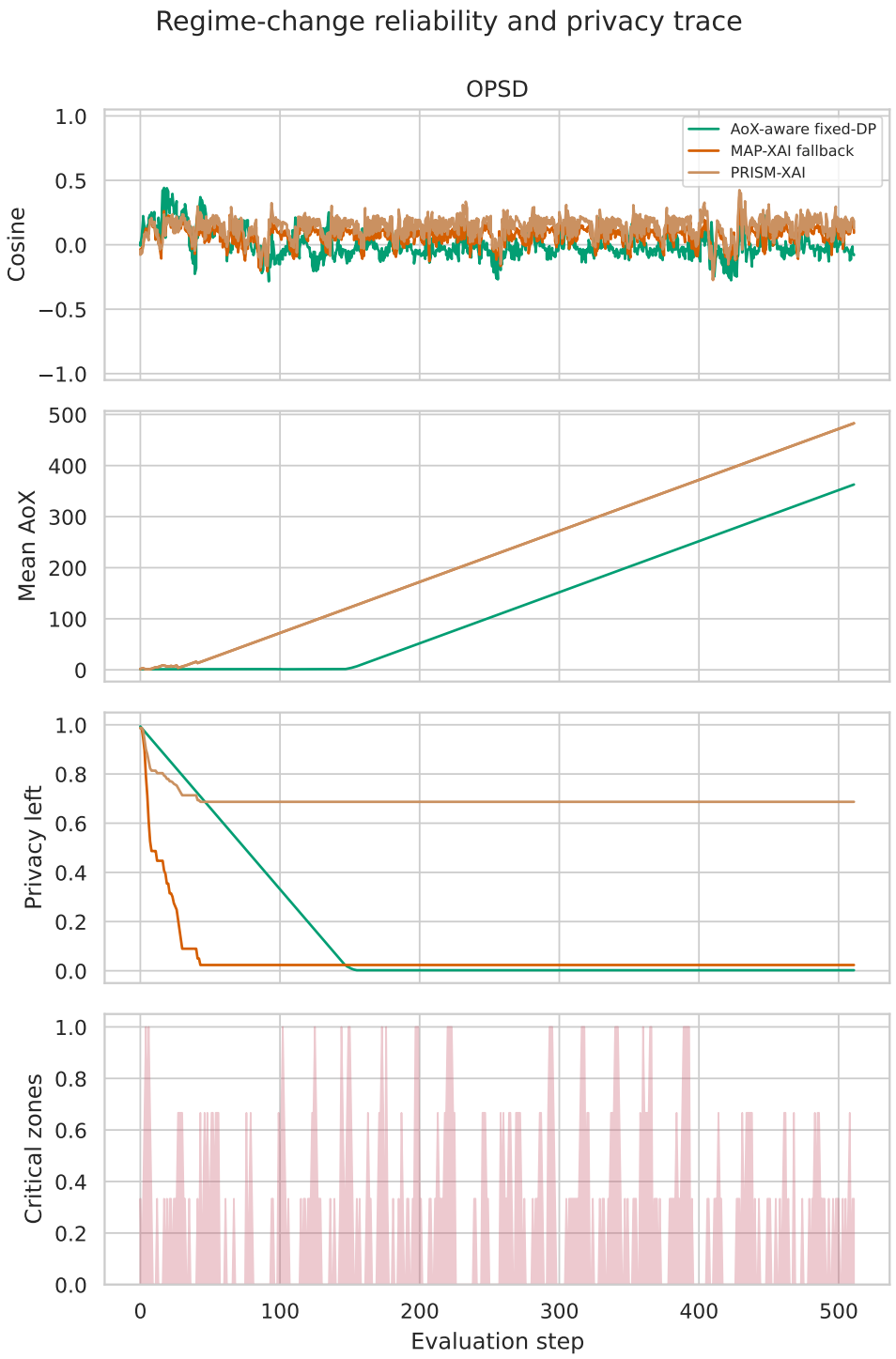


Fig. 130. Public-score threshold $\tau = 3$, release cap $M = 32$, OPSD: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

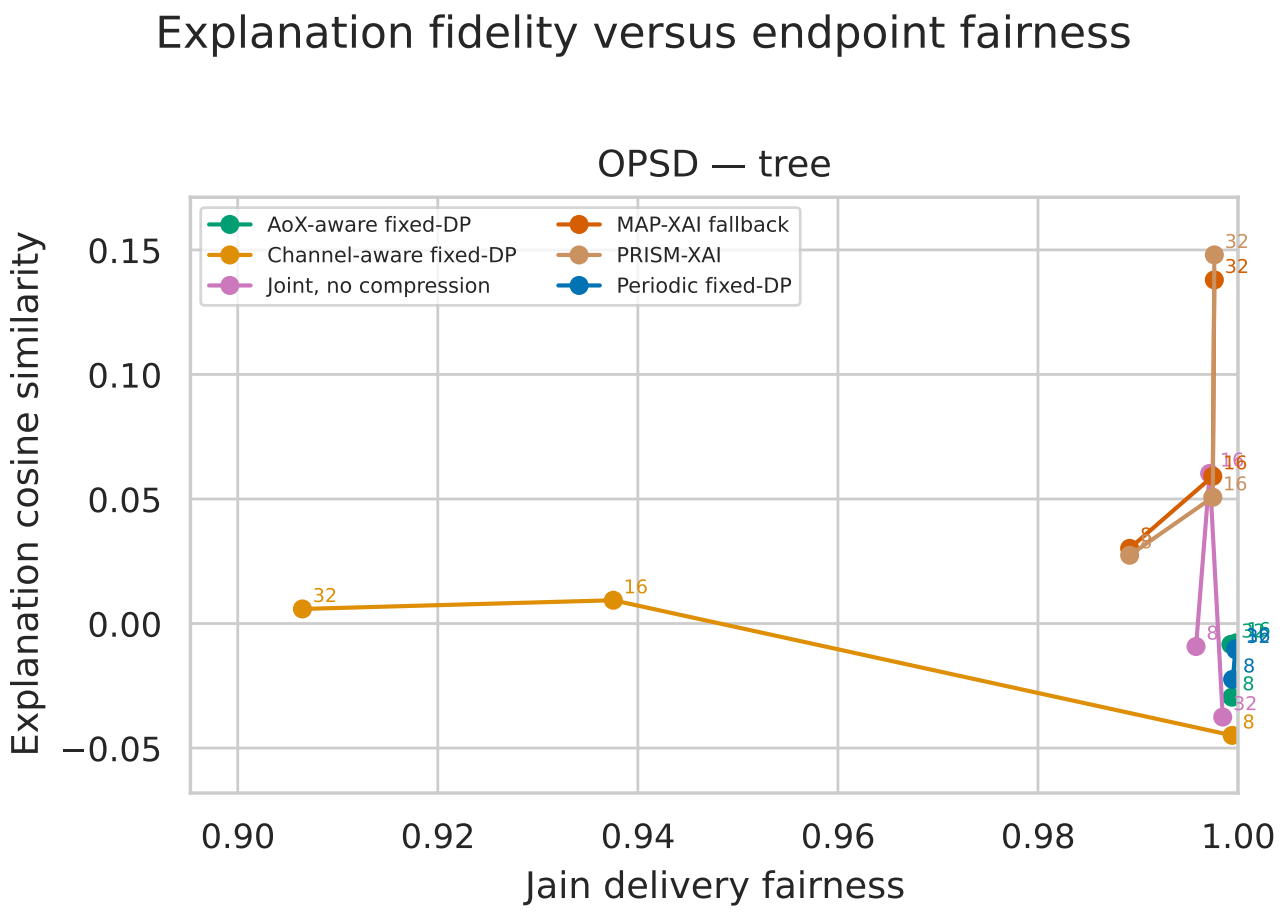


Fig. 131. Public-score threshold $\tau = 3$, release cap $M = 32$, OPSD: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

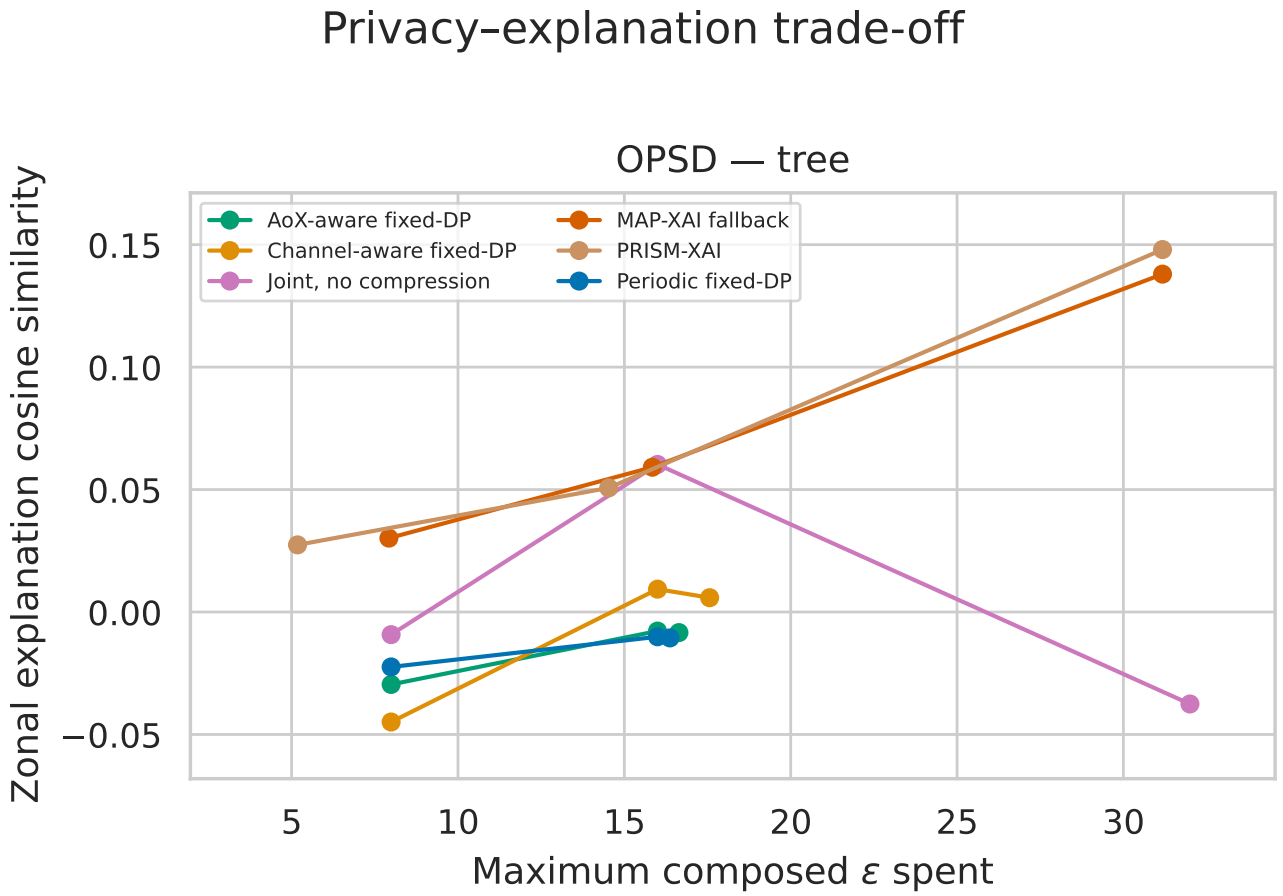


Fig. 132. Public-score threshold $\tau = 3$, release cap $M = 32$, OPSD: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

Communication–operation trade-off (labels show privacy budget)

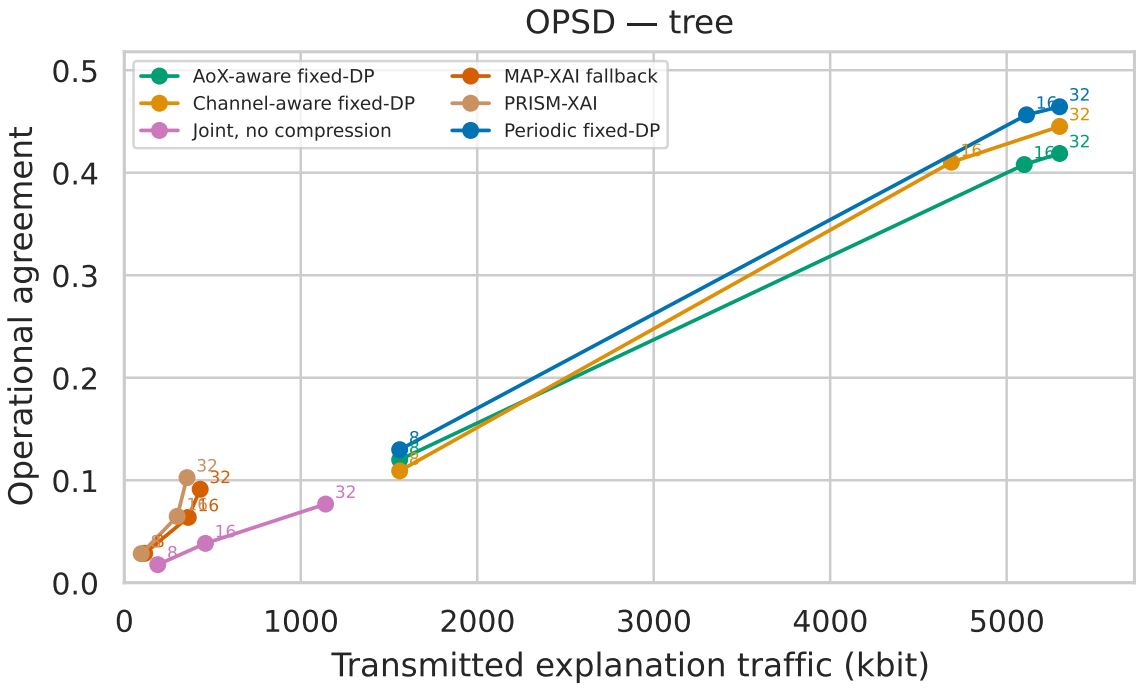


Fig. 133. Public-score threshold $\tau = 3$, release cap $M = 32$, OPSD: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

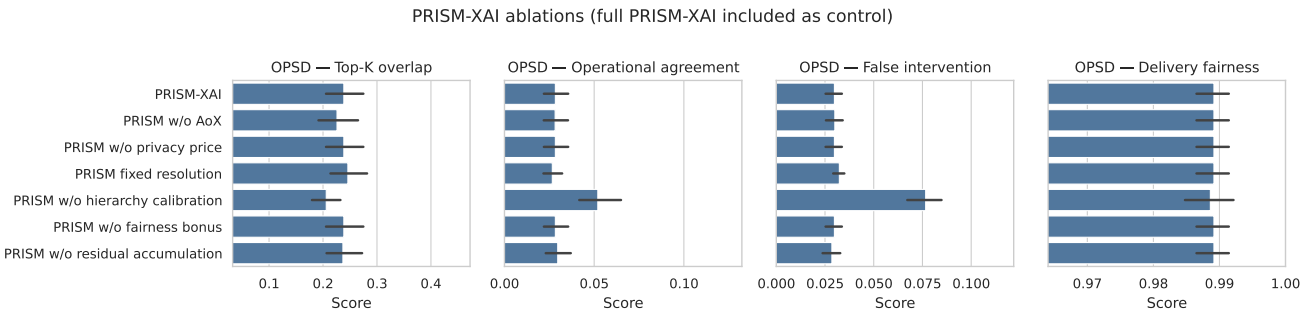


Fig. 134. Public-score threshold $\tau = 3$, release cap $M = 32$, OPSD: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

c) UCI.: The three generated tables below report the full method means, all condition-level values, and paired differences. The five figures that follow provide the corresponding temporal, fairness, privacy, Pareto, and ablation views. Aggregate panels support the setting-level summary above; dataset panels reveal whether that summary is shared by OPSD and UCI rather than being driven by their different endpoint counts.

TABLE XCIV
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, UCI: COMPLETE GENERATED METHOD TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	Top-K	Cosine	Decision	Recall	Prec.	Abst.	P95 ms	kbit	Spent ϵ	False int.	bit/correct	Fairness
UCI	Tree	8	AoX-aware fixed-DP	0.263 ± 0.041	-0.007 ± 0.009	0.457 ± 0.031	0.457 ± 0.031	0.362 ± 0.024	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.025	14002.896 ± 1040.243	0.999 ± 0.000
UCI	Tree	8	AoX-aware fixed-DP	0.260 ± 0.022	0.010 ± 0.055	0.455 ± 0.047	0.455 ± 0.047	0.361 ± 0.038	0.003 ± 0.001	7.500 ± 0.000	5300.224 ± 0.000	7.990 ± 0.000	0.557 ± 0.048	14198.303 ± 1508.274	0.999 ± 0.000
UCI	Tree	8	Channel-aware fixed-DP	0.248 ± 0.031	0.004 ± 0.012	0.001 ± 0.001	0.001 ± 0.001	0.150 ± 0.151	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	4632196.500 ± 2102650.252	0.991 ± 0.001
UCI	Tree	8	Channel-aware fixed-DP	0.239 ± 0.030	-0.020 ± 0.055	0.000 ± 0.000	0.000 ± 0.000	0.117 ± 0.225	0.999 ± 0.001	4.609 ± 1.781	5295.177 ± 3.740	7.990 ± 0.000	0.001 ± 0.001	5296989.000 ± 41104.572	0.991 ± 0.001
UCI	Tree	8	Joint, no compression	0.286 ± 0.045	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.875 ± 0.003	5.123 ± 1.792	672.300 ± 0.452	7.990 ± 0.000	0.053 ± 0.004	15676.175 ± 1475.621	0.994 ± 0.003
UCI	Tree	8	Joint, no compression	0.229 ± 0.052	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.875 ± 0.003	5.131 ± 1.787	672.300 ± 0.452	7.990 ± 0.000	0.052 ± 0.004	14702.786 ± 2334.488	0.994 ± 0.003
UCI	Tree	8	Periodic fixed-DP	0.259 ± 0.060	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.013 ± 0.002	1215103.896 ± 390671.319	0.999 ± 0.000
UCI	Tree	8	Periodic fixed-DP	0.260 ± 0.044	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.989 ± 0.001	4.596 ± 1.788	5299.318 ± 1.451	7.786 ± 0.000	0.015 ± 0.002	1555837.203 ± 955472.503	0.999 ± 0.000
UCI	Tree	8	PRISM-XAI	0.207 ± 0.052	0.004 ± 0.012	0.046 ± 0.010	0.046 ± 0.010	0.469 ± 0.081	0.901 ± 0.005	5.035 ± 1.860	341.133 ± 1.293	6.779 ± 0.208	0.035 ± 0.005	9780.191 ± 2585.599	0.988 ± 0.003
UCI	Tree	8	PRISM-XAI	0.256 ± 0.043	0.000 ± 0.081	0.047 ± 0.008	0.047 ± 0.008	0.503 ± 0.086	0.902 ± 0.005	5.524 ± 1.825	338.064 ± 2.047	6.657 ± 0.214	0.032 ± 0.007	9159.950 ± 1670.303	0.989 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.193 ± 0.046	0.014 ± 0.010	0.046 ± 0.007	0.046 ± 0.007	0.479 ± 0.053	0.901 ± 0.005	5.106 ± 1.806	395.052 ± 0.314	7.924 ± 0.000	0.035 ± 0.004	10596.392 ± 1554.907	0.988 ± 0.003
UCI	Tree	8	MAP-XAI fallback	0.246 ± 0.061	0.064 ± 0.067	0.049 ± 0.007	0.049 ± 0.007	0.510 ± 0.076	0.902 ± 0.005	5.409 ± 1.720	395.226 ± 1.353	7.924 ± 0.000	0.033 ± 0.007	10216.728 ± 2063.280	0.989 ± 0.003

TABLE XCV
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, UCI: COMPLETE CONDITION-LEVEL GENERATED TABLE.

Dataset	Model	$\epsilon_{\max}$	Method	kbit	Top-K	Random Top-K	Cosine	Decision	Recall	Precision	Coverage	Abstain	False int.	Spent ϵ
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.286 ± 0.045	0.250 ± 0.000	0.004 ± 0.020	0.052 ± 0.005	0.052 ± 0.005	0.406 ± 0.037	0.125 ± 0.003	0.875 ± 0.003	0.053 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Joint, no compression	672.300 ± 0.452	0.229 ± 0.052	0.250 ± 0.000	-0.032 ± 0.050	0.057 ± 0.007	0.057 ± 0.007	0.428 ± 0.041	0.125 ± 0.003	0.875 ± 0.003	0.052 ± 0.004	7.990 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	8	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	8	PRISM-XAI	341.133 ± 1.293	0.207 ± 0.052	0.250 ± 0.000	0.004 ± 0.012	0.046 ± 0.010	0.046 ± 0.010	0.469 ± 0.081	0.099 ± 0.005	0.901 ± 0.005	0.035 ± 0.005	6.779 ± 0.208
UCI	Tree	8	PRISM-XAI	338.064 ± 2.047	0.256 ± 0.043	0.250 ± 0.000	0.000 ± 0.081	0.047 ± 0.008	0.047 ± 0.008	0.503 ± 0.086	0.098 ± 0.005	0.902 ± 0.005	0.032 ± 0.007	6.657 ± 0.214
UCI	Tree	16	Joint, no compression	1632.150 ± 0.339	0.216 ± 0.041	0.250 ± 0.000	0.001 ± 0.012	0.094 ± 0.005	0.094 ± 0.005	0.330 ± 0.024	0.228 ± 0.005	0.772 ± 0.005	0.133 ± 0.010	15.997 ± 0.000
UCI	Tree	16	Joint, no compression	1632.600 ± 0.554	0.204 ± 0.011	0.250 ± 0.000	-0.021 ± 0.033	0.099 ± 0.016	0.099 ± 0.016	0.347 ± 0.054	0.228 ± 0.005	0.772 ± 0.005	0.128 ± 0.011	15.997 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	16	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	16	PRISM-XAI	1110.802 ± 3.293	0.269 ± 0.046	0.250 ± 0.000	0.018 ± 0.010	0.126 ± 0.013	0.126 ± 0.013	0.407 ± 0.047	0.285 ± 0.006	0.715 ± 0.006	0.127 ± 0.012	15.063 ± 0.065
UCI	Tree	16	PRISM-XAI	1099.581 ± 2.815	0.240 ± 0.048	0.250 ± 0.000	-0.063 ± 0.054	0.115 ± 0.016	0.115 ± 0.016	0.379 ± 0.064	0.282 ± 0.007	0.718 ± 0.007	0.132 ± 0.019	14.998 ± 0.062
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.246 ± 0.031	0.250 ± 0.000	-0.002 ± 0.015	0.228 ± 0.013	0.228 ± 0.013	0.361 ± 0.028	0.537 ± 0.009	0.463 ± 0.009	0.279 ± 0.017	31.999 ± 0.000
UCI	Tree	32	Joint, no compression	4034.550 ± 5.034	0.241 ± 0.033	0.250 ± 0.000	-0.012 ± 0.030	0.228 ± 0.028	0.228 ± 0.028	0.360 ± 0.047	0.530 ± 0.010	0.470 ± 0.010	0.280 ± 0.026	31.999 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.259 ± 0.060	0.250 ± 0.000	-0.005 ± 0.008	0.006 ± 0.001	0.006 ± 0.001	0.244 ± 0.049	0.011 ± 0.001	0.989 ± 0.001	0.013 ± 0.002	7.786 ± 0.000
UCI	Tree	32	Periodic fixed-DP	5299.318 ± 1.451	0.260 ± 0.044	0.250 ± 0.000	0.022 ± 0.070	0.005 ± 0.001	0.005 ± 0.001	0.195 ± 0.050	0.011 ± 0.001	0.989 ± 0.001	0.015 ± 0.002	7.786 ± 0.000
UCI	Tree	32	PRISM-XAI	1317.135 ± 2.878	0.286 ± 0.041	0.250 ± 0.000	0.016 ± 0.014	0.150 ± 0.019	0.150 ± 0.019	0.423 ± 0.062	0.338 ± 0.007	0.662 ± 0.007	0.143 ± 0.016	31.175 ± 0.000
UCI	Tree	32	PRISM-XAI	1302.454 ± 3.178	0.249 ± 0.044	0.250 ± 0.000	-0.087 ± 0.048	0.148 ± 0.011	0.148 ± 0.011	0.437 ± 0.045	0.335 ± 0.007	0.665 ± 0.007	0.132 ± 0.013	31.175 ± 0.000

TABLE XCVI
PUBLIC-SCORE THRESHOLD $\tau = 3$, RELEASE CAP $M = 32$, UCI: COMPLETE GENERATED PAIRED COMPARISONS.

Dataset	Baseline	Delta kbit	Delta Top-K	Delta recall	Delta fairness
UCI	AoX-aware fixed-DP	-4959.091 ± 1.293	-0.056 ± 0.063	-0.411 ± 0.026	-0.011 ± 0.003
UCI	AoX-aware fixed-DP	-4189.422 ± 3.293	0.005 ± 0.058	-0.332 ± 0.035	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-3983.089 ± 2.878	0.023 ± 0.064	-0.307 ± 0.043	-0.001 ± 0.001
UCI	AoX-aware fixed-DP	-4962.160 ± 2.047	-0.004 ± 0.042	-0.409 ± 0.046	-0.010 ± 0.003
UCI	AoX-aware fixed-DP	-4200.643 ± 2.815	-0.020 ± 0.050	-0.340 ± 0.048	-0.002 ± 0.001
UCI	AoX-aware fixed-DP	-3997.770 ± 3.178	-0.012 ± 0.032	-0.307 ± 0.052	-0.001 ± 0.001
UCI	Channel-aware fixed-DP	-4954.044 ± 4.047	-0.041 ± 0.066	0.046 ± 0.010	-0.002 ± 0.003
UCI	Channel-aware fixed-DP	-4189.422 ± 3.293	0.001 ± 0.058	0.125 ± 0.013	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-3983.089 ± 2.878	0.018 ± 0.049	0.150 ± 0.019	0.743 ± 0.001
UCI	Channel-aware fixed-DP	-4957.113 ± 4.745	0.017 ± 0.045	0.046 ± 0.008	-0.002 ± 0.003
UCI	Channel-aware fixed-DP	-4200.643 ± 2.815	-0.020 ± 0.052	0.115 ± 0.016	0.667 ± 0.001
UCI	Channel-aware fixed-DP	-3997.770 ± 3.178	-0.011 ± 0.043	0.148 ± 0.011	0.743 ± 0.001
UCI	Joint, no compression	-331.167 ± 1.523	-0.079 ± 0.085	-0.006 ± 0.011	-0.006 ± 0.003
UCI	Joint, no compression	-521.348 ± 3.369	0.052 ± 0.038	0.031 ± 0.015	-0.000 ± 0.001
UCI	Joint, no compression	-2717.415 ± 6.000	0.040 ± 0.049	-0.077 ± 0.014	-0.001 ± 0.001
UCI	Joint, no compression	-334.236 ± 2.296	0.026 ± 0.066	-0.010 ± 0.011	-0.005 ± 0.003
UCI	Joint, no compression	-533.019 ± 2.619	0.037 ± 0.048	0.016 ± 0.022	-0.000 ± 0.001
UCI	Joint, no compression	-2732.096 ± 5.519	0.007 ± 0.058	-0.080 ± 0.032	-0.001 ± 0.001
UCI	Periodic fixed-DP	-4958.185 ± 2.040	-0.053 ± 0.077	0.040 ± 0.010	-0.011 ± 0.003
UCI	Periodic fixed-DP	-4188.516 ± 3.590	0.009 ± 0.089	0.120 ± 0.014	-0.002 ± 0.001
UCI	Periodic fixed-DP	-3982.184 ± 3.108	0.027 ± 0.068	0.145 ± 0.020	-0.002 ± 0.001
UCI	Periodic fixed-DP	-4961.254 ± 2.973	-0.005 ± 0.068	0.041 ± 0.007	-0.011 ± 0.003
UCI	Periodic fixed-DP	-4199.738 ± 3.048	-0.020 ± 0.078	0.109 ± 0.015	-0.002 ± 0.001
UCI	Periodic fixed-DP	-3996.864 ± 3.261	-0.012 ± 0.073	0.142 ± 0.011	-0.002 ± 0.001
UCI	MAP-XAI fallback	-53.919 ± 1.196	0.014 ± 0.036	-0.000 ± 0.005	0.000 ± 0.000
UCI	MAP-XAI fallback	-182.879 ± 3.287	0.006 ± 0.059	-0.002 ± 0.010	0.000 ± 0.000
UCI	MAP-XAI fallback	-216.465 ± 2.875	0.076 ± 0.044	0.002 ± 0.015	0.000 ± 0.000
UCI	MAP-XAI fallback	-57.162 ± 1.160	0.010 ± 0.051	-0.002 ± 0.008	-0.000 ± 0.001
UCI	MAP-XAI fallback	-193.593 ± 2.996	0.006 ± 0.034	-0.010 ± 0.011	-0.000 ± 0.000
UCI	MAP-XAI fallback	-230.577 ± 3.247	0.018 ± 0.059	0.003 ± 0.011	-0.000 ± 0.000

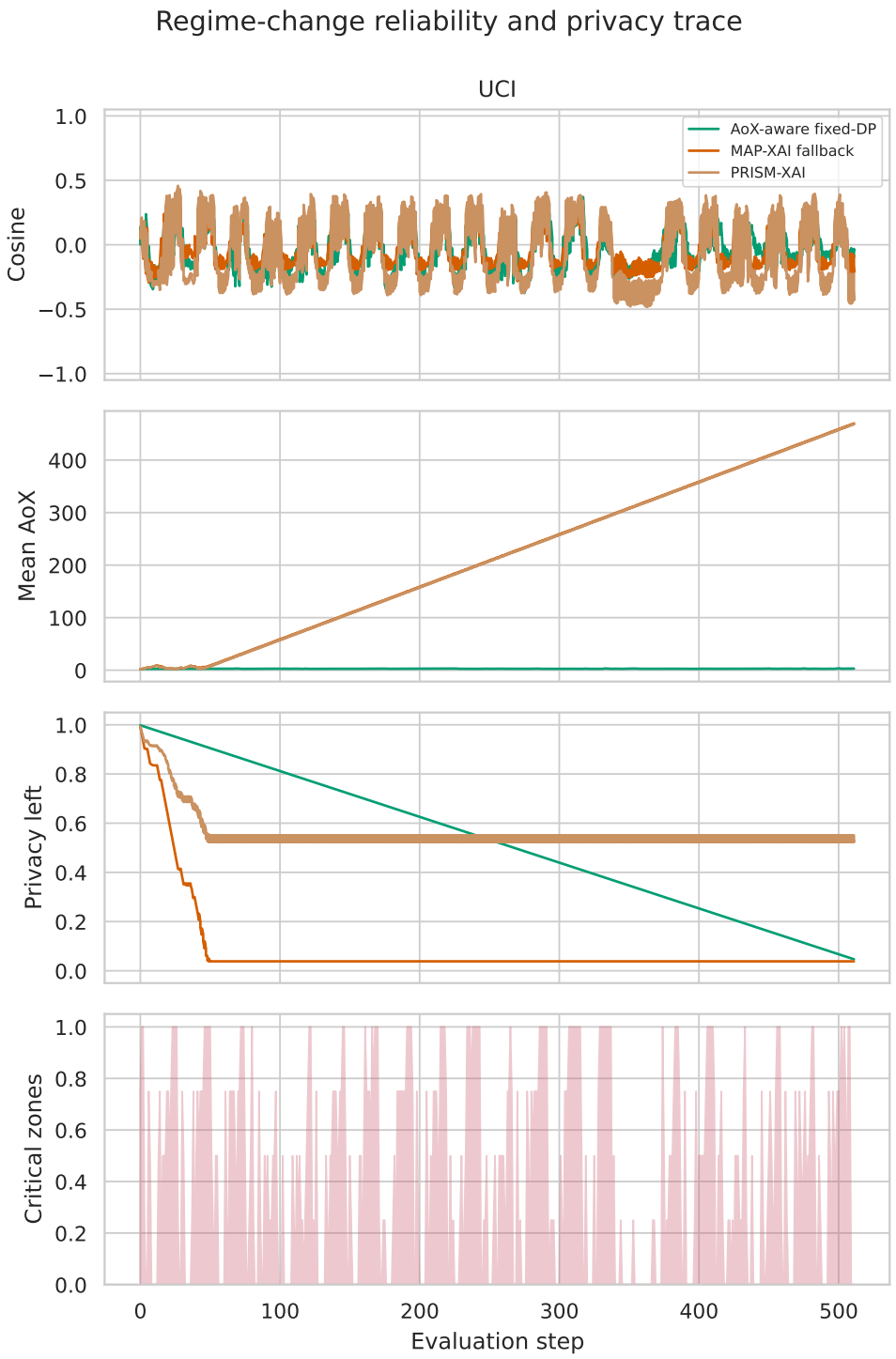


Fig. 135. Public-score threshold $\tau = 3$, release cap $M = 32$, UCI: representative regime trace. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

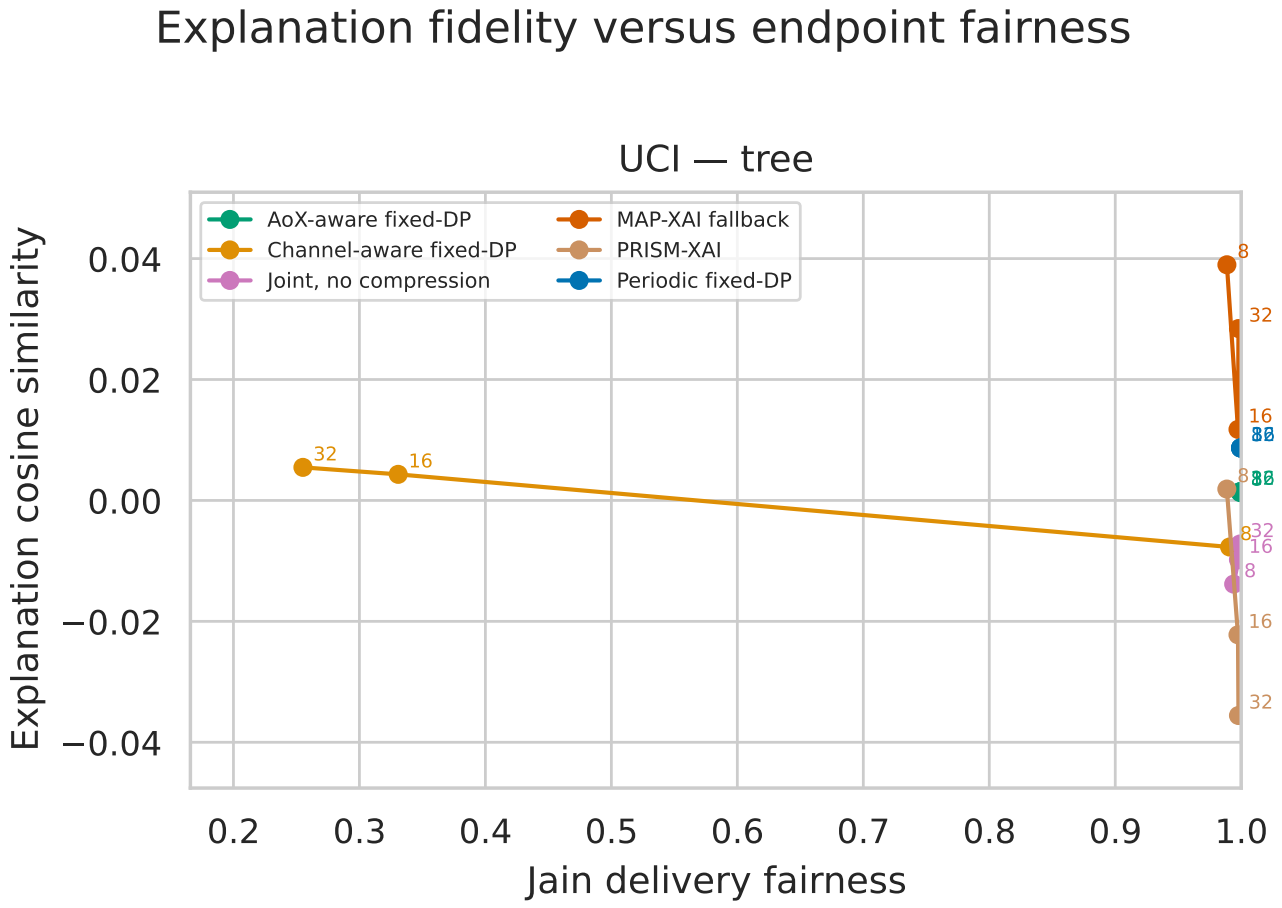


Fig. 136. Public-score threshold $\tau = 3$, release cap $M = 32$, UCI: fairness–reliability plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

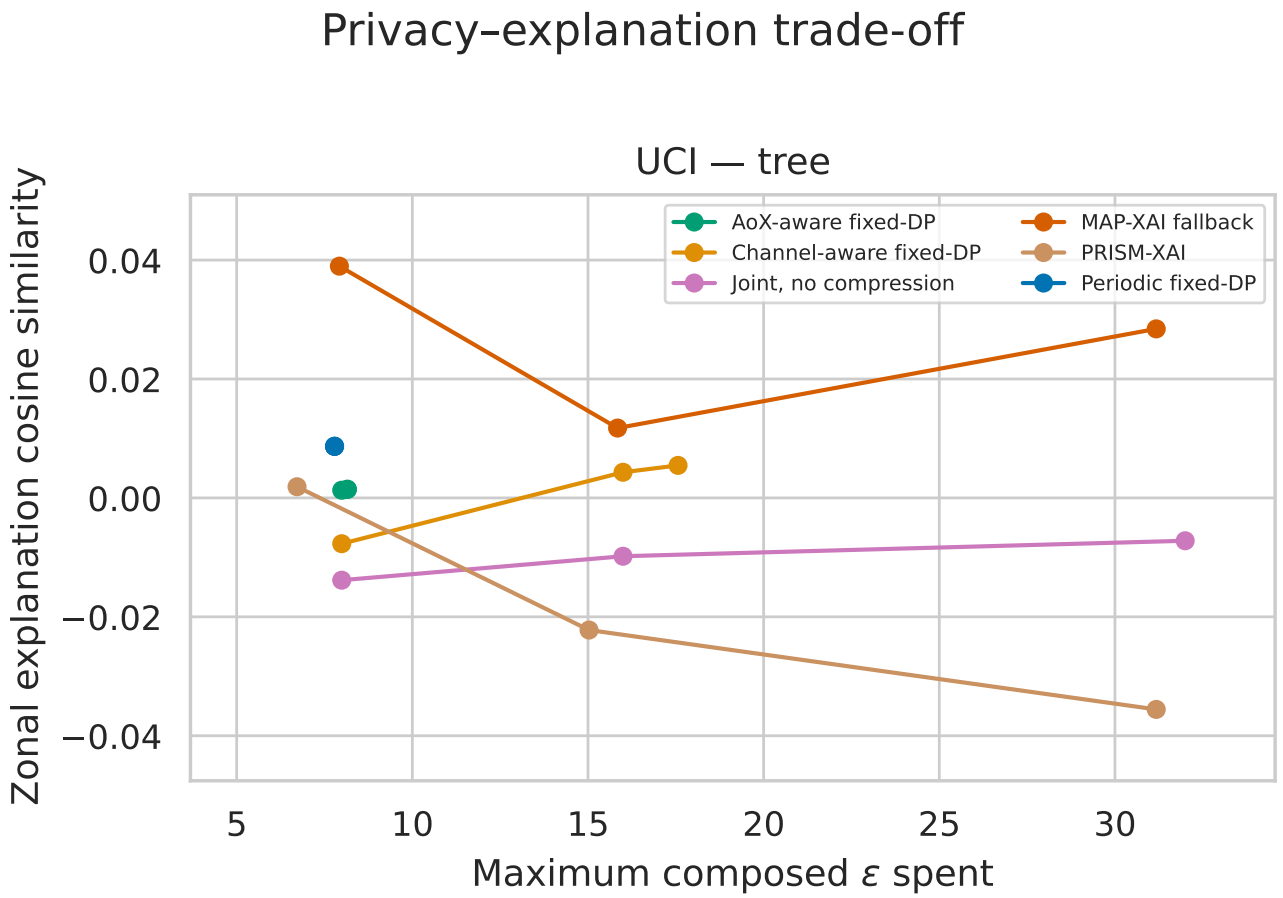


Fig. 137. Public-score threshold $\tau = 3$, release cap $M = 32$, UCI: privacy–fidelity plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

Fig. 138. Public-score threshold $\tau = 3$, release cap $M = 32$, UCI: traffic–fidelity Pareto plane. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

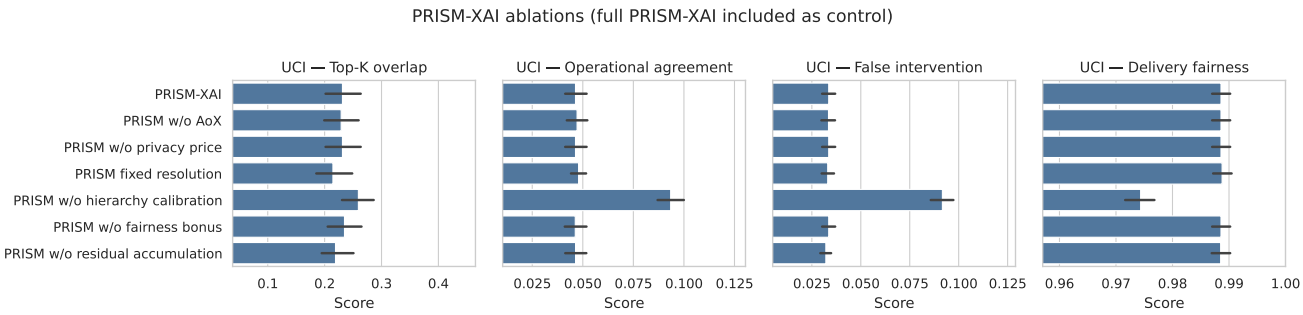


Fig. 139. Public-score threshold $\tau = 3$, release cap $M = 32$, UCI: component ablation. This panel should be read with the setting-level quantitative interpretation and the complete tables immediately preceding the figure block.

C. What the complete sweep establishes

The threshold and release cap have different roles. At fixed $M = 8$, raising τ primarily moves coverage and recall while traffic remains near 205–209 kbit. Raising M changes the available service region more strongly and increases traffic. PRISM-XAI remains below MAP-XAI-only traffic throughout the observed grid, with aggregate-mean ratio savings between 14.35% and 16.56%. The complete matched-region condition, however, is preserved across all cells only at $M = 8$; conclusions at larger caps are therefore descriptive trade-off results. The fidelity and fairness panels add diagnostic context but do not convert the service proxies into evidence of operational grid benefit. Together, the sweep supports a configurable communication scheduler whose operating point must be chosen from an explicit traffic–service requirement rather than from traffic alone.

VIII. Complete Result-Figure Atlas

This section reproduces every vector result figure generated by the three validated experimental matrices. For each configuration, it includes the combined-dataset, OPSD, and UCI versions of the representative regime trace, fairness–reliability analysis, privacy–fidelity analysis, traffic–fidelity Pareto analysis, and scheduler ablation results. The corresponding PNG files are duplicate raster exports and remain in the reproducibility package; the vector PDF versions are used here for print quality. Thus the atlas contains 3 configurations $\times$ 3 scopes $\times$ 5 figure families $= 45$ generated result figures.

A. How the five figure families should be read

a) Representative regime trace.: These panels show the temporal mechanism behind the aggregate metrics: prediction or explanation-quality variation, cumulative traffic, remaining privacy allowance, and transmission activity over the evaluation horizon. They reveal whether a policy communicates steadily, exhausts its allowance early, or uses sparse opportunities. A single trace is illustrative and is not used for statistical inference; the paired aggregate tables provide that evidence.

b) Fairness–reliability plane.: These plots place explanation fidelity or service reliability against Jain delivery fairness. A point near the right boundary serves endpoints evenly, but fairness alone does not establish adequate coverage or explanation quality. The relevant question is therefore whether traffic savings move the proposed method into a worse fairness–reliability region, not simply which point is farthest right.

c) Privacy–fidelity plane.: These plots compare maximum composed privacy expenditure with agreement-based explanation fidelity. They test whether a larger privacy allowance visibly improves the measured proxy. Flat or noisy trends indicate that clipping, Gaussian perturbation, compression, or the dataset may dominate the available budget increase; they do not establish semantic usefulness by themselves.

d) Traffic–fidelity Pareto plane.: These plots expose the communication cost of each operating point. Policies with much larger traffic can also occupy different coverage and recall regions, so geometric dominance in this plane is not sufficient for the paper's matched-service claim. Only the paired PRISM-XAI versus MAP-XAI-only contrast is interpreted as an incremental scheduler comparison.

e) Ablation panels.: The ablations remove one design element at a time and report the resulting metric changes. They identify whether savings arise from packet compression, adaptive mode choice, the release cap, or the candidate score. Because several metrics and ablations are tested, the numerical tables and BH-adjusted tests remain the inferential reference.

B. Residual-score $M = 8$: combined datasets

Across 120 paired conditions, residual-score PRISM-XAI transmits 242.52 kbit on average versus 243.71 kbit for MAP-XAI-only. The 1.19 ± 0.41 kbit reduction is statistically detectable but operationally small (0.50%). Coverage changes by only $+0.021$ percentage points and critical recall by -0.039 points. This configuration therefore establishes reproducibility and near-equivalence of the two proposed layers, not a large scheduling gain.

The five figures explain that small aggregate difference. The traces show similar sparse-release behavior for PRISM-XAI and MAP-XAI-only; the fairness panels place both in the same high-fairness region. The privacy–fidelity and Pareto plots show that fixed-DP and full-packet policies use much more traffic but also occupy different service regions, so they cannot support a matched comparison. Top-K overlap is above random in the combined analysis, driven by OPSD rather than UCI. The ablation panels show only a 1.46% traffic gain from adaptive resolution in this configuration.

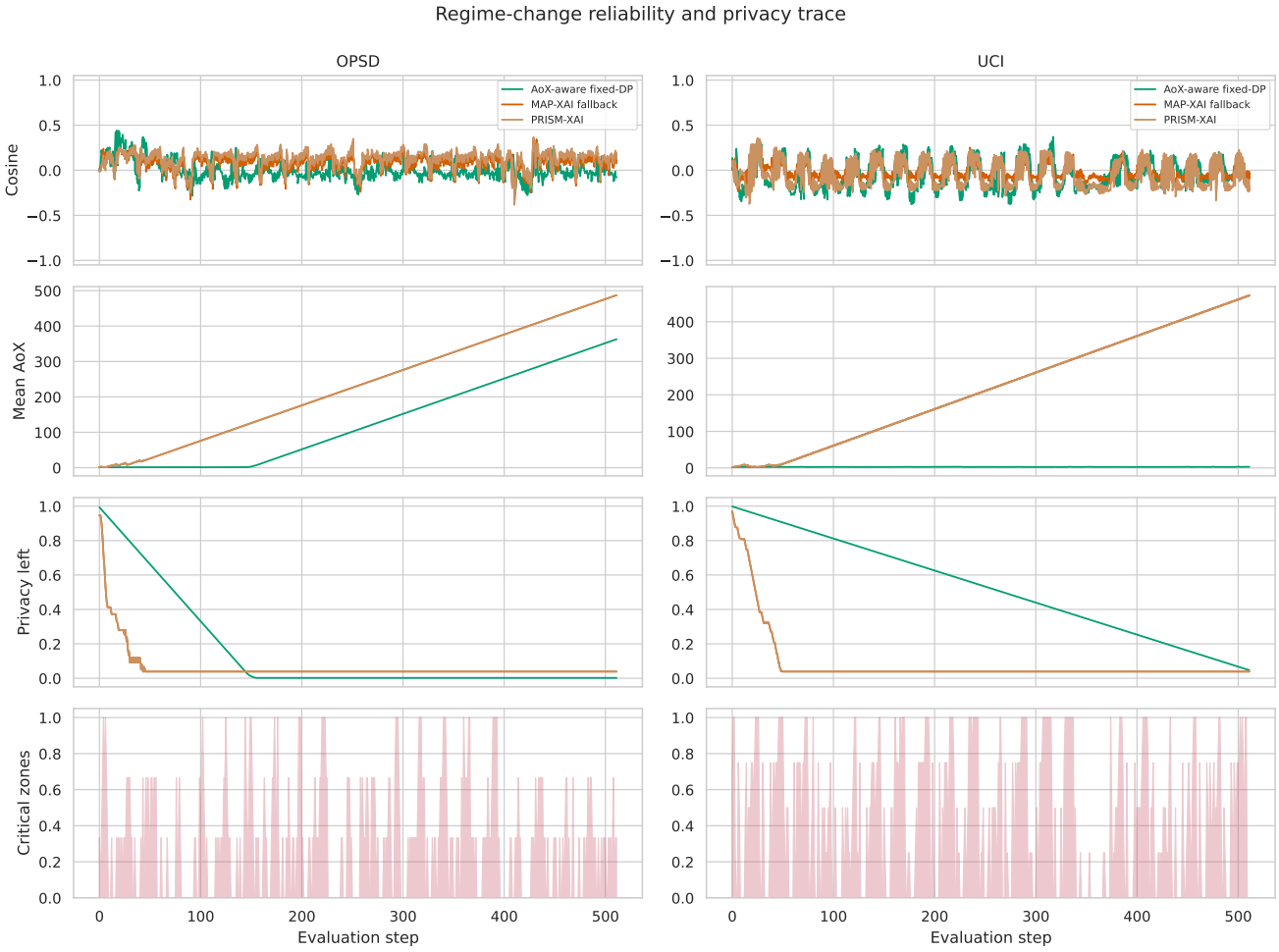


Fig. 140. Residual-score $M = 8$, combined datasets: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

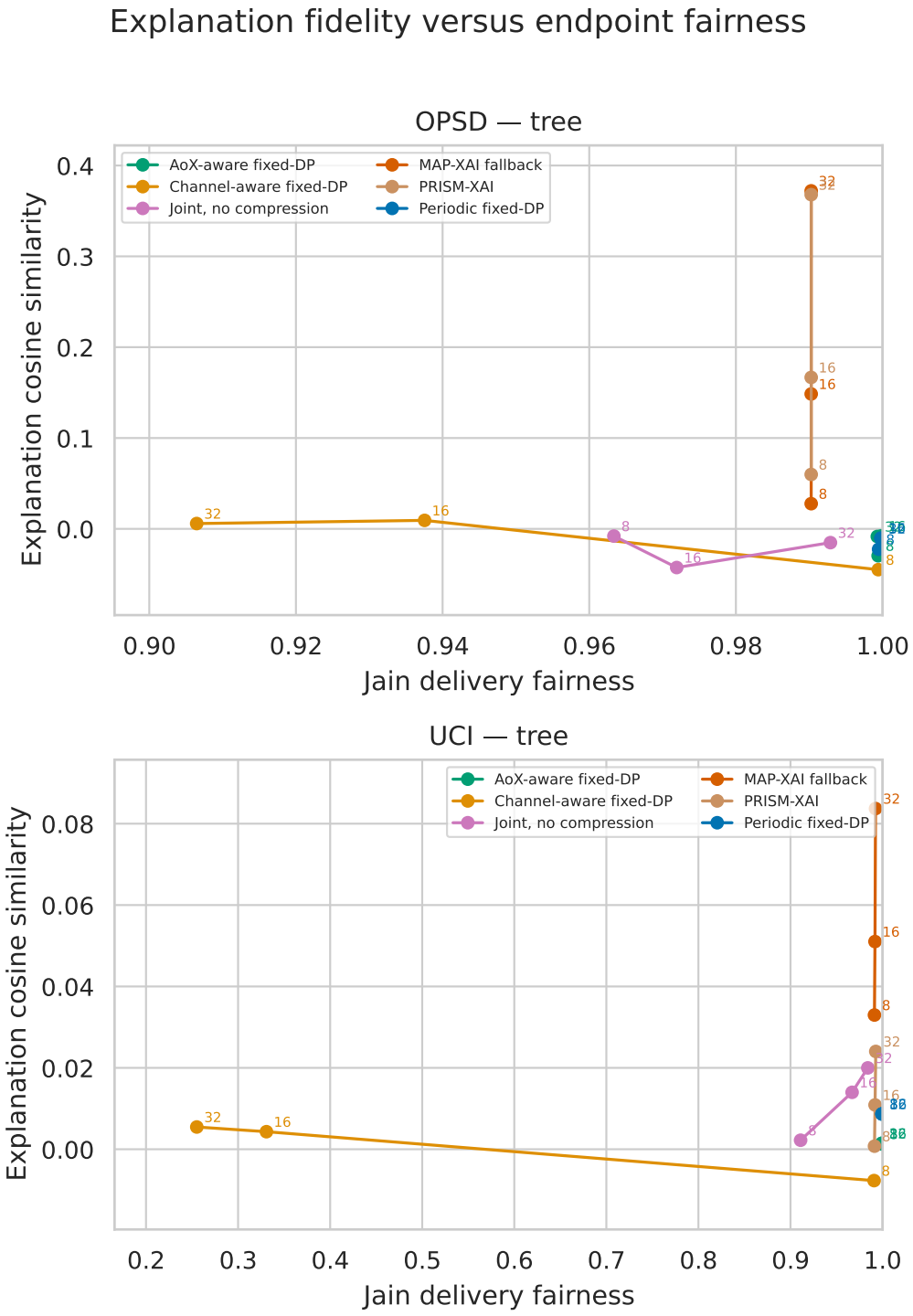


Fig. 141. Residual-score $M = 8$, combined datasets: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

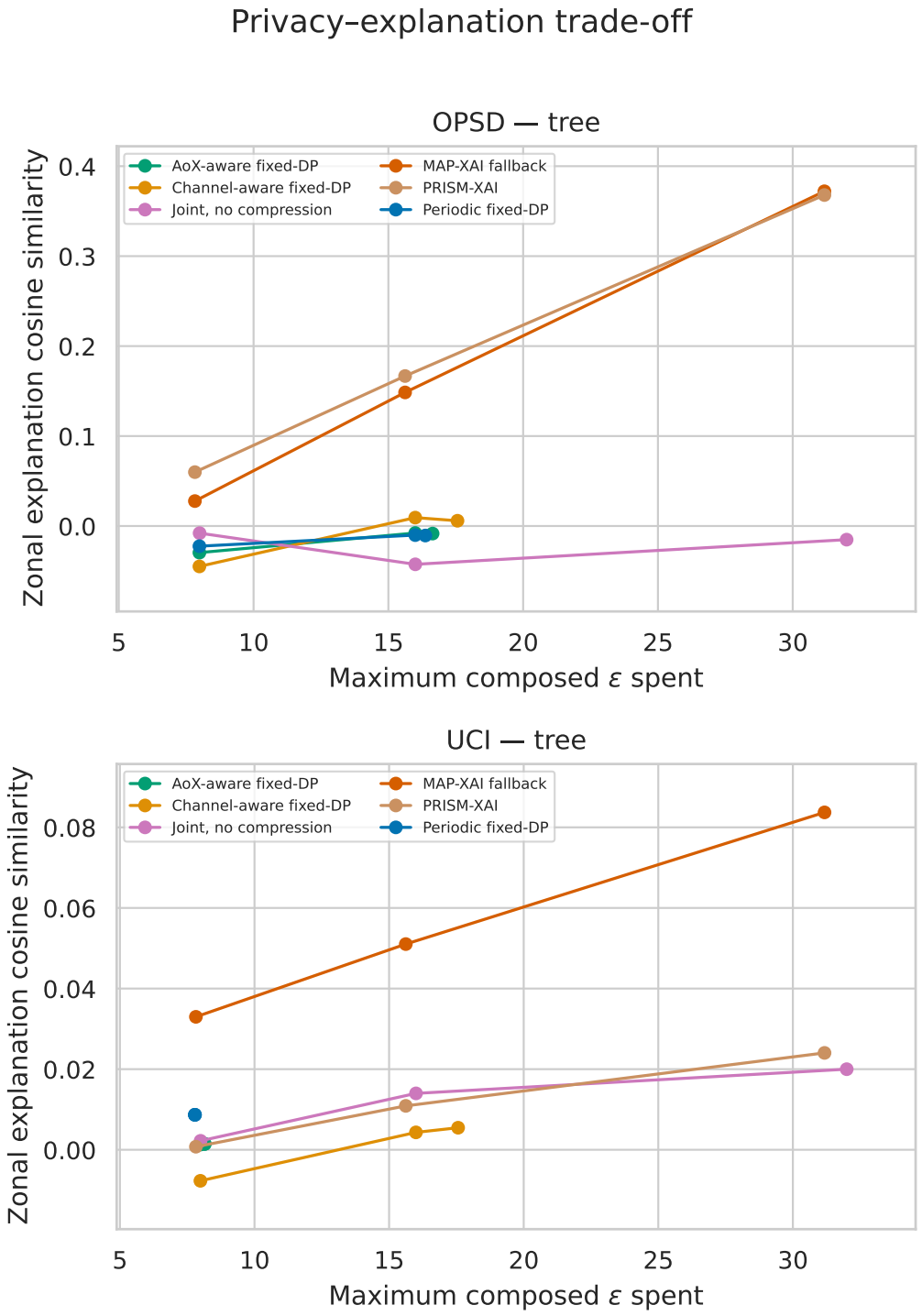


Fig. 142. Residual-score $M = 8$, combined datasets: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

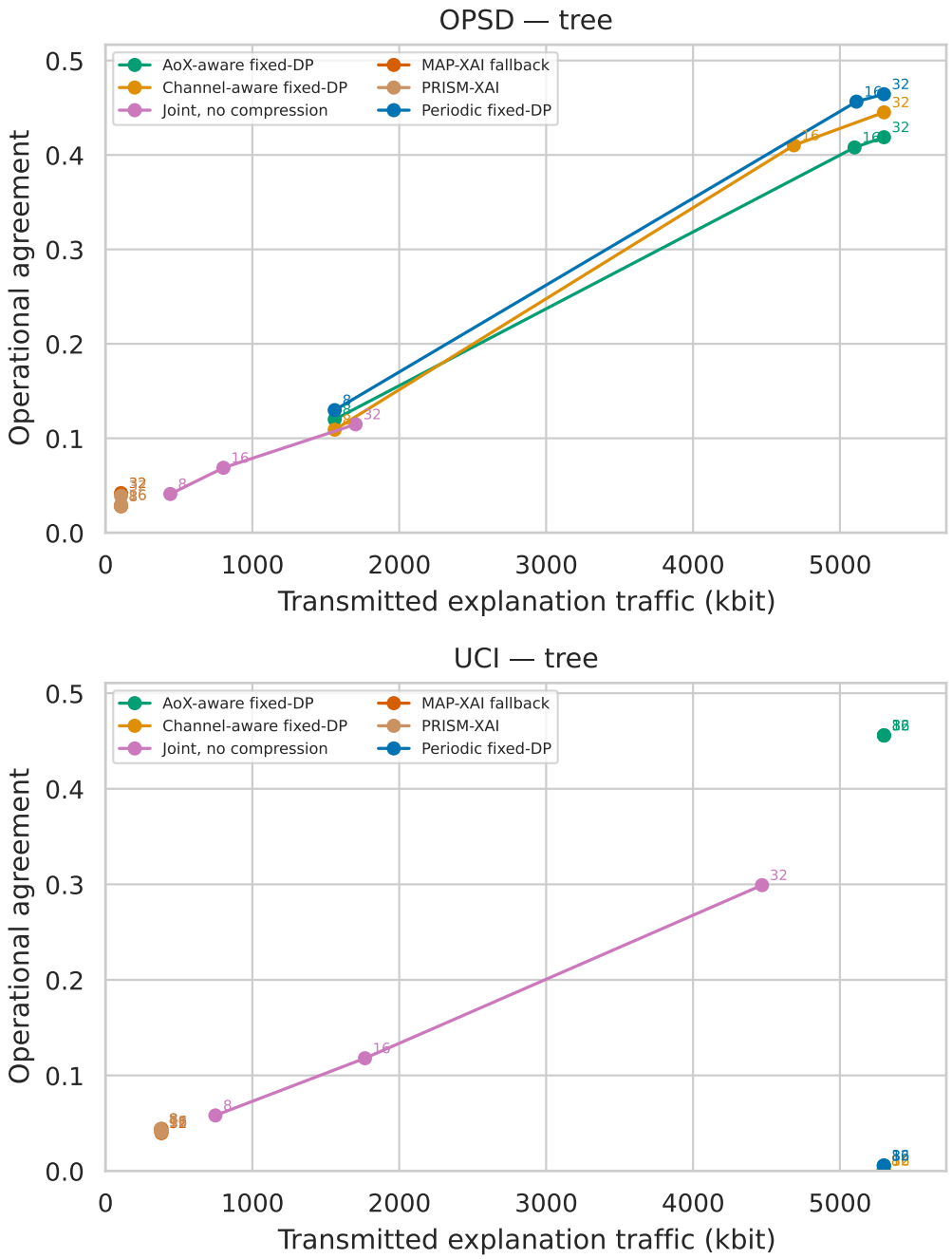


Fig. 143. Residual-score $M = 8$, combined datasets: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

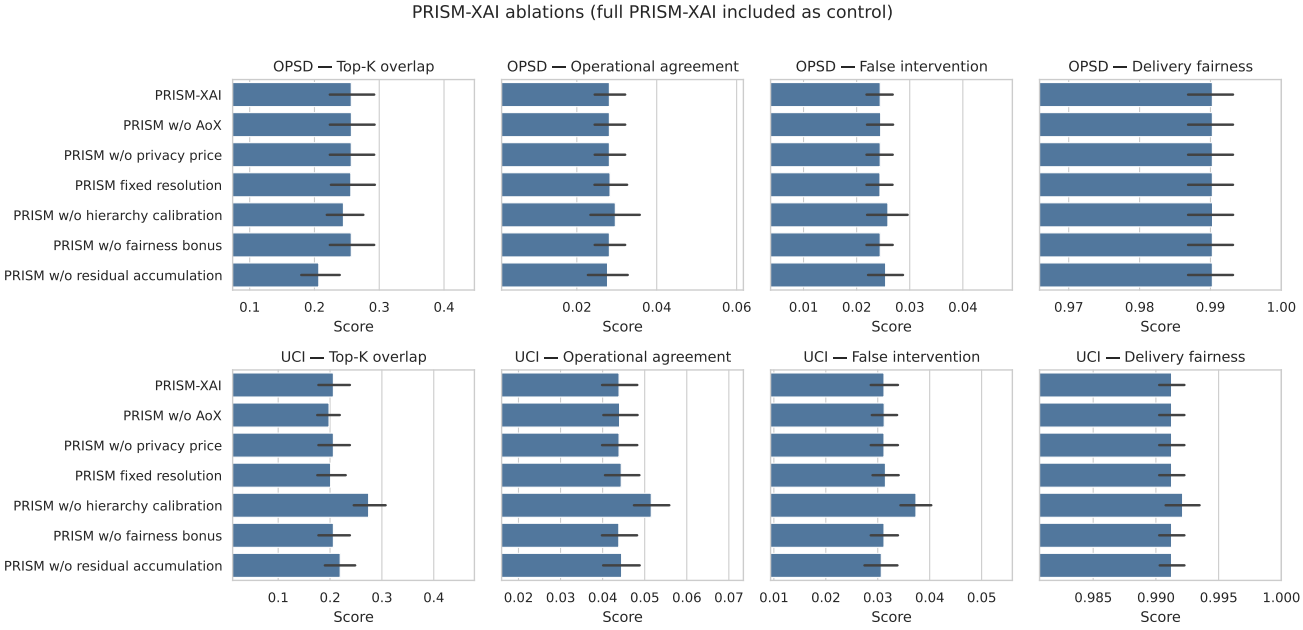


Fig. 144. Residual-score $M = 8$, combined datasets: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

C. Residual-score $M = 8$: OPSD

On OPSD, PRISM-XAI uses about 104–106 kbit across the displayed budgets, while periodic fixed-DP rises from roughly 1.56 Mbit at $\epsilon_{\max} = 8$ to 5.11 Mbit at $\epsilon_{\max} = 16$. This is not an equal-service comparison: PRISM coverage is about 0.071, whereas the periodic operating point increases from about 0.278 to 0.985. The figures should therefore be read as a traffic–service frontier, not as evidence that the lowest-traffic point universally dominates.

OPSD provides the clearest positive fidelity signal. Across the paired study, Top-K overlap exceeds the random reference by 0.0668 ± 0.0265, with multiplicity-adjusted support. The privacy–fidelity panels nevertheless show substantial uncertainty and no monotone guarantee for every explainer and budget. The high Jain values indicate that selective release is distributed relatively evenly across endpoints, while the trace and ablation panels show that compression and abstention, rather than channel opportunism alone, create the low-traffic operating point.

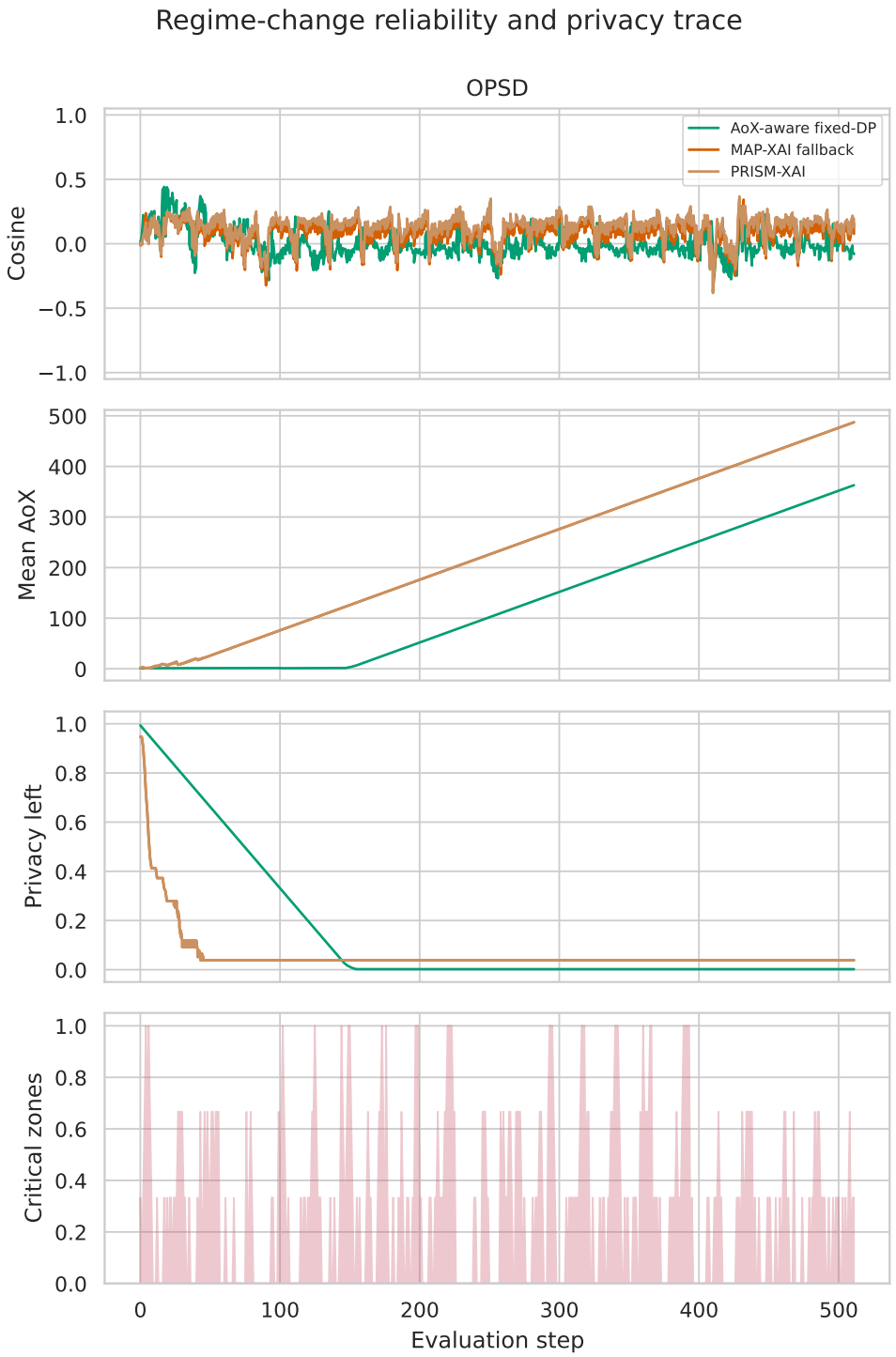


Fig. 145. Residual-score $M = 8$, OPSD: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

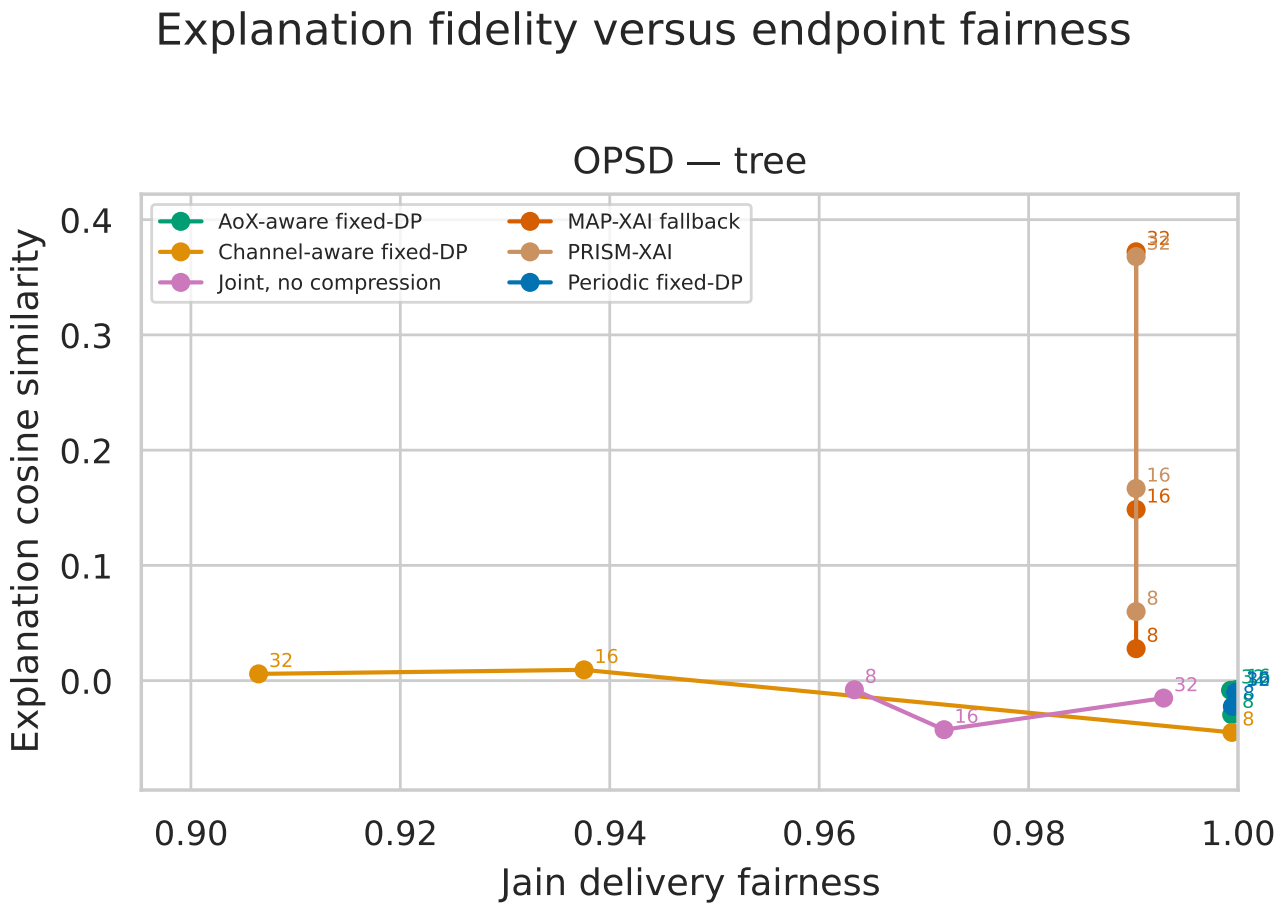


Fig. 146. Residual-score $M = 8$, OPSD: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

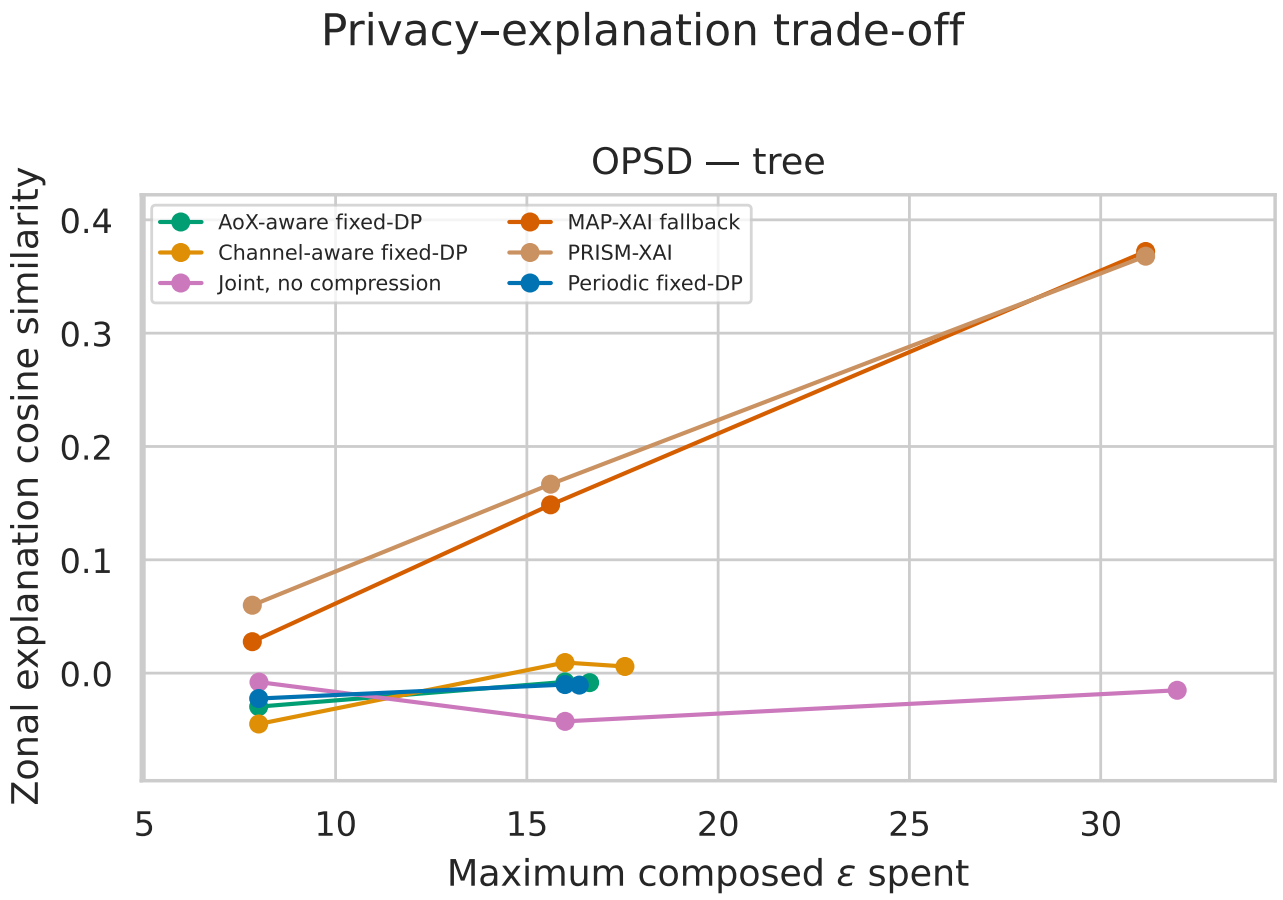


Fig. 147. Residual-score $M = 8$, OPSD: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

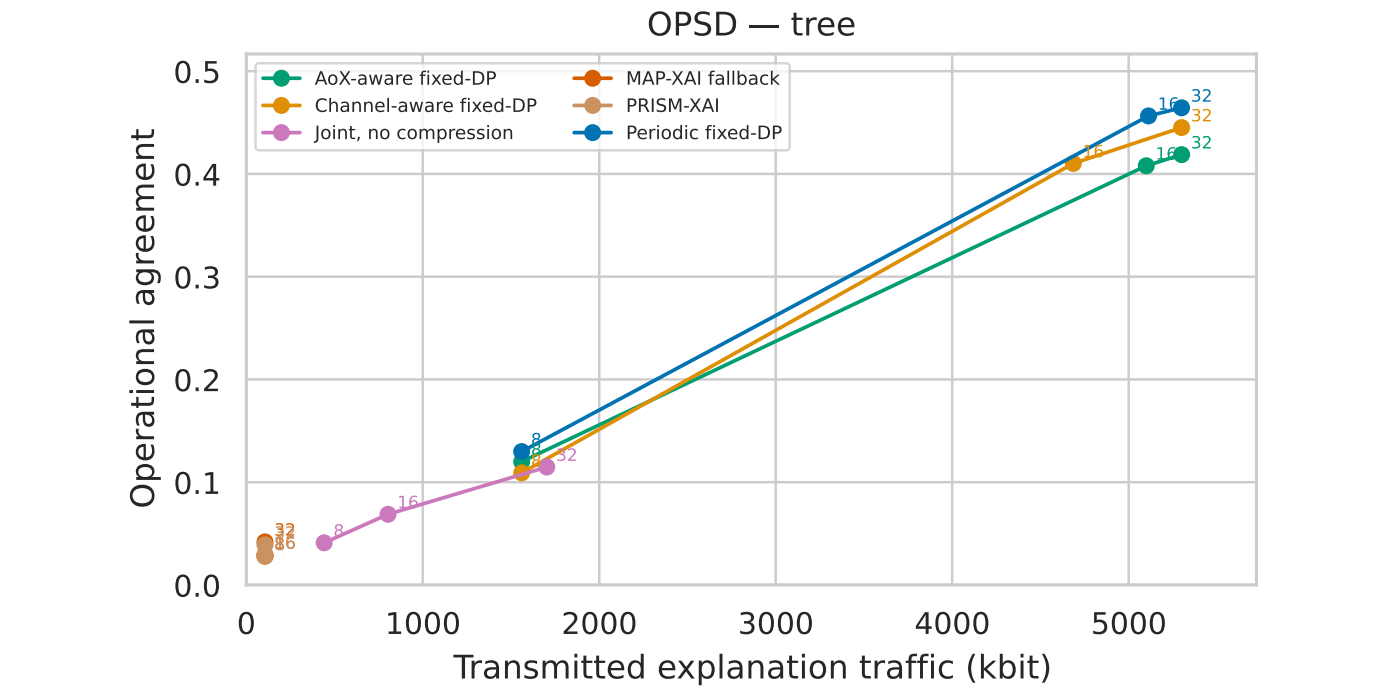


Fig. 148. Residual-score $M = 8$, OPSD: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

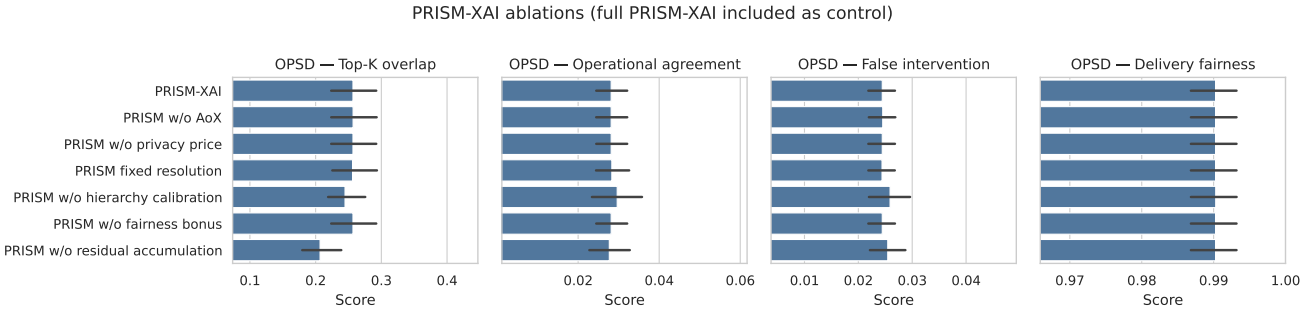


Fig. 149. Residual-score $M = 8$, OPSD: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

D. Residual-score $M = 8$: UCI

UCI has more endpoints and therefore higher absolute PRISM traffic: approximately 373–381 kbit in the displayed conditions. Periodic fixed-DP is near 5.30 Mbit, but it operates at very different coverage and recall. PRISM maintains a sparse selective-release regime with coverage near 0.09 in the shown conditions; this low service level must remain visible when interpreting its communication efficiency.

The UCI fidelity conclusion is weaker than OPSD. The paired Top-K difference from random is -0.0097 ± 0.0234 and is not detected by either adjusted test. Consequently, the fairness and Pareto figures support a communication-allocation interpretation, but not a claim that the transmitted feature sets are reliably more informative than random selection on UCI. The regime traces also show that additional privacy allowance does not automatically repair this limitation.

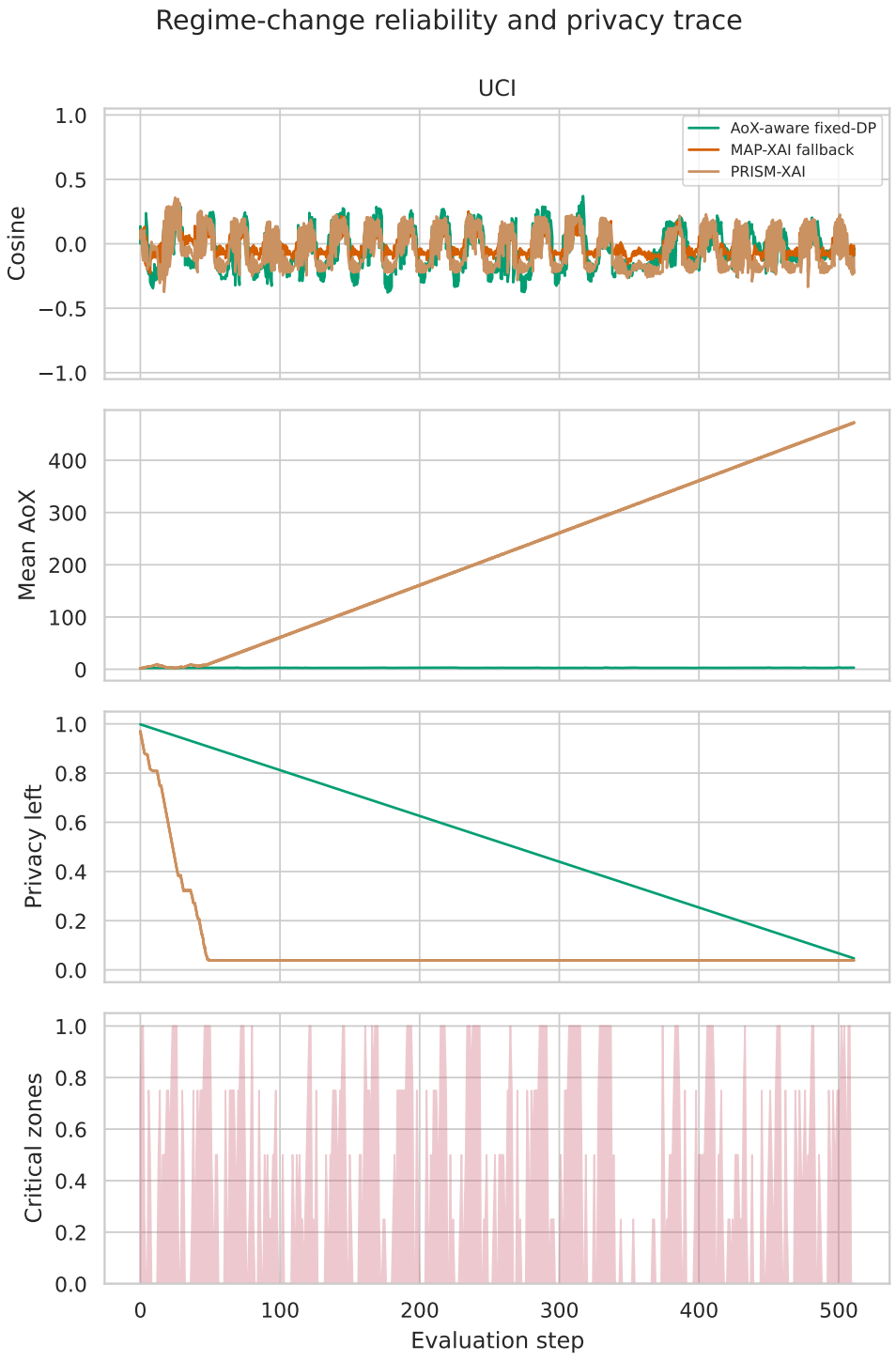


Fig. 150. Residual-score $M = 8$, UCI: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

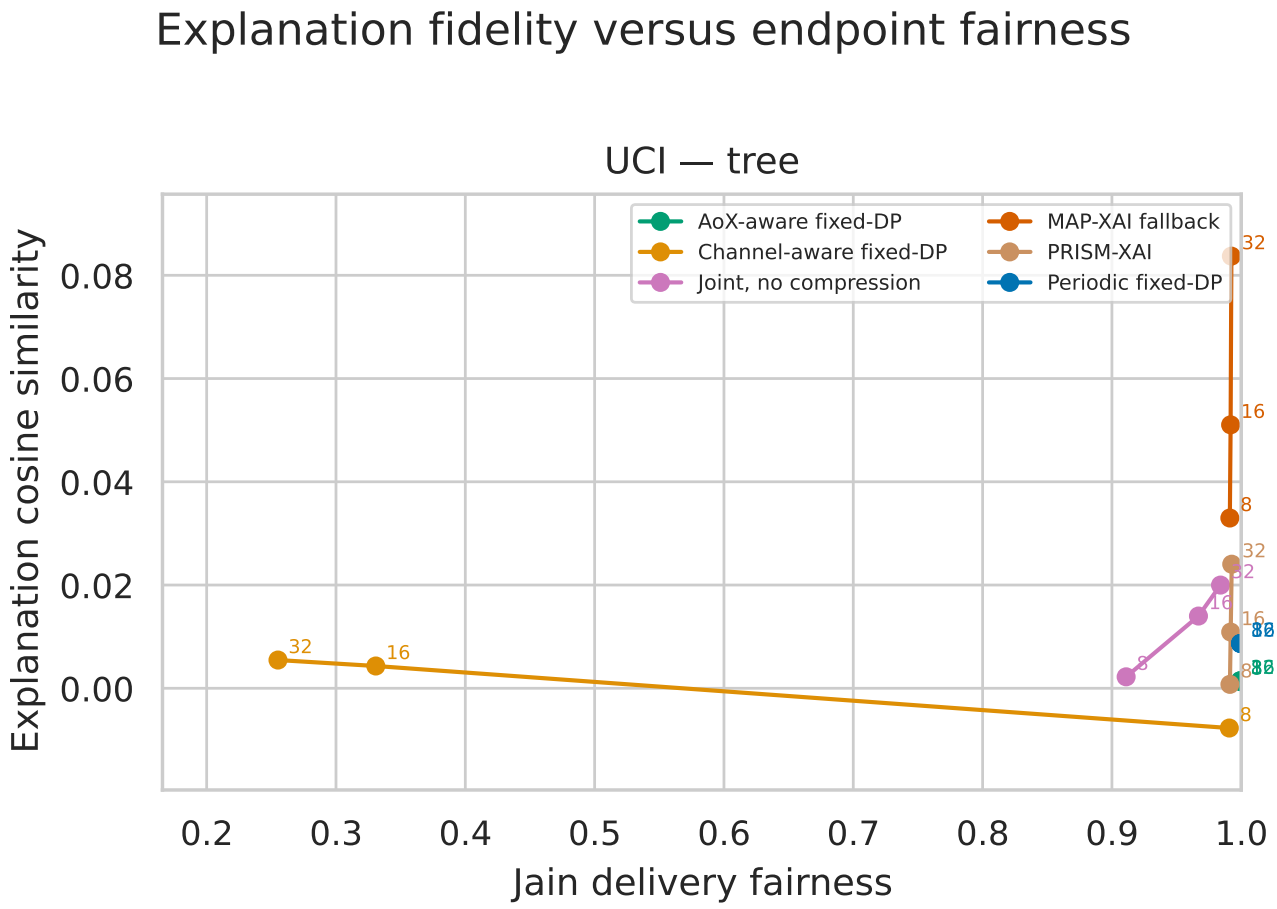


Fig. 151. Residual-score $M = 8$, UCI: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

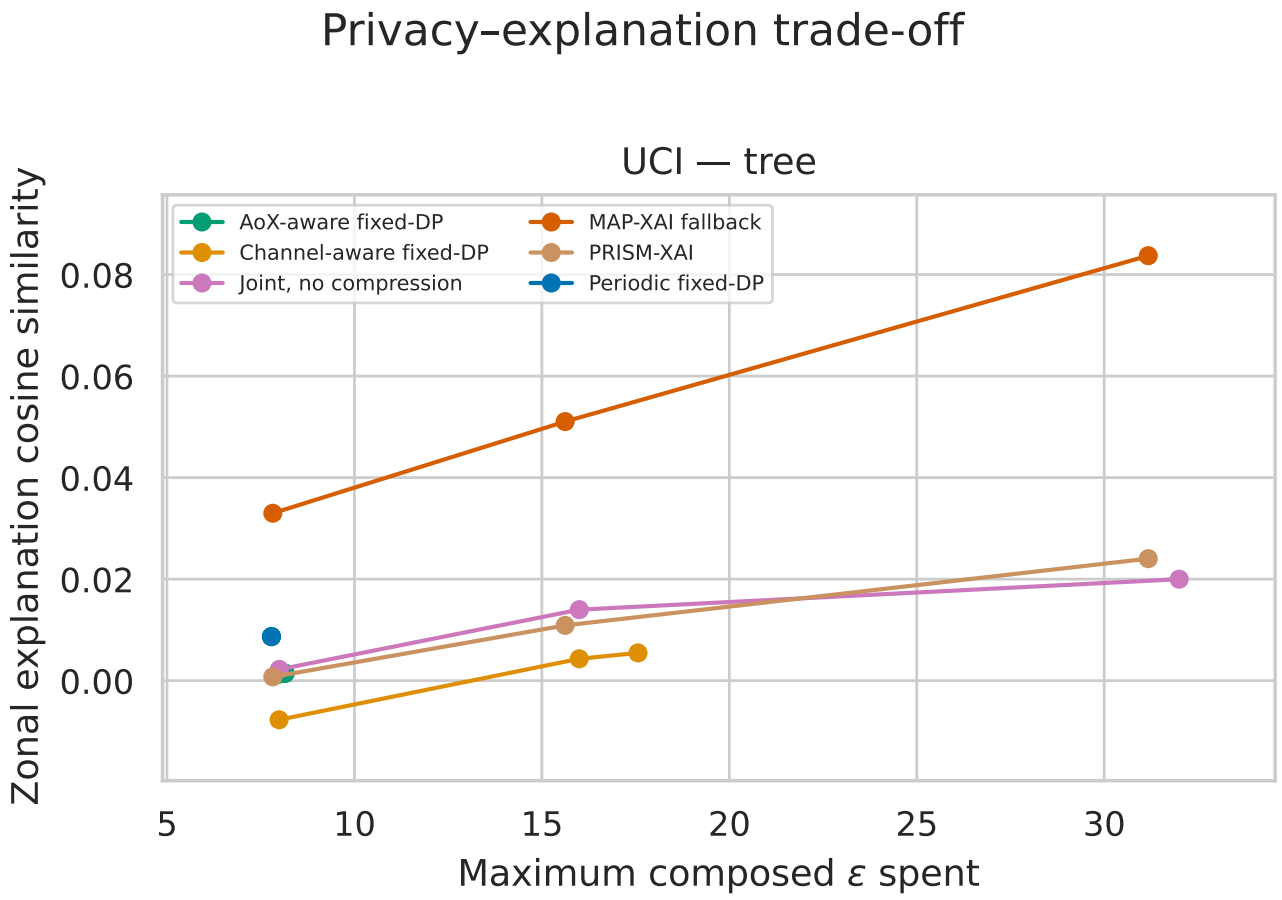


Fig. 152. Residual-score $M = 8$, UCI: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

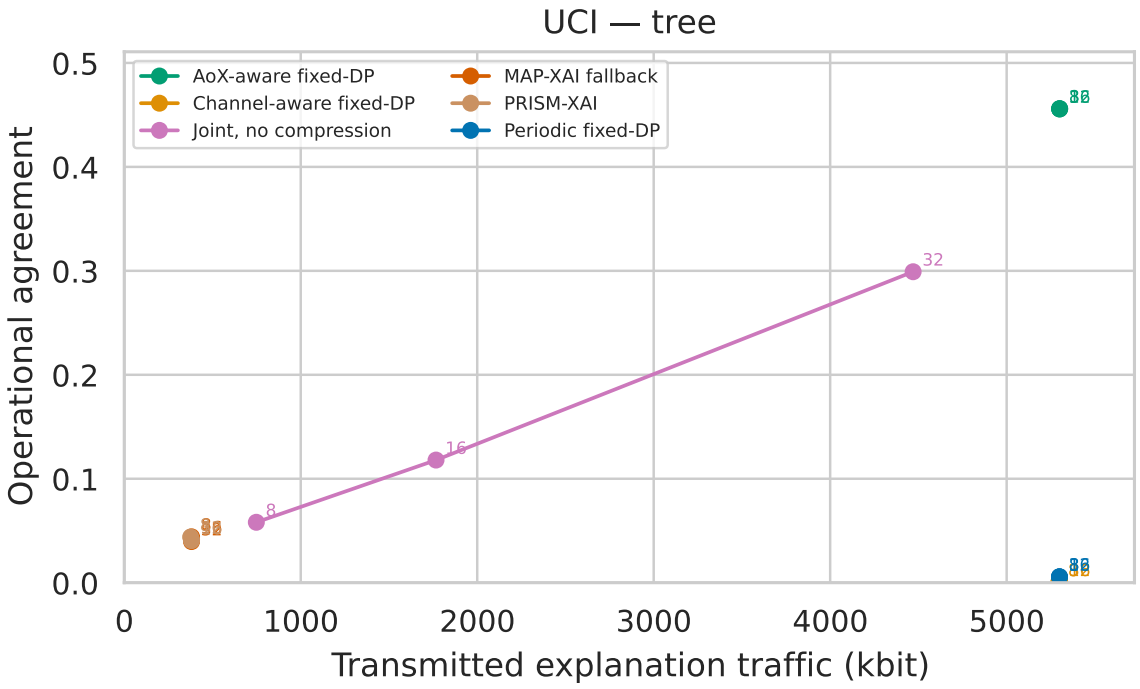


Fig. 153. Residual-score $M = 8$, UCI: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

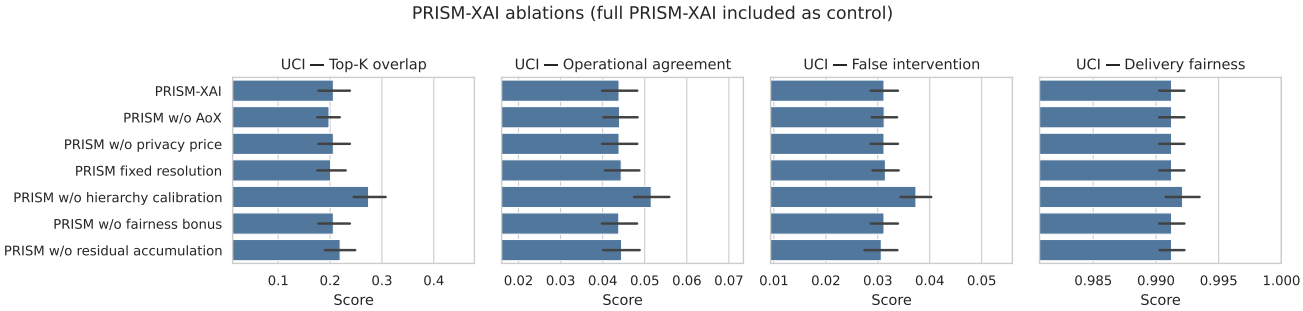


Fig. 154. Residual-score $M = 8$, UCI: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

E. Residual-score $M = 4$: combined datasets

Reducing the release cap from $M = 8$ to $M = 4$ approximately halves MAP-XAI-only and PRISM-XAI traffic to 120.54 and 120.52 kbit, respectively. It also lowers mean coverage to about 0.042 and critical recall to roughly 0.022–0.024. The paired traffic difference between the two proposed layers is only -0.022 ± 0.028 kbit and is not detected after multiplicity adjustment. Thus the release cap changes the operating region, but does not by itself create PRISM's headline incremental gain.

The figure family makes the mechanism visible: traces contain fewer admitted transmissions, fairness remains high because the cap is applied across endpoints, and the Pareto point moves toward lower traffic and lower service. Combined Top-K overlap is above random by 0.0599 ± 0.0203, but this is again dataset dependent. The adaptive-resolution ablation saves only 0.052% traffic and is not significant after correction, confirming that the cap, rather than mode adaptation, dominates this configuration.

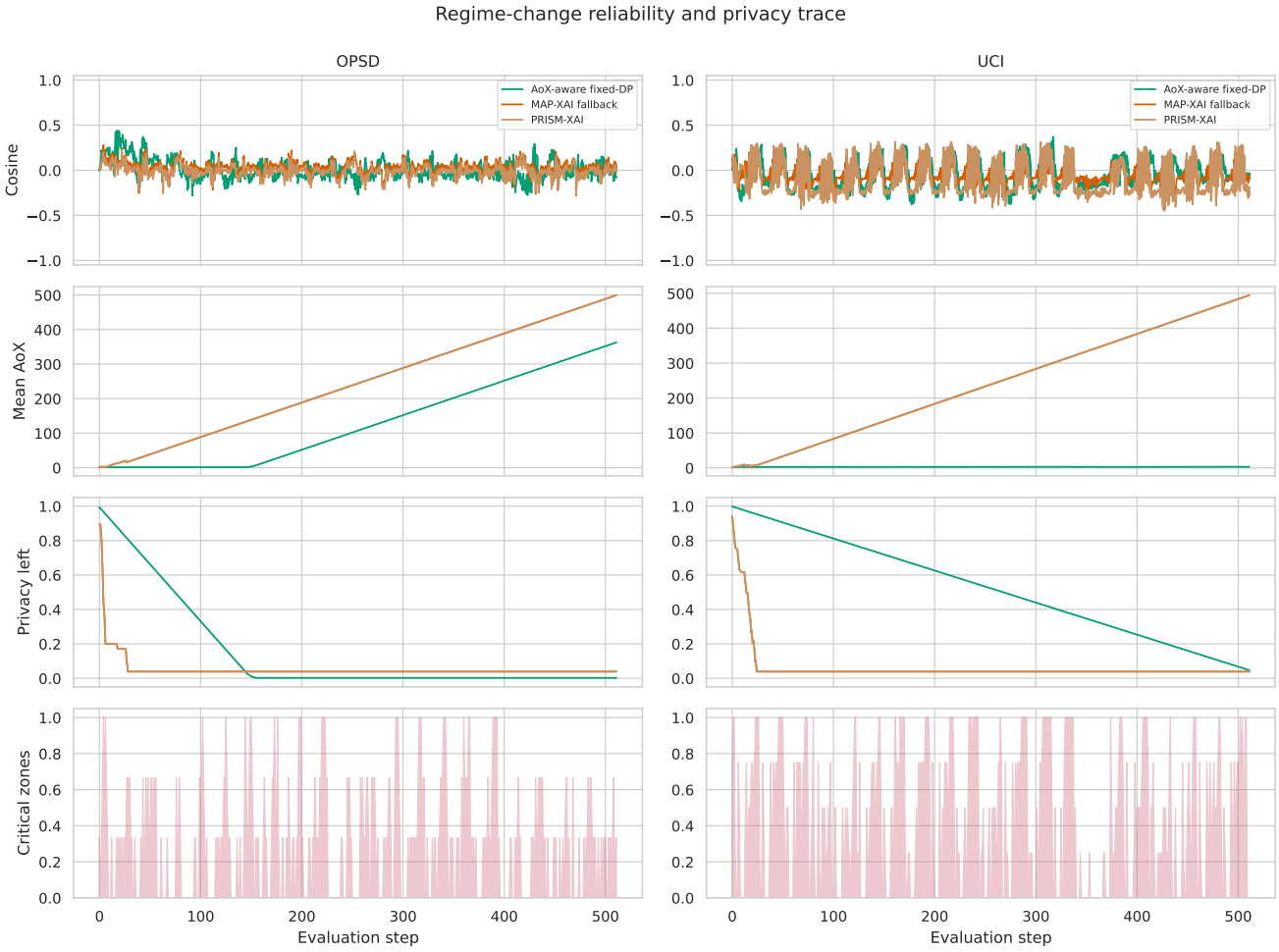


Fig. 155. Residual-score $M = 4$, combined datasets: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

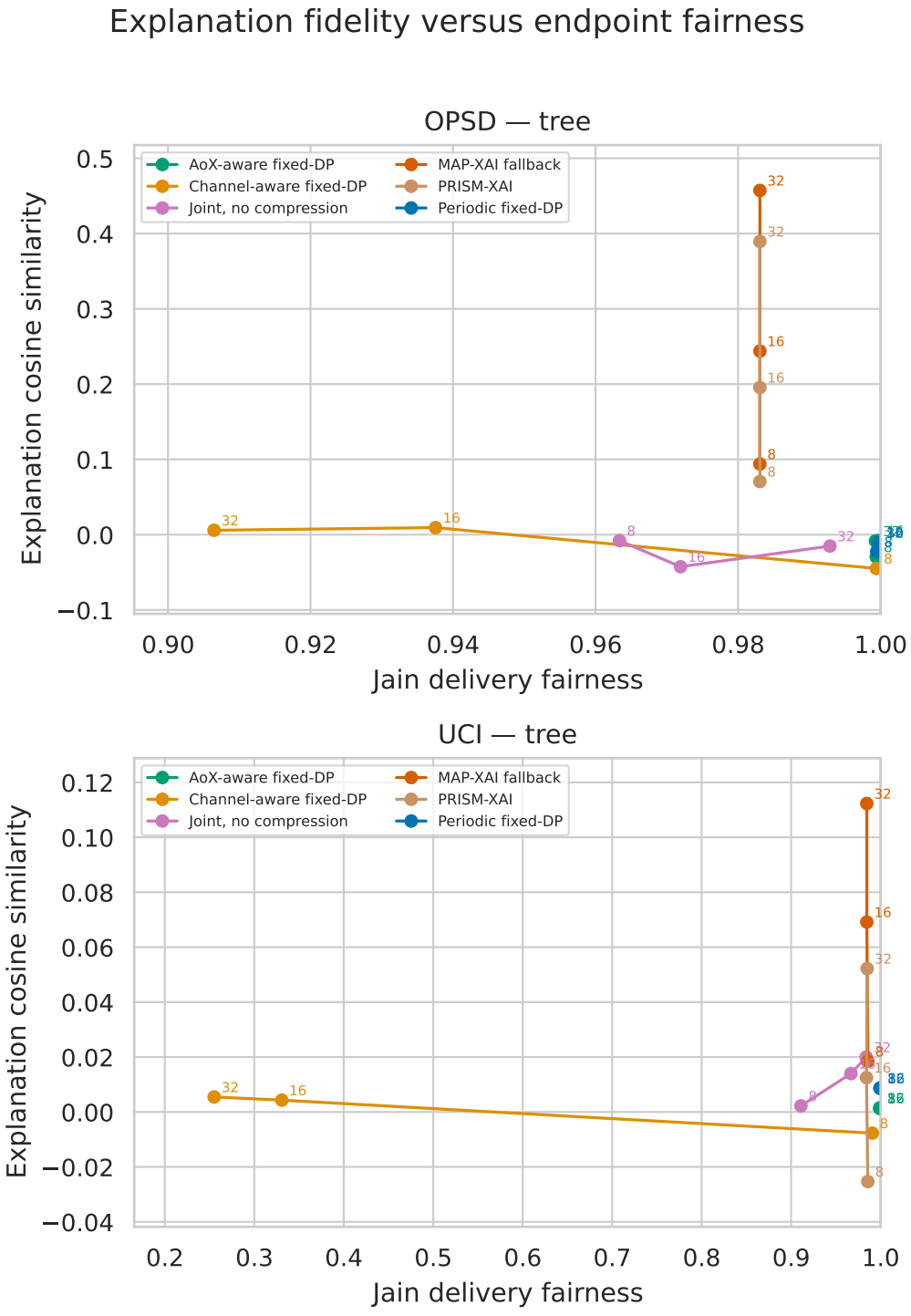


Fig. 156. Residual-score $M = 4$, combined datasets: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

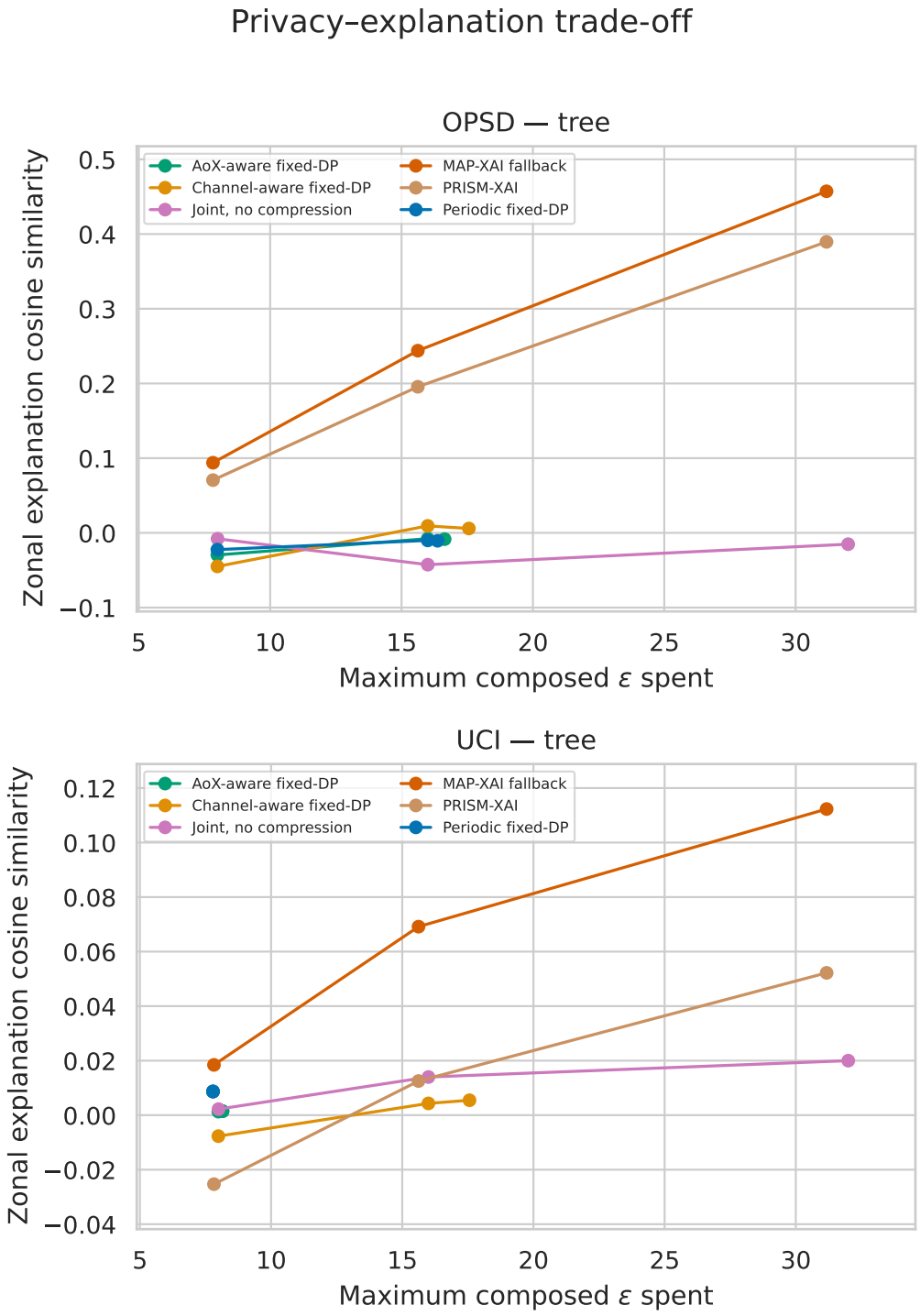


Fig. 157. Residual-score $M = 4$, combined datasets: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

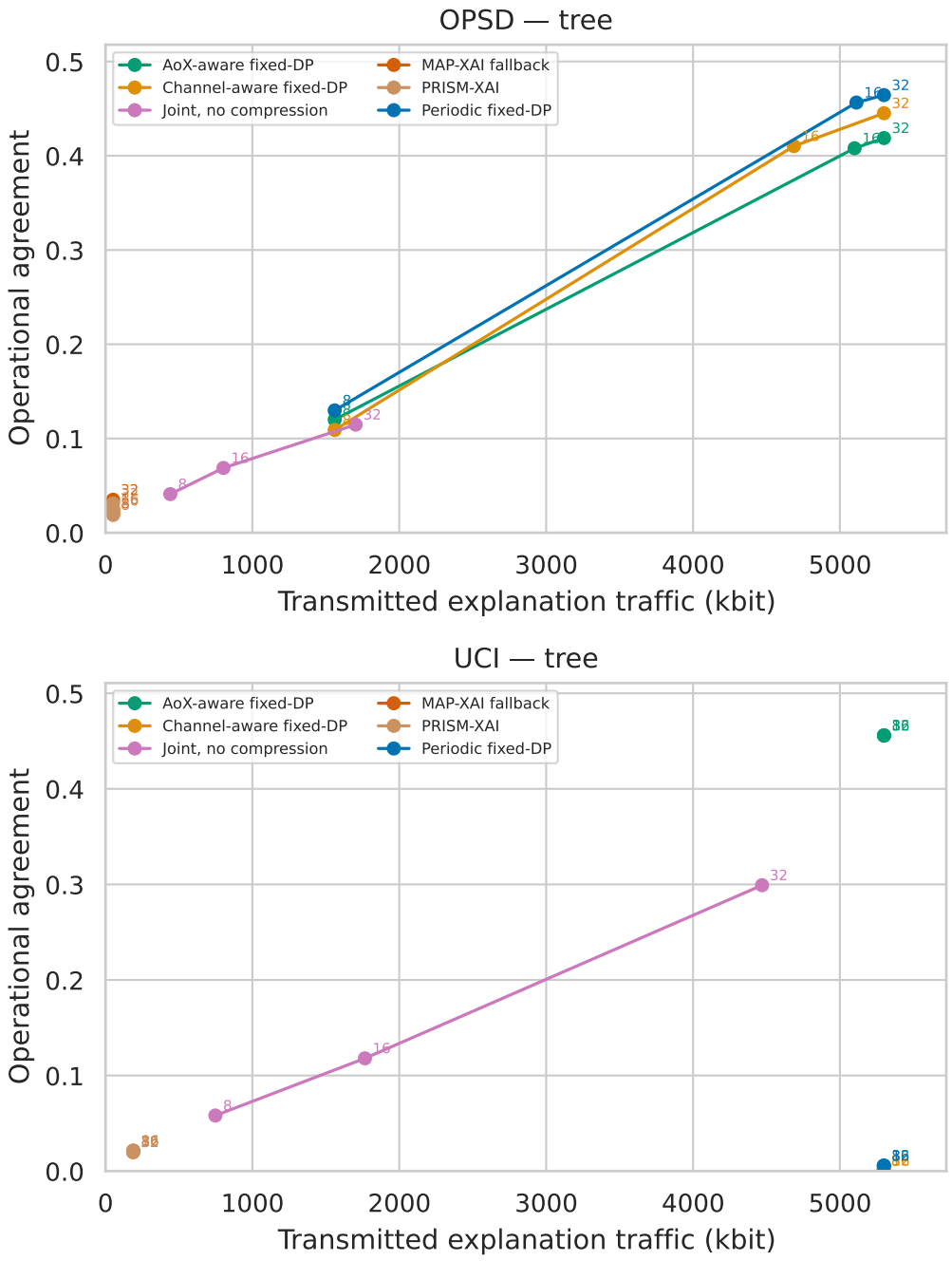


Fig. 158. Residual-score $M = 4$, combined datasets: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

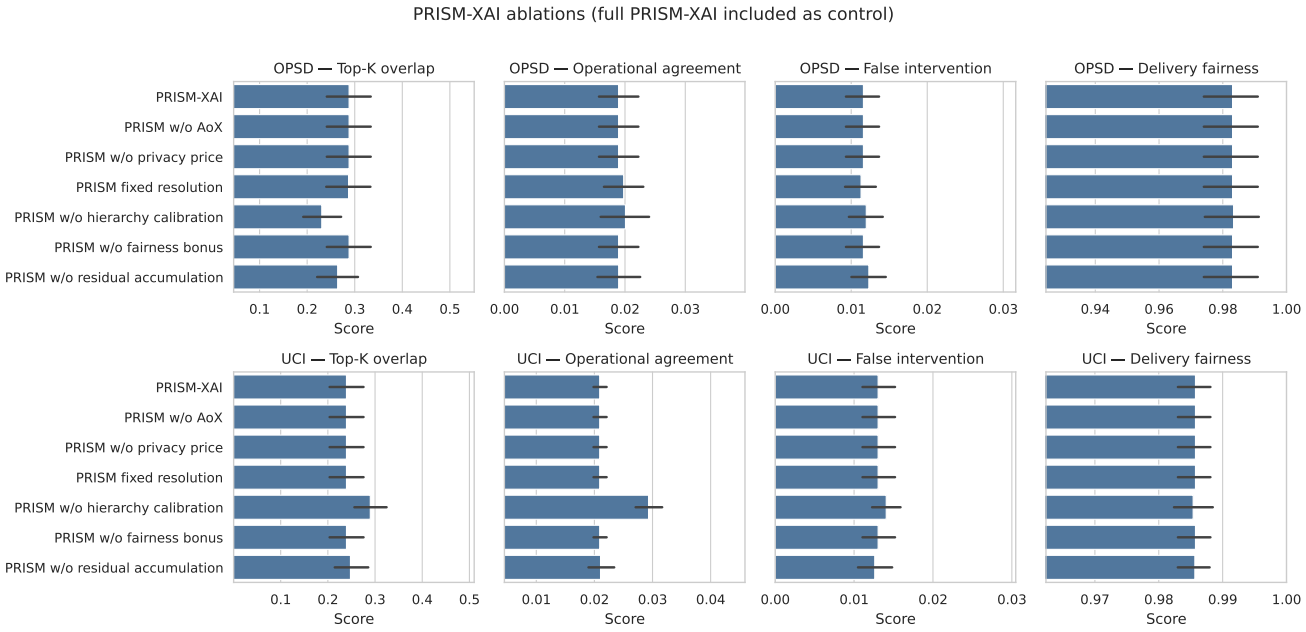


Fig. 159. Residual-score $M = 4$, combined datasets: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

F. Residual-score $M = 4$: OPSD

For OPSD, the capped PRISM policy transmits approximately 52.1 kbit in the displayed conditions and abstains in about 95.5% of opportunities. Coverage is about 0.045, lower than in the $M = 8$ configuration, while the maximum composed expenditure remains close to the admitted horizon allowance. The low traffic therefore follows directly from stricter release scarcity and should not be presented independently of the reduced service rate.

The fidelity plots are comparatively favorable on OPSD: the paired Top-K margin above random is 0.1000 ± 0.0304. The fairness panels show that the scarce releases are not concentrated on only a few endpoints, but high Jain fairness cannot compensate for limited coverage. The ablation panels show no robust incremental traffic benefit from adaptive resolution once the four-release cap is binding.

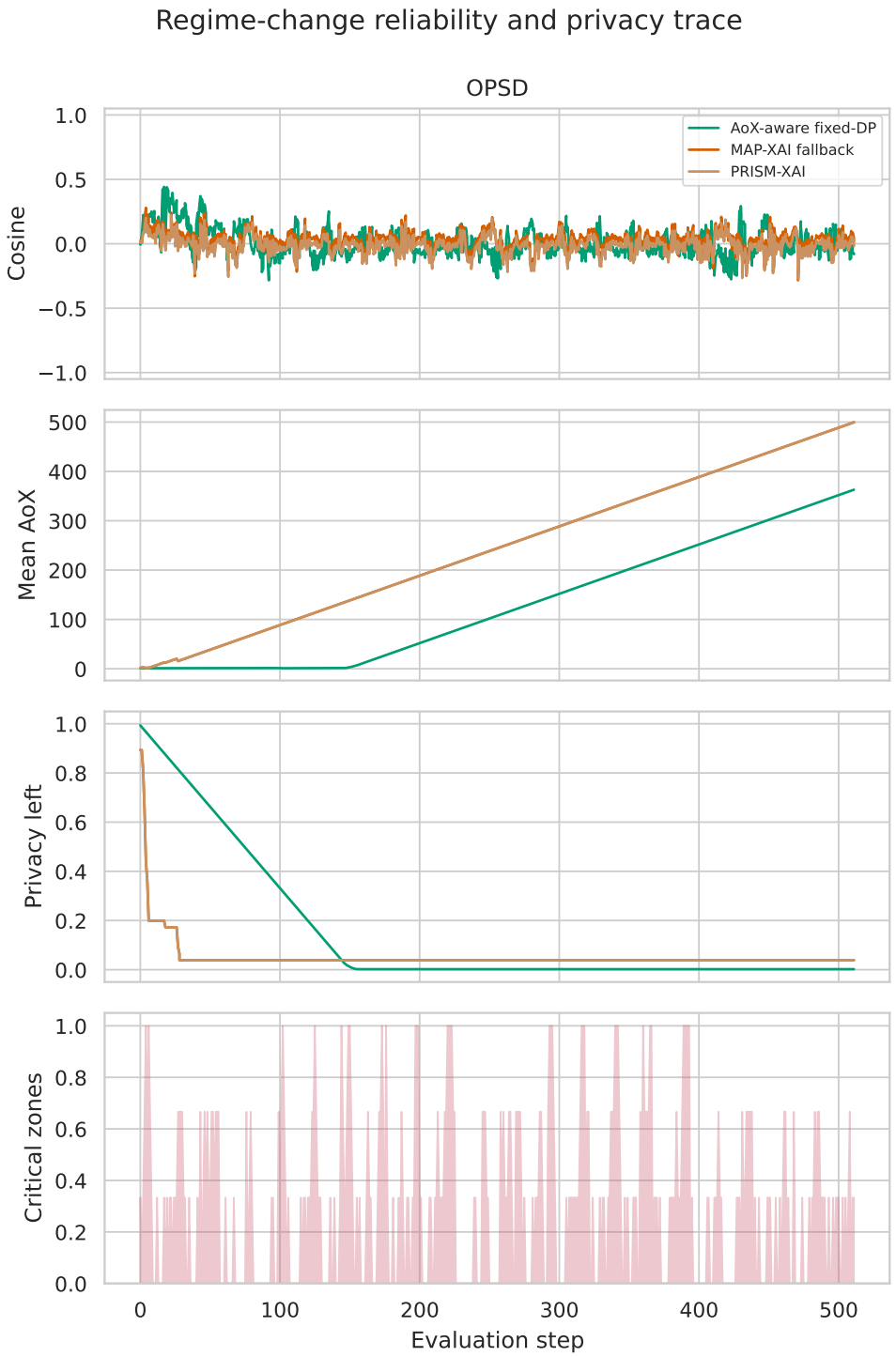


Fig. 160. Residual-score $M = 4$, OPSD: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

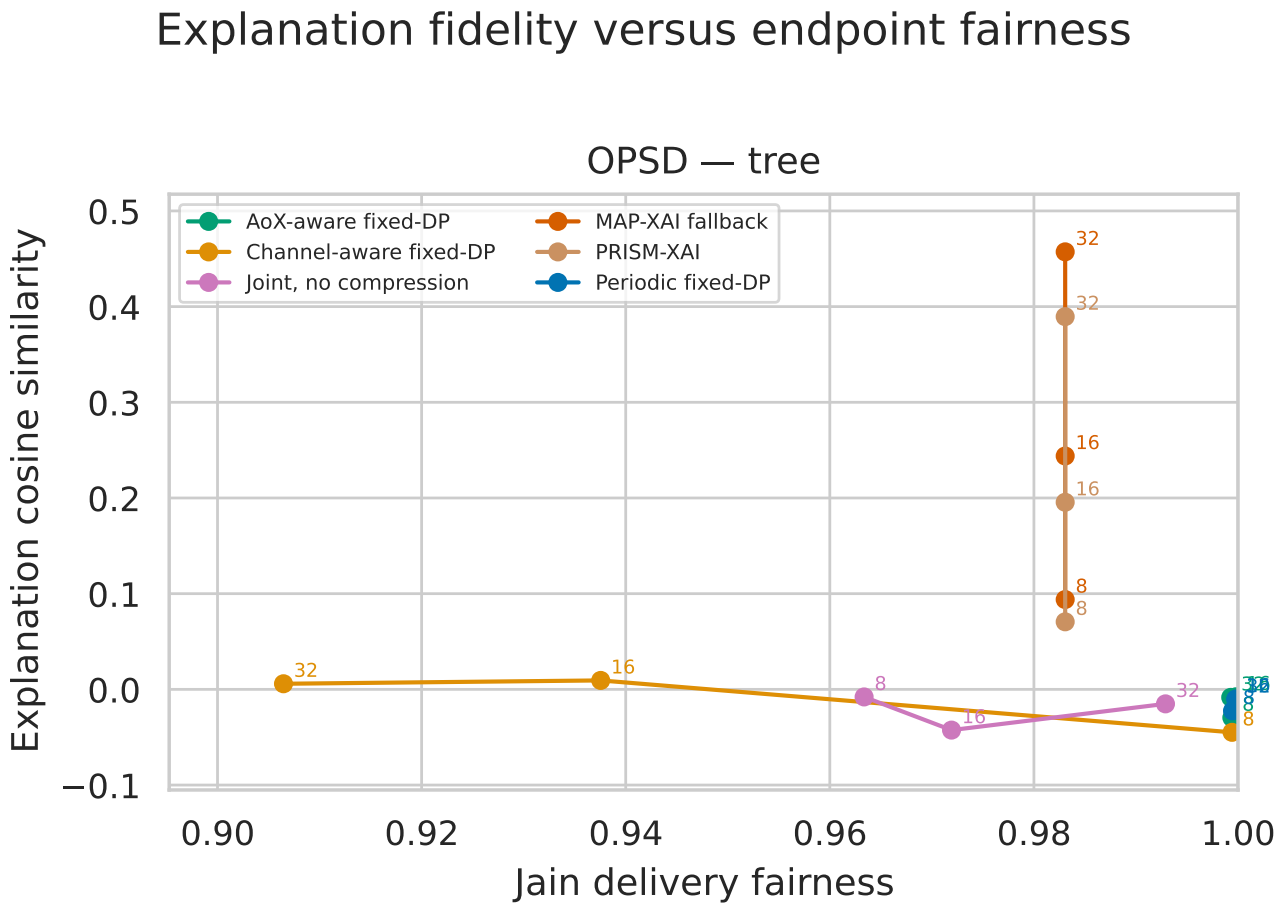


Fig. 161. Residual-score $M = 4$, OPSD: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

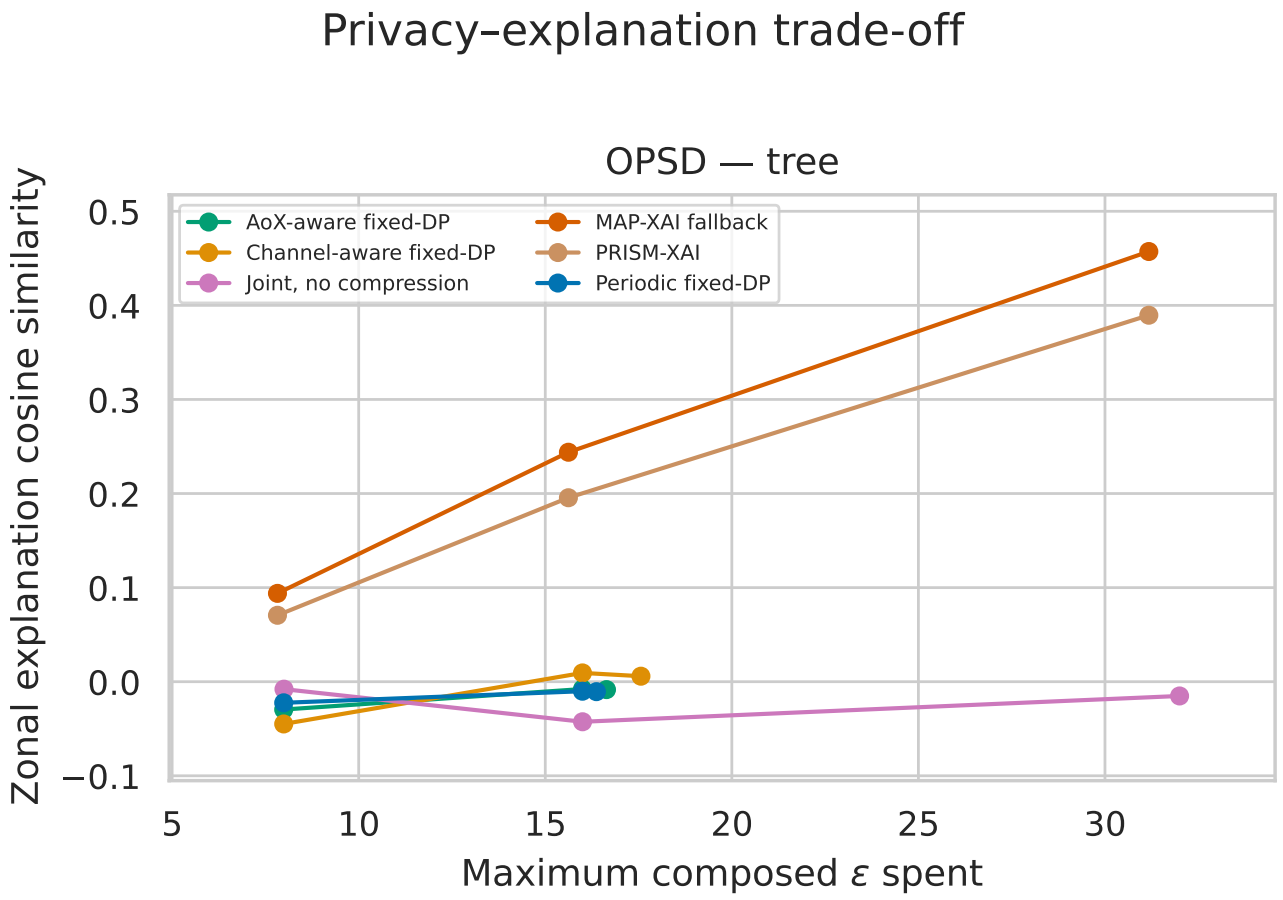


Fig. 162. Residual-score $M = 4$, OPSD: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

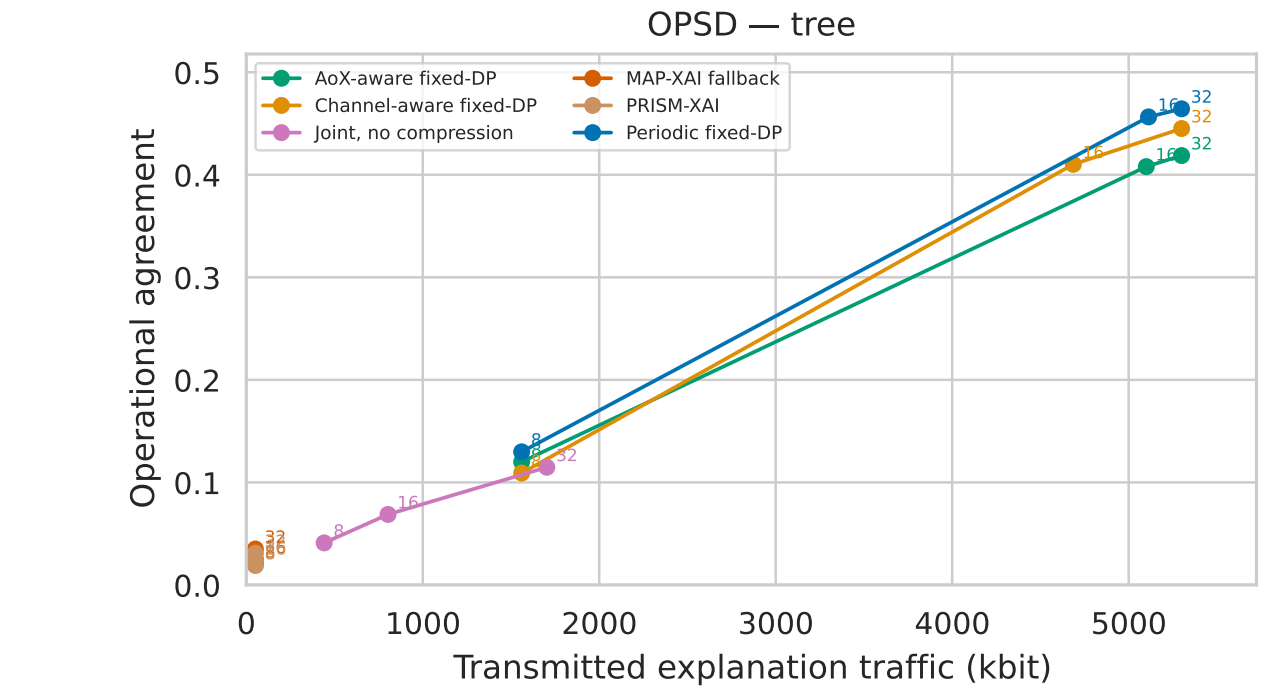


Fig. 163. Residual-score $M = 4$, OPSD: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

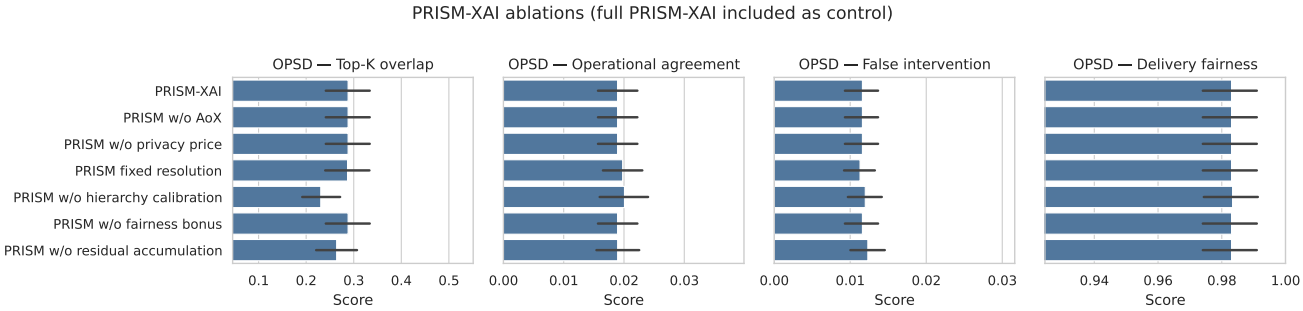


Fig. 164. Residual-score $M = 4$, OPSD: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

G. Residual-score $M = 4$: UCI

For UCI, capped PRISM traffic is about 189 kbit, again roughly half of the residual-$M = 8$ level. Coverage falls to approximately 0.042–0.044 and abstention approaches 0.96. These plots characterize an intentionally sparse explanation channel rather than a high-availability monitoring service.

UCI Top-K overlap is only 0.0197 ± 0.0237 above random and is not statistically detected after correction. The privacy–fidelity plane therefore shows no reliable fidelity benefit from the capped policy on this dataset. The fairness and trace panels are still useful systems diagnostics: they indicate even allocation and controlled expenditure, but do not establish that the received explanations are operationally useful.

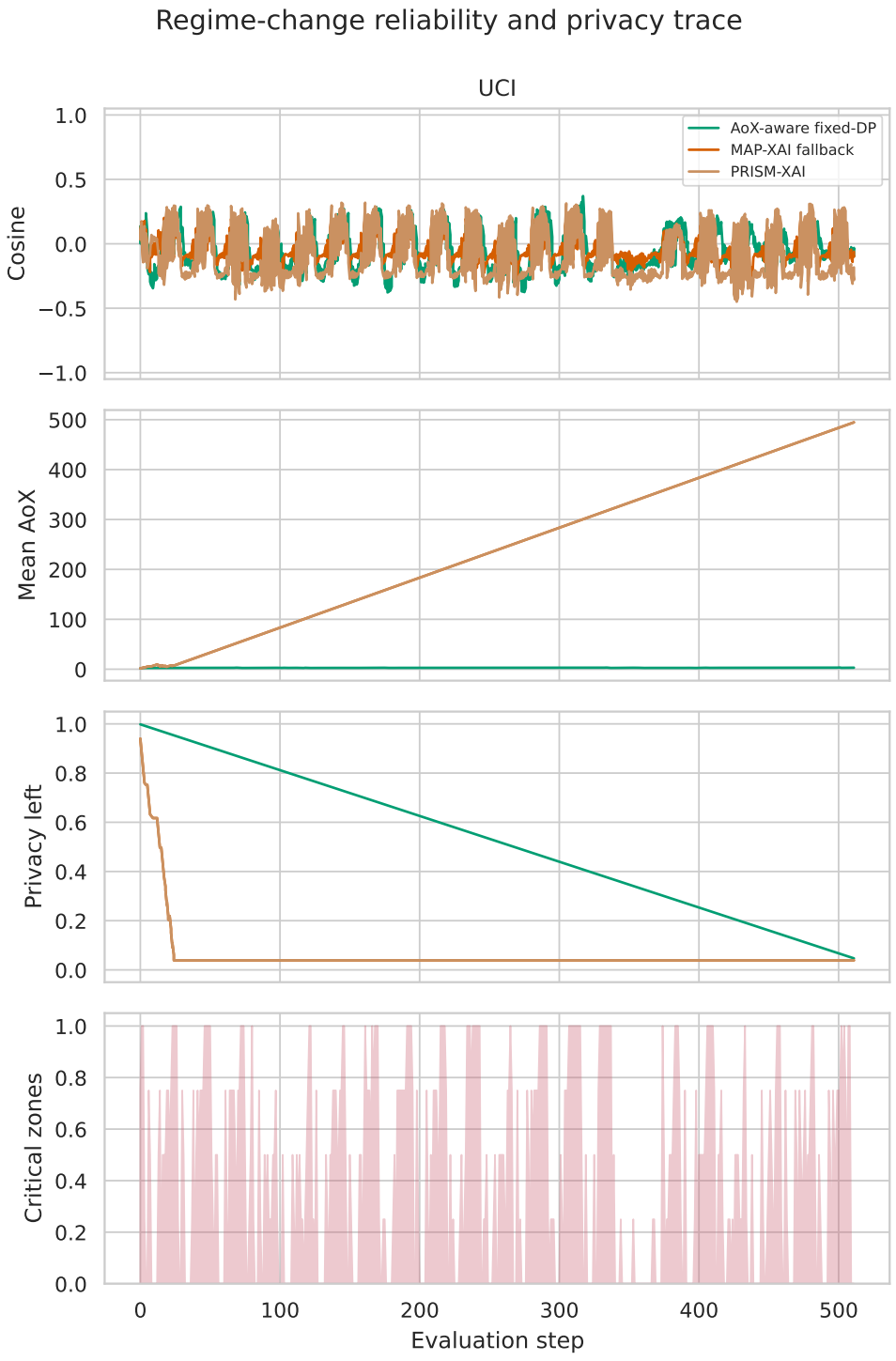


Fig. 165. Residual-score $M = 4$, UCI: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

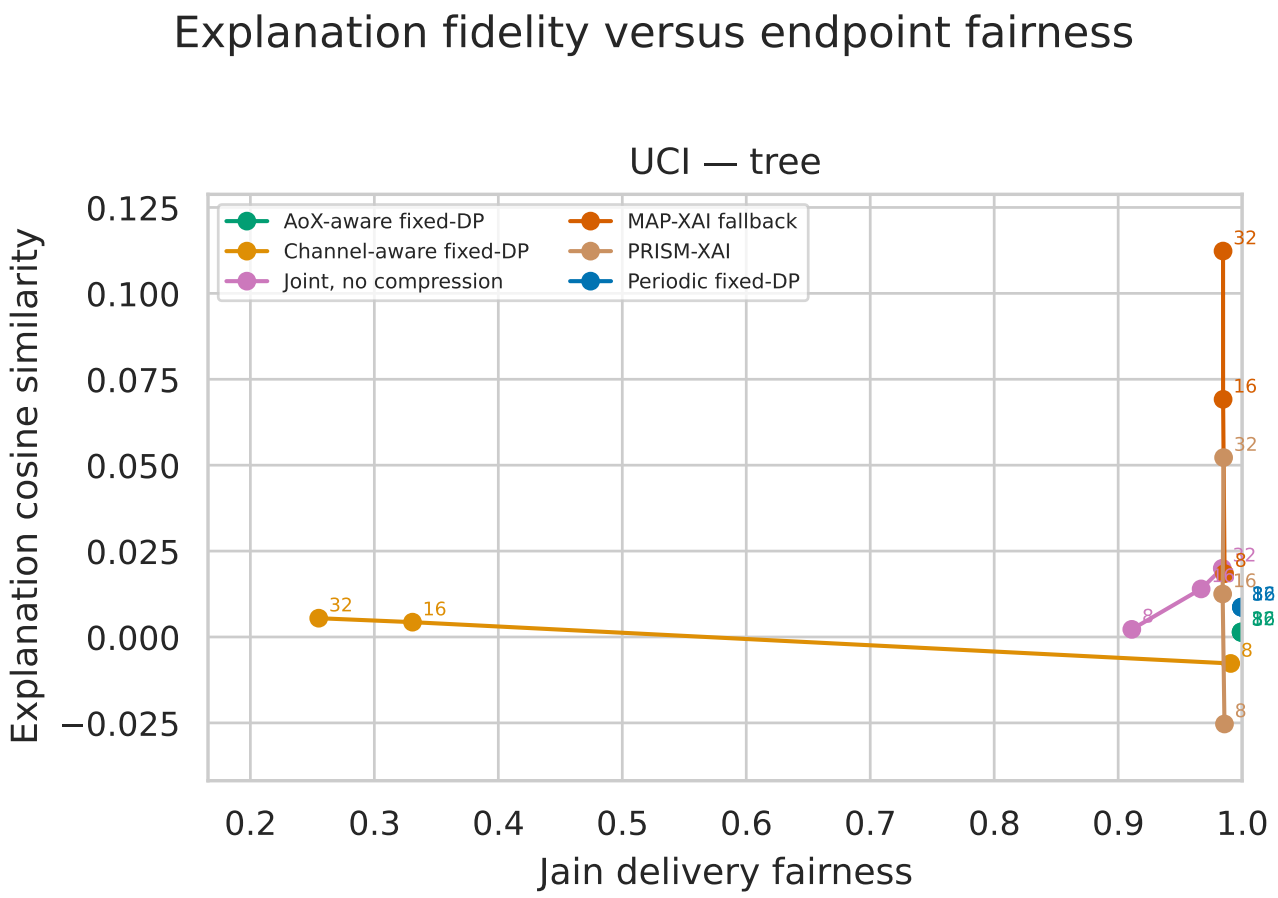


Fig. 166. Residual-score $M = 4$, UCI: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

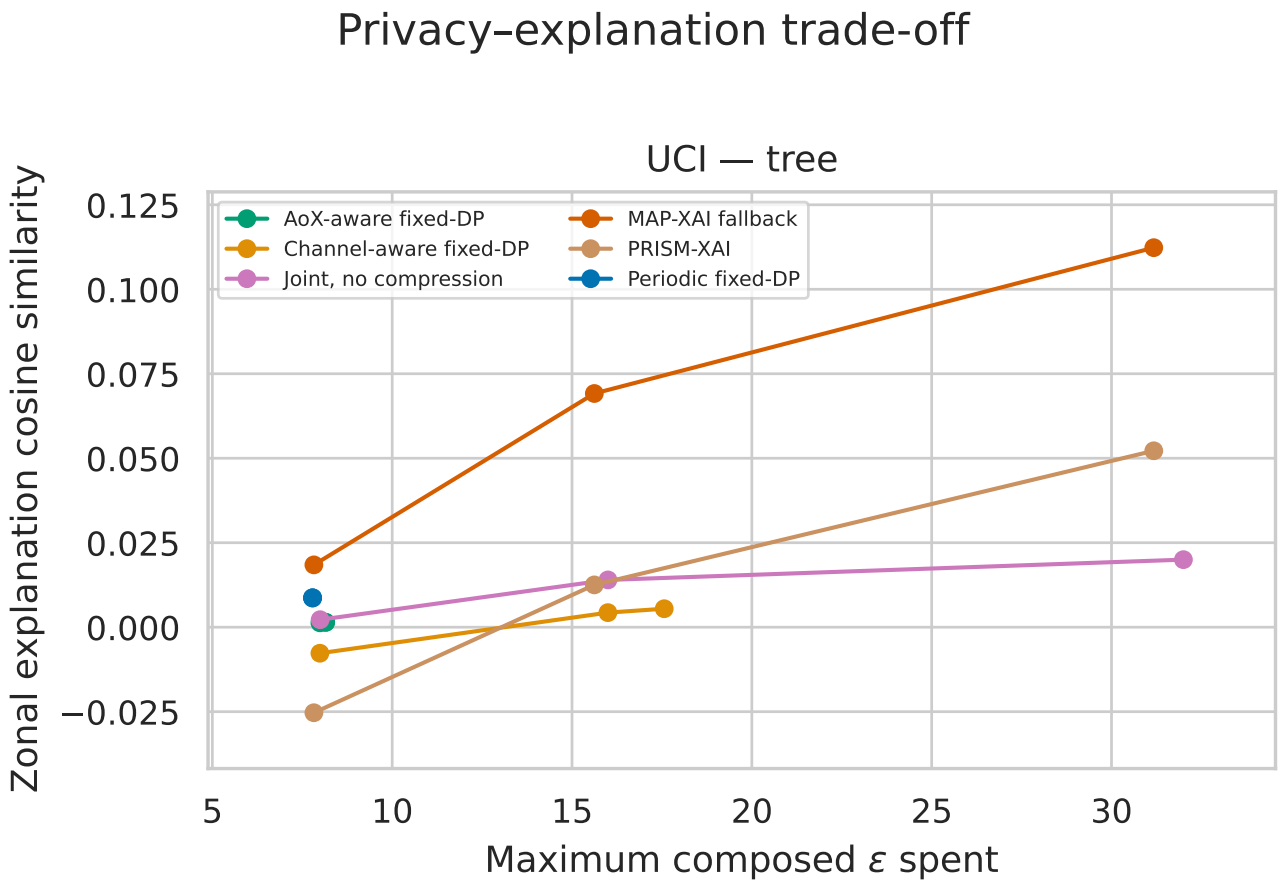


Fig. 167. Residual-score $M = 4$, UCI: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

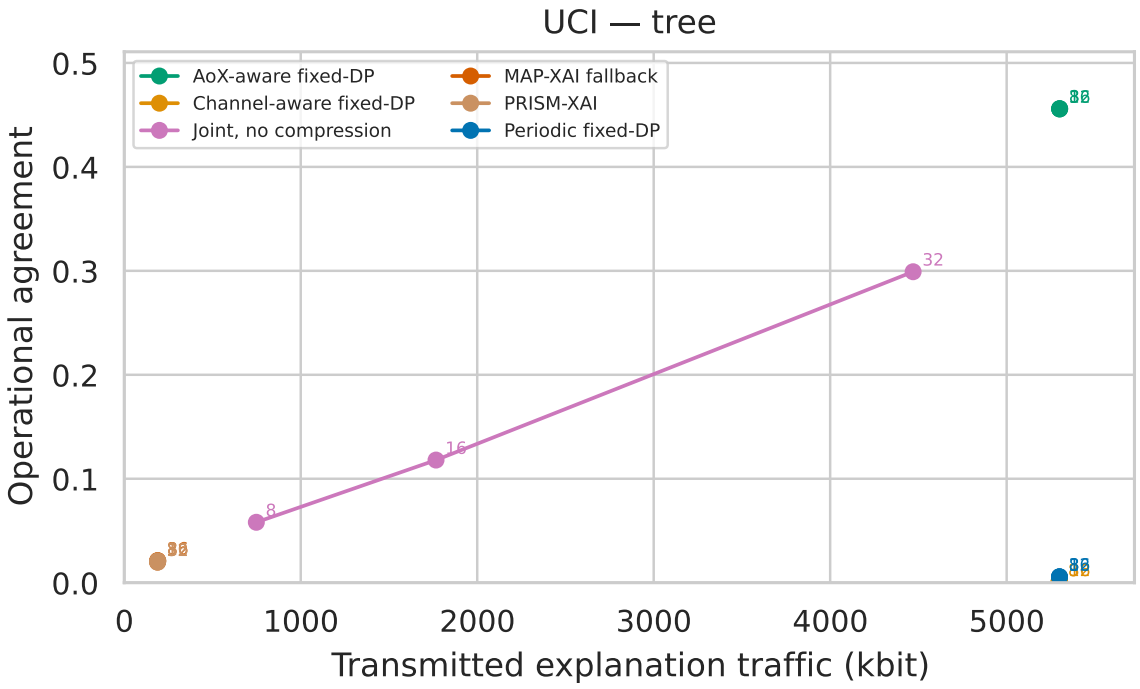


Fig. 168. Residual-score $M = 4$, UCI: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

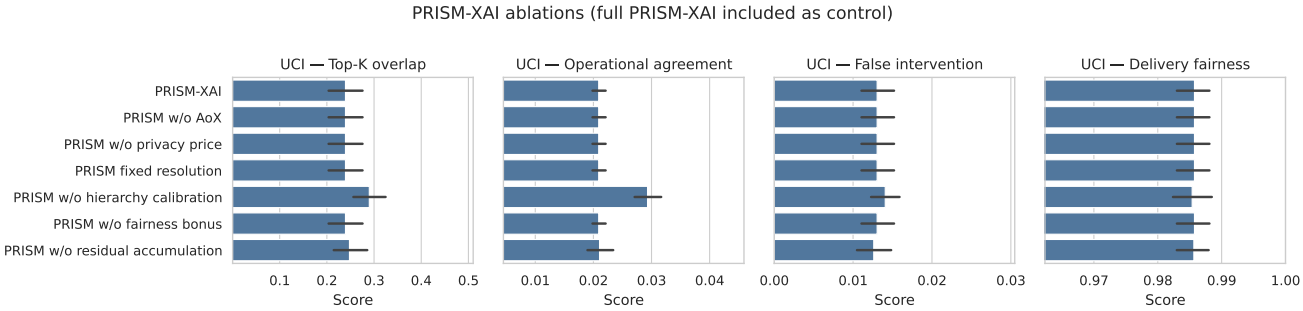


Fig. 169. Residual-score $M = 4$, UCI: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

H. Public-score $M = 8$: combined datasets

This is the primary configuration. Across 120 paired runs, PRISM-XAI transmits 207.31 kbit versus 243.71 kbit for MAP-XAI-only, a reduction of 36.40 ± 3.32 kbit or 15.72%. Coverage changes by -0.010 percentage points, critical recall by -0.083 points, and recorded maximum privacy expenditure is unchanged. These are the conditions that support the paper's matched component claim.

The figure family shows why the public score matters. The regime traces schedule fewer packets without introducing a visible collapse in the paired service measures; the fairness panels keep the proposed method in the high-Jain region. The Pareto plot separates the matched MAP-XAI-only contrast from external fixed-DP policies that deliver different service. The adaptive-mode ablation accounts for a 35.33 ± 5.63 kbit saving (15.43%), whereas fidelity, recall, and intervention-precision changes are not detected after BH correction. The combined Top-K parametric result is positive, but its adjusted Wilcoxon result is not, so fidelity is interpreted by dataset below.

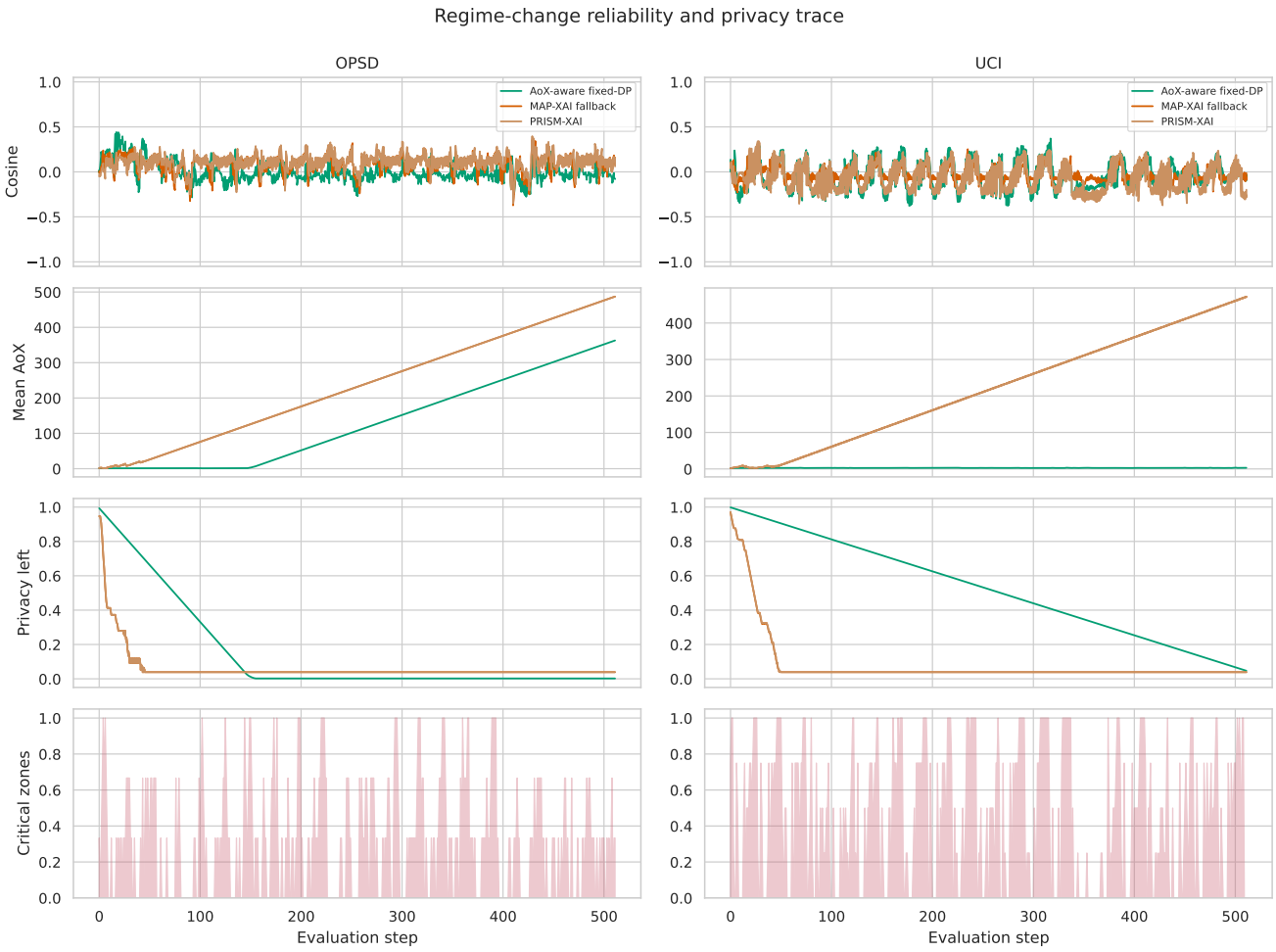


Fig. 170. Public-score $M = 8$, combined datasets: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

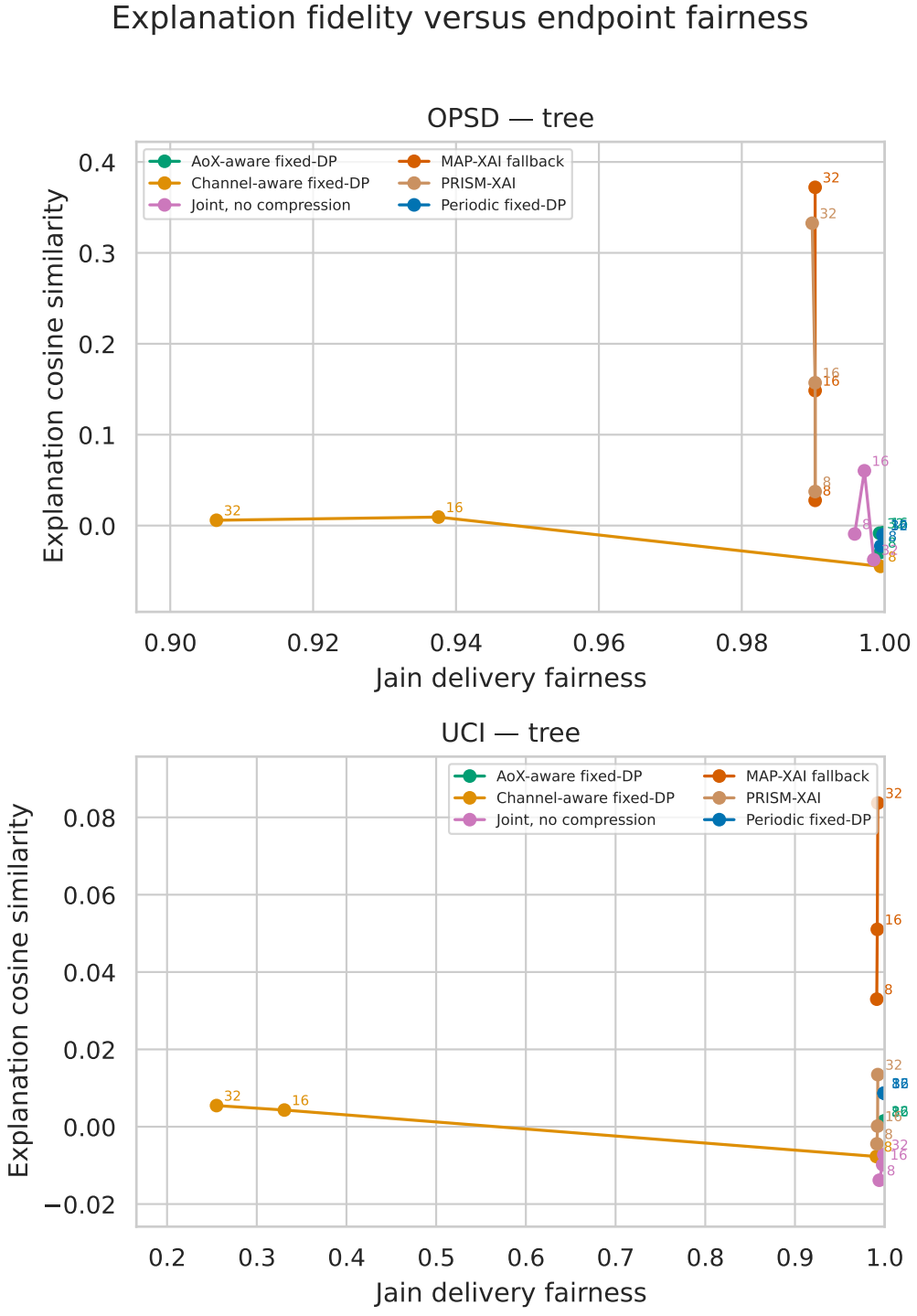


Fig. 171. Public-score $M = 8$, combined datasets: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

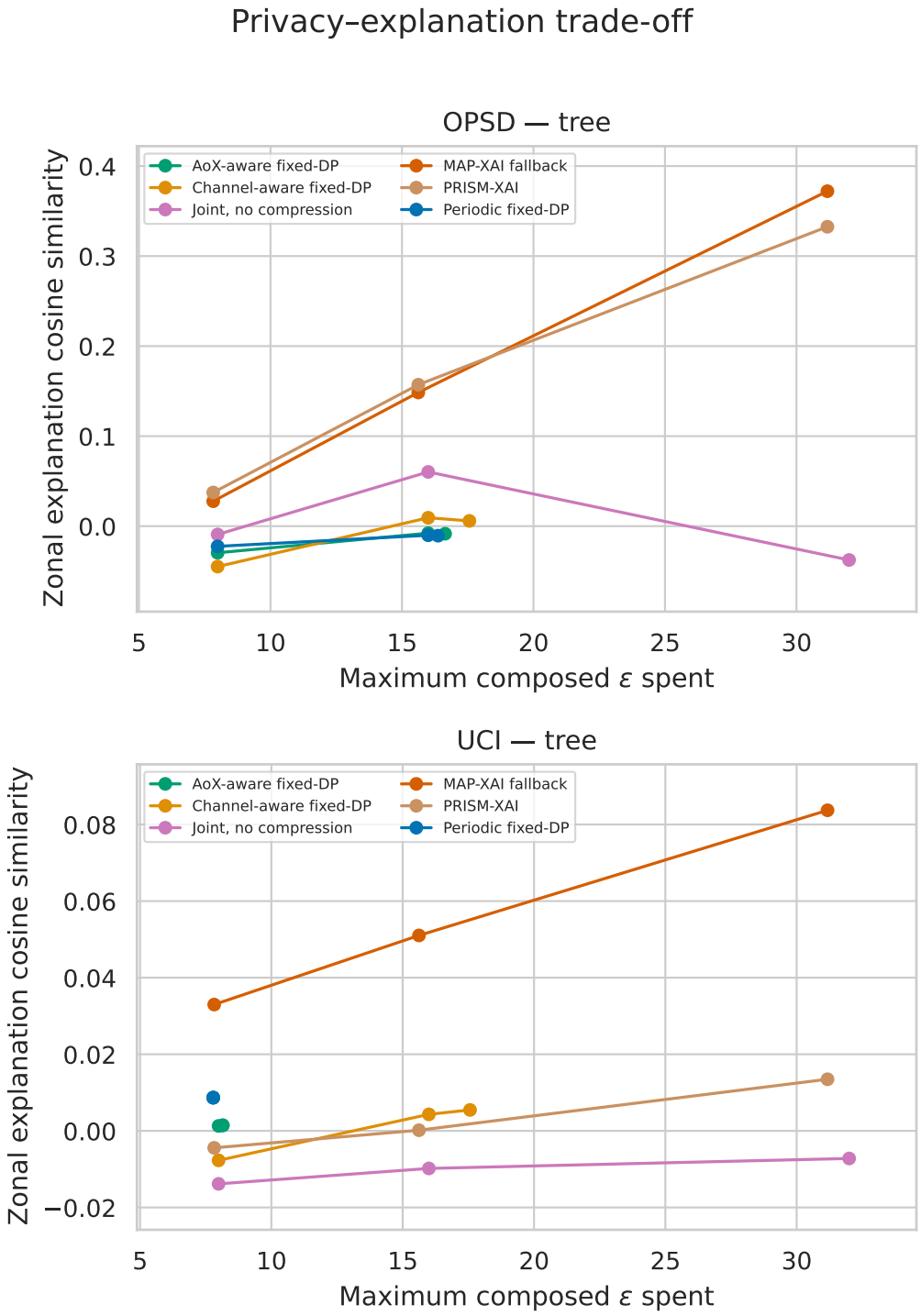


Fig. 172. Public-score $M = 8$, combined datasets: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

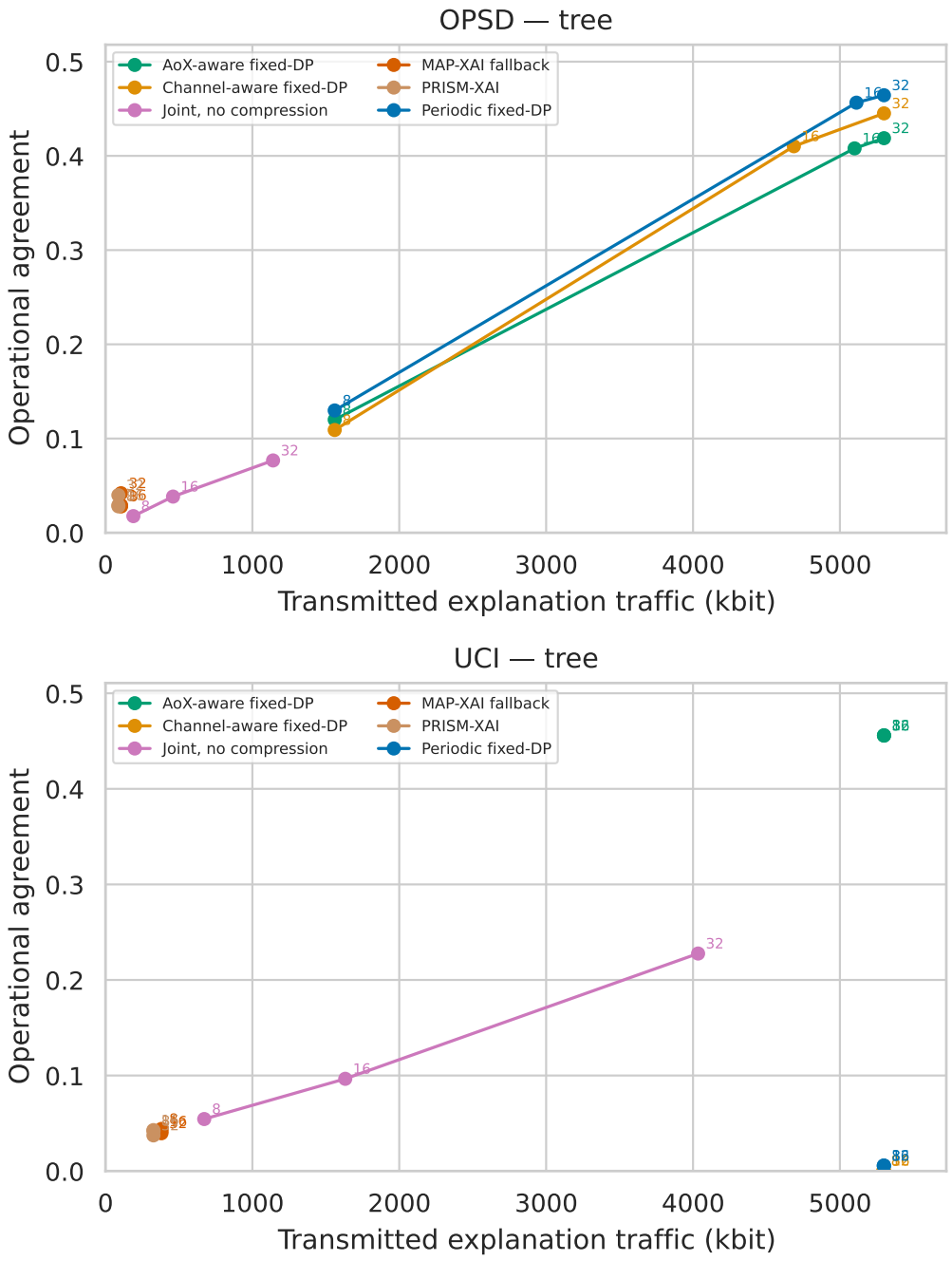


Fig. 173. Public-score $M = 8$, combined datasets: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

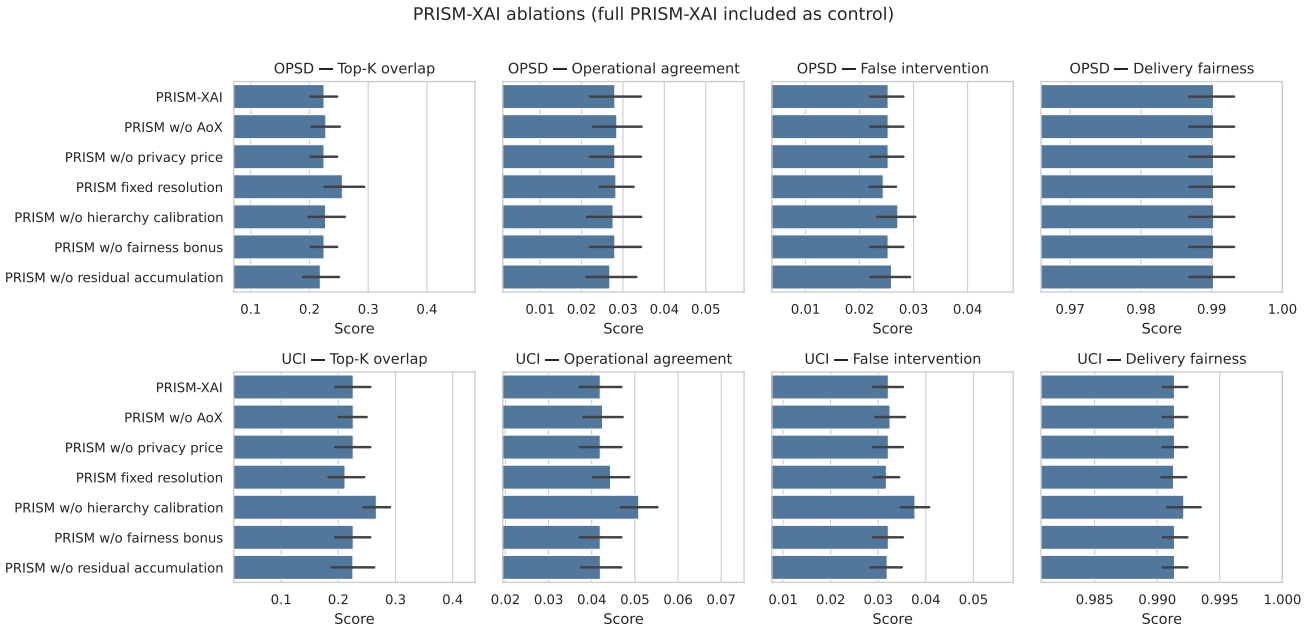


Fig. 174. Public-score $M = 8$, combined datasets: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

I. Public-score $M = 8$: OPSD

On OPSD, the primary PRISM configuration averages 87.98 kbit compared with 106.15 kbit for MAP-XAI-only in the dataset-level summary. Coverage remains about 0.071 and critical recall changes only slightly. This establishes that a meaningful portion of the aggregate traffic reduction occurs even in the smaller nine-endpoint setting.

OPSD also supplies the positive fidelity evidence: Top-K overlap exceeds random by 0.0435 ± 0.0243, with adjusted Wilcoxon $q = 0.0089$. The privacy–fidelity panel nevertheless shows that agreement varies across explainers and budgets, so the result is a measured proxy rather than a universal semantic guarantee. The traces demonstrate sparse public-state scheduling, and the ablations show that the gain is associated with adaptive public-score mode selection rather than a lower release cap.

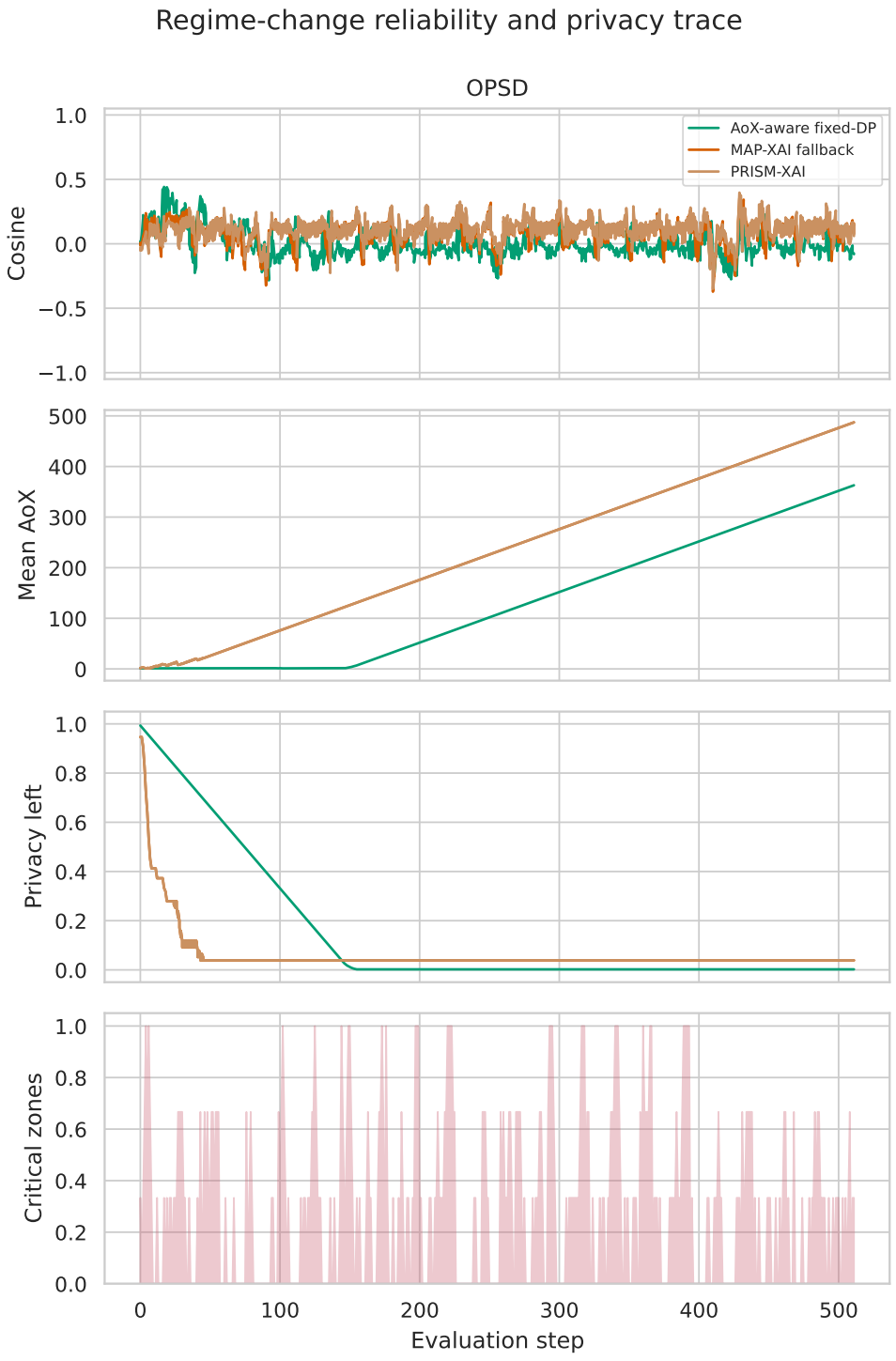


Fig. 175. Public-score $M = 8$, OPSD: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

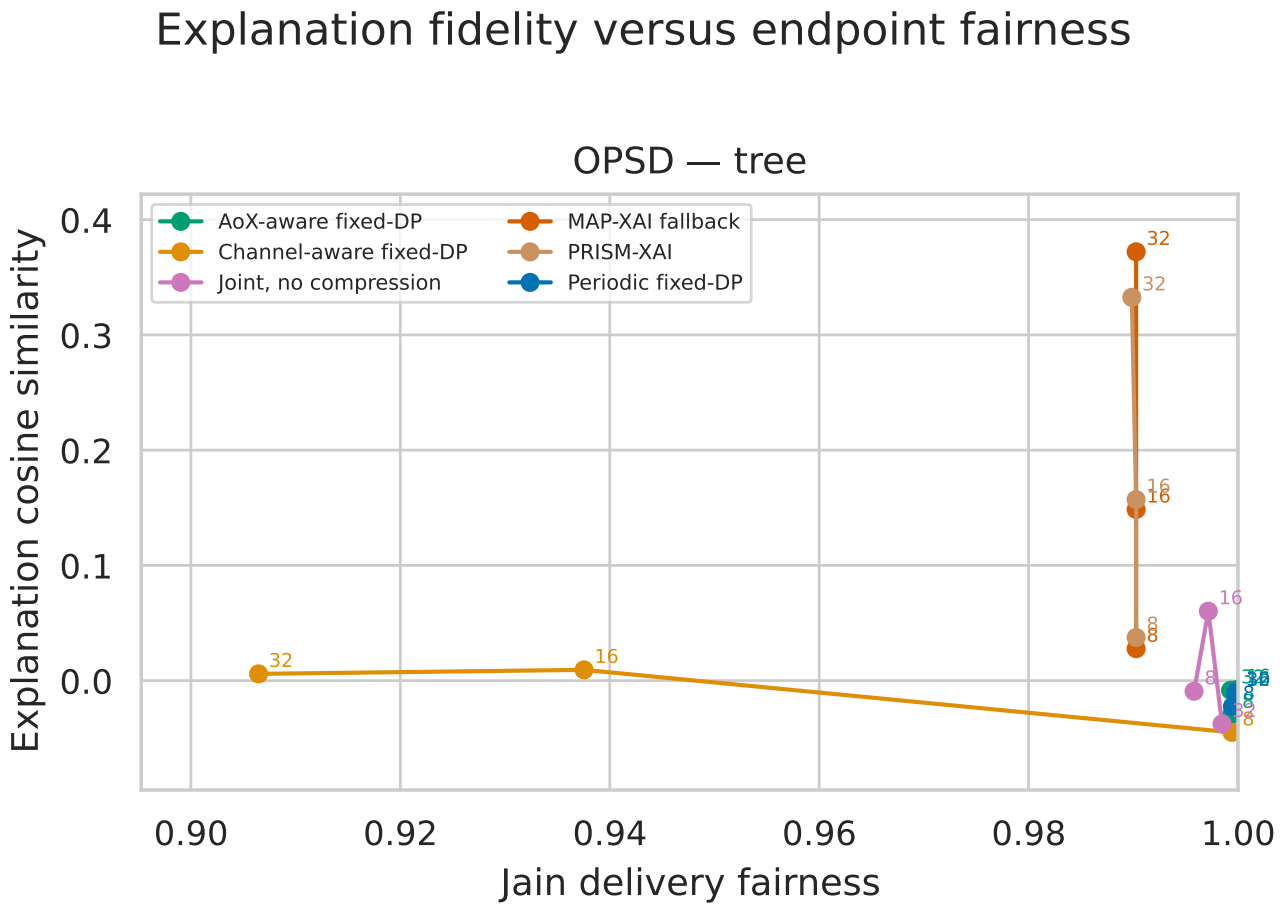


Fig. 176. Public-score $M = 8$, OPSD: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

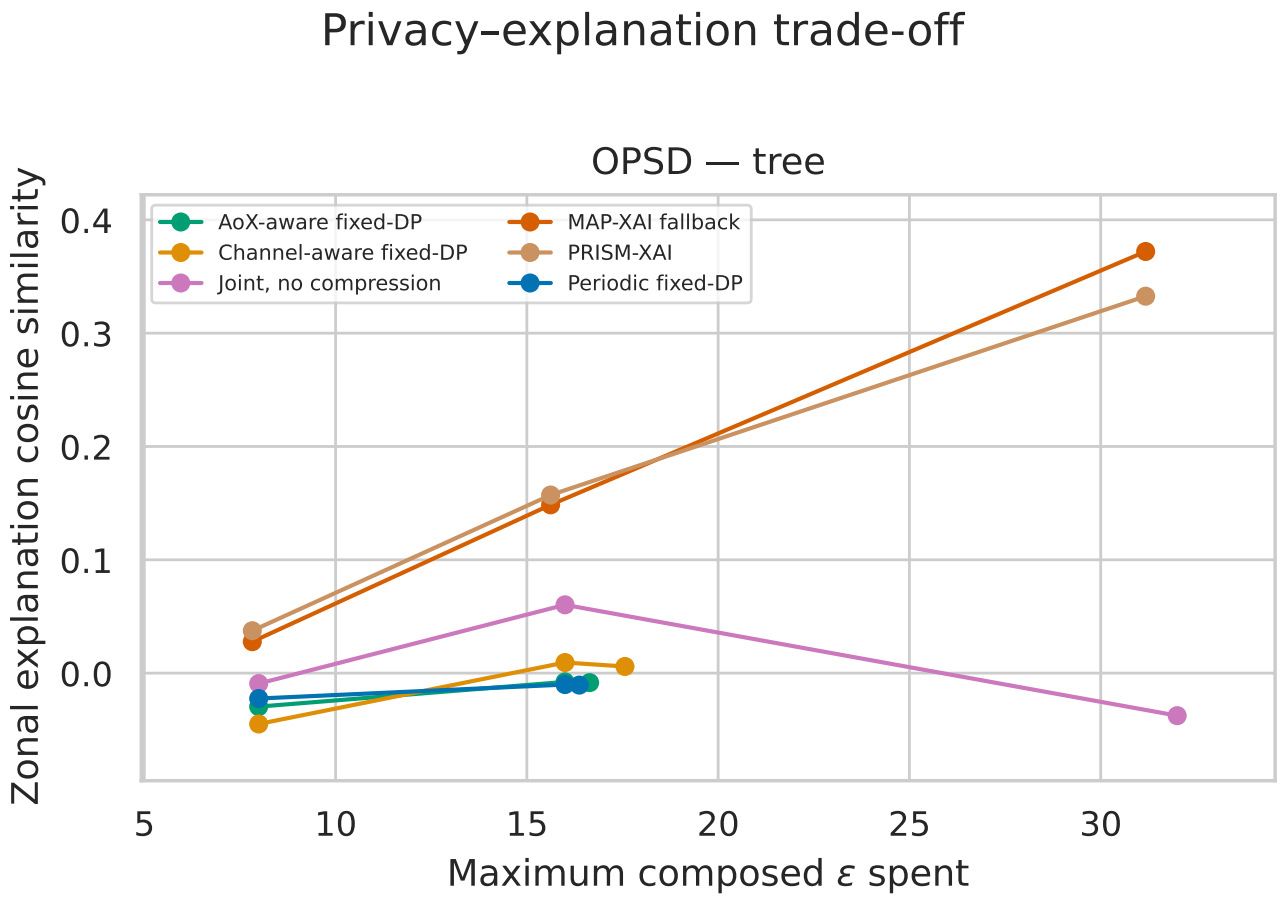


Fig. 177. Public-score $M = 8$, OPSD: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

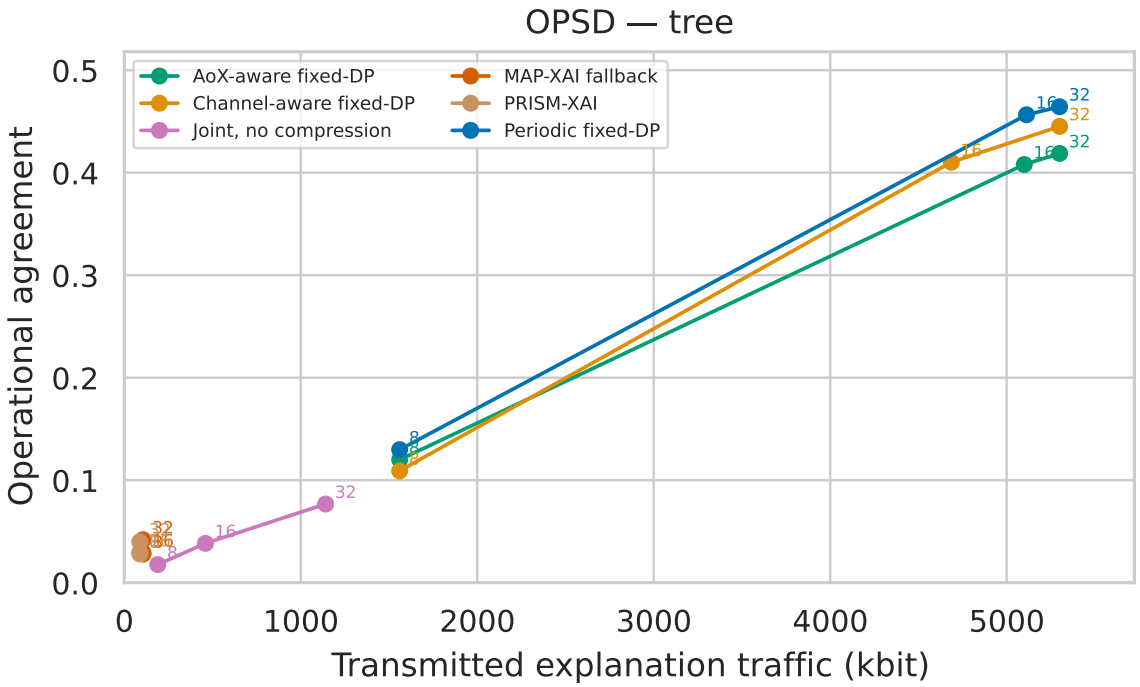


Fig. 178. Public-score $M = 8$, OPSD: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

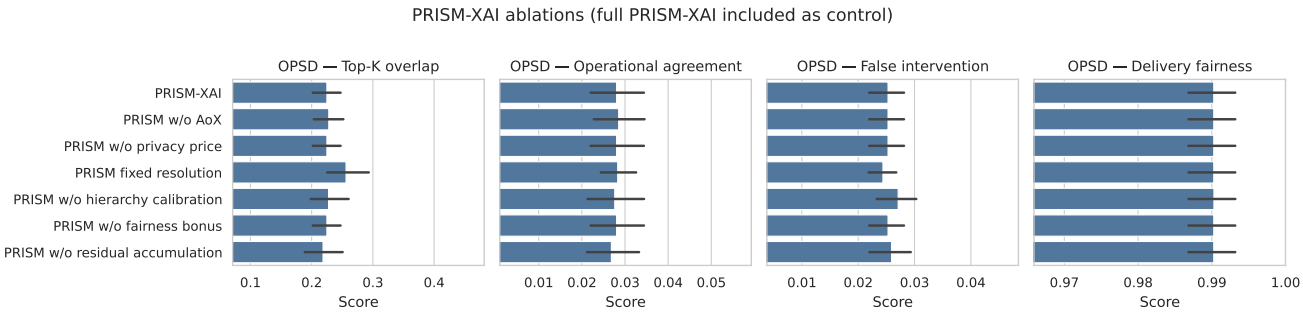


Fig. 179. Public-score $M = 8$, OPSD: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

J. Public-score $M = 8$: UCI

On UCI, PRISM-XAI averages 326.63 kbit compared with 381.26 kbit for MAP-XAI-only. Because UCI contains 32 endpoints, its absolute traffic is higher than OPSD, and it contributes the larger absolute share of the combined reduction. Coverage remains near 0.085–0.090 and critical recall near 0.041, so the comparison remains within the declared matched region.

The UCI fidelity result remains the principal caution. Top-K overlap differs from random by -0.0050 ± 0.0216, with no adjusted statistical support. The traffic and fairness figures therefore establish resource-allocation behavior, not explanation usefulness. The Pareto and ablation panels show that public-score scheduling can remove traffic while preserving the tested service proxies, but a task-grounded or operator study is still required before treating these packets as actionable explanations.

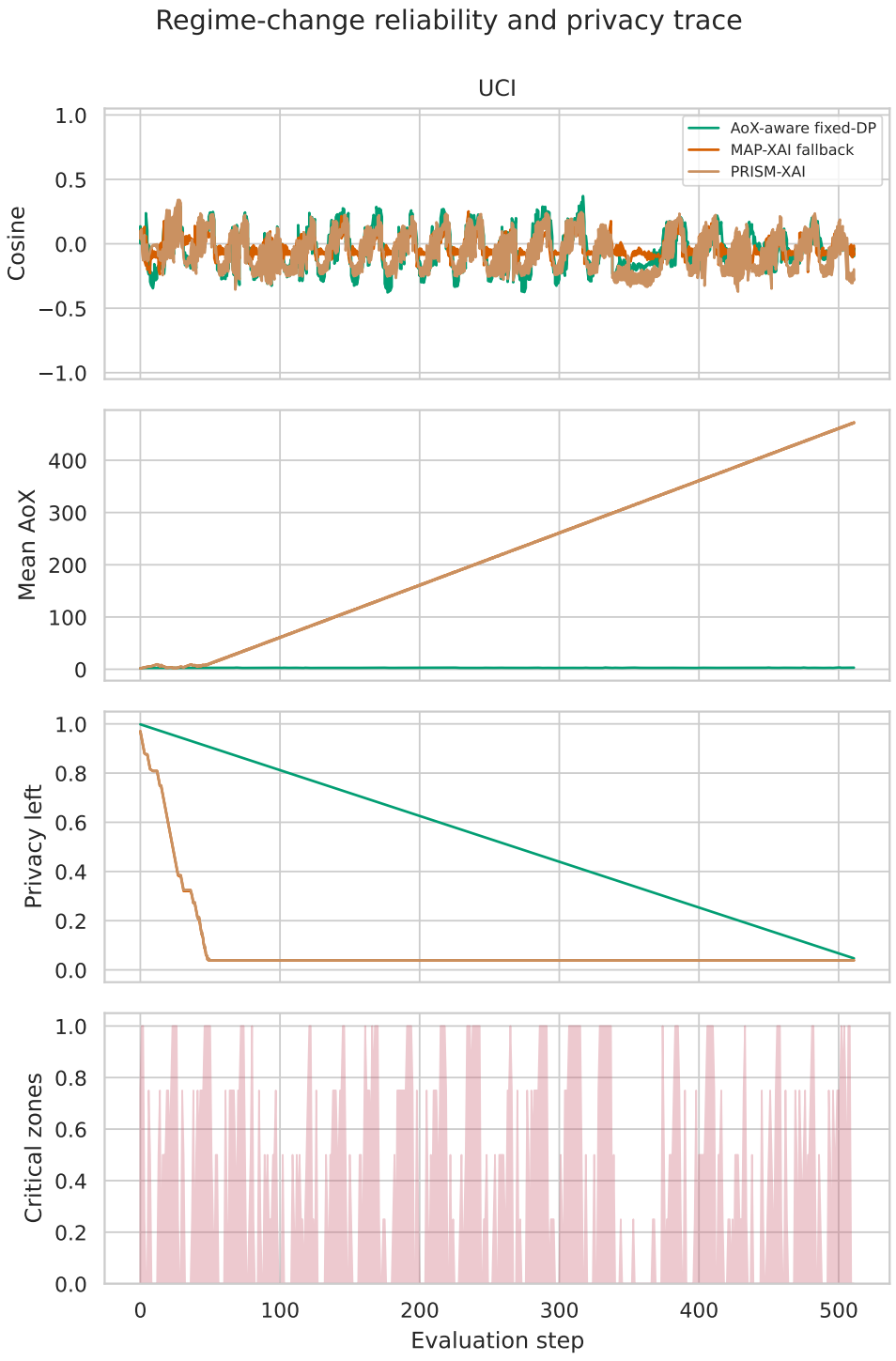


Fig. 180. Public-score $M = 8$, UCI: representative regime trace. The interpretation and claim boundary are given in the preceding subsection discussion.

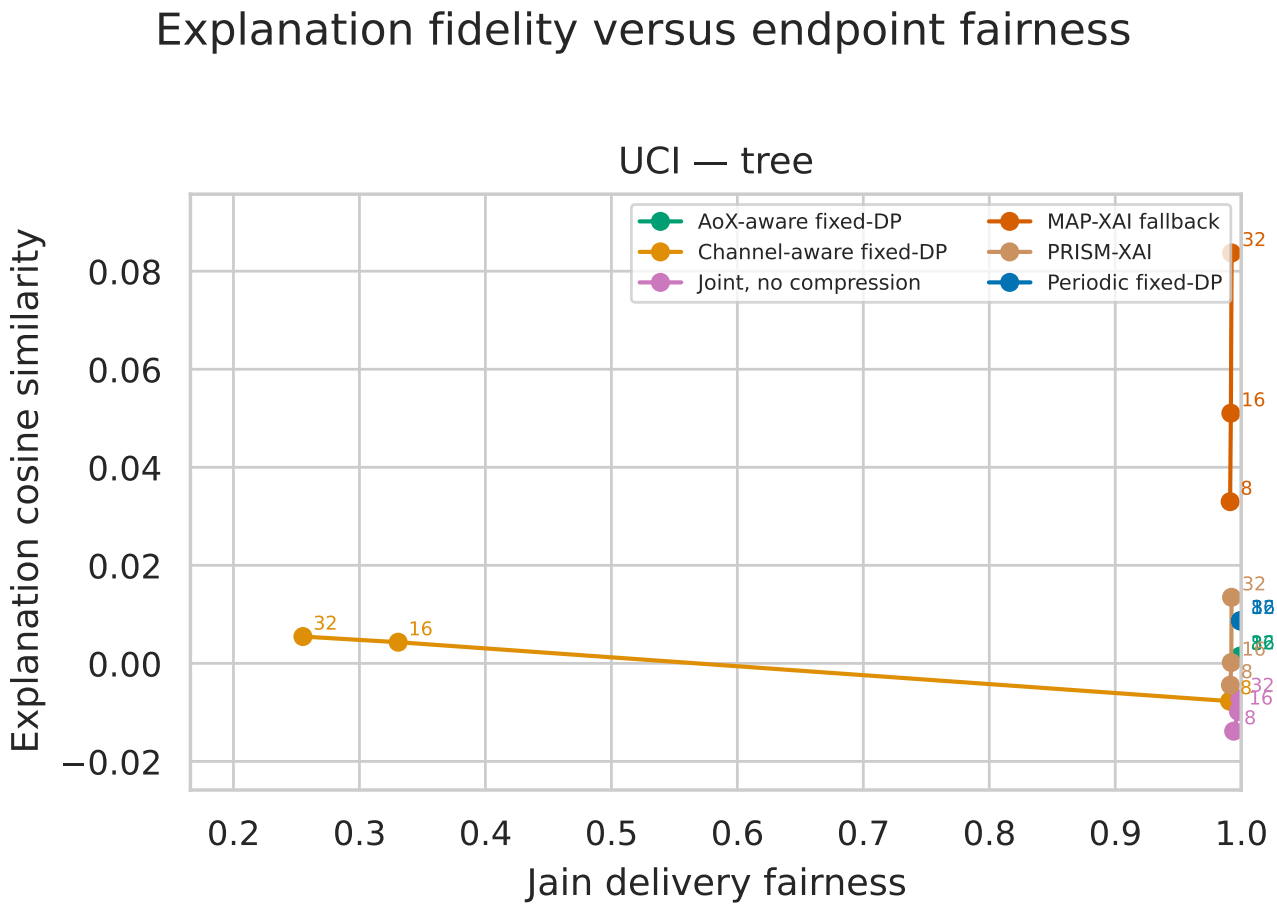


Fig. 181. Public-score $M = 8$, UCI: fairness and reliability. The interpretation and claim boundary are given in the preceding subsection discussion.

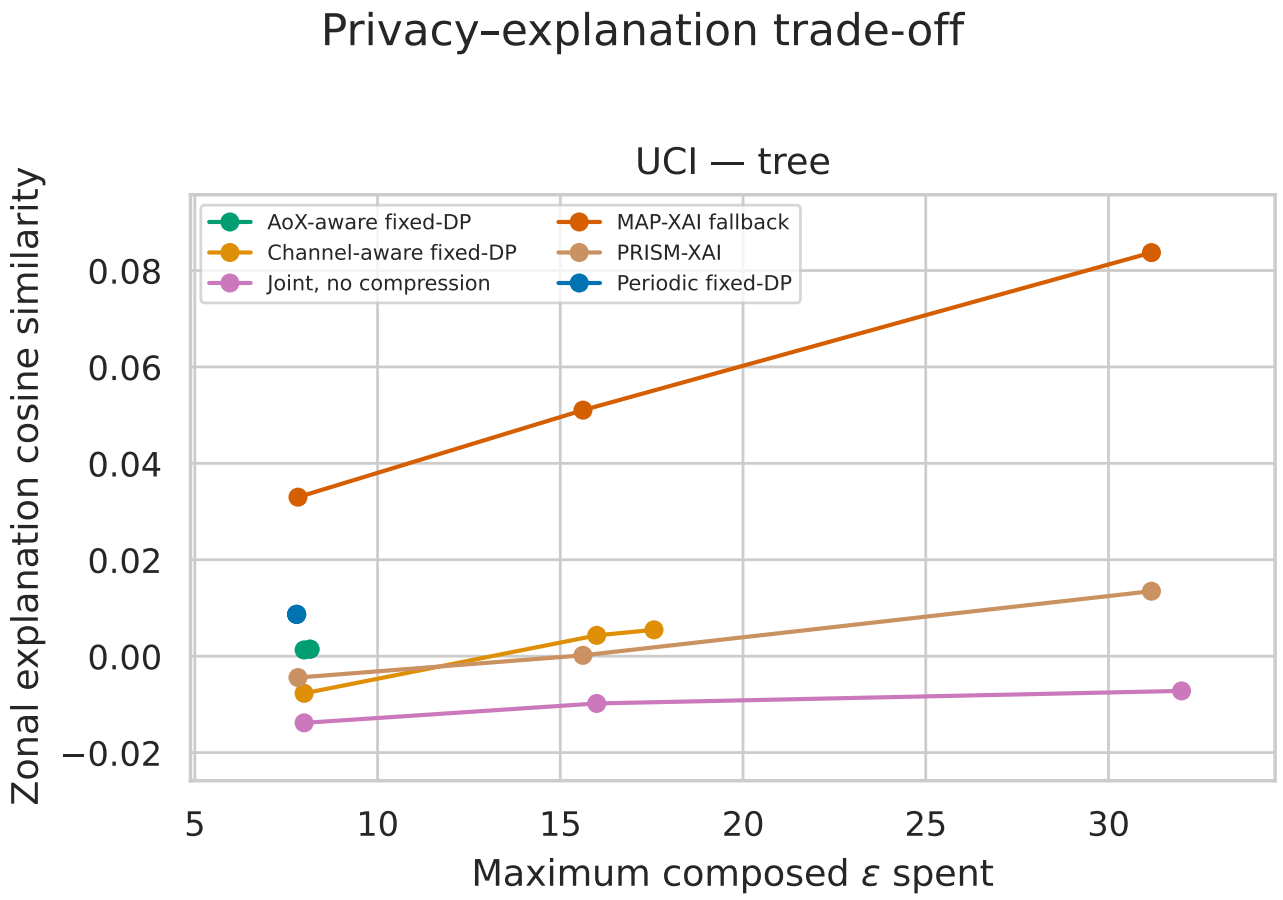


Fig. 182. Public-score $M = 8$, UCI: privacy and fidelity. The interpretation and claim boundary are given in the preceding subsection discussion.

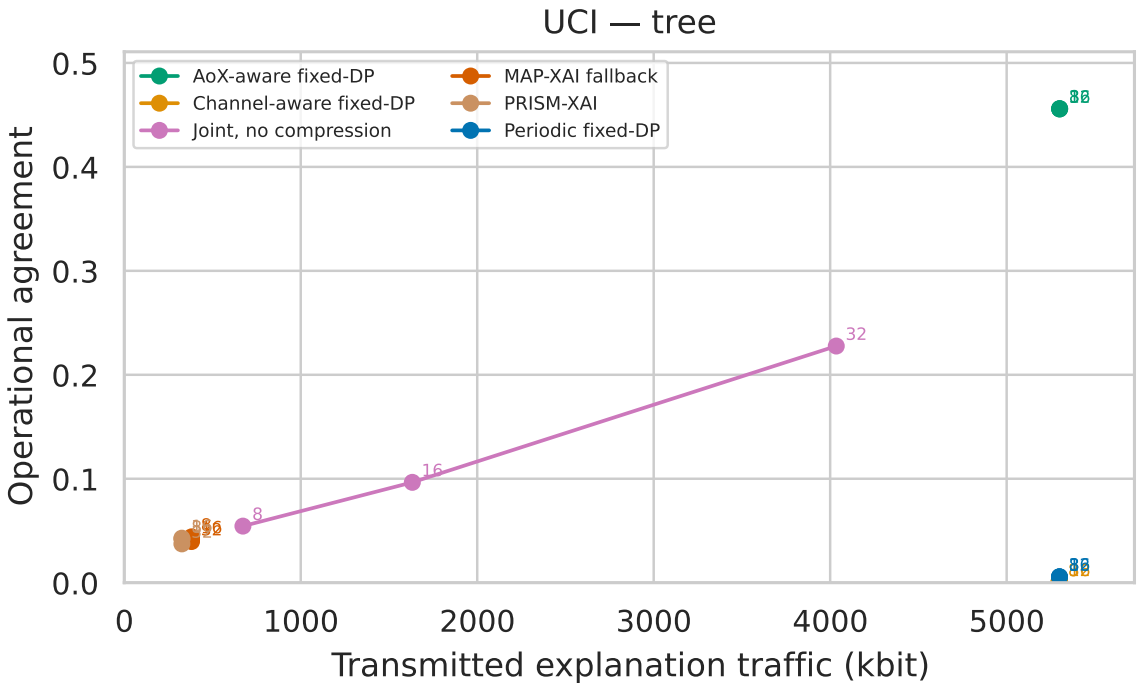


Fig. 183. Public-score $M = 8$, UCI: traffic–fidelity Pareto analysis. The interpretation and claim boundary are given in the preceding subsection discussion.

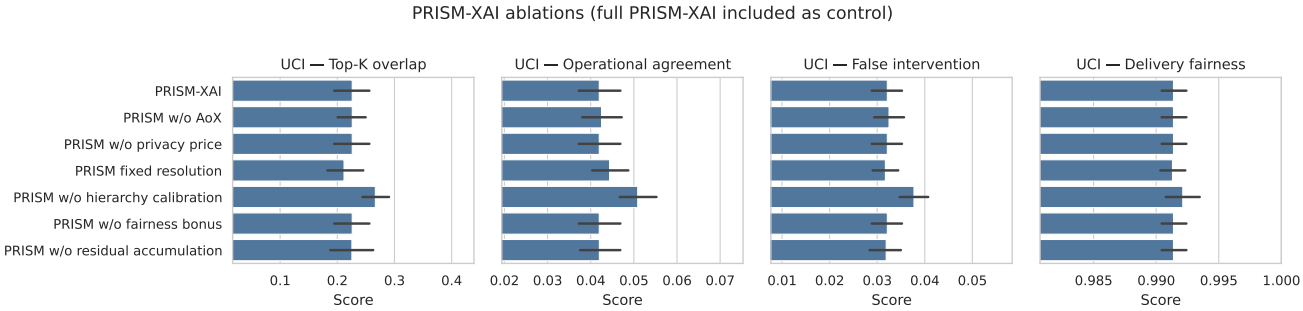


Fig. 184. Public-score $M = 8$, UCI: ablation results. The interpretation and claim boundary are given in the preceding subsection discussion.

IX. Interpretation of the Extended Evidence

The complete results support three focused conclusions. First, public-score PRISM-XAI reduces traffic relative to the MAP-XAI-only component while meeting the declared matched-service and recorded-privacy rule. Second, the stronger reduction is specific to the public-score formulation: the residual-score configuration yields only a small saving, and the stricter $M = 4$ cap alone does not produce a detectable traffic difference. Third, explanation fidelity is dataset dependent. OPSD exceeds the random Top-K reference, whereas UCI does not. The communication result therefore does not imply universal explanation usefulness.

All 720 runs in each matrix pass the recorded validation checks, with no duplicate experimental keys and no observed accountant overspend. The study remains a scheduler-level evaluation with synthetic zones and a reproducible radio trace. A field deployment should predefine required service and fidelity levels, evaluate real interference and retransmissions, and add a transcript-level mechanism when communication metadata is considered private.